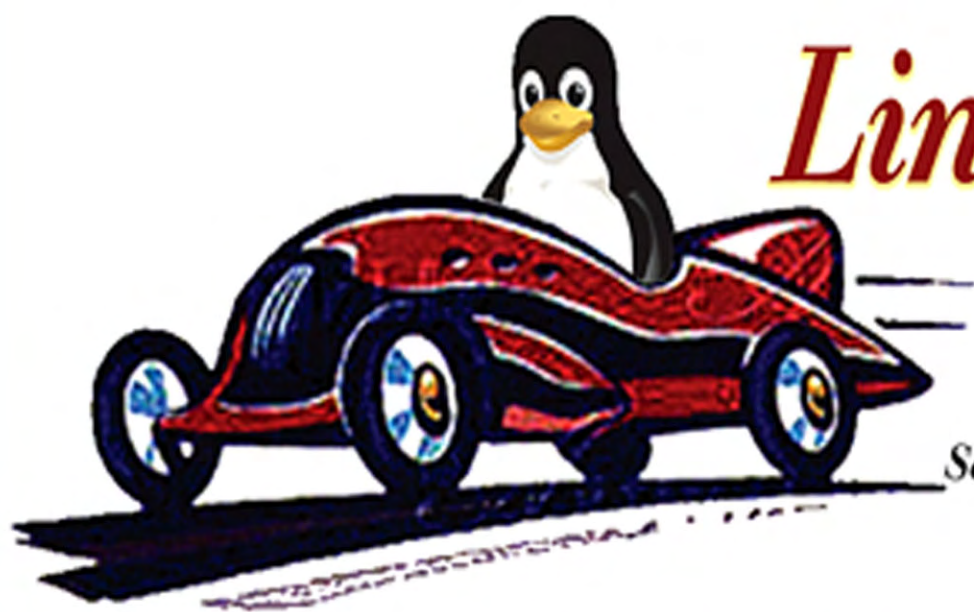


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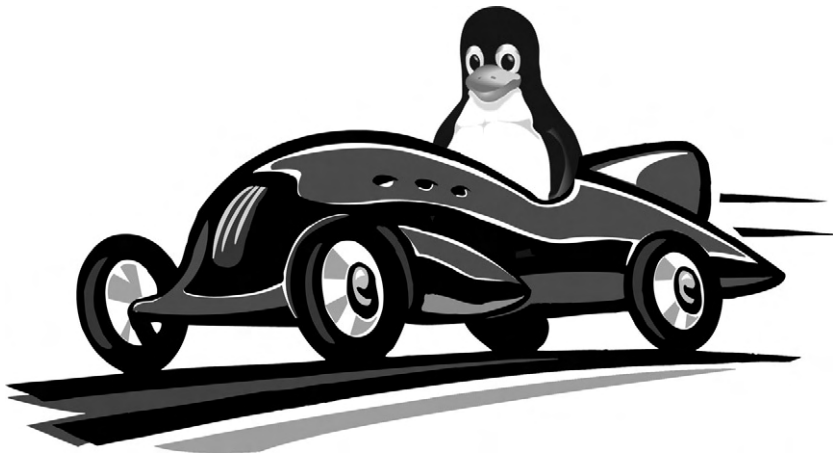
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Publisher: Denise E. M. Penrose
Publishing Services Manager: George Morrison
Senior Project Manager: Brandy Lilly
Editorial Assistant: Kimberlee Honjo
Cover Design: Alisa Andreola and Hannus Design
Composition: diacriTech
Technical Illustration: diacriTech
Copyeditor: Denise Moore
Proofreader: Katherine Antonsen
Indexer: Steve Rath
Interior Printer: The Maple-Vail Book Manufacturing Group, Inc.
Cover Printer: Phoenix Color

Morgan Kaufmann Publishers is an imprint of Elsevier.
500 Sansome Street, Suite 400, San Francisco, CA 94111

This book is printed on acid-free paper.

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Library of Congress Cataloging-in-Publication Data
Application Submitted

ISBN 13: 978-0-12-088421-6

ISBN 10: 0-12-088421-6

For information on all Morgan Kaufmann publications,
visit our Web site at www.mkp.com or www.books.elsevier.com

Printed in the United States of America
06 07 08 09 10 5 4 3 2 1

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Foreword

The MIPS architecture was born in the early 1980s from the work done by John Hennessy and his students at Stanford University. They were exploring the architectural concept of RISC (Reduced Instruction Set Computing), which theorized that relatively simple instructions, combined with excellent compilers and hardware that used pipelining to execute the instructions, could produce a faster processor with less die area. The concept was so successful that MIPS Computer Systems was formed in 1984 to commercialize the MIPS architecture.

Over the course of the next 14 years, the MIPS architecture evolved in a number of ways and its implementations were used very successfully in workstation and server systems. Over that time, the architecture and its implementations were enhanced to support 64-bit addressing and operations, support for complex memory-protected operating systems such as UNIX, and very high performance floating point. Also in that period, MIPS Computer Systems was acquired by Silicon Graphics and MIPS processors became the standard for Silicon Graphics computer systems. With 64-bit processors, high-performance floating point, and the Silicon Graphics heritage, MIPS processors became the solution of choice in high-volume gaming consoles.

In 1998, MIPS Technologies emerged from Silicon Graphics as a stand-alone company focused entirely on intellectual property for embedded markets. As a result, the pace of architecture development has increased to address the unique needs of these markets: high-performance computation, code compression, geometry processing for graphics, security, signal processing, and multi-threading. Each architecture development has been matched by processor core implementations of the architecture, making MIPS-based processors the standard for high-performance, low-power applications.

The MIPS legacy in complex systems such as workstations and servers directly benefits today's embedded systems, which have, themselves, become very complex. A typical embedded system is composed of multiple processing elements, high-performance memory, and one or more operating systems.

When compared with other embedded architectures, which are just now learning what is required to build a complex system, the MIPS architecture provides a proven base on which to implement such systems.

In many ways, the first edition of *See MIPS Run* was a ground-breaking book on the MIPS architecture and its implementations. While other books covered similar material, *See MIPS Run* focused on what the programmer needed to understand of the architecture and the software environment in order to effectively program a MIPS chip.

Increasing complexity of embedded systems has been matched by enhancements to the MIPS architecture to address the needs of such systems. The second edition of this book is required reading for any current developer of MIPS-based embedded systems. It adds significant new material, including the architectural standardization of the MIPS32 and MIPS64 architectures, brand new application-specific extensions such as multithreading, and a very nice treatment of the implementation of the popular Linux operating system on the MIPS architecture. Short of the MIPS architecture specifications, the second edition of *See MIPS Run* is the most current description of the state of the art of the architecture and is, bar none, the most readable.

I hope that you will find this as worthwhile and as entertaining to read as I did.

Michael Uhler,
Chief Technology Officer, MIPS Technologies, Inc.
Mountain View, CA
May 2006

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Preface

This book is about MIPS, the cult hit from the mid-1980s' crop of RISC CPU designs. These days MIPS is not the highest-volume 32-bit architecture, but it is in a comfortable second place. Where it wins, hands down, is its range of applications. A piece of equipment built around a MIPS CPU might have cost you \$35 for a wireless router or hundreds of thousands of dollars for an SGI supercomputer (though with SGI's insolvency, those have now reached the end of the line). Between those extremes are Sony and Nintendo games machines, many Cisco routers, TV set-top boxes, laser printers, and so on.

The first edition of this book has sold close to 10,000 English copies over the years and has been translated into Chinese. I'm pleased and surprised; I didn't know there were so many MIPS programmers out there.

This second edition is *See MIPS Run... Linux*. The first edition struggled to motivate some features of the MIPS architecture, because they don't make sense unless you can see how they help out inside an OS kernel. But now a lot of you have some sense of how Linux works, and I can quote its source code; more importantly, I can refer to it knowing that those of you who get interested can read the source code and find out how it's *really* done.

So this is a book about the MIPS architecture, but the last three chapters stroll through the Linux kernel and application-programming system to cast light on what those weird features do. I hope Linux experts will forgive my relative ignorance of Linux details, but the chance to go for a description of a real OS running on a real architecture was too good to pass up.

MIPS is a RISC: a useful acronym, well applied to the common features of a number of computer architectures invented in the 1980s, to realize efficient pipelined implementation. The acronym CISC is vaguer. I'll use it in a narrow sense, for the kind of features found in x86 and other pre-1982 architectures, designed with microcoded implementations in mind.

Some of you may be up in arms: He's confusing implementation with architecture! But while computer architecture is supposed to be a contract with the

programmer about what programs will run correctly, it's also an engineering design in its own right. A computer architecture is designed to make for good CPUs. As chip design becomes more sophisticated, the trade-offs change.

This book is for programmers, and that's the test we've used to decide what gets included—if a programmer might see it, or is likely to be interested, it's here. That means we don't get to discuss, for example, the strange system interfaces with which MIPS has tortured two generations of hardware design engineers. And your operating system may hide many of the details we talk about here; there is many an excellent programmer who thinks that C is quite low level enough, portability a blessing, and detailed knowledge of the architecture irrelevant. But sometimes you do need to get down to the nuts and bolts—and human beings are born curious as to how bits of the world work.

A result of this orientation is that we'll tend to be rather informal when describing things that may not be familiar to a software engineer—particularly the inner workings of the CPU—but we'll get much more terse and technical when we're dealing with the stuff programmers have met before, such as registers, instructions, and how data is stored in memory.

We'll assume some familiarity and comfort with the C language. Much of the reference material in the book uses C fragments as a way of compressing operation descriptions, particularly in the chapters on the details of the instruction set and assembly language.

Some parts of the book are targeted at readers who've seen some assembly language: the ingenuity and peculiarity of the MIPS architecture shows up best from that viewpoint. But if assembly is a closed book to you, that's probably not a disaster.

This book aims to tell you everything you need to know about programming generic MIPS CPUs. More precisely, it describes the architecture as it's defined by MIPS Technologies' MIPS32 and MIPS64—specifically, the second release of those specifications from 2003. We'll shorten that to “MIPS32/64.” But this is not just a reference manual: To keep an architecture in your head means coming to understand it in the round. I also hope the book will interest students of programming (at college or enrolled in the school of life) who want to understand a modern CPU architecture all the way through.

If you plan to read this book straight through from front to back, you will expect to find a progression from overview to detail, and you won't be disappointed. But you'll also find some progression through history; the first time we talk about a concept we'll usually focus on its first version. Hennessy and Patterson call this “learning through evolution,” and what's good enough for them is certainly good enough for me.

We start in Chapter 1 with some history and background, and set MIPS in context by discussing the technological concerns and ideas that were uppermost in the minds of its inventors. Then in Chapter 2 we discuss the characteristics of the MIPS machine language that follow from their approach.

To help you see the big picture, we leave out the details of processor control until Chapter 3, which introduces the ugly but eminently practical system that allows MIPS CPUs to deal with their caches, exceptions and startup, and memory management. Those last three topics, respectively, become the subjects of Chapters 4 through 6.

The MIPS architecture has been careful to separate out the part of the instruction set that deals with floating-point numbers. That separation allows MIPS CPUs to be built with various levels of floating-point support, from none at all through partial implementations to somewhere near the state of the art. So we have also separated out the floating-point functions, and we keep them back until Chapter 7.

Up to this point, the chapters follow a reasonable sequence for getting to know MIPS. The following chapters change gear and are more like reference manuals or example-based tutorials.

In Chapter 8, we go through the whole machine instruction set; the intention is to be precise but much more terse than the standard MIPS reference works—we cover in 10 pages what takes a hundred in other sources.¹ Chapter 9 is a brief introduction to reading and writing assembly, and falls far short of an assembly programming manual.

Chapter 10 is a checklist with helpful hints for those of you who have to port software between another CPU and a MIPS CPU. The longest section tackles the troublesome problem of endianness in CPUs, software, and systems.

Chapter 11 is a bare-bones summary of the software conventions (register use, argument passing, etc.) necessary to produce interworking software with different toolkits. Chapter 12 introduces the debug and profiling features standardized for MIPS CPUs.

Then we're on to seeing how MIPS runs GNU/Linux. We describe relationship between the Linux kernel and a computer architecture in Chapter 13; then Chapters 14 and 15 dig down into some of the detail as to how the MIPS architecture does what the Linux kernel needs. Chapter 16 gives you a quick look at the dynamic linking magic that makes GNU/Linux applications work.

Appendix A covers the MIPS MT (multithreading) extension, probably the most important addition to the architecture in many years. And Appendix B describes the more important add-ons: MIPS16, the new MIPS DSP extensions, and MDMX.

You will also find at the end of this book a glossary of terms—a good place to look for specialized or unfamiliar usage and acronyms—and a list of books, papers, and online references for further reading.

1. I have taken considerable care in the generation of these tables, and they are mostly right. But if your system depends on it, be sure to cross-check this information. An excellent source of fairly reliable information can be found in the behavior and source code of the GNU tool collection—but I referred to that too, so it's not completely independent.

Style and Limits

Every book reflects its author, so we'd better make a virtue of it.

Since some of you will be students, I wondered whether I should distinguish general use from MIPS use. I decided not to; I aim to be specific except where it costs the reader nothing to be general. I also try to be concrete rather than abstract. I don't worry overmuch about whatever meaning terms like "TLB" have in the wider industry, but I explain them in a MIPS context. Human beings are great generalizers, and this is unlikely to damage your learning much.

It's 20 years since I started working with MIPS CPUs in the fall of 1986. Some of the material in this book goes back as far as 1988, when I started giving training courses on MIPS architecture. In 1993, I gathered them together to make a software manual focused on IDT's R3051 family CPUs. It took quite a lot of extra material to create the first edition, published in 1999.

A lot has happened since 1999. MIPS is now at the very end of its life in servers with SGI but has carved out a significant niche in embedded systems. Linux has emerged as the most-used OS for embedded MIPS, but there's still a lot of diversity in the embedded market. The MIPS specifications have been reorganized around MIPS32 and MIPS64 (which this edition regards as the baseline). This second edition has been in the works for about three years.

The MIPS story continues; if it did not, we'd only be writing this book for historians, and Morgan Kaufmann wouldn't be very interested in publishing it. MIPS developments that weren't announced by the end of 2005 are much too late for this edition.

Conventions

A quick note on the typographical conventions used in this book:

- Type in this font (Minion) is running text.
- Type in this font (Futura) is a sidebar.
- **Type in this font (Courier bold) is used for assembly code and MIPS register names.**
- Type in this font (Courier) is used for C code and hexadecimals.
- *Type in this font (Minion italic, small) is used for hardware signal names.*

Acknowledgments

The themes in this book have followed me through my computing career. Mike Cole got me excited about computing, and I've been trying to emulate his skill in picking out good ideas ever since. In the brief but exciting life of Whitechapel Workstations (1983–1988), many colleagues taught me something about computer architecture and about how to design hardware—Bob Newman and Rick Filipkiewicz probably the most. I also have to thank Whitechapel's salesperson, Dave Gravell, for originally turning me on to MIPS. My fellow engineers during the lifetime of Algorithmics Ltd. (Chris Dearman, Rick Filipkiewicz, Gerald Onions, Nigel Stephens, and Chris Shaw) have to be doubly thanked, both for all I've learned through innumerable discussions, arguments, and designs and for putting up with the book's competition for my time.

Many thanks are due to the reviewers who've read chapters over a long period of time: Phil Bourekas of Integrated Device Technology, Inc.; Thomas Daniel of the LSI Logic Corporation; Mike Murphy of Silicon Graphics, Inc.; and David Nagle of Carnegie Mellon University.

On the second edition: I've known Paul Cobb for a long time, as we both worked around MIPS companies. Paul contributed material updating the historical survey of MIPS CPUs and the programming chapter and cleaning up the references. In all cases, though, I've had a final edit—so any errors are mine.

During the preparation of this edition I've been employed by MIPS Technologies Inc. It's dangerous to pick out some colleagues and not others, but I'll do it anyway.

Ralf Baechle runs the `www.linux-mips.org` site, which coordinates MIPS work on the Linux kernel. He's been very helpful in dispelling some of the illusions I'd formed about Linux: I started off thinking it was like other operating systems . . . (Robert Love's *Linux Kernel Development* book helped too; I warmly recommend it to anyone who wants a more educated guidebook to the kernel). Thanks to MIPS Technologies and my various managers for being flexible about my time, and to many colleagues at MIPS Technologies (too many to name) who have read and commented on drafts.

Todd Bezenek has been my most persistent colleague/reviewer of this edition. Reviewers outside MIPS Technologies did it for their love and respect for the field: notable contributors were Steven Hill (Reality Diluted, Inc.), Jun Sun (DoCoMo USA Labs), Eric DeVolder, and Sophie Wilson.

Denise Penrose is easily the Best Editor Ever. Not many people in Finsbury Park (my home in North London) can say they're just flying to San Francisco for brunch with their publisher.

Last but not least, thanks to Carol O'Brien, who was rash enough to marry me in the middle of this rewrite.

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RISCs and MIPS Architectures

MIPS is the most elegant among the effective RISC architectures; even the competition thought so, as evidenced by the strong MIPS influence to be seen in later architectures like DEC's Alpha and HP's Precision. Elegance by itself doesn't get you far in a competitive marketplace, but MIPS microprocessors have generally managed to be among the most efficient of each generation by remaining among the simplest.

Relative simplicity was a commercial necessity for MIPS Computer Systems Inc., which spun off in 1985 from an academic project to make and market the chips. As a result, the architecture had (and perhaps still has) the largest range of active manufacturers in the industry—working from ASIC cores (MIPS Technologies, Philips) through low-cost CPUs (IDT, AMD/Alchemy) to the only 64-bit CPUs in widespread embedded use (PMC-Sierra, Toshiba, Broadcom).

At the low end the CPU has practically disappeared from sight in the “system on a chip”; at the high end Intrinsity's remarkable processor ran at 2 GHz—a speed unmatched outside the unlimited power/heat budget of contemporary PCs.

ARM gets more headlines, but MIPS sales volumes remain healthy enough: 100 M MIPS CPUs were shipped in 2004 into embedded applications.

The MIPS CPU is one of the RISC CPUs, born out of a particularly fertile period of academic research and development. RISC (Reduced Instruction Set Computing) is an attractive acronym that, like many such, probably obscures reality more than it reveals it. But it does serve as a useful tag for a number of new CPU architectures launched between 1986 and 1989 that owe their remarkable performance to ideas developed a few years earlier in a couple of seminal research projects. Someone commented that “a RISC is any computer architecture defined after 1984”; although meant as a jibe at the industry's use

of the acronym, the comment is also true for a technical reason—no computer defined after 1984 can afford to ignore the RISC pioneers’ work.

One of these pioneering projects was the MIPS project at Stanford. The project name MIPS (named for the key phrase “microcomputer without interlocked pipeline stages”) is also a pun on the familiar unit “millions of instructions per second.” The Stanford group’s work showed that pipelining, although a well-known technique, had been drastically underexploited by earlier architectures and could be much better used, particularly when combined with 1980 silicon design.

1.1 Pipelines

Once upon a time in a small town in the north of England, there was Evie’s fish and chip shop. Inside, each customer got to the head of the queue and asked for his or her meal (usually fried cod, chips, mushy peas,¹ and a cup of tea). Then each customer waited for the plate to be filled before going to sit down.

Evie’s chips were the best in town, and every market day the lunch queue stretched out of the shop. So when the clog shop next door shut down, Evie rented it and doubled the number of tables. But they couldn’t fill them! The queue outside was as long as ever, and the busy townsfolk had no time to sit over their cooling tea.

They couldn’t add another serving counter; Evie’s cod and Bert’s chips were what made the shop. But then they had a brilliant idea. They lengthened the counter and Evie, Bert, Dionysus, and Mary stood in a row. As customers came in, Evie gave them a plate with their fish, Bert added the chips, Dionysus spooned out the mushy peas, and Mary poured the tea and took the money. The customers kept walking; as one customer got the peas, the next was already getting chips and the one after that fish. Less hardy folk don’t eat mushy peas—but that’s no problem; those customers just got nothing but a vacant smile from Dionysus.

The queue shortened and soon they bought the shop on the other side as well for extra table space.

That’s a pipeline. Divide any repetitive job into a number of sequential parts and arrange them so that the work moves past the workers, with each specialist doing his or her part for each unit of work in turn. Although the total time any customer spends being served has gone up, there are four customers being served at once and about three times as many customers being served in that market day lunch hour. Figure 1.1 shows Evie’s organization, as drawn by her son Einstein in a rare visit to nonvirtual reality.²

Seen as a collection of instructions in memory, a program ready to run doesn’t look much like a queue of customers. But when you look at it from

1. Non-English readers should probably not inquire further into the nature of this delicacy.

2. It looks to me as if Einstein has been reading books on computer science.

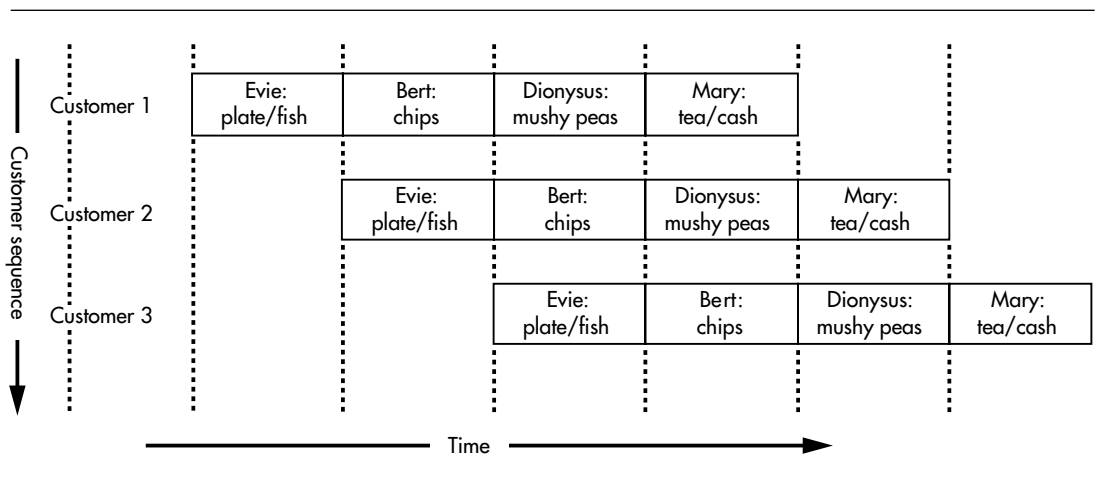


FIGURE 1.1 Evie’s fish and chip shop pipeline.

the CPU’s point of view, things change. The CPU fetches each instruction from memory, decodes it, finds any operands it needs, performs the appropriate action, and stores any results—and then it goes and does the same thing all over again. The program waiting to be run is a queue of instructions waiting to flow through the CPU one at a time.

The various different jobs required to deal with each instruction already require different specialized chunks of logic inside the CPU, so building a pipeline doesn’t even make the CPU much more complicated; it just makes it work harder.

The use of pipelining is not new with RISC microprocessors. What makes the difference is the redesign of everything—starting with the instruction set—to make the pipeline more efficient.³ So how do you make a pipeline efficient? Actually, that’s probably the wrong question. The right question is this: What makes a pipeline inefficient?

1.1.1 What Makes a Pipeline Inefficient?

It’s not good if one stage takes much longer than the others. The organization of Evie’s shop depends on Mary’s ability to pour tea with one hand while giving change with the other—if Mary takes longer than the others, the whole queue will have to slow down to match her.

3. The first RISC in this sense was probably the CDC6600, designed by Seymour Cray in the 1970s, but the idea didn’t catch on at that time. However, this is straying into the history of computer architecture, and if you like this subject you’ll surely want to read Hennessy and Patterson, 1996.

In a pipeline, you try to make sure that every stage takes roughly the same amount of time. A circuit design often gives you the opportunity to trade off the complexity of logic; against its speed, and designers can assign work to different stages: with care, the pipeline is balanced.

The hard problem is not difficult actions, it's awkward customers. Back in the chip shop Cyril is often short of cash, so Evie won't serve him until Mary has counted his money. When Cyril arrives, he's stuck at Evie's position until Mary has finished with the three previous customers and can check his pile of old bent coins. Cyril is trouble, because when he comes in he needs a resource (Mary's counting) that is being used by previous customers. He's a *resource conflict*.

Daphne and Lola always come in together (in that order) and share their meals. Lola won't have chips unless Daphne gets some tea (too salty without something to drink). Lola waits on tenterhooks in front of Bert until Daphne gets to Mary, and so a gap appears in the pipeline. This is a *dependency* (and the gap is called a *pipeline bubble*).

Not all dependencies are a problem. Frank always wants exactly the same meal as Fred, but he can follow him down the counter anyway—if Fred gets chips, Frank gets chips.

If you could get rid of awkward customers, you could make a more efficient pipeline. This is hardly an option for Evie, who has to make her living in a town of eccentrics. Intel is faced with much the same problem: The appeal of its CPUs relies on the customer being able to go on running all that old software. But with a new CPU you get to define the instruction set, and you can define many of the awkward customers out of existence. In section 1.2 we'll show how MIPS did that, but first we'll come back to computer hardware in general with a discussion of caching.

1.1.2 *The Pipeline and Caching*

We said earlier that efficient pipeline operation requires every stage to take the same amount of time. But a 2006 CPU can add two 64-bit numbers 50 to 100 times quicker than it can fetch a piece of data from memory.

So effective pipelining relies on another technique to speed most memory accesses by a factor of 50—the use of *caches*. A cache is a small, very fast, local memory that holds copies of memory data. Each piece of data is kept with a record of its main memory address (the *cache tag*) and when the CPU wants data the cache gets searched and, if the requisite data is available, it's sent back quickly. Since we've no way to guess what data the CPU might be about to use, the cache merely keeps copies of data the CPU has had to fetch from main memory in the recent past; data is discarded from the cache when its space is needed for more data arriving from memory.

Even a simple cache will provide the data the CPU wants more than 90 per centage of the time, so the pipeline design needs only to allow enough time to

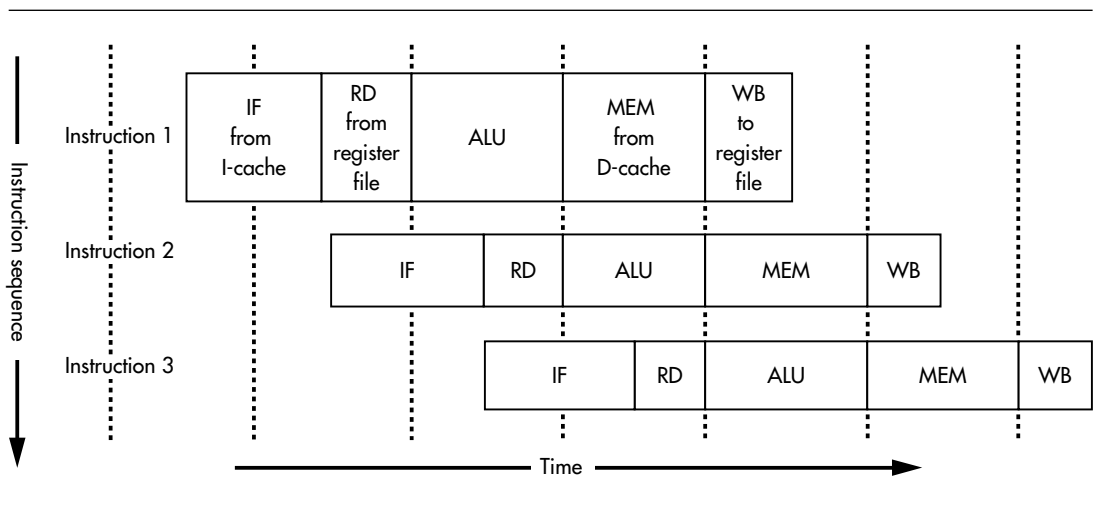


FIGURE 1.2 MIPS five-stage pipeline.

fetch data from the cache; a cache miss is a relatively rare event and we can just stop the CPU when it happens (though cleverer CPUs find more useful things to do).

The MIPS architecture was planned with separate instruction and data caches, so it can fetch an instruction and read or write a memory variable simultaneously.

CISC architectures have caches too, but they're most often afterthoughts, fitted in as a feature of the memory system. A RISC architecture makes more sense if you regard the caches as very much part of the CPU and tied firmly into the pipeline.

1.2 The MIPS Five-Stage Pipeline

The MIPS architecture is made for pipelining, and Figure 1.2 is close to the earliest MIPS CPUs and typical of many. So long as the CPU runs from the cache, the execution of every MIPS instruction is divided into five phases, called *pipestages*, with each pipestage taking a fixed amount of time. The fixed amount of time is usually a processor clock cycle (though some actions take only half a clock, so the MIPS five-stage pipeline actually occupies only four clock cycles).

All instructions are rigidly defined so they can follow the same sequence of pipestages, even where the instruction does nothing at some stage. The net result is that, so long as it keeps hitting the cache, the CPU starts an instruction every clock cycle.

Let's look at Figure 1.2 and consider what happens in each pipestage.

- IF* (instruction fetch) Gets the next instruction from the instruction cache (*I-cache*).
- RD* (read registers) Fetches the contents of the CPU registers whose numbers are in the two possible source register fields of the instruction.
- ALU* (arithmetic/logic unit) Performs an arithmetical or logical operation in one clock cycle (floating-point math and integer multiply/divide can't be done in one clock cycle and are done differently, but that comes later).
- MEM* Is the stage where the instruction can read/write memory variables in the data cache (*D-cache*). On average, about three out of four instructions do nothing in this stage, but allocating the stage for each instruction ensures that you never get two instructions wanting the data cache at the same time. (It's the same as the mushy peas served by Dionysus.)
- WB* (write back) Stores the value obtained from an operation back to the register file.

You may have seen other pictures of the MIPS pipeline that look slightly different; it has been common practice to simplify the picture by drawing each pipestage as if it takes exactly one clock cycle. Some later MIPS CPUs have longer or slightly different pipelines, but the pipeline with five stages in four cycles is where the architecture started, and something very like it is still used by the simpler MIPS CPUs.

The tyranny of the rigid pipeline limits the kinds of things instructions can do. First, it forces all instructions to be the same length (exactly one machine word of 32 bits), so that they can be fetched in a constant time. This itself discourages complexity; there are not enough bits in the instruction to encode really complicated addressing modes, for example. And the fixed-size instructions directly cause one problem; in a typical program built for an architecture like x86, the average size of instructions is only just over three bytes. MIPS code will use more memory space.

Second, the pipeline design rules out the implementation of instructions that do any operation on memory variables. Data from cache or memory is obtained only in phase 4, which is much too late to be available to the ALU. Memory accesses occur only as simple load or store instructions that move the data to or from registers (you will see this described as a *load/store architecture*).

The RISC CPUs launched around 1987 worked because the instruction sets designed around those restrictions prove just as useful (particularly for compiled code) as the complicated ones that give so much more trouble to the

hardware. A 1987 or later RISC is characterized by an instruction set designed for efficient pipelining and the use of caches.

The MIPS project architects also attended to the best thinking of the time about what makes a CPU an easy target for efficient optimizing compilers. Many of those requirements are quite compatible with the pipeline requirements, so MIPS CPUs have 32 general-purpose registers and three-operand arithmetical/logical instructions. Happily, the complicated special-purpose instructions that particularly upset pipelines are often those that compilers are unwilling to generate.

The RISC pioneers' judgment has stood the test of time. More recent instruction sets have pushed the hardware/software line back even further; they are called VLIW (very long instruction word) and/or EPIC (explicitly parallel instruction computing). The most prominent is Intel's IA64 architecture, but it has not succeeded despite massive investment; it appears to have got the hardware/software boundary wrong.

1.3 RISC and CISC

We can now have a go at defining what we mean by these overused terms. For me, RISC is an adjective applied to machine architectures/instruction sets. In the mid-1980s, it became attached to a group of relatively new architectures in which the instruction set had been cunningly and effectively specified to make pipelined implementations efficient and successful. It's a useful term because of the great similarity of approach apparent in SPARC, MIPS, PowerPC, HP Precision, DEC Alpha, and (to a lesser extent) in ARM.

By contrast to this rather finely aimed description, CISC (Complex Instruction Set Computing) is used negatively to describe architectures whose definition has not been shaped by those insights about pipelined implementations. The RISC revolution was so successful that no post-1985 architecture has abandoned the basic RISC principles;⁴ thus, CISC architectures are inevitably those born before 1985. In this book you can reasonably assume that something said about CISC is being said to apply to both Intel's x86 family and Motorola's 680x0.

Both terms are corrupted when they are applied not to instruction sets but to implementations. It's certainly true that Intel accelerated the performance of its far-from-RISC x86 family by applying implementation tricks pioneered by RISC builders. But to describe these implementations as having a RISC core is misleading.

4. Even Intel's complex and innovation-packed IA64 shares some RISC pipeline-friendly features. But the adjective EPIC—as used by Intel—nicely captures both their boundless ambition and the possibility of a huge flop.

1.4 Great MIPS Chips of the Past and Present

It's time to take a tour through the evolution of MIPS processors and the systems that use them, over the span of the past 20 years or so. We'll look at events in the order they occurred, roughly speaking, with a few scenic detours. Along the way, we'll see that although the MIPS architecture was originally devised with UNIX workstations in mind, it has since found its way into all sorts of other applications—many of which could hardly have been foreseen during the early years. You'll get to know some of these names much better in the chapters that follow.

And although much has happened to the instruction set as well as the silicon, the user-level software from a 1985 R2000 would run perfectly well and quite efficiently on any modern MIPS CPU. That's possibly the best backward-compatibility achievement of any popular architecture.

1.4.1 *R2000 to R3000 Processors*

MIPS Becomes a Corporation

MIPS Computer Systems Inc. was formed in 1984 to commercialize the work of Stanford University's MIPS CPU group; we'll abbreviate the name to "MIPS Inc." Stanford MIPS was only one of several U.S. academic projects that were bringing together chip design, compiler optimization, and computer architecture in novel ways with great success. The commercial MIPS CPU was enhanced with memory management hardware and first appeared late in 1985 as the R2000.

Chip fabrication plants were very expensive to set up even during the mid-1980s; they were certainly beyond the means of a small start-up company. MIPS got its designs into production by licensing them to existing semiconductor vendors who'd already committed the sizable investments required. Early licensees included Integrated Device Technology (IDT), LSI Logic, Performance Semiconductor, and NEC.

An ambitious external math coprocessor chip (the R2010 floating-point accelerator, or FPU) first shipped in mid-1987. Since MIPS was intended to serve the vigorous market for engineering workstations, good floating-point performance was important, and the R2010 delivered it.

MIPS itself bought some of the devices produced by those vendors, incorporating them into its own small servers and workstations. The vendors were free under their licensing agreements to supply the devices to other customers.

The R3000 Processor

First shipped in 1988–1989, this took advantage of a more advanced manufacturing process along with some well-judged hardware enhancements, which

combined to give a substantial boost to performance. From the programmer's point of view, the R3000 was almost indistinguishable from the R2000, which meant the speed of this new design could be unleashed immediately on the rapidly growing base of MIPS software. It was soon joined by the R3010 FPU—a similarly improved version of its predecessor.

By the beginning of the 1990s, a few pioneers were using the R3000 in embedded applications, beginning with high-performance laser printers and typesetting equipment.

The R2000/R3000 chips include cache controllers—to get a cache, just add commodity static RAMs. The FPU shared the cache buses to read instructions (in parallel with the integer CPU) and to transfer operands and results. At 1986 speeds, this division of function was ingenious, practical, and workable; importantly, it held the number of signal connections on each device within the pin count limitations of the pin-grid array packages commonly used at the time. This made it possible to produce the devices at reasonable cost and also to assemble them into systems using existing manufacturing equipment.

The Challenges of Raising the Clock Rate

Although it made good sense at the time of its introduction, difficulties eventually arose centering on the partitioning of functions among the R3000, the R3010 FPU, and the external caches.

First, the R3000's external cache implementation led indirectly to some tricky problems for system designers. To squeeze as much performance as possible from the external cache RAMs, their control signals had to be switched at very short, very precisely defined time intervals. The responsibility for implementing the precision delays was passed along to the system designer: the R3000 required four externally generated copies of the input clock, separated by phase shifts that defined the time intervals essential to correct management of the cache control signals. At 20 MHz that was manageable, but as clock speeds rose through 30 MHz and above, the relentless tightening of the accuracy requirements made the task much harder.

Second, the pressure to increase system clock rates also led to problems for the RAM vendors: To keep pace with shrinking cycle times at the processor pipeline, they had to find ways to achieve corresponding improvements in the access time of the memory devices.

All these difficulties became increasingly apparent as the 1980s drew to a close and limited the designs of this generation to a modest rate of improvement. Starting at 25 MHz in 1988, R3000 systems eventually reached 40 MHz in 1991—and they weren't going any faster.

1.4.2 *The R6000 Processor: A Diversion*

The late 1980s saw lively debates among processor designers about the best way to increase microprocessor clock rates. Two subjects in particular came

to the fore. First: Would it be better for future processor designs to keep the cache implementation external, or to bring the caches on-chip? Second: Which logic technology would be the most advantageous choice for future designs?

The first-generation RISC CPUs were built using CMOS chips. They ran cool and packed a lot of logic into a small space, and all low-cost (pre-RISC) microprocessors used CMOS. CMOS proponents thought they had an advantage for many years to come. Yes, CMOS logic was not the fastest, but that would get fixed—the necessary investment would certainly be forthcoming from companies like Intel. And they argued that CMOS would get even better at the things it already did well—packing even more logic into a given silicon area and switching at even higher frequency within a given power budget.

Other designers knew how compelling speed was for CPUs, and they concluded that high-end processors would be better off using ECL chips like those that were already used for mainframe and supercomputer CPUs. Simple ECL logic gates were faster, and it was much faster at sending signals between chips. But you got less logic into a single chip, and it ran much hotter.

Since the two technologies faced such different challenges, it was very difficult to predict which one was the more likely to emerge as the eventual winner. Among the ECL champions was Bipolar Integrated Technology (BIT), and in 1988 it started work on a MIPS CPU called R6000. The project was ambitious, and BIT hoped to redefine the performance of “super-minicomputers” in the same way that CMOS RISC microprocessors had redefined workstation performance.

There were problems. Because of ECL’s density limitations, the processor design had to be partitioned into multiple devices. And customers were anxious about a complete shift to ECL’s chip-to-chip signaling standards. BIT built BiCMOS hybrids that sought to mix the best of both worlds.

In the end, the problems overwhelmed the project. The R6000 was delayed by one problem after another, and slipped to be later than the R4000: the first of a new generation of CMOS processors that used their greater density to move the caches on-chip, gaining clock rate by a different route.

BiCMOS CPUs didn’t die along with BIT: A few years later, a company named Exponential Technology made another valiant attempt, creating a PowerPC implementation around 1996 that achieved a very impressive clock rate for its time of over 500 MHz. Like BIT, however, the company was eventually thwarted by a combination of technical and contractual difficulties and went out of business.

In a really big argument, both sides are often wrong. In the end, several more years were to pass before on-chip implementation of the caches became essential to achieving the highest clock rate. Hewlett Packard stuck with CMOS chips and a (large) external primary cache for its rather MIPS-like Precision architecture. HP eventually pushed its clock rate to around 120 MHz—three times the fastest R3000—*without* using ECL or BiCMOS. HP was its own

customer for these processors, using them in its own workstations; the company felt this market was best served by an evolutionary approach and could bear the costs of high pin-count packages and careful high-speed system-level design. This strategy put HP at the top of the performance stakes for a long, long time; the winner is not always the most revolutionary.

1.4.3 *The First CPU Cores*

In the early 1980s, LSI Logic pioneered the idea of adapting high-volume chip design and manufacturing techniques so that systems companies could create devices specifically tailored to the needs of their own products. Those chips were called Application-Specific Integrated Circuits (ASICs); by around 1990, they could contain up to several thousand gates, equivalent to a large board full of 1970s-era logic devices. The unit cost was very low, and development costs were manageable.

We've seen already that LSI took a very early interest in MIPS and made some R2000/R3000 chips. A couple of years later, it was a natural move for the company to create an implementation of the MIPS architecture that used its own in-house ASIC technology; that move opened the door for customers to include a MIPS processor within a chip that also incorporated other logic. Other MIPS licensees, such as IDT, also began to offer products that integrated simple peripheral functions alongside a MIPS CPU.

Even at the very beginning of the 1990s, you could easily put the basic logic of an R3000-class CPU on an ASIC; but ASICs didn't have very efficient RAM blocks, so integrating the caches was a problem. But ASIC technology progressed rapidly, and by 1993 it was becoming realistic to think of implementing an entire microprocessor system on a chip—not just the CPU and caches, but also the memory controllers, the interface controllers, and any small miscellaneous blocks of supporting logic.

The ASIC business depended on customers being able to take a design into production in a relatively short time—much less than that needed to create a chip using “custom” methods. While it was obviously attractive to offer customers the idea of integrating a complete system on a chip, ASIC vendors had to strike a balance: How could the inevitable increase in complexity still be accommodated within the design cycles that customers had come to expect?

The ASIC industry's answer was to offer useful functional elements—such as an entire MIPS processor—in the form of cores: ready-made building blocks that conveniently encapsulated all the necessary internal design work and verification, typically presented as a set of machine-readable files in the formats accepted by ASIC design software. Systems designs of the future would be created by connecting several ASIC cores together on a chip; in comparison with existing systems—created by connecting together devices on a circuit board—the new systems implemented as core-based ASICs would be smaller, faster, and cheaper.

Until this time, ASIC designers had naturally thought in terms of combining fairly small logic blocks—state machines, counters, decoders, and so forth. With the advent of ASIC cores, designers were invited to work with a broader brush on a much larger canvas, bringing together processors, RAMs, memory controllers, and on-chip buses.

If you suspect that it can't have been that easy, you have good instincts. It sounded compelling—but in practice, creating cores and connecting them together both turned out to be very difficult things to do well. Nevertheless, these early ASIC cores are of great historical significance; they're the direct ancestors of the system-on-a-chip (SoC) designs that have become pervasive during the early 2000s. We'll take up the SoC story again a bit later on, after we've followed several threads of MIPS development through the 1990s.

1.4.4 *The R4000 Processor: A Revolution*

The R4000, introduced in 1991, was a brave and ground-breaking development. Pioneering features included a complete 64-bit instruction set, the largest possible on-chip caches (dual 8 KB), clock rates that seemed then like science fiction (100 MHz on launch), an on-chip secondary cache controller, a system interface running at a fraction of the internal CPU clock, and on-chip support for a shared-memory multiprocessor system. The R4000 was among the first devices to adopt a number of the engineering developments that were to become common by around 1995, though it's important to note that it didn't take on the complexity of superscalar execution.

The R4000 wasn't perfect. It was an ambitious chip and the design was hard to test, especially the clever tricks used for multiprocessor support. Compared with the R3000, it needs more clock cycles to execute a given instruction sequence—those clock cycles are so much shorter that it ends up well in front, but you don't like to give performance away. To win on clock speed the primary caches are on-chip: To keep the cost of each device reasonable, the size of the caches had to be kept relatively small. The R4000 has a longer pipeline, mainly to spread the cache access across multiple clock cycles. Longer pipelines are less efficient, losing time when the pipeline is drained by a branch.

1.4.5 *The Rise and Fall of the ACE Consortium*

Around the time of the R4000's introduction, MIPS had high hopes that the new design would help it to become an important participant in the market for workstations, desktop systems, and servers.

This wasn't mere wishful thinking on the part of MIPS. During the early 1990s, many observers predicted that RISC processors would take an increasing portion of the market away from their CISC competitors; the bolder

prognosticators even suggested that the CISC families would die away entirely within a few years.

In 1991, a group of about 20 companies came together to form a consortium named the Advanced Computing Environment (ACE) initiative. The group included DEC (minicomputers), Compaq (PCs), Microsoft, and SCO (then responsible for UNIX System V). ACE's goal was to define specifications and standards to let future UNIX or Windows software drop straight onto any of a range of machines powered by either Intel x86 or MIPS CPUs. Even in 1991, a small percentage of the PC business would have meant very attractive sales for MIPS CPUs and MIPS system builders.

If hype could create a success, ACE would have been a big one. But looking back on it, Microsoft was more interested in proving that its new Windows NT system was portable (and perhaps giving Intel a fright) than in actually breaking up their PC market duopoly. For MIPS, the outcome wasn't so good; chip volumes wouldn't sustain it and its systems business entered into a decline, which before long became so serious that the future of the company was called into question.

1.4.6 *SGI Acquires MIPS*

As 1992 progressed, the hoped-for flock of new ACE-compliant systems based on MIPS processors was proving slow to materialize, and DEC—MIPS's highest-profile workstation user—decided that future generations of its systems would instead use its own Alpha processor family.

That left workstation company Silicon Graphics, Inc. (SGI) as by far the leading user of MIPS processors for computer systems. So in early 1993, SGI was the obvious candidate to step in and rescue MIPS Inc., as a way of safeguarding the future of the architecture on which its own business depended. By the end of 1994, late-model R4400 CPUs (a stretched R4000 with bigger caches and performance tuning) were running at 200–250 MHz and keeping SGI in touch with the RISC performance leaders.

1.4.7 *QED: Fast MIPS Processors for Embedded Systems*

Some of MIPS Inc.'s key designers left to start a new company named Quantum Effect Design (QED). The QED founders had been deeply involved in the design of MIPS processors from the R2000 through R4000.

With IDT as a manufacturing partner and early investor, QED's small team set out to create a simple, fast 64-bit MIPS implementation. The plan was to create a processor that would offer good performance for a reasonable selling price, so that the device could find a home in many applications, ranging from low-end workstations, through small servers, to embedded systems like top-of-the-range laser printers and network routers.

There were determined people who'd applied R4000 chips to embedded systems, but it was a fight. QED made sure that the R4600 was much more appealing to embedded systems designers, and the device soon became a success. It went back to a simple five-stage pipeline and offered very competitive performance for a reasonable price. Winning a place in Cisco routers as well as SGI's Indy desktops led to another first: The R4600 was the first RISC CPU that plainly turned in a profit.

The QED design team continued to refine its work, creating the R4650 and R4700 during the mid-1990s. We'll take up the QED story again a little further on, when we talk about the R5000.

1.4.8 *The R10000 Processor and Its Successors*

During the mid-1990s, SGI placed very high importance on high-end workstations and supercomputers. Absolute performance was a very important selling point, and the MIPS division was called upon to meet this challenge with its next processor design.

The SGI/MIPS R10000 was launched in early 1996. It was a major departure for MIPS from the traditional simple pipeline; it was the first CPU to make truly heroic use of out-of-order execution, along with multiple instruction issue. Within a few years, out-of-order designs were to sweep all before them, and all really high-end modern CPUs are out-of-order. But the sheer difficulty of verifying and debugging the R10000 convinced both participants and observers to conclude that it had been a mistake for SGI to undertake such an ambitious design in-house.

SGI's workstation business began to suffer during the latter half of the 1990s, leading inevitably to a decline in its ability to make continuing investments in the MIPS architecture. Even as this took place, the market for mainstream PCs continued to expand vigorously, generating very healthy revenue streams to fund the future development of competing architectures—most notably Intel's Pentium family and its descendants and, to a lesser extent, the PowerPC devices designed by Motorola and IBM.

Against this backdrop, SGI started work on MIPS CPUs beyond the R10000; but, because of mounting financial pressures, the projects were canceled before the design teams were able to complete their work. In 1998, SGI publicly committed itself to using the Intel IA-64 architecture in its future workstations, and the last MIPS design team working on desktop/server products was disbanded. In 2006 (as I write) some SGI machines are still dependent on the R16000 CPU; while it takes advantage of advances in process technology to achieve higher clock rates, the internal design has scarcely been enhanced beyond that of the 1996 R10000. Meanwhile, IA-64 CPUs have sold well below Intel's most pessimistic projections, and the fastest CPUs in the world are all variants of the x86. SGI seems to be unlucky when it comes to choosing CPUs!

1.4.9 *MIPS Processors in Consumer Electronics*

LSI Logic and the Sony PlayStation

In 1993, Sony contracted with LSI Logic for the development of the chip that was to form the heart of the first PlayStation. Based on LSI's CW33000 processor core, it was clocked at 33 MHz and incorporated a number of peripheral functions, such as a DRAM controller and DMA engine. The PlayStation's highly integrated design made it cheap to produce and its unprecedented CPU power made for exciting gaming. Sony rapidly overtook more established vendors to become the leading seller of video game consoles.

The Nintendo64 and NEC's Vr4300 Processor

Nintendo game consoles lost considerable market share to Sony's PlayStation. In response, Nintendo formed an alliance with Silicon Graphics and decided to leapfrog 32-bit CPU architectures and go straight for a 64-bit chip—in a \$199 games machine.

The chip at its heart—the NEC Vr4300—was a cut-down R4000, but not *that* cut-down. It did have a 32-bit external bus, to fit in a cheaper package with fewer pins, and it shared logic between integer and floating-point maths. But it was a lot of power for a \$199 box.

The Vr4300's computing power, low price, and frugal power consumption made it very successful elsewhere, particularly in laser printers, and helped secure another niche for the MIPS architecture in “embedded” applications.

But the Vr4300 was the last really general-purpose CPU to storm the games market; by the late 1990s, the CPU designs intended for this market had become increasingly specialized, tightly coupled with dedicated hardware accelerators for 3D rendering, texture mapping, and video playback. When Sony came back with the PlayStation 2, it had a remarkable 64-bit MIPS CPU at its heart. Built by Toshiba, it features a floating-point coprocessor whose throughput wouldn't have disgraced a 1988 supercomputer (though its accuracy would have been a problem). It has proven too specialized to find applications outside the games market, but a version of the same CPU is in Sony's PSP handheld games console, which will certainly be with us for a few years to come.

Cumulative sales of these video game consoles worldwide is well into the tens of millions, accounting for a larger volume of MIPS processors than any other application—and also causing them to outsell a good many other CPU architectures.

1.4.10 *MIPS in Network Routers and Laser Printers*

The R5000 Processor

Following the success of the R4600 and its derivatives, QED's next major design was the R5000. Launched in 1995—the same year as SGI's R10000—this

was also a superscalar implementation, though in terms of general design philosophy and complexity, the two designs stood in stark contrast to each other.

The R5000 used the classic five-stage pipeline and issued instructions in-order. It was capable, however, of issuing one integer instruction and one floating-point instruction alongside each other. The MIPS architecture makes this scheme relatively easy to implement; the two instruction categories use separate register sets and execution units, so the logic needed to recognize opportunities for dual issue doesn't have to be very complicated.

Of course, the other side of the same coin is that the performance gain is relatively modest. Unless the R5000 is used in a system that runs a significant amount of floating-point computation, the superscalar ability goes unused. Even so, the R5000 incorporated other improvements that made it appealing to system designers as an easy upgrade from the R4600 generation.

QED Becomes a Fabless Semiconductor Vendor

During the first few years of its life, QED had operated purely as a seller of intellectual property, licensing its designs to semiconductor device vendors who then produced and sold the chips. In 1996, the company decided it could do better by selling chips under its own name. The manufacturing was still carried out by outside partners—the company remained “fabless” (that is, it didn't have any fabrication plants under its direct ownership)—but now QED took charge of testing the chips and handled all of its own sales, marketing, and technical support.

Around this time, QED embarked on a project to develop a PowerPC implementation in the same lean, efficient style as the R4600. Unfortunately, business and contractual difficulties with the intended customer reared their heads, with the result that the device was never brought to market. After this brief excursion into PowerPC territory, QED resumed its exclusive focus on the MIPS architecture.

QED's RM5200 and RM7000 Processors

QED's first “own-brand” CPU was the RM5200 family, a direct descendant of the R5000. With a 64-bit external bus it played well in network routers, while a 32-bit bus and cheaper package was good for laser printers.

QED built on the RM5200's success, launching the RM7000 in 1998. This device marked several important milestones for MIPS implementations: It was the first to bring the (256 Kbyte) secondary cache on-chip.⁵ RM7000 was also a serious superscalar design, which could issue many pairings of integer instructions besides the integer/floating-point combination inherited from the R5000.

5. QED originally hoped to use a DRAM-like memory to make the RM7000's secondary cache very small, but it turned out that an adequate compromise between fast logic and dense DRAM on a single chip was not then possible. It still isn't.

The RM5200 and RM7000 processor families sold well during the mid to late 1990s into many high-end embedded applications, finding especially widespread use in network routers and laser printers. QED wisely ensured that the RM7000 was very closely compatible with the RM5200, both from the programmer's and system designer's points of view. This made it fairly straightforward to give aging RM5200 systems a quick midlife boost by upgrading them to the RM7000, and many customers found it useful to follow this path.

SandCraft

Around 1998, the design team that had created the Vr4300 for Nintendo incorporated as SandCraft, and set out to produce embedded CPUs intended for the high-end embedded applications then served by QED's RM5200 and RM7000 families.

SandCraft's designs were architecturally ambitious and took time to bring to market. Despite several years of continued efforts to build a large enough customer base, the company eventually went out of business. Its assets were acquired by Raza Technologies, and it remains to be seen whether any significant portion of SandCraft's legacy will eventually find its way into production.

1.4.11 MIPS Processors in Modern Times

Alchemy Semiconductor: Low-Power MIPS Processors

By 1999, the markets for cellphones, personal organizers, and digital cameras were growing rapidly. The priority for such processors is low power consumption: Since these appliances need to be small, light, and have to run from internal batteries, they must be designed to operate within very tight power budgets. At the same time, competitive pressures require each generation of appliances to offer more features than its predecessor. Manufacturers sought 32-bit computing power to meet the growing applications' hungry demands.

Taken together, these requirements present a moving target for processor designers: Within a power budget that grows only gradually (with advances in battery chemistry and manufacturing), they're called upon to deliver a significant boost in performance with every design generation.

It is really just a matter of historical accident that nobody had implemented a fast, low-power MIPS processor. But DEC had built a 200-MHz low-power ARM ("StrongARM") and the ARM company was willing to build less exalted machines that would eke out your battery life even longer. When DEC engaged in unwise litigation with Intel over CPU patents, they lost, big-time. Among the things Intel picked up was the StrongARM development. Amazingly, it seems

to have taken Intel a couple of years to notice this jewel, and by that time all the developers had left.

In 1999, Alchemy Semiconductor was founded precisely to exploit this opportunity. With backing from Cadence, a vendor of chip design tools, some members of the design team that had created StrongARM now turned their ingenuity to the design of a very low power 32-bit MIPS CPU. It works very well, but their designs were too high-end, and perhaps just a bit late, to break the ARMlock on cellphones.

Alchemy pinned its hopes on the market for personal organizers, which certainly needed faster CPUs than the phones did. But the market didn't boom in the same way. Moreover, the organizer market seemed to be one in which every innovator lost money; and finally Microsoft's hot-then-cold support of MIPS on Windows CE made the MIPS architecture a handicap in this area.

SiByte

This company was also founded in 1999 around an experienced design team, again including some members who had worked on DEC's Alpha and StrongARM projects.⁶

SiByte built a high-performance MIPS CPU design—it aimed for efficient dual-issue at 1 GHz. Moreover, this was to be wrapped up for easy integration into a range of chip-level products; some variants were to emphasize computational capacity, featuring multiple CPU cores, while others laid the stress on flexible interfacing by integrating a number of controllers.

SiByte's design found considerable interest from networking equipment makers who were already using QED's RM5200 and RM7000 devices; as the company began to put the device into production, however, manufacturing difficulties caused long delays, and the 1-GHz target proved difficult.

Consolidation: PMC-Sierra, Broadcom, AMD

The last years of the 1990s saw the infamous "dotcom bubble." Many small technology companies went public and saw their stock prices climb to dizzying heights within weeks.

Networking companies were among the darlings of the stock market and with their market capitalizations rising into tens of billions, they found it easy to buy companies providing useful technology—and that sometimes meant MIPS CPUs.

This was the climate in which Broadcom acquired SiByte, and PMC-Sierra acquired QED—both in mid-2000. It seemed that the future of high-end MIPS

6. In the U.S. market, a canceled project can have as seminal an effect as a successful one.

designs for embedded systems was now doubly safeguarded by the deep pockets of these two new parent companies.

The collapse of the technology bubble came swiftly and brutally. By late 2001, the networking companies saw their stock prices showing unexpected weakness, and orders from their customers slowing alarmingly; by 2002, the entire industry found itself in the grip of a savage downturn. Some companies saw their market capitalizations drop to less than a tenth of their peak values over one year.

The resulting downdraft inevitably affected the ability of PMC-Sierra and Broadcom to follow through with their plans for the QED and SiByte processor designs. It wasn't just a matter of money; it became extremely difficult for these companies even to find a reasonable strategic direction, as sales for many established product lines slowed to a trickle.

Alchemy Semiconductor also felt the cold wind of change, and despite the impressively low power consumption of its designs, the company had difficulty finding high-volume customers. Finally, in 2002, Alchemy was acquired by Advanced Micro Devices (AMD), which continued to market the Au1000 product line for a couple of years. As we go to press, we hear that the Alchemy product line has been acquired by Raza Technologies.

Highly Integrated Multiprocessor Devices

Broadcom had initially announced plans for an ambitious evolution of SiByte's 1250 design from two CPU cores to four, along with an extra memory controller and much faster interfaces. This project became a casualty of the downturn, and the evolutionary future of the 1250 product line fell into uncertainty.

Meanwhile, the QED design team—now operating as PMC-Sierra's MIPS processor division—created its own dual-CPU device, the RM9000x2. This also integrated an SDRAM controller and various interfaces. Due in part to the chilly market conditions, the RM9000 family was slow to find customers, though it did surpass the 1-GHz milestone for CPU clock rate. Subsequent derivatives added further interfaces, including Ethernet controllers, but the difficulties in securing large design wins persisted.

In 2006, the future for such highly integrated devices appears doubtful. As the transistors in chips shrink, the amount of logic you can get for the production-cost dollar increases. But the one-off cost of getting a new design into production keeps going up: For the most recent 90-nanometer generation, it's \$1 M or more. If you fill the space available with logic to maximize what your chip will do, design and test costs will be tens of millions.

To get that back, you need to sell millions of units of each chip over its product lifetime. A particular version of a chip like the RM9000x2 or Broadcom's 1250 can sell tens or hundreds of thousands: It isn't enough. It's not clear what sort of device may attract enough volume to fund full-custom embedded-CPU design in future.

Intrinsity: Taking a MIPS Processor to 2 GHz

Alert readers will have noticed the overall arc of the MIPS story to date: The early R2000/R3000 implementations were performance leaders among micro-processors, but competing families eventually caught up and overtook MIPS.

So you might be wondering: Has anyone tried to make a really fast MIPS processor in the last few years? The answer is yes: Intrinsity Semiconductor announced in 2002 its FastMath processor. Using careful design techniques to extract high performance from essentially standard process technologies, Intrinsity was able to produce a lean 32-bit MIPS design with an impressive clock rate of 2 GHz.

While this was widely recognized as a fine technical achievement, the device has struggled to find a market. It's still not nearly as fast as a modern PC processor, and its power consumption and heat dissipation is relatively high by consumer standards.

1.4.12 *The Rebirth of MIPS Technologies*

In 1998, SGI—facing mounting cash-flow problems—decided to spin off its CPU design group, restoring it to independence as MIPS Technologies. The new company was chartered to create core CPUs to be used as part of a system-on-a-chip (SoC). You might recall that we encountered the idea of an SoC much earlier in this section, when we described the appearance of the first ASIC cores.

In the early days of SoCs, CPU vendors found that it was very difficult to guarantee a core's performance—for example, the CPU clock rate—unless they provided their customers with a fixed silicon layout for the core internals, predefined for each likely target chip “foundry”—a “hard core.”

MIPS Technologies originally intended to build high-performance hard cores and built and shipped fast 64-bit designs (20 Kc and later 25 Kf). But that was the wrong horse. During the last few years, the market has increasingly preferred its cores to be synthesizable (originally called “soft core”).

A synthesizable core is a set of design files (usually in Verilog) that describes the circuit and can be compiled into a real hardware design. A synthesizable core product consists of a lot more than a Verilog design, since the customer must be able to incorporate it in a larger SoC design and validate the CPU and its connections well enough that the whole chip will almost certainly work.

MIPS Technologies' first synthesizable core was the modest 32-bit 4-K family; since then, it has added the 64-bit 5 K, the high-performance 32-bit 24 K, and (launched in early 2006) the multithreading 34 K.

1.4.13 *The Present Day*

MIPS CPUs in use today come in four broad categories:

- *SoC cores*: MIPS CPUs still benefit in size and power consumption from the simplicity that came from the Stanford project, and an architecture with a long history spanning handhelds to supercomputers is an attractive alternative to architectures tailored for the low end. MIPS was the first “grown up” CPU to be available as an ASIC core—witness its presence in the Sony PlayStation games console. The most prominent supplier is MIPS Technologies Inc., but Philips retains their own designs.
- *Integrated embedded 32-bit CPUs*: From a few dollars upward, these chips contain CPU, caches, and substantial application-oriented blocks (network controllers are popular). There’s considerable variation in price, power consumption, and processing power. Although AMD/Alchemy has some very attractive products, this market seems to be doomed, with devices in the target marketplace finding that an SoC heart does a better job of maximizing integration, saving vital dollars and milliwatts.
- *Integrated embedded 64-bit CPUs*: These chips offer a very attractive speed/power-consumption trade-off for high-end embedded applications: Network routers and laser printers are common applications. But it doesn’t look as though they can sell in sufficient numbers to go on paying for chip development costs.
But somewhere in this category are companies that are trying radically new ideas, and it’s a tribute to the MIPS architecture’s clean concepts that it often seems the best base for leading-edge exploration. Raza’s XLR series of multicore, multithreaded processors represent a different kind of embedded CPU, which aims to add more value (and capture more revenue per unit) than a “traditional” embedded CPU. Cavium’s Octium is also pretty exciting.
- *Server processors*: Silicon Graphics, the workstation company that was the adoptive parent of the MIPS architecture, continued to ship high-end MIPS systems right up to its insolvency in 2006, even though that was seven years after it committed to a future with Intel IA-64. But it’s the end of the road for these systems: MIPS is destined to be “only” in the vast consumer and embedded markets.

The major distinguishing features of some milestone products are summarized in Table 1.1. We haven’t discussed the instruction set revision levels from MIPS I through MIPS64, but there’ll be more about them in section 2.7, where you’ll also find out what happened to MIPS II.

TABLE 1.1 Milestones in MIPS CPUs

<i>Year</i>	<i>Designer/model/ clock rate (MHz)</i>	<i>Instruction set</i>	<i>Cache (I+D)</i>	<i>Notes</i>
1987	MIPS R2000-16	MIPS I	External; 4 K+4 K to 32 K+32 K	External (R2010) FPU.
1990	IDT R3051-20		4 K+1 K	The first embedded MIPS CPU with on-chip cache and progenitor of a family of pin-compatible parts.
1991	MIPS R4000-100	MIPS III	8 K+8 K	Integrates FPU and L2 cache controller with pinout option. Full 64-bit CPU—but five years later, few MIPS CPUs were exploiting their 64-bit instruction set. Long pipeline and half-speed interface help achieve high clock rates.
1993	IDT/QED R4600-100		16 K+16 K	QED's brilliantly tuned redesign is much faster than R4000 or R4400 at the same clock rate—partly because it returned to the classic MIPS five-stage pipeline. Important to SGI's fast and affordable low-end Indy workstation and Cisco's routers.
1995	NEC/MIPS Vr4300-133		16 K+8 K	Low cost, low power but full-featured R4000 derivative. Initially aimed at Nintendo 64 games console, but embedded uses include HP's LJ4000 laser printers.
1996	MIPS R10000-200	MIPS IV	32 K+32 K	Bristling with microprocessor innovations, the R10000 is not at all simple. The main MIPS tradition it upholds is that of taking a principle to extremes. The result was hot, unfriendly, but with unmatched performance/MHz.
1998	QED RM7000		16 K+16 K+256 K L2	The first MIPS CPU with on-chip L2 cache, this powered generations of high-end laser printers and Internet routers.
2000	MIPS 4 K core family	MIPS32	16 K+16 K (typ)	The most successful MIPS core to date—synthesizable and frugal.
2001	Alchemy AU-1000		16 K+16 K	If you wanted 400 MHz for 500 mW, this was the only show in town. But it lost markets.
2001	Broadcom BCM1250	MIPS64	32 K+32 K+256 K L2	Dual-CPU design at 600 MHz+ (the L2 is shared).
2002	PMC-Sierra RM9000x2	MIPS64	16 K+16 K+256 K L2	Dual-CPU design at 1 GHz (the L2 is NOT shared; each CPU has its own 256 K). First MIPS CPU to reach 1 GHz.
2003	Intrinsity FastMath	MIPS32	16 K+16 K+1 M L2	Awesome 2-GHz CPU with vector DSP did not find a market.
2003	MIPS 24 K core	MIPS32 R2	At 500 MHz in synthesizable logic, a solidly successful core design.	
2005	MIPS 34 K core	MIPS32+MT ASE	32 K+32 K (typ)	MIPS multithreading pioneer.

1.5 MIPS Compared with CISC Architectures

Programmers who have some assembly-language-level knowledge of earlier architectures—particularly those brought up on x86 or 680x0 CISC instruction sets—may get some surprises from the MIPS instruction set and register model. We'll try to summarize them here, so you don't get sidetracked later into doomed searches for things that don't quite exist, like a stack with push/pop instructions!

We'll consider the following: constraints on MIPS operations imposed to make the pipeline efficient; the radically simple load/store operations; possible operations that have been deliberately omitted; unexpected features of the instruction set; and the points where the pipelined operation becomes visible to the programmer.

The Stanford group that originally dreamed up MIPS was paying particular attention to the short, simple pipeline it could afford to build. But it's a testament to the group's judgment that many of the decisions that flowed from that have proven to make more ambitious implementations easier and faster, too.

1.5.1 *Constraints on MIPS Instructions*

- *All instructions are 32 bits long:* That means that no instruction can fit into only two or three bytes of memory (so MIPS binaries are typically 20 percent to 30 percent bigger than for 680x0 or 80x86) and no instruction can be bigger.

It follows that it is impossible to incorporate a 32-bit constant into a single instruction (there would be no instruction bits left to encode the operation and the target register). The MIPS architects decided to make space for a 26-bit constant to encode the target address of a jump or jump to subroutine; but that's only for a couple of instructions. Other instructions find room only for a 16-bit constant. It follows that loading an arbitrary 32-bit value requires a two-instruction sequence, and conditional branches are limited to a range of 64-K instructions.

- *Instruction actions must fit the pipeline:* Actions can only be carried out in the right pipeline phase and must be complete in one clock. For example, the register write-back phase provides for just one value to be stored in the register file, so instructions can only change one register.

Integer multiply and divide instructions are too important to leave out but can't be done in one clock. MIPS CPUs have traditionally provided them by dispatching these operations into a separately pipelined unit we'll talk about later.

- *Three-operand instructions:* Arithmetical/logical operations don't have to specify memory locations, so there are plenty of instruction bits to define two independent sources and one destination register. Compilers love

three-operand instructions, which give optimizers much more scope to improve code that handles complex expressions.

- *The 32 registers:* The choice of the number of registers is largely driven by software requirements, and a set of 32 general-purpose registers is easily the most popular in modern architectures. Using 16 would definitely not be as many as modern compilers like, but 32 is enough for a C compiler to keep frequently accessed data in registers in all but the largest and most intricate functions. Using 64 or more registers requires a bigger instruction field to encode registers and also increases context-switch overhead.
- *Register zero: \$0* always returns zero, to give a compact encoding of that useful constant.
- *No condition codes:* One feature of the MIPS instruction set that is radical even among the 1985 RISCs is the lack of any condition flags. Many architectures have multiple flags for “carry,” “zero,” and so on. CISC architectures typically set these flags according to the result written by any or a large subset of machine instructions, while some RISC architectures retain flags (though typically they are only set explicitly, by compare instructions).

The MIPS architects decided to keep all this information in the register file: Compare instructions *set* general-purpose registers and conditional branch instructions *test* general-purpose registers. That does benefit a pipelined implementation, in that whatever clever mechanisms are built in to reduce the effect of dependencies on arithmetical/logical operations will also reduce dependencies in compare/branch pairs.

We’ll see later that efficient conditional branching (at least in one favorite simple pipeline organization) means that the decision about whether to branch or not has to be squeezed into only half a pipeline stage; the architecture helps out by keeping the branch decision tests very simple. So MIPS conditional branches test a single register for sign/zero or a pair of registers for equality.

1.5.2 *Addressing and Memory Accesses*

- *Memory references are always plain register loads and stores:* Arithmetic on memory variables upsets the pipeline, so it is not done. Every memory reference has an explicit load or store instruction. The large register file makes this much less of a problem than it sounds.
- *Only one data-addressing mode:* Almost all loads and stores select the memory location with a single base register value modified by a 16-bit signed displacement (a limited register-plus-register address mode is available for floating-point data).

- *Byte-addressed*: Once data is in a register of a MIPS CPU, all operations always work on the whole register. But the semantics of languages such as C fit badly on a machine that can't address memory locations down to byte granularity, so MIPS gets a complete set of load/store operations for 8- and 16-bit variables (we will say *byte* and *halfword*). Once the data has arrived in a register it will be treated as data of full register length, so partial-word load instructions come in two flavors—sign-extend and zero-extend.

- *Load/stores must be aligned*: Memory operations can only load or store data from addresses aligned to suit the data type being transferred. Bytes can be transferred at any address, but halfwords must be even-aligned and word transfers aligned to four-byte boundaries. Many CISC microprocessors will load/store a four-byte item from any byte address, but the penalty is extra clock cycles.

However, the MIPS instruction set architecture (ISA) does include a couple of peculiar instructions to simplify the job of loading or storing at improperly aligned addresses.

- *Jump instructions*: The limited 32-bit instruction length is a particular problem for branches in an architecture that wants to support very large programs. The smallest opcode field in a MIPS instruction is 6 bits, leaving 26 bits to define the target of a jump. Since all instructions are four-byte aligned in memory, the two least significant address bits need not be stored, allowing an address range of $2^{28} = 256$ MB. Rather than make this branch PC relative, this is interpreted as an absolute address within a 256-MB segment. That's inconvenient for single programs larger than this, although it hasn't been much of a problem yet!

Branches out of segment can be achieved by using a jump register instruction, which can go to any 32-bit address.

Conditional branches have only a 16-bit displacement field—giving a 2^{18} -byte range, since instructions are four-byte aligned—which is interpreted as a signed PC-relative displacement. Compilers can only code a simple conditional branch instruction if they know that the target will be within 128 KB of the instruction following the branch.

1.5.3 *Features You Won't Find*

- *No byte or halfword arithmetic*: All arithmetical and logical operations are performed on 32-bit quantities. Byte and/or halfword arithmetic requires significant extra resources and many more opcodes, and it is rarely really useful. The C language's semantics cause most calculations to be carried out with `int` precision, and for MIPS `int` is a 32-bit integer.

However, where a program explicitly does arithmetic as `short` or `char`, a MIPS compiler must insert extra code to make sure that the

results wrap and overflow as they would on a native 16- or 8-bit machine.

- *No special stack support:* Conventional MIPS assembly usage does define one of the registers as a stack pointer, but there's nothing special to the hardware about **sp**. There is a recommended format for the stack frame layout of subroutines, so that you can mix modules from different languages and compilers; you should almost certainly stick to these conventions, but they have no relationship to the hardware.

A stack pop wouldn't fit the pipeline, because it would have two register values to write (the data from the stack and the incremented pointer value).

- *Minimal subroutine support:* There is one special feature: jump instructions have a jump and link option, which stores the return address into a register. **\$31** is the default, so for convenience and by convention **\$31** becomes the return address register.

This is less sophisticated than storing the return address on a stack, but it has some significant advantages. Two examples will give you a feeling for the argument: First, it preserves a pure separation between branch and memory-accessing instructions; second, it can aid efficiency when calling small subroutines that don't need to save the return address on the stack at all.

- *Minimal interrupt handling:* It is hard to see how the hardware could do less. It stashes away the restart location in a special register, modifies the machine state just enough to let you find out what happened and to disallow further interrupts, then jumps to a single predefined location in low memory. Everything else is up to the software.
- *Minimal exception handling:* Interrupts are just one sort of exception (the MIPS word *exception* covers all sorts of events where the CPU may want to interrupt normal sequential processing and invoke a software handler). An exception may result from an interrupt, an attempt to access virtual memory that isn't physically present, or many other things. You go through an exception, too, on a deliberately planted trap instruction like a system call that is used to get into the kernel in a protected OS. All exceptions result in control passing to the same fixed entry point.⁷

On any exception, a MIPS CPU *does not* store anything on a stack, write memory, or preserve any registers for you.

By convention, two general-purpose registers are reserved so that exception routines can bootstrap themselves (it is impossible to do anything

7. I exaggerate slightly; these days there are quite a few different entry points, and there were always at least two. Details will be given in section 5.3.

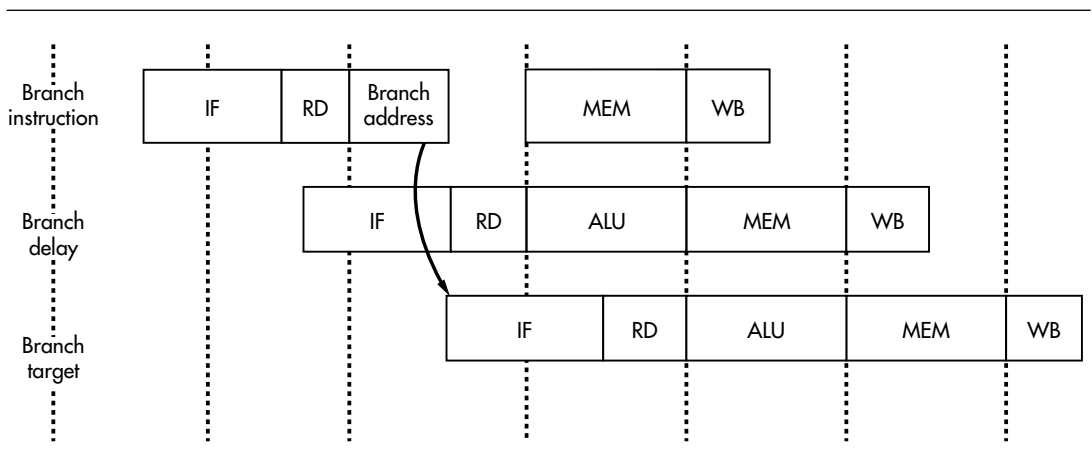


FIGURE 1.3 The pipeline and branch delays.

on a MIPS CPU without using some registers). For a program running in any system that takes interrupts or traps, the values of these registers may change at any time, so you'd better not use them.

1.5.4 Programmer-Visible Pipeline Effects

So far, this has all been what you might expect from a simplified CPU. However, making the instruction set pipeline friendly has some stranger effects as well, and to understand them we're going to draw some pictures.

- Delayed branches:** The pipeline structure of the MIPS CPU (Figure 1.3) means that when a jump/branch instruction reaches the execute phase and a new program counter is generated, the instruction after the jump will already have been started. Rather than discard this potentially useful work, the architecture dictates that the instruction after a branch must always be executed before the instruction at the target of the branch. The instruction position following any branch is called the *branch delay slot*. If nothing special was done by the hardware, the decision to branch or not, together with the branch target address, would emerge at the end of the ALU pipestage—by which time, as Figure 1.3 shows, you're too late to present an address for an instruction in even the next-but-one pipeline slot.

But branches are important enough to justify special treatment, and you can see from Figure 1.3 that a special path is provided through the ALU to make the branch address available half a clock cycle early. Together with the odd half-clock-cycle shift of the instruction fetch stage, that means that the branch target can be fetched in time to become the next but one,

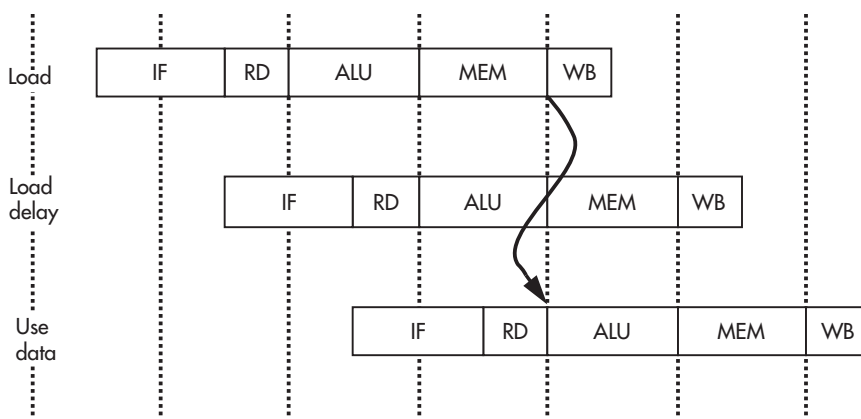


FIGURE 1.4 The pipeline and load delays.

so the hardware runs the branch instruction, then the branch delay slot instruction, and then the branch target—with no other delays.

It is the responsibility of the compiler system or the assembly programming wizard to allow for and even to exploit the branch delay; it turns out that it is usually possible to arrange that the instruction in the branch delay slot does useful work. Quite often, the instruction that would otherwise have been placed before the branch can be moved into the delay slot.

This can be a bit tricky on a conditional branch, where the branch delay instruction must be (at least) harmless on both paths. Where nothing useful can be done, the delay slot is filled with a **nop** instruction.

Many MIPS assemblers will hide this odd feature from you unless you explicitly ask them not to.

- *Late data from load (load delay slot):* Another consequence of the pipeline is that a load instruction's data arrives from the cache/memory system *after* the next instruction's ALU phase starts—so it is not possible to use the data from a load in the following instruction. (See Figure 1.4 for how this works.)

The instruction position immediately after the load is called the *load delay slot*, and an optimizing compiler will try to do something useful with it. The assembler will hide this from you but may end up putting in a **nop**.

On modern MIPS CPUs the load result is interlocked: If you try to use the result too early, the CPU stops until the data arrives. But on early MIPS CPUs, there were no interlocks, and the attempt to use data in the load delay slot led to unpredictable results.

MIPS Architecture

The rather grandiose word *architecture* is used in computing to describe the abstract machine you program, rather than the actual implementation of that machine. That's a useful distinction—and one worth defending from the widespread misuse of the term in marketing hype. The abstract description may be unfamiliar, but the concept isn't. If you drive a stick-shift car you'll find the gas pedal on the right and the clutch on the left, regardless of whether the car is front-wheel drive or rear-wheel drive. The architecture (which pedal is where) is deliberately kept the same, although the implementation is different.

Of course, if you're a rally driver concerned with going very fast along slippery roads, it's suddenly going to matter a whole lot which wheels are driven. Computers are like that too—once your performance needs are extreme or unusual, the details of the implementation may become important to you.

In general, a CPU architecture consists of an instruction set and some knowledge about registers. The terms *instruction set* and *architecture* are pretty close to synonymous, and you can get both at once in the acronym ISA (Instruction Set Architecture).

These days the MIPS ISA is best defined by the MIPS32 and MIPS64 architecture specifications published by MIPS Technologies Inc. MIPS32 is a subset of MIPS64 for CPUs with 32-bit general-purpose registers. “The MIPS32 and MIPS64 architecture specifications” is a bit of a mouthful, so we'll abbreviate it to “MIPS32/64.”

Most of the companies making MIPS CPUs are now making them compatible with these specifications—and those that are not strictly compatible are generally reducing differences as they can.

Before MIPS32/64, a number of versions of the MIPS architecture were written down. But those older definitions formally applied only to the instructions and resources used by higher-level software—they regarded the CPU control mechanisms necessary to operating systems as implementation dependent. That meant that portable operating system work depended on a

gentleman’s agreement to keep the unspecified areas of the MIPS architecture stable, too. The way that worked out was that each architecture version was naturally associated with a “parent” implementation:

- MIPS I: The instruction set used by the original 32-bit processors (R2000/3000); every MIPS CPU there ever was will run these instructions, so it is still common as a *lingua franca*.
- MIPS II: A minor upgrade defined for the MIPS R6000, which didn’t get beyond preproduction. But something very like it has been widely used for 32-bit MIPS CPUs for the embedded market. MIPS II is the immediate parent of MIPS32.
- MIPS III: The 64-bit instruction set introduced by the R4000.
- MIPS IV: Adds a few useful instructions (mostly floating point) to MIPS III, appearing in two different implementations (R10000 and R5000).
- MIPS V: Adds some surprising two operations at once (“SIMD”) floating-point operations—but no MIPS V CPU was built. Most of it reappeared as an optional part of MIPS64 called “paired-single.”
- MIPS32, MIPS64: Standards promulgated by MIPS Technologies Inc. after its demerger from Silicon Graphics in 1998. For the first time, the standards encompassed the “CPU control” functions known as coprocessor 0. MIPS32 is a superset of MIPS II, while MIPS64—which includes 64-bit instructions—is a superset of MIPS IV (and contains much of MIPS V as an optional extension).
Most MIPS CPUs designed since 1999 are compatible with these standards, so we’ll use MIPS32/64 as the base architectures in this book. To be more precise, our base is Release 2 of the MIPS32/64 specifications, as published in 2003. Earlier MIPS32/64 CPUs might be referred to as Release 1.

The architecture levels define everything the original company documentation chose to define; up to MIPS V, that was typically rather more than enough to ensure the ability to run the same UNIX application and less than enough to ensure complete portability of code that uses OS or low-level features. By contrast, if a CPU conforms to MIPS32/64, it should run a reasonable portable operating system.

Early MIPS CPUs (starting with the R3000) had CPU control instructions and registers that are fairly different from MIPS32; they’re very old now, and these features aren’t detailed in this book.

Quite a few implementations add some of their own new instructions and interesting features. It’s not always easy to get software or tools (particularly compilers) that take advantage of implementation-specific features. We’ll describe most that I find interesting.

There are two levels of detail at which we can describe the MIPS architecture. The first (this chapter) is the kind of view you'd get if you were writing a user program for a workstation but chose to look at your code at the assembly level. That means that the whole normal working of the CPU would be visible.

In the later chapters we'll take on everything, including all the gory details that a high-level operating system hides—CPU control registers, interrupts, traps, cache manipulation, and memory management. But at least we can cut the task into smaller pieces.

CPUs are often much more compatible at the user level than when everything is exposed. All known MIPS CPUs can run MIPS I user-level code.

Bringing Order to Instruction Set Extensions—ASEs

We've kept stressing that being RISC has very little to do with keeping the instruction set small. In fact, RISC simplicity makes it temptingly easy to extend the instruction set with new twists on three-operand register-to-register calculations.

When MIPS CPUs began to be used in embedded systems, new instructions aimed at helping out some particular application began to mushroom. MIPS32/64 has included some of those inventions. But it also provides some regulations, in the form of recognized ASEs (Application-Specific instruction set Extensions). ASEs are optional extensions to MIPS32/64 whose presence is marked in a standard way through configuration registers. There are already quite a lot of them:

- **MIPS16e:** This extension long predates MIPS32 and MIPS Technologies. It was pioneered by LSI Logic in the 1990s, with a view to reducing the size of MIPS binary programs. MIPS16 encodes a subset of user-privilege MIPS instructions (and a few special ones added in) in 16-bit op-codes. Such an instruction set is evidently “too small” and programs compile into significantly more MIPS16 than regular MIPS instructions; however, the half-size instructions lead to much smaller code. The MIPS16 instruction set was organized and slightly augmented to be an extension of MIPS32, and the augmented version is called MIPS16e. This book has few details about MIPS16e: It's a large subject and rarely met up with even at assembly level, but there are some words on the subject in section B.1.
- **MDMX:** Another old extension, this one championed by SGI. It adds a large set of SIMD arithmetic operations using the FP (coprocessor 1) register set. They're SIMD because each operation runs in parallel on each of a short array of integer values packed into the 64-bit registers. Many of the operations are “saturating”; when a result would otherwise overflow, the destination is filled with the nearest representable value. Saturating

small-integer SIMD operations accelerates a variety of audio and video “media stream” computations. They’re also broadly characteristic of the specialized CPUs called DSPs (Digital Signal Processors).

MDMX resembles the early versions of Intel’s MMX extension. But MDMX was never implemented by SGI, and few implementations survive: Broadcom’s CPUs seem to be the only ones.

- SmartMIPS: A very small extension to the architecture aimed at improving the performance of encryption operations believed to be key to the smart card market; it is coupled to a bunch of other security- and size-related tweaks to the CPU control system. One day the 32-bit smart card market will be huge, but it hasn’t happened yet.
- MT: A rather small extension in terms of instructions but a hugely significant one: This adds hardware multithreading to MIPS cores. It was first seen in MIPS Technologies’ 34-K family, launched in 2005. Appendix A gives an overview of the MIPS MT system.
- DSP: Like MDMX, this is a set of instructions held to be useful for audio/video processing, with saturating and SIMD arithmetic on small integers to the fore—but it already looks more useful in practice than MDMX. It was also new in 2005, available in MIPS Technologies’ 24-K and 34-K families. The DSP ASE was made public just in time to include a sketchy description here in section B.2.

There are some other optional parts of the MIPS32/64 specification that may not be regarded as primarily instruction set extensions:

- Floating point: The oldest are the best. Floating point has been an optional part of the MIPS instruction set since the earliest days. But it fits within the confines of the “coprocessor 1” encodings. It’s well described in this book.
- CP2: The second coprocessor encoding region is free for brave customers. But it’s a lot of design and testing work, and few try it. There’s nothing more about it here.
- CorExtend: MIPS Technologies’ best shot at a relatively easy-to-use user-definable instruction set. The idea was much hyped in 2002/2003, with announcements from companies such as ARM and Tensilica.
- EJTAG: An optional system to improve the debugging facilities, described in section 12.1.
- Paired single: An extension to the floating-point unit that provides SIMD operations, each of which works simultaneously on two single-precision (32-bit) sets of values.

- MIPS-3D: Usually associated with paired single, adds a handful of instructions to help out with the floating-point matrix operations heavily used by 3D scene rendering.

2.1 A Flavor of MIPS Assembly Language

Assembly language is the human-writable (and readable) version of the CPU's raw binary instructions, and Chapter 9 tells you more about how to understand it. Readers who have never seen any assembly language will find some parts of this book mystifying—but there's no time like now to start.

For readers familiar with assembly language, but not the MIPS version, here are some examples of what you might see:

```
# this is a comment

entrypoint:          # that's a label
    addu $1, $2, $3   # (registers) $1 = $2 + $3
```

Like most assembly languages, it is line oriented. The end of a line delimits instructions, and the assembly's native comment convention is that it ignores any text on a line beyond a “#” character. But it is possible to put more than one instruction on a line, separated by semicolons.

A label is a word followed by a colon “:”—*word* is interpreted loosely, and labels can contain all sorts of strange characters. Labels are used to define entry points in code and to name storage locations in data sections.

MIPS assembly programs interpret a rather stark language, full of register numbers. Most programmers use the C preprocessor and some standard header files so they can write registers by name; the names of the general-purpose registers reflect their conventional use (which we'll talk about in section 2.2). If you use the C preprocessor, you can also use C-style comments.

A lot of instructions are three-operand, as shown. The destination register is on the left (watch out, that's opposite to the Intel x86 convention). In general, the register result and operands are shown in the same order you'd use to write the operation in C or any other algebraic language, so:

```
subu $1, $2, $3
```

means exactly:

```
$1 = $2 - $3;
```

That should be enough for now.

2.2 Registers

There are 32 general-purpose registers for your program to use: **\$0** to **\$31**. Two, and only two, behave differently than the others:

- \$0** Always returns zero, no matter what you store in it.
- \$31** Is always used by the normal subroutine-calling instruction (**jal**) for the return address. Note that the call-by-register version (**jalr**) can use *any* register for the return address, though use of anything except **\$31** would be eccentric.

In all other respects, all these registers are identical and can be used in any instruction (you can even use **\$0** as the destination of instructions, though the resulting data will disappear without a trace).

In the MIPS architecture the program counter is not a register, and it is probably better for you not to think of it that way—in a pipelined CPU there are multiple candidates for its value, which gets confusing. The return address of a **jal** is the next instruction *but one* in sequence:

```
...
jal printf
mov $4, $6
xxx # return here after call
```

That makes sense, because the instruction immediately after the call is the call's delay slot—remember, the rules say it must be executed before the branch target. The delay slot instruction of the call is rarely wasted, because it is typically used to set up a parameter.

There are no condition codes; nothing in the status register or other CPU internals is of any consequence to the user-level programmer.

There are two register-sized result ports (called **hi** and **lo**) associated with the integer multiplier. They are not general-purpose registers, nor are they useful for anything except multiply and divide operations. However, there are instructions defined that insert an arbitrary value back into these ports—after some reflection, you may be able to see that this is required when restoring the state of a program that has been interrupted.

The floating-point math coprocessor (floating-point accelerator, or FPU), if available, adds 32 floating-point registers; in simple assembly language, they are called **\$f0** to **\$f31**.

Actually, for early 32-bit machines (conforming to MIPS I), only the 16 even-numbered registers were usable for math. But each even-numbered register can be used for either a single-precision (32-bit) or double-precision (64-bit) number; when you do double-precision arithmetic, register **\$f1** holds the remaining bits of the register identified as **\$f0**, and so on. In MIPS I programs

you only see odd-numbered math registers when you move data between integer and FPU registers or load/store floating-point register values—and even then the assembler helps you forget that complication.

MIPS32/64 CPUs have 32 genuine FP registers. But you may still meet software that avoids using the odd-numbered registers, preferring to maintain software compatibility with old CPUs. There's a mode bit in one of the control registers to make a modern FPU behave like an old one.

2.2.1 *Conventional Names and Uses of General-Purpose Registers*

We're a couple of pages into an architecture description and here we are talking about software. But I think you need to know this now.

Although the hardware makes few rules about the use of registers, their practical use is governed by a forest of conventions. The hardware cares nothing for these conventions, but if you want to be able to use other people's subroutines, compilers, or operating systems, then you had better fit in.

With the conventional uses of the registers go a set of conventional names. Given the need to fit in with the conventions, use of the conventional names is pretty much mandatory. The common names are listed in Table 2.1.

Somewhere about 1996 Silicon Graphics began to introduce compilers that use new conventions. The new conventions can be used to build programs that use 32-bit or 64-bit addressing, and in those two cases they are called, respectively, "n32" and "n64." We'll ignore them for now, but there is a terse but fairly complete description of them in section 11.2.8.

Conventional Assembly Names and Usages for Registers

- **at**: This register is reserved for the synthetic instructions generated by the assembler. Where you must use it explicitly (such as when saving or restoring registers in an exception handler), there's an assembly directive to stop the assembly from using it behind your back—but then some of the assembler's macro instructions won't be available.
- **v0, v1**: Used when returning non-floating-point values from a subroutine. If you need to return anything too big to fit in two registers, the compiler will arrange to do it in memory. See section 11.2.1 for details.
- **a0–a3**: Used to pass the first four non-FP parameters to a subroutine. That's an occasionally false oversimplification—see section 11.2.1 for the grisly details.
- **t0–t9**: By convention, subroutines may use these values without preserving them. This makes them a good choice for "temporaries" when evaluating expressions—but the compiler/programmer must remember that the values stored in them may be destroyed by a subroutine call.

TABLE 2.1 Conventional Names of Registers with Usage Mnemonics

<i>Register number</i>	<i>Name</i>	<i>Used for</i>
0	zero	Always returns 0
1	at	(assembly temporary) Reserved for use by assembly
2–3	v0, v1	Value returned by subroutine
4–7	a0–a3	(arguments) First few parameters for a subroutine
8–15	t0–t7	(temporaries) Subroutines can use without saving
24, 25	t8, t9	
16–23	s0–s7	Subroutine register variables; a subroutine that writes one of these must save the old value and restore it before it exits, so the <i>calling</i> routine sees the values preserved
26, 27	k0, k1	Reserved for use by interrupt/trap handler; may change under your feet
28	gp	Global pointer; some runtime systems maintain this to give easy access to (some) <i>static</i> or <i>extern</i> variables
29	sp	Stack pointer
30	s8/fp	Ninth register variable; subroutines that need one can use this as a frame pointer
31	ra	Return address for subroutine

- **s0–s8**: By convention, subroutines must guarantee that the values of these registers on exit are the same as they were on entry, either by not using them or by saving them on the stack and restoring them before exit. This makes them eminently suitable for use as register variables or for storing any value that must be preserved over a subroutine call.
- **k0, k1**: Reserved for use by an OS trap/interrupt handlers, which will use them and not restore their original value; they are of little use to anyone else.
- **gp**: This is used for two distinct purposes. In the kind of position-independent code (PIC) used by Linux applications, all out-of-module code and data references go through a table of pointers known as the GOT (for Global Offset Table). The **gp** register is maintained to point to the GOT. See Chapter 16 for details.

In regular non-PIC code (as is used by simpler embedded systems) **gp** is sometimes used to point to a link-time-determined location in the

midst of your static data. This means that loads and stores to data lying within 32 KB of either side of the **gp** value can be performed in a single instruction using **gp** as the base register.

Without the global pointer, loading data from a static memory area takes two instructions: one to load the most significant bits of the 32-bit constant address computed by the compiler and loader and one to do the data load.

To create **gp**-relative references, a compiler must know at compile time that a datum will end up linked within a 64-KB range of memory locations. In practice it can't know; it can only guess. The usual practice is to put small global data items (eight bytes and less in size) in the **gp** area and to get the linker to complain if it still gets too big.

- **sp**: It takes explicit instructions to raise and lower the stack pointer, so MIPS code usually adjusts the stack only on subroutine entry and exit; it is the responsibility of the subroutine being called to do this. **sp** is normally adjusted, on entry, to the lowest point that the stack will need to reach at any point in the subroutine. Now the compiler can access stack variables by a constant offset from **sp**. Once again, see section 11.2.1 for conventions about stack usage.
- **fp**: Also known as **s8**, a frame pointer will be used by a subroutine to keep track of the stack if for any reason the compiler or programmer was unable or unwilling to compute offsets from the stack pointer. That might happen, in particular, if the program does things that involve extending the stack by an amount that is determined at run time. Some languages may do this explicitly; assembly programmers are always welcome to experiment; and C programs that use the `alloca()` library routine will find themselves doing so.

If the stack bottom can't be computed at compile time, you can't access stack variables from **sp**, so **fp** is initialized by the function prologue to a constant position relative to the function's stack frame. Cunning use of register conventions means that this behavior is local to the function and doesn't affect either the calling code or any nested function calls.

- **ra**: On entry to any subroutine, return address holds the address to which control should be returned—so a subroutine typically ends with the instruction `jr ra`. In theory, any register could be used, but some sophisticated CPUs use optimization (branch prediction) tricks, which work better if you use a `jr ra`.

Subroutines that themselves call subroutines must first save **ra**, usually on the stack.

There is a corresponding set of standard uses for floating-point registers too, which we'll summarize in section 7.5. We've described here the original conventions promulgated by MIPS; some evolution has occurred in recent times and

the old conventions are called *o32*—but we won’t discuss newer conventions until section 11.2.8.

2.3 Integer Multiply Unit and Registers

The MIPS architects decided that integer multiplication was important enough to deserve a hardwired instruction. This was not universal in 1980s, RISCs. One alternative was to implement a multiply step that fits in the standard integer execution pipeline but mandates software routines for every nontrivial multiplication; early SPARC CPUs did just that.

Another way of avoiding the complexity of the integer multiplier would be to perform integer multiplication in the floating-point unit—a good solution used in Motorola’s short-lived 88000 family—but that would compromise the optional nature of the MIPS floating-point coprocessor.

Instead, a MIPS CPU has a special-purpose integer multiply unit, which is not quite integrated with the main pipeline. The multiply unit’s basic operation is to multiply two register-sized values together to produce a twice-register-sized result, which is stored inside the multiply unit. The instructions **mfhi**, **mflo** retrieve the result in two halves into specified general registers.

Since multiply results are not returned so fast as to be automatically available for any subsequent instruction, the multiply result registers are and always were *interlocked*. An attempt to read out the results before the multiplication is complete results in the CPU being stopped until the operation completes.

The integer multiply unit will also perform an integer division between values in two general-purpose registers; in this case the **lo** register stores the result (quotient) and the **hi** register stores the remainder.

You don’t get a multiply unit answer out in one clock: multiply takes 4–12 clock cycles and division 20–80 clock cycles (it depends on the implementation, and for some implementations it depends on the size of the operands). Some CPUs have fully or partially pipelined multiply units—that is, they can start a multiply operation every one or two clocks, even though the result will not arrive for four or five clocks.

MIPS32/64 includes a three-operand **mul** instruction, which returns the low half of the multiply result to a general-purpose register. But that instruction must stall until the operation is finished; highly tuned software will still use separate instructions to start the multiply and retrieve the results. MIPS32/64 CPUs (and most other CPUs still on the market) also have multiply-accumulate operations, where products from successive multiply operations are accumulated inside the **lo/hi** pair.

Integer multiply and divide operations never produce an exception: not even divide by zero (though the result you get from that is unpredictable). Compilers will often generate additional instructions to check for and trap on errors, particularly on divide by zero.

Instructions **mt_{hi}**, **mt_{lo}** are defined to set up the internal registers from general-purpose registers. They are essential to restore the values of **hi** and **lo** when returning from an exception but probably not for anything else.

2.4 Loading and Storing: Addressing Modes

As mentioned previously, there is only one addressing mode.¹ Any load or store machine instruction can be written:

lw \$1, offset(\$2)

You can use any registers for the destination and source. The offset is a signed, 16-bit number (and so can be anywhere between -32768 and 32767); the program address used for the load is the sum of **\$2** and the offset. This address mode is normally enough to pick out a particular member of a C structure (offset being the distance between the start of the structure and the member required). It implements an array indexed by a constant; it is enough to reference function variables from the stack or frame pointer and to provide a reasonable-sized global area around the **gp** value for static and extern variables.

The assembler provides the semblance of a simple direct addressing mode to load the values of memory variables whose address can be computed at link time.

More complex modes such as double-register or scaled index must be implemented with sequences of instructions.

2.5 Data Types in Memory and Registers

MIPS CPUs can load or store between one and eight bytes in a single operation. Naming conventions used in the documentation and to build instruction mnemonics are as follows.

2.5.1 *Integer Data Types*

Byte and halfword loads come in two flavors. Sign-extending instructions **lb** and **lh** load the value into the least significant bits of the 32-bit register but fill the high-order bits by copying the sign bit (bit 7 of a byte, bit 15 of a halfword). This correctly converts a signed integer value to a 32-bit signed integer, as shown in the following chart.

1. Not entirely true any more; now there is a register+register mode available for floating-point load and store.

<i>C name</i>	<i>MIPS name</i>	<i>Size (bytes)</i>	<i>Assembly mnemonic</i>
long long	dword	8	“d” as in ld
int	word	4	“w” as in lw
long ²			
short	halfword	2	“h” as in lh
char	byte	1	“b” as in lb

The unsigned instructions **lb** and **lbu** zero-extend the data; they load the value into the least significant bits of a 32-bit register and fill the high-order bits with zeros.

For example, if the byte-wide memory location whose address is in **t1** contains the value `0xFE` (−2, or 254 if interpreted as unsigned), then:

```

lb      t2, 0(t1)
lbu     t3, 0(t1)

```

will leave **t2** holding the value `0xFFFF.FFFE` (−2 as signed 32-bit value) and **t3** holding the value `0x0000.00FE` (254 as signed or unsigned 32-bit value).

The above description relates to MIPS machines considered as 32-bit CPUs, but many have 64-bit registers. It turns out that *all* partial-word loads (even unsigned ones) *sign-extend* into the top 32 bits; this behavior looks bizarre but is helpful, as is explained in section 2.7.3.

Subtle differences in the way shorter integers are extended to longer ones are a historical cause of C portability problems, and the modern C standards have very definite rules to remove possible ambiguity. On machines like the MIPS, which cannot do 8- or 16-bit precision arithmetic directly, expressions involving `short` or `char` variables require the compiler to insert extra instructions to make sure things overflow when they should; this is nearly always undesirable and rather inefficient. When porting code that uses small integer variables to a MIPS CPU, you should consider identifying which variables can be safely changed to `int`.

2.5.2 *Unaligned Loads and Stores*

Normal loads and stores in the MIPS architecture must be aligned; halfwords may be loaded only from two-byte boundaries and words only from four-byte boundaries. A load instruction with an unaligned address will produce a trap.

2. Nothing is simple. Recent MIPS compilers offering 64-bit pointers interpret the `long` data type as 64 bits (it’s good practice for a C compiler that a `long` should be big enough to hold a pointer).

Because CISC architectures such as the MC680x0 and Intel x86 do handle unaligned loads and stores, you may come across this as a problem when porting software; in extremity, you may even decide to install a trap handler that will emulate the desired load operation and hide this feature from the application—but that’s going to be horribly slow unless the references are very rare.

All data items declared by C code will be correctly aligned.

Where you know in advance that you want to code a transfer from an address whose alignment is unknown and that may turn out to be unaligned, the architecture does allow for a two-instruction sequence (much more efficient than a series of byte loads, shifts, and adds). The operation of the constituent instructions is obscure and hard to grasp, but they are normally generated by the macro-instruction **ulw** (unaligned load word). They’re described fully in section 8.5.1.

A macro-instruction **ulh** (unaligned load half) is also provided and is synthesized by two loads, a shift, and a bitwise “or” operation.

By default, a C compiler takes trouble to align all data correctly, but there are occasions (e.g., when importing data from a file or sharing data with a different CPU) when being able to handle unaligned integer data efficiently is a requirement. Most compilers permit you to flag a data type as potentially unaligned and will generate (reasonably efficient) special code to cope—see section 11.1.5.

2.5.3 *Floating-Point Data in Memory*

Loads into floating-point registers from memory move data without any interpretation—you can load an invalid floating-point number (in fact, an arbitrary bit pattern) and no FP error will result until you try to do arithmetic with it.

On 32-bit processors, this allows you to load single-precision values by a load into an even-numbered floating-point register, but you can also load a double-precision value by a macro instruction, so that on a 32-bit CPU the assembly instruction:

```
l.d      $f2, 24(t1)
```

is expanded to two loads to consecutive registers:

```
lwc1     $f2, 24(t1)
lwc1     $f3, 28(t1)
```

On a 64-bit CPU, **l.d** is the preferred alias for the machine instruction **ldc1**, which does the whole job.

Any C compiler that complies with the MIPS/SGI rules aligns eight-byte-long double-precision floating-point variables to eight-byte boundaries. The 32-bit hardware does not require this alignment, but it’s done for forward compatibility: 64-bit CPUs will trap if asked to load a double from a location that is not eight-byte aligned.

2.6 Synthesized Instructions in Assembly Language

MIPS machine code might be rather dreary to write; although there are excellent architectural reasons why you can't load a 32-bit constant value into a register with a single instruction, assembly programmers don't want to think about it every time. So the GNU assembler (and other good MIPS assemblers) will synthesize instructions for you. You just write an `li` (load immediate) instruction and the assembler will figure out when it needs to generate two machine instructions.

This is obviously useful, but it can be abused. Some MIPS assemblers end up hiding the architecture to an extent that is not really necessary. In this book, we will try to use synthetic instructions sparingly, and we will tell you when it happens. Moreover, in the instruction tables, we will consistently distinguish between synthetic and machine instructions.

It is my feeling that these features are there to help human programmers and that serious compilers should generate instructions that are one-for-one with machine code.³ But in an imperfect world many compilers will in fact generate synthetic instructions.

Helpful things the assembler does include the following:

- *A 32-bit load immediate:* You can code a load with any value (including a memory location that will be computed at link time), and the assembler will break it down into two instructions to load the high and low half of the value.
- *Load from memory location:* You can code a load from a memory-resident variable. The assembler will normally replace this by loading a temporary register with the high-order half of the variable's address, followed by a load whose displacement is the low-order half of the address. Of course, this does not apply to variables defined inside C functions, which are implemented either in registers or on the stack.
- *Efficient access to memory variables:* Some C programs contain many references to `static` or `extern` variables, and a two-instruction sequence to load/store any of them is expensive. Some compilation systems, with the help of some runtime support, get around this. Certain variables are selected at compile/assemble time (most commonly the assembler selects variables that occupy eight or less bytes of storage) and are kept together in a single section of memory that must end up smaller than 64 KB. The runtime system then initializes one register—`$28` or `gp` by convention—to point to the middle of this section.

3. This principle was behind the MIPS back end being reworked for the GNU C compiler in 2003/2004.

Loads and stores to these variables can now be coded as a single **gp**-relative load or store.

- *More types of branch conditions:* The assembler synthesizes a full set of branches conditional on an arithmetic test between two registers.
- *Simple or different forms of instructions:* Unary operations such as **not** and **neg** are produced as a **nor** or **sub** with the zero-valued register **\$0**. You can write two-operand forms of three-operand instructions and the assembler will put the result back into the first-specified register.
- *Hiding the branch delay slot:* In normal coding the assembler will move the instruction you wrote before the branch around into the delay slot if it can see it is safe to do so. The assembler can't see much, so it is not very good at filling branch delays. An assembly directive **.set noreorder** is available to tell the assembler that you're in control and it must not move instructions about.
- *Hiding the load delay:* Some assemblers may detect an attempt to use the result of a load in the immediately succeeding instruction and may move an instruction up or back in sequence if it can.
- *Unaligned transfers:* The unaligned load/store instructions (**ulh**, **ulw**, etc.) will fetch halfword and word quantities correctly, even if the target address turns out to be unaligned.
- *Other pipeline corrections:* Some instructions (such as those that use the integer multiply unit) have additional constraints on some old CPUs. If you have such an old CPU, you may find that your assembler helps out.

In general, if you really want to correlate assembly source language (not enclosed by a **.set noreorder**) with instructions stored in memory, you need help. Use a disassembler utility.

2.7 MIPS I to MIPS64 ISAs: 64-Bit (and Other) Extensions

The MIPS architecture has grown since its invention—notably, it's grown from 32 to 64 bits. That growth has been done so neatly that it would be quite possible to describe contemporary MIPS as a 64-bit architecture with a well-defined 32-bit subset for lower-cost implementations. We haven't quite done that, for several reasons. First, that is not how it happened, so such a description is in danger of mystifying you. Second, one of the lessons that MIPS has to offer the world is the art of extending an architecture nicely. And third, the material in this book was in fact written about 32-bit MIPS before it was extended to encompass 64 bits.

So the approach is a hybrid one. We will usually introduce the 32-bit version first, but once we get down to the details we'll handle both versions together. We'll use the acronym *ISA* for the long-winded term *instruction set architecture*.

Once MIPS started to evolve, the ISA of the original 32-bit MIPS CPUs (the R2000, R3000, and descendants) was retrospectively called MIPS I.⁴ The next variant to be widely used is a substantial enhancement that leads to a complete 64-bit ISA for the R4000 CPU and its successors; this was called MIPS III.

Although there has been much ISA evolution, at least at the application level (all the code that you can see when writing applications on a workstation), the newer instruction sets are generally backward compatible, fully supporting programs written for old ones. 32-bit ISAs obviously don't run 64-bit programs; otherwise the only ISA version that could cause you trouble is MIPS V, some of which is not available in MIPS64. But then it was never implemented, either.

There was a MIPS II, but it came to nothing because its first implementation (the R6000) ended up being overtaken by the MIPS III R4000. However, MIPS II was very close to being the same as the subset of MIPS III that you get by leaving out the 64-bit integer operations. This interpretation of the MIPS II ISA made a bit of a comeback in the 1990s as the ISA of choice for new implementations of 32-bit MIPS CPUs.

As we mentioned previously, the different ISA levels define whatever they define; at a minimum, they define all the instructions usable by a user-level program in a protected operating system—which includes the floating-point operations.⁵ To go with the instructions, the ISA defines and describes the integer, floating-point data, and floating-point control register.

Prior to MIPS32/64, ISA definitions were careful to exclude the CPU control (coprocessor 0) registers and the whole CPU control instruction set. I don't know how much this helps, though it does create employment for MIPS consultants by concealing information; a book called *The MIPS IV Instruction Set* is no good if you want to know how to program the cache on an R5000.

In practice, coprocessor 0 has evolved in step with the formal ISA, and like the formal ISA there were two major variants: One associated with the R3000—now pretty much obsolete—and the other deriving from the very first MIPS III CPU, the R4000. The R4000 family approach has now been standardized as MIPS64. I'll refer to these family groups as “R3000-style” and “MIPS32/64,” respectively. But you won't find much material about R3000-style in this edition of the book.

4. This is similar to a film fan asking whether you've seen *Terminator 1*, even though there never was a film called that. Even Beethoven's Symphony No. 1 was once called *Beethoven's Symphony*.

5. But it's always been possible to make a CPU that doesn't implement floating point.

2.7.1 *To 64 Bits*

With the introduction of the R4000 CPU in 1990, MIPS became the first 64-bit RISC architecture to reach production. The MIPS64 ISA⁶ defines 64-bit general-purpose registers, and some of the CPU control registers are wider than 32 bits. Moreover, all operations produce 64-bit results, though some of the instructions carried forward from the 32-bit instruction set do not do anything useful on 64-bit data. New instructions are added where the 32-bit operation can't be compatibly extended to do the corresponding job for 64-bit operands.

Although a MIPS32 CPU is permitted to have a 32-bit-only FPU, so far none has done so. MIPS32 and MIPS64 CPUs have FPUs with real 64-bit FP registers, so you don't need a pair of them to hold a double-precision value any more. This extension is incompatible with the old MIPS I model (which had 32 32-bit registers but used in pairs so you seemed to have 16 64-bit registers), so a mode switch in a CPU control register can be set to make the registers behave like a MIPS I CPU and allow the use of old software.

2.7.2 *Who Needs 64 Bits?*

By 1996, 32 bits was no longer a big enough address space for the very largest workstation and server applications. Pundits seem to agree that programs grow bigger exponentially, doubling every 18 months or so. So long as this goes on, demand for address space is expanding at about $\frac{3}{4}$ of a bit per year. Genuine 32-bit CPUs (68020, i386) appeared to replace 16/20-bit machines somewhere around 1984—and indeed, 32 bits really began to look inadequate around 2002. If this makes MIPS's 1991 move seem premature, that's probably true—big MIPS proponent Silicon Graphics did not introduce its first 64-bit-capable OS into general use until 1995.

MIPS's early move was spurred by research interest in operating systems using large sparse virtual address spaces, which permit objects to be named by their virtual address over a long period of time. MIPS was by no means the most prestigious organization to be deceived about the rate at which operating systems would evolve; Intel's world-dominating 32-bit CPU range had to wait *11 years* before Windows 95 brought 32-bit operation to the mass market.

A side effect of the 64-bit architecture is that such a computer can handle more bits at once, which can speed up some data-intensive applications in graphics and imaging. It's not clear, though, whether this is really preferable to the multimedia instruction set extensions exemplified by Intel's MMX, which features wide data paths with a new set of wide registers, and some way of operating simultaneously on multiple one-byte or 16-bit chunks of that wide data. MIPS's DSP ASE is somewhat similar—see section B.2.

6. The R4000 is not 100 percent compatible with the later MIPS64 standard, but it's close, and we're going to focus on the difference between MIPS32 and MIPS64 in this section.

By 1996, any architecture with pretensions to longevity needed a 64-bit implementation. Maybe getting there early was not a bad thing.

The nature of the MIPS architecture—committed to a flat address space and the use of general-purpose registers as pointers—means that 64-bit addressing and 64-bit registers go together. Even where the long addresses are irrelevant, the increased bandwidth of the wide registers and ALU may be useful for routines that shovel a lot of data, which are often found in graphics or high-speed communication applications.

It's one of the signs of hope for the MIPS architecture (and certain other simpler RISC architectures) that the move to 64 bits makes segmentation (which still burden even 64-bit versions of x86 and PowerPC architectures) totally pointless.

2.7.3 *Regarding 64 Bits and No Mode Switch: Data in Registers*

The standard way to extend a CPU to new areas is to do what DEC did long ago when taking the PDP-11 up to the VAX and Intel did when going from the 8086 to the i286 and i386: They defined a mode switch in the new processor that, when turned on, makes the processor behave exactly like its smaller ancestor.

But mode switches are kludges and in any case are difficult to implement in a machine that is not microcoded. MIPS64 uses a different approach:

- All instructions from the 32-bit architecture are preserved.
- So long as you only run MIPS32 instructions you get 100 percent compatibility: Programs behave identically, and the low 32 bits of each MIPS64 64-bit register hold the same values as would have filled the corresponding MIPS32 register at this point in your program.
- As many as possible of the MIPS32 instructions are defined so as to be both compatible and still to be useful MIPS64 instructions.

The crucial decision (and an easy one, once you ask the right question) is: What shall be in the high-order 32 bits of a register when we're being MIPS32 compatible? There are a number of choices, but only a few are simple, and only one really makes sense.

We could simply decide that the high bits should be undefined; when you're being MIPS32 compatible, the high bits of registers can contain any old garbage. This is easy to achieve but fails the third test above: We would need separate MIPS32 and MIPS64 versions of test instructions and conditional branches (they test registers for equality, or for being negative, by looking at the top bit—and that's either bit 31 or bit 63).

A second and more promising option would be to decide that the high-register bits should remain zero while we're running MIPS32 instructions; but again, this means we'll have to double up tests for negatives and for comparisons

of negative numbers. Also, a MIPS64 **nor** instruction between two top-half-zero values doesn't naturally produce a top-half-zero value.

The third, and best, solution is to maintain the top half of the register full of copies of bit 31. If (when running only MIPS32 instructions) we ensure that each register contains the correct low 32 bits and the top half flooded with copies of bit 31, then all MIPS64 comparisons and tests are compatible with their MIPS32 versions—so we can go on using the MIPS32 encodings for those instructions. All bitwise logical instructions must work too (anything that works on bit 31 works the same on bits 32–63).

The successful candidate can be described by saying that you keep 32-bit values in registers by sign-extending them to 64 bits; but note that this is done without regard to whether the 32-bit value is being interpreted as signed or unsigned.

With that decided, MIPS64 needs new 64-bit versions of simple arithmetic (the MIPS32 **addu** instruction, when confronted by 32-bit overflow, has to produce the overflow value in the low half of the register, and bit 31 copies in the top half—not the same as a 64-bit add!). It also needs a load-64-bits and new shift instructions, but it's a modest enough set. Where new instructions are needed for 64-bit data they get a “d” for double in the instruction mnemonic, generating names like **daddu**, **dsub**, **dmult**, and **ld**.

Slightly less obvious is that the existing 32-bit load instruction **lw** is now more precisely described as load word signed, so a new zero-extending **lwu** appears. The number of instructions added is fattened by the need to support existing variants of load and store and (in the case of shift-by-a-constant) the need to use a different opcode to escape the limits of MIPS32's 5-bit shift amount field.

All MIPS64 instructions are listed in horrible detail in Chapter 8.

2.8 Basic Address Space

The way MIPS processors use and handle addresses is subtly different from that of traditional CISC CPUs, and we know that it causes confusion. Read the first part of this section carefully. We'll start off with the original 32-bit picture and then describe the 64-bit version—if you'll bear with me you'll see why.

Here are some guidelines. With a MIPS CPU the addresses you put in your programs are *never* the same as the physical addresses that come out of the chip (sometimes they're rather simply related, but they're not the same). We'll refer to them as *program addresses*⁷ and *physical addresses*, respectively.

A MIPS CPU runs at one of two privilege levels: user and kernel.⁸ We'll often talk about *user mode* and *kernel mode* for brevity, but it's a feature of the MIPS

7. A *program address* in this sense is exactly the same as a *virtual address*—but for many people *virtual address* suggests a lot of operating system complications that aren't relevant here.

8. MIPS CPUs after R4000 have a third *supervisor mode*; however, since all MIPS OSs so far have ignored it, we will mostly do so too.

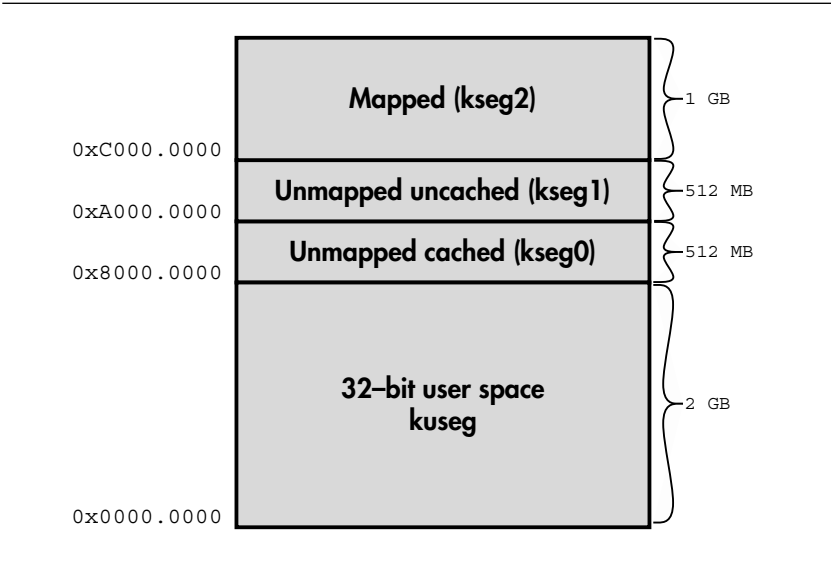


FIGURE 2.1 MIPS memory map: the 32-bit view.

architecture that the change from kernel to user never makes anything work differently; it just sometimes makes it illegal. At the user level, any program address with the most significant bit of the address set is illegal and causes a trap. Also, some instructions cause a trap in user mode.

In the 32-bit view (Figure 2.1), the program address space is divided into four big areas with traditional (and thoroughly meaningless) names; different things happen according to the area an address lies in, as follows:

kuseg `0x0000.0000–7FFF.FFFF` (low 2 GB): These are the addresses permitted in user mode. In machines with an MMU, they will always be translated (see Chapter 6). You should not attempt to use these addresses unless the MMU is set up. Some documentation calls this region “useg,” particularly when describing the address space seen by a user program. This book doesn’t use “useg” again.

For machines without an MMU, what happens is implementation defined; your particular CPU’s manual may tell you about something useful you could do with them. But if you want your code to be portable to and between MMU-less MIPS processors, avoid this area.

kseg0 `0x8000.0000–9FFF.FFFF` (512MB): These addresses are translated into physical addresses by merely stripping off the top bit and mapping them contiguously into the low 512 MB of physical memory. Since this is a trivial translation, these addresses are often called “untranslated,” but now you know better!

Addresses in this region are almost always accessed through the cache, so they may not be used until the caches are properly initialized. They will be used for most programs and data in systems not using the MMU and will be used for the OS kernel for systems that do use the MMU.

kseg1 0xA000.0000–BFFF.FFFF (512MB): These addresses are mapped into physical addresses by stripping off the leading 3 bits, giving a duplicate mapping of the low 512 MB of physical memory. But this time, access will not use the cache.

The kseg1 region is the only chunk of the memory map that is guaranteed to behave properly from system reset; that’s why the after-reset starting point (0xBFC0.0000) lies within it. The *physical* address of the starting point is 0x1FC0.0000—tell your hardware engineer.⁹

You will therefore use this region to access your initial program ROM, and most people use it for I/O registers. If your hardware designer proposes to map such things outside the low 512 MB of physical memory, apply persuasion.

kseg2 0xC000.0000–FFFF.FFFF (1GB): This area is only accessible in kernel mode but is once again translated through the MMU. Don’t access it before the MMU is set up. Unless you are writing a serious operating system, you will probably never have cause to use kseg2.

You’ll sometimes see this region divided into two halves called kseg2 and kseg3, to emphasize that the lower half (kseg2) is accessible to programs running in supervisor mode. If you ever use supervisor mode...

2.8.1 Addressing in Simple Systems

MIPS program addresses are never simply the same as physical addresses, but simple embedded software will probably use addresses in kseg0 and kseg1, where the program address is related in an obvious way to physical addresses.

Physical memory locations from 0x2000.0000 (512 MB) upward are not mapped anywhere in that simple picture, and most simple systems map everything below 512 MB. But if you need to, you can reach higher addresses by putting translation entries in the memory management unit (the TLB) or by using some of the extra spaces available in 64-bit CPUs.

2.8.2 Kernel versus User Privilege Level

With kernel privileges (where the CPU starts up) it can do anything. In user mode, program addresses above 2 GB (top bit set) are illegal and will cause a

9. The engineer wouldn’t be the first to have put the ROM at physical address 0xBFC0.0000 and found that the system wouldn’t bootstrap.

trap. Note that if the CPU has an MMU, this means that all user addresses must be translated by the MMU before reaching physical memory, giving an OS the power to prevent a user program from running amok. That means, though, that the user privilege level is redundant for a MIPS CPU running without a memory-mapped OS.

Also, in user mode some instructions—particularly the CPU control instructions an OS needs—become illegal.

Note that when you change the kernel/user privilege mode bit, it does not change the interpretation of anything—it just means that some things cease to be allowed in user mode. At kernel level, the CPU can access low addresses just as if they were in user mode, and they will be translated in the same way.

Note also that, though it can sound as if kernel mode is for operating systems writers and user mode is the simple everyday mode, the reverse is the truth. Simple systems (including many real-time operating systems) never leave MIPS kernel mode.

2.8.3 *The Full Picture: The 64-Bit View of the Memory Map*

MIPS addresses are always formed by adding a 16-bit offset to a value in a register. In 64-bit MIPS CPUs, the register always holds a 64-bit value, so there are 64 bits of program address. Such a huge space permits a rather cavalier attitude to chopping up the address space, and you can see how it's done in Figure 2.2.

The first thing to notice is that the 64-bit memory map is packed inside of the 32-bit map. That's an odd trick—like Doctor Who's TARDIS, the inside is much bigger than the outside—and it depends upon the rule we described in section 2.7.3: When emulating the 32-bit instruction set, registers always contain the 64-bit sign extension of the 32-bit value. As a result, a 32-bit program gets access to the lowest and highest 2 GB of the 64-bit program space. So the extended map assigns those lowest and highest regions to the same purpose as in the 32-bit version, and extension spaces are defined in between.

In practice, the vastly extended user space and supervisor-accessible spaces are not likely to be of much significance unless you're implementing a virtual memory operating system; hence, many MIPS64 users continue to define pointers as 32-bit objects. The large unmapped windows onto physical memory might be useful to overcome the 512-MB limit of `kseg0` and `kseg1`, but you can achieve the same effect by programming the memory manager unit (the TLB).

2.9 Pipeline Visibility

Any pipelined CPU hardware is always subject to timing delays for operations that won't fit into a strict one-clock-cycle regime. The designers of the architecture, though, get to choose which (if any) of these delays is visible to the programmer. Hiding timing foibles simplifies the programmer's model of what

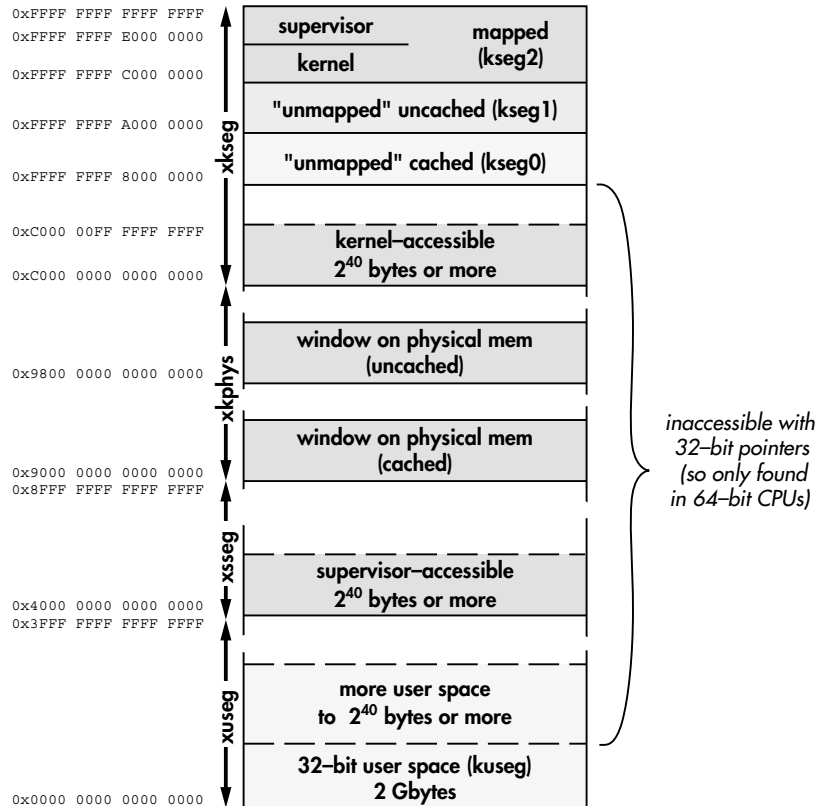


FIGURE 2.2 A 64-bit view of the memory map.

the CPU is doing, but it also loads complexity onto the hardware implementer. Leaving the scheduling problem to programmers and their software tools simplifies the hardware but can create development and porting problems.

As we've said several times already, the MIPS architecture leaves some pipeline artifacts visible and makes the programmer or compiler responsible for making the system work. The following points summarize where the pipeline shows up:

- **Branch delay:** In all MIPS CPUs, the instruction following any branch instruction (in the branch delay slot) is executed even though the branch is taken. The programmer or compiler should find a useful or at least harmless instruction for the branch delay slot—at worst, use a **nop**. But even the assembler will hide the branch delay from you unless you specify otherwise.

In the *branch likely* variant instructions, introduced as an option by the MIPS II instruction set, the delay slot instruction is executed *only* if the branch is taken; see section 8.5.4 for when this is useful.

- *Load delay:* I can't think of any MIPS CPU where the instruction immediately following a load could use the loaded data without causing a delay.¹⁰

But optimizing compilers and programmers should always be aware of how much time a particular CPU needs to get data ready to use—the load-to-use delay. Long load-to-use delays sap performance, and hardware designers will do a lot of work to ensure data is ready for the next-but-one instruction after the load.

- *Floating-point (coprocessor 1) foibles:* Floating-point computations nearly always take multiple clock cycles to complete, and typical MIPS FPU hardware has several somewhat independent pipelined units. MIPS hardware must hide the FPU pipeline; FP computations are allowed to proceed in parallel with the execution of later instructions, and the CPU is stalled if an instruction reads a result register before the computation finishes. Really heavyweight optimization requires the compiler to have tables of instruction repeat rates and latencies for each target CPU type, but you won't want to depend on those for the program to work at all.
- *CPU control instruction problems:* This is where life gets tricky. When you change CP0 fields like those in the CPU status register, you are potentially affecting things that happen at all pipeline stages.

With MIPS32/64 (at least Release 2) things are better. CP0 interactions have been divided into two. Those where the previous CP0 operation might affect the instruction fetch of a later instruction are the most troublesome and are called *instruction hazards*; the rest are *execution hazards*. Then you are provided with two flavors of *hazard barrier* instructions: There's a barrier suitable for execution hazards and a choice of souped-up branch instructions, which make you safe against instruction hazards too. You're guaranteed freedom from half-done CP0 side effects if you put an appropriate hazard barrier between the producer and consumer. See section 3.4.

Prior to the second revision of MIPS32/64, CP0 interactions were firmly consigned to being machine dependent. No hazard barrier instructions were formally available, and programmers must read the CPU manual to discover how to add enough padding instructions to make sure side effects are given time to propagate.

10. In very old MIPS I CPUs it was the programmer's or compilation tool's job to put in an explicit no-op, but you probably don't need to worry about that any more.

Coprocessor 0: MIPS Processor Control

In addition to its normal computational functions, any CPU needs units to handle interrupts, configuration options, and some way of observing or controlling on-chip functions like caches and timers. But it's difficult to do this in the neat implementation-independent way that the ISA does for the computational instruction set.

It would be desirable and easier for you to follow if we could introduce this through some chapters that separate out the different functions. We're going to do that. But we have to describe the common mechanisms used to implement these features first. You should read the first part of this chapter before tackling the next three chapters of this book; take particular note of the use of the word *coprocessor*, as explained on the next page.

So what jobs does CP0 on a MIPS CPU do?

- *CPU configuration*: MIPS hardware is often very flexible, and you may be able to select major features of the CPU (such as its endianness; see Chapter 10) or alter the way the system interface works. One or more internal registers provide control and visibility of these options.
- *Cache control*: MIPS CPUs have always integrated cache controllers, and all but the oldest integrate caches too. The CP0 **cache** instruction is used—in multiple different flavors—to manipulate cache entries. We'll talk about caches in Chapter 4.
- *Exception/interrupt control*: What happens on an interrupt or any exception, and what you do to handle it, are defined and controlled by CP0 registers and a few special instructions. This is described in Chapter 5.
- *Memory management unit control*: This is discussed in Chapter 6.

Special MIPS Use of the Word *Coprocessor*

The word *coprocessor* is normally used to mean an optional part of a processor that takes responsibility for some extension to the instruction set. The MIPS standard instruction set omits many features needed in any real CPU, but op-codes are reserved and instruction fields defined for up to four coprocessors.

One of these (coprocessor 1) is the floating-point coprocessor, which really is a coprocessor in anyone's language.

Another (coprocessor 0 or CP0) is described by MIPS as the *system control coprocessor*, and its instructions are essential to handle all those functions outside the responsibility of a user-mode program; they are the subject of this chapter.

Coprocessor 0 has no independent existence and is certainly not optional—you can't possibly make a MIPS CPU without a CPU status register, for example. But it does provide a standard way of coding the instructions that access the status register, so that, although the definition of the status register has evolved through different MIPS implementations, you can at least use the same assembly program for both CPUs.

The OS-only coprocessor 0 functions are deliberately corralled off from the MIPS ISA. When MIPS was young, system suppliers invariably shipped a customized operating system kernel, so changes in CP0 required changes in the kernel—but only in the kernel; it didn't affect application compatibility. So for MIPS I through MIPS V, the CP0 functions were regarded as implementation dependent.

Not any more. Multiple bodies now build OS code for MIPS, and it's a major headache if OS porting work is required between different MIPS CPUs. So the newer standards, MIPS32 and MIPS64, define the CP0 registers and functions in enough detail that you can build a portable OS.

Getting back to the architecture: Of the four coprocessor encoding spaces, CP3 has been invaded by floating-point instructions from MIPS32/64 and is now only usable where you are sure you'll never want to implement floating point. CP2 is available and occasionally used for custom ISA extensions or to provide application-specific registers in a few SoC applications. CP1 is the floating-point unit itself.

-
- *Miscellaneous:* There's always more: timers, event counters, parity error detection. Whenever additional functions are built into the CPU too tightly to be conveniently accessed as I/O devices, this is where they get attached.

We'll summarize everything found in MIPS32/64 CPUs in the second half of this chapter. But first, we'll leave aside what we're trying to do and look at the mechanisms we use to do it. There are relatively few CP0 instructions—wherever possible, low-level control over the CPU involves reading and writing bitfields within special CP0 registers.

Table 3.1 introduces the functions of CPU control registers. It lists every register that is required for compliance with MIPS32/64, and a few that are optional but common.

This is not a complete list; other registers are associated with optional extensions to the instruction set (ASEs) or with other optional features of MIPS32/64.

In addition, MIPS CPUs may have implementation-specific registers—this is a preferred way to add features to the MIPS architecture. Refer to your particular CPU's manuals.

To avoid burying you in detail at this stage, we’ve banished the bit-by-bit description of the CP0 registers required by MIPS32/64 to a separate section, section 3.3. You can skip over that section for now if you’re interested in going on to the following chapters.

While we’re listing registers, **k0** and **k1** (general purpose registers **\$26–27**) are worth a mention. These are two general-purpose registers reserved (by software convention) for use in exception handler code. It’s pretty much essential to reserve at least one register; the choice of which register is arbitrary, but it must be one that is embedded in all extant MIPS toolkits and binaries.

3.1 CPU Control Instructions

There are several special CPU control instructions used in the memory management implementation, but we’ll leave those until Chapter 6. MIPS32/64 defines a set of **cache** instructions that do everything required to manage caches, described in Chapter 4.

But those aside, MIPS CPU control requires very few instructions. Let’s start with the ones that give you access to all the registers we just listed:

```
mtc0    s, <n>    # Move to coprocessor 0
```

This instruction loads coprocessor 0 register number **n** from CPU general register **s**, with 32 bits of data (even in 64-bit CPUs many of the CP0 registers are only 32 bits long, but for the few long CP0 registers there’s the **dmtc0** instruction). This is the only way of setting bits in a CPU control register.

When MIPS was new, there could be up to 32 CP0 registers. But MIPS32/64 cater for up to 256, and for instruction backward-compatibility that’s been done by appending a 3-bit *select* field to the CP0 number (which is in fact encoded in a previously zero part of the instruction). So **mtc0 s, \$12, 1** is interpreted as accessing “register 12, select 1.” We’ll write that as **12.1**.

It is not good practice to refer to CPU control registers by their number in assembly programs; normally, you use the mnemonic names shown in Table 3.1. Most toolchains define these names in a C-style *include* file and arrange for the C preprocessor to be run as a front end to the assembler; see your toolkit documentation for guidance on how to do this. Although there’s a fair amount of influence from original MIPS standards, there is some variation in the names used for these registers. We’ll stick to the mnemonics shown in Table 3.1.

Getting data out of CP0 registers is the opposite:

```
mfc0    d, $n      # Move from coprocessor 0
```

d is loaded with the values from CPU control register number **n**, and that’s the only way of inspecting bits in a control register (though again, there’s a **dmfc0** variant for the few wide CP0 registers in 64-bit machines). So if you want to

TABLE 3.1 MIPS CPU Control Registers

<i>Register mnemonic</i>	<i>CP0 register No.</i>	<i>Description</i>
SR	12	The <i>Status Register</i> , which, perversely, consists mostly of writable control fields. Fields determine the CPU privilege level, which interrupt pins are enabled, and other CPU modes.
Cause	13	What caused that exception or interrupt?
EPC	14	<i>Exception Program Counter</i> : where to restart after exception /interrupt.
Count	9	Together, these form a simple but useful high-resolution interval timer, ticking at (usually) half the CPU pipeline clock rate.
Compare	11	
BadVAddr	8	The program address that caused the last address-related exception. Set by address errors of all kinds, even if there is no MMU.
Context	4	Registers for programming the memory management/translation hardware (the TLB), described in Chapter 6.
EntryHi	10	
EntryLo0-1	2-3	
Index	0	
PageMask	5	
Random	1	
Wired	6	
PRId	15	CPU type and revision level. The type number is managed by MIPS Technologies and should (at least) change when the coprocessor 0 register set changes. There's a list of values up to mid-2004 in Table 3.3.
Config	16	CPU setup parameters, usually system determined; some writable here, some read-only. Some CPUs have higher-numbered registers for implementation-specific purposes.
Config1-3	16.1-3	
EBase	15.1	Exception entry point base address and—for multi-CPU systems—CPU ID.
IntCtl	12.1	Setup for interrupt vector and interrupt priority features.
SRSCtl	12.2	Shadow register control, see section 5.8.6.
SRSMap	12.3	A map of eight shadow register numbers to be used with each of eight possible interrupt causes, when the CPU is using the “vectored interrupt” feature.
CacheErr	27	Fields for analyzing (and possibly recovering from) a memory error, for CPUs using error-correcting code on the data path.
ECC	26	See section 4.9.3 for more details.
ErrorEPC	30	

TABLE 3.1 *continued*

<i>Register mnemonic</i>	<i>CP0 register No.</i>	<i>Description</i>
TagLo	28.0	Registers for cache manipulation, described in section 4.9.
DataLo	28.1	
TagHi	29.0	
DataHi	29.1	
Debug	23.0	Registers for the EJTAG debug unit, described in section 12.1.7.
DEPC	24.0	
DESAVE	31.0	
WatchLo	18.0	Data watchpoint facility, which can cause an exception when the CPU attempts to load or store at this address—potentially useful for debugging. See section 12.2.
WatchHi	19.0	
PerfCtl	25.0	Performance counter registers (more control/count pairs at subsequent odd/even select numbers). See section 12.4.
PerfCnt	25.1	
LLAddr	17.0	Some CPUs (notably those with coherent caches or the multithreading extension) store an address associated with an ll (“load-linked”) instruction; when they do, it’s visible here, even though only diagnostic software will ever read it. See section 8.5.2.
HWREna	7.0	A writable bit map that determines which <i>hardware registers</i> will be accessible to user-privilege programs—see section 8.5.12.

update a single field inside—for example—the status register **SR** you’re usually going to have to code something like:

```

mfc0    t0, SR
and     t0, <complement of bits to clear>
or      t0, <bits to set>
mtc0    t0, SR

```

The last crucial component of the control instruction set is a way of undoing the effect of an exception. We’ll discuss exceptions in detail in Chapter 5, but the basic problem is shared by any CPU that can implement any kind of secure OS; the problem is that an exception can occur while running user (low-privilege) code, but the exception handler runs at high privilege.¹ So when returning from

1. Almost universally, CPUs use a software-triggered exception—a *system call*—as the only mechanism that user code can employ to invoke a service from the OS kernel (which runs at a higher privilege level).

the exception back to the user program, the CPU needs to steer between two dangers: On the one hand, if the privilege level is lowered before control returns to the user program, you'll get an instant and fatal second exception caused by the privilege violation; on the other hand, if you return to user code before lowering the privilege level, a malicious nonprivileged program might get the chance to run an instruction with kernel privileges. The return to user mode and the change of privilege level must be indivisible from the programming viewpoint (or *atomic*, in architecture jargon).

On all but the oldest MIPS CPUs the instruction **eret** does this job (on those long-lost CPUs you needed a jump instruction with an **rfe** in its delay slot). We'll go into the details in Chapter 5.

3.2 Which Registers Are Relevant When?

These are the registers you will need to consult in the following circumstances:

- *After power-up:* You'll need to set up **SR** to get the CPU into a workable state for the rest of the bootstrap process. It's common practice for the hardware to leave many register bits undefined following reset.

MIPS32 CPUs other than the earliest have one or more configuration registers: **Config** and **Config1–3**. Some CPUs may have more, with CPU-specific fields (**Config7** is sometimes used for particularly CPU-specific fields).

The first **Config** register has a few writable fields, which you may need to set before very much will work. Consult your hardware engineer about making sure that the CPU and system agree enough about configuration to get to the point of writing these registers!

- *Handling any exception:* In the early MIPS CPUs any exception (apart from one particular MMU event) invoked a single common “general exception handler” routine at a fixed address. But in the years since then many reasons have been found to add separate handlers for different purposes; see section 5.3.

On entry, no program registers have been saved, only the restart address in **EPC**. The MIPS hardware knows nothing about stacks. In any case, in a secure OS the privileged exception handler can't assume anything about the integrity of the user-level code—so, in particular, it can't assume that the stack pointer is valid or that stack space is available.

You need to use at least one of **k0** and **k1** to point to some memory space reserved to the exception handler. Now you can save things, using

Encoding of Control Registers

A note about reserved fields is in order here. Many unused control register fields are marked “0.” Bits in such fields are guaranteed to read zero, and it is harmless to write them (though the value written is ignored). Other reserved fields are

marked “reserved” or “x”; you should take care to always write such a field as either zero or a value you previously read from it. But you should not assume that you will get back zero or any other particular value.

the other **k0** or **k1** register to stage data from control registers where necessary.

Consult the **Cause** register to find out what kind of exception it was and dispatch accordingly.

- *Returning from exception:* Control must eventually be returned to the value stored in **EPC** on entry. Whatever kind of exception it was, you will have to adjust **SR** back when you return, restoring the user-privilege state, enabling interrupts, and generally unwinding the exception effect. Finally, the return-from-exception instruction **eret** combines the return to user space and resetting of **SR (EXL)**.
- *Interrupts:* **SR** is used to adjust the interrupt masks, to determine which (if any) interrupts will be allowed higher priority than the current one. The hardware offers no interrupt prioritization, but the software can do whatever it likes.
- *Instructions just there to cause exceptions:* These are often used (for system calls, debug breakpoints, and to emulate some kinds of instruction). All MIPS CPUs have implemented instructions called **break** and **syscall**; some implementations have added extra ones.

3.3 CPU Control Registers and Their Encoding

This section tells you about the format of the control registers, with a sketch of the function of each field. In most cases, more information about how things work is to be found in separate sections below. However, we’ve left the registers that are specific to the memory management system to Chapter 6, and those that are just for cache management in Chapter 4, where we’re dealing with caches in general.

Note that individual CPUs may define extra fields in some registers. The fields described here are those that are either compulsory in MIPS32/64 or that seem to be very commonly used.

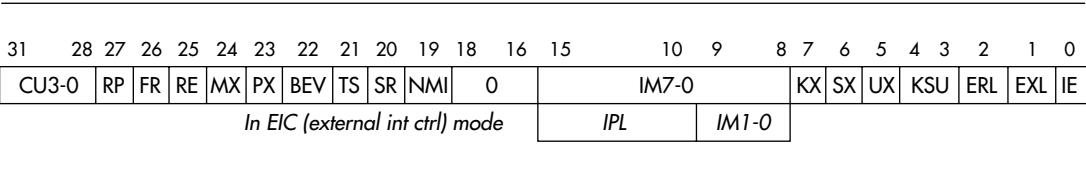


FIGURE 3.1 Fields in the **SR** register (status register).

3.3.1 *Status Register (SR)*

The MIPS CPU has remarkably few mode bits; those that exist are defined by fields in the very packed CPU status register **SR**, as shown in Figure 3.1. These are all the fields recognized by the MIPS32/64 standards; some of the spare fields may be used for implementation-dependent purposes.

We emphasize again that there are no nontranslated or uncached modes in MIPS CPUs; all translation and caching decisions are made on the basis of the program address.

Here are the critical shared fields; it would be very bad form for a new implementation to recycle any of them for any purpose, and they are probably now nailed down for the foreseeable future.

- CU3-0
- Each represents “coprocessor enable” for coprocessors 3–0, respectively.
- Coprocessor 1 is the FPU—so **CU1** is set 1 to use the FPU if you have it and 0 to disable it. When 0, all FPU instructions cause an exception. While it’s obviously a bad idea to enable FPU instructions if your CPU lacks FPU hardware, it can be useful to turn off an FPU even when you have one.²
- Setting the **CU0** bit has the unexpected effect of enabling user-privilege programs to run CP0 instructions. You probably don’t want to do that.
- CU3–2** control the usability of coprocessors 3 and 2, respectively. Cores from MIPS Technologies allow system builders to implement a CP2—that could be to add interesting new instructions, but it’s quite likely to be done just to get 32 additional, easily accessible registers.
- The coprocessor 3 decode space has been invaded by instructions from the standard MIPS32/64 floating-point instruction set, so is really no longer compatible with a standard CP1.

2. Why turn off a perfectly good FPU? Some operating systems disable FP instructions for every new task; if the task attempts some floating point, it will trap, and the FPU will be enabled for that task. But now we can distinguish tasks that never use floating-point instructions, and when such a task is suspended and restored we don’t need to save or restore the FP registers; that may save some time in crucial context-saving code.

RP	Reduced Power—but whether and how it does anything is CPU dependent. It might, for example, reduce the CPU's operating frequency, voltage, or both. Consult your CPU's manual and talk to your system designer.
FR	A mode switch: Set 1 to expose all 32 double-sized floating-point registers to software; set 0 to make them act as pairs of 32-bit registers, as found in MIPS I.
RE	(reverse endianness in user mode): The MIPS processors can be configured, at reset time, with either endianness (see section 10.2 if you don't know what that means). Since human beings are perverse, there are now two universes of MIPS implementation: DEC and Windows NT led off with little-endian, while SGI and their UNIX world was big-endian. Embedded applications originally showed a strong big-endian bias but are now thoroughly mixed. It could be a useful feature in an operating system to be able to run software from the opposite universe; the RE bit makes it possible. When RE is active, user-privilege software runs as if the CPU had been configured with the opposite endianness. However, achieving cross-universe running would require a large software effort as well, and to date nobody has done it.
MX	Enable for either the DSP or MDMX ASE (instruction set extensions)—you can't have both in the same CPU. At the time of writing, the DSP ASE is quite new, and MDMX appears to lack toolchain and middleware support.
PX	See description of SR (UX) below.
BEV	Boot exception vectors: When BEV == 1, the CPU uses the ROM (kseg1) space exception entry point (described in section 5.3). BEV is usually set to 0 in running systems.
TS	TLB shutdown: See Chapter 6 for details. On some CPUs, TS gets set if a program address simultaneously matches two TLB entries, which is certainly a sign of something horribly wrong in the OS software. Prolonged operation in this state, in some implementations, could cause internal contention and damage to some chips, so the TLB just switches off and ceases to match anything. TLB shutdown is terminal and can be cleared only by a hardware reset. Some MIPS CPUs have foolproof TLB hardware and may not implement this bit, or it may indicate an attempt to write a duplicate entry that was suppressed. Consult your CPU manual.
SR, NMI	Soft reset or nonmaskable interrupt occurred: MIPS CPUs offer several different grades of reset, distinguished by hardware signals. In particular, the configuration register Config retains its values across a soft reset but must be reprogrammed after a hard

reset. MIPS resets are somewhat like exceptions—though a soft or hard reset is an exception from which no CPU ever returns—and reset reuses a lot of the exception machinery.

The field **SR (SR)** is clear following a hard reset (one where all operating parameters are reloaded from scratch) but set following a soft reset or NMI. The field **SR (NMI)** is set only after an NMI exception.

IM7-0 Interrupt mask: An 8-bit field defining which interrupt sources, when active, will be allowed to cause an exception. Six of the interrupt sources are generated by signals from outside the CPU core (one may be used by the FPU, which, although it lives on the same chip, is logically external); the other two are the software-writable interrupt bits in the **Cause** register.

If you set your modern CPU to use EIC interrupts, then the interpretation of **SR (IM)** changes; see section 5.8.5.

Unless you're using the EIC system, no interrupt prioritization is provided for you: the hardware treats all interrupt bits the same. See section 5.8 for details.

UX, SX, KX Broadly speaking, these enable the much larger address space available with a 64-bit CPU. There are separate bits for the three different (user, supervisor, kernel) privilege levels; when the appropriate one is set, the most common memory translation exceptions (TLB misses) are redirected to a different entry point where the software will expect to deal with 64-bit addresses.

In addition, when **SR (UX)** is zero the CPU won't run 64-bit instructions from the MIPS64 ISA in user mode. This allows an OS to construct a user-mode “sandbox” within which a 32-bit program—even a defective program that executes what ought to be illegal instructions for a 32-bit CPU—will behave exactly as it would on a MIPS32 CPU. Combining these features may not always be good: If you want to use 64-bit instructions but still stick with 32-bit addressing in user mode, you can set **SR (PX)**.

KSU CPU privilege level: 0 for kernel, 1 for supervisor, 2 for user. Regardless of this setting, the CPU is in kernel mode whenever the EXL or ERL bits are set following an exception. The supervisor privilege level was introduced with the R4x00 but has never been used; see the sidebar for some background.

Some manuals document this field as two separate bits, with the top bit called **UM**.

ERL Error level: This gets set when the CPU discovers it has received bad data. MIPS CPUs can optionally receive and check extra parity or ECC (error-correcting code) bits with each chunk of data

Why Is There a Supervisor Mode?

The R3000 CPU offered only two privilege levels, which are all that is required by most UNIX implementations and all that has ever been used in any MIPS OS. So why did the R4000's designers go to considerable trouble to add a feature that has never been used?

In 1989–1990, one of the biggest successes for MIPS was the use of the R3000 CPU in DEC's DECstation product line, and MIPS wanted the R4000 to be selected as DEC's future workstation CPU. The competition was an in-house development that evolved into DEC's Alpha architecture, but they were coming from behind; R4000 was usable about 18 months before Alpha.

Whichever CPU was chosen had to run not only UNIX but DEC's minicomputer operating system, VMS.

Alpha's basic instruction set is almost identical to MIPS's; its biggest difference was the attempt to do

without any partial-word loads or stores (and that turned out to be DEC's mistake, fixed later).

In the end, it appears that the VMS software team was decisive in choosing Alpha over the R4000—they insisted that R4000's much simpler CPU control system would make VMS too insecure, or the port too long. Somewhere along the way they cited MIPS's two-level security system as a particular problem; the R4000's supervisor mode was MIPS's response.

I am deeply skeptical about the arguments; I think this was more down to good old-fashioned prejudice. Behind the technical smokescreen, perhaps DEC was right to believe that control over its microprocessor development was essential, but it's interesting to speculate how things might have turned out differently if DEC had stayed on board with the R4000 and exploited that 18-month leadership.

from cache or memory. Parity errors are generally fatal (unless there's a known good replacement for the data available). But ECC errors can be software-corrected, so long as no more than 1 (perhaps 2) bits have gone wrong.

When that happens, the CPU takes a parity/ECC error exception with this special bit set. This is handled separately from standard exceptions, because a correctable ECC error can happen anywhere—even in the most sensitive part of an ordinary exception routine—and if the system is aiming to patch up ECC errors and keep running, it must be able to fix them regardless of when they occur. That's challenging, since the ECC error exception routine has no registers it can safely use; with no registers to use as pointers, it can't start saving register values.

To get us out of this hole, **SR(ERL)** has drastic effects; all access to normal user-space-translated addresses disappears, and program addresses from 0 through `0x7FFF.FFFF` become uncached windows onto the same physical addresses. The intention is that the cache error exception handler can use base + offset addressing off the **zero** register to access some uncached memory space (reserved by the OS for this purpose), save some registers, and make itself room to run.

- EXL

Exception level: Set by any exception, this forces kernel mode and disables interrupts; the intention is to keep EXL on long enough for software to decide what the new CPU privilege level and interrupt mask are to be.
- IE

Global interrupt enable: Note that either ERL or EXL inhibit all interrupts, regardless.

3.3.2 Cause Register

Figure 3.2 shows the fields in the **Cause** register, which you consult to find out what kind of exception happened and use to decide which exception routine to call. **Cause** is a key register in exception handling and is little changed since the very earliest MIPS CPUs.

- BD

Branch delay: **EPC** is committed to being the address to which control should return after an exception. Normally, this also points at the exception victim instruction.

But when the exception victim is an instruction in the delay slot following a branch, **EPC** has to point to the branch instruction; it is harmless to re-execute the branch, but if you returned from the exception to the branch delay instruction itself, the branch would not be taken and the exception would have broken the interrupted program.

Cause (BD) is set whenever an exception occurs on an instruction in a delay slot and **EPC** points to the branch. You need only look at **Cause (BD)** if you want to analyze the exception victim instruction (if **Cause (BD)** == 1, then the instruction is at **EPC** + 4).
- TI

(Newer MIPS32/64 CPUs only)—this exception was caused by an interrupt from the internal timer.
- CE

Coprocesor error: If the exception is taken because a coprocesor format instruction was not enabled by the corresponding **SR (CUx)** field, then **Cause (CE)** has the coprocesor number from that instruction.

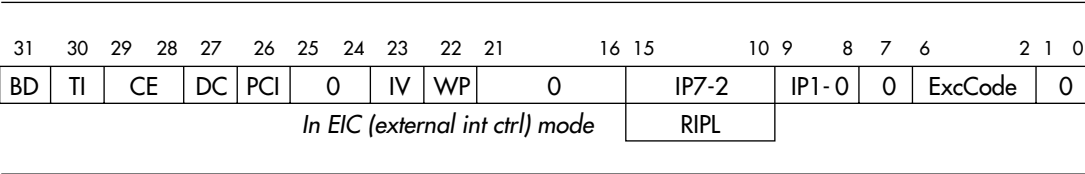


FIGURE 3.2 Fields in the **Cause** register.

DC	(Newer MIPS32/64 CPUs only)— <i>write</i> this bit to 1 to stop the Count register counting, which you might sometimes do to save power.
PCI	(Newer MIPS32/64 CPUs only)—a CP0 performance counter overflowed and generated this interrupt.
IV	<i>Write</i> this bit to 1 to use a special exception entry point for interrupts, as described in section 5.8.5.
WP	Reads 1 to remember that a watchpoint triggered when the CPU was already in exception mode (which suppresses the watchpoint exception). This bit will cause the CPU to take the watchpoint exception as soon as it returns from exception mode to normal operation, and it must be cleared by your watchpoint exception handler. <i>Write</i> this bit to 1 to use a special exception entry point for interrupts, as described in section 5.8.5.
IP7-0	Interrupt pending: Shows you the interrupts that want to happen. Cause (IP7-2) follow the CPU hardware input signals, whereas Cause (IP1-0) (the software interrupt bits) are readable/writable and contain whichever value you last wrote to them. Any of these 8 bits that are active when enabled by the appropriate SR (IM) bit (and subject to all the other conditions that inhibit interrupts) will cause an interrupt. Cause (IP) is subtly different from the rest of the Cause register fields: It doesn't tell you what happened when the exception took place; instead, it tells you what is happening now. Note that the interpretation of Cause (IP7-2) changes when you use the EIC interrupt system, described in section 5.8.5.
ExcCode	This is a 5-bit code that tells you what kind of exception happened, as detailed in Table 3.2. This table needs to be here, but many of the conditions it describes have not been mentioned in this book up to this point. For now, please regard this as reference material; it may all make sense later!

3.3.3 *Exception Restart Address (EPC) Register*

This is just a register that holds the address of the return point for this exception. The instruction causing (or suffering) the exception is at **EPC**, unless **Cause (BD)**, in which case **EPC** points to the previous (branch) instruction. **EPC** is 64 bits wide if the CPU is.

TABLE 3.2 ExcCode Values: Different Kinds of Exceptions

<i>ExcCode value</i>	<i>Mnemonic</i>	<i>Description</i>
0	Int	Interrupt.
1	Mod	Store, but page marked as read-only in the TLB.
2	TLBL	No TLB translation (read or store, respectively). That is, no valid entry in the TLB matches the program address used.
3	TLBS	When there is no matching entry at all (not even an invalid one) and the CPU is not already in exception mode—i.e., SR (EXL) set—this is a TLB miss, which is handled through a special exception entry point to streamline handling this common event.
4	AdEL	Address error (on load/I-fetch or store, respectively): This is either an attempt to get outside kuseg when in user mode or an attempt to read a doubleword, word, or halfword at a misaligned address.
5	AdES	
6	IBE	Bus error (instruction fetch or data read, respectively): External hardware has signaled an error of some kind; what you have to do about it is system dependent. A bus error on a store can only come about indirectly, as a result of a cache read to obtain the cache line to be written.
7	DBE	
8	Syscall	Executed a syscall instruction.
9	Bp	Executed a break instruction, used by debuggers.
10	RI	Instruction code not recognized (or not legal).
11	CpU	Tried to run a coprocessor instruction, but the appropriate coprocessor is not enabled in SR (CU3–0) . In particular, this is the exception you get from a floating-point operation if the FPU usable bit, SR (CU1) , is not set; hence, it is where floating-point emulation starts.
12	Ov	Overflow from trapping form of integer arithmetic instructions—possible, for example, with add but not with addu . C programs don't use overflow-trapping instructions.
13	TRAP	Condition met on one of the conditional trap instructions teq , etc.
14		Now unused. On some older CPUs with L2 caches this was used when hardware detected a possible cache alias, explained in section 4.12.
15	FPE	Floating-point exception. (In some very old CPUs, floating-point exceptions are signaled as interrupts.)

TABLE 3.2 *continued*

<i>ExcCode value</i>	<i>Mnemonic</i>	<i>Description</i>
16–17	–	Custom exception types, implementation dependent.
18	C2E	Exception from coprocessor 2 (which, if fitted, will be a custom extension to the instruction set).
19–21	–	Reserved for future expansion.
22	MDMX:	Tried to run an MDMX instruction, but SR (MX) wasn't set (most likely, the CPU doesn't do MDMX).
23	Watch	Physical address of load/store matched enabled value in WatchLo/WatchHi registers.
24	MCheck	Machine check—CPU detected some disastrous error in the CPU control system. Some MIPS Technologies cores take this exception if you load a second translation matching the same program address into the TLB.
25	Thread	Thread-related exception, as described in Appendix A. There's another register field, VPEControl (EXCPT) , that tells you more details about a thread-related exception.
26	DSP	Tried to run a DSP ASE instruction, but either this CPU does not support DSP instructions or SR (MMX) isn't set to enable DSP.
27–29	–	Reserved for future expansion.
30	CacheErr	Parity/ECC error somewhere in the core, on either instruction fetch, load, or cache refill. Such errors have their own (uncached-space) exception entry point. In fact you never see this value in Cause (ExcCode) ; but some of the codes in this table, including this one, can be visible in the debug mode of the EJTAG debug unit—see section 12.1, and in particular the notes on the Debug register.
31	–	Now unused, but historical use like bit 14, above.

3.3.4 *Bad Virtual Address (BadVAddr) Register*

This register holds the address whose use led to an exception; it is set on any MMU-related exception, on an attempt by a user program to access addresses outside kuseg, or if an address is wrongly aligned. After any other exception it is undefined. Note in particular that it is not set after a bus error. **BadVAddr** is 64 bits wide if the CPU is.

3.3.5 *Count/Compare Registers: The On-CPU Timer*

These registers provide a simple general-purpose interval timer that runs continuously and that can be programmed to interrupt. The interrupt usually comes out of the CPU and is wired back to an interrupt by some system-dependent mechanism—but see the `IntCtl` register for how to find out.

`Count` is a 32-bit counter that counts up continually, at the CPU’s pipeline clock rate, half the rate or (rarely) some other divider. You can find out the counter rate by reading a hardware register, as described in section 8.5.12.

When `Count` reaches the maximum 32-bit unsigned value, it overflows quietly back to zero. You can read `Count` to find the current “time.” You can also write `Count` at any time—but it’s normal practice not to do so.

`Compare` is a 32-bit read/write register. When `Count` increments to a value equal to `Compare`, the interrupt is raised. The interrupt remains asserted until cleared by a subsequent write to `Compare`.

To produce a periodic interrupt, the interrupt handler should always increment `Compare` by a fixed amount (not an increment to `Count`, because the period would then get stretched slightly by interrupt latency). The software needs to check for the possibility that a late interrupt response might lead it to set `Compare` to a value that `Count` has already passed; typically, it rereads `Count` after writing `Compare`.

3.3.6 *Processor ID (PRId) Register*

Figure 3.3 shows the layout of the `PRId` register, a read-only register to be consulted to identify your CPU. `CPU Id` should change—at least—when there’s a change in either the instruction set or the CPU control register definitions. `Revision` is strictly manufacturer dependent and wholly unreliable for any purpose other than helping a CPU vendor to keep track of silicon revisions.

The `Company ID` field—values are available from MIPS Technologies—is relatively recent, so historical CPUs have it set to zero. The `Company Options` field is defined only in your CPU manual.

Some `CPU Id` settings we know about are listed in Table 3.3.

If you want to print out the values, it is conventional to print them out as “x.y,” where x and y are the decimal values of `CPU ID` and `Revision`,

31	24	23	16	15	8	7	0
Company Options		Company ID		CPU ID		Revision	

FIGURE 3.3 `PRId` register fields.

TABLE 3.3 MIPS CPU Implementation Numbers in **PRId (Imp)**

<i>Id</i>	<i>CPU type</i>	<i>Id</i>	<i>CPU type</i>
1	R2000, R3000	128	MIPS 4KC
2	IDT R305x family	129	MIPS 5KC
3	R6000	130	MIPS 20KC
4	R4000, R4400	131	MIPS 4KMP
5	Early LSI Logic 32-bit CPUs	132	MIPS 4KEc
6	R6000A	133	MIPS 4KEmp
7	IDT R3041	134	MIPS 4KSc
9	R10000	135	MIPS M4K
10	NEC Vr4200	136	MIPS 25Kf
11	NEC Vr4300	137	MIPS 5KE
12	NEC Vr41xx family	144	MIPS 4KEc (MIPS32R2 compliant)
16	R8000	145	MIPS 4KEmp (MIPS32R2 compliant)
32	R4600	146	MIPS 4KSd
33	IDT R4700	147	MIPS 24K
34	Toshiba R3900 family	149	MIPS 34K
35	R5000	150	MIPS 24KE
40	QED RM5230, RM5260		

respectively. Avoid using the contents of this register to establish parameters (like cache size, speed, and so on) or to establish the presence or absence of particular features; some features have a standard encoding in one of the **Config** registers, or you can design code sequences to probe for the existence of individual features.

3.3.7 *Config Registers: CPU Resource Information and Configuration*

The MIPS32/64 standard defines four configuration registers for initialization software to use: **Config** and **Config1–3**. Most of the register fields are read-only fields, which software interrogates to discover relevant information about the CPU hardware; but (particularly in the original **Config** register—which for historical reasons is *not* called “Config0”) there are also some writable fields, which select some CPU options that are likely to be made just one way for any system.

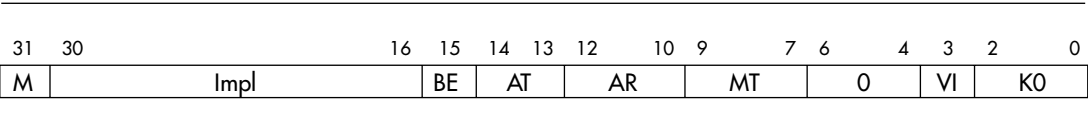


FIGURE 3.4 Fields in MIPS32/64 **Config** register.

All MIPS32/64-compliant CPUs have the **Config** register, as shown in Figure 3.4, with the following fields:

- MContinuation bit—reads 1 if there’s at least one more configura-
tion register (i.e. **Config1**) available.
- ImplImplementation-dependent configuration flags. Most CPUs will
have some custom fields here, both read-only for information
and writable for system setup. Look at your CPU’s manual.
- BEReads 1 for big-endian, 0 for little-endian.
- ATMIPS32 or MIPS64? Encoding:
0 MIPS32
1 MIPS64 instruction set but MIPS32 address map
2 MIPS64 instruction set with full address map
- ARArchitecture revision level:
0 MIPS32/MIPS64 Release 1
1 MIPS32/MIPS64 Release 2
This book is based on MIPS32/MIPS64 Release 2.
- MTMMU type:
0 None
1 MIPS32/64-compliant TLB
2 BAT type
3 MIPS32-standard FMT fixed mapping
The BAT type is for historical compatibility with some old CPUs
and is perhaps unlikely to be met with.
- VISet 1 if the L1 I-cache is indexed and tagged with virtual (pro-
gram) addresses. A virtual I-cache requires special care by oper-
ating systems.
- K0Writable field determining whether the fixed kseg0 region is cach-
ed or uncached. And if cached, how exactly does it behave? This
field is encoded just like the cache options field of a TLB entry, as
seen in **EntryLo0–1 (C)** and described in the notes to Figure 6.3.

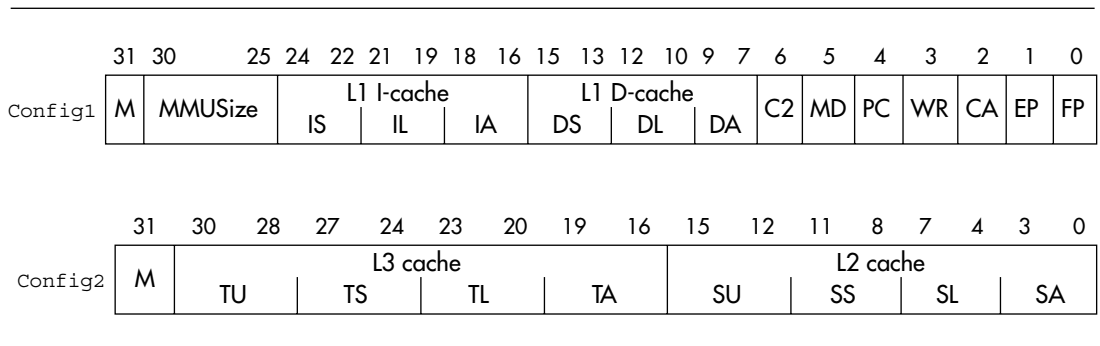


FIGURE 3.5 Fields in the MIPS32/64 **Config1–2** registers.

Figure 3.5 has the following fields:

- | | |
|------------------------|--|
| Config1(M) | Continuation bit, 1 if Config2 is implemented. |
| Config1(MMUSize) | The size of the TLB array (the array has MMUSize+1 entries). |
| L1 I-cache, L1 D-cache | For each cache, this reports three values: <ul style="list-style-type: none"> S Number of cache index positions is 64×2^S. Multiply by associativity to get the number of lines in the cache. L Zero means no cache at all; otherwise, tells you that the cache line size is 2×2^L bytes. A Associativity—this cache is $(A + 1)$-way set-associative. So if (IS, IL, IA) is (2, 4, 3), the cache has 256 sets/way, 32 bytes/line, and is four-way set-associative: That's a 32-Kbyte cache. |
| Config1(C2) | 1 if there's a coprocessor 2 fitted (that would be likely to be some very application-specific coprocessor). |
| Config1(MD) | 1 if the old MDMX ASE is implemented in the floating-point unit (see section B.3). |
| Config1(PC) | There is at least one performance counter implemented; see section 12.4. |
| Config1(WR) | Reads 1 if your CPU has at least one watchpoint register; see section 12.2. |
| Config1(CA) | Reads 1 when the MIPS16e compressed-code instruction set is available; see section 12.1. |
| Config1(EP) | Reads 1 if an EJTAG debug unit is provided; see section 12.1. |
| Config1(FP) | A floating-point unit is attached. |

Config2(M)	Continuation bit, 1 if Config3 is implemented.
Config2(TU)	Implementation-specific bits related to L3 cache, if fitted. Might even be writable.
Config2(TS,TL,TA)	L3 cache size and shape—encoded just like Config1 (IS, IL, IA) .
Config2(SU)	Implementation-specific bits for secondary cache, if fitted. Can be writable.
Config2(SS, SL, SA)	Secondary cache size and shape, encoded like Config1 (IS, IL, IA) , above.

Fields shown in Figure 3.6 include:

Config3(M)	Continuation bit, zero if—as is likely—there is no Config4 .
LPA	Reads 1 when large physical address (LPA) support exists, which allows for a physical address range larger than 2 ³⁶ bytes. When there is LPA support, there’s an extra CP0 register called Page-Grain , and the layout of the fields in EntryLo0–1 and EntryHi change. No MIPS32 core (to date) implements LPA, so it’s not described in this book. Refer to manufacturer’s manuals as required.
DSPP	Reads 1 if the MIPS DSP extension is implemented, as described in section B.2.
VEIC	Read-only bit, which indicates the availability of an EIC-compatible interrupt controller; see section 5.8.5. Note that this is not part of the core, so it is typically wired as an input to the CPU or CPU core by the system designer.
VInt	Reads 1 if your CPU can handle vectored interrupts.
SP	Reads 1 only if your CPU supports sub-4-Kbyte page sizes. Most general-purpose MIPS CPUs don’t.
MT	Reads 1 if your CPU does multithreading, implementing the MIPS MT extension described in Appendix A.
SM	Reads 1 if your CPU handles instructions from the “SmartMIPS” ASE. This extension provides some help to encryption routines on slower CPUs and is mostly aimed at CPUs built for smart cards.

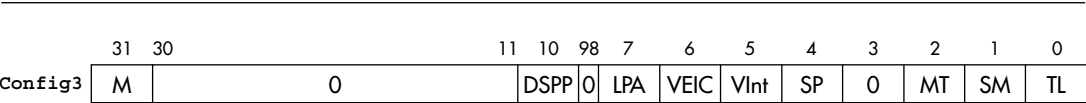


FIGURE 3.6 Fields in the MIPS32/64 **Config3** registers.

TL Reads 1 if your core can record and output instruction traces. Instruction tracing is an advanced optional feature of the EJTAG debug unit, described in section 12.1.

3.3.8 *EBase and IntCtl: Interrupt and Exception Setup*

These registers (new in Release 2 of MIPS32/64) give you control over the new interrupt capabilities added there.

EBase was added to allow you to relocate all the exception entry points for a CPU; it's primarily there for multiprocessor systems that share memory, so that different CPUs are not obliged to use the same exception handlers.

The fields of **EBase** are shown in Figure 3.7:

- 1, 0 Read-only bits prefixed to the base address bits to make sure the exception vector ends up in the kseg0 region, conventionally used for OS code.
- ExceptionBase Is the base address for the exception vectors, adjustable to a resolution of 4 Kbytes. See Table 5.1 for where that leaves all the exception entry points.
That means any of your CPUs (even the “virtual” CPUs of a multithreaded CPU) can have its own unique exception handlers.
- CPUNum A number to distinguish this CPU from others in the same multiprocessor system. The contents of this field are most likely hardwired by the designer of your system.

Then in **IntCtl**, shown in Figure 3.8:

- IPTI, IPPCI Are read-only fields, telling you how timer and performance counter interrupts (generated inside the CPU) are wired up in your system. It's relevant in nonvectored and simple-vectored (VI) interrupt modes.
Each is a 3-bit binary number identifying which CPU interrupt input is shared by the internal timer interrupt (**IPTI**) or the performance counter overflow interrupt (**IPPCI**).
The interrupt is specified by giving the number of the **Cause (IPx)** bit where the resulting interrupt is seen. Because **Cause**

31	30	29		12	11	10	9		0
1	0		ExceptionBase		0			CPUNum	

FIGURE 3.7 Layout of the **EBase** register.

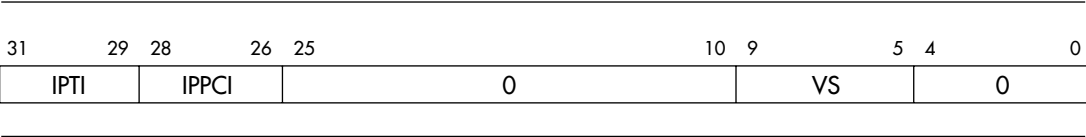


FIGURE 3.8 Layout of the **IntCtl** register.

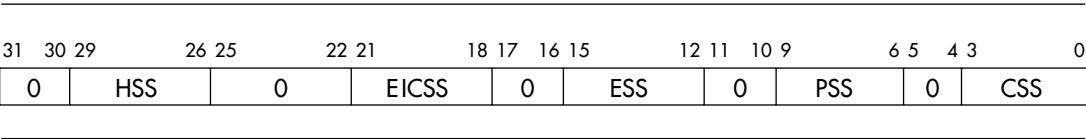


FIGURE 3.9 Layout of the **SRSCtl** register.

(**IP0–1**) are software interrupt bits, unconnected to any input, the legal values for **IntCtl** (**IPTI**) and **IntCtl** (**IPPCI**) are between 2 and 7.

The timer and performance counter interrupts are taken out to the CPU interface, where they are generally sent back again down one of the interrupt signals. So this information is not determined inside the CPU itself and is provided by the system designer.

VS Is writable to give you software control of the vector spacing; the spacing you get between consecutive entries is **IntCtl** (**VS**) × 32 bytes. Only values of 1, 2, 4, 8, and 16 work (to give spacings of 32, 64, 128, 256, and 512 bytes, respectively). A value of zero does give a zero spacing, so all interrupts arrive at the same address.

3.3.9 **SRSCtl and SRSMap: Shadow Register Setup**

Shadow registers are a new feature in Release 2 of MIPS32/64. CPUs are equipped with one or more extra sets of general-purpose registers and switch to a different set on an exception—and in particular, on an interrupt.

In **SRSCtl** :, shown in Figure 3.9:

- HSS** The highest-numbered register set available on this VPE/CPU (i.e., the number of available register sets minus one).
On multithreading CPUs this field may be changed by software that assigns shadow register sets. But for most purposes, this field is read-only.
- CSS** The register set currently in use. It’s read-only here; set on any exception, replaced by the value in **SRSCtl** (**PSS**) on an **eret**.
- ESS** This writable field is the software-selected register set to be used for “all other” exceptions; that’s other than an interrupt in VI or

	EIC mode (both have their own special ways of selecting a register set). It's probably quite unusual for it to be anything other than zero.
PSS	The “previous” register set, which will be used following the next eret. You can get at the values of registers in this set using rdpgpr and wrpgpr. SRSCt1 (PSS) is writable, allowing the OS to dispatch code in a new register set; load this value and then execute an eret.
EICSS	In EIC mode (see section 5.8.5), an external interrupt controller proposes a shadow register set number with each requested interrupt (nonzero IPL). When the CPU takes an interrupt, the externally supplied set number determines the next set and is made visible here, until the next interrupt.

Just a note: **SRSCt1 (PSS)** and **SRSCt1 (CSS)** are not updated by *all* exceptions, but only those which write a new return address to **EPC** (or equivalently, those occasions where the exception level bit **SR (EXL)** goes from zero to one). Exceptions where **EPC** is *not* written include:

- Exceptions occurring with **SR (EXL)** already set
- Cache error exceptions, where the return address is loaded into **ErrorEPC**
- EJTAG debug exceptions, where the return address is loaded into **DEPC**

3.3.10 *Load-Linked Address (LLAddr) Register*

This register holds the physical address of the last-run load-linked operation, which is kept to monitor accesses that may cause a future store conditional to fail; see section 5.8.4. Software access to **LLAddr** is for diagnostic use only.

3.4 CP0 Hazards—A Trap for the Unwary

Since the CPU is pipelined, the effects of CP0 operations may not reach their target until after the later stages of the CP0 instruction—and even then may take some number of clocks to filter out to the hardware. But even before our CP0 instruction reaches its later stages, other instructions will have been fetched and started. How can we be sure that those instructions will be executed in the light of our CP0 change?

It would in theory be possible for the hardware engineers to find and interlock every possible interaction, producing a CPU where the software engineer had nothing to worry about on this score. SGI's R10000 approaches this ideal.

But CP0 operations are obscure, many happen rarely, and all are under the control of OS software, which must already be trusted. So it has seemed reasonable to MIPS architects and designers to push some of this trouble onto the shoulders of the software authors.

Historically, they were supposed to do this by analyzing the flow of their program, detect places where the software might malfunction, and apply fixes. The weapons at hand were **nop** instructions, which safely do nothing while changes propagate, and branch/jump instructions, which “invalidate” sequentially prefetched instructions, forcing them to be refetched and re-executed. If you look at the back of a manual of a MIPS CPU that doesn't conform to the 2003 (second) revision of MIPS32/64, you will usually find a table of CP0 hazards, detailing how many clocks various changes take to propagate before you can guarantee they will be seen as done by subsequent instructions.

Even with early CPUs with simple pipelines, this was a bit of a challenge. As the pipelines grew longer and more complicated, it became a real nuisance for portable software. Once there were CPUs that executed instructions in parallel (simple dual-issue or even out-of-order), it became difficult to count how many **nops** were required. For a while, we even got the **ssnop** (“superscalar no-op”) instruction, guaranteed to burn a whole clock time by issuing alone.

But MIPS32/64 CPUs now have a more sensible approach: hazard barrier instructions. These are special instructions, which need only be placed where needed, where they have the effect of delaying subsequent instructions until all the side effects of preceding CP0 instructions have propagated.

MIPS32/64 distinguishes two flavors of CP0 hazard, depending on which stage of a dependent instruction's operation may be affected. *Execution* hazards are those where the dependent instruction will not be affected until it reads a CP0 register value; *instruction* hazards are those where the dependent instruction is affected in its very earliest stages—at worst, at the time it is fetched from cache or memory.

Another interesting distinction is between hazards that strictly affect only other CP0 instructions (which must be part of the OS) and those (we might call them *user* hazards) that can affect ordinary instructions.

3.4.1 *Hazard Barrier Instructions*

Before we get started, note that any exception clears all hazards (so nothing can go wrong because of something incomplete at the start of an exception handler), and so does an **eret**—so nothing done by the OS can cause trouble back in the user program.

There are three explicit hazard barrier instructions. You can clear execution hazards with **ehb**—which older CPUs will see as a no-op. Instruction hazards

are cleared by special jump-register instructions **jr.hb** and **jalr.hb**, which are most often substituted for a normal subroutine return or subroutine call, respectively.

MIPS architects were being smart when they chose special jump-register instructions, which will just be decoded as **jr** or **jalr** on older CPUs. On those CPUs such instructions clear the CPU pipeline (the jump-to-register is inherently “unpredictable”), and in most cases will provide the necessary delay on CPUs that don’t comply with later MIPS32/64 specifications.

3.4.2 *Instruction Hazards and User Hazards*

These typically happen when we make a change to CP0 state (a register, a TLB entry, or perhaps a cache line) that will affect the way we fetch ordinary instructions (or, in a few cases, will affect the way load/store instructions access memory). Such hazards must be safely resolved before we return to any kind of “uncontrolled” code.

In these cases you should put the hazard barrier after the CP0 operation that changed the state. Most often you should put it immediately afterwards—but you might have some other work to do that you know is safe and can run while the hazardous operation’s effects percolate out. But that’s on your own head!

These sort of hazards include:

Change of TLB entry	⇒	fetch, load, or store in affected page
Change of EntryHi (ASID field)	⇒	any non-globally-mapped fetch, load, or store
Change to ERL mode	⇒	fetch, load, or store from kuseg
cache instruction altering cache line	⇒	fetch, load, or store in affected line
Change to watchpoint register	⇒	fetch, load, or store that matches
Change of shadow register setting	⇒	any use of GPR (an execution hazard)
CP0 register change that disables interrupt	⇒	instruction that could still be interrupted (an execution hazard)

Most of these are instruction hazards—and where there’s no **eret** to act as an adequate barrier for these instructions, you should use a **jr.hb** or **jalr.hb**. The execution hazards can be cleared with an **ehb**.

3.4.3 *Hazards between CP0 Instructions*

Any **mfc0** instruction is explicitly dependent on the value in a CP0 register, but because all TLB information is staged through registers, so are **tlbwi**, **tlbwr**,

and **tlbr**. Similarly, those **cache** instructions that read data out of CP0 registers are dependent.

Less obviously, **tlbp** depends on **EntryHi** because of its **ASID** field.

All these are execution hazards and can be made safe with an **ehb**, placed before the consuming CP0 instruction. If you have the chance to put it a few instructions early in the sequence, so much the better (some CPUs may inhibit all CP0 operations for a few clocks after an **ehb**).

How Caches Work on MIPS Processors

A MIPS CPU without a cache isn't really a RISC. Perhaps that's not fair; for special purposes you might be able to build a MIPS CPU with a small, tightly coupled memory that can be accessed in a fixed number of pipeline stages (preferably one). But MIPS CPUs have always had cache hardware built tightly into the pipeline.

This chapter will describe the way MIPS caches work and what the software has to do to make caches useful and reliable. From reset, almost everything about the cache state is undefined, so bootstrap software must be careful to initialize the caches correctly before relying on them. You might also benefit from some hints and tips for use when sizing the caches (it would be bad software practice to assume you know how big the cache is). For the diagnostics programmer, we discuss how to test the cache memory and probe for particular entries.

Some real-time applications writers may want to control exactly what will get cached at run time. We discuss how to do that, even though I am skeptical about the wisdom of using such tricks.

There's also some evolution to contend with. In early 32-bit MIPS processors, cache management functions relied upon putting the cache into a special state and then using ordinary reads and writes whose side effects could initialize or invalidate cache locations. But we won't dwell on that here; even CPUs that do not fully comply with MIPS32/64 generally use something close to the mechanism described in the following text.

4.1 Caches and Cache Management

The cache's job is to keep a copy of memory data that has been recently read or written, so it can be returned to the CPU quickly. For L1 caches, the read must complete in a fixed period of time to keep the pipeline running.

MIPS CPUs always have separate L1 caches for instructions and data (I-cache and D-cache, respectively) so that an instruction can be read and a load or store done simultaneously.

Cached CPUs in old-established families (such as the x86) have to be compatible with code that was written for CPUs that didn't have any caches. Modern x86 chips contain ingeniously designed hardware to make sure that software doesn't have to know about the caches at all (if you're building a machine to run MS-DOS, this is essential to provide backward compatibility).

But because MIPS machines have always had caches, there's no absolute requirement for the cache to be so clever. The caches must be transparent to application software, apart from the increased speed. But in a MIPS CPU, which has always had cache hardware, there is no attempt to make the caches invisible to system software or device drivers—cache hardware is installed to make the CPU go fast, not to help the system programmer. A UNIX-like operating system hides the cache from applications, of course, but while a more lightweight OS might hide the details of cache manipulation from you, you will still probably have to know when to invoke the appropriate subroutine.

4.2 How Caches Work

Conceptually, a cache is an associative memory, a chunk of storage where data is deposited and can be found again using an arbitrary data pattern as a key. In a cache, the key is the full memory address. Produce the same key back to an associative memory and you'll get the same data back again. A real associative memory will accept a new item using an arbitrary key, at least until it's full; however, since a presented key has to be compared with every stored key simultaneously, a genuine associative memory of any size is either hopelessly resource hungry, slow, or both.

So how can we make a useful cache that is fast and efficient? Figure 4.1 shows the basic layout of the simplest kind of cache, the direct-mapped cache used in most MIPS CPUs up to the 1992 generation.

The direct-mapped arrangement uses a simple chunk of high-speed memory (the *cache store*) indexed by enough low address bits to span its size. Each *line* inside the cache store contains one or more words of data and a *cache tag* field, which records the memory address where this data belongs.

On a read, the cache line is accessed, and the tag field is compared with the higher addresses of the memory address; if the tag matches, we know we've got the right data and have "hit" in the cache. Where there's more than one word in the line, the appropriate word will be selected based on the very lowest address bits.

If the tag doesn't match, we've missed and the data will be read from memory and copied into the cache. The data that was previously held in the cache is simply discarded and will need to be fetched from memory again if the CPU references it.

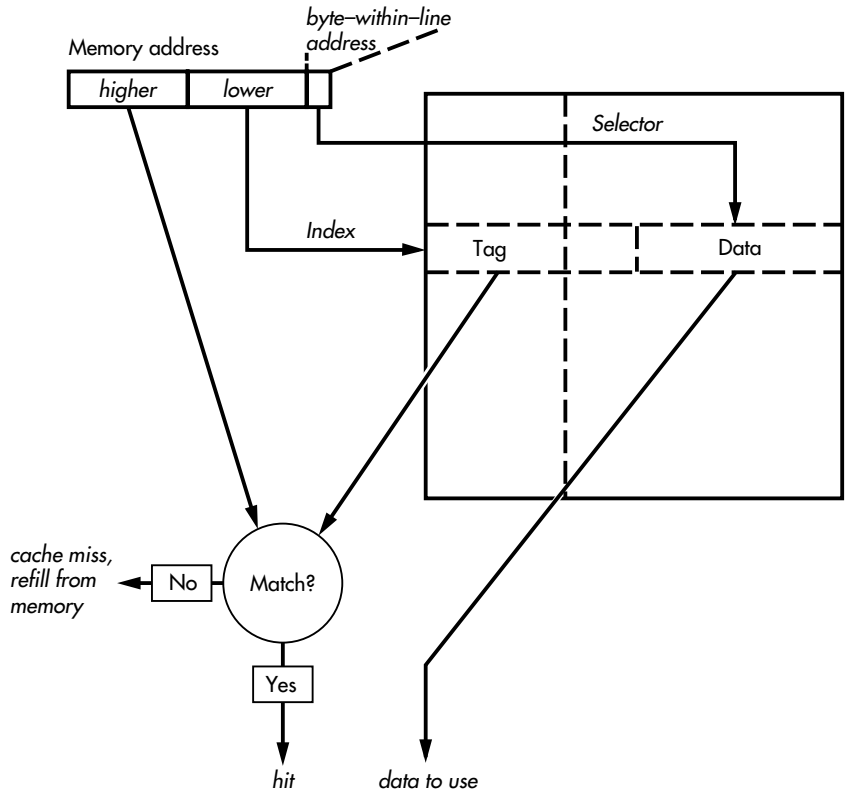


FIGURE 4.1 Direct-mapped cache.

A direct-mapped cache like this one has the property that, for any given memory address, there is only *one* line in the cache store where that data can be kept.¹ That might be good or bad; it's good because such a simple structure will be fast and will allow us to run the whole CPU faster. But simplicity has its bad side too: If your program makes repeated reference to two data items that happen to share the same cache location (presumably because the low bits of their addresses happen to be close together), then the two data items will keep pushing each other out of the cache and efficiency will fall drastically.

A real associative memory wouldn't suffer from this kind of thrashing, but it is too slow and expensive.

A common compromise is to use a two-way set-associative cache—which is really just a matter of running two direct-mapped caches in parallel and looking up memory locations in both of them, as shown in Figure 4.2.

1. In a fully associative memory, data associated with any given memory address (key) can be stored anywhere; a direct-mapped cache is as far from being content addressable as a cache store can be.

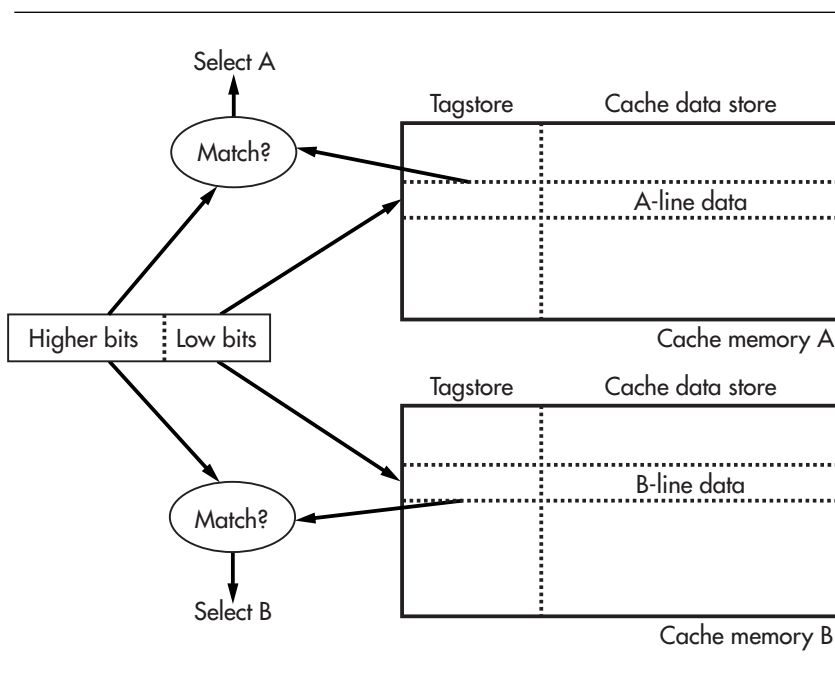


FIGURE 4.2 Two-way set-associative cache.

Now we've got two chances of getting a hit on any address. Four-way set-associative caches (where there are effectively four direct-mapped subcaches) are also common in on-chip caches.

In a multiway cache there's more than one choice of the cache location to be used in fixing up a cache miss, and thus more than one acceptable choice of the cache line to be discarded. The ideal solution is probably to keep track of accesses to cache lines and pick the "least recently used" ("LRU") line to be replaced, but maintaining strict LRU order means updating LRU bits in every cache line every time the cache is read. Moreover, keeping strict LRU information in a more-than-four-way set-associative cache becomes impractical. Real caches often use compromise algorithms like "least recently filled" to pick the line to discard.

There are penalties, however. Compared with a direct-mapped cache, a set-associative cache requires many more bus connections between the cache memory and controller. That means that caches too big to integrate onto a single chip are much easier to build direct mapped. More subtly, because the direct-mapped cache has only one possible candidate for the data you need, it's possible to keep the CPU running ahead of the tag check (so long as the CPU does not do anything irrevocable based on the data). Simplicity and running ahead can translate to a faster clock rate.

Once the cache has been running for awhile it will be full, so capturing new memory data usually means discarding some previously cached data. If you know that the data in the cache is already safely in memory, you can just discard the cached copy; if the data in the cache is more up-to-date than memory, you need to write it back first.

That brings us to how the cache handles writes.

4.3 Write-Through Caches in Early MIPS CPUs

CPUs don't just read data (as the above discussion seems to be assuming)—they write it too. Since a cache is intended to be a local copy of some data from main memory, one obvious way of handling the CPU's writes is the use of what is called a *write-through* cache.

In a write-through cache, the CPU's data is always written to main memory; if a copy of that memory location is resident in the cache, the cached copy is updated too. If we always do this, then any data in the cache is known to be in memory too, so we can discard the contents of a cache line anytime without losing any data.

We would slow the processor down drastically if we waited for the memory write-through to complete, but we can fix that. Writes (address and data together) destined for main memory can always be kept in a queue while the memory controller gets itself ready and completes the write. The place where writes are queued is organized as a first-in, first-out (FIFO) store and is called a *write buffer*.

Early MIPS CPUs had a direct-mapped write-through cache and a write buffer. So long as the memory system can happily absorb writes at the average rate produced by a CPU, running a particular program this way works very well.

But CPU speeds have grown much faster than memory speeds, and somewhere around the time that the 32-bit MIPS generation was giving way to the 64-bit R4000, MIPS speeds passed the point where a memory system could reasonably hope to absorb every write.²

4.4 Write-Back Caches in MIPS CPUs

While early MIPS CPUs use simple write-through data caches, later CPUs are too fast for this approach—they would swamp their memory systems with writes and slow to a (relative) crawl.

2. A very rough rule of thumb for programs is that there is one store per 10 instructions, so a write-through solution may work until the memory cycle time reaches about 5–7 instruction times. With DRAM cycle times then around 180 ns, the simple solution ran out of steam at about 30–40 MHz.

The solution is to retain write data in the cache. Write data is stored only to the cache, and the cache line is marked to make sure we don't forget to write it back to memory sometime later (a line that needs writing back is called *dirty* or *modified*).³

There's a subvariant here: If the addressed data is not currently in the cache, we can either write it to main memory and ignore the cache, or we can bring the data in specially just so we can write it—this is called *write-allocate*. Write-allocate is not always the best approach: Data that is being written for some I/O device to read (and that the CPU will not read or write again) is ideally kept out of the cache. But that only works if the programmers can be relied upon to tell the operating system.

All except the very lowest-end modern MIPS CPUs have on-chip caches that are write-back and have line sizes of 16 or 32 bytes.

The R4000 and some later CPUs found uses in large computer servers from Silicon Graphics and others. Their cache design choices are influenced by the needs of multiprocessor systems. There's a short description of typical multiprocessor systems in section 15.3.

4.5 Other Choices in Cache Design

The 1980s and 1990s have seen much work and exploration of how to build caches. So there are yet more choices:

- *Physically addressed/virtually addressed*: While the CPU is running a grown-up OS, data and instruction addresses in your program (the *program address* or *virtual address*) are translated before appearing as physical addresses in the system memory.

A cache that works purely on physical addresses is easier to manage (we'll explain why below), but raw program (virtual) addresses are available to start the cache lookup earlier, letting the system run that little bit faster.

So what's wrong with program addresses? They're not unique; many different programs running in different address spaces on a CPU may share the same program address for different data. We could reinitialize the entire cache every time we switch contexts between different address spaces; that used to be done some years ago and may be a reasonable solution for very small caches. But for big caches it's ridiculously inefficient,

3. You might ask: Since we're going to have to write it back sometime, surely we might as well do it now? But actually programs often write many times in quick succession to the same small patch of memory, and the write-back cache allows many individual write operations to be combined into just one write to memory.

and we'll need to include a field identifying the address space in the cache tag to make sure we don't mix them up.

Many MIPS CPUs use the program (virtual) address to provide a fast index for their L1 caches. But rather than using the program address plus an address space identifier to tag the cache line, they use the physical address. The physical address is unique to the cache line and is efficient because the scheme allows the CPU to translate program addresses to physical addresses at the same time as it is looking up the cache.

There's another, more subtle problem with program addresses, which the physical tag does not solve: The same physical location may be described by different addresses in different tasks. In turn, that might lead to the same memory location being cached in two different cache entries (because they were referred to by different virtual addresses that selected different cache indexes). Many MIPS CPUs do not have hardware to detect or avoid these *cache aliases* and leave them as a problem to be worked around by the OS's memory manager; see section 4.12 for details.

- *Choice of line size:* The line size is the number of words of data stored with each tag. Early MIPS caches had one word per tag, but it's usually advantageous to store multiple words per tag, particularly when your memory system will support fast burst reads (most do). Modern MIPS caches tend to use four- or eight-word line sizes, but large L2 and L3 caches may have bigger lines.

When a cache miss occurs, the whole line must be filled from memory.

- *Split/unified:* MIPS L1 caches are always separated into an I- and a D-cache; the selection is done purely by function, in that instruction fetches look in the I-cache and data loads/stores in the D-cache. (This means, by the way, that if you try to execute code that the CPU just copied into memory, you must both write back those instructions out of the D-cache and invalidate the I-cache locations, to make sure you really execute the new instructions.)

However, L2 caches are rarely divided up this way—it's complex, more costly, and generally performs poorly.

4.6 Managing Caches

I hope you recall from section 2.8 that a MIPS CPU has two fixed 512-MB windows onto physical memory, one cached ("kseg0") and one uncached ("kseg1"). Typically, OS code runs in kseg0 and uses kseg1 to build uncached references.

Physical addresses higher than 512 MB are not accessible here: A 64-bit CPU will have direct access through another window, or you can set up the TLB (memory management/translation hardware) to map an address. Each TLB entry can be marked to make accesses either cached or uncached.

The cache hardware, with the help of system software, must be able to ensure that any application gets the same data as it would have in an uncached system and that any direct memory access (DMA) I/O controller (getting data directly from memory) obtains the data that the program thinks it has written.

We've said before that in PCs and some other integrated systems the assistance of system software is often not required; it's worth spending the money, silicon area, and extra cycles to get the hardware to make the cache genuinely transparent. Such a hardware-managed cache is said to be *coherent*.

The contents of the cache arrays of your CPU are typically random following power-up. Bootstrap software is responsible for initializing the caches; this can be quite an intricate process, and there's some advice about it below. But once the system is up and running, there are only three circumstances in which the CPU must intervene:

- *Before DMA out of memory:* If a device is taking data out of memory, it's vital that it gets the right data. If the data cache is write back and a program has recently written some data, some of the correct data may still be held in the D-cache but not yet be written back to main memory. The CPU can't see this problem, of course; if it looks at the memory locations it will get the correct data back from its cache.
So before the DMA device starts reading data from memory, any data for that range of locations that is currently held in the D-cache must be written back to memory if necessary.
- *DMA into memory:* If a device is loading data into memory, it's important to invalidate any cache entries purporting to hold copies of the memory locations concerned; otherwise, the CPU reading these locations will obtain stale cached data. The cache entries must be invalidated before the CPU uses any data from the DMA input stream, but it's quite common to invalidate them before starting the DMA.
- *Writing instructions:* When the CPU itself is storing instructions into memory for subsequent execution, you must first ensure that the instructions are written back to memory and then make sure that the corresponding I-cache locations are invalidated; the MIPS CPU has no connection between the D-cache and the I-cache.
In the most modern MIPS CPUs the **synci** instruction does everything that is necessary to make the instructions you've just stored usable for execution—and it's a user-privilege instruction.

Why Not Manage Caches in Hardware?

Caches managed with hardware are described as “coherent” (or, more informally, “snoopy.”) When another CPU or some DMA device accesses memory, the cache control logic is notified. With a CPU attached to a shared bus, this is pretty straightforward; the address bus contains most of the information you need. The cache control logic watches (snoops) the address bus even when the CPU is not using it and picks out relevant cycles. It does that by looking up its own cache to see whether it holds a copy of the location being accessed.

If someone is *writing* data that is inside the cache, the controller could pick up the data and update the cache line but is more likely to just invalidate its own, now stale, copy. If someone is reading data for which updated information is held in the cache, the controller may be able to intervene on the bus, telling the memory controller that it has a more up-to-date version.

One major problem with doing this is that it works only within a system designed to operate that way. Not all systems have a single bus where all transactions appear, and bought-in I/O controllers are unlikely to conform to the right protocols.

Also, that’s a lot of snooping going on. Most of the locations that CPUs work with are the CPU’s private areas; they will never be read or written by any other CPU or device. We’d like not to build hardware ingenuity into the cache, loading every cache location and bus cycle with complexity that will only sometimes

be used. It’s easy to suppose that a hardware cache control mechanism must be faster than software, but that’s not necessarily so. A snoopy cache controller must look at the cache tags on every external cycle, which could shut the CPU out of its cache and slow it down; snoopy cache controllers usually fix this by keeping two copies of the cache tags (a private-to-CPU and public version). Software management can operate on blocks of cache locations in a single fast loop; hardware management will interleave invalidations or write-backs with CPU accesses at I/O speed, and that usually implies more arbitration overhead.

Historically, MIPS designers took the radical RISC position: MIPS CPUs either had no cache management hardware or, where designed for multiprocessors, they had everything.

From a 21st-century perspective, the trade-off looks different. For most classes of CPU, it looks more worthwhile to accept some hardware complexity to avoid the system software bugs that occur when programmers miss a place where the cache needs attention. At the time of writing (2006), there’s a shift in progress toward implementing “invisible” caches with some level of hardware cache management on all but the smallest MIPS CPUs. If your MIPS CPU is fully coherent, you may not need this chapter at all (but very few do the whole job). And many MIPS cores that will be in production for years to come still don’t have coherent cache controllers, so the shift will take a long time to complete.

If your software is going to fix these problems, it needs to be able to do two distinct operations on a cache entry.

The first operation is called *write-back*. When the data of interest is present in the cache and is dirty (marked as having been written by the CPU since it was last obtained from memory or written back to memory), then the CPU copies the data from the cache into main memory.

The second is *invalidate*. When the data of interest is in the cache, the CPU marks it invalid so that any subsequent access will fetch fresh data from memory.

It's tempting to use the more colorful and evocative word “flush” in this context, but it is ambiguously used to mean write-back, invalidate, or the combination of the two—so we'll avoid it.

There are some much more complicated issues involved when two or more processors share memory. Most shared-memory systems are too complex for one CPU to know which locations the other will read or write, so the software can't figure out exactly which memory regions need to be invalidated or written back. Either any shared memory must always be accessed uncached (very slow unless the amount of interaction is very limited), or the caches must have special hardware that keeps the caches (and memory) coherent. Multi-CPU cache coherency got less scary after it was systematized by a group of engineers working on the ambitious FutureBus standard in the mid-1980s. There's a discussion of multiprocessor mechanisms in section 15.3.

4.7 L2 and L3 Caches

In larger systems, there's often a nested hierarchy of caches. A small and fast *L1* or *primary* cache is close to the CPU. Accesses that miss in the L1 cache are looked up not directly in memory but in an *L2* or *secondary* cache—typically several times bigger and several times slower than the L1 (but still several times faster than the main memory). The number of levels of hierarchy that might be useful depends on how slow main memory is compared with the CPU's fastest access; with CPU cycle times falling much faster than memory access times, 2006 desktop systems commonly have L3 cache. 2006 embedded systems, trailing desktop speeds by several years (and different constraints, particularly for power consumption and heat dissipation) are only just getting around to using L2 caches outside a few specialist high-end applications.

4.8 Cache Configurations for MIPS CPUs

We can now classify some landmark MIPS CPUs, ancient and modern, by their cache implementations and see how the cache hierarchy has evolved (Table 4.1).

As clock speeds get higher, we see more variety in cache configurations as designers try to cope with a CPU that is running increasingly ahead of its memory system. To earn its keep, a cache must improve performance by supplying data significantly faster than the next outer memory and must usually succeed in supplying the data (*hitting*).

CPUs that add a second level of cache reduce the miss penalty for the L1 cache—which may then be able to be smaller or simpler. CPUs with on-chip

TABLE 4.1 Cache Evolution in MIPS CPUs

CPU (MHz)	L1				L2			L3		
	Size		Direct/ n-way	On- chip?	Size	Direct/ n-way	On- chip?	Size	Direct/ n-way	On- chip?
	I-cache	D-cache								
R3000-33	32 K	32 K	Direct	Off						
R3052-33	8 K	2 K	Direct	On						
R4000-100	8 K	8 K	Direct	On	1 M	Direct	Off			
R10000-250	32 K	32 K	Two-way	On	4 M	Two-way	Off			
R5000-200	32 K	32 K	Two-way	On	1 M	Direct	Off			
RM7000-250	16 K	16 K	Four-way	On	256 K	Four-way	On	8 M	Direct	Off

L2 caches typically have smaller L1s, with dual 16-KB L1 caches a favored “sweet spot.” Until recently, most MIPS CPUs fitted the L1 cache access into one clock cycle. It seems reasonable that, as chip performance grows and clock rates increase, the size of memory that can be accessed in one clock period should be more or less constant. But in fact, on-chip memory speed is lagging behind logic. Modern CPU designs often lengthen the pipeline mainly to allow for a two-clock cache access.⁴

Off-chip caches are often direct mapped because a set-associative cache needs much wider interfaces to carry multiple tags, so they can be matched in parallel.

Amidst all this evolution, there have been two main generations of the software interface to a MIPS cache. There is one style founded by the R3000 and followed by most early 32-bit MIPS CPUs; there is another starting with the R4000 and used by all 64-bit CPUs to date. The R4000 style has now become a part of the MIPS32 standard too and is now commonplace—and that’s all we will describe.

Most modern MIPS CPUs have L1 caches that are write-back, virtually indexed, physically tagged, and two- or four-way set-associative. Cache management is done using a special **cache** instruction.

The first R4000 CPUs had on-chip L2 cache control, and QED’s RM7000 (around 1998) introduced on-chip L2 cache, which is now commonplace in high-end designs. The **cache** instruction as defined by MIPS32/64 extends to L2 and L3 caches.

4. This is not a recent invention. Early systems built with the RISC HP-8x00 CPU family accepted a 2-clock-cycle L1 cache latency in return for a huge external L1 cache, and performed very well.

CAUTION! Some system implementations have L2 caches that are not controlled by hardware inside the MIPS CPU but built onto the memory bus. The software interface to caches like that is going to be system specific and may be quite different from the CPU-implemented or CPU-controlled caches described in this chapter.

4.9 Programming MIPS32/64 Caches

The MIPS32/64 specifications define a tidy way of managing caches (following the lead of the R4000, which fixed the unseemly cache maintenance of the earlier CPUs).

MIPS32/64 CPUs often have much more sophisticated caches than early MIPS CPUs—write-back, and with longer lines. Because it's a write-back cache, each line needs a status bit that marks it as dirty when it gets written by the CPU (and hence becomes different from the main memory copy of the data). To manage the caches, there are a number of primitive operations we'd really like. The actions we need to achieve are:

- *Invalidate range of locations:* Removes any data from this address range from the cache, so that the next reference to it will acquire fresh data from memory.

The instruction **cache HitInvalidate** has the form of a load/store instruction, providing a virtual address. It invalidates any single cache line containing the data referenced by that virtual address. Repeat the instruction at line-size-separated addresses across the range.

- *Write-back range of locations:* Ensures that any CPU-written data in this range currently held in dirty cache lines is sent back to main memory.

The instruction **cache HitWritebackInvalidate** writes back any single cache line containing the data referenced by that virtual address, and then makes it invalid as a bonus.

- *Invalidate entire cache:* Discards all cached data. This is rarely required except at initialization, but is sometimes the last resort of operating system code that is not sure which cached data needs to be invalidated.

The instruction **cache IndexInvalidate** is used here. Its address argument is interpreted only as an index into the memory array underlying the cache—successive lines are accessed at index values from 0 upward, in cache-line-size increments.

- *Initialize caches:* Whatever is required to get the caches usable from an unknown state. Setting up the cache control fields (the “tag”) usually involves zeroing the CP0 **TagLo** register and using a **cache Index-StoreTag** operation on each line-sized chunk of the cache. Caches with data check bits may also need to be prefilled with data—even though the lines are invalid, they should not have bad check bits. Prefilling an I-cache would be tricky, but you can usually use a **cache Fill** instruction.

4.9.1 *The Cache Instruction*

The **cache** instruction has the general form of a MIPS load or store instruction (with the usual register plus 16-bit signed displacement address)—but where the data register would have been encoded there’s a 5-bit operation field, which must encode the cache to be operated on, determine how to find the line, and figure out what to do with the line when you find it. You write a cache line in assembly as **cache OP, addr**, where OP is just a numeric value for the operation field.

The best thing to do is to use the C preprocessor to define text “names” representing the numeric values for the operations. There are no standard names; I’ve arbitrarily used the names of C preprocessor definitions found in header files from the MIPS Technologies toolkit package.

Of the 5-bit field, the upper 2 bits select which cache to work on:

- 0 = L1 I-cache
- 1 = L1 D-cache
- 2 = L3 cache, if fitted
- 3 = L2 cache, if fitted

Before we list the individual commands, note that they come in three flavors, which differ in how they select the cache entry (the “cache line”) they will work on:

- *Hit-type cache operation:* Presents an address (just like a load/store), which is looked up in the cache. If this location is in the cache (if it “hits”), the cache operation is carried out on the enclosing line. If this location is not in the cache, nothing happens.
- *Address-type cache operation:* Presents an address of some memory data, which is processed just like a cached access. That is, if the line you addressed was not previously in cache, the data is fetched from memory before the cache operation.
- *Index-type cache operation:* Uses as many low bits of the virtual address as are needed to select the byte within the cache line, then the cache

line address inside one of the cache ways, and then the way.⁵ You have to know the size of your cache (discoverable from **Config1-2**, see section 3.3.7 for details) to know exactly where the field boundaries are, but your address is used something like this:

31			5	4	0
Unused		Way1-0	Index	byte-within-line	

Once you’ve picked your cache and cache line, you have a choice of operations you can perform on it, as shown in Table 4.2. Three operations must be supported by a CPU to claim MIPS32/64 compatibility: that’s index invalidate, index store tag, and hit write-back invalidate. Other operations are optional—if you use them, check your CPU manual carefully.

The **synci** instruction (new to the MIPS32 Release 2 update) provides a clean mechanism for ensuring that instructions you’ve just written are correctly presented for execution (it combines a D-cache write-back with an I-cache invalidate). If your CPU supports it, you should use **synci** in preference to the traditional alternative (a pair of **cache** instructions, to do D-cache write-back followed by an I-cache invalidate).

4.9.2 Cache Initialization and Tag/Data Registers

For diagnostic and maintenance purposes it’s good to be able to read and write the cache tags; MIPS32/64 defines a pair of 32-bit registers **TagLo** and **TagHi**⁶ to stage data between the tag part of the cache and management software. **TagHi** is often not necessary: Until your physical address space is more than 36 bits, there is usually room for the whole tag in **TagLo**.

The fields in the two registers reflect the tag field in the cache and are CPU dependent. Only one thing is guaranteed: An all-zero value in the tag register(s) corresponds to a legal, properly formed tag for a line that contains no valid data. The CPU implements a **cache IndexStoreTag** instruction, which copies the contents of the tag registers into the cache line. So by setting the registers to zero, and storing the tag value, you can start to initialize a cache from an unknown starting state.

There are “store data” and “load data” cache operations defined by MIPS32/64, but they are optional and should only be used by code that is already known to be CPU-specific. You can always access the data by “hitting” on it.

5. Some still-in-use MIPS CPUs use the least significant address bits to select the “way.” Those CPUs may need special initialization. I am not aware of any MIPS32/64-compliant CPUs that do this: but it’s another thing to be careful about.

6. Some CPU manuals use different registers to talk to the I- and D-caches, and then call them **ITagLo**, and so on.

TABLE 4.2 Operations on a Cache Line Available with the Cache Instruction

Value	Command	What it does
0	Index invalidate	Sets the line to “invalid.” If it’s a D-cache line that is valid and dirty (has been written by CPU since fetched from memory), then write the contents back to memory first. This is the best and simplest way to invalidate an I-cache when initializing the CPU—though if your cache is parity protected, you also need to fill it with good-parity data; see Fill below. And this is not suitable for D-caches, where it might cause random write-backs: see IndexStoreTag type below. All MIPS32/64 CPUs must provide this operation.
1	Index Load Tag	Read the cache line tag bits and addressed doubleword data into TagLo/TagHi , DataLo/DataHi . This command is obscure, for diagnostics and geeks only.
2	Index Store Tag	Set the cache tag from the TagLo/TagHi registers. To initialize a D-cache from an unknown state, set the TagLo/TagHi registers to zero and then do this to each line. All MIPS32/64 CPUs must provide this operation.
4	Hit invalidate	Invalidate, but do not write-back the data even if dirty. All MIPS32/64 CPUs implement this operation on the I-cache. May cause data loss unless you know the line is not dirty.
5	Hit Writeback invalidate	Invalidate the line—but only after writing it back, if dirty. This is the recommended way of invalidating a D-cache line in a running cache. All MIPS32/64-compatible CPUs implement this on the D-cache.
	<i>On a D-cache</i>	
5	Fill	Address-type operation—fill a suitable cache line from the data at the supplied address—it will be selected just as if you were processing an I-cache miss at this address. Used to initialize an I-cache line’s data field, which should be done when setting up the CPU when the cache is parity protected.
	<i>On an I-cache</i>	
6	Hit writeback	If a line is dirty, write it back to memory but leave it valid in the cache. Used in a running system where you want to ensure that data is pushed into memory for access by a DMA device or other CPU.
7	Fetch and Lock	An address-type operation. Get the addressed data into the same line as would be used on a regular cached reference (if the data wasn’t already cached, that might involve writing back the previous occupant of the cache line). Then lock the line. Locked lines are not replaced on a cache miss. It stays locked until explicitly invalidated with a cache “invalidate” of some kind.

A cache tag must hold all the address bits that are not implicitly known from the cache index (look back at Figure 4.1 if this does not seem obvious).

So the L1 cache tag field length is the difference between the number of bits of physical address supported and the number of bits used to index the L1 cache—12 bits for a 16-K four-way set-associative cache. **TagLo (PTagLo)** has room for 24 bits, which would support up to a 36-bit physical address range for such a cache.

The field **TagLo (PState)** contains the state bits. Your CPU manual will tell you more about them; however, for all cache management and initialization, it suffices to know that an all-zero tag is always a legitimate code representing an invalid cache entry.

4.9.3 *CacheErr, ERR, and ErrorEPC Registers: Memory/Cache Error Handling*

The CPU's caches form a vital part of the memory system, and high-availability or trustworthy systems find it worthwhile to use some extra bits to monitor the integrity of the data stored there.

Data integrity checks should ideally be implemented end to end; check bits should be computed as soon as data is generated or introduced to the system, stored with the data, and checked just before it's used. That way the check catches faults not just in the memory array but in the complex buses and gizmos that data passes through on its way to the CPU and back.

For this reason MIPS CPUs that may be used to implement high-reliability systems often choose to provide error checking in the caches. As in a main memory system, you can use either simple parity or an error-correcting code (ECC).

The components used to build memory systems tend to come in multiples of 8-bit widths these days, and memory modules that allow for checking provide 64 data and 8 check bits. So whatever else we use should follow that lead.

Parity is simple to implement as an extra bit for each byte of memory. A parity error tells the system that data is unreliable and allows a somewhat-controlled shutdown instead of creeping random failure. A crucial role of parity is that it can be an enormous help during system development, because it unambiguously identifies problems as being due to memory data integrity.

But a byte of complete garbage has a 50 percent chance of having correct parity, and random rubbish on the 64-bit bus and its parity bits will still escape detection 1 time in 256. Some systems want something better.

An *error-correcting code* (ECC) is more complex to calculate, because it involves the whole 64-bit word with 8 check bits used together. It's more thorough: a 1-bit error can be uniquely identified and corrected, and no

2-bit error can escape detection. ECC is a powerful tool for weeding out random errors in very large memory arrays.

Because the ECC bits check the whole 64-bit word at once, ECC memories can't perform a partial-word write by just selecting which part of the word to operate on but must always merge the new data and recompute the ECC. MIPS CPUs running uncached require their memory system to implement partial-word writes, making things complicated. Memory system hardware must transform partial-word writes into a read-merge-recalculate-write sequence.

For simpler systems, the choice is usually parity or nothing. It can be valuable to make parity optional, to get the diagnostic benefits during design development without paying the price in production.

Ideally, whichever check mechanism is implemented in the memory system will be carried inside the caches. In different CPUs, you may be able to use parity, a per-doubleword 8-bit ECC field, or no protection at all.

If possible, the data check bits are usually carried straight from the system interface into the cache store: They need not be checked when the data arrives from memory and is loaded into the cache. The data is checked when it's used, which ensures that any cache parity exception is delivered to the instruction that causes it, not just to one that happens to share the same cache line.

A data check error on the system bus for an *uncached* fetch is handled by the same mechanism, which means it will be reported as a “cache parity error”—which can confuse you.

Note that it's possible for the system interface to mark incoming data as having no valid check bits. In this case, the CPU will generate check bits for its internal cache.

If an error occurs, the CPU takes the special error trap. This vectors through a dedicated location in uncached space (if the cache contains bad data, it would be foolish to execute code from it). If a system uses ECC, the hardware will either correct the error or present enough information for the software to fix the data.

The fields in the **CacheErr** register are implementation dependent, and you'll need to consult your CPU manual. You may be able to get sample cache error management routines from your CPU supplier.

4.9.4 *Cache Sizing and Figuring Out Configuration*

In MIPS32/64-compliant CPUs cache size, cache organization, and line size are reliably reported to you as part of the CP0 **Config1–2** registers, described in section 3.3.7.

But for portability it makes sense to write or recycle initialization software, which works robustly across a large range of MIPS CPUs. The next section has some tried-and-tested solutions.

4.9.5 Initialization Routines

Here's one good way to do it:

1. Set up some memory you can fill the cache from—it doesn't matter what data you store in it, but it needs to have correct check bits if your system uses parity or ECC. A good trick is to reserve the bottom 32 K of system memory for this purpose, at least until after the cache has been initialized. If you fill it with data using uncached writes, it will contain correct parity.
A buffer that size is not going to be big enough to initialize a large L2 cache, and you may need to do something more complicated.
2. Set **TagLo** to zero, which makes sure that the valid bit is unset and the tag parity is consistent (we'd set **TagHi** on a CPU that needed it).
The **TagLo** register will be used by the **cache IndexStoreTag** cache instructions to forcibly invalidate a line and clear the tag parity.
3. Disable interrupts if they might otherwise happen.
4. Initialize the I-cache first, then the D-cache. Following is C code for I-cache initialization. (You have to believe in the functions or macros like `Index_Store_Tag_I()`, which do low-level functions; they're either trivial assembly code subroutines that run the appropriate machine instructions or—for the brave GNU C user—macros invoking a C `asm` statement.)

```
for (addr = KSEG0; addr < KSEG0 + size; addr += lnsz)
    /* clear tag to invalidate */
    Index_Store_Tag_I (addr);
for (addr = KSEG0; addr < KSEG0 + size; addr += lnsz)
    /* fill once, so data field parity is correct */
    Fill_I (addr);
for (addr = KSEG0; addr < KSEG0 + size; addr += lnsz)
    /* invalidate again---prudent but not strictly necessary */
    Index_Store_Tag_I (addr);
```

We did the fill operation because some CPUs may detect and trap on parity errors, even on apparently invalid cache lines. Unfortunately the **Fill.I** operation is not mandated by MIPS32/64. You can reasonably expect that any CPU that implements parity or ECC protection will include it: CPUs that don't protect cache data need only the first store tag loop.

Moreover, we use three separate loops rather than combining them, because you have to be careful about the tags; with a two-way cache, a single loop would initialize half the cache twice, since the “index store

tag” with a zero tag register will reset the LRU bit, which determines which set of the cache line is to be used on the next cache miss.

5. D-cache initialization is slightly more awkward, because there is no D-side equivalent of the “fill” operation; we have to load through the cache and rely on normal miss processing. Here’s how it’s done:

```
/* clear all tags */
for (addr = KSEG0; addr < KSEG0 + size; addr += lnsz)
    Index_Store_Tag_D (addr);
/* load from each line (in cached space) */
for (addr = KSEG0; addr < KSEG0 + size; addr += lnsz)
    junk = *addr;
/* clear all tags */
for (addr = KSEG0; addr < KSEG0 + size; addr += lnsz)
    Index_Store_Tag_D (addr);
```

4.9.6 *Invalidating or Writing Back a Region of Memory in the Cache*

The parameters for an invalidate or write-back will invariably be a range of program or physical addresses corresponding to some I/O buffer.

You will nearly always do this using the hit-style operations, which invalidate or write back only the locations that need it. If you needed to invalidate or write back a huge area of memory, it might be faster to use index operations to invalidate or write back the entire cache, but this is an optimization you may well choose to ignore.

It’s sufficient to do this:

```
PI_cache_invalidate (void *buf, int nbytes)
{
    char *s;

    for (s = (char *)buf; s < buf+nbytes; s += lnsz)
        Hit_Invalidate_I (s);
}
```

There’s no need to generate a special address so long as `buf` is a program address. If you had to invalidate based on a physical address `p`, then so long as `p` was in the first 512 MB of physical space, you would just add a constant to generate the corresponding `kseg0` region address:

```
PI_cache_invalidate (p + 0x80000000, nbytes);
```

4.10 Cache Efficiency

Ever since the move to on-chip caches in the early 1990s, the performance of high-end CPUs has been to a large extent determined by the efficiency of their cache systems. In many current systems (particularly embedded systems, where there's a need to economize on cache sizes and memory performance) the CPU is waiting for a cache refill for 50 percent to 65 percent of its time. At this point, doubling the performance of the CPU core will deliver only a 15 percent to 25 percent increase in application performance.

Cache efficiency depends on the amount of time the system is waiting for a cache refill. You can define it as the product of two numbers:

- *Cache misses per instruction*: The number of cache misses divided by the number of instructions executed. Scale to “cache misses per thousand instructions” for a more useful measure.
- *Cache miss/refill penalty*: The time it takes for the memory system to refill the cache and restart the CPU.

You might think it would be better to consider the cache miss rate—the number of misses per CPU memory access. But cache miss rates are affected by many factors, some of them rather unexpected. For example, x86 CPUs are rather short of registers, so a program compiled for x86 will generate many more data load and store events than the same program compiled for MIPS. But the extra loads and stores will be of the stack locations that the x86 compiler uses as surrogates for registers; this is a very heavily used area of memory and will be very effectively cached. The number of misses per thousand instructions is not so affected by this kind of difference.

Even such a trivial analysis is useful in pointing out the following obvious ways of making a system go faster.

- *Reduce the number of cache misses*:
 - Make the cache bigger. This is always effective, but expensive. A cache of 64 KB occupies roughly the same space as the whole of the rest of a simple CPU (excluding floating-point hardware).
 - Increase the set associativity of the cache. It's worth going up to four-way but after that the gains in cache performance are too small to notice.⁷

7. There are systems with eight-way or more caches, but that's usually done for some reason other than reducing cache misses. Generous provision of cache ways can be good for power reduction (whole ways can be powered down when not being used) and can sometimes avoid cache aliases, as described in section 4.12.

- Add another level of cache. That makes the calculation much more complicated, of course. Apart from the complication of yet another subsystem, the miss rate in an L2 cache is often depressingly high; the L1 cache has already “skimmed the cream”⁸ of the repetitive data access behavior of the CPU. To make it worthwhile, the L2 cache must be much larger (typically eight times or greater) than the L1 cache, and an L2 cache hit must be much faster (two times or better) than a memory reference.
 - Reorganize your software to reduce the number of cache misses. It’s not clear whether this works in practice: It’s easy to reorganize a small or trivial program to great effect, but so far nobody has succeeded in building a general tool that has any useful effect on an arbitrary program. See section 4.11.
- *Decrease the cache refill penalty:*
- Get the first word back to the CPU faster. DRAM memory systems have to do a lot of work to start up, then tend to provide data quite fast. The closer the memory is to the CPU and the shorter the data path between them, the sooner the data will arrive.
Increasing bandwidth costs money. Decreasing latency, on the other hand, is usually achieved by simplifying the system: So this is the only point in this list where you can save money and gain performance at the same time. Paradoxically, it’s had the least attention, probably because it requires more integration between the CPU interface and memory system design. CPU designers are loath to deal with system issues when they decide the interface of their chips, perhaps because their job is too complicated already!
 - Increase the memory burst bandwidth. This was traditionally approached by the expensive technique of *bank interleaving*, where two or more memories are used to store alternate words; after the start-up delay, you can take words from each memory bank alternately, doubling the available bandwidth. However, while memory latency has reduced only very slowly as the chips shrink, the bandwidth available from a single standard memory component has exploded. And within a DRAM bank, chips expose separate internal banks where accesses can be pipelined to reduce overall latency. As a result, physical bank interleaving is now rare.
- *Restart the CPU earlier:* A naive CPU will remain stalled until the whole cache line is filled with new data. But you can arrange that data from memory is passed both to the cache and directly to the waiting CPU, and have the CPU restart as soon as the data it is waiting for arrives. The rest of the cache refill continues in parallel with CPU activity.

8. Thanks to Hennessy and Patterson for this evocative metaphor.

Many MIPS CPUs enhance this by passing the address of the word whose request triggered the cache miss to the memory controller and having the critical word of the block returned first (“critical word first”).

- *Don’t stop the CPU until it must have the data:* More radically, you can just let execution continue through a load; the load operation is handed off to a bus interface unit and the CPU runs on until such time as it actually refers to the register data that was loaded. This is called a nonblocking load and is now common practice.

In normal CPUs, and with most software, you really won’t go very far before something wants the value from the load and the CPU stops anyway. But it effectively shaves a few cycles off the latency, which can be particularly useful when there’s an L2 cache.

Most drastically, you can just keep running any code that isn’t actually waiting for data to be loaded, as is done by the out-of-order (OOO) execution R10000. OOO designs use this technique quite generally, not just for loads but for computational instructions and branches. All the fastest high-end CPUs are now implemented with OOO pipelines. It’s true that OOO is not just complicated but relatively power hungry, which may slow its arrival in more power-sensitive embedded applications. But it probably is just a matter of time.

- *Multithread the CPU:* The detrimental effect of cache-miss latency on program performance can be mitigated but never avoided. So a truly radical approach is to run multiple threads on the CPU, which can take advantage of idle CPU resources during each others’ waiting times. See Appendix A for a brief introduction to the MIPS MT system.

4.11 Reorganizing Software to Influence Cache Efficiency

In systems running an unpredictable mix taken from a very large number of possible applications, cache behavior can only be estimated. But where an embedded system runs a single application, the pattern of cache misses is likely to be very characteristic of a particular build of a particular piece of software. It’s tempting to wonder whether we can massage the application code in a systematic manner to improve caching efficiency. To see how this might work, you can classify cache misses by their cause:

- *First-time accesses:* Everything has to be read from memory once.
- *Replacement:* The cache has a finite size, and soon after your program starts, every cache miss and refill will be displacing some other valid data, some of which would have been worth keeping. As the program runs, it will repeatedly lose data and have to load it again. You can minimize

replacement misses by using a bigger cache or a smaller program (it's the ratio of program size to cache size that matters).

- *Thrashing*: In a four-way set-associative cache (more ways are uncommon for MIPS CPUs) there are only four positions in the cache that can keep any particular memory location. (In a direct-mapped cache, there's just one).

If your program happens to make heavy use of a number of pieces of data whose low-order addresses are close enough that they use the same cache line, then once the number of pieces is higher than the set associativity of the cache you can get periods of very high cache misses as the different chunks of data keep pushing each other out of the cache.

Thrashing losses diminish rapidly with set associativity; most research suggests that going beyond four ways makes little difference to performance (though there are other reasons to build eight-way and more caches).

With this background, what kind of changes to a program will make it behave better in a cache?

- *Make it smaller*: A good idea if you can do it. You can use modest compiler optimization (exotic optimization often makes programs larger).
- *Make the heavily used portion of the program smaller*: Access density in programs is not at all uniformly distributed. There's often a significant amount of code that is almost never used (error handling, obscure system management), or used only once (initialization code). If you can separate the rarely used code, you might be able to get better cache hit rates for the remainder.

An approach that has been tried with qualified success is to use a profiler to establish the most heavily used functions in a program while running a representative workload, then to arrange the functions in memory in decreasing order of execution time. That means at least that the very most frequently used functions won't fight each other for cache locations.

- *Force some important code/data to be cache resident*: Some vendors provide a mechanism to allow part of the cache to be loaded and then to protect those contents from replacement. This seems like a way of obtaining deterministic performance for interrupt handlers or other crucial pieces of software. This is usually implemented by consuming a set from a multiway set-associative cache.

I am very skeptical about the viability of this approach, and I don't know of any research that backs up its usefulness. The loss in performance to the rest of the system is likely to outweigh the performance gain of the critical code. Cache locking has been used as a rather dubious marketing tool to tackle customer anxiety about the heuristic nature of caches. The

anxiety is understandable, but faster, more complex, larger systems seem to be inherently unpredictable, and most developers should probably learn to live with that—caches are only one part of this issue.

- *Lay out the program to avoid thrashing:* Beyond making the active part of the program smaller (see above) this seems to me to cause too much maintenance hassle to be a good idea. And a set-associative cache (even just two-way) usually makes it quite pointless.
- *Make some rarely used data or code uncacheable:* It seems appealing to just reserve the cache for important code, leaving used-once or used-rarely code out.

This is almost always a mistake. If the data is really rarely used, it will almost never get into the cache in the first place. And because caches usually read data in lines of 4–16 words, they often produce a huge speedup even when traversing data or code that is used only once; the burst refill from memory takes little longer than a single-word access and gives you the next 3–15 words free.

In short, we warmly recommend the following approach as a starting point (to be abandoned only after much measurement and deep thought). To start with, allow everything to be cacheable except I/O registers and lightly used remote memory. See what the cache heuristics do for your application before you try to second-guess them. Second, fix hardware problems in hardware. There's no software bandage that will regain performance lost to excessive cache refill latency or low memory bandwidth. The attempt to lower cache miss rates by reorganizing software is bound to be lengthy and complicated; be aware at the start that the gains will be small and hard-won. Try to get the hardware fixed too!

4.12 Cache Aliases

This problem afflicts caches where the address used to generate the *cache index* is different from the address stored in the *cache tag*. It's common practice for the L1 caches of MIPS CPUs to be indexed using the virtual address and tagged using the physical address. This is good for performance: If we used a physical index, we couldn't even start a cache lookup until the address had been translated through the TLB. But it can lead to *aliases*. (See Figure 4.3.)

Most of these CPUs can translate addresses in 4-KB pages. That means the low 12 bits of the virtual address need no translation. So long as your cache is 4 KB or less in size, a virtual index is the same as the physical index would be, and you're OK. It's better than that: So long as your cache index spans 4 KB or less, you're OK. In a set-associative cache, each index accesses a number of cache slots (one for each way)—so even in a 16-KB four-way set-associative cache, the virtual index is the same as the physical index, and no alias can occur.

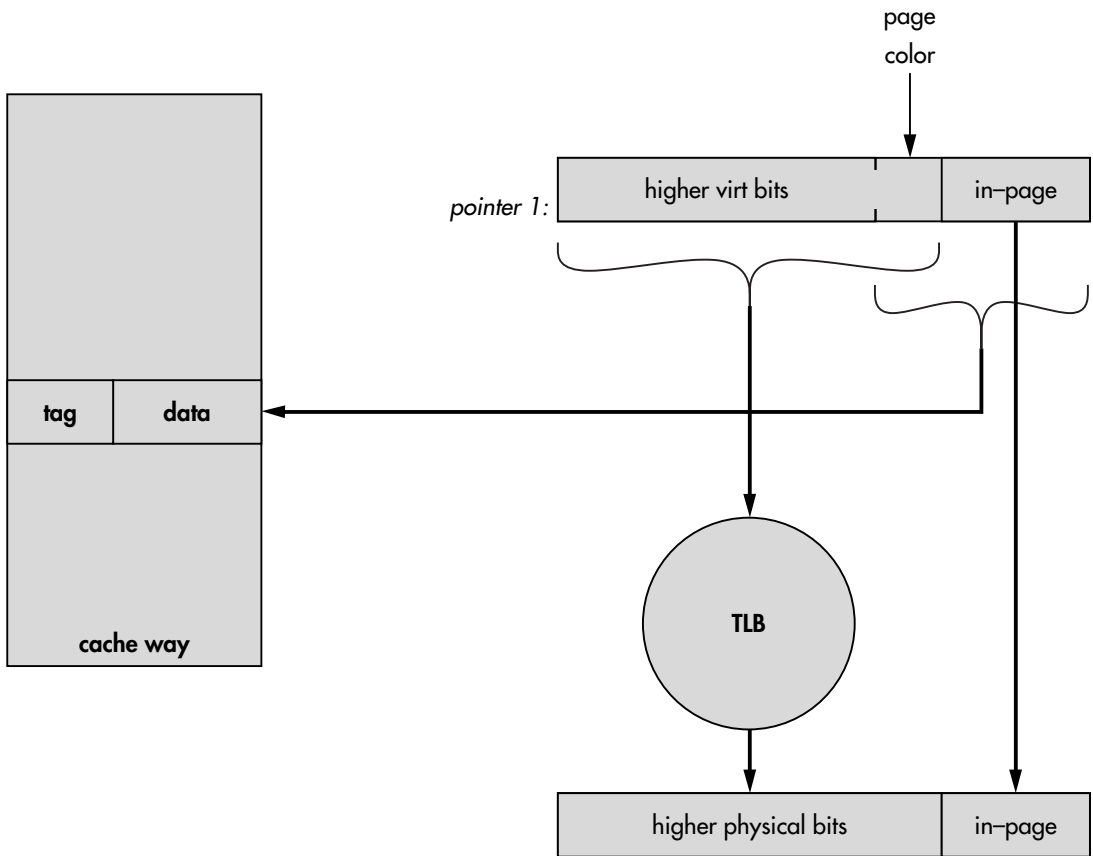


FIGURE 4.3 Cache aliases.

But suppose your cache index spans 8 KB (as it does for a 32-KB four-way set-associative cache). You might have a single physical page accessible at two different virtual addresses: They might be, for example, sequential pages—let's say those starting at 0 and 4 KB. If the program accesses data at 0, it will be loaded into one of the cache slots at index 0. If it accesses the same data at the alternate address of 4 KB, it will be fetched again from memory into the cache at the different index of 4 KB. Now there are two copies of the same cache line, and modifications made at one address will not find their way to the other one. This is a cache alias. Aliases on read-only data are confusing and may lead to errors, but aliases on data you're writing are a time-bomb waiting to explode.

MIPS L2 caches are always physically indexed and tagged, so they don't suffer from aliases.⁹

Cache aliases crept into the MIPS world by accident. The original R4000 CPU was—as described above—alias-safe so long as you used the external L2 cache. But it turned out that the R4000 was a valuable component in smaller, cheaper systems that did without the L2. That exposed the aliases, but the desire for 64-bit MIPS CPUs to conquer the world overruled caution, and the alias problem was redefined as something for system software to work around.

Aliases can't arise between any pair of translations where the alternative virtual addresses produce the same cache index. With 4-KB pages, the low 12 bits of the cache index are guaranteed to be equal; it's only necessary to ensure that any two alternative program addresses for any physical page are separated by a multiple of the largest likely L1 cache set size. If your OS is careful when creating multiple virtual mappings to the same page and makes sure that the virtual page addresses are a multiple of (say) 64 KB apart, it's hard to imagine that you'll ever have any trouble.¹⁰

So it's possible to avoid aliases, so long as the OS is careful about where it puts multiple views of the same physical data. Unfortunately, Linux is a portable OS being actively developed by people who don't know (or much care) about problems that afflict MIPS CPUs but not any common desktop or server system. So the Linux kernel avoids most aliases, but attempts to live with others.

Simple operating systems either won't have multiply mapped pages or will be able to work within such a constraint. But cache aliases are a source of lurking trouble and bugs for operating systems, and it would be good if hardware designers remembered that this is really a bug that was made into a feature for pragmatic reasons 15 years ago.

9. CPUs with on-chip L2 cache controllers can use some bits in the L2 cache to keep track of cache fetches into the L1 cache; R4000 and R4400 CPUs used this to detect cache aliases and take a special exception to allow system software to resolve the problem. But this tradition doesn't seem to have been carried on in later MIPS CPUs.

10. Although CPUs get relentlessly bigger and faster with every year that passes, it's unlikely that L1 cache set sizes will get bigger than the current 16-KB record seen in 64-KB four-way set-associative caches. L1 caches run at the full CPU clock rate, and smaller is faster; in the future, more highly integrated CPUs will go for on-chip L2 caches instead.

Exceptions, Interrupts, and Initialization

In the MIPS architecture interrupts, traps, system calls, and everything else that can disrupt the normal flow of execution are called *exceptions* and are handled by a single mechanism. What sort of events are they?

- *External events*: Some event outside the CPU core—that is, from some real “wire” input signal. These are interrupts.¹ Interrupts are used to direct the attention of the CPU to some external event: an essential feature of an OS that attends to more than one different event at a time.

Interrupts are the only exception conditions that arise from something independent of the CPU’s normal instruction stream. Since you can’t avoid interrupts just by being careful, there have to be software mechanisms to inhibit the effect of interrupts when necessary.

- *Memory translation exceptions*: These happen when an address needs to be translated, but no valid translation is available to the hardware or perhaps on a write to a write-protected page.

The OS must decide whether such an exception is an error or not. If the exception is a symptom of an application program stepping outside its permitted address space, it will be fixed by terminating the application to protect the rest of the system. The more common benign memory translation exceptions can be used to initiate operating system functions as complex as a complete demand-paged virtual memory system or as simple as extending the space available for a stack.

- *Other unusual program conditions for the kernel to fix*: Notable among these are conditions resulting from floating-point instructions, where the

1. There are some more obscure noninterrupt external events like bus errors reported on a read—for now, just assume that they are a special sort of interrupt.

hardware is unable to cope with some difficult and rare combination of operation and operands and is seeking the services of a software emulator. This category is fuzzy, since different kernels have different ideas about what they're willing to fix. An unaligned load may be an error on one system and something to be handled in software on another.

- *Program or hardware-detected errors:* This includes nonexistent instructions, instructions that are illegal at user-privilege level, coprocessor instructions executed with the appropriate **SR** flag disabled, integer overflow, address alignment errors, and accesses outside kuseg in user mode.
- *Data integrity problems:* Many MIPS CPUs continually check data on the bus or data coming from the cache for a per-byte parity or for word-wide error-correcting code. Cache or parity errors generate an exception in CPUs that support data checking.
- *System calls and traps:* These are instructions whose whole purpose is to generate recognizable exceptions; they are used to build software facilities in a secure way (system calls, conditional traps planted by careful code, and breakpoints).

Some things do not cause exceptions, though you'd expect them to. For example, you will have to use other mechanisms to detect bus errors on write cycles. That's because most modern systems queue up writes: Any write-associated error would happen much too late to relate to the instruction that caused it. Systems that do inform you about errors on writes must use some outside-the-CPU hardware, probably signaled with an interrupt.

In this chapter, we'll look at how MIPS CPUs decide to take exceptions and what the software has to do to handle them cleanly. We'll explain why MIPS exceptions are called "precise," discuss exception entry points, and discuss some software conventions.

Hardware interrupts from outside the CPU are the most common exceptions for embedded applications, the most time critical, and the ones most likely to cause subtle bugs. Special problems can arise with a *nested exception*, one that happens before you have finished handling an earlier exception.

The way that a MIPS CPU starts up after system reset is implemented as a kind of exception and borrows functions from exceptions—so that's described in this chapter too.

At the end of the chapter, we'll look at a couple of related subjects: how to emulate an instruction (as needed by an instruction set extension mechanism) and how to build semaphores to provide robust task-to-task communication in the face of interrupts. Chapter 14 describes in some detail how interrupts are handled in a real, grown-up OS for MIPS.

5.1 Precise Exceptions

You will see the phrase *precise exceptions* used in the MIPS documentation. It is a useful feature, but to understand why, you need to meet its alternative.

In a CPU tuned for the best performance by pipelining (or by more complicated tricks for overlapping instruction execution), the architecture's sequential model of execution is an illusion maintained by clever hardware. Unless the hardware is designed cleverly, exceptions can cause this illusion to unravel.

When an exception suspends its thread of execution, a pipelined CPU has several instructions in different phases of completion. Since we want to be able to return from the exception and carry on without disruption to the interrupted flow of execution, each instruction in the pipeline must be either completed or made as though we never saw it.² Moreover, we need to be able to remember which instruction falls in each of those categories.

A CPU architecture features precise exceptions when it prescribes a solution to this problem that makes life as easy as possible for the software. In a precise-exception CPU, on any exception we are pointed at one instruction (the *exception victim*). All instructions preceding the exception victim in execution sequence are complete; but it's as if the exception victim and any successors were never started.³ When exceptions are precise, the software that handles exceptions can ignore all the timing effects of the CPU's implementation.

The MIPS architecture comes close to prescribing that all exceptions are precise. Here are the ingredients:

- *Unambiguous proof of guilt*: After any exception, the CPU control register **EPC** points to the correct place to restart execution after the exception is dealt with. In most cases, it points to the exception victim, but if the victim was in a branch delay slot, **EPC** points to the preceding branch instruction: Returning to the branch instruction will re-execute the victim instruction, but returning to the victim would cause the branch to be ignored. When the victim is in a branch delay slot, the cause register bit **Cause (BD)** is set.

It may seem obvious that it should be easy to find the victim, but on some heavily pipelined CPUs it may not be possible.

- *Exceptions appear in instruction sequence*: This would be obvious for a nonpipelined CPU, but exceptions can arise at several different stages of

2. You can do something more complicated, where the CPU stores intermediate results and the exception handler unpicks the tangle and reweaves it, but who needs such complexity?

3. This is not quite the same as saying that the exception victim and subsequent instructions haven't done anything. But it does require that, when re-executed after the exception, those instructions will behave exactly as they would have done if the exception hadn't happened. Computer architects say that any side effect must be *idempotent*—doing it twice is the same as doing it once.

execution, creating a potential hazard. For example, if a load instruction suffers an address exception, this won't happen until a pipeline stage where the address translation would have been complete—and that's usually late. If the next instruction hits an address problem on an instruction fetch (right at the start of the pipeline), the exception event affecting the second-in-sequence instruction will actually happen first.

To avoid this problem, an exception that is detected early is not acted on immediately; the event is just noted and passed along the pipeline. In most CPU designs, one particular pipeline stage is designated as the place you detect exceptions. If an older instruction's late-detected event reaches this finish line while our exception note is making its way down the pipeline, the exception note just gets discarded. In the case above, the instruction-fetch address problem is forgotten—it will likely happen again when we finish handling the victim instruction's problem and re-execute the victim and subsequent instructions.

- *Subsequent instructions nullified:* Because of the pipelining, instructions lying in sequence after the victim at **EPC** have been started. But you are guaranteed that no effects produced by these instructions will be visible in the registers or CPU state, and no effect at all will occur that will prevent execution, properly restarted at **EPC**, from continuing just as if the exception had not happened.

The MIPS implementation of precise exceptions is quite costly, because it limits the scope for pipelining. That's particularly painful in the FPU, because floating-point operations often take many pipeline stages to run. The instruction following a MIPS FP instruction can't be allowed to commit state (or reach its own exception-determination point) until the hardware can be sure that the FP instruction won't produce an exception.

5.1.1 *Nonprecise Exceptions—The Multiplier in Historic MIPS CPUs*

MIPS's original multiply/divide instructions are started by instructions like **mult** or **div**, which take two register operands and feed them into the multiplier machine. The program then issues an **mflo** instruction (and sometimes also **mghi**, for a 64-bit result or to obtain the remainder) to get the results back into a general-purpose register. The CPU stalls on **mflo** if the computation is not finished.

In MIPS implementations, a multiply takes 4–10 cycles, but divide may take 15–30.

By dividing these long-latency instructions into two stages (“launch” and “get results”), the instruction set encourages the use of a multiply machine separately pipelined from the regular integer unit. Later CPUs provided a richer set of instructions, including multiply-accumulate instructions, which give software different ways of exploiting the pipeline—see section 8.5.5 for details.

On modern CPUs conforming to MIPS32/64, the instructions behave themselves. But older CPUs can show a problem. To make the hardware simpler, the original architecture allowed a multiply/divide operation to be unstoppable, even by an exception. That's not normally a problem, but suppose we have a code sequence like the following, in which we're retrieving one multiply unit result and then immediately firing off another operation:

```
mflo    $8
mult    $9, $10
```

If we take an exception whose restart address is the **mflo** instruction, then the first execution of **mflo** will be nullified under the precise-exception rules, and the register **\$8** will be left as though the **mflo** had never happened. Unfortunately, the **mult** will already have been started too, and since the multiply unit knows nothing of the exception it will continue to run. Before the exception returns, the computation will most likely have finished and the **mflo** will now deliver the result of the **mult** that should have followed it.

We can avoid this problem (inherent in most pre-MIPS32 CPUs), by interposing at least two⁴ nonmultiply instructions between the **mflo/mfhi** on the one hand and the **mult** (or any other instruction that will write new values to **hi/lo**) on the other.

5.2 When Exceptions Happen

Since exceptions are precise, the programmer's view of when an exception happens is unambiguous: the last instruction executed before the exception was the one before the exception victim. And, if the exception wasn't an interrupt, the victim is the instruction that caused it.

On an interrupt in a typical MIPS CPU, the last instruction to be completed before interrupt processing starts will be the one that has just finished its MEM stage when the interrupt is detected. The exception victim will be the one that has just finished its ALU stage. However, take care: MIPS architects don't make promises about exact interrupt latencies, and signals may be resynchronized through one or more clock stages before reaching the CPU core.

5.3 Exception Vectors: Where Exception Handling Starts

Most CISC processors have hardware (or concealed microcode) that analyzes an exception, dispatching the CPU to different entry points according to what kind of exception happened. When even interrupts are handled at different entry

4. Why two? It just happens to be enough to avoid the **mult** being started on all pre-MIPS32 CPUs.

points according to the interrupt input signal(s) activated, that's called *vectored interrupts*. Historically, MIPS CPUs did very little of this. If that seems a serious omission, consider the following.

First, vectored interrupts are not as useful in practice as we might hope. In most operating systems, interrupt handlers share code (for saving registers and such like), and it is common for CISC microcode to spend time dispatching to different interrupt entry points, where OS software loads a code number and spends a little more time jumping back to a common handler.

Second, it's difficult to envision much exception analysis being done by pure hardware rather than microcode; on a RISC CPU, ordinary code is fast enough to be used in preference.

Here and elsewhere, you should bear in mind just how fast CPUs of the RISC generation are compared with their peripherals. A useful interrupt routine is going to have to read/write some external registers, and on an early 21st-century CPU, that external bus cycle is likely to take 50–200 internal clock cycles. It's easy to write interrupt dispatch code on a MIPS CPU that will be faster than a single peripheral access, so this is unlikely to be a performance bottleneck. That's probably emphasized by the fact that a vectored interrupt option in the 2003 revision of MIPS32 has found little use yet.

However, even in MIPS not all exceptions were ever equal, and differences have grown as the architecture has developed. So we can make some distinctions:

- *TLB refill of user-privilege address*: There is one particularly frequent exception in a protected OS, related to the address translation system (see Chapter 6). The TLB hardware only holds a modest number of address translations, and in a heavily used system running a virtual memory OS, it's common for the application program to run on to an address whose translation is not recorded in the TLB—an event called a *TLB miss* (because the TLB is used as a software-managed cache).

The use of software to handle this condition was controversial when RISC CPUs were introduced, and MIPS CPUs provide significant support for a preferred scheme for TLB refill. The hardware helps out enough that the exception handler for the preferred refill scheme can run in as few as 13 clock cycles.

As part of this, common classes of TLB refill are given an entry point different from all other exceptions, so that the finely tuned refill code doesn't have to waste time figuring out what kind of exception has happened.

- *TLB refill for 64-bit address spaces*: Memory translation for tasks wanting to take advantage of the larger program address space available on 64-bit CPUs uses a slightly different register layout and a different TLB refill routine; MIPS calls this an XTLB refill ("X" for extended, I guess). Again, a desire to keep this very efficient makes a separate entry point useful.

- *Uncached alternative entry points:* For good performance on exceptions, the interrupt entry point must be in cached memory, but this is highly undesirable during system bootstrap; from reset or power-up, the caches are unusable until initialized. If you want a robust and self-diagnosing start-up sequence, you have to use uncached read-only memory entry points for exceptions detected in early bootstrap. In MIPS CPUs there is no uncached “mode”—there are uncached program memory regions instead—so there’s a mode bit **SR (BEV)** that reallocates the exception entry points into the uncached, start-up-safe kseg1 region.
- *Parity/ECC error:* MIPS32 CPUs may detect a data error (usually in data arriving from main memory, but often not noticed until it’s used from cache) and take a trap. It would be silly to vector through a cached location to handle a cache error, so regardless of the state of **SR (BEV)** the *cache error exception* entry point is in uncached space.
- *Reset:* For many purposes, it makes sense to see reset as another exception, particularly when many CPUs use the same entry point for *cold reset* (where the CPU is completely reconfigured; indistinguishable from power-up) and *warm reset* (where the software is completely reinitialized). In fact, *nonmaskable interrupt* (NMI) turns out to be a slightly weaker version of warm reset, differing only in that it waits for the current instruction and any pending load/store to finish before taking effect.
- *Interrupt:* As an option in MIPS32 (and some earlier CPUs from IDT and PMC-Sierra), you can set the CPU to dispatch interrupt exceptions to a separate entry point. This is convenient, though little used: Perhaps software authors can’t bring themselves to special-case their OS for a feature that is not universally available.

Further, in some of these CPUs you can enable vectored interrupt operation—multiple entry points to be used by different interrupts. This is a more substantial change; as explained elsewhere in this chapter, the MIPS tradition was that interrupts were only prioritized in software. But if you have two active interrupts and have to choose an interrupt entry point, the hardware must decide which has the higher priority. This change is therefore significantly more disruptive to software, since the software loses control over interrupt priority; your OS maintainer and hardware engineers will have to liaise closely.

All exception entry points lie in untranslated regions of the MIPS memory map, in kseg1 for uncached entry points and in kseg0 for cached ones. The uncached entry points used when **SR (BEV)** is set are fixed, but when **SR (BEV)** is clear, the **EBase** register can be programmed to shift all the entry points—together—to some other block. It’s particularly useful to be able to move the interrupt base when your CPU is part of a multiprocessor system sharing the

TABLE 5.1 Exception Entry Points

<i>Memory region</i>	<i>Entry point</i>	<i>Exceptions handled here</i>
On-chip debug	0xFF20.0200	EJTAG debug, when mapped to “probe” memory. See section 12.1 for some notes on the EJTAG on-chip debug system.
	0xBFC0.0480	EJTAG debug, when using normal ROM memory.
Reset (ROM-only)	0xBFC0.0000	Reset and NMI entry point.
ROM entry points (when SR (BEV) is one)	0xBFC0.0400	Dedicated to interrupts—only used when Cause (IV) is set, not available in all CPUs.
	0xBFC0.0380	All others.
	0xBFC0.0300	Cache Error.
	0xBFC0.0200	Simple TLB Refill (SR (EXL) is zero).
“RAM” entry points (SR (BEV) is zero)	BASE+0x200+...	Multiple interrupt entry points—seven more in VI mode, 62 in EIC mode.
	BASE+0x200	Interrupt special (Cause (IV) is one).
	BASE+0x180	All others.
	BASE+0x100	Cache error—in RAM but always through uncached kseg1 window.
	BASE+0x000	Simple TLB Refill (SR (EXL) is zero).

kseg0 memory but wants to have separate exception entry points from the other CPUs in the system.⁵

In these areas the nominal 32-bit addresses given in Table 5.1 extend to a 64-bit memory map by sign extension: The program address 0x8000.0000 in the 32-bit view is the same as 0xFFFF.FFFF.8000.0000 in the 64-bit view.

Table 5.1 describes the entry points with just 32-bit addresses—you need to accept that **BASE** stands for the exception base address programmed by the **EBase** register.

Presumably the default 128-byte (0x80) gap between the original exception vectors occurs because the original MIPS architects felt that 32 instructions would be enough to code the basic exception routine, saving a branch instruction without wasting too much memory! Modern programmers are rarely so frugal.

5. The **EBase** register was introduced in Release 2 of the MIPS32/64 specification.

Here's what a MIPS CPU does when it decides to take an exception:

1. It sets up **EPC** to point to the restart location.
2. It sets **SR (EXL)**, which forces the CPU into kernel (high-privilege) mode and disables interrupts.
3. **Cause** is set up so that software can see the reason for the exception. On address exceptions, **BadVAddr** is also set. Memory management system exceptions set up some of the MMU registers too; more details are given in Chapter 6.
4. The CPU then starts fetching instructions from the exception entry point, and everything else is up to software.

Very short exception routines can run entirely with **SR (EXL)** set (in exception mode, as we'll sometimes say) and need never touch the rest of **SR**. For more conventional exception handlers, which save state and pass control over to more complex software, exception level provides a cover under which system software can save essential state—including the old **SR** value—in safety.

With a couple of tweaks this mechanism can allow a minimal nested exception within the primitive TLB miss handler, but we'll talk more about how that's done when we get to it.

5.4 Exception Handling: Basics

Any MIPS exception handler routine has to go through the same stages:

- *Bootstrapping*: On entry to the exception handler, very little of the state of the interrupted program has been saved, so the first job is to make yourself enough room to do whatever it is you want without overwriting something vital to the software that has just been interrupted.

Almost inevitably, this is done by using the **k0** and **k1** registers (which are conventionally reserved for the use of low-level exception handling code) to reference a piece of memory that can be used for other register saves.

- *Dispatching different exceptions*: Consult **Cause (ExcCode)**, whose possible values are listed in Table 3.2. It tells you why the exception happened, and allows an OS to define separate functions for the different causes.
- *Constructing the exception processing environment*: Complex exception-handling routines will probably be written in a high-level language, and you will want to be able to use standard library routines. You will have to provide a piece of stack memory that isn't being used by any other piece of software and save the values of any CPU registers that might be both

important to the interrupted program and that called subroutines are allowed to change.

Some operating systems may do this before dispatching different exceptions.

- *Processing the exception:* You can do whatever you like now.
- *Preparing to return:* The high-level function is usually called as a subroutine and therefore returns into the low-level dispatch code. Here, saved registers are restored, and the CPU is returned to its safe (kernel mode, exceptions off) state by changing **SR** back to its postexception value.
- *Returning from an exception:* The end-of-exception processing is another area where the CPU has changed, and its description follows in section 5.5.

5.5 Returning from an Exception

The return of control to the exception victim and the change (if required) back from kernel to a lower-privilege level must be done at the same time (“atomically,” in the jargon of computer science). It would be a security hole if you ran even one instruction of application code at kernel-privilege level; on the other hand, the attempt to run a kernel instruction with user privileges would lead to a fatal exception.

MIPS CPUs have an instruction, **eret**, that does the whole job; it both clears the **SR (EXL)** bit and returns control to the address stored in **EPC**.⁶

5.6 Nesting Exceptions

In many cases, you will want to permit (or will not be able to avoid) further exceptions occurring within your exception processing routine; these are called *nested* exceptions.

Naïvely done, this would cause chaos; vital state from the interrupted program is held in **EPC** and **SR**, and you might expect another exception to overwrite them. Before you permit anything but a very peculiar nested exception, you must save those registers’ contents. Moreover, once exceptions are re-enabled, you can no longer rely on the reserved-for-exception-handler general-purpose registers **k0** and **k1**.

6. Very early MIPS I CPUs had a more complex scheme relying on a two-level stack to save and restore preexception values of the **SR (IE, KU)** fields. The switch back to preexception mode was done by an instruction called **rfe**, which had to be run in the delay slot of the **jr** that took you back to the interrupted program.

An exception handler that is going to survive a nested exception must use some memory locations to save register values. The data structure used is often called an *exception frame*; multiple exception frames from nested exceptions are usually arranged on a stack.

Stack resources are consumed by each exception, so arbitrarily deep nesting of exceptions cannot be tolerated. Most systems award each kind of exception a priority level and arrange that while an exception is being processed, only higher-priority exceptions are permitted. Such systems need have only as many exception frames as there are priority levels.

You can avoid all exceptions; interrupts can be individually masked by software to conform to your priority rules, masked all at once with the **SR (IE)** bit, or implicitly masked (for later CPUs) by the exception-level bit. Other kinds of exceptions can be avoided by appropriate software discipline. For example, privilege violations can't happen in kernel mode (used by most exception processing software), and programs can avoid the possibility of addressing errors and TLB misses. It's essential to do so when processing higher-priority exceptions.

5.7 An Exception Routine

The following MIPS32 code fragment is as simple as an exception routine can be. It does nothing except increment a counter on each exception:

```
.set noreorder
.set noat
xcptgen:
    la      k0,xcptcount    # get address of counter
    lw      k1,0(k0)        # load counter
    addu    k1,1            # increment counter
    sw      k1,0(k0)        # store counter
    eret                    # return to program
.set at
.set reorder
```

This doesn't look very useful: Whichever condition caused the exception will still probably be active on its return, so it might just go round and round. And the counter `xcptcount` had better be in `kseg0` so that you can't get a TLB Miss exception when you read or write it.

5.8 Interrupts

The MIPS exception mechanism is general purpose, but democratically speaking there are two exception types that happen far more often than all the

rest put together. One is the TLB miss when an application running under a memory-mapped OS like UNIX steps outside the (limited) boundaries of the on-chip translation table; we mentioned that before and will come back to it in Chapter 6. The other popular exceptions are interrupts, occurring when a device outside the CPU wants attention. Since we're dealing with an outside world that won't wait for us, interrupt service time is often critical.

Embedded-system MIPS users are going to be most concerned about interrupts, which is why they get a special section. We'll talk about the following:

- *Interrupt resources in MIPS CPUs:* This describes what you've got to work with.
- *Implementing interrupt priority:* All interrupts are equal to MIPS CPUs, but in your system you probably want to attend to some of them before the others.
- *Critical regions, disabling interrupts, and semaphores:* It's often necessary to prevent an interrupt from occurring during critical operations, but there are particular difficulties about doing so on MIPS CPUs. We look at solutions.

5.8.1 *Interrupt Resources in MIPS CPUs*

MIPS CPUs have a set of eight independent⁷ interrupt bits in their **Cause** register. On most CPUs you'll find five or six of these are signals from external logic into the CPU, while two of them are purely software accessible. The on-chip counter/timer (made of the **Count** and **Compare** registers, described in section 3.3.5) will be wired to one of them; it's sometimes possible to share the counter/timer interrupt with an external device, but rarely a good idea to do so.

An active level on any input signal is sensed in each cycle and will cause an exception if enabled.

The CPU's willingness to respond to an interrupt is affected by bits in **SR**. There are three relevant fields:

- The global interrupt enable bit **SR(IE)** must be set to 1, or no interrupt will be serviced.
- The **SR(EXL)** (exception level) and **SR(ERL)** (error level) bits will inhibit interrupts if set (as one of them will be immediately after any exception).
- The status register also has eight individual interrupt mask bits **SR(IM)**, one for each interrupt bit in **Cause**. Each **SR(IM)** bit should be set to 1 to enable the corresponding interrupt so that programs can determine exactly which interrupts can happen and which cannot.

7. Not so independent if you're using EIC mode; see section 5.8.5.

What Are the Software Interrupt Bits For?

Why on earth should the CPU provide 2 bits in the **Cause** register that, when set, immediately cause an interrupt unless masked?

The clue is in “unless masked.” Typically this is used as a mechanism for high-priority interrupt routines to flag actions that will be performed by lower-priority interrupt routines once the system has dealt with all high-priority business. As the high-priority processing completes, the software will open up the inter-

rupt mask, and the pending software interrupt will occur.

There is no absolute reason why the same effect should not be simulated by system software (using flags in memory, for example) but the soft interrupt bits are convenient because they fit in with an interrupt-handling mechanism that already has to be provided, at very low hardware cost.

To discover which interrupt inputs are currently active, you look inside the **Cause** register. Note that these are exactly that—current levels—and do not necessarily correspond to the signal pattern that caused the interrupt exception in the first place. The active input levels in **Cause (IP)** and the masks in **SR (IM)** are helpfully aligned to the same bit positions, in case you want to “and” them together. The software interrupts are at the lowest positions, and the hardware interrupts are arranged in increasing order.

In architectural terms, all interrupts are equal.⁸ When an interrupt exception is taken, an older CPU uses the “general” exception entry point—though MIPS 32/64 CPUs and some other modern CPUs offer an optional distinct exception entry point reserved for interrupts, which can save a few cycles. You can select this with the **Cause (IV)** register bit.

Interrupt processing proper begins after you have received an exception and discovered from **Cause (ExcCode)** that it was a hardware interrupt. Consulting **Cause (IP)**, we can find which interrupt is active and thus which device is signaling us. Here is the usual sequence:

- Consult the **Cause** register IP field and logically “and” it with the current interrupt masks in **SR (IM)** to obtain a bit map of active, enabled interrupt requests. There may be more than one, any of which would have caused the interrupt.
- Select one active, enabled interrupt for attention. Most OSs assign the different inputs to fixed priorities and deal with the highest priority first, but it is all decided by the software.
- You need to save the old interrupt mask bits in **SR (IM)**, but you probably already saved the whole **SR** register in the main exception routine.
- Change **SR (IM)** to ensure that the current interrupt and all interrupts your software regards as being of equal or lesser priority are inhibited.

8. That’s not quite true in vectored interrupt and “EIC mode,” described in section 5.8.5, but they’re not widely used.

- If you haven't already done it in the main exception routine, save the state (user registers, etc.) required for nested exception processing.
- Now change your CPU state to that appropriate to the higher-level part of the interrupt handler, where typically some nested interrupts and exceptions are permitted.

In all cases, set the global interrupt enable bit **SR(IE)** to allow higher-priority interrupts to be processed. You'll also need to change the CPU privilege-level field **SR(KSU)** to keep the CPU in kernel mode as you clear exception level and, of course, clear **SR(EXL)** itself to leave exception mode and expose the changes made in the status register.

- Call your interrupt routine.
- On return you'll need to disable interrupts again so you can restore the preinterrupt values of registers and resume execution of the interrupted task. To do that you'll set **SR(EXL)**. But in practice you're likely to do this implicitly when you restore the just-after-exception value of the whole **SR** register, before getting into your end-of-exception sequence.

When making changes to **SR**, you need to be careful about changes whose effect is delayed due to the operation of the pipeline—"CP0 hazards." See section 3.4 for more details and how to program around the hazards.

5.8.2 *Implementing Interrupt Priority in Software*

The MIPS CPU (until you use the new vectored interrupt facilities) has a simple-minded approach to interrupt priority; all interrupts are equal.

If your system implements an interrupt priority scheme, then:

- At all times the software maintains a well-defined *interrupt priority level* (IPL) at which the CPU is running. Every interrupt source is allocated to one of these levels.
- If the CPU is at the lowest IPL, any interrupt is permitted. This is the state in which normal applications run.
- If the CPU is at the highest IPL, then all interrupts are barred.

Not only are interrupt handlers run with the IPL set to the level appropriate to their particular interrupt cause, but there's provision for programmers to raise and lower the IPL. Those parts of the application side of a device driver that communicate with the hardware or the interrupt handler will often need to prevent device interrupts in their critical regions, so the programmer will temporarily raise the IPL to match that of the device's interrupt input.

In such a system, high-IPL interrupts can continue to be enabled without affecting the lower-IPL code, so we've got the chance to offer better interrupt

response time to some interrupts, usually in exchange for a promise that their interrupt handlers will run to completion in a short time.

Most UNIX systems have between four and six IPLs.

While there are other ways of doing it, the simplest schemes have the following characteristics:

- *Fixed priorities*: At any IPL, interrupts assigned to that and lower IPLs are barred, but interrupts of higher IPLs are enabled. Different interrupts at the same IPL are typically scheduled first come, first served.
- *IPL relates to code being run*: Any given piece of code always executes at the same IPL.
- *Simple nested scheduling (above IPL 0)*: Except at the lowest level, any interrupted code will be returned to as soon as there are no more active interrupts at a higher level. At the lowest level there's quite likely a scheduler that shares the CPU out among various tasks, and it's common to take the opportunity to reschedule after a period of interrupt activity.⁹

On a MIPS CPU a transition between interrupt levels must (at least) be accompanied by a change in the status register **SR**, since that register contains all the interrupt control bits. On some systems, interrupt level transitions will require doing something to external interrupt control hardware, and most OSs have some global variables to change, but we don't care about that here; for now we'll characterize an IPL by a particular setting of the **SR** interrupt fields.

In the MIPS architecture **SR** (like all coprocessor registers) is not directly accessible for bit setting and clearing. Any change in the IPL, therefore, requires a piece of code that reads, modifies, and writes back the **SR** in separate operations:

```

        mfc0    t0, SR
1:
        or      t0, things_to_set
        and     t0, ~(things_to_clear)
2:
        mtc0    t0, SR
        ehb

```

(The last instruction, **ehb**, is a *hazard barrier* instruction; see section 3.4.)

In general, this piece of code may itself be interrupted, and a problem arises: Suppose we take an interrupt somewhere between label **1** and **2** and that interrupt routine itself causes a change in **SR**? Then when we write our

9. Linux systems that reschedule after an interrupt (even when the interrupted task was working in the kernel) are called *pre-emptive*, and with the 2004 v2.6 version pre-emption became standard for Linux.

own altered value of **SR** at label **2**, we'll lose the change made by the interrupt routine.

It turns out that we can only get away with the code fragment above—some version of which is pretty much universal in MIPS implementations of OSs—in systems where we can rely on the IPL being constant in any particular piece of code (and where we don't make any other running changes to **SR**). With those conditions, even if we get interrupted in the middle of our read-modify-write sequence, it will do no harm; when the interrupt returns it will do so with the same IPL, and therefore the same **SR** value, as before.

Where this assumption breaks down, we need the following discussion.

5.8.3 *Atomicity and Atomic Changes to SR*

In systems with more than one thread of control—including a single application with interrupt handlers—you will quite often find yourself doing something during which you don't want to be caught halfway. In more formal language, you may want a set of changes to be made *atomically*, so that some cooperating task or interrupt routine in the system will see either none of them made or all of them, but never anything in between.¹⁰ The code implementing the atomic change is sometimes called a *critical region*.

On a uniprocessor that runs multiple threads in software, a thread switch that surprises the current thread can only happen as the consequence of some interrupt or other. So in a uniprocessor system, any critical region can be simply protected by disabling all interrupts around it; this is crude but effective.

But as we saw above, there's a problem: The interrupt-disabling sequence (requiring a read-modify-write sequence on **SR**) is itself not guaranteed to be atomic. I know of four ways of fixing this impasse and one way to avoid it.

You can fix it if you know that all CPUs running your software implement MIPS32 Release 2: In that case, you can use the **di** instruction instead of the **mfc0**. **di** atomically clears the **SR (IE)** bit, returning the original value of **SR** in a general-purpose register.¹¹ But until MIPS32 Release 2 compliance becomes the rule rather than the exception, you may need to look further.

A more general fix is to insist that no interrupt may change the value of **SR** held by any interruptible code; this requires that interrupt routines always restore **SR** before returning, just as they're expected to restore the state of all the user-level registers. If so, the nonatomic RMW sequence above doesn't matter; even if an interrupt gets in, the old value of **SR** you're using will still be correct. This approach is generally used in UNIX-like OS kernels for MIPS and goes well with the interrupt priority system in which every piece of code is associated with a fixed IPL.

10. An old saying goes: "Never show fools and children things half done"—I take it as read that computers and their software are foolish and childish.

11. There's an **ei** too for enabling interrupts, but don't use it—restore the **di**-returned value of **SR** instead, and your "disable interrupts" code will not malfunction if you accidentally invoke it when interrupts were already disabled.

But sometimes this restriction is too much. For example, when you’ve sent the last waiting byte on a byte-at-a-time output port, you’d like to disable the ready-to-send interrupt (to avoid eternal interrupts) until you have some more data to send. And again, some systems like to rotate priorities between different interrupts to ensure a fair distribution of interrupt service attention.

Another solution is to use a system call to disable interrupts (probably you’d define the system call as taking separate bit-set and bit-clear parameters and get it to update the status register accordingly). Since a **syscall** instruction works by causing an exception, it disables interrupts atomically. Under this protection your bit-set and bit-clear can proceed cheerfully. When the system call exception handler returns, the global interrupt enable status is restored (once again atomically).

A system call sounds pretty heavyweight, but it actually doesn’t need to take long to run; however, you will have to untangle this system call from the rest of the system’s exception-dispatching code.

The third solution—which all substantial systems should use for at least some critical regions—is to use the load-linked and store-conditional instructions to build critical regions without disabling interrupts at all, as described below. Unlike anything described above, that mechanism extends correctly to multiprocessor or hardware-multithreading systems.

5.8.4 *Critical Regions with Interrupts Enabled: Semaphores the MIPS Way*

A semaphore¹² is a coding convention to implement critical regions (though extended semaphores can do more tricks than that). The semaphore is a shared memory location used by concurrently running processes to arrange that some resource is only accessed by one of them at once.

Each atomic chunk of code has the following structure:¹³

```
wait(sem);
/* do your atomic thing */
signal(sem);
```

Think of the semaphore as having two values: 1 meaning “in use” and 0 meaning “available.” The `signal()` is simple; it sets the semaphore to 0.¹⁴

12. A simple semaphore is also called a *mutex* and informally just called a “lock.”

13. Dijkstra formulated these ideas back in the 1970s and named it a “semaphore” from a railway-signaling analogy. Hoare set them in the wider context of “cooperating parallel processes” and called essentially the same functions `wait()` and `signal()`—and that’s what we’ve used. Dijkstra’s original names are `p()` and `v()`, respectively. You can understand why he called them “p” and “v” quite easily, if you speak Dutch.

14. For a thread-to-thread semaphore in an OS, `signal()` also has to do something to “wake up” any other thread that was waiting on the semaphore.

`wait()` checks for the variable to have the value 0 and won't continue until it does. It then sets the variable to 1 and returns. That should be easy, but you can see that it's essential that the process of checking the value of `sem` and setting it again is itself atomic. High-level atomicity (for threads calling `wait()`) is dependent on being able to build low-level atomicity, where a test-and-set operation can operate correctly in the face of interrupts (or, on a multiprocessor, in the face of access by other CPUs).

Most mature CPU families have some special instruction for this: 680x0 CPUs—and many others—have an atomic test-and-set instruction; x86 CPUs have an “exchange register with memory” operation that can be made atomic with a prefix “lock” instruction.

For large multiprocessor systems this kind of test-and-set process becomes expensive; essentially, all shared memory access must be stopped while first the semaphore user obtains the value and completes the test-and-set operation, and, second, the set operation percolates through to every cached copy in the system. This doesn't scale well to large multiprocessors, because you're holding up lots of probably unrelated traffic just to make sure you held up the occasional thing that mattered.

It's much more efficient to allow the test-and-set operation to run without a prior guarantee of atomicity so that the “set” succeeds only if it was atomic. Software needs to be told whether the set succeeded: Then unsuccessful test-and-set sequences can be hidden inside the `wait()` function and retried as necessary.¹⁵

This is what MIPS has, using the **ll** (load-linked) and **sc** (store-conditional) instructions in sequence. **sc** will only write the addressed location if the hardware confirms that there has been no competing access since the last **ll** and will leave a 1/0 value in a register to indicate success or failure.

In most implementations this information is pessimistic: Sometimes **sc** will fail even though the location has not been touched; CPUs will fail the **sc** when there's been any exception serviced since the **ll**,¹⁶ and most multiprocessors will fail on any write to the same “cache line”-sized block of memory. It's only important that the **sc** should usually succeed when there's been no competing access and that it *always* fails when there has been one such.

Here's `wait()` for the binary semaphore `sem`:

```
wait:
    la      t0, sem
TryAgain:
    ll      t1, 0(t0)
    bne     t1, zero, WaitForSem
```

15. Of course, you'd better make sure that there are no circumstances where it ends up retrying forever—but with other kinds of semaphores, you always have to make sure a waiting task comes back sometime.

16. To be precise, if an **eret** has been executed.

```

li      t1, 1
sc      t1, 0(t0)
beq     t1, zero, TryAgain
/* got the semaphore... */
jr      ra

```

`ll/sc` was invented for multiprocessors, but even in a uniprocessor system, this kind of operation can be valuable, because it does not involve shutting out interrupts. It avoids the interrupt-disabling problem described above and contributes to a coordinated effort to reduce worst-case interrupt latency, very desirable in embedded systems.

5.8.5 *Vectored and EIC Interrupts in MIPS32/64 CPUs*

Release 2 of the MIPS32 specification—first seen in the 4KE and 24K family CPU cores from MIPS Technologies—adds two new features that can make interrupt handling more efficient. The savings are modest and probably wouldn't be important in a substantial OS, but MIPS CPUs are also used in very low level embedded environments where these kinds of improvements are very welcome. Those features are vectored interrupts and a way of providing a large number of distinguishable interrupts to the CPU, called *EIC mode*.

If you switch on the vectored interrupt feature, an interrupt exception will start at one of eight addresses according to the interrupt input signal that caused it. That's slightly ambiguous: There's nothing to stop two interrupts being active at once, so the hardware uses the highest-numbered interrupt signal, which is both active and enabled. Vectored interrupts are set up by programming **IntCtl (VS)**, which gives you a few choices as to the spacing between the different entry points (a zero causes all the interrupts to use the same entry point, as was traditional).

Embedded systems often have a very large number of interrupt events to signal, far beyond the six hardware inputs of traditional MIPS CPUs. In EIC mode, those six formerly independent signals become a 6-bit binary number: Zero means no interrupt, but that leaves 63 distinct interrupt codes. In EIC mode, each nonzero code has its own interrupt entry point, allowing an appropriately designed interrupt controller to dispatch the CPU directly to handle up to 63 events.

Vectored interrupts (whether with traditional or EIC signaling) are likely to be most helpful in circumstances where one or two interrupt events in your system are particularly frequent or time-critical. The small number of cycles saved are likely to be lost in fitting into the interrupt-handling discipline of a more sophisticated OS, so don't be surprised to find that your favorite OS does not use these features.

5.8.6 *Shadow Registers*

Even with interrupt vectors, an interrupt routine is burdened with the need to avoid trashing the register values of the code it interrupted, and must load addresses for itself before it can do any useful work.

Release 2 of the MIPS32/64 specification permits CPUs to provide one or more distinct sets of general-purpose registers: The extra register sets are called *shadow registers*. Shadow registers can be used for any kind of exception but are most useful for interrupts.

An interrupt handler using a shadow register set has no need to save the interrupted program's register values and may keep its own state between invocations in its own registers (if more than one interrupt handler uses the shadow set, they'd better agree who gets to use which register).

An interrupt handler using vectored interrupts and a shadow register set can be unburdened by housekeeping and can run phenomenally fast. But again, that advantage can be lost by the discipline of an OS (in particular because the OS is likely to disable all interrupts for periods of time that far exceed our superfast interrupt handler's run time). Some applications that would benefit from shadow registers might get the same kind of benefit more cleanly by using a multithreading CPU, but that's a much bigger story—see Appendix A.

5.9 Starting Up

In terms of its software-visible effect on the CPU, reset is almost the same as an exception, though one from which we're not going to return. In the original MIPS architecture, this is mostly a matter of economy of implementation effort and documentation, but later CPUs have offered several different levels of reset, from a cold reset through to a nonmaskable interrupt. In MIPS, reset and exception conditions shade imperceptibly into each other.

We're recycling mechanisms from regular exceptions, following reset **EPC** points to the instruction that was being executed when reset was detected, and most register values are preserved. However, reset disrupts normal operation, and a register being loaded or a cache location being stored to or refilled at the moment reset occurred may be trashed.

It is possible to use the preservation of state through reset to implement some useful postmortem debugging, but your hardware engineer needs to help; the CPU cannot tell you whether reset occurred to a running system or from power-up. But postmortem debugging is an exercise for the talented reader; we will focus on starting up the system from scratch.

The CPU responds to reset by starting to fetch instructions from `0xBFC0.0000`. This is physical address `0x1FC0.0000` in the uncached kseg1 region.

Following reset, enough of the CPU's control register state is defined that the CPU can execute uncached instructions. "Enough state" is interpreted minimally; note the following points:

- Only three things are guaranteed in **SR**: The CPU is in kernel mode; interrupts are disabled; and exceptions will vector through the uncached entry points—that is, **SR (BEV)** = 1. In modern CPUs, the first two conditions (and more beside) are typically guaranteed by setting the exception-mode bit **SR (EXL)**, and this is implied by treating reset as an exception.
- The caches will be in a random, nonsensical state, so a cached load might return rubbish without reading memory.
- The TLB will be in a random state and *must not be accessed* until initialized (in some CPUs the hardware has only minimal protection against the possibility that there are duplicate matches in the TLB, and the result could be a TLB shutdown, which can be amended only by a further reset).

The traditional start-up sequence is as follows:

1. Branch to the main ROM code. Why do a branch now?
 - The uncached exception entry points start at `0xBFC0.0100`, which wouldn't leave enough space for start-up code to get to a "natural break"—so we're going to have to do a branch soon, anyway.
 - The branch is a very simple test to see if the CPU is functioning and is successfully reading instructions. If something terrible goes wrong with the hardware, the MIPS CPU is most likely to keep fetching instructions in sequence (and next most likely to get permanent exceptions).

If you use test equipment that can track the addresses of CPU reads and writes, it will show the CPU's uncached instruction fetches from reset; if the CPU starts up and branches to the right place, you have strong evidence that the CPU is getting mostly correct data from the ROM.

By contrast, if your ROM program plows straight in and fiddles with **SR**, strange and undiagnosable consequences may result from simple faults.

2. Set the status register to some known and sensible state. Now you can load and store reliably in uncached space.
3. You will probably have to run using registers only until you have initialized and (most likely) run a quick check on the integrity of some RAM memory. This will be slow (we're still running uncached from ROM), so you will probably confine your initialization and check to a chunk of memory big enough for the ROM program's data.

4. You will probably have to make some contact with the outside world (a console port or diagnostic register) so you can report any problem with the initialization process.
5. You can now assign yourself some stack and set up enough registers to be able to call a standard C routine.
6. Now you can initialize the caches and run in comfort. Some systems can run code from ROM cached and some can't; on most MIPS CPUs, a memory supplying the cache must be able to provide 4/8-word bursts, and your ROM subsystem may or may not oblige.

5.9.1 *Probing and Recognizing Your CPU*

You can identify your CPU implementation number and a manufacturer-defined revision level from the **PRId(Imp)** and **PRId(Rev)** fields. However, it's best to rely on this information as little as possible; if you rely on **PRId**, you guarantee that you'll have to change your program for any future CPU, even though there are no new features that cause trouble for your program.

Whenever you can probe for a feature directly, do so. Where you can't figure out a reliable direct probe, then for CPUs that are MIPS32/64-compliant, there's a good deal of useful information encoded in the various **Config** registers described in section 3.3.7. Only after exhausting all other options should you consider relying on **PRId**.

Nonetheless, your boot ROM and diagnostic software should certainly display **PRId** in some readable form. And should you ever need to include a truly unpleasant software workaround for a hardware bug, it's good practice to make it conditional on **PRId**, so your workaround is not used on future, fixed, hardware.

Since we've recommended that you probe for individual features, here are some examples of how you could do so:

- *Have we got FP hardware?* Your MIPS32/64 CPU should tell you through the **Config1(FP)** register bit. For older CPUs one recommended technique is to set **SR(CU1)** to enable coprocessor 1 operations and to use a **cfc1** instruction from coprocessor 1 register 0, which is defined to hold the revision ID. A nonzero value in the **ProcessorID** field (bits 8–15) indicates the presence of FPU hardware—you'll most often see the same value as in the **PRId(Imp)** field. A skeptical programmer¹⁷ will probably follow this up by checking that it is possible to store and retrieve data from the FPU registers. Unless your system supports unconditional use of the floating-point unit, don't forget to reset **SR(CU1)** afterward.

17. I assume, Gentle Reader, that this is you.

- *Cache size:* On a MIPS32/64-compliant CPU, it probably is better to rely on the values encoded in the configuration registers **Config1-2**. But you can also deduce the size by reading and writing cache tags and looking for when the cache size wraps around.
- *Have we got a TLB?* That's memory translation hardware. The **Config (MT)** bit tells you whether you have it. But you could also read and write values to **Index** or look for evidence of a continuously counting **Random** register. If it looks promising, you may want to check that you can store and retrieve data in TLB entries.
- *CPU clock rate:* It is often useful to work out your clock rate. You can do this by running a loop of known length, cached, that will take a fixed large number of CPU cycles and then comparing with “before” and “after” values of a counter outside the CPU that increments at known speed. Do make sure that you are really running cached, or you will get strange results—remember that some hardware can't run cached out of ROM.

The Linux kernel does this when it boots and reports a number called *BogoMIPS*, to emphasize that any relationship between the number and CPU performance is bogus.

If your CPU is compliant to MIPS32/64 Release 2 (the 2003 specification), the **rdhwr** instruction gives you access to the divider between the main CPU clock and the speed with which the **Count** register increments. It's simpler to compare the counter speed with some external reference.

Some maintenance engineer will bless you one day if you make the CPU ID, clock rate, and cache sizes available, perhaps as part of a sign-on message.

5.9.2 *Bootstrap Sequences*

Start-up code suffers from the clash of two opposing but desirable goals. On the one hand, it's robust to make minimal assumptions about the integrity of the hardware and to attempt to check each subsystem before using it (think of climbing a ladder and trying to check each rung before putting your weight on it). On the other hand, it's desirable to minimize the amount of tricky assembly code. Bootstrap sequences are almost never performance sensitive, so an early change to a high-level language is desirable. But high-level language code tends to require more subsystems to be operational.

After you have dealt with the MIPS-specific hurdles (like setting up **SR** so that you can at least perform loads and stores), the major question is how soon you can make some read/write memory available to the program, which is essential for calling functions written in C.

Sometimes diagnostic suites include bizarre things like the code in the original PC BIOS, which tests each 8086 instruction in turn. This seems to me like

chaining your bicycle to itself to foil thieves. If any subsystem is implemented inside the same chip as the CPU, you don't lose much by trusting it.

5.9.3 *Starting Up an Application*

To be able to start a C application (presumably with its instructions coming safely from ROM) you need writable memory, for three reasons.

First, you need stack space. Assign a large enough piece of writable memory and initialize the `sp` register to its upper limit (aligned to an eight-byte boundary). Working out how large the stack should be can be difficult: A generous guess will be more robust.

Then you may need some initialized data. Normally, the C data area is initialized by the program loader to set up any variables that have been allocated values. Most compilation systems that purport to be usable for embedded applications permit read-only data (implicit strings and data items declared `const`) to be managed in a separate segment of object code and put into ROM memory.

Initialized writable data can be used only if your compilation system and runtime system cooperate to arrange to copy writable data initializations from ROM into RAM before calling `main()`.

Last, C programs use a different segment of memory for all `static` and `extern` data items that are not explicitly initialized—an area sometimes called “BSS,” for reasons long lost. Such variables should be cleared to zero, which is readily achieved by zeroing the whole data section before starting the program.

If your program is built carefully, that's enough. However, it can get more complicated: Take care that your MIPS program is not built to use the global pointer register `gp` to speed access to nonstack variables, or you'll need to do more initialization.

5.10 Emulating Instructions

Sometimes an exception is used to invoke a software handler that will stand in for the exception victim instruction, as when you are using software to implement a floating-point operation on a CPU that doesn't support FP in hardware. Debuggers and other system tools may sometimes want to do this too.

To emulate an instruction, you need to find it, decode it, and find its operands. A MIPS instruction's operands were in registers, and by now those preexception register values have been put away in an exception frame somewhere. Armed with those values, you do the operation in software and patch the results back into the exception frame copy of the appropriate result register. You then need to fiddle with the stored exception return address so as to step over the emulated instruction and then return. We'll go through these step by step.

Finding the exception-causing instruction is easy; it's usually pointed to by **EPC**, unless it was in a branch delay slot, in which case **Cause (BD)** is set and the exception victim is at the address **EPC + 4**.

To decode the instruction, you need some kind of reverse-assembly table. A big decode-oriented table of MIPS instructions is part of the widely available GNU debugger `gdb`, where it's used to generate disassembly listings. So long as the GNU license conditions aren't a problem for you, that will save you time and effort.

To find the operands, you'll need to know the location and layout of the exception frame, which is dependent on your particular OS (or exception-handling software, if it's too humble to be called an OS).

You'll have to figure out for yourself how to do the operation, and once again you need to be able to get at the exception frame, to put the results back in the saved copy of the right register.

There's a trap for the unwary in incrementing the stored **EPC** value to step over the instruction you've emulated. If your emulated instruction was in a branch delay slot, you can't just return to the "next" instruction—that would be as if the branch was not taken, and you don't know that.

So when your emulation candidate is in a branch delay slot, you also have to emulate the branch instruction, testing for whether the branch should be taken or not. If the branch should be taken, you need to compute its target and return straight there from the exception.

Fortunately, all MIPS branch instructions are simple and free of side effects.

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Low-level Memory Management and the TLB

We’ve tended to introduce most topics in this book from the bottom, which is perhaps natural in a book about low-level computer architecture. But—unless you’re already pretty familiar with a virtual memory system and how it’s used by an OS—you’re recommended to turn now to section 14.4, which describes how virtual memory management works in Linux. Once that makes some sense to you, come back here for hardware details and to see how the hardware can be made to work in other contexts.

When we’re thinking top-down, the hardware is often called a memory-management unit or MMU. When we’re looking bottom-up, we focus on the main hardware item, known as the TLB (for “translation lookaside buffer,” which is unlikely to help much).

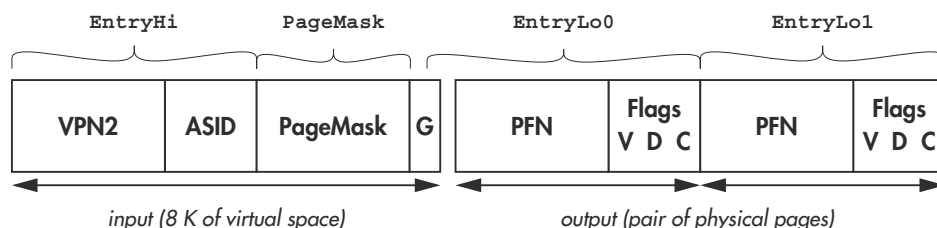
6.1 The TLB/MMU Hardware and What It Does

The TLB is the hardware that translates an address used by your program (*program* addresses or *virtual* addresses) to the physical address, which accesses memory. The OS’s control over memory translation is the key to all software security features.

On MIPS CPUs (and all modern CPUs), translation operates on 4-Kbyte¹ chunks called *pages*. The 12-bit address within the page is simply passed on from virtual address to physical address.

Each entry in the table has the virtual address of a page (the VPN for *virtual page number*), and a physical page address (PFN for *page frame number*). When a program presents a virtual address, it is compared with

1. Page sizes bigger than 4 Kbytes are supported by MIPS hardware and, with memory so cheap, a bigger basic page size might be advantageous to software, but the 4-Kbyte habit is proving very hard to kick. Much larger page sizes are occasionally used for special translations.

**FIGURE 6.1** A TLB entry.

each VPN in the TLB, and if it matches an entry the corresponding PFN is delivered. The TLB is what is called an associative or content-addressable memory—an entry is selected not by an index but by its contents. It's a fairly complex piece of logic, where each entry has a built-in comparator, and complexity and performance scale badly. So the TLB typically has between 16 and 64 entries.

A set of flag bits is stored and returned with each PFN and allows the OS to designate a page as read-only or to specify how data from that page might be cached.

Most modern MIPS CPUs (and all MIPS32/64 CPUs) double up, with each TLB entry holding two independent physical addresses corresponding to an adjacent pair of virtual pages.

Figure 6.1 shows one TLB entry. The fields are labeled with the names of the CP0 registers used when loading and reading TLB entries in software, as described in the next section.

When you're running a real, complicated OS, the software rapidly covers more ground than can be accommodated by the TLB's modest set of translations. That's fixed by maintaining the TLB as a software-managed cache of recently used translations. When a needed translation isn't in the TLB, that generates an exception and the exception handler should figure out and install the correct translation. It's hard to believe this can work efficiently, but it does.

6.2 TLB/MMU Registers Described

Like everything else in a MIPS CPU, MMU control is effected by a rather small number of extra instructions and a set of registers in the coprocessor 0 set. Table 6.1 lists the control registers, and we'll get around to the instructions in section 6.3. All TLB entries being written or read are staged through the registers **EntryHi**, **EntryLo0-1**, and **PageMask**.

TABLE 6.1 CPU Control Registers for Memory Management

<i>Register mnemonic</i>	<i>CP0 register number</i>	<i>Description</i>
EntryHi	10	Together these registers hold everything needed for a TLB entry. All reads and writes to the TLB must be staged through them. EntryHi holds the VPN and ASID; in MIPS32/64 CPUs, each entry maps two consecutive VPNs to different physical pages, so the PFN and access permission flags are specified independently for the two pages by registers called
EntryLo0-1	2-3	EntryLo0 and EntryLo1 .
PageMask	5	The field EntryHi (ASID) does double duty, since it remembers the currently active ASID. EntryHi grows to 64 bits in 64-bit CPUs but in such a way as to preserve the illusion of a 32-bit layout for software that doesn't need long addresses. PageMask can be used to create entries that map pages bigger than 4 KB; see section 6.2.1. Some CPUs support smaller page sizes, but that's barely mentioned in this book.
Index	0	This determines which TLB entry will be read/written by appropriate instructions.
Random	1	This pseudorandom value (actually a free-running counter) is used by a tlbwr to write a new TLB entry into a randomly selected location. Saves time when processing TLB refill traps for software that likes the idea of random replacement (there is probably no viable alternative).
Context	4	These are convenience registers, provided to speed up the processing of TLB refill traps. The high-order bits are read/write; the low-order bits are taken from the VPN of the address that couldn't be translated.
XContext	20	The register fields are laid out so that, if you use the favored arrangement of memory-held copies of memory translation records, then following a TLB refill trap Context will contain a pointer to the page table record needed to map the offending address. See section 6.2.4. XContext does the same job for traps from processes using more than 32 bits of effective address space; a straightforward extension of the Context layout to larger spaces would be unworkable because of the size of the resulting data structures. Some 64-bit CPU software is happy with 32-bit virtual address spaces; but for when that's not enough, 64-bit CPUs are equipped with "mode bits" SR (UX) , SR (KX) , which can be set to invoke an alternative TLB refill handler; in turn that handler can use XContext to support a huge but manageable page table format.

6.2.1 TLB Key Fields—*EntryHi* and *PageMask*

Figure 6.2 shows these registers, which, with **EntryLo** (described below), are the programmer's only view of a TLB entry. The diagram shows both the MIPS32 and MIPS64 versions of **EntryHi**, with these fields:

- **VPN2** (*virtual page number*): These are the high-order bits of a program address (with the low, in-page bits 0–13 omitted). Bits 0–12 would be the in-page address, but bit 13 of the program address isn't looked up either: Each entry is going to map a pair of 4-KB pages, and bit 13 will automatically select between the two possible output values.

Following a refill exception, this field is set up automatically to match the program address that could not be translated. When you want to write a different TLB entry, or attempt a TLB probe, you have to set it up by hand.

The MIPS64 version of **EntryHi** allows for bigger virtual address regions than any implemented so far—to as large as 2^{62} bits, though current CPUs typically only implement 2^{40} bits. The implemented bits of **VPN2** go up far enough to cope, and, in fact, you find out how much virtual space you've got through this register. If you write all-ones to **EntryHi** (**Fill/VPN2**) and then read it back, the valid bits of **EntryHi** (**VPN2**) will be those that read back as 1.

In use, higher bits of **VPN2** than are used by your CPU *must* be written as all ones or all zeros, matching the most significant bit of the **R** field. Equivalently, the higher bits are all 1 when accessing kernel-only address spaces and all 0 otherwise.

If you are only using the 32-bit instruction set, this will happen automatically, because when you work this way all register values contain the 64-bit sign extension of a 32-bit number. Therefore, 32-bit software running on 64-bit hardware can pretend it has a 32-bit **EntryHi** layout.

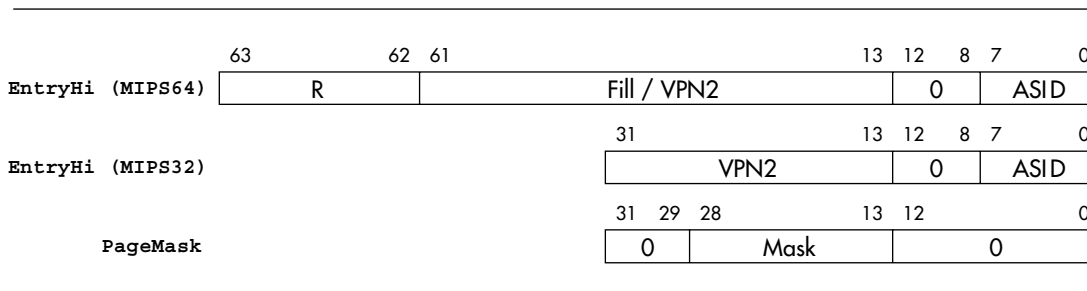


FIGURE 6.2 **EntryHi** and **PageMask** register fields.

Note also there are a few unused zero bits below the VPN2 field; in fact they're not always unused—some CPUs can be configured to support a 1-KB page size, in which case VPN2 and the corresponding mask field grow downward by 2 bits.

- *ASID (address space identifier)*: This is normally left holding the operating system's idea of the current address space. This is not changed by exceptions, so after a refill exception, this will still have the right value in it for the currently running process.

An OS using multiple address spaces will maintain this field to the current address space. But because it is tucked into **EntryHi**, you have to be careful when using **tlbr** to inspect TLB entries; that operation overwrites the whole of **EntryHi**, so you will have to restore the correct current ASID value afterward.

- *R*: (64-bit version only) This is an address region selector. But you can consistently regard this field as just more bits of **EntryHi (VPN2)**; it's just the highest-order bits of the 64-bit MIPS virtual address. However, if you remember the 64-bit extended-memory map (see Figure 2.2 in section 2.8), you can see that these high-order bits select memory regions with different access privileges:

R value	Region name	Brief description
0	xuseg	User-mode accessible space in low virtual memory
1	xsseg	Supervisor-mode accessible space (supervisor mode is optional)
2	xkphys	Kernel-only large windows on physical memory (cached and uncached)
3	xkse	Kernel-mode space (includes MIPS32 compatibility segments)

The **R** bits are unlike the high bits of VPN2 because they can indeed take on different values—an implementation-defined number of high-order bits of **EntryHi (VPN2)** may only be all ones or all zeros.

The **PageMask** register allows you to set up TLB fields that map larger pages. The **PageMask (Mask)** field represents part of the TLB entry, and 1 bits have the effect of causing the corresponding bit of the virtual address to be ignored when matching the TLB entry (and causing that bit to be carried unchanged to the resulting physical address), effectively matching a larger page size.

No MIPS CPU permits arbitrary bit patterns in **PageMask (Mask)** . Most allow page sizes between 4 KB and 16 MB in $\times 4$ steps:

<i>PageMask bits</i>			
<i>24–21</i>	<i>20–17</i>	<i>16–13</i>	<i>Page size</i>
0000	0000	0000	4 KB
0000	0000	0011	16 KB
0000	0000	1111	64 KB
0000	0011	1111	256 KB
0000	1111	1111	1 MB
0011	1111	1111	4 MB
1111	1111	1111	16 MB

At least one older MIPS CPU supports only 4-KB and 16-MB pages but uses the standard encodings for those sizes; you’re recommended to do some probing to see which values “stick” in **PageMask**.

If your CPU is able to support 1-KB pages, you’ll have 2 extra bits on the bottom of **PageMask**, whose setting follows the same pattern and is an exercise for the reader.

6.2.2 *TLB Output Fields—EntryLo0-1*

Figure 6.3 shows both the 64- and 32-bit versions of **EntryLo**, whose fields are as follows:

- *PFN*: These are the high-order bits of the physical address to which values matching the entry field corresponding to **EntryHi (VPN2)** s will be translated. The active width of this field depends on the physical memory space supported by your CPU. MIPS32 CPUs are quite often attached to external interfaces limited to 2^{32} bytes range, but the MIPS32 version of **EntryLo** can potentially support as much as a 2^{38} bytes physical range (the 26-bit **PFN** provides 2^{26} pages, each 4 KB or 2^{12} bytes in size).

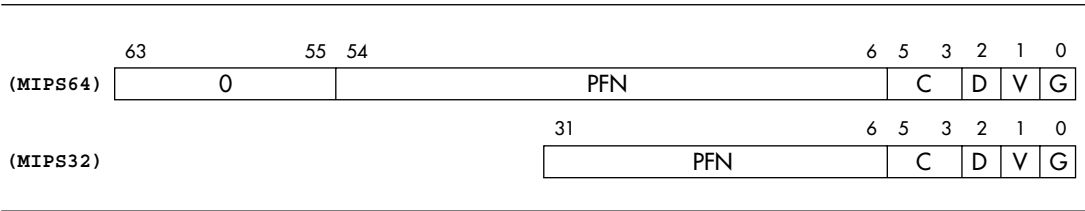


FIGURE 6.3 **EntryLo0-1** register fields.

- **C**: A 3-bit field originally defined for cache-coherent multiprocessor systems to set the “cache algorithm” (or “cache coherency attribute”—some manuals call this field **CCA**). An OS will typically know that some pages will not need to have changes tracked automatically through multiple caches—pages known to be used only by one CPU, or known to be read-only, don’t need so much care. It can make the system more efficient to turn off the cache snooping and interaction for accesses to these pages, and this field is used by the OS to note that the page is, for example, cacheable but doesn’t need coherency management (“cacheable noncoherent”).

But the field has also been used in CPUs aimed at embedded applications, when it selects how the cache works—for example, marking some particular page to be managed “write-through” (that is, all writes made there are sent straight to main memory as well as to any cached copy).

The only universally supported values of this field denote “uncached” (2) and “cacheable noncoherent” (3).

- **D** (*dirty*): This functions as a write-enable bit. Set 1 to allow writes, 0 to cause any store using this translation to be trapped. See section 14.4.7 for an explanation of the term *dirty*.
- **V** (*valid*): If set 0, any use of an address matching this entry will cause an exception. Used either to mark a page that is not available for access (in a true virtual memory system) or to mark one **EntryLo** part of a paired translation as not available.
- **G** (*global*): When the G bit in a TLB entry is set, that TLB entry will match solely on the VPN field, regardless of whether the TLB entry’s ASID field matches the value in **EntryHi**. That provides an efficient mechanism to implement parts of the address space that are shared between all processes. Note that there is really only one “G” bit in a TLB entry, although there are two **EntryLo** registers: bad things will happen if you have different values in **EntryLo0** (**G**) and **EntryLo1** (**G**).
- *Fields called 0 and unused PFN bits*: These fields always return zero, but unlike many reserved fields, they do not need to be written as zero (nothing happens regardless of the data written). This is important; it means that the memory-resident data used to generate **EntryLo** when refilling the TLB can contain some software-interpreted data in these fields, which the TLB hardware will ignore without the need to spend precious CPU cycles masking it out.

6.2.3 *Selecting a TLB Entry—Index, Random, and Wired Registers*

The **Index** register is used to pick a particular TLB entry—they run from zero up to the number of entries less one. You set **Index** when you deliberately want

to read or write a particular entry, and **Index** is also set automatically when you do a software search for a TLB entry using **tlbp**.

Its low bits hold the TLB index.² Not many bits are needed, since no MIPS CPU has yet had a TLB with more than 128 entries. The high bit (bit 31) is magic, being set by **tlbp** when the probe fails to find a matching entry. Bit 31 is a good choice—because it makes the value appear negative, it's easy to test for.

Random holds an index into the TLB that counts (downward, if that's important to you) with each instruction the CPU executes. It acts as an index into the TLB for the write-entry instruction **tlbwr**, facilitating a random replacement strategy when you need to write a new TLB entry.

You never have to read or write the **Random** register in normal use. The hardware sets the **Random** field to its maximum value—the highest-numbered entry in the TLB—on reset, and it decrements every clock period until it reaches a floor value, when it wraps back to its maximum and starts again.

TLB entries from 0 upward whose index is less than the floor value are therefore immune from random replacement, and an OS can use those slots for permanent translation entries—they are referred to as *wired* in MIPS OS documentation. The **Wired** register allows you to specify that floor and thus to determine the number of wired entries. When you write **Wired**, the **Random** register is reset to point to the top of the TLB.

6.2.4 *Page-Table Access Helpers—Context and XContext*

When the CPU takes an exception because a translation isn't in the TLB, the virtual address whose translation wasn't available is already in **BadVAddr**. The page-resolution address is also reflected in **EntryHi (VPN2)**, which is thereby preset to exactly the value needed to create a new entry to translate the missed address.

But to further speed the processing of this exception, the **Context** or **XContext** register repackages the page-resolution address in a format that can act as a ready-made pointer to a memory-based page table.

MIPS32 CPUs have just the **Context** register, which helps out the TLB refill process for 32-bit virtual address spaces; MIPS64 CPUs add the **XContext** register, to be used when using a larger address space (up to 40 bits).

The registers (both MIPS32 and MIPS64 versions) are shown in Figure 6.4.

Note that **XContext** is the only register in which the MIPS64 definition does not exactly define field boundaries: the **XContext (BadVPN2)** field grows on CPUs supporting virtual address regions bigger than 2^{40} bits and pushes the **R** and **PTEBase** fields left (the latter is squashed to fit).

2. Ancient R2000-compatible CPUs had a different register layout, with the index offset, starting from bit 8 upward. You're unlikely to meet one of those.

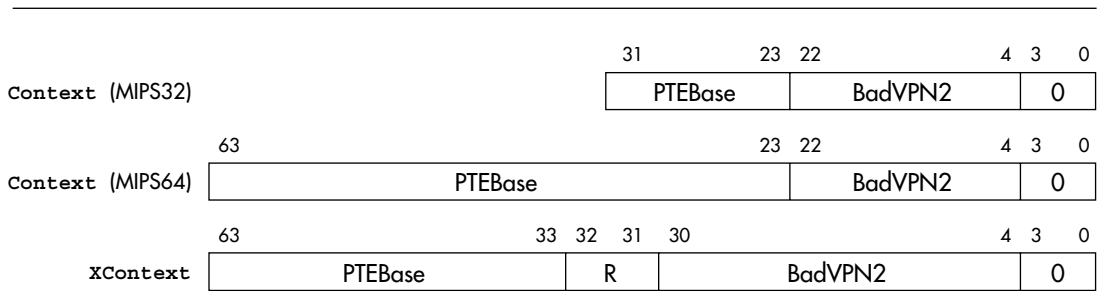


FIGURE 6.4 Fields in the MIPS32 and MIPS64 **Context** registers.

Section 6.2.1, which describes Figure 6.2, explains how to find out how many bits are used in your CPU. The fields are:

- **Context (PTEBase)**: This field just stores what you put in it. To exploit the register as its designers intended, write in the high-order bits of an (appropriately aligned) starting address of a memory-resident page table. The starting address must be picked to have zeros in bits 22 and downward—that’s a 4-MB boundary. While it would be grossly inefficient to provide that alignment in physical or unmapped memory, the intention is that this table should be put in the kseg2 mapped region. See below for how this works.
- **Context (BadVPN2) / XContext (BadVPN2)**: Following a TLB-related exception, this holds the page address, which is just the high-order bits of **BadVAddr**.

Why the “2” in the name? Recall that in the MIPS32/MIPS64 TLB, each entry maps an adjacent pair of virtual-address pages onto two independent physical pages.

The **BadVPN2** value starts at bit 4, so as to precalculate a pointer into a table of 16-byte entries whose base address is in **PTEBase**. If the OS maintains this table so that the entry implicitly accessed by a particular virtual page address contains exactly the right **EntryLo0–1** data to create a TLB entry translating that page, then you minimize the work a TLB miss exception handler has to carry out; you can see that in section 6.5. If you’re only translating 32-bit addresses and don’t need too many bits of software-only state in the page table, you could get by with an eight-byte page-table entry. This turns out to be one of the reasons why Linux doesn’t use the **Context** registers in the prescribed manner.

The **XContext (BadVPN2)** field may be larger than is shown in Figure 6.4 if your CPU can handle more than 2^{40} bits of user virtual address space. When that happens, the **R** and **PTEBase** fields are pushed along to make space.

- **XContext (PTEBase)**: The page table base for 64-bit address regions must be aligned so that all the bits below those specified by **XContext (PTEBase)** are zero: That's 8 GB aligned. That sounds intolerable, but there is a suitable large, mapped, kernel-only-accessible region ("xkseg") in the basic MIPS64 memory map.
- **XContext (R)**: TLB misses can come from any mapped region of the CPU's memory map, not just from user space. All regions lie within one overarching 64-bit space, but are much smaller than is required to pack it full (you might like to refer to Figure 2.2 in section 2.8). Usable 64-bit virtual addresses are divided into four "xk..." segments within which you can use a 62-bit in-segment address.

So as to save space in **XContext**, the miss address as shown here is kept as a separate 40-bit in-region page address (**BadVPN2**) and a 2-bit mapped-region selector **XContext (R)**, defined as follows:

R <i>value</i>	<i>Region name</i>	<i>Brief description</i>
0	xuseg	User-mode accessible space in low virtual memory
1	xsseg	Supervisor-mode accessible space (supervisor mode is optional)
2		Would correspond to unmapped segments, not used
3	xkseg	Kernel-mode mapped space (including old kseg2)

Note that not all operating systems use **Context/XContext** as originally envisaged—notably, Linux doesn't. We'll discuss why later.

6.3 TLB/MMU Control Instructions

The instructions:

```
tlbr    # read TLB entry at index
tlbwi   # write TLB entry at index
```

move MMU data between the TLB entry selected by the **Index** register and the **EntryHi** and **EntryLo0-1** registers.

You won't often read a TLB entry; when you do, remember that you'll have overwritten the **EntryHi (ASID)** field, which the OS should maintain to select the address space of the currently running process. So put it back again.

The instruction:

```
tlbwr    # write TLB entry selected by Random
```

copies the contents of **EntryHi** (including its ASID field, of course), **EntryLo**, and **PageMask** into the TLB entry indexed by the **Random** register—this saves time if you are adopting a random replacement policy. In practice, **tlbwr** will be used to write a new TLB entry in a TLB refill exception handler, but **tlbwi** will be used anywhere else.

The instruction:

```
tlbp     # TLB lookup
```

searches the TLB for an entry whose virtual page number and ASID matches those currently in **EntryHi** and stores the index of that entry in the **Index** register. Bit 31 of **Index** is set if nothing matches—this makes the value look negative, which is easy to test.

If more than one entry matches, anything might happen. This is a horrible error and is never supposed to happen: Software should be very careful never to install a second translation for any address.

Note that **tlbp** does not fetch data from the TLB; you have to run a subsequent **tlbr** instruction to do that.

In most CPUs the TLB is internally pipelined so that translation can proceed efficiently. Management/diagnostic operations like this may not be able to fit in with the standard pipeline flow, so there may be hazards if instructions using translated addresses are run too soon after TLB maintenance instructions. See section 3.4 on CP0 hazards, but to avoid tricky corner cases you normally perform TLB maintenance with software executed in an untranslated region such as **kseg0**.

6.4 Programming the TLB

TLB entries are set up by writing the required fields into **EntryHi** and **EntryLo**, then using a **tlbwr** or **tlbwi** instruction to copy that entry into the TLB proper.

When you are handling a TLB refill exception, you will find that **EntryHi** has been set up for you already.

Be very careful not to create two entries that will match the same program address/ASID pair. If the TLB contains duplicate entries, an attempt to translate such an address, or probe for it, has the potential to damage the CPU chip. Some CPUs protect themselves in these circumstances by a TLB shutdown, which shows up as the **SR(TS)** bit being set. The TLB will now match nothing until a hardware reset.

System software often won't need to read TLB entries at all. But if you need to read them, you can find the TLB entry matching some particular program address using **tlbp** to set up the **Index** register. Don't forget to save **EntryHi** and restore it afterward—the **EntryHi (ASID)** field is likely to be important.

Use a **tlbr** to read the TLB entry into **EntryHi** and **EntryLo0-1**.

You'll see references in the CPU documentation to separate "ITLB" and "DTLB" (or sometimes collectively "uTLB"—the "u" is for "micro") structures. The micro-TLBs perform translation for instruction and data addresses, respectively; these are tiny hardware-managed caches of translations, whose operation is completely transparent to software—they are automatically invalidated whenever you write an entry to the main TLB.

6.4.1 *How Refill Happens*

When a program makes an access in any of the translated address regions (normally kuseg for application programs under a protected OS and kseg2 for kernel-privilege mappings), and no translation record is present, the CPU takes a TLB refill exception.

The TLB can only map a fraction of the physical memory range of a modern server or workstation. Large OSs maintain some kind of memory-held page table that holds a large number of page translations and uses the TLB as a cache of recently used translations. For efficiency, it's common to arrange that the page table will be an array of ready-to-use TLB entries; for even more efficiency, you can set locate and structure the table so that you can use the **Context** or **XContext** register as a pointer into it.

Since MIPS systems usually run OS code in the untranslated kseg0 memory region, the common situation will be a miss by a user-privilege program. Several hardware features are provided, with the aim of speeding up the exception handler in this common case. First, these refill exceptions are vectored through a low-memory address used for no other exception.³ Second, a series of cunning tricks allows the memory-held page table to be located in kernel virtual memory (the kseg2 region or its 64-bit alternative) so that physical memory space is not needed for the parts of the page table that map "holes" in the process's address map.

And to top it off, the **Context** or **XContext** register can be used to give immediate access to the right entry from a memory-held page table.

We'll work through this process in section 6.5. But before we get too far into it, we should note that use of all these features is *not compulsory*. In a smaller system the TLB can be used to produce a fixed or rarely changing translation from program (virtual) to physical addresses; in these cases it won't even need to be a cache.

Even some big virtual memory OSs implemented for MIPS have not used the "standard" page table—notably Linux. It would be quite against the spirit of

3. On the original MIPS architecture, this is the *only* event deemed worthy of its own entry point.

the way the Linux kernel uses memory, because the Linux kernel address map is the same for all processes. See section 14.4.8 for how it's done.

6.4.2 *Using ASIDs*

By setting up TLB entries with a particular ASID setting and with the **EntryLo0-1 (G)** bit set 0, those entries will only ever be used when the CPU's then-current **EntryHi (ASID)** register field matches the TLB entry's value. Since **EntryHi (ASID)** is an 8-bit field, that allows you to map up to 256 different address spaces simultaneously, without requiring that you clear out the TLB on a context change. If you do run out of ASIDs, you will have to pick some process that you can completely throw out of the TLB. Once you've discarded all its mappings, you can revoke its ASID value and reapply to some other process.

6.4.3 *The Random Register and Wired Entries*

The hardware offers you no way of finding out which TLB entries have been used most recently. When you are using the TLB as a cache and you need to install a new mapping, the only practicable strategy is to replace an entry at random. The CPU makes this easy for you by maintaining the **Random** register, which counts (down, actually) with every processor cycle.

Random replacement sounds horribly inefficient; you may end up discarding the translation entry that has been in heaviest use recently and that will almost certainly be needed again very soon. But in fact this doesn't happen so often as to be a real problem when you have a reasonable number of possible victims to choose from.

However, it can be very useful to have some TLB entries that are guaranteed to stay there until you choose to remove them. These may be useful to map pages that you know will be required very often, but they are really important because they allow you to map pages and *guarantee* that no refill exception will be generated on them.

The stable TLB entries are described as *wired*: They're the entries whose index is lower than whatever value you programmed into the **Wired** register. The TLB itself does nothing special about these entries; the magic is in the **Random** register, which never takes values 0 through "wired-1"; it counts down but then steps directly from "wired" to its maximum value. So conventional random replacement leaves TLB entries 0 through "wired-1" unaffected, and entries written there will stay until explicitly removed.

6.5 Hardware-Friendly Page Tables and Refill Mechanism

There's a particular translation mechanism that the MIPS architects undoubtedly had in mind for user addresses in a UNIX-like OS. It relies upon building a page table in memory for each address space. The page table consists of a linear

array of entries, indexed by the VPN, whose format is matched to the bitfields of the **EntryLo** register. The paired TLBs need 2×64 -bit entries, 16 bytes per entry.

That minimizes the load on the critical refill exception handler but opens up other problems. Since each 8 KB of user address space takes 16 bytes of table space, the entire 2 GB of user space needs a 4-MB table, which is an embarrassingly large chunk of data.⁴ Of course, most user address spaces are only filled at the bottom (with code and data) and at the top (with a downward growing stack) with a huge gap in between. The solution MIPS adopted is inspired by DEC's VAX architecture, and is to locate the page table itself in virtual memory in a kernel-mapped (kseg2 or xkseg) region. This neatly solves two problems at once:

- It saves physical memory: Since the unused gap in the middle of the page table will never be referenced, no physical memory need actually be allocated for those entries.
- It provides an easy mechanism for remapping a new user page table when changing context, without having to find enough virtual addresses in the OS to map all the page tables at once. Instead, you have a different kernel memory map for each different address space, and when you change the **ASID** value, the kseg2 pointer to the page table is now automatically remapped onto the correct page table. It's nearly magic.

The MIPS architecture supports this kind of linear page table in the form of the **Context** register (or **XContext** for extended addressing in 64-bit CPUs).

If you make your page table start at a 4-MB boundary (since it is in virtual memory, any gap created won't use up physical memory space) and set up the **Context** PTEBase field with the high-order bits of the page table starting the address, then, following a user refill exception, the **Context** register will contain the address of the entry you need for the refill with no further calculation needed.

So far so good: But this scheme seems to lead to a fatal vicious circle, where a TLB refill exception handler may itself get a TLB refill exception, because the kseg2 mapping for the page table isn't in the TLB. But we can fix that, too.

If a nested TLB refill exception happens, it happens with the CPU already in exception mode. In MIPS CPUs, a TLB refill from exception mode is directed to the general exception entry point, where it will be detected and can be handled specially.

Moreover, an exception from exception mode behaves strangely: It does not change the restart location **EPC**, so when the "inner" exception returns, it returns straight to the nonexception TLB miss point. It's rather as if the hardware

4. And of course this is for a 32-bit virtual address space; 64-bit CPUs using larger address spaces need vast tables.

started processing one exception, then changed its mind and processed another: but the second exception is no longer nested, it has usurped the first one. We'll see how that works when we look at an example.

6.5.1 *TLB Miss Handling*

A TLB miss exception always uses a dedicated entry point unless the CPU is already handling an exception—that is, unless **SR (EXL)** is set.

Here is the code for a TLB miss handler for a MIPS32 CPU (or a MIPS64 CPU handling translations for a 32-bit address space):

```
.set    noreorder
.set    noat
TLBmiss32:
    mfc0    k1, C0_CONTEXT    # (1)
    lw      k0, 0(k1)         # (2)
    lw      k1, 8(k1)         # (3)
    mtc0    k0, C0_ENTRYLO0    # (4)
    mtc0    k1, C0_ENTRYLO1    # (5)
    ehb                      # (6)
    tlbwr                    # (7)
    eret                      # (8)
.set     at
.set     reorder
```

Following is a line-by-line analysis of the code:

- (1) The **k0–1** general-purpose registers are conventionally reserved for the use of low-level exception handlers; we can just go ahead and use them.
- (2–5) There are a pair of physical-side (**EntryLo**) descriptions in each TLB entry (you might like to glance back at the TLB entry diagram, Figure 6.1). The layout of the MIPS32/64 **Context** register shown in Figure 6.4 reserves 16 bytes for each paired entry (eight bytes of space for each physical page), even though MIPS32's **EntryLo0–1** are 32-bit registers. This is for compatibility with the 64-bit page table and to provide some spare fields in the page table to keep software-only information.

Interleaving the **lw/mtc0** sequences here will save time: Few MIPS CPUs can keep going without pause if you use loaded data in the very next instruction.

These loads are vulnerable to a nested TLB miss if the **kseg2** address's translation is not in the page table. We'll talk about that later.

- (6) It's no good writing the entry with **tlbwr** until it will get the right data from **EntryLo1**. The MIPS32 architecture does not guarantee this will be ready for the immediately following instruction, but it does guarantee that the sequence will be safe if the instructions are separated by an **ehb** (execution hazard barrier) instruction—see section 3.4 for more information about hazard barriers.
- (7) This is random replacement of a translation pair as discussed.
- (8) MIPS32 (and all MIPS CPUs later than MIPS I) have the **eret** instruction, which returns from the exception to the address in **EPC** and unsets **SR(EXL)**.

So what happens when you get another TLB miss? The miss from exception level invokes not the special high-speed handler but the general-purpose exception entry point. We're already in exception mode, so we don't alter the exception return register **EPC**.

The **Cause** register and the address-exception registers (**BadVAddr**, **EntryHi**, and even **Context** and **XContext**) will relate to the TLB miss on the page table address in **kseg2**. But **EPC** still points back at the instruction that caused the original TLB miss.

The exception handler will fix up the **kseg2** page table miss (so long as this was a legal address) and the general exception handler will return into the user program. Of course, we haven't done anything about the translation for the user address that originally caused the user-space TLB miss, so it will immediately miss again. But the second time around, the page table translation will be available and the user miss handler will complete successfully. Neat.

6.5.2 *XTLB Miss Handler*

MIPS64 CPUs have two special entry points. One—shared with MIPS32 CPUs—is used to handle translations for processes using only 32 bits of address space; an additional entry point is provided for programs marked as using the bigger address spaces available with 64-bit pointers.

The status register has three fields, **SR(UX)**, **SR(SX)**, and **SR(KX)**, that select which exception handler to use based on the CPU privilege level at the time of the failed translation.⁵

With the appropriate status bit set (**SR(UX)** for user mode), a TLB miss exception uses a different vector, where we should have a routine that will reload translations for a huge address space. The handler code (of an XTLB miss handler for a CPU with 64-bit address space) looks much like the 32-bit version,

5. **SR(UX)** doubles up as something of a “64-bit mode” bit for user programs; when it's zero, 64-bit integer instructions are not available to a user program. But **SR(SX, KX)** are used only to select the TLB refill routine.

except for the 64-bit-wide registers and the use of the **XContext** register in place of **Context**:

```

        .set      noreorder
        .set      noat
TLBmissR4K:
        dmfc0     k1, C0_XCONTEXT
        ld        k0, 0(k1)
        ld        k1, 8(k1)
        dmtc0     k0, C0_ENTRYLO0
        dmtc0     k1, C0_ENTRYLO1
        ehb
        tlbwr
        eret
        .set      at
        .set      reorder

```

Note, though, that the resulting page table structure in kernel virtual memory is far bigger and will undoubtedly be in the giant xkseg region.

I should remind you again that this system is not compulsory, and in fact is not used by the MIPS version of Linux (which is overwhelmingly the most popular translated-address OS for MIPS applications). It's a rather deeply ingrained design choice in the Linux kernel that the kernel's own code and data are not remapped by a context switch, but exactly that is required for the kseg2/xkseg page table trick described here. See section 14.4.8 for how it's done.

6.6 Everyday Use of the MIPS TLB

If you're using a full-scale OS like Linux, then it will use the TLB behind your back, and you'll rarely notice. With a less ambitious OS or runtime system, you may wonder whether it's useful. But, because the MIPS TLB provides a general-purpose address translation service, there are a number of ways you might take advantage of it.

The TLB mechanism permits you to translate addresses (at page granularity) from any mapped address to any physical address and therefore to relocate regions of program space to any location in your machine's address map. There's no need to support a TLB refill exception or a separate memory-held page table if your mapping requirements are modest enough that you can accommodate all the translations you need in the TLB.

The TLB also allows you to define some address as temporarily or permanently unavailable, so that accesses to those locations will cause an exception that can be used to run some operating system service routine. By using

user-privilege programs you can give some software access only to those addresses you want it to have, and by using address space IDs in the translation entries, you can efficiently manage multiple mutually inaccessible user programs. You can write-protect some parts of memory.

The applications for this are endless, but here's a list to indicate the range:

- *Accessing inconvenient physical address ranges:* Hardware registers for a MIPS system are most conveniently located in the physical address range 0–512 MB, where you can access them with a corresponding pointer from the `kseg1` region. But where the hardware can't stay within this desirable area, you can map an arbitrary page of higher physical memory into a convenient mapped area, such as `kseg2`. The TLB flags for this translation should be set to ensure uncached access, but then the program can be written exactly as though the address was in the convenient place.
- *Memory resources for an exception routine:* Suppose you'd like to run an exception handler without using the reserved `k0/k1` registers to save context. If so, you'd have trouble, because a MIPS CPU normally has nowhere to save any registers without overwriting at least one of these. You can do loads or stores using the **zero** register as a base address, but with a positive offset these addresses are located in the first 32 KB of `kuseg`, and with a negative offset they are located in the last 32 KB of `kseg2`. Without the TLB, these go nowhere. With the TLB, you could map one or more pages in this region into read/write memory and then use zero-based stores to save context and rescue your exception handler.
- *Extendable stacks and heaps in a non-VM system:* Even when you don't have a disk and have no intention of supporting full demand paging, it can still be useful to grow an application's stack and heap on demand while monitoring its growth. In this case you'll need the TLB to map the stack/heap addresses, and you'll use TLB miss events to decide whether to allocate more memory or whether the application is out of control.
- *Emulating hardware:* If you have hardware that is sometimes present and sometimes not, then accessing registers through a mapped region can connect directly to the hardware in properly equipped systems and invoke a software handler on others.

The main idea is that the TLB, with all the ingenuity of a specification that fits so well into a big OS, is a useful, straightforward general resource for programmers.

6.7 Memory Management in a Simpler OS

An OS designed for use off the desktop is generally called a real-time OS (RTOS), hijacking a term that once meant something about real time.⁶ The UNIX-like system outlined in the first part of this chapter has all the elements you're likely to find in a smaller OS, but many RTOSs are much simpler.

Some OS products you might meet up with are VxWorks from Wind River Systems, Thread/X from Express Logic, and Nucleus from Mentor (following their acquisition of Accelerated Technology). All provide multiple threads running in a single address space. There is no task-to-task protection—software running on these is assumed to be a single tightly integrated application. In many cases the OS run time is really quite small, and much of the supplier's effort is devoted to providing developers with build, debug, and profiling tools.

The jury is still out on whether it's worth using a more sophisticated OS such as Linux for many different kinds of embedded systems. You get a richer programming environment, task-to-task protection that can be very valuable when integrating a system, and probably cleaner interfaces. Is that worth devoting extra memory and CPU power to, and losing a degree of control over timing, for the benefits of the cleverer system? Builders of TV set-top boxes, DVD players, and domestic network routers have found Linux worthwhile: Other systems (not necessarily of very different complexity) are still habitually using simpler systems.

And of course Linux is open source. Sometimes it's just good that there are no license fees; perhaps, more importantly, if your system doesn't work because of an OS bug, open source means you can fix it yourself or commission any of a number of experts to fix it for you—right away. It's paradoxical, but the more successful a commercial OS becomes, the harder it is to find someone to fix it on a reasonable schedule.

But for now, as a developer, you may be faced with almost anything. When you're trying to understand a new memory management system, the first thing is to figure out the memory maps, both the virtual map presented to application software and the physical map of the system. It's the simple-minded virtual address map that makes UNIX memory management relatively straightforward to describe. But operating systems targeted at embedded applications do not usually have their roots in hardware with memory management, and the process memory map often has the fossils of unmapped memory maps hidden inside it. The use of a pencil, paper, and patience will sort it out.

6. Real “real-time” involves programming to deadlines and is hard and interesting but very much a minority pursuit.

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Floating-Point Support

In 1987, the MIPS FPU set a new benchmark for microprocessor math performance in affordable workstations. Unlike the CPU, which was mostly a rather straightforward implementation relying on its basic architecture for its performance, the FPU was a heroic silicon design bristling with innovation and ingenuity.

Later on, the MIPS FPU was pulled onward by Silicon Graphics's need for math performance that would once have been the preserve of supercomputers. The use of floating-point computations in embedded systems has grown fairly slowly; but the trend toward FP is a classic trade-off of clever hardware that makes software simpler and more maintainable, and that trade-off (over time) only goes one way. Some day all "main application" processors will require floating point.

7.1 A Basic Description of Floating Point

Floating-point math retains a great deal of mystery. You probably have a very clear idea of what it is for, but you may be hazy about the details. This section describes the various components of the data and what they mean. In so doing we are bound to tell most of you things you already know; please skip ahead but keep an eye on the text!

People who deal with numbers that may be very large or very small are used to using exponential (scientific) notation; for example, the distance from the earth to the sun is:

$$93 \times 10^6 \text{ miles}$$

The number is defined by 93, the *mantissa*, and 6, the *exponent*. The same distance can be written:

$$9.3 \times 10^7 \text{ miles}$$

Numerical analysts like to use the second form; a decimal exponent with a mantissa between 1.0 and 9.999... is called *normalized*.¹ The normalized form is useful for computer representation, since we don't have to keep separate information about the position of the decimal point.

Computer-held floating-point numbers are an exponential form, but in base 2 instead of base 10. Both mantissa and exponent are held as binary fields. Just changing the exponent into a power of 2, the distance quoted above is:

$$1.38580799102783203125 \times 2^{26} \text{ miles}$$

The mantissa can be expressed as a binary “decimal,” which is just like a real decimal; for example:

$$1.38580799102783203125 = 1 + 3 \times \frac{1}{10} + 8 \times \frac{1}{100} + 5 \times \frac{1}{1000} + \dots$$

is the same value as binary:

$$1.01100010110001000101 = 1 + 0 \times \frac{1}{2} + 1 \times \frac{1}{4} + 1 \times \frac{1}{8} + \dots$$

However, neither the mantissa nor the exponent are stored just like this in standard formats—and to understand why, we need to review a little history.

7.2 The IEEE 754 Standard and Its Background

Because floating point deals with the approximate representations of numbers (in the same way as decimals do), computer implementations used to differ in the details of their behavior with regard to very small or large numbers. This meant that numerical routines, identically coded, might behave differently. In some sense these differences shouldn't have mattered: You only got different answers in circumstances where no implementation could really produce a “correct” answer.

The use of calculators shows the irritating consequences of this: If you take the square root of a whole number and square it, you will rarely get back the whole number you put in, but rather something with lots of nines.

Numerical routines are intrinsically hard to write and hard to prove correct. Many heavily used functions (common trigonometric operations, for example) are calculated by repeated approximation. Such a routine might reliably converge to the correct result on one CPU and loop forever on another when fed a difficult value.

The ANSI/IEEE 754 Standard—1985 IEEE Standard for Binary Floating Point Arithmetic (usually referred to simply as IEEE 754)—was introduced to bring order to this situation. The standard defines exactly which result will be

1. In this form the mantissa may also be called “the fractional part” or “fraction”—it's certainly easier to remember.

produced by a small class of basic operations, even under extreme situations, ensuring that programmers can obtain identical results from identical inputs regardless of which machine they are using. Its approach is to require as much precision as is possible within each supported data format.

Perhaps IEEE 754 has too many options, but it is a huge improvement on the chaos that motivated it; since it became a real international standard in 1985, it has become the basis for all new implementations.

The operations regulated by IEEE 754 include every operation that MIPS FPU's can do in hardware, plus some that must be emulated by software. IEEE 754 legislates the following:

- *Rounding and precision of results:* Even results of the simplest operations may not be representable as finite fractions; for example, in decimals:

$$\frac{1}{3} = 0.33333 \dots$$

is infinitely recurring and can't be written precisely. IEEE 754 allows the user to choose between four options: round up, round down, round toward zero, or round to nearest. The rounded result is what would have been achieved by computing with infinite precision and then rounding. This would leave an ambiguity in round to nearest when the infinite-precision result is exactly halfway between two representable forms; the rules provide that in this case you should pick the value whose least significant bit is zero.

- *When is a result exceptional?* IEEE 754 has its own meaning for the word *exception*. A computation can produce a result that is:
 - Nonsense, such as the square root of -1 (“invalid”)
 - “Infinite,” resulting from an explicit or implicit division by zero
 - Too big to represent (“overflow”)
 - So small that its representation becomes problematic and precision is lost (“underflow”)
 - Not perfectly represented, like $\frac{1}{3}$ (“inexact”)—needless to say, for most purposes the nearest approximation is acceptable

All of these are bundled together and described as exceptional results.

- *Action taken when an operation produces an exception result:* For each class of exceptional results listed above, the user can choose between the following:
 - The user can have the computation interrupted and the user program signaled in some OS- and language-dependent manner. Partly because the standard doesn't actually define a language binding for user exceptions, they're pretty much never used. Some FORTRAN compiler systems are wired to cause a fatal error invalid or infinite results.

- Most often, the user program doesn't want to know about the IEEE exception. In this case, the standard specifies which value should then be produced. Overflows and division by zero generate infinity (with a positive and negative type); invalid operations generate NaN (not a number) in two flavors called “quiet” and “signaling.” Very small numbers get a “denormalized” representation that loses precision and fades gradually into zero.

The standard also defines the result when operations are carried out on exceptional values. Infinities and NaNs necessarily produce further NaNs and infinities, but while a quiet NaN as operand will not trigger the exception-reporting mechanism, a signaling NaN causes a new exception whenever it is used.

Most programs leave the IEEE exception reporting off but do rely on the system producing the correct exceptional values.

7.3 How IEEE Floating-Point Numbers Are Stored

IEEE recommends a number of different binary formats for encoding floating-point numbers, at several different sizes. But all of them have some common ingenious features, which are built on the experience of implementers in the early chaotic years.²

The first thing is that the exponent is not stored as a signed binary number, but *biased* so that the exponent field is always positive: The exponent value 1 represents the tiniest (most negative) legitimate exponent value; for the 64-bit IEEE format the exponent field is 11 bits long and can hold numbers from 0 to 2,047. The exponent values 0 and 2,047 (all ones, in binary) are kept back for special purposes we'll come to in a moment, so we can represent a range of exponents from $-1,022$ to $+1,023$.

For a number:

$$\text{mantissa} \times 2^{\text{exponent}}$$

we actually store the binary representation of:

$$\text{exponent} + 1,023$$

in the exponent field.

The biased exponent (together with careful ordering of the fields) has the useful effect of ensuring that FP comparisons (equality, greater than, less than, etc.) have the same result as is obtained from comparing two signed integers

2. IEEE 754 is a model of how good standardization should be done; a fair period of chaotic experimentation allowed identifiably good practice to evolve, and it was then standardized by a small committee of strong-minded users (numerical programmers in this case), who well understood the technology. However, the ecology of standards committees, while a fascinating study, is a bit off the point.

composed of the same bits. FP compare operations can therefore be provided by cheap, fast, and familiar logic.

7.3.1 *IEEE Mantissa and Normalization*

The IEEE format uses a single sign bit separate from the mantissa (0 for positive, 1 for negative). So the stored mantissa only has to represent positive numbers. All properly represented numbers in IEEE format are normalized, so:

$$1 \leq \text{mantissa} < 2$$

This means that the most significant bit of the mantissa (the single binary digit before the point) is always a 1, so we don't actually need to store it. The IEEE standard calls this the *hidden bit*.

So now the number 93,000,000, whose normalized representation has a binary mantissa of 1.01100010110001000101 and a binary exponent of 26, is represented in IEEE 64-bit format by setting the fields:

$$\begin{aligned}\text{mantissafield} &= 01100010110001000101000\dots \\ \text{exponentfield} &= 1049 = 10000011001\end{aligned}$$

Looking at it the other way, a 64-bit IEEE number with an exponent field of E and a mantissa field of m represents the number f , where:

$$f = 1.m \times 2^{E-1023}$$

(provided that you accept that $1.m$ represents the binary fraction with 1 before the point and the mantissa field contents after it).

7.3.2 *Reserved Exponent Values for Use with Strange Values*

The smallest and biggest exponent field values are used to represent otherwise-illegal quantities.

$E == 0$ is used to represent zero (with a zero mantissa) and denormalized forms, for numbers too small to represent in the standard form. The denormalized number with E zero and mantissa m represents f , where:

$$f = 0.m \times 2^{-1022}$$

As denormalized numbers get smaller, precision is progressively lost. No MIPS FPU built to date is able to cope with either generating denormalized results or computing with them, and operations creating or involving them will be punted to the software exception handler. Modern MIPS FPUs can be configured to replace a denormalized result with a zero and keep going.

$E == 111 \dots 1$ (i.e., the binary representation of 2,047 in the 11-bit field used for an IEEE double) is used to represent the following:

- With the mantissa zero, it is the illegal values `+inf`, `-inf` (distinguished by the usual sign bit).
- With the mantissa nonzero, it is a NaN. For MIPS, the most significant bit of the mantissa determines whether the NaN is quiet (MS bit 0) or signaling (MS bit 1). This choice is not part of the IEEE standard and is opposite to the convention used by most other IEEE-compatible architectures.³

7.3.3 MIPS FP Data Formats

The MIPS architecture uses two FP formats recommended by IEEE 754:

- *Single precision*: These are fitted into 32 bits of storage. Compilers for MIPS use single precision for `float` variables.
- *Double precision*: These use 64 bits of storage. C compilers use double precision for `C double` types.

The memory and register layout is shown in Figure 7.1, with some examples of how the data works out. Note that the `float` representation can't hold a number as big as 93,000,000 exactly.

The way that the two words making up a double are ordered in memory (most significant bits first, or least significant bits first) is dependent on the CPU's endianness—discussed to the point of exhaustion in section 10.2. It is always consistent with the endianness of the integer unit.

The C structure definition below defines the fields of the two FP types for a MIPS CPU (this works on most MIPS toolchains, but note that, in general, C structure layout is dependent on a particular compiler and not just on the target CPU):

```
#if BYTE_ORDER == BIG_ENDIAN

struct ieee754dp_konst {
    unsigned sign:1;
    unsigned bexp:11;
    unsigned manthi:20; /* cannot get 52 bits into ... */
    unsigned mantlo:32; /* ... a regular C bitfield */
};
```

3. I believe only HP Precision makes the same choice as MIPS, suggesting some intellectual inheritance. Arbitrary 1/0 choices are the mitochondrial DNA of computer technology.

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FIGURE 7.1 Floating-point data formats.

```

struct ieee754sp_konst {
    unsigned    sign:1;
    unsigned    bexp:8;
    unsigned    mant:23;
};

#else /* little-endian */

struct ieee754dp_konst {
    unsigned    mantlo:32;
    unsigned    manthi:20;
    unsigned    bexp:11;
    unsigned    sign:1;
};

struct ieee754sp_konst {
    unsigned    mant:23;
    unsigned    bexp:8;
    unsigned    sign:1;
};

#endif

```

7.4 MIPS Implementation of IEEE 754

IEEE 754 is quite demanding and sets two major problems. First, building in the ability to detect exceptional results makes pipelining harder. You might want to do this to implement the IEEE exception signaling mechanism, but the deeper reason is to be able to detect certain cases where the hardware cannot produce the correct result and needs help.

If the user opts to be told when an IEEE exceptional result is produced, then to be useful this should happen synchronously;⁴ after the trap, the user will want to see all previous instructions complete and all FP registers still in the preinstruction state and will want to be sure that no subsequent instruction has had any effect.

In the MIPS architecture, hardware traps (as noted in section 5.1) were traditionally like this. This does limit the opportunities for pipelining FP operations, because you cannot commit the following instruction until the hardware can be sure that the FP operation will not produce a trap. To avoid adding to the execution time, an FP operation must decide to trap or not in the first clock phase after the operands are fetched. For most kinds of exceptional results, the FPU can guess reliably and stop the pipeline for any calculation that might trap;⁵ however, if you configure the FPU to signal IEEE inexact exceptional results, all FP pipelining is inhibited and everything slows down. You probably won't do that.

The second big problem regarding IEEE 754 is the use of exceptional results, particularly with denormalized numbers—which are legitimate operands. Chip designs like the MIPS FPU are highly structured pieces of logic, and the exceptional results don't fit in well. Where correct operation is beyond the hardware, it traps with an unimplemented operation code in the **Cause (ExcCode)** field. This immediately makes an exception handler compulsory for FP applications. Modern MIPS CPUs often include one or more implementation-dependent option bits in the **FCSR** control/status register, which you can set when you're prepared to trade strict IEEE 754 compliance either for performance or—more likely—to avoid “unimplemented operation” traps in corner cases.

4. Elsewhere in this book and in the MIPS documentation you will see exactly this condition referred to as a “precise exception.” But since both “precise” and “exception” are used to mean different things by the IEEE standard, we will instead talk about a “synchronous trap.” Sorry for any confusion.

5. Some CPUs may use heuristics for this that sometimes stop the pipeline for an operation that in the end does not trap; that's only a performance issue and is not important if they don't do it often.

7.4.1 *Need for FP Trap Handler and Emulator in All MIPS CPUs*

The MIPS architecture does not prescribe exactly which calculations will be performed without software intervention. A complete software floating-point emulator is mandatory for serious FP code.

In practice, the FPU traps only on a very small proportion of the calculations that your program is likely to produce. Simple uses of floating point are quite likely never to produce anything that the hardware can't handle.

A good rule of thumb, which seems to cover the right cases, follows:

- MIPS FPUs take the unimplemented trap whenever an operation should produce *any* IEEE exception or exceptional result other than inexact and overflow. For overflow, the hardware will generate an infinity or a largest-possible value (depending on the current rounding mode). The FPU hardware will not accept or produce denormalized numbers or NaNs.
- MIPS FPUs (other than the very earliest ones) offer you a non-IEEE optional mode for underflow, where a denormalized (tiny) result can be automatically written as zero.

The unimplemented trap is a MIPS architecture implementation trick and is quite different from the IEEE exceptions, which are standardized conditions. You can run a program and ignore IEEE exceptions, and offending instructions will produce well-defined exceptional values; but you can't ignore the unimplemented trap without producing results that are nonsense.

7.5 Floating-Point Registers

MIPS CPUs have 32 floating-point registers, usually referred to as **\$f0–\$f31**. With the exception of some really old (MIPS I) CPUs, each is a 64-bit register capable of holding a double-precision value.

The very first MIPS CPUs had only 16 registers. Well, in a sense they had 32 32-bit registers, but each even/odd-numbered pair made up a unit for math (including double-precision math, of course). The odd-numbered registers are only referenced when doing loads, stores, and moves between floating-point and integer registers.⁶ If you tell the assembler you're building code for an old CPU, it synthesizes double-width move and load/store operations from a pair of machine instructions; you need never see the odd-numbered registers when writing MIPS I-compatible code.

6. It may be worth stressing that the role of the odd-numbered registers is not affected by the CPU's endianness.

TABLE 7.1 FP Register Usage Conventions

	ABI		
	o32	n32	n64
	\$f0, \$f2		
	\$f12, \$f14	\$f12–\$f19	
	evens \$f20–\$f30		\$f24–\$f31
function return values			
argument registers			
saved over function call (suitable for register variables)			
temporaries (not saved over function call, or “caller-saved”)	evens \$f4–\$f10, \$f16, \$f18	evens \$f4–\$f10, \$f16, \$f18, all odds \$f1–\$f31	\$f1, \$f3–\$f11, \$f20–\$f23

MIPS I has been gone a long time, but later CPUs are fitted with a “compatibility bit” in **SR(FR)**—leave it zero and you get MIPS I operation. There is still quite a lot of software out there that works that way. It seems like a no-brainer to prefer more FP registers—but this is not a personal choice; you need to check what your compiler will support, and the entire system (including all libraries and other imported code) needs to be consistent in its register usage.

It’s also worth pointing out that MIPS FP registers sometimes get used for storing and manipulating signed integer data (32 or 64 bits). In particular, when a program does conversion between integer and FP data, those conversion operations operate entirely within the FPU—integer data in FP registers is converted to floating-point data in FP registers.

7.5.1 Conventional Names and Uses of Floating-Point Registers

Like the general-purpose registers, the MIPS calling conventions add a whole bunch of rules about register use that have nothing to do with the hardware; they tell you which FP registers are used for passing arguments, which register values are expected to be preserved over function calls, and so on. Table 7.1 shows these for the three most common ABIs. An ABI (Application Binary Interface) is a comprehensive statement of conventions that allow modules—perhaps compiled with different toolchains—to be successfully glued together into a program and run under a conforming operating system. We’ll talk more about ABIs in section 11.2; for now, it’s enough to say that the o32 convention is stuck with the 16-register arrangement of old MIPS I CPUs, and the n32 and n64 conventions (despite the former name) are usable only on 64-bit CPUs.

The division of functions is much the same as for the integer registers, without the special cases. But it’s much messier, because of the history of non-existent odd-numbered registers.

7.6 Floating-Point Exceptions/Interrupts

When a floating-point operation can't produce the right result, or has been asked to trap on some or all of the IEEE exceptional results, it takes a MIPS exception, as described in Chapter 5, and the **Cause** register descriptions discussed in section 3.3.2.

The floating-point exceptions are always precise: At the point you arrive at the exception handler, the exception-causing instruction and any instruction later in sequence will appear never to have happened. **EPC** will point to the correct place to restart the instruction. As described in Chapter 5, **EPC** will either point to the offending instruction or to a branch instruction immediately preceding it. If it is the branch instruction, the status register **SR (BD)** will be set.

7.7 Floating-Point Control: The Control/Status Register

Information about the FPU hardware and the controls to change optional behavior are provided as coprocessor control registers, accessed with **ctc1** and **cfcl** instructions, which move data to and from the control registers, respectively (into and out of GP registers, of course).

TABLE 7.2 FP Control Register Summary

<i>Conventional name</i>	<i>CP1 ctrl reg No.</i>	<i>Description</i>
FCSR	31	Extensive control register—the only FPU control register on historical MIPS CPUs. Contains <i>all</i> the control bits. But, in practice, some of them are more conveniently accessed through FCCR , FEXR , and FENR ; see below.
FIR	0	FP implementation register: read-only information about the capability of this FPU, described in the next section.
FCCR	25	These partial views of FCSR are better structured and allow you to update fields without interfering with the operation of independent bits. However, they're unlikely to be available on any CPU not claiming full compliance with MIPS32/64. FCCR has FP condition codes, FEXR contains the IEEE exceptional-condition information (cause and flag bits) you read, and FENR has the (writable) IEEE exceptional-condition enables.
FEXR	26	
FENR	28	

The most commonly seen of them, the floating-point control/status register **FCSR**,⁷ brings together informational and control fields relating to user options about floating-point operations. Information about the capabilities of the hardware and who built it are found in the implementation/revision register **FCR0**, described in the next section.

Using **FCSR** had become a bit of a nightmare because of the mix of readable and writable fields: So in MIPS32/64 CPUs there are three auxiliary registers providing more digestible views: **FCCR**, **FEXR**, and **FENR**.

That makes a fair number of registers, so refer to Table 7.2.

The following are notes regarding Figure 7.2. The field marked 0 will read, and should be written, as zero.

- **FCC7-1, FCC0**: These are condition bits, set by FP compare instructions and tested by conditional branches. **FCC0** was there first and used to be called **FCSR(C)**—but all modern MIPS FPUs provide all 8 bits. The **FCCR** register—if your CPU provides it—does you the service of bringing all the condition bits together.

Note that here, as elsewhere, the floating-point implementation cuts across the RISC principles we talked about in Chapter 1. There are a number of reasons for this:

- The original FPU was a separate chip. The conditional branches that tested FP conditions had to execute inside the integer unit (it was responsible for finding the address of the branch target), so they were remote from the FP registers. A single condition bit corresponds to a single hardware signal.

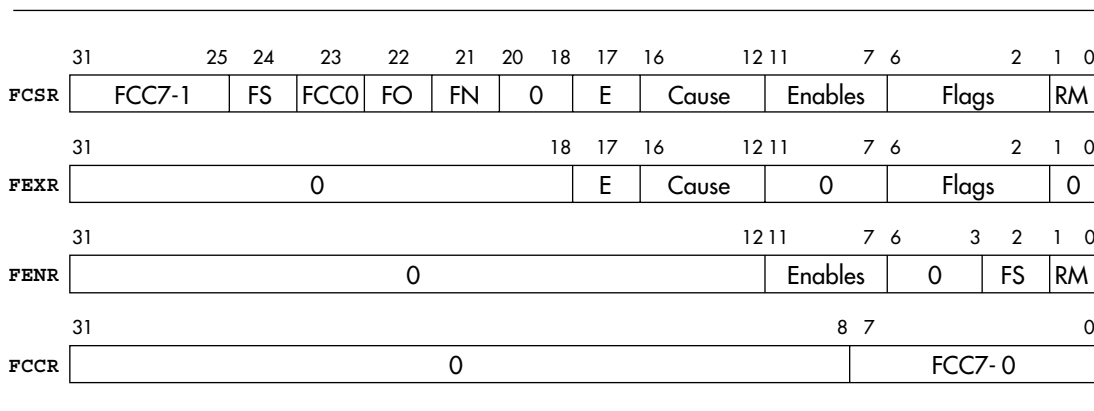


FIGURE 7.2 FPU control/status register fields.

7. In times past **FCSR** was referred to as **FCR31**, but I prefer the mnemonic acronym.

- FP operations are just too computationally demanding to be carried out in one clock cycle, so the kind of simple pipeline that works so well on the integer side won't deliver the best performance.

Both branch and test instructions have an additional 3-bit field that specifies which of 8 possible condition bits they will set or test. On a MIPS III or earlier CPU, there was only one condition bit, and all such fields will be zero. The extra condition bits are valuable for software-pipelining conditional FP code; see section 8.5.7.

- **FS/FO/FN:** These settings affect how the CPU behaves in the face of a number too small for the standard floating-point representation (a denormalized number). If you set any of them, your system is no longer IEEE754-compliant, but you will avoid some “unimplemented” traps.

If your CPU is MIPS32/64-compliant, it will have all of these. If not, it will implement **FS**, but you should check your CPU manual for the others.

FS (flush to zero) causes a result that would be too small for the standard representation (a denormalized result) to be quietly replaced with zero. Since no known MIPS FPU hardware can generate the right answer for you, this avoids a (slow) trap to emulation software.

FO (flush override) in conjunction with **FS** detects denormalized operands and internally replaces them with zero. Setting **FS** alone will not avoid a trap if the inputs are denormalized.

FN (flush to nearest) in conjunction with **FS**, improves accuracy somewhat by forcing denormalized numbers to the closest normalized value (which will sometimes be the smallest representable number, instead of zero).

- **E:** Following an FPU trap, this bit will be set to mark an unimplemented instruction exception.⁸

This bit will be set and an interrupt raised whenever there really is no instruction like this that the FPU will perform (but the instruction is a coprocessor 1 encoding) *or* whenever the FPU is not confident that it can produce an IEEE 754 correct result and/or exception signaling on this operation, using these operands.

For whatever reason, when **E** is set, you should arrange for the offending instruction to be re-executed by a software emulator.

Any FPU operation that takes an “unimplemented” exception will leave the destination register unaffected and the FP **Cause** bits undefined.

- **Cause/Enables/Flags:** Each of these is a 5-bit field, one bit for each IEEE exception type:

- V** bit 4 Invalid operation (e.g., square root of -1)
- Z** bit 3 Divide by zero

8. The MIPS documentation looks slightly different because it treats this as part of the **Cause** field.

- bit 2 Overflow, result is too large to represent
- U bit 1 Underflow, result is too small to represent
- I bit 0 Inexact (rarely used—even 1/div 3 is inexact in binary)

The three different fields work as follows:

- **Cause:** Bits are set (by hardware or emulation software) according to the result of the last completed FP instruction. It's easiest to read them in **FEXR**.
 - **Enables:** If one of these bits is set when an operation produces an exceptional result that would have set the corresponding **Cause** bit, then the CPU will trap so that software can do whatever is necessary to report the exceptional result. You can set up these bits in either **FCSR** or **FENR**.
 - **Flags:** Bits are “sticky” versions of the **Cause** bits and are the logical “or” of the exceptional results that have occurred since the register was last cleared. The **Flags** bits can only be zeroed again by writing **FCSR** or **FEXR**.
- **RM (rounding mode):** This is required by IEEE 754. The values are as shown in Table 7.3.
- Most systems define RN as the default behavior. You'll probably never use anything else.

The architecture promises you that if an operation doesn't set the **FCSR (E)** bit but does set one of the **Cause** bits, then both the **Cause** bit setting and the result produced (if the corresponding **Enable** bit is off) are in accordance with the IEEE 754 Standard.

TABLE 7.3 Rounding Modes Encoded in FP Control Registers

<i>RM value</i>	<i>Description</i>
0	RN (round to nearest): Round a result to the nearest representable value; if the result is exactly halfway between two representable values, round to zero.
1	RZ (round toward zero): Round a result to the closest representable value whose absolute value is less than or equal to the infinitely accurate result.
2	RP (round up, or toward +infinity): Round a result to the next representable value up.
3	RN (round down, or toward –infinity): Round a result to the next representable value down.

MIPS FPUs rely on software emulation (i.e., use the unimplemented trap) for several purposes:

- Any operation that is given a denormalized operand or underflows (produces a denormalized result) will trap to the emulator. The emulator itself must test whether the enable underflow bit is set and either cause an IEEE-compliant exception or produce the correct result.
- Operations that should produce the invalid trap are correctly identified, so if the IEEE exception is enabled, the emulator need do nothing. But if the IEEE invalid exception is disabled, the software emulator is invoked, because the hardware is unable to generate the appropriate result (usually a quiet NaN).
Exactly the same is done with a signaling NaN operand.
- Most FP hardware can handle overflow on regular arithmetic (producing either the extreme finite value or a signed infinity, depending on the rounding mode). But the software emulator is needed to implement a convert-to-integer operation that overflows.

The **Cause** bits are undefined after an unimplemented operation traps to the emulator.

It is normal practice to provide a full emulator (capable of delivering IEEE-compatible arithmetic on a CPU with no FPU fitted) to back up the FPU hardware. If your system provides less than this, it is hard to figure out where it's safe to leave functions out.

7.8 Floating-Point Implementation Register

Consult the read-only **FIR** register after you check **Config1 (FP)** to find out whether you really have a floating-point unit: **FIR**'s fields show you what it can do, and (occasionally) what revision it is—see Figure 7.3.

Then the fields are:

- **FC** (full convert range): The hardware will complete *any* conversion operation (between floating-point types or between floating-point and

	31		25	24	23	22	21	20	19	18	17	16	15		8	7		0
FIR	0				FC	0	F64	L	W	3D	PS	D	S	ProcessorID		Revision		

FIGURE 7.3 FPU implementation/revision register.

integer types) without running out of bits and causing an “unimplemented” exception.

- **F64:** This indicates that there are 32 full double-precision FP registers—that is, this is not an old MIPS I style floating-point unit.
- **L/W/D/S:** Individual bit flags show whether this CPU implements 64-bit integer (**L**), 32-bit integer (**W**), 64-bit FP double (**D**), and 32-bit FP single (**S**) operations. If it doesn’t run all of these, you’re going to need some unusual software support.
- **3D/PS:** Indicate the availability of two optional instruction set extensions. **PS** means you have paired-single SIMD versions of all arithmetic operations, marked with a **.ps** suffix: Each does two operations simultaneously on a pair of single-precision values packed into a register. See section 7.10.

3D indicates that you have the MIPS-3D instruction set extension available to you. This is a small extension and only useful if you already have paired-single, so it’s also described in section 7.10.

- **ProcessorID:** Typically this returns the same value as the integer-side processor ID in the coprocessor zero **PRID** register. Perhaps more usefully, it should only ever read nonzero if there really is FPU hardware available to your program. Before you read it or succeed in running any floating-point operation, you or your OS needs to switch on the “coprocessor 1” enable **SR(CU1)** in the CP0 “status” register.
- **Revision:** This field is at the whim of implementers; it is probably useful to make this field visible to commissioning or test engineers, and you might test it when implementing some bug workaround. Otherwise, software should studiously ignore it.

7.9 Guide to FP Instructions

This section mentions all standard FP instructions, categorized by their function. Optional instructions in the “paired-single” and MIPS-3D extensions are in the next section.

FP instructions are listed in mnemonic order in Table 8.4, and their binary encodings are listed with all other MIPS instructions in Table 8.6.

We’ve divided the instructions up into the following categories:

- *Load/store:* Moving data directly between FP registers and memory.
- *Move between registers:* Data movement between FP and general-purpose registers.
- *Three-operand arithmetic operations:* The regular add, multiply, and so on.

- *Multiply-add operations*: Fancy (and distinctly non-RISC) high-performance instructions, not available in MIPS III and earlier.
- *Sign changing*: Simple operations, separated out because their dumb implementation means no IEEE exceptions.
- *Conversion operations*: Conversion between single, double, and integer values.
- *Conditional branch and test instructions*: Where the FP unit meets the integer pipeline again.

7.9.1 *Load/Store*

These operations load or store 32 or 64 bits of memory in or out of an FP register. On loads and stores, note the following points:

- The data is unconverted and uninspected, so no FP exception will occur even if it does not represent a valid FP value.
- MIPS I style 32-bit FPUs only do math operations on the even-numbered registers. Load/store of an odd-numbered register gives you access to the other half of 64-bit values.
- MIPS I and MIPS II CPUs are permitted to take an extra clock before FP load data is valid; they are not allowed to interlock any use of that data (the same rule applies to data being loaded into general-purpose registers). The compiler and/or assembler will usually take care of this for you, but on these old CPUs it is invalid for an FP load to be immediately followed by an instruction using the loaded value.
- When writing assembly, the “synthetic” form of the instruction (e.g., `l.d`) is preferred to the native machine instruction `ldc1`; the “synthetic” form is easier to read and can be used for all CPUs. The assembly may use multiple instructions for CPUs that don’t implement the machine instruction. You can use a synthetic load/store operation with any addressing mode that the assembler can understand (as described in section 9.4).
- The address for an FP load/store operation must be aligned to the size of the object being loaded—on a four-byte boundary for single-precision or word values or an eight-byte boundary for double-precision or 64-bit integer types.

In machine instruction descriptions, (`disp` is a signed 16-bit quantity, and “*” is the C operator that dereferences a pointer):

```
lwcl fd, disp(s)      fd = *(s + disp);
swcl fs, disp(s)      *(s + disp) = fs;
```

All MIPS64/MIPS32-compliant FPUs known to me have 64-bit loads/stores:

```
ldc1 fd, disp(s)    fd = (double)*(s + disp);
sdc1 fs, disp(s)    *(s + disp) = (double)fs;
```

MIPS32/64 adds indexed addressing, with two registers:

```
lwxcl fd, i(s)      fd = *(s + i);
swxcl fs, i(s)      *(s + i) = fs;
ldxc1 fd, i(s)      fd = (double)*(s + i);
sdxcl fs, i(s)      *(s + i) = (double)fs;
```

But, in fact, you don't have to remember any of these when you're writing assembly. Instead, "addr" can be any address mode the assembler understands:

```
l.d fd, addr        fd = (double)*addr;
l.s fd, addr        fd = (float)*addr;
s.d fs, addr        (double)*addr = fs;
s.s fs, addr        (float)*addr = fs;
```

The assembler will generate the appropriate instructions, including allowing a choice of valid address modes. Double-precision loads on a 32-bit CPU will assemble to two load instructions.

7.9.2 *Move between Registers*

When data is copied between integer and floating-point registers, no data conversion is done and no exception results from any value. But when data is moved between floating-point registers, you need to specify the particular floating-point data type, and an attempt to move a value that has no sense in that type—while it won't produce an exception—will not necessarily copy all the bits correctly. This makes it possible for FPUs that use a different (perhaps higher-precision) internal representation for FP data to implement move instructions without the overhead of back-converting the data.

Even on a MIPS I FPU these instructions can specify the odd-numbered FP registers:

Between integer and FP registers:

```
mtc1 s, fd          fd = s; /* 32b uninterpreted */
mfc1 d, fs          d = fs;
dmtc1 s, fd         fd = (long long) s; /* 64 bits */
dmfc1 d, fs         d = (long long) fs;
```

Between FP registers:

```
mov.d fd, fs          fd = fs;
                        /* move 64b between register pairs */
mov.s fd, fs          fd = fs; /* 32b between registers */
```

Conditional moves (missing from MIPS III and earlier)—the **.s** versions are omitted to save space:

```
movt.d fd, fs, cc     if (fpcondition(cc)) fd = fs;
movf.d fd, fs, cc     if (!fpcondition(cc)) fd = fs;
movz.d fd, fs, t      if (t == 0) fd = fs;
                        /* t is an integer register */
movn.d fd, fs, t      if (t != 0) fd = fs;
```

The FP condition code called `fpcondition(cc)` is a hard-to-avoid forward reference; you'll see more in section 7.9.7. If you want to know why conditional move instructions are useful, see section 8.5.3.

7.9.3 *Three-Operand Arithmetic Operations*

Note the following points:

- All arithmetic operations can cause any IEEE exception type and may result in an unimplemented trap if the hardware is not happy with the operands.
- All these instructions come in single-precision (32-bit, C `float`) and double-precision (64-bit, C `double`) versions; the instructions are distinguished by “**.s**” or “**.d**” on the op-code. We'll only show the double-precision version. Note that you can't mix formats: Both source values and the result will all be either single or double. To mix singles and doubles you need to use explicit conversion operations.

In all ISA versions:

```
add.d fd, fs1, fs2    fd = fs1 + fs2;
div.d fd, fs1, fs2    fd = fs1 / fs2;
mul.d fd, fs1, fs2    fd = fs1 × fs2;
sub.d fd, fs1, fs2    fd = fs1 - fs2;
```

Not in MIPS III and earlier CPUs, fast but not quite IEEE accurate—the error may be as much as that represented by the lowest 2 bits of the mantissa (2 ULPs).

```
recip.dfd, fs         fd = 1/fs;
rsqrt.dfd, fs         fd = 1/(squarerootof(fs));
```

7.9.4 *Multiply-Add Operations*

These were added after MIPS III in response to Silicon Graphics' interest in achieving supercomputer-like performance in very high end graphics systems (related to the 1995 SGI acquisition of Cray Research, Inc.). IBM's PowerPC chips seemed to get excellent FP performance out of their multiply-add, too. Although it's against RISC principles to have a single instruction doing two jobs, a combined multiply-add is widely used in common repetitive FP operations (typically the manipulation of matrices or vectors). Moreover, it saves a significant amount of time by avoiding the intermediate rounding and renormalization step that IEEE mandates when a result gets written back into a register.

Multiply-add comes in various forms, all of which take three register operands and an independent result register:

madd.d	fd, fs1, fs2, fs3	$fd = fs2 \times fs3 + fs1;$
msub.d	fd, fs1, fs2, fs3	$fd = fs2 \times fs3 - fs1;$
nmadd.d	fd, fs1, fs2, fs3	$fd = -(fs2 \times fs3 + fs1);$
nmsub.d	fd, fs1, fs2, fs3	$fd = -(fs2 \times fs3 - fs1);$

IEEE 754 does not rule specifically for multiply-add operations, so to conform to the standard the result produced should be identical to that coming out of a two-instruction multiply-then-add sequence—tricky! Since every FP operation may involve some rounding, this also means that IEEE 754 mandates somewhat poorer precision for multiply-add than could be achieved.

7.9.5 *Unary (Sign-Changing) Operations*

Although nominally arithmetic functions, these operations only change the sign bit and so can't produce most IEEE exceptions. They can produce an invalid trap if fed with a signaling NaN value. They are as follows:

abs.d	fd, fs	$fd = \text{abs}(fs)$
neg.d	fd, fs	$fd = -fs$

7.9.6 *Conversion Operations*

Note that “convert from single to double” is written “**cvt.d.s**”—and, as usual, the destination register is specified first. Conversion operators work between data in the FP registers: When converting data from CPU integer registers, the move from FP to CPU registers must be coded separately from the conversion operation. Conversion operations can result in any IEEE exception that makes sense in the context.

Originally, all this was done by the one family of instructions:

cvt.x.y fd, fs

where **x** and **y** specify the destination and source format, respectively, as one of the following:

- s** C float, MIPS/IEEE single, 32-bit floating point
- d** C double, MIPS/IEEE double, 64-bit floating point
- w** C int, MIPS “word,” 32-bit integer
- l** C long long, MIPS “long,” 64-bit integer (64-bit CPUs only)

The instructions are as follows:

```
cvt.s.d fd, fs    /* double fs -> float, leave in fd */
cvt.w.s fd, fs    /* float fs -> int, leave in fd */
cvt.d.l fd, fs    /* long long fs -> double, leave in fd */
```

There’s more than one reasonable way of converting from floating point to integer formats, and the result depends on the current rounding mode (as set up in the **FCSR** register, described in section 7.7). But FP calculations quite often want to round to the integer explicitly (for example, the ceiling operator rounds upward), and it’s a nuisance trying to generate code to modify and restore **FCSR**. So except for the earliest MIPS I CPUs, conversions that do explicit rounding are available and often more useful:

Conversions to integer with explicit rounding:

```
round.x.y fd, fs      /* round to nearest */
trunc.x.y fd, fs      /* round toward zero */
ceil.x.y fd, fs       /* round up */
floor.x.y fd, fs      /* round down */
```

These instructions are only valid with **x** representing an integer format.

7.9.7 *Conditional Branch and Test Instructions*

The FP branch and test instructions are separate. We’ll discuss the test instructions below—they have names like **c.le.s**, and they compare two FP values and set the FPU condition bit accordingly.

The branch instructions, therefore, just have to test whether the condition bit is true (set) or false (zero). They can optionally specify which condition bit to use:

```
bc1t label         if (fpcondition(0)) branch-to-label;
bc1t cc, label     if (fpcondition(cc)) branch-to-label;
bc1f label         if (!fpcondition(0)) branch-to-label;
bc1f cc, label     if (!fpcondition(cc)) branch-to-label;
```


Like all MIPS branch instructions, each of these has a “branch-likely” variant:⁹

```
bclt1 label          /* branch-likely form of bclt ... */
bclf1 label
```

Like the CPU’s other instructions called branch, the target **label** is encoded as a 16-bit signed word displacement from the next instruction plus one (pipelining works in strange ways). If **label** was more than 128 KB away, you’d be in trouble and you would have to resort to a **jx** instruction.

MIPS CPUs up to and including MIPS III had only one FP condition bit, called “C,” in the FP control/status register **FCSR**. Mainstream CPUs have 7 extra condition bits, called FCC7-1. If you leave the **cc** specification out of branch or compare instructions, you implicitly pick the old “C” bit, which has the honorary title of FCC0. That’s compatible with older instruction set versions. (See section 8.5.7 if you’re interested in why this extension was introduced.) In all the instruction sets, **cc** is optional.

But before you can branch, you have to set the condition bit appropriately. The comparison operators are as follows:

```
c.cond.d fs1,fs2 /* compare fs1 and fs2 and set FCC(0) */
c.cond.d cc,fs1,fs2 /* compare fs1 and fs2; set FCC(cc) */
```

In these instructions, **cond** can be a mnemonic for any of 16 conditions. The mnemonic is sometimes meaningful (**eq**) and sometimes more mysterious (**ult**). Why so many? It turns out that when you’re comparing FP values there are four mutually incompatible outcomes:

```
fs1 < fs2
fs1 == fs2
fs1 > fs2
unordered (fs1, fs2)11
```

The IEEE standard defines “unordered” as true when either of the operands is an IEEE NaN value.

It turns out we can always synthesize “greater than” from “less than or equal to” by reversing the order of the operands and/or inverting the test, so we have three outcomes to allow for. MIPS provides instructions to test for any “or” combination of the three conditions. On top of that, each test comes in two flavors, one that takes an “invalid” trap if the operands are unordered and one that never takes such a trap.

9. See section 8.5.4 for more on “likely” branches.

TABLE 7.4 FP Test Instructions

“C” bit is set if...	Mnemonic	
	No trap	Trap
always false	f	sf
unordered(fs1,fs2)	un	ngle
fs1 == fs2	eq	seq
fs1 == fs2 unordered(fs1,fs2)	ueq	ngl
fs1 < fs2	olt	lt
fs1 < fs2 unordered(fs1,fs2)	ult	nge
fs1 < fs2 fs1 == fs2	ole	le
fs1 < fs2 fs1 == fs2 unordered(fs1,fs2)	ule	ngt

We don’t have to provide tests for conditions like “not equal”; we test for equality, but then use a **bc1f** rather than a **bc1t** branch. Table 7.4 tabulates all the available instruction names:

In many CPU implementations, the compare instruction produces its result too late for a branch instruction to run without delay in the immediately following instruction. In MIPS III and earlier CPUs the branch instruction may misfire if run directly after the test. A compiler or assembler supporting older CPUs should generate an appropriate delay, inserting a **nop** if required.

7.10 Paired-Single Floating-Point Instructions and the MIPS-3D ASE

A floating-point unit compliant with MIPS32/64 may choose to implement *paired-single* instructions. The base of this extension is a set of arithmetic operations that work on pairs of IEEE single-precision values, each packed into a single 64-bit register, and do two operations in parallel on each half.

That sounds like it means building a whole new FPU. But it turns out that it really isn’t too costly to tinker with the double-precision FPU to provide paired-single operations—at least for the “one-pass” operations like add, multiply, multiply-add, tests, moves, and so on.

We’ll use **fs** and **ft** as source registers and **fd** as a destination. We’ll find it convenient to say that if **fs** is an FP register containing a paired-single value, then **fs.upper** and **fs.lower** stand for the single-precision values in the higher-numbered and lower-numbered bits, respectively.

To find out if your CPU implements paired-single instructions, see whether the **FIR(PS)** bit is set.

7.10.1 *Exceptions on Paired-Single Instructions*

Usually, in the circumstances where you want to use paired-single instructions, you probably will take pains to prevent exceptions. But if either of the two floating-point instructions causes an exception, then the instruction trips an exception: If either of the two instructions needs a MIPS trap, then you take a MIPS “exception” and both halves of the instruction are rolled back as if they’d never started. An exception is delivered with no indication of whether it applies to the upper or lower computation, or both.

7.10.2 *Paired-Single Three-Operand Arithmetic, Multiply-Add, Sign-Changing, and Nonconditional Move Operations*

These all do exactly the same as the corresponding single-precision (`.s`) version, except they do it twice. That is, an `add.ps fd, fs, ft` instruction is:

```
fd.upper = fs.upper + ft.upper;
fd.lower = fs.lower + ft.lower;
```

These are *SIMD* instructions (single-instruction, multiple-data) and are very useful for vector-type operations: They get two operations done in the time of one.

However, not all instructions are available in `.ps` form:

- Implemented, same mnemonic as single-precision operation: **add**, **sub**, **mul**, **abs**, **mov**, and **neg**.
- Not available in paired-single: all the round-in-a-particular-way operators **round**, **trunc**, **ceil**, **floor**. They probably didn’t seem important enough in vectorizable applications.

Divide, reciprocal, and square-root—corresponding to **div.s**, **recip.s**, **sqrt.s**, and **rsqrt**—are not available in paired form (those functions are typically implemented with iterative algorithms, which can’t be simply done two-at-once).

The MIPS-3D extension, where available, adds instructions that can calculate a reciprocal or reciprocal square-root in just two steps. Those instructions are available for PS values too. See section 7.10.5.

The integer-conversion instructions **round**, **trunc**, **ceil**, and **floor** are not provided. The result of such an operation would be a paired-word, double-integer format, and inventing a whole new data type for such purposes seems excessive.

7.10.3 *Paired-Single Conversion Operations*

To build a paired-single value from two single-precision values use **cvt.ps.s fd, fs, ft**, which does:

```
fd.upper = fs;
fd.lower = ft;
```

To extract a single-precision value from either the upper or lower position of a pair use **cvt.s.pl fd, fs** ($fd = fs.lower;$) or **cvt.s.pu fd, fs** ($fd = fs.upper;$).

To repack paired-single values into a new paired single, you have a choice of four instructions. One of these instructions picks one single value from each of two paired-single registers, and repacks them into a new pair:

```
pll.ps fd, fs  fd.upper = fs.lower;
                fd.lower = ft.lower;

plu.ps fd, fs  fd.upper = fs.lower;
                fd.lower = ft.upper;

pul.ps fd, fs  fd.upper = fs.upper;
                fd.lower = ft.lower;

puu.ps fd, fs  fd.upper = fs.upper;
                fd.lower = ft.upper;
```

alnv.ps fd, fs, ft, s packs together two single-precision values into a paired-single. Its main role—in conjunction with the unaligned load instruction **luxc1**—is to help software load and pack an array of single-precision values two at a time, even though the memory alignment of the array is such as to get the pairs the wrong way round.

For the purposes of this instruction, it's helpful to think of paired-single values as packing together singles called “a” and “b,” where “a” has the lowest address when the data is stored in memory (the association of “a” and “b” with bit numbers in the register changes with endianness).

The value of the low 3 bits of the general-purpose register **s** may only be 0 or 4 (**s** is typically the pointer used to load a pair of values from memory)—any value other than 4 will lead to an exception. So it's like this:

```
if ((s & 7) == 0) {
    fd.a = fs.a; fd.b = fs.b;
}
else {
    /* s & 7 == 4 */
    fd.a = fs.b; fd.b = ft.a;
}
```

7.10.4 *Paired-Single Test and Conditional Move Instructions*

Since there are two values in each paired-single register, each compare instruction sets two condition bits. You specify an even-numbered condition bit, and the instruction will use it and the next-up odd-numbered condition bit. So if you write `c.eq.ps 2, fs, ft`, that implies:

```
fcc2 = (fs.upper == ft.upper) ? 1: 0;
fcc3 = (fs.lower == ft.lower) ? 1: 0;
```

If you can't remember what all the mysterious tests are, see Table 7.4.

MIPS-3D (see below) includes some branch instructions that test more than one condition code at a time, which may be useful.

The conditional move instructions, `movf.ps` and `movt.ps`, test two FP condition codes and do two *independent* move-if-true operations on the upper and lower halves of the destination, so:

```
movt.ps fd, fs, 2    if (fcc2) fd.upper = fs.upper;
                    if (fcc3) fd.lower = fs.lower;
```

Contrastingly, `movn.ps` and `movz.ps` do their move conditionally on the value of an integer register and treat both halves the same.

7.10.5 *MIPS-3D Instructions*

The MIPS-3D ASE (instruction set extension) adds a handful of instructions that are judged to make the paired-single floating-point extension more efficient at high-quality 3D graphics coordinate calculations.

- Floating-point “reduction add/multiply” (add/multiply pair):
Adds or multiplies the pairs inside a register. In true SIMD style, it will add or multiply the pairs inside two different registers at once:

`addr.ps fd, fs, ft` does:

```
fd.upper = fs.upper + fs.lower;
fd.lower = ft.upper + ft.lower;
```

And `mulr.ps fd, fs, ft` does:

```
fd.upper = fs.upper * fs.lower;
fd.lower = ft.upper * ft.lower;
```

- Comparison operations on absolute values (i.e., ignoring sign bit):
There are a complete set of **cabs.xx.x** operations:

cabs.eq.d	cabs.nge.d	cabs.ole.s	cabs.ueq.s
cabs.eq.ps	cabs.nge.ps	cabs.olt.d	cabs.ule.d
cabs.eq.s	cabs.nge.s	cabs.olt.ps	cabs.ule.ps
cabs.f.d	cabs.ngl.d	cabs.olt.s	cabs.ule.s
cabs.f.ps	cabs.ngl.ps	cabs.seq.d	cabs.ult.d
cabs.f.s	cabs.ngl.s	cabs.seq.ps	cabs.ult.ps
cabs.le.d	cabs.ngle.ps	cabs.seq.s	cabs.ult.s
cabs.le.ps	cabs.ngt.d	cabs.sf.d	cabs.un.d
cabs.le.s	cabs.ngt.ps	cabs.sf.ps	cabs.un.ps
cabs.lt.d	cabs.ngt.s	cabs.sf.s	cabs.un.s
cabs.lt.ps	cabs.ole.d	cabs.ueq.d	
cabs.lt.s	cabs.ole.ps	cabs.ueq.ps	

- Branch instructions testing multiple condition codes:
bc1any2f, **bc1any2t**, **bc1any4f**, and **bc1any4t** test the OR of some number of conditions before branching. You might write **bc1any2f N, offset** with **N** being the condition code number between 0 and 6.
The tests performed are:

<i>Instruction</i>	<i>Branch test</i>
bc1any2f 2,target	if (!fcc2 !fcc3) goto target
bc1any2t 2,target	if (fcc2 fcc3) goto target
bc1any4f 4,target	if (!fcc4 !fcc5 !fcc6 !fcc7) goto target
bc1any4t 4,target	if (fcc4 fcc5 fcc6 fcc7) goto target

- Reciprocal and square-root calculations: This is motivated mostly by the lack of divide, reciprocal, and square-root operations for paired-single (although single- and double-precision versions of these instructions are included).
recip1 fd,fs is a quick-and-dirty approximation to a reciprocal. It's as good as can be done reasonably cheaply without an "under-the-hood" iterative process—iterative processes don't convert well to paired-single SIMD. It's a *much* worse approximation than you could get with the standard **recip.s** or **recip.d** instructions, even though those aren't IEEE accurate. The operation of **recip1** is unaffected by the current rounding

mode (though it does respect the flags that control generation and use of tiny “denormalized” values).

recip2 *fd, fs, ft* is not anything like a reciprocal—it’s a custom multiply-add that computes $1 - (fs * ft)$ without you needing to load a register with a constant 1. That function is picked because it means that this three-instruction sequence refines **recip1**’s guess:

```
recip1.s f1, f0           # f1 = ~ 1/f0
recip2.s f2, f1, f0       # f2 = f0 * (error in f1)
madd.s f1, f1, f1, f2     # f1 = f1 - f2*f1, a better guess
```

That’s enough to produce a good result in single precision (you can’t refine it further and reach IEEE accuracy without handling some least significant bits separately, which is a much longer computation). A further **recip2/madd** sequence is necessary to get the best available result for double precision.

But the really nice thing about this is that the corresponding paired-single calculation just works, producing an excellent pair of reciprocals for three instructions:

```
recip1.ps f1, f0
recip2.ps f2, f1, f0
madd.ps f1, f1, f1, f2
```

The **rsqrt1/rsqrt2** instructions use a similar approach. In this case the error function computed by **rsqrt2** *fd, fs, ft* is $(1 - fs * ft) / 2$.

This time you need an additional multiplication to produce an excellent paired-single value (though again, not IEEE exact):

```
rsqrt1.ps f1, f0
mul.ps f2, f1, f0
rsqrt2.ps f3, f2, f1
madd.ps f1, f1, f1, f3
```

Again, getting a good-quality double-precision value requires an additional refinement step.

- Convert between paired-integer and paired-single values:

Since integer/floating-point conversions have always been done strictly in floating-point registers, once you’ve got paired-single it seems to make sense to have paired-32-bit-integer support too: But it isn’t in the original paired-single instruction set.

To convert a packed pair of 32-bit integers to a paired-single (converting both fields at once) use **cvt.ps.pw** *fd, fs*; the reverse operation converts a paired-single to a packed pair of words and is **cvt.pw.ps** *fd, fs*. In both cases, the current rounding mode determines how approximate results are rounded.

7.11 Instruction Timing Requirements

Normal FP arithmetic instructions are interlocked, and there is no need to interpose **nops** or to reorganize code for correctness. To get the best performance, the compiler should lay out FP instructions to make the best use of overlapped execution of integer instructions and of the FP pipeline—but that needs to be sensitive to the details of a particular CPU implementation.

In some older CPUs (not compliant to MIPS32/64) some other interactions are not interlocked, and for correct operation, programs must avoid certain sequences of instructions. In this case, your compiler, your assembler, or (in the end) you, the programmer, must take care about the timing of the following:

- *Operations on the FP control and status register:* When altering **FCSR**, take care with the pipeline. Its fields can affect any FP operation, which might be running in parallel. Make sure that at the point you write **FCSR** there are no FP operations live (started, but whose results have not yet been collected). The register is probably written late, too, so it's wise to allow one or two instructions to separate the **ctc1 rs, FCSR** from an affected computational instruction.
- *Moves between FP and general-purpose registers:* These complete late, and if the subsequent instruction uses the value, you'll normally lose a cycle or two. On MIPS II and earlier CPUs you are required to avoid the dependency.
- *FP register loads:* Like integer loads, these take effect late. The value can't be used in the following instruction.
- *Test condition and branch:* The test of the FP condition bit using the **bc1t**, **bc1f** instructions must be carefully coded, because the condition bit is tested a clock cycle earlier than you might expect. So the conditional branch cannot immediately follow a test instruction.

7.12 Instruction Timing for Speed

All MIPS FPU take more than one clock cycle for most arithmetic instructions; hence, the pipelining becomes visible. The pipeline can show up in three ways:

- *Hazards:* These occur where the software must ensure the separation of instructions to work correctly. There are no hazards of any kind visible to user-level FP code on a MIPS32/64 CPU.
Even in older CPUs there are no hazards between FP instructions caused by FP data.

- *Interlocks*: These occur where the hardware will protect you by delaying use of an operand until it is ready. Knowledgeable rearrangement of the code will improve performance.
- *Visible pipelining*: This occurs where the hardware is prepared to start one operation before another has completed (provided there are no data dependencies). Compilers, and determined assembly programmers, can write code that works the hardware to the limit by keeping the pipeline full.

Modern MIPS FPUs are often fully pipelined, allowing a new FP multiply, add, or multiply-add operation to start on every clock (it may be every second clock for double-precision operations). But if you use the result of one of those instructions in the next instruction in sequence, you'll hold up your program: With 2006-era CPUs, that's likely to be by four or five clock times. In other language, FP multiplication has a one- or two-clock *repeat rate* but a four- or five-clock *latency*. If you can avoid using FP results until they're ready, your program will go faster.

With floating point, multiply is generally as fast as addition and subtraction. Divide and square-root instructions are much slower; if you need to repeatedly divide by the same number, it makes sense to compute a reciprocal and multiply by it instead.

7.13 Initialization and Enabling on Demand

From reset, you will normally have initialized the CPU's **SR** register to disable all optional coprocessors, which includes the FPU (coprocessor 1). The **SR (CU1)** bit has to be set for the FPU to work. All modern CPUs have 32 64-bit FP registers, but have a MIPS I compatibility mode, when **SR (FR)** is zero, where only even-numbered registers can be used for math.

You should read the FPU implementation register; if it reads zero, no FP is fitted and you should run the system with CU1 off.

Once CU1 is switched on, you should set up the control/status register **FCSR** with your choice of rounding modes and trap enables. Anything except round to nearest and all traps disabled is uncommon. There's also the choice of setting the **SR (FS)** bit to cause very small results to be returned as zero, saving a trap to the emulator. This is not IEEE compatible, but the hardware can't produce the specified denormalized result.

Once the FPU is operating, you need to ensure that the FP registers are saved and restored during interrupts and context switches. Since this is (relatively) time consuming, you can optimize this, as some operating systems do, by doing "lazy context switching," like this:

- Leave the FPU disabled by default when running a new task. Since the task cannot now access the FPU, you don't have to save and restore registers when scheduling or parking it.
- On a CU1-unusable trap, mark the task as an FP user and enable the FP before returning to it.
- Disable FP operations while in the kernel or in any software called directly or indirectly from an interrupt routine. Then you can avoid saving FP registers on an interrupt; instead, FP registers need to be saved and restored only when you are context-switching to or from an FP-using task.

7.14 Floating-Point Emulation

Some low-cost MIPS CPUs and cores omit the FPU. Floating-point functions for these processors are provided by software and are perhaps 50–300 times slower than the hardware. Software FP is useful for systems where floating point is employed in some rarely used routines.

There are two approaches:

- *Soft float*: Some compilers can be requested to implement floating-point operations with software. FP arithmetic operations are likely to be implemented with a hidden library function, but housekeeping tasks such as moves, loads, and stores can be handled in line.
- *Trap and emulate*: The compiler can produce the regular FP instruction set. The CPU will then take a trap on each FP instruction that is caught by the FP emulator. The emulator decodes the instruction and performs the requested operation in software. Part of the emulator's job will be emulating the FP register set in memory.

As described here, a runtime emulator is also required to back up FP hardware for very small operands or obscure operations; since the architecture is deliberately vague about the limits of the hardware's responsibility, the emulator is usually complete. However, it will be written to ensure exact IEEE compatibility and is only expected to be called occasionally, so it will probably be coded for correctness rather than speed.

Compiled-in floating point is much more efficient; the emulator has a high overhead on each instruction from the trap handler, instruction decoder, and emulated register file. But of course you have to recompile the world to take advantage of it.

There may still be some compilers that don't offer soft float operation: The history of the MIPS architecture is in workstations, where FP hardware was mandatory.

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Complete Guide to the MIPS Instruction Set

Chapters 8 and 9 are written for the programmer who wants to understand or generate assembly code (whether in person or indirectly because you're writing or fixing a compiler). While Chapter 9 discusses real assembly language programming, this chapter only concerns itself with assembly language instructions; broadly speaking, you can skip Chapter 9 if you only want to read disassembly listings. We begin with a simple piece of MIPS code and an overview.

8.1 A Simple Example

This is an implementation of the C library function `strcmp(1)`, which compares two character strings and returns zero on equal, a positive value if the first string is greater (in string order) than the second, and a negative value otherwise. Here's a naïve C algorithm:

```
int strcmp (char *str1, char *str2)
{
    char c1, c2;

    do {
        c1 = *str1++;
        c2 = *str2++;
    } while (c1 != 0 && c2 != 0 && c1 == c2);

    return c1 - c2; /* clever: 0, +ve or -ve as required */
}
```

In assembly code the two arguments of the C function arrive in the registers called **a0** and **a1**. (See Table 2.1 if you’ve forgotten about the naming conventions for registers; the MIPS standard calling convention is described in detail in section 11.2.1.) A simple subroutine like this one is free to use the temporary registers **t0** and so on without saving and restoring their values, so they’re the obvious choices for temporaries. The function returns a value, which by convention needs to be in the register **v0** at the time we return. So let’s have a go at it:

```

strcmp:
1:
    lbu    t0, 0(a0)
    addu   a0, a0, 1
    lbu    t1, 0(a1)
    addu   a1, a1, 1

    beq    t0, zero, .t01    # end of first string?
    beq    t1, zero, .t01    # end of second string?
    beq    t0, t1, 1b

.t01:
    subu   v0, t0, t1
    j      ra

```

We will examine it from the top:

- *Labels:* **strcmp** is a familiar named label, which in assembly can define a function entry point, an intermediate branch, or even a data storage location.
 .t01 is a legitimate label; the full-stop “.” character is legal in labels and must not be confused with a C name elsewhere in your program.
 1: is a numeric label, which many assemblers will accept as a local label. You can have as many labels called “1” as you like in a program; “1f” refers to the next one in sequence and “1b” the previous one. This is very useful, since it avoids inventing unique names for labels only used locally.
- *Register names:* The “dollar-free” names shown here are common usage, but they require that the assembly code be passed through some kind of macro processor before getting to the real MIPS assembler; typically, the C preprocessor is used and most toolkits have ways to make this straightforward.

It would hardly be worth writing a function such as this in assembly; the compiler will probably do a better job. But we’ll see later (in section 9.1) how much more clever we could have been.

8.2 Assembly Instructions and What They Mean

This section consists of a long list of all legal assembly instruction names (mnemonics) in most MIPS assemblers, up to and including the MIPS64 (Release 2) instruction set. After some agonizing and experimentation, I decided that this table should contain a mixture of real machine operations and the assembler's synthesized instructions. So for each instruction we'll list the following:

- *Assembly format*: How the instruction is written.
- *Machine instructions generated*: For assembly instructions that are aliases for machine code or expanded into a sequence of machine instructions, we'll put a "⇒" to show a macro expansion and list typical instructions in an expansion.
- *Function*: A description of what the instruction does, in pseudo-C code, which is meant to combine precision with brevity. C typecasts, where used, are necessary.

Not every possible combination of instruction and operands is listed, because it gets too long. So we won't list the following:

- *Two-operand forms of three-operand instructions*: For example, MIPS assemblers allow you to write:

```
addu $1, $2           # $1 = $1 + $2
```

which would otherwise have to be written as:

```
addu $1, $1, $2
```

You can do that pretty much anywhere it makes sense.

- *All possible load/store address formats (addr)*: MIPS machine instructions always generate addresses for load/store operations using just the contents of a register plus a 16-bit signed displacement,¹ written, for example, **lw \$1, 14(\$2)**. MIPS assemblers support quite a few other addressing mode formats; notably **lw \$1, thing**, which loads data from the location whose assembly code label (or external C name) is "thing." See section 9.4 for details; note that all of these modes are quietly available to any assembly instruction that specifies a memory address. We'll just write **lw t, addr** for the assembly instruction and the base+displacement format for the machine code.

1. Someone always has to break things; there are some register+register address formats but only for load/stores with floating-point registers. This is done in deference to the importance of multidimensional-array organizations in floating-point codes.

The **la** (load address) instruction provided by the assembler uses the same addressing-mode syntax, even though it loads or stores nothing—it just generates the address value in the destination register.

When synthesizing some address formats (particularly on stores) the assembler needs a scratch register and quietly uses **at**. Programmers working in code where the implicit use of a register might matter (in an interrupt handler that hasn't yet saved all preinterrupt register values, for instance) need to take care: The GNU assembler has a **.set noat** directive that prevents it.

- *Immediate versions of instructions:* A constant value embedded within an instruction is, by ancient convention, called an immediate value. MIPS assembly language lets you specify the last source operand as a constant. MIPS CPUs offer some real hardware instructions supporting immediates of up to 16 bits in size for some operations, but you're recommended to write the "root" mnemonic instead. Names like **addui** are recognized by the assembler as legal mnemonics; but probably only compilers generating assembly intermediate code should ever do so, and you'll see these "immediate" forms when we're discussing machine instructions (Table 8.6, for example) and in disassembly listings.

In assembly language you're not limited to 16 bits, and you don't have to remember which of the "immediate" forms uses signed or unsigned values; if you write an arbitrary constant, the assembler will synthesize away, as described in section 9.3.2.

Once again, the assembler may need to use the temporary register **at** for some complicated cases.

8.2.1 *U and Non-U Mnemonics*

Before we get started, there's a particularly confusing thing about the way instruction mnemonics are written. A "**u**" suffix on the assembly mnemonic is usually read as "unsigned." But that's not always what it means (at least, not without a big stretch of your powers of imagination). There are a number of subtly different meanings for a "**u**" suffix, depending on context:

- *Overflow trap versus no trap:* In most arithmetic operations, **U** denotes "no overflow test." Unsuffixes arithmetic operations like **add** cause a CPU exception if the result overflows into bit 31 (the sign bit when we're thinking of integers as signed). The suffixed variant **addu** produces exactly the same result for all combinations of operands but never takes an exception.

C and C++ do not support integer overflow exceptions and always use the "**u**" form—it has nothing to do with whether a variable is

signed or unsigned. You should always use **addu** and so on unless you really know why not.

- *Set if:* The universal test operations **slt** (set if less than) and **sltu** (set if less than, unsigned) have to produce genuinely different results when operands are interpreted as negative numbers.
- *Multiply and divide:* Integer multiply operations produce a result with twice the precision of the operands, and that means that they need to produce genuinely different results for signed and unsigned inputs; hence, there are two instructions: **mult** and **multu**. Note that the low part of the result, left in the **lo** register, will be the same for both the signed and the unsigned version; it's the way that overflows into **hi** are handled that differs.

Integer divide instructions are also sign dependent (think about dividing `0xFFFF.FFFE` by 2), so there's a **div** and a **divu**. The same variation exists for shift-right instructions (shift-right by one is really just divide by two), but this was obviously a U too far; the shift instructions are called **sra** (shift-right arithmetic, suitable for signed numbers) and **srl** (shift-right logical). The world is indeed a wonderful place.

- *Partial-register loads:* Loads of less-than-register-size chunks of data must decide what to do with the excess bits in the register. For the unsigned instructions, such as **lbu**, the byte value is loaded into the register and the remaining bits are cleared to zero (we say that the value has been *zero-extended*). If the byte value represented a signed number, its top bit would tell us if it was negative. In this case, we'll translate to the corresponding register-sized representation by filling the remaining bits of the register with copies of the sign bit, using the instruction **lb**. That's called *sign-extending*.

8.2.2 Divide Mnemonics

In machine code for integer multiply and divide, there are separate initiation and result-collecting instructions. The assembler likes to cover this up, generating macro expansions for a three-operand format and doing a divide-by-zero check at the same time. This would be OK except that unfortunately the assembly macro name for divide is **div**, which is also the name for the basic machine code instruction. That means you need a trick to write a machine code divide instruction in assembly; a three-operand assembly divide with **zero** as the destination should just produce the machine start-divide operation and nothing else.

Some toolchains have offered a better way out of this mess, by defining new mnemonics **divd** (divide direct) to mean just the hardware operation and **divo** (divide with overflow check) for the complicated macro. This didn't catch on, but you may see it in some codes.

8.2.3 *Inventory of Instructions*

In the assembly descriptions we use the conventions given in Table 8.1. Table 8.2 gives a full inventory of the instruction descriptions in mnemonic order.

TABLE 8.1 Conventions Used in Instruction Tables

<i>Word</i>	<i>Used for</i>
s, t	CPU registers used as operands.
d	CPU register that receives the result.
j	“Immediate” constant.
label	The name of an entry point in the instruction stream.
shf	For shift/rotate/extract/insert instructions, this is the amount of the implied shift operation, in bits.
sz	For extract/insert instructions, this is the size of the field being manipulated, in bits.
offs	A signed 16-bit PC-relative word offset representing the distance in words (one word per instruction) to a label.
addr	One of a number of different legitimate data address expressions usable when writing load/store (or load address) instructions in assembly. (See section 9.4 for a description of how the assembler implements the various options.)
at	The assembly temporary register, which is really \$1 .
zero	This register, \$0 , always contains a zero value.
ra	The return address register \$31 .
hilo	The double-precision integer multiply result formed by concatenating hi and lo . Each of hi and lo holds the same number of bits as a machine register, so hilo can hold a 64-bit integer on a 32-bit machine and a 128-bit result on a 64-bit machine.
MAXNEG32BIT MAXNEG64BIT	The most negative number representable in twos complement arithmetic, 32- and 64-bit, respectively. It’s a feature of twos complement numbers that the positive number $-\text{MAXNEG32BIT}$ is not representable in 32 bits.
cd	Coprocessor register that is written by instruction.
cs	Coprocessor register that is read by instruction.

TABLE 8.1 *continued*

<i>Word</i>	<i>Used for</i>
exception(CAUSE, code) exception(CAUSE)	Take a CPU trap; CAUSE determines the setting of the Cause (ExcCode) register field. “code” is a value not interpreted by the hardware, but rather one encoded in a don’t-care field of the instruction, where system software can find it by reading the instruction. Not every such instruction sets a “code” value, so sometimes we’ll leave it out.
const31..16	Denotes the number obtained by just using bits 31 through 16 of the binary number “const.” The MIPS books use a similar convention.

TABLE 8.2 Assembly Instructions in Alphabetical Order

<i>Assembly/Machine Code</i>	<i>Description</i>
abs d,s ⇒ sra \$at,s,31 xor d,s,\$at subu d,d,\$at	$d = s < 0 ? -s : s;$
add d,s,j ⇒ addi d,s,j	Traps on overflow, rare $d = s + (\text{signed})j;$
add d,s,t	Traps on overflow, rare $d = s + t;$
addu d,s,j ⇒ addiu d,s,j	You can also write this with j outside the range $-32768 \leq j < 32768$, but the code generated gets more complicated. $d = s + (\text{signed})j;$
addu d,s,t	$d = s + t;$
and d,s,j ⇒ andi d,s,j	For $0 \leq j < 65535$ —for larger numbers, extra instructions will be generated. $d = s \& (\text{unsigned})j;$
and d,s,t	$d = s \& t;$
b label ⇒ beq \$zero,\$zero,offs	goto label;
bal label ⇒ bgezal \$zero,offs	Function call (limited range but PC-relative addressing). Note that the return address that is left in ra is that of the next instruction but one: The next instruction in memory order is in the branch delay slot and gets executed before the function is invoked.

TABLE 8.2 *continued*

<i>Assembly/Machine Code</i>	<i>Description</i>
bc0f <i>label</i> bc0f1 <i>label</i> bc0t <i>label</i> bc0t1 <i>label</i>	Branch on coprocessor 0 condition. On old CPUs, this tested the state of a CPU input pin. No longer part of MIPS32 or MIPS64.
bclf <i>\$fccN, label</i> bclf1 <i>\$fccN, label</i> bclt <i>\$fccN, label</i> bclt1 <i>\$fccN, label</i>	Branch on floating-point (coprocessor 1) condition set/true (t) or clear/false (f); described in section 7.9.7. Modern FPUs have multiple FP condition bits, selected by <i>N</i> = 0..7. Older code only uses condition bit 0, and then the “ \$fccN, ” can be omitted. The suffix “ 1 ” in bclf1 and so on indicates a branch-likely instruction; see section 8.5.4 for details, but note that the MIPS32/64 specifications deprecate branch-likely, and recommend programmers and compilers not to generate the instruction on code that may be ported to more than one implementation of the MIPS architecture.
bclany2f <i>\$fccN, label</i> bclany2t <i>\$fccN, label</i> bclany4f <i>\$fccN, label</i> bclany4t <i>\$fccN, label</i>	MIPS-3D instructions, which branch on the “OR” of two or four conditions. See section 7.10.4.
bc2f <i>label</i> bc2f1 <i>label</i> bc2t <i>label</i> bc2t1 <i>label</i>	Branch on coprocessor 2 condition. Useful only if a CPU uses the CP2 instruction set or offers an external pin. See bc1f and so on above for details.
beq <i>s, t, label</i>	if (<i>s</i> == <i>t</i>) goto <i>label</i> ;
beql <i>s, t, label</i>	Deprecated branch-likely variants of conditional branch above. The delay slot instruction is only executed if the branch is taken; see section 8.5.4.
beqz <i>s, label</i> ⇒ beq <i>s, \$zero, offs</i>	if (<i>s</i> == 0) goto <i>label</i> ;
beqz1	Branch-likely variant of beqz ; see section 8.5.4.
bge <i>s, t, label</i> ⇒ slt <i>at, s, t</i> beq <i>at, \$zero, offs</i>	if ((signed) <i>s</i> ≥ (signed) <i>t</i>) goto <i>label</i> ;
bge1 <i>s, t, label</i> ⇒ slt <i>at, s, t</i> beql <i>at, \$zero, offs</i>	“Likely” form of bge , even though that instruction is itself a macro. Deeply deprecated, even though the assembler supports it. See section 8.5.4.

TABLE 8.2 *continued*

<i>Assembly/Machine Code</i>	<i>Description</i>
bgeu <i>s,t,label</i> ⇒ sltu <i>at,s,t</i> beq <i>at,\$zero,offs</i>	if ((unsigned) <i>s</i> ≥ (unsigned) <i>t</i>) goto label;
bgez <i>s,label</i>	if (<i>s</i> ≥ 0) goto label;
bgezal <i>s,label</i>	If (<i>s</i> ≥ 0) call to "label()". But note that the "return address" is <i>unconditionally</i> saved in the register ra (\$31).
bgezall <i>s,label</i>	Deprecated branch-likely variant; see section 8.5.4. It's hard to see what this instruction is good for.
bgezl <i>s,label</i>	Deprecated branch-likely variant; see section 8.5.4.
bgt <i>s,t,label</i> ⇒ slt <i>at,t,s</i> bne <i>at,\$zero,offs</i>	if ((signed) <i>s</i> > (signed) <i>t</i>) goto label;
bgtu <i>s,t,label</i> ⇒ slt <i>at,t,s</i> beq <i>at,\$zero,offs</i>	if ((unsigned) <i>s</i> > (unsigned) <i>t</i>) goto label;
bgtz <i>s,label</i>	if (<i>s</i> > 0) goto label;
bgtzl <i>s,label</i>	Deprecated branch-likely version of bgtz ; see section 8.5.4.
ble <i>s,t,label</i> ⇒ sltu <i>at,t,s</i> beq <i>at,\$zero,offs</i>	if ((signed) <i>s</i> ≤ (signed) <i>t</i>) goto label;
bleu <i>s,t,label</i> ⇒ sltu <i>at,t,s</i> beq <i>at,\$zero,offs</i>	if ((unsigned) <i>s</i> ≤ (unsigned) <i>t</i>) goto label;
blez <i>s,label</i>	if (<i>s</i> ≤ 0) goto label;
blezl <i>s,label</i>	Deprecated branch-likely variant of blez ; see section 8.5.4.
blt <i>s,t,label</i> ⇒ slt <i>at,s,t</i> bne <i>at,\$zero,offs</i>	if ((signed) <i>s</i> < (signed) <i>t</i>) goto label;
bltl <i>s,t,label</i>	Deprecated branch-likely macro; see section 8.5.4.
bltu <i>s,t,label</i> ⇒ sltu <i>at,s,t</i> bne <i>at,\$zero,offs</i>	if ((unsigned) <i>s</i> < (unsigned) <i>t</i>) goto label;
bltz <i>s,label</i>	if (<i>s</i> < 0) goto label;

TABLE 8.2 *continued*

<i>Assembly/Machine Code</i>	<i>Description</i>
bltzal <i>s</i> , <i>label</i>	If <i>s</i> < 0 call function at <i>label</i> . But the “return address” is <i>unconditionally</i> saved in the register ra (\$31), whether the function call is taken or not. <code>if (s < 0) label();</code>
bltzall <i>s</i> , <i>label</i>	Bizarre branch-likely variant; see section 8.5.4.
bltzl <i>s</i> , <i>label</i>	Deprecated branch-likely variant; see section 8.5.4.
bne <i>s</i> , <i>t</i> , <i>label</i>	<code>if (s != t) goto label;</code>
bnel <i>s</i> , <i>t</i> , <i>label</i>	Deprecated branch-likely variant; see section 8.5.4.
bnez <i>s</i> , <i>label</i>	<code>if (s != 0) goto label;</code>
bnezl <i>s</i> , <i>t</i> , <i>label</i>	Deprecated branch-likely variant; see section 8.5.4.
break <i>code</i>	Breakpoint instruction for debuggers. The value code has no hardware effect, but the breakpoint exception routine can retrieve it by reading the exception-causing instruction.
cache <i>k</i> , <i>addr</i>	Do something to a cache line, as described in section 4.9. Not implemented on very old MIPS CPUs, where cache management relies on CPU-dependent tricks.
cfcl <i>t</i> , <i>cs</i> cfc2 <i>t</i> , <i>cs</i>	Move data from coprocessor control register cs to general-purpose register t . Only useful for a coprocessor that uses the auxiliary control register set, as the floating-point unit (coprocessor 1) does. The cfc0 instruction is not part of the MIPS32 specification.
ctcl <i>t</i> , <i>cs</i> ctc2 <i>t</i> , <i>cs</i>	Move data from general-purpose register t to coprocessor control register cs .
clo <i>d</i> , <i>s</i>	Count leading (high-order) one bits in <i>s</i> considered as a 32-bit word.
clz <i>d</i> , <i>s</i>	Count leading (high-order) zero bits in <i>s</i> considered as a 32-bit word.
dabs <i>d</i> , <i>s</i> ⇒ dsra <i>at</i> , <i>s</i> , 31 xor <i>d</i> , <i>s</i> , <i>at</i> dsub <i>d</i> , <i>d</i> , <i>at</i>	64-bit version <code>d = s < 0: -s: s;</code>
dadd <i>d</i> , <i>s</i> , <i>t</i>	64-bit version; but this one traps on overflow, and is used rarely. <code>d = s + t;</code>
daddi <i>d</i> , <i>s</i> , <i>j</i>	64-bit add with overflow trap, rare <code>d = s + j;</code>
daddiu <i>d</i> , <i>s</i> , <i>j</i>	64-bit add immediate, more often written with a daddu mnemonic. <code>d = s + j;</code>

TABLE 8.2 *continued*

<i>Assembly/Machine Code</i>	<i>Description</i>
daddu d,s,t	64-bit $d = s + t;$
dclo d,s	Count leading ones in s starting at bit 63 of register.
dclz d,s	Count leading zeros in s starting at bit 63 of register.
ddiv \$zero,s,t $\Rightarrow$ ddiv s,t	Plain 64-bit hardware divide instruction, because we specified \$zero as the destination. $lo = (\text{long long})\ s / (\text{long long})\ t;$ $hi = (\text{long long})\ s \% (\text{long long})\ t;$
ddiv d,s,t $\Rightarrow$ teq t,\$zero,0x7 ddiv \$zero,s,t daddiu \$at,\$zero,-1 bne t,\$at,1f daddiu \$at,\$zero,1 dsll32 \$at,\$at,31 teq \$t1,\$at,0x6 1: mflo d	64-bit signed divide with divide-by-zero and overflow check. $lo = (\text{long long})\ s / (\text{long long})\ t;$ $hi = (\text{long long})\ s \% (\text{long long})\ t;$ if ($t == 0$) exception (BREAK, 7); if ($t == -1 \ \&\& \ s = \text{MAXNEG64BIT}$) /* result overflows */ exception (BREAK, 6); $d = lo;$
ddivu \$zero,s,t $\Rightarrow$ ddivu s,t	Plain unsigned 64-bit hardware divide instruction. $lo = (\text{unsigned long long})\ s / (\text{unsigned long long})\ t;$ $hi = (\text{unsigned long long})\ s \% (\text{unsigned long long})\ t;$
ddivu d,s,t $\Rightarrow$ teq t,\$zero,0x7 ddivu s,t mflo d	64-bit unsigned divide with divide-by-zero check. $lo = (\text{unsigned long long})\ s / (\text{unsigned long long})\ t;$ $hi = (\text{unsigned long long})\ s \% (\text{unsigned long long})\ t;$ if ($t == 0$) exception (BREAK, 7); $d = lo;$
deret	Return from EJTAG debug exception. Control passes to the instruction at the location found in the CP0 register DEPC , and the debug mode bit is cleared. The instruction sequentially after deret is not run (there is no delay-slot instruction). There's more about EJTAG in section 12.1.
dext d,s,shf,sz	Extract bitfield from 64-bit register. shf is the distance the field needs to be shifted so it starts at bit 0 in s , and sz is the number of bits in the field. $s = d(sz+shf) .. shf;$
dext d,s,shf,sz $\Rightarrow$ dextm d,s,shf,sz dext d,s,shf,sz $\Rightarrow$ dextu d,s,shf,sz	The assembler gives you a dextm or dextu machine code as required when shf or sz is more than 32 bits. In all normal circumstances, just code dext and let the assembler handle it.

TABLE 8.2 *continued*

<i>Assembly/Machine Code</i>	<i>Description</i>
dextm d, s, shf, sz	Machine instruction to encode dext when sz is 32 or more.
dextu d, s, shf, sz	Machine instruction to encode dext when shf is 32 or more.
di d	Disable interrupts. Clears the global interrupt enable bit in the status register (SR (IE) , see section 3.3.1), leaving the original value of SR in d . This operation is atomic; the alternative read/modify/write sequence can be interrupted in the middle with confusing results.
dins d, s, shf, sz	Insert bitfield into 64-bit register. The data to insert forms the low bits of s . shf is the distance the data needs to be shifted left, and sz is the width of the field in bits. $d = d63..(shf+sz) \mid s(sz)..0 \mid (shf > 0 ? d(shf-1)..0 : 0);$
dins d, s, shf, sz $\Rightarrow$ dinsm d, s, shf, sz dins d, s, shf, sz $\Rightarrow$ dinsu d, s, shf, sz	The assembler gives you a dinsm or dinsu machine code as required when shf or sz is more than 32 bits. In all normal circumstances, just code dins and let the assembler handle it.
dinsm d, s, shf, sz	Machine instruction to encode dins when sz is 32 or more.
dinsu d, s, shf, sz	Machine instruction to encode dins when shf is 32 or more.
div \$zero, s, t $\Rightarrow$ div s, t	Plain signed 32-bit hardware divide; the assembler doesn't insert any divide-by-zero or overflow checks when the destination register is \$zero . $lo = s / t;$ $hi = s \% t;$
div d, s, t $\Rightarrow$ teq t, \$zero, 0x7 div \$zero, s, t li \$at, -1 bne t, \$at, 1f lui \$at, 0x8000 teq s, \$at, 0x6 1f: mflo d	Signed 32-bit division with exceptions generated by divide-by-zero and overflow conditions: $\text{if } (t == 0) \quad \text{exception}(\text{BREAK}, 7); \text{ /* divide by zero */}$ $lo = s/t; hi = s\%t;$ $\text{if } (t == -1 \ \&\& \ s == \text{MAXNEG32BIT}) \quad \text{exception}(\text{BREAK}, 6); \text{ /* result overflows */}$ $d = lo;$
divu \$zero, s, t $\Rightarrow$ divu s, t	$\text{/* \$zero as destination means no checks */}$ $lo = s/t;$ $hi = s \% t;$

TABLE 8.2 *continued*

<i>Assembly/Machine Code</i>	<i>Description</i>
divu d,s,t ⇒ teq t,\$zero,0x7 divu s,t mflo d	Unsigned divide, but take an exception if you divide by zero: <pre>if (t == 0) exception(BREAK, 7); lo = (unsigned) s / (unsigned) t; hi = (unsigned) s % (unsigned) t; d = lo;</pre>
dla t,addr ⇒ # various ...	Load 64-bit address; see section 9.4.
dli t,const ⇒ # biggest case: lui t,const63..48 ori t,const47..32 dsll t,16 ori t,const31..16 dsll t,16 ori t,const15..0	Load 64-bit constant. Separate mnemonic from li required only for values between 0x8000.0000 and 0xFFFF.FFFF, where 32⇒64-bit transition rules require li to flood the high-order 32 bits with ones.
dmadd16 s,t	Found only on NEC's Vr41xx family CPUs. <pre>(long long)lo = (long long)lo + ((short)s * (short)t);</pre>
dmfc1 t,fs dmfc2 t,fs	Move 64 bits from coprocessor register cs to general-purpose register t . Only needed and implemented, of course, for coprocessors with 64-bit registers. dmfc1 is for floating-point unit registers; dmfc2 is exceedingly rare.
dmtc1 t,cs dmtc2 t,cs	Move 64 bits from general-purpose register t to coprocessor register cs . Comments as for dmfc1 above.
dmul d,s,t ⇒ dmultu s,t mflo d	64-bit signed multiply instruction; the product of s and t is computed as a 128-bit value, with no overflow possible. There's no machine-level single instruction that does a three-register 64-bit multiply, although there is such an instruction (mul) for 32-bit operands. <pre>hilo = s * t; /* with 128-bit precision */ d = lo;</pre>
dmulo d,s,t ⇒ dmult s,t mflo d dsra d,d,63 mfhi \$at tne d,\$at,0x6 mflo d	Signed multiply, take an exception on overflow. <pre>hilo = s * t; /* with 128-bit precision */ if ((lo ≥ 0 && hi != 0) (lo < 0 && hi != -1)) exception(BREAK, 6); d = lo;</pre>

TABLE 8.2 *continued*

<i>Assembly/Machine Code</i>	<i>Description</i>
dmulou d,s,t ⇒ dmultu s,t mfhi \$at mflo d tne \$at,\$zero,0x6	Unsigned multiply, take an exception on overflow. <code>hilo = (long long) s * (long long) t;</code> <code>if (hi != 0)</code> <code>exception(BREAK, 6);</code> <code>d = lo;</code>
dmult s,t	Real machine-level 64-bit signed integer multiply instruction, result in hilo . <code>hilo = (long long) s * (long long) t;</code>
dmultu s,t	Unsigned version of real machine-level 64-bit integer multiply. <code>hilo = (unsigned long long) s * (unsigned long long) t;</code>
dneg d,s ⇒ dsub d,\$zero,s	Unitary negate, which traps on overflow—you probably want dnegu , below. <code>(long long) d = -(long long) s; /* trap on overflow */</code>
dnegu d,s ⇒ dsubu d,\$zero,s	<code>(long long) d = -(long long) s;</code>
drem d,s,t ⇒ teq t,\$zero,0x7 ddiv \$zero,s,t daddiu \$at,\$zero,-1 bne t,\$at,1f daddiu \$at,\$zero,1 dsll \$at,\$at,63 teq s,\$at,0x6 1f: mfhi d	64-bit signed integer remainder, with overflow checks. <code>if (t == 0) exception(BREAK, 7);</code> <code>if (s == MAXNEG64BIT && t == -1)</code> <code>exception(BREAK, 6); /* overflow */</code> <code>d = (long long) s % (long long) t;</code>
dremu d,s,t ⇒ teq t,\$zero,0x7 ddivu \$zero,s,t mfhi d	64-bit unsigned integer remainder, with overflow check. <code>if (t == 0) exception(BREAK, 7);</code> <code>d = (unsigned long long)s % (unsigned long long)t;</code>
dret	Out-of-date return-from-exception instruction, used on now-obsolete R6000 CPU and some “MIPS II” followers.
drol d,s,t ⇒ dnegu \$at,t drotrv d,s,\$at	64-bit rotate left, where rotate amount is a variable. <code>d = (s <<t) ((unsigned long long)s >>(64-t));</code>

TABLE 8.2 *continued*

<i>Assembly/Machine Code</i>	<i>Description</i>
drol d,s,j ⇒ drotr d,s,64-j	64-bit rotate left by a constant amount. <code>d = (s <<j) ((unsigned long long)s >>(64-j));</code>
dror d,s,t ⇒ drotrv d,s,t	64-bit rotate right, where rotate amount is a variable. <code>d = ((unsigned long long)s >>t) (s <<(64-t));</code>
dror d,s,j ⇒ drotr d,s,j	64-bit rotate right by a constant amount less than 32 positions. <code>d = ((unsigned long long)s >>j) (s <<(64-j));</code>
dror d,s,j ⇒ drotr32 d,s,j	64-bit rotate right by a constant amount of 32 positions or more. <code>d = ((unsigned long long)s >>j) (s <<(64-j));</code>
dsbh d,t	Swap each pair of bytes in the register (there are four pairs in a 64-bit register).
dshd d,t	Swap each pair of halfwords (16-bit chunks) in the register (there are two pairs in a 64-bit register).
dsll d,s,t ⇒ dsllv d,s,t	64-bit shift left by variable shift amount. <code>d = (long long)s << (t % 64);</code>
dsllv d,s,t	You can write the name of the machine-code shift-left-by-amount-in-register instruction like this, but it's better to just write dsll .
dsll d,s,shf	64-bit shift-left by a constant less than 32. <code>d = (long long) s <<shf;</code>
dsll d,s,shf ⇒ dsll32 d,s,shf-32	64-bit shift-left by a constant of 32 or more. <code>d = (long long) s <<shf /* 32 ≤ shf < 63 */</code>
dsra d,s,t ⇒ drsav d,s,t	64-bit shift-right: C semantics for shift-right signed, or an <i>arithmetic</i> shift—as bits shift down, the top of the register is filled with copies of bit 63, which gives you a correct implementation of signed divide by a power of two. <code>d = (signed long long) s >>(t%64);</code>
dsra d,s,shf	64-bit shift-right arithmetic by a constant (for “arithmetic” see dsra d,s,t above). Where the constant is less than 32, this is a machine instruction with the same name. <code>d = (signed long long) s >>(t%64);</code>
dsra d,s,shf ⇒ dsra32 d,s,shf-32	As above, for $32 \leq \text{shf} < 63$. You probably don't want to ever write dsra32 .

TABLE 8.2 *continued*

<i>Assembly/Machine Code</i>	<i>Description</i>
dsrl d,s,t ⇒ dsrlv d,s,t	Shift-right <i>logical</i> —that means zeros are drawn in at top, consistent with C semantics for unsigned integers. This is the variable-shift-amount version. $d = (\text{long long unsigned})\ s \gg (t\%64)$
dsrl d,s,shf	Shift-right by a constant, a constant less than 32. $d = (\text{long long unsigned})\ s \gg \text{shf}\%32;$
dsrl d,s,shf ⇒ dsrl32 d,s,shf-32	As above, when shf is 32 or more.
dsub d,s,t	64-bit subtract, which takes exception on overflow: rare. $d = s - t;$
dsubu d,s,t	$d = s - t; /*\ 64\text{-bit}\ */$
ehb	Execution hazard barrier—an instruction when you need to be sure that any coprocessor zero side effects of previous instructions have completed before any subsequent instructions get to do anything. See section 8.5.10.
ei d	Enable interrupts—well, at least unconditionally set the global interrupt enable bit in the status register (SR(IE) , see section 3.3.1). The old value of SR is left in d . This operation is atomic. It’s an analog of the (more useful) di instruction above.
eret	Return from exception: a privileged-mode instruction. Clears the SR(EXL) bit and branches to the location saved in EPC . See section 5.5.
ext d,s,shf,sz	Extract bitfield from 32-bit register. shf is the distance the field needs to be shifted so it starts at bit 0 in s , and sz is the number of bits in the field. $\text{mask} = (2^{**sz} - 1) \ll \text{shf};$ $d = (s \& \text{mask}) \gg \text{shf};$
ins d,s,shf,sz	Insert bitfield into 32-bit register. The data to insert forms the low bits of s . shf is the distance the data needs to be shifted left, and sz is the width of the field in bits. $\text{mask} = (2^{**sz} - 1) \ll \text{shf};$ $d = (d \& \sim\text{mask}) ((s \ll \text{shf}) \& \text{mask});$
j label	The basic “go-to” instruction. Note that it’s limited to reaching instructions within a 2^{28} -byte “page.” <code>goto label;</code>
j r ⇒ jr r	Jump to the instruction pointed to by register r . This is the only way of transferring control to an arbitrary address, since all the address-in-instruction formats span less than 32 bits.

TABLE 8.2 *continued*

<i>Assembly/Machine Code</i>	<i>Description</i>
jal label	Subroutine call, with return address in \$ra (\$31). Note that the return address is the next instruction but one—as is usual with MIPS branches, the immediately following instruction position is the branch delay slot, and the instruction there is always executed before you reach the subroutine.
jal d,addr ⇒ la \$at,addr jalr d,\$at	Like function call, but leaves the return address in the register d instead of the usual \$31 . Synthesized with jalr . It's cheating to use the instruction la in the machine code expansion, as la is itself a macro—but it means we can avoid explaining addressing modes here (see section 9.4 instead).
jalr d,s	Variant of jal d,addr above, when the address syntax is just another register s . You can write jal or jalr .
jal s ⇒ jalr \$ra,s	If you specify just one register, that's the address to call, and the return address is put in the usual \$ra .
la d,addr ⇒ # many options	Load address—always a synthesized instruction, which may produce very different code sequences according to how addr is written. More about this in section 9.4.
lb d,addr	8-bit load, sign-extend to fill register. For this and all load/store instructions, you can write addr in many ways—see section 9.4—but the load/store instruction can only compute an address with a register and signed 16-bit offset. <code>d = *((signed char *) addr);</code>
lbu d,addr	8-bit load, zero-extend to fill register. <code>d = *((unsigned char *) addr);</code>
ld d,addr	64-bit load: exception if address is not eight-byte-aligned. <code>d = *((long long *) addr);</code>
ldc1 fd,addr	64-bit load of coprocessor 1 (floating-point) register. More often written as l.d , see section 8.3.
ldc2 fd,addr	64-bit load of coprocessor 2 register, if coprocessor 2 is used and is 64 bits wide.
ldl d,addr ldr d,addr	Load double “left/right”—used as a pair, these instructions implement a 64-bit unaligned load uld ; see below and section 2.5.2.
ldxc1 fd,s(t)	64-bit load of coprocessor 1 (floating-point) register, with two-register “indexed” addressing. More often written as l.d , see section 8.3. <code>fd = *((double *) (t+b));</code>

TABLE 8.2 *continued*

<i>Assembly/Machine Code</i>	<i>Description</i>
lh d, addr	16-bit load, sign-extend to fill register. <code>t = *((signed short *)addr);</code>
lhu d, addr	16-bit load, zero-extend to fill register. <code>t = *((unsigned short *)addr);</code>
li d, j ⇒ ori d, \$zero, j	Load register with constant value (an “immediate”). This expansion is for $0 \leq j \leq 65535$.
li d, j ⇒ addiu d, \$zero, j	This one is for $-32768 \leq j < 0$.
li d, j ⇒ lui d, hi16(j) ori d, d, lo16(j)	This one is for any other value of j that is representable as a 32-bit integer.
ll t, addr lld t, addr	Load-linked. Load 32 bits/64 bits, respectively, with link side effects; used together with sc or scd to implement a lockless semaphore (see section 8.5.2).
lui t, u	Load upper immediate (constant u is sign-extended into 64-bit registers). <code>t = u << 16;</code>
lw t, addr	32-bit load, sign-extended for 64-bit CPUs. <code>t = *((int *) (addr));</code>
lwc1 fd, addr	Load FP single to FP register file—more often written l.s . See section 8.3.
lwc2 cd, addr	32-bit load to coprocessor2 register, if implemented. Rare.
lwl t, addr lwr t, addr	Load word left/right. See ulw below and section 2.5.2 for how these instructions work together to perform an unaligned 32-bit load operation.
lwu t, addr	32-bit zero-extending load, only found on 64-bit CPUs. <code>t = (unsigned long long)*((unsigned int *)addr);</code>
lwxcl fd, t(b)	Load 32-bit FP value using indexed (register+register) address. More often written as l.s , see section 8.3. <code>fd = *((float *) (t+b));</code>
mad s, t	32-bit integer multiply-accumulate, standard in MIPS32. The two registers are multiplied with full precision and accumulated: <code>hilo = hilo + ((long long) s * (long long) t);</code>
madu s, t	Same but unsigned: <code>hilo = hilo+((unsigned long long)s * (unsigned long long)t);</code>

TABLE 8.2 *continued*

<i>Assembly/Machine Code</i>	<i>Description</i>
madd d, s, t maddu d, s, t	Integer multiply-accumulate with integral write of result to a general register (signed and unsigned versions). The general-register write is only implemented by Toshiba 3900 series cores, with this unique name. But note that in the case where d is actually zero , this is compatible with the MIPS32 standard mad/madu . $\text{hilo} += (\text{long long})\ s * (\text{long long})\ t;$ $d = \text{lo};$
madd16 s, t	NEC Vr4100-specific integer multiply-accumulate; handles only 16-bit operands: $\text{lo} = \text{lo} + ((\text{short})s * (\text{short})t);$
mfc0 t, cs mfc1 t, fs mfc2 t, cs	Move 32 bits of data from coprocessor register cs into general-purpose register t —if cs is 64 bits wide, that will be the low-numbered bits. mfc0 is vital for access to the CPU control registers, mfc1 for putting floating-point unit data back into integer registers. mfc2 is only useful if coprocessor 2 is implemented, which is rare.
mfhc1 t, cs mfhc2 t, fs	Move the higher 32 bits of the 64-bit coprocessor register cs or fs into general-purpose register t . Provided only when a MIPS32 integer unit is partnered by a 64-bit, otherwise MIPS64-compatible coprocessor. For example, MIPS Technologies' 24Kf core has a 64-bit floating-point unit.
mfhi d mflo d	Move integer multiply unit results to general-purpose register d . lo contains the result of a division, the least significant 32 bits of the result of a mul , or the least significant 64 bits of the result of a dmul . hi contains the remainder of a division or the most significant bits of a multiplication. These instructions are always interlocked; even on the earliest CPUs, the hardware waits for any incomplete multiply/divide to finish.
move d, s => or d, s, \$zero	$d = s;$
movf d, s, \$fccN	A variety of conditional-move instructions—more about them in section 8.5.3. $\text{if } (!\text{fcc}(N))\ d = s;$
movn d, s, t	$\text{if } (t)\ d = s;$
movt d, s, \$fccN	$\text{if } (\text{fcc}(N))\ d = s;$
movt.d fd, fs, N movt.s fd, fs, N	Double- and single-precision versions. $\text{if } (\text{fcc}(N))\ fd = fs;$
movz d, s, t	$\text{if } (!t)\ d = s;$

TABLE 8.2 *continued*

<i>Assembly/Machine Code</i>	<i>Description</i>
msub s,t msubu s,t	The negative version of integer multiply/accumulate, in signed and unsigned versions. <code>hilo = hilo - ((long long) s * (long long) t);</code>
mtc0 t,cd mtc1 t,fd mtc2 t,cd	Move 32 bits from general-purpose register t to coprocessor register cd . Note that this instruction doesn't obey the usual convention of writing the destination register first. mtc0 is for the CPU control registers, mtc1 is for putting integer data into floating-point registers (although they're more often loaded directly from memory), and mtc2 is implemented only if the CPU uses coprocessor 2 instructions (very rare). If the coprocessor register is 64 bits wide, the data is loaded into the low bits, but the state of the high 32 bits is not defined.
mtbc1 t,cd mtbc2 t,fd	Move 32 bits from general-purpose register t to the higher bits of 64-bit coprocessor register cd , while leaving the low bits unchanged. Provided only when a MIPS32 integer unit is partnered by a 64-bit, otherwise MIPS64-compatible coprocessor—for example, MIPS Technologies' 32-bit 24Kf core has a 64-bit floating-point unit.
mthi s mtlo s	Move contents of general-purpose register s into the multiply-unit result registers hi and lo , respectively. This may not seem useful, but they are required to restore the CPU state when returning from an exception.
mul d,s,t	Genuine three-register 32-bit integer multiply, defined in MIPS32 and available on some earlier CPUs. The full-precision result is still delivered to the internal hilo register. There is no unsigned version. <code>hilo = (long long) s * (long long) t;</code> <code>d = lo;</code>
mul d,s,t ⇒ mult s,t mflo d	Three-register multiply can be synthesized when the assembler is generating a pre-MIPS32 instruction set.
mulo d,s,t ⇒ mult s,t mflo d sra d,d,31 mfhi \$at tne d,\$at,0x6 mflo d	32-bit signed multiply with overflow check. Overflow is detected by the case where hi does not simply contain the sign-extension of lo . <code>hilo = (signed)s * (signed)t;</code> <code>if ((lo ≥ 0 && hi != 0) (s < 0) && hi != -1)</code> <code>exception(BREAK, 6);</code>
mulou d,s,t ⇒ multu s,t mfhi \$at mflo d tne \$at,\$zero,0x6	32-bit unsigned multiply with overflow check: <code>hilo = (unsigned)s * (unsigned)t;</code> <code>if (hi != 0)</code> <code>exception(BREAK, 6);</code>

TABLE 8.2 *continued*

<i>Assembly/Machine Code</i>	<i>Description</i>
mult <i>s,t</i>	<code>hilo = (signed)s * (signed)t;</code>
multu <i>s,t</i>	<code>hilo = (unsigned)s * (unsigned)t;</code>
neg <i>d,s</i> $\Rightarrow$ sub <i>d,\$zero,s</i>	This version traps on overflow and is very rarely used. <code>d = -s;</code>
negu <i>d,s</i> $\Rightarrow$ subu <i>d,\$zero,s</i>	No overflow: C always generates this. <code>d = -s;</code>
nop $\Rightarrow$ sll <i>\$zero,\$zero,\$zero</i>	No-op, instruction code == 0.
nor <i>d,s,t</i>	Like all bitwise operations, there's no need for a separate op-code for 64-bit CPUs. <code>d = ~(s t);</code>
not <i>d,s</i> $\Rightarrow$ nor <i>d,s,\$zero</i>	<code>d = ~s;</code>
nudge <i>addr</i> nudgex <i>s(t)</i>	Shortcuts for prefetch instructions pref nudge and prefx nudge , see below and section 8.5.8
or <i>d,s,t</i>	<code>d = s t;</code>
ori <i>t,r,j</i>	OR with a constant. Machine instruction, but more often written as or d,s,j : <code>d = s (unsigned) j;</code>
pref <i>hint,addr</i> prefx <i>hint,t(b)</i>	Prefetch instruction for memory reference optimization. A program that knows in advance it may need data can arrange for it to be brought into the cache early, with no real side effects. Implementations are always entitled to treat pref as a no-op. hint defines which sort of prefetch this is; see section 8.5.8. prefx is only available with floating-point, where it matches the register+register address mode available for floating-point load/store instructions.
r2u <i>s</i>	LSI ATMizer-II only; converts to strange floating-point format. Result appears in lo .
radd <i>s,t</i>	LSI ATMizer-II only; strange floating-point add. Result appears in lo .
rdhwr <i>d,\$cs</i>	Read hardware register: allows unprivileged user-mode software to read one of a set of CPU registers. See section 8.5.12.
rdpgpr <i>d,s</i>	Read register from previous shadow set—see section 5.8.6.

TABLE 8.2 *continued*

<i>Assembly/Machine Code</i>	<i>Description</i>
rem d,s,t ⇒ teq t,\$zero,0x7 div \$zero,s,t li \$at,-1 bne t,\$at,1f lui \$at,0x8000 teq s,\$at,0x6 1f: mfhi d	Signed integer remainder, with divide-by-zero and overflow checks: if (t==0) exception(BREAK, 7); if (t==−1 && s==MAXNEG32BIT) exception(BREAK, 6); d = s % t;
remu d,s,t ⇒ teq t,\$zero,0x7 divu \$zero,s,t mfhi d	Unsigned integer remainder, with divide-by-zero check: if (t==0) exception(BREAK, 7); d = (unsigned)s % (unsigned)t;
rfe	Pre-MIPS32 (in fact, MIPS I-only) instruction to restore CPU state when returning from exception. Now obsolete and not described further here.
rmul s,t	LSI ATMizer-II only; strange floating-point multiply. Result appears in lo .
rol d,s,shf ⇒ rotr d,s,32-shf	Rotate left by constant. rotr was new with MIPS32R2; for older ISAs, this will be synthesized to something longer. d = (s<<shf) ((unsigned)s>>(32-shf));
rol d,s,t ⇒ negu \$at,t rotrv d,s,\$at	Rotate left. Instruction sets prior to MIPS32R2 lacked the rotate-right machine instruction; for older instruction sets, it will be synthesized to a longer sequence. d = (s<<t) ((unsigned)s>>(32-t));
ror d,s,shf ⇒ rotr d,s,shf	Rotate right by constant. Will be synthesized for instruction sets earlier than MIPS32R2. d = ((unsigned)s>>shf) (s<<(32-shf));
ror d,s,t ⇒ rotrv d,s,t	Rotate right by variable amount. Will be synthesized for instruction sets earlier than MIPS32R2. d = ((unsigned)s>>t) (s<<(32-t));
rotr d,s,shf	Rotate-right by immediate: machine code, only available in MIPS32R2.
rotrv d,s,t	Rotate-right by variable amount: machine code, only available in MIPS32R2.
rsub s,t	LSI ATMizer-II only; strange floating-point multiply. Result appears in lo .
sb t,addr	*((char *)addr) = t;
sc t,addr	Store word/double conditional; explained in section 8.5.2.
scd t,addr	

TABLE 8.2 *continued*

<i>Assembly/Machine Code</i>	<i>Description</i>
sd t, addr	Will take an exception if addr is not eight-byte aligned. <code>*((long long *)addr) = t;</code>
sdbbp c	Debug breakpoint instruction—different from a break because it drops straight into debug mode, not just exception mode. The optional c is just encoded into the instruction, where a debugger could read it. Note that there have been two different encodings used for this instruction—the obsolete one is now associated only with the Toshiba R3900 core and its descendants. See description of EJTAG debug unit in section 12.1.
sdcl ft, addr	Store floating-point double register to memory; more often called s.d .
sdcl cs, addr	Store contents of 64-bit coprocessor 2 register to memory.
sdll t, addr	Store double left/right; see section 2.5.2 for an explanation.
sdr t, addr	
sdxcl fs, t(b)	Indexed FP store double (both t and b are registers), usually written s.d . <code>*((double *) (t+b)) = fs;</code>
seb d, s	In-register sign-extend, byte to register: <code>d = (long long) (signed char) (s & 0xff);</code>
seh d, s	In-register sign-extend, halfword to register: <code>d = (long long) (signed short) (s & 0xffff);</code>
seq d, s, t ⇒ xor d, s, t sltiu d, d, 1	First of set of “set if” assembly mnemonics, built by analogy to the real machine instruction slt . <code>d = (s == t) ? 1 : 0;</code>
sge d, s, t ⇒ slt d, s, t xori d, d, 1	<code>d = ((signed)s ≥ (signed)t) ? 1 : 0;</code>
sgeu d, s, t ⇒ sltu d, s, t xori d, d, 1	<code>d = ((unsigned)s ≥ (unsigned)t) ? 1 : 0;</code>
sgt d, s, t ⇒ slt d, t, s	<code>d = ((signed)s > (signed)t) ? 1 : 0;</code>
sgtu d, s, t ⇒ sltu d, t, s	<code>d = ((unsigned)s > (unsigned)t) ? 1 : 0;</code>
sh t, addr	Store halfword: <code>*((short *)addr) = t;</code>

TABLE 8.2 *continued*

<i>Assembly/Machine Code</i>	<i>Description</i>
sle d,s,t ⇒ slt d,t,s xori d,d,1	$d = ((\text{signed})s \leq (\text{signed})t) ? 1 : 0;$
sleu d,s,t ⇒ sltu d,t,s xori d,d,1	$d = ((\text{unsigned})s \leq (\text{unsigned})t) ? 1 : 0;$
sll d,s,shf	$d = s \ll \text{shf}; \text{ /* } 0 \leq \text{shf} < 32 \text{ */}$
sll d,t,s ⇒ sllv d,t,s sllv d,t,s	$d = t \ll (s \% 32);$
slt d,s,t	$d = ((\text{signed}) s < (\text{signed}) t) ? 1 : 0;$
slt d,s,j ⇒ slti d,s,j slti d,s,j	$\text{/* } j \text{ constant */}$ $d = ((\text{signed}) s < (\text{signed}) j) ? 1 : 0;$
sltiu d,s,j	$\text{/* } j \text{ constant */}$ $d = ((\text{unsigned}) s < (\text{unsigned}) j) ? 1 : 0;$
sltu d,s,t	$d = ((\text{unsigned}) s < (\text{unsigned}) t) ? 1 : 0;$
sne d,s,t ⇒ xor d,s,t	$d = (s \neq t) ? 1 : 0;$
sltu d,\$zero,d	
sra d,s,shf	32-bit shift-right by a constant. C semantics for shift-right signed, an <i>arithmetic</i> shift—as bits shift down the top of the register is filled with copies of bit 31, which gives you a correct implementation of signed divide by a power of 2. $d = (\text{signed}) s \gg \text{shf};$
sra d,s,t ⇒ srav d,s,t	32-bit shift-right arithmetic, by a variable shift amount: $d = (\text{signed}) s \gg (t \% 32)$
srl d,s,shf	32-bit shift-right logical: like a C shift of an unsigned quantity, where zeros are shifted in from the top: $d = (\text{unsigned}) s \gg \text{shf};$
srl d,s,t ⇒ slrv d,s,t	32-bit shift-right logical, shift amount in register. $d = (\text{unsigned}) s \gg (t \% 32);$

TABLE 8.2 *continued*

<i>Assembly/Machine Code</i>	<i>Description</i>
ssnop $\Rightarrow$ sll \$zero, \$zero, 1	“Superscalar” no-op; a no-op, but one that no CPU should issue in the same clock cycle as other instructions. Used for mystical timing purposes.
standby	Enter one of the power-down modes, for NEC Vr4100 family CPU only; wait —see below—is a more widely used instruction.
sub d, s, t	Trap on overflow, little used. $d = s - t;$
subu d, s, j $\Rightarrow$ addiu d, s, -j	$d = s - j;$
subu d, s, t	$d = s - t;$
suspend	Enter a Vr4100 CPU’s power-down modes.
sw t, addr	Store word. $*((\text{int } *)\text{addr}) = t;$
swc1 ft, addr	Floating-point store single; more often written s.s .
swc2 ft, addr	Store 32-bit data from coprocessor 2 register, rare.
swl t, addr swr t, addr	Store word left/right; see section 2.5.2.
swxcl fs, t(b)	Store floating-point single using indexed (two-register) addressing; usually written with s.s . $*((\text{float } *) (t + b)) = fs;$
sync	Load/store barrier, mainly for multiprocessors; see section 8.5.9.
synci addr	Synchronize I-cache with D-cache: Run instruction for each cache-line-sized block after writing instructions but before executing them. See section 8.5.11.
syscall B	Cause a “system call” exception. $\text{exception}(\text{SYSCALL}, B);$
teq s, t	Conditional trap instruction: generate a TRAP exception if the appropriate condition is satisfied; this one is. $\text{if } (s == t) \text{ exception}(\text{TRAP});$
teq s, j $\Rightarrow$ teqi s, j	$\text{if } (s == j) \text{ exception}(\text{TRAP});$
tge s, t	$\text{if } ((\text{signed}) s \geq (\text{signed}) t) \text{ exception}(\text{TRAP});$
tge s, j $\Rightarrow$ tgei s, j	$\text{if } ((\text{signed}) s \geq (\text{signed}) j) \text{ exception}(\text{TRAP});$

TABLE 8.2 *continued*

<i>Assembly/Machine Code</i>	<i>Description</i>
tgeu s,t	if ((unsigned) s ≥ (unsigned) t) exception(TRAP);
tgeu s,j ⇒ tgeiu s,j	if ((unsigned) s ≥ (unsigned) j) exception(TRAP);
tlbp	TLB maintenance; see Chapter 6. If the virtual page number currently in EntryLo matches a TLB entry, sets Index to that entry. Otherwise sets Index to the illegal value 0x8000.0000 (top bit set).
tlbr	TLB maintenance; see Chapter 6. Copies information from the TLB entry selected by Index into the registers EntryLo , EntryHi1 , EntryHi0 , and PageMask .
tlbwi tlbwr	TLB maintenance; see Chapter 6. Writes the TLB entry selected by Index (instruction tlbwi) or Random (instruction tlbwr), respectively, using data from EntryLo , EntryHi1 , EntryHi0 , and PageMask .
tlt s,t	More conditional traps: if ((signed) s < (signed) t) exception(TRAP);
tlt s,j ⇒ tlti s,j	if ((signed) s < (signed) j) exception(TRAP);
tltu s,t	if ((unsigned) s < (unsigned) t) exception(TRAP);
tltu s,j ⇒ tltiu s,j	if ((unsigned) s < (unsigned) j) exception(TRAP);
tne s,t	if (t != s) exception(TRAP);
tne s,j ⇒ tnei s,j	if (t != j) exception(TRAP);
u2r s	LSI ATMizer-II only; converts unsigned to strange floating point. Result appears in lo .
udi0 d,r,s,uc through udi15 d,r,s,uc	Builds an instruction in that corner of the instruction encoding reserved for user-defined instructions. Such instructions can use three general-purpose instructions, and have a subsidiary 5-bit op-code uc , which is available to the user logic.
uld d,addr ⇒ ldl d,addr ldr d,addr+7	Unaligned load double, synthesized from load-left and load-right as detailed in section 2.5.2 (shown for big-endian only).

TABLE 8.2 *continued*

<i>Assembly/Machine Code</i>	<i>Description</i>
ulh <i>d</i>,<i>addr</i> ⇒ lb <i>d</i>,<i>addr</i> lbu <i>at</i>,<i>addr</i>+1 sll <i>d</i>,<i>d</i>, 8 or <i>d</i>,<i>d</i>,<i>at</i>	Unaligned load halfword and sign-extend. This is the big-endian expansion (the little-endian version is left as an exercise for the reader). The expansion may be more complex, depending on addressing mode.
ulhu <i>d</i>,<i>addr</i> ⇒ lbu <i>d</i>,<i>addr</i> lbu <i>at</i>,<i>addr</i>+1 sll <i>d</i>,<i>d</i>, 8 or <i>d</i>,<i>d</i>,<i>at</i>	Unaligned load halfword and zero-extend.
ulw <i>d</i>,<i>addr</i> ⇒ lwl <i>d</i>,<i>addr</i> lwr <i>d</i>,<i>addr</i>+3	Load word unaligned; sign-extend if 64 bits (shown for big-endian only). See section 2.5.2.
usd <i>d</i>,<i>addr</i> ⇒ sdl <i>d</i>,<i>addr</i> sdr <i>d</i>,<i>addr</i>+7	Unaligned store double.
ush <i>addr</i> ⇒ sb <i>d</i>,<i>addr</i>+1 srl <i>d</i>,<i>d</i>, 8 sb <i>d</i>,<i>addr</i>	Unaligned store half.
usw <i>s</i>,<i>addr</i> ⇒ swl <i>s</i>,<i>addr</i> swr <i>s</i>,<i>addr</i>+3	Store word unaligned; see section 2.5.2.
wait	MIPS32 instruction to enter some kind of power-down state. Usually implemented by suspending execution until an interrupt is detected. Software should not assume that the suspension will always happen or that waking up from wait necessarily indicates an unmasked interrupt— wait should be called from an idle loop.
wrpgpr <i>cd</i> , <i>t</i>	Write to a register in the previous shadow register set; see section 5.8.6 for details.
wsbh	32-bit byte-swap within each of the two halfwords. This is a 32-bit instruction; on a 64-bit CPU the top half of the register is left filled with the sign extension of bit 31. wsbh works well together with bit-rotates to perform many forms of byte reorganization in a small number of instructions.
xor <i>d</i> , <i>s</i> , <i>t</i>	$d = s \oplus t;$
xor <i>d</i>,<i>s</i>,<i>j</i> ⇒ xori <i>d</i>,<i>s</i>,<i>j</i>	$d = s \oplus j;$

8.3 Floating-Point Instructions

There's a reasonable set of MIPS floating-point instructions (see Tables 8.3 and 8.4), but they quickly develop their own complications. Note the following points:

- Every FP instruction comes in a single-precision version and a double-precision version, distinguished by **.s** or **.d** in the mnemonic.

If you have the paired-single extension, as described in section 7.10, there will even be **.ps** versions of instructions, which do exactly the same thing as the single-precision version, but twice. Paired-single instructions that aren't just **.ps** versions of instructions in the big table are listed there, but will just refer you to the descriptions in section 7.10.

To save space and avoid making your eyes water, Table 8.4 only lists single-precision versions, so long as exactly the same description serves for both versions.

- The basic FP encodings and instruction results conform to the IEEE 754 standard. Where the hardware can't produce the value required by IEEE 754, the default behavior is to take an exception so a software emulator can fill in the gaps to produce complete compliance to the standard.
- FP computational and type conversion instructions can cause exceptions. This is true both in the IEEE sense, where they detect conditions that a programmer may be interested in, and in a low-level architecture sense: MIPS FP hardware, if faced with a combination of operands and an operation it can't do correctly, will take an FP "unimplemented" exception with the aim of getting a software emulator to carry out the FP operation for it.

Data movement instructions (loads, stores, and moves between registers) don't ever cause exceptions. The **neg.s**, **neg.d**, **abs.s**, or **abs.d** instructions just flip the sign bit without inspecting the contents: The only condition that leads to a trap is that a "signaling NaN" operand to these instructions produces an IEEE "invalid" exception.

TABLE 8.3 Floating-Point Register and Identifier Conventions

<i>Word</i>	<i>Used for</i>
fs, ft	Floating-point register operands.
fd	Floating-point register that receives the result.
fdhi, fdlo	Pair of adjacent FP registers in a 32-bit processor, used together to store an FP double. Use of the high-order (odd-numbered) register is implicit in normal arithmetic instructions.
\$fccN	One of the floating-point condition bits, found in the FCSR register and that is tested by instructions like bc1t . There's been evolution here; the MIPS I–III ISAs have only 1 condition bit, but modern FPUs have 8. An instruction that omits to specify which condition bit to use will quietly use the original “zeroth” one.
fcc(N)	The same thing, as seen in C code.
upper, lower	The higher and lower values in an FP register holding a paired-single value, as discussed in section 7.10.

TABLE 8.4 Floating-Point Instruction Descriptions in Mnemonic Order

<i>Assembly code</i>	<i>Function</i>
abs.s fd, fs	$fd = (fs < 0) ? -fs : fs$
add.s fd, fs, ft	$fd = fs + ft;$
addr.ps fd, fs, ft	/* MIPS 3D "reduction add", see Section 7.10 */ $fd.upper = fs.upper + fs.lower;$ $fd.lower = ft.upper + ft.lower;$
alnv.ps fd, fs, ft, rs	Pack single-precision values into a paired-single, in either of the two possible ways as directed by the value of rs .
bc1f \$fccN, label bc1fl \$fccN, label bc1t \$fccN, label bc1tl \$fccN, label	Several branch-on-FP-condition instructions, all found in Table 8.2.
bclany2f \$fccN, label bclany2t \$fccN, label bclany4f \$fccN, label bclany4t \$fccN, label	MIPS-3D instructions that branch on the “OR” of two or four conditions. See section 7.10.4.

TABLE 8.4 *continued*

<i>Assembly code</i>	<i>Function</i>
c.eq.s \$fccN, fs, ft c.f.s \$fccN, fs, ft c.le.s \$fccN, fs, ft c.lt.s \$fccN, fs, ft c.nge.s \$fccN, fs, ft c.ngl.s \$fccN, fs, ft c.ngt.s \$fccN, fs, ft c.ole.s \$fccN, fs, ft c.olt.s \$fccN, fs, ft c.seq.s \$fccN, fs, ft c.sf.s \$fccN, fs, ft c.ueq.s \$fccN, fs, ft c.ule.s \$fccN, fs, ft c.ult.s \$fccN, fs, ft c.un.s \$fccN, fs, ft	FP compare instructions, which compare fs and ft and store a result in FP condition bit \$fccN . They are described at length in section 7.9.7.
cabs.xx.s \$fccN, fs, ft	MIPS-3D extension instructions to compare the absolute value of two FP values and store the result. The possible tests expressed by “ xx ” are the same as for the c.xx.s instructions listed above.
ceil.l.d fd, fs ceil.l.s fd, fs	Convert FP to equal or next-higher signed 64-bit integer value.
ceil.w.d fd, fs ceil.w.s fd, fs	Convert FP to equal or next-higher signed 32-bit integer value.
cfcl rt, fs ctcl rs, fd	Copy data between a floating-point control register and a general-purpose register (“f” is from FP, “t” is to FP). Used for the FCSR register and so on described in section 7.7.
cvt.d.l fd, fs cvt.d.s fd, fs cvt.d.w fd, fs	Floating-point type conversions, where the types d , l , s , and w (double, long, float, and int, respectively) are the destination and source type in that order.
cvt.l.d fd, fs cvt.l.s fd, fs cvt.s.d fd, fs cvt.s.l fd, fs	Where the conversion is losing precision, the current rounding mode from FCSR (RM) is used to determine how the approximation is done. For integer conversions where the desired approximation is specific to the algorithm, you’re better off writing instructions like floor.w.s and so on.
cvt.s.w fd, fs cvt.w.d fd, fs cvt.w.s fd, fs	

TABLE 8.4 *continued*

<i>Assembly code</i>	<i>Function</i>
cvt.ps.s <i>fd, fs, ft</i>	Packs two single-precision values into a paired-single, see section 7.10.
cvt.ps.pw <i>fd, fs, ft</i>	MIPS-3D instruction that converts two halves, each representing a 32-bit integer, to paired-single; see section 7.10.4.
cvt.pw.ps <i>fd, fs, ft</i>	MIPS-3D instruction that converts both halves of a paired-single to integers at once; see section 7.10.4.
cvt.s.pl <i>fd, fs, ft</i> cvt.s.pu <i>fd, fs, ft</i>	Unpacks half of a paired-single into a conventional single-precision value, see section 7.10.
div.s <i>fd, fs, ft</i>	$fd = fs/ft;$
dmfc1 <i>rd, fs</i>	Move 64-bit value from floating point (coprocessor 1) to integer register with no conversion.
dmtc1 <i>rs, fd</i>	Move 64-bit value from integer to floating-point (coprocessor 1) register with no conversion or validity check.
floor.l.d <i>fd, fs</i> floor.l.s <i>fd, fs</i>	Convert FP to equal or next-lower 64-bit integer value.
floor.w.d <i>fd, fs</i> floor.w.s <i>fd, fs</i>	Convert FP to equal or next-lower 32-bit integer value.
l.d <i>fd, addr</i> $\Rightarrow$ ldc1 <i>fd, addr</i>	Load FP double, must be eight-byte aligned. $fd = *((double *) (o+b));$
l.s <i>fd, addr</i> $\Rightarrow$ lwc1 <i>fd, addr</i>	Load FP single, must be four-byte aligned. $fd = *((float *) (o+b));$
ldc1 <i>fd, disp(b)</i>	Deprecated equivalent of l.d .
l.d <i>fd, t(b)</i> $\Rightarrow$ ldxc1 <i>fd, t(b)</i>	Indexed load to floating-point register. Best to write with the l.d form. Note that the role of the two registers is not quite symmetrical— b is expected to hold an address and t an offset, and it's an offense for (b+t) to end up in a different section of the overall MIPS address map than b (defined by the top 2 bits of the 64-bit address). $fd = *((double *) (b+t));$
li.s <i>fd, const</i> li.d <i>fd, const</i>	Load floating-point constant, commonly synthesized by placing the constant in a memory location and loading it.
luxc1 <i>fd, i(b)</i>	Double-indexed load-double, exactly like ldxc1 except that if the address turns out to be misaligned, no exception is taken and the load happens from the address obtained by zeroing the bottom 3 bits of the effective address. Used together with alnv to handle misaligned pairs of single-precision values—see the description of alnv in section 7.10.

TABLE 8.4 *continued*

<i>Assembly code</i>	<i>Function</i>
lwc1 fd, disp(b)	Deprecated equivalent of l.s .
lwxcl fd, i(b)	Explicit double-indexed load instruction; usually better to use l.s with the appropriate address mode. See note on ldxc1 , above.
madd.s fd, fr, fs, ft	$fd = fr + fs * ft;$
mfcl rs, fd	Move 32-bit value from floating point (coprocessor 1) to integer register with no conversion.
mfhcl rs, fd	Move 32-bit value from the high 32 bits of a floating-point (coprocessor 1) register to an integer register. Useful for 32-bit integer CPUs with 64-bit FPUs.
mov.s fd, fs	$fd = fs;$
movf.s fd, fs, N	$if (!fcc(N)) \quad fd = fs;$
movn.s fd, fs, t	$if (t \neq 0) \quad fd = fs; \quad /* \text{ t is a GPR } */$
movt.s fd, fs, N	$if (fcc(N)) \quad fd = fs;$
movz.s fd, fs, t	$if (t == 0) \quad fd = fs; \quad /* \text{ t is a GPR } */$
msub.s fd, fr, fs, ft	$fd = fs * ft - fr;$
mtcl rs, fd	Move 32-bit value from integer to floating-point (coprocessor 1) register with no conversion or validity check.
mtlcl rs, fd	Move 32-bit value from integer to the high 32 bits of a floating-point (coprocessor 1) register. Mainly useful for a 32-bit integer CPU with a full 64-bit FPU (the great majority of MIPS CPUs are 64-bit ones).
mul.s fd, fs, ft	$fd = fs * ft;$
mulr.ps fd, fs	$/* \text{ MIPS-3D "Reduction Add", see Section 7.10.4 } */$ $fd.upper = fs.upper * fs.lower;$ $fd.lower = ft.upper * ft.lower;$
neg.s fd, fs	$fd = -fs;$
nmadd.s fd, fr, fs, ft	$fd = -(fs * ft + fr);$
nmsub.s fd, fr, fs, ft	$fd = -(fr - fs * ft);$

TABLE 8.4 *continued*

<i>Assembly code</i>	<i>Function</i>
p1l.ps fd, fs, ft plu.ps fd, fs, ft pul.ps fd, fs, ft puu.ps fd, fs, ft	Repack paired-single, see section 7.10.
prefx hint, i(b)	Register/register address mode cache prefetch instruction. Only available with floating point (where the same address mode is used for ldxc1/sdxc1). But it's also listed in the integer instruction table. See section 8.5.8 for how it works.
pul.ps fd, fs, ft puu.ps fd, fs, ft	Repeated in alphabetical sequence—see p1l above.
recip.s fd, fs	Fast reciprocal. Not IEEE accurate, but only wrong by one unit in the least significant place. $fd = 1/fs;$
recip1.s fd, fs recip2.s fd, fs, ft	MIPS-3D— recip1 is a quick-and-dirty reciprocal approximation, and recip2 is a special multiply-add that does a reciprocal refinement step. See section 7.10.4
round.l.d fd, fs round.l.s fd, fs	Convert FP to equal or closest 64-bit integer value.
round.w.d fd, fs round.w.s fd, fs	Convert FP to equal or closest 32-bit integer value.
rsqrt.s fd, fs	Fast and fairly accurate (not wrong by more than 2 bits in the least significant place) but not IEEE accurate. $fd = \text{sqrt}(1/fs);$
rsqrt1.s fd, fs rsqrt2.s fd, fs, ft	MIPS-3D— recip1 is a quick-and-dirty square-root approximation, and rsqrt2 is a special multiply-add that does a square root refinement step. See section 7.10.4.
s.d ft, addr ⇒ sdcl ft, addr	FP store double; address must be eight-byte aligned. Will be synthesized to two swc1 instructions on a CPU with a 32-bit FPU. $*((\text{double } *)\text{addr}) = ft;$

TABLE 8.4 *continued*

<i>Assembly code</i>	<i>Function</i>
s.s <i>ft, addr</i> ⇒ swc1 <i>ft, addr</i>	FP store single; address must be four-byte aligned. <code>*((float *)addr) = ft;</code>
sdc1 <i>fd, disp(b)</i>	Deprecated equivalent to s.d .
sdxcl <i>fd, i(b)</i>	Explicit double-indexed store double—see notes on ldxc1 , above; usually better to write s.d with an appropriate addressing mode.
sqrts <i>fd, fs</i>	<code>fd = sqrt(fs); /* IEEE compliant */</code>
sub.s <i>fd, fs, ft</i>	<code>fd = fs - ft;</code>
suxcl <i>fd, i(b)</i>	Double-indexed store, exactly like sdxcl except that no exception results from a misaligned address. The store proceeds, but to the address rounded down to an eight-byte boundary.
swc1 <i>fd, disp(b)</i>	Deprecated equivalent to s.s .
swxc1 <i>fd, i(b)</i>	Explicit double-indexed store of 32-bit FP value; usually better to write s.s with an appropriate addressing mode. See notes on ldxc1 above.
trunc.l.d <i>fd, fs</i> trunc.l.s <i>fd, fs</i>	Convert FP to equal or next-nearest-to-zero 64-bit integer value.
trunc.w.d <i>fd, fs</i> trunc.w.s <i>fd, fs</i>	Convert FP to equal or next-nearest-to-zero 32-bit integer value.

8.4 Differences in MIPS32/64 Release 1

The MIPS32/64 specifications were updated in 2003 to “Revision 2,” which is what is described here. So all this section needs to do is to provide a summary of what *isn’t* in CPUs that are compliant to Revision 1 of the specifications.

8.4.1 *Regular Instructions Added in Release 2*

The instructions listed here include a few defined for the second release of the MIPS32/64 specifications in 2003. They’re listed here mostly for the benefit of those using CPUs compliant to the *first* release of the specifications, which won’t have these instructions.

Extract and Insert bitfield

The **ext** and **ins** instructions (and 64-bit variants **dext**, **dextm**, **dextu**, **dins**, **dinsm**, and **dinsu**) make access to fixed bitfields more efficient (the field size and offset are built into these two-register instructions).

Word Reorganization—Rotate, Byte-extend, and Swap Assistance

In response to requirements to do more register-to-register data reorganization (a particular concern in networking applications), there are new:

- Bit rotate instructions: Prior to MIPS32/64R2, MIPS had only shift instructions. Now there are **rotr** (rotate amount encoded in instruction) and **rotrv** (rotate amount specified by a source register), with 64-bit **drot** and **drotv** variants.
- Sign-extend byte or halfword: You can always sign-extend by a shift-left followed by a shift-right-arithmetic, but the instructions **seb**, **seh** are register-to-register operations that sign-extend a byte/halfword, respectively.
- Byte swap within halfwords: **wsbh** seems a bit of an odd instruction, but in combination with a rotate, it allows most useful byte reorganizations within a 32-bit word to be accomplished in one or two instructions.

Provide for 64-bit FPU (and CP2) on 32-bit CPUs

It had become evident that when a 32-bit CPU was equipped with floating-point hardware, it was often sensible to make that a full 64-bit FPU. While MIPS32/64 release 1 left that possible, there was no way to get full 64-bit values between general-purpose registers and FP registers without passing them through memory. The **mfhc1/mt hc1** instructions do that job, copying the high half of a value from and to FP registers—the existing **mf c1/mt c1** instructions already handle the low half.

Revision 2 also defines **mfhc2/mt hc2**, which do the same job for any 64-bit coprocessor 2 a CPU may define.

Read Hardware Register

rdhwr provides user-mode read-only access to some CPU-specific information, see section 8.5.12.

Make Newly Written Instructions Visible

The **synci** instruction does all the cache manipulation necessary to make any instructions you just wrote into memory reliably visible for instruction fetch through the I-cache. Unlike older cache instructions, it is not privileged and is available for user-mode programs. See section 8.5.11.

8.4.2 *Privileged Instructions Added in Release 2*

There are just a couple of changes in the privileged (kernel-only) instruction set.

Atomic Interrupt Disable/Enable

Prior to Release 2, there was no atomic way of disabling all interrupts in a MIPS CPU: The required RMW sequence on the **SR** register could itself be interrupted. You could make that safe by OS discipline (see section 5.8.3), but now you have a genuine atomic instruction.

di disables interrupts (clearing **SR (IE)**) in a single atomic step. It returns the old value of **SR** in its target register, so you generally enable interrupts again by using an **mtc0** with the saved value. But for reasons of instruction set symmetry, R2 also defines an **ei** instruction, which atomically sets **SR (IE)**.

Shadow Register Support

Shadow registers are one or more duplicate sets of GP registers, which you may choose to use in an exception handler, most often in an interrupt handler. The R2 specification provides different ways you can use them, as described in section 5.8.6.

The new instructions involved are **rdpgpr/wrpgpr**, which, respectively, read from and write to a register in some register set other than the current one.

8.5 Peculiar Instructions and Their Purposes

MIPS has never avoided innovation, and the instruction set contains features whose ingenuity might go unheeded (and unused) because they are hard to understand and have not been well explained. This section discusses those features.

8.5.1 *Load Left/Load Right: Unaligned Load and Store*

Any CPU is going to be more efficient if frequently used data as arranged in memory is aligned on memory boundaries that fit the hardware. For a machine with a 32-bit bus, this favors 32-bit data items that are stored on an aligned 32-bit boundary; similarly, a 64-bit bus favors 64-bit data items stored on an aligned 64-bit boundary.

If a CPU must fetch or store unaligned data that crosses the normal storage-width boundary, it must do a double operation. RISC pipeline simplicity will not let the CPU perform two operations for one instruction, so an unaligned transfer will take at least two instructions.

The ultimate RISC attitude is that we’ve got byte-sized operations and that any unaligned operation you like can be built out of those. If a piece of data (formatted as a four- or eight-byte integer value) might be unaligned, the programmer/compiler can always read it as a sequence of byte values and then use shift/mask operations to build it up in a register. The sequence for a word-sized load looks something like this (assuming a big-endian CPU, and without optimizing for the load delay in the CPU pipeline):

```

lbu      rt, o(b)
sll      rt, rt, 24
lbu      rtmp, o+1(b)
sll      rtmp, rtmp, 16
or       rt, rt, rtmp
lbu      rtmp, o+2(b)
sll      rtmp, rtmp, 8
or       rt, rt, rtmp
lbu      rtmp, o+3(b)
or       rt, rt, rtmp

```

That’s 10 instructions, four loads, and requires a temporary register. It is likely to be quite a performance hit if you do it a lot.

The MIPS solution to this is a pair of instructions, each of which can obtain as much of the unaligned word as fits into an aligned word-sized chunk of memory. The instructions invented for the MIPS instruction set have names like *load left* and *load right*: A pair of them is enough to do an unaligned load/store (word or double size) operation. They were mentioned in section 2.5.2.

The hardware that accesses the memory (or cache) transfers four or eight bytes of aligned data. Partial-word stores are implemented either by a hardware signal that instructs the memory controller to leave certain bytes unchanged or by a read-modify-write (RMW) sequence on the entire word/doubleword. MIPS CPUs mostly have RMW hardware available for writes to the data cache, but don’t do that for memory—the memory controller must implement partial-word writes for itself.

We said that there need to be two instructions, because there are two bus cycles. The 32-bit instructions are **lwl** and **lwr**, for “load word left” and “load word right”; the 64-bit instructions are **ldl** and **ldr**, for “load double left” and “load double right.” The “left” instruction deals with the high-order bits of the unaligned integer, and the “right” instruction fetches the low-order bits (“left” is used in the same sense as in “shift-left”). Because the instructions are defined in terms of more significant and less significant bits, but must deal with a byte-addressed memory, their detailed use depends on the endianness of the CPU (see section 10.2). A big-endian CPU keeps more significant bits earlier, in lower byte addresses, and a little-endian CPU keeps more significant bits later, in higher addresses.

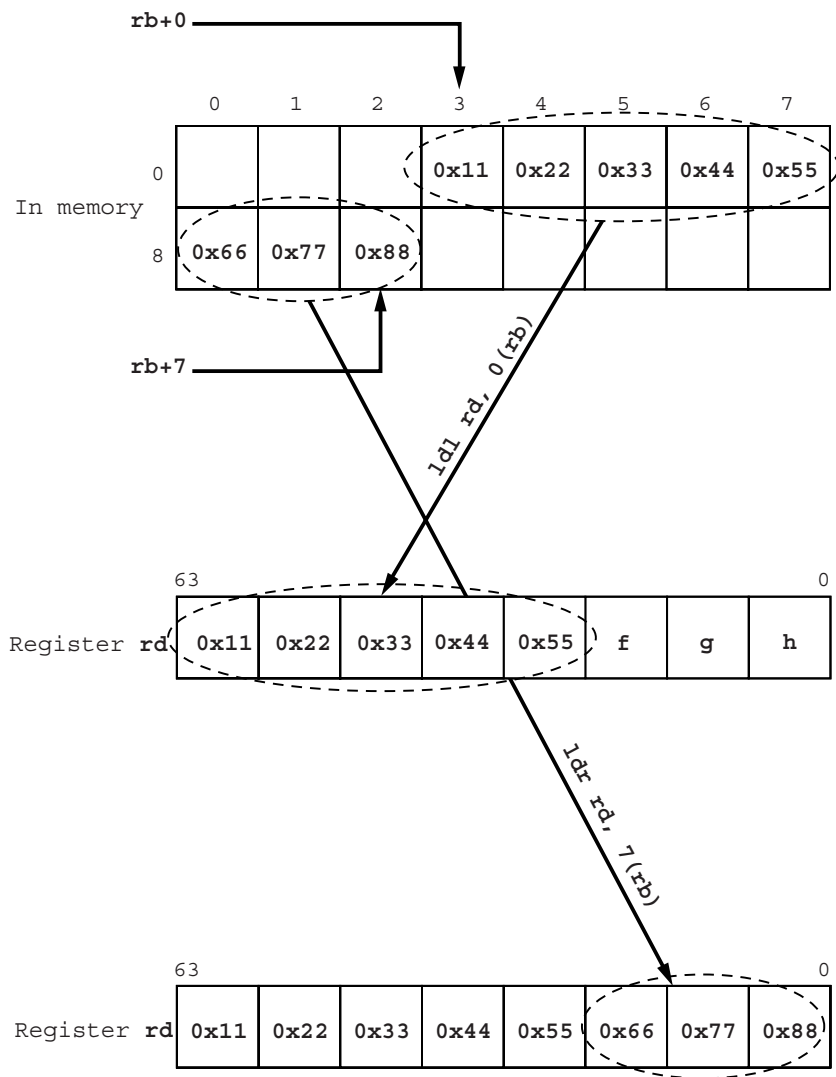
**FIGURE 8.1** Unaligned load double on a big-endian CPU.

Figure 8.1 is an attempt to show what's happening for a big-endian CPU when the unaligned pseudo-operation `ulld d, 0(b)` is coded as:

```
ldl    d, 0(b)
ldr    d, 7(b)
```

The address value in *b* can, of course, be arbitrarily aligned (an unaligned value would cause an exception in the conventional `ld` instruction). So what's going on in Figure 8.1?

- **ldl d, 0(b)**: The 0 offset marks the lowest-addressed byte of the unaligned doubleword, and since we’re big-endian that’s the 8 most significant bits. **ldl** is looking for bits to load into the left (most significant bits) of the register, so it takes the addressed byte and then the ones after it in memory to the end of the word. They’re going up in memory address, so they’re going down in significance; they want to be butted up against the high-numbered end of the register, as shown.
- **ldr d, 7(b)**: The 7 is a bit odd, but it points at the highest-addressed byte of the doubleword—**b+8** would point at the first byte of the next doubleword, of course. **ldr** is concerned with the rightmost, least significant bits; it takes the remaining bytes of our original data and butts them against the low-numbered bits of the register, and the job’s done.

If you’re skeptical about whether this works for words in any alignment, go ahead and try it. Note that in the case where the address is, in fact, correctly aligned (so the data could have been loaded with a conventional **ld** instruction), **uld** loads the same data twice; this is not particularly interesting but usually harmless.

The situation can get more confusing for people who are used to little-endian integer ordering, because they often write data structures with the least significant bits to the left. Once you’ve done that, the “left” in the instruction name becomes “right” on the picture (though it’s still movement toward more significant bits).

On a little-endian CPU the roles of **ldl/ldr** are exchanged, and the code sequence is:

```
ldr    d, 0(b)
ldl    d, 7(b)
```

Figure 8.2 shows you what happens: the most significant bits are reluctantly kept on the left, so it’s the mirror image of the diagram I’d naturally have drawn.

With these figures in front of us, we can try to formulate an exact description of what the instructions do:

- *Load/store left*: Find the addressed byte and enclosing word (or doubleword, for 64-bit operations). Operate on the addressed byte and any more bytes between it and the least significant end of that memory word (higher-byte addresses for big-endian and lower-byte addresses for little-endian).
Load: Grab all those bytes and shift them to higher bit numbers until they’re up against the top of the register. Leave any lower-bit-numbered byte positions within the register unchanged.

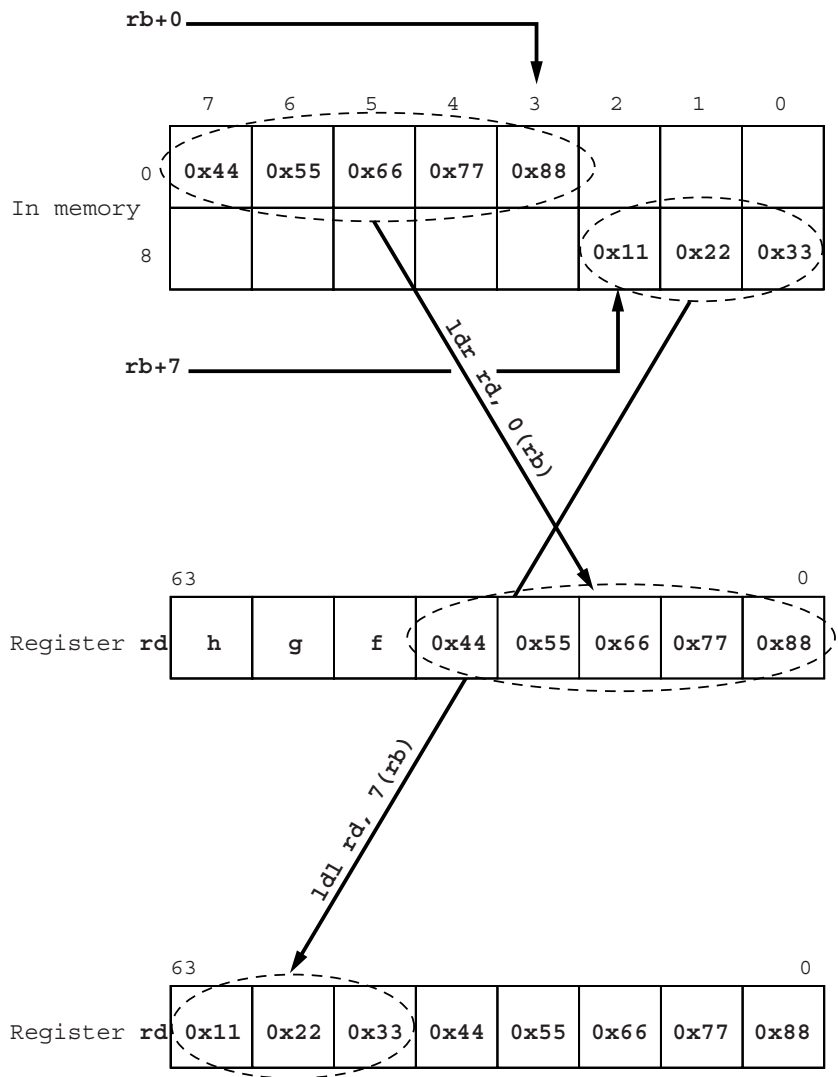


FIGURE 8.2 Unaligned load double on a little-endian CPU.

Store: Replace those bytes with as many bytes of the register as there's room for, starting at the most significant byte in the register.

- *Load/store right:* Find the addressed byte and enclosing word/double-word. Operate on the addressed byte and any more bytes between it and the most significant end of that memory word (lower-byte addresses for big-endian and higher-byte addresses for little-endian).

Load: Grab all those bytes and shift them to lower bit numbers until they're down against the bottom of the register. Leave any higher-bit-numbered byte positions within the register unchanged.

Store: Replace those bytes with as many bytes of the register as there's room for, starting with the least significant byte in the register.

The load/store left/right instructions do not require the memory controller to offer selective operations on arbitrary groups of bytes within a word; the active byte lanes are always together at one end of a word or doubleword.

Note that these instructions do not perform all possible realignments; there's no special support for unaligned load half (which has to be implemented with byte loads, shifts, masks, and combines).

8.5.2 *Load-Linked/Store-Conditional*

The instructions **ll** (load-linked) and **sc** (store-conditional) provide an alternative to the atomic test-and-set sequence that is part of most traditional instruction sets. They provide a test-and-set sequence that operates without any guarantee of atomicity but that succeeds (and tells you it's succeeded) only if it turned out to be atomic. That seems more complicated, but **ll/sc** scale well with increasingly large multiprocessors with relatively "distant" shared memory (while atomic operations scale badly).

See section 5.8.4 for what they're for and how they're used. But here's how they work. The instruction **ll d, o(b)** performs a 32-bit load from the usual base+offset address. But as a side effect, it remembers that a load-link has happened (setting an otherwise invisible linked status bit inside the CPU). It also keeps the address of the load in the register **LLAddr**.

A subsequent **sc t, o(b)** first checks whether it can be sure that the read-modify-write sequence that began with the last-executed **ll** will complete atomically. If it can be sure, then the value of **t** is stored into the location, and the "true" value 1 is returned in **t**. If it can't be sure that the operation was atomic, no store happens and **t** is set 0.

CAUTION! The test for atomicity is unlikely to be exhaustive. The instruction may fail when the memory location has not, in fact, been changed, because the CPU detected some condition where it might have been changed.

There are two reasons why **sc** could fail. The first is that the CPU took an exception somewhere between executing the **ll** and the **sc**. Its exception handler, or a task switch triggered by the exception, might have done something nonatomic.

The second type of failure happens only in a multiprocessor, when another CPU has written the memory location or one near it (commonly in the same

line, but some implementations may monitor the whole memory translation page). For efficiency reasons, this detector is only enabled when both participating CPUs have agreed to map this data as a shared area—strictly, if the other CPU has completed a coherent store to the sensitive block.

Let’s emphasize again: The failure of the **sc** is *not* proof that some other task or CPU has in fact written the variable, it just says that might have happened and you should check; implementations are encouraged to trade off a fair number of false warnings against simplicity or performance.

Multiprocessor CPUs must keep track of the address used by the last **ll**, and they keep it in the coprocessor 0 register **LLAddr**, where software can read and write it. But the only reasons to read and write this register are diagnostic; on a CPU without multiprocessor features, it is redundant. You’re recommended not to rely on its existence.

Here’s a brief example: an “atomic” increment routine that matches the Linux kernel’s `atomic_inc(&mycount)` call. Concurrent threads—possibly on different CPUs—can call this knowing that each call will increment the value by 1:

```
atomic_inc:
    ll    v0, 0(a0)           # a0 has pointer to 'mycount'
    addu  v0, 1
    sc    v0, 0(a0)
    beq   v0, zero, atomic_inc # retry if sc fails
    nop
    jr    ra
    nop
```

NEC omitted **ll/sc** instructions from its Vr41xx CPU family, probably unaware that uniprocessors can benefit from these instructions too.

8.5.3 *Conditional Move Instructions*

A conditional move instruction copies data from one register to another, but only if some condition is satisfied—otherwise, it does nothing. They were featured in other RISC architectures (ARM may have been first) before making a MIPS debut with the MIPS IV instruction set (first implemented in the R8000, R10000, and R5000 in 1995–1996). Conditional moves allow compilers to generate code with fewer conditional branches—which is good, because conditional branches are bad for pipeline efficiency.

CPUs built with the simple five-stage pipeline described in Chapter 1 don’t have much trouble with branches; the branch delay slot instruction is usually executed, and the CPU then moves straight to the branch target. With these simple CPUs, most branches are free (provided the branch delay slot contains a useful instruction) and the others cost only one clock cycle.

But more extravagant implementations of the MIPS ISA may lose many instruction-execution opportunities while waiting for the branch condition to be resolved and the target instruction to be fetched. The long-pipeline R4400, for example, always pays a two-clock-cycle penalty on every taken branch. In the highly superscalar R10000 (which can issue four instructions per clock cycle) you might lose seven instruction issue opportunities waiting for the branch condition to be resolved. To reduce the effect of this, the R10000 has special branch prediction circuits that guess the branch outcome and run ahead accordingly, while keeping the ability to back off from those speculative instructions. This is quite complicated: If the compiler can reduce the frequency with which it relies on the complicated features, it will run faster.

How do conditional move instructions get rid of branches? Consider a piece of code generating the minimum of two values:

```
n = (a < b) ? a : b;
```

Assuming that the compiler has managed to get all the variables into registers, this would normally compile to a sequence like the following (this is logical assembly language sequence, before making pipeline adjustments for delay slots):

```
slt t0, a, b
move n, a
bne zero, t0, 1f
move n, b
1:
```

can be replaced with:

```
slt t0, a, b
move n, a
movz n, b, t0
```

Although the conditional move instruction **movz** looks strange, its role in the pipeline is exactly like any other register/register computational instruction. A branch has been removed and our highly pipelined CPU will go faster.

8.5.4 *Branch-Likely*

Another pipeline optimization, this one introduced with MIPS II, is branch-likely.

Note that these instructions are hard to implement efficiently on machines with particularly sophisticated pipelines, so the MIPS32 specification deprecates their use in portable code.

Compilers are generally reasonably successful in filling branch delay slots, but they have the hardest time at the end of small loops. Such loop-closing branches are the most frequently executed, so **nops** in their delay slots are significant; however, the loop body is often full of dependent code that can't be reorganized.

The branch-likely instruction nullifies the branch delay slot instruction when the branch is not taken. An instruction is nullified by preventing its write-back stage from happening—and in MIPS, that's as if the instruction had never been executed. By executing the delay slot instruction only when the branch is taken, the delay slot instruction becomes part of the next go around the loop.

So any loop:

```
loop:
    first
    second
    ...
    blez t0, loop
    nop
```

can be transformed to:

```
loop:
    first
loop2:
    second
    ...
    blezl t0, loop2
    first
```

This means we can fill the branch delay slot on loops almost all the time, greatly reducing the number of **nops** actually executed.

You'll see it implied in some manufacturers' documentation that branch-likely instructions, by eliminating **nops**, make programs smaller; this is a misunderstanding. You can see from the example that the **nop** is typically replaced by a duplicated instruction, so there's no gain in program size. The gain is in speed.

8.5.5 *Integer Multiply-Accumulate and Multiply-Add Instructions*

Many multimedia algorithms include calculations that are basically a sum of products. In the inner loops of something like a JPEG image decoder, the

calculation is intensive enough to keep the CPU's arithmetic units fully utilized. The calculations break down into a series of multiply-accumulate operations, each of which looks like this:

```
a = a + b*c;
```

Although the RISC principles described in Chapter 1 suggest that it might be better to build this calculation out of simple, separate functions, this may be a deserved exception. Multiply is a multiple-clock-cycle operation, leaving a simple RISC with the problem of scheduling the subsequent (quick) add. If you attempt the add too early, the machine stalls; if you leave it until too late, you don't keep the critical arithmetic units busy. In a floating-point unit, there's an additional advantage in that some housekeeping associated with every instruction can be shared between the multiply and add stages.

Before they were defined by MIPS32/64, multiply-accumulate operations were vendor-specific extensions. MIPS32/64 was able to bless the compatible operations already implemented on IDT, Toshiba, and QED CPUs. They operate in the independently clocked integer multiply unit and so are all multiply-accumulate operations,² accumulating in the multiply unit output registers **lo** and **hi**. Confusingly, all vendors have called their instructions **mad** or **madd** rather than "mac."

8.5.6 *Floating-Point Multiply-Add Instructions*

All the arguments above apply to floating-point calculations too, though the critical applications here are 3D graphics transformations and heavy-duty matrix math. In a floating-point unit, there's an additional benefit—some housekeeping (normalization and rounding) associated with every nontrivial FP instruction can be shared between the multiply and add stages.

The multiply-add at the heart of most PowerPC floating-point units has certainly produced some very impressive figures, which was obviously an influence on MIPS's adoption of these operations.

The floating-point operations **madd**, **msub**, **nmadd**, and **nmsub** are genuine four-operand multiply-add instructions, performing operations such as:

```
a = b + c*d;
```

They're aimed at large graphics/numeric-intensive applications on SGI workstations and heavyweight numerical processing in SGI's range of supercomputers. But they do not always produce exactly the same result as is

2. Toshiba's R3900 and some other CPUs have a three-operand multiply-add, but even there the addend is constrained to come from **lo/hi**. The IDT and QED CPUs offer a two-operand instruction that is identical to Toshiba's in the special case where the destination register is **\$0**.

mandated by the IEEE 754 standard for a sequence of multiply-then-add (because the combined instructions do not round the product before doing the add, the combined instructions are excessively precise).

8.5.7 *Multiple FP Condition Bits*

Prior to MIPS IV, all tests on floating-point numbers communicated with the main instruction set through a single condition bit, which was set explicitly by compare instructions and tested explicitly by special conditional branch instructions. The architecture grew like this because in the early days the floating-point unit was a separate chip, and the FP condition bit was implemented with a signal wire that passed into the main CPU.

The trouble with the single bit is that it creates dependencies that reduce the potential for launching multiple instructions in parallel. There is an unavoidable write-to-read dependency between the compare instruction that creates a condition and the branch instruction that tests it, while there's an avoidable read-to-write interaction where a subsequent compare instruction must be delayed until the branch has seen and acted on its previous value.

FP array calculations benefit from a compilation technique called *software pipelining*, where a loop is unrolled and the computations of successive loop iterations are deliberately interleaved to make maximum use of multiple FP units. But if something in the loop body requires a test and branch, the single condition unit will make this impossible; hence, multiple conditions can make a big difference.

Modern FPU's provide 8 bits, not just 1; compare and FP conditional branch instructions specify which condition bit should be used. Older compilers set reserved fields to zero, so old code will run correctly using just condition code zero.

8.5.8 *Prefetch*

pref provides a way for a program to signal the cache/memory system that data is going to be needed soon. Implementations that take advantage of this can prefetch the data into a cache. It's not really clear how many applications can foresee what references are likely to cause cache misses; prefetch is useful for large-array arithmetic functions, however, where chunks of data can be prefetched in one loop iteration so as to be ready for the next go-around.

The first argument to **pref** is a small-integer coded "hint" about how the program intends to use the data. A range of values that have been implemented in some MIPS32/64 CPUs is shown in Table 8.5.

If the CPU does not understand a hint, it is likely to treat it either as a prefetch with a "load" hint or ignore it altogether. In general, CPUs are free to ignore **pref** completely, treating it as a **nop**, so although an optimization using hints is aimed at one particular CPU, it should not break any other.

TABLE 8.5 Prefetch Hint Codes

<i>Value / MIPS name</i>	<i>What might happen</i>	<i>When to use it</i>
0 - load 1 - store	Read the cache line into the D-cache if not present.	When you expect to read the data soon. Use “store” hint if you also expect to modify it.
4 - load_streamed 5 - store_streamed	Do something to avoid overwriting whole cache with a stream of data items, each used only “once.” Perhaps use just one cache way.	For data that you expect to process sequentially and can afford to discard from the cache once processed.
6 - load_retained 7 - store_retained	Opposite of streamed (so perhaps use any other cache way).	For data that you expect to use more than once and that may be subject to competition from streamed data.
25 - writeback_invalidate/ nudge	If the line is in the cache, invalidate it (writing it back first if it was dirty).	When you know you’ve finished with the data and want to make sure it loses in any future competition for cache resources.
30 - PrepareForStore	If the line is not in the cache, create a cache line—but instead of reading it from memory, fill it with zeros and mark it as dirty. If the line is already in the cache, do nothing— <i>this operation cannot be relied upon to zero the line.</i>	When you know you will overwrite the whole line, so reading the old data from memory is unnecessary. A recycled line is zero-filled only because its former contents could have belonged to a sensitive application—allowing them to be visible to the new owner would be a security breach.

Some MIPS CPUs implement a nonblocking load in which execution continues after a load cache miss, just so long as the load target register is not referenced. However, the **pref** instruction is more useful for longer-range prediction of memory accesses (and becoming a no-op in CPUs that don’t implement it is more benign than blocking on a load a program has deliberately performed early).

8.5.9 *Sync: A Memory Barrier for Loads and Stores*

Suppose we have a program that consists of a number of cooperating sequential tasks, each running on a different “processor” and sharing memory. We’re

probably talking about a multiprocessor using sophisticated cache coherency algorithms, but the cache management is not relevant right now. And the same issues can arise when the other “processor” is an I/O controller using DMA.

Any task’s robust shared memory algorithm will be dependent on when shared data is accessed by other tasks: Did they read that data before I changed it? Have they changed it yet?

Since each task is strictly sequential, why is this a problem? It turns out that the problem occurs because CPU tuning features often interfere with the logical sequence of memory operations; by definition, this interference must be invisible to the program itself, but it will show up when viewed from outside. There can be good reasons for breaking natural sequence. For optimum memory performance, reads—where the CPU is stalled waiting for data—should overtake pending writes. As long as the CPU stores both the address and data on a write, it may defer the write for a while. If a CPU does that, it had better check that the overtaking read is not for a location for which a write is pending; that can be done.

Another example is when a CPU that implements nonblocking loads ends up with two reads active simultaneously; for best performance the memory system should be allowed to choose which one to complete first.

CPUs that allow neither of these changes of sequence, performing all reads and writes in program order, are called *strongly ordered*. Many MIPS CPUs, when configured as uniprocessors, are strongly ordered. But there are exceptions: Even some early R3000 systems would allow reads to overtake pending writes (after checking that the read was not for any pending-write location).

A **sync** instruction defines a load/store barrier. You are guaranteed that all load/stores initiated before the **sync** will be seen before any load/store initiated afterward.

Note that in a multiprocessor, we have to insist that the phrase “be seen” means “be seen by any task in the system that correctly implements the shared memory caching system.” This is usually done by ensuring that **sync** produces a reordering barrier for transactions between the CPU and the cache/memory/bus subsystem.

There are limitations. There is no guarantee about the relative timing of load/stores and the execution of the **sync** itself; it merely separates load/stores before the instruction from those after. **sync** is not guaranteed to help out with the problem of ensuring some timing relationship between the CPU’s program execution and external writes, which we mention in section 10.4.

And inside a system **sync** works only on certain access types (uncached and *coherent* cached accesses). Much “normal” cached memory is noncoherent; any data space that is known not to be shared is safe, and so is anything that is read-only to the sharing tasks.

sync does not need to do anything on CPUs that are strongly ordered; in such cases it may be a **nop**.

However, it’s not unusual for **sync** to be much stronger than it needs to be; consult your CPU’s manual.

8.5.10 Hazard Barrier Instructions

It's commonplace for one instruction to use a result obtained by its immediate predecessor. In a pipelined processor you'd normally read the second instruction's operand from the register file before its predecessor's value had been written back. That's not a problem, because MIPS instructions only pass data in registers, and data going to and from the general-purpose and floating-point registers is extensively bypassed—that is, the hardware detects the dependency and arranges to forward the data directly to the second instruction, just as it needs it.

However, this isn't done (in general) for CP0 registers. If you write a field of a CP0 register, it may affect subsequent instructions, and the MIPS rules do not guarantee how long that might take.

There are two kinds of hazards. The most obvious is where the dependent instruction uses the value provided by the first: That's called an *exception hazard*.

But a more troublesome case is where the write changes some CPU state so as to affect even the fetch of subsequent instructions: That's the case, for example, with a CP0 write that changes the memory translation setup. Those are called *instruction hazards*.

Traditionally, MIPS CPUs left the kernel/low-level software engineer with the job of designing sequences that are guaranteed to run correctly, usually by adding a sufficient number of **nop** or **ssnop**³ instructions between the write and the dependent instruction.

From Release 2 of the MIPS32 specification, though, this is replaced by *hazard barrier* instructions. **eret**, **jr.hb**, and **jalr.hb** are barriers to all side effects, including execution hazards, while **ehb** (a kind of no-op on steroids) deals, with less overhead, with exception hazards.

For a longer discussion of hazards and barriers, see section 3.4.

Porting Software to Use the New Instructions

If you know your software will only ever run on a MIPS32 Release 2 or higher CPU, then that's great. But to maintain software that has to continue running on older CPUs, you're guaranteed that:

- *ehb is a no-op*: On all previous CPUs. You can create software that continues to work on old CPUs but is also safe on all future MIPS32/64-compliant CPUs: Just substitute an **ehb** for the last no-op in your sequence of “enough no-ops.”
- *jr.hb and jalr.hb*: Are decoded as plain jump-register and call-by-register instructions on earlier CPUs. Again, provided you already had enough no-ops for your worst-case older CPU, your system should now be safe on MIPS32/64 Release 2 CPUs.

3. **ssnop** is a special sort of no-op that is guaranteed to occupy a whole issue cycle by itself on a CPU that is capable of issuing more than one instruction per clock cycle.

8.5.11 *Synci: Cache Management for Instruction Writers*

A program that loads another program into memory is actually writing the D-side cache. The instructions it has loaded can't be executed until they reach the I-cache. MIPS CPUs have no logic to join up the two caches (both caches are performance-critical, so it probably doesn't make much sense to add logic for this purpose, which would be rarely used).

After the instructions have been written, the loader should arrange to write back any containing D-cache line and invalidate any locations already in the I-cache. You can certainly do that with **cache** instructions, as described in section 4.6—but those instructions are only available in kernel mode, and a loader writing instructions for the use of its own process need not be privileged software.

So, in the latest MIPS32/64 CPUs, MIPS provides the **synci** instruction, which does the whole job for a cache-line-sized chunk of the memory you just loaded: That is, it arranges a D-cache write-back and an I-cache invalidate.

To employ **synci** at user level, you need to know the size of a cache line, and that can be obtained with a **rdhwr SYNCI.Step** from one of the standard “hardware registers” described in the next section.

8.5.12 *Read Hardware Register*

rdhwr provides useful information about the hardware directly to unprivileged (user-mode) software.

The MIPS32/64 specification defines four registers so far. The OS can control access to each register individually through a bitmask in the CP0 register **HWREna** (set bit 0 to enable register 0 and so on.) **HWREna** is cleared to all zeros on reset, so software has to explicitly enable user access. Privileged code can always read everything, irrespective of zeros in **HWREna**.

The four registers are:

- **CPUNum (0)**: Number of the CPU on which the program is currently running. This comes directly from the coprocessor 0 **EBase (CPUNum)** field.
- **SYNCI.Step (1)**: The effective size of an L1 cache line.⁴

The line size is important to user programs, because they can now do things to the caches using the **synci** instruction to make instructions you've written visible for execution. Then **SYNCI.Step** tells you the “step size”—the address increment between successive **synci**s required to cover all the instructions in a range.

If **SYNCI.Step** returns zero, that means that you don't need to use **synci** at all.

4. Strictly, it's the lesser of the I-cache and D-cache line size, but it's most unusual to make them different.

- *CC* (2): User-mode read-only access to the CP0 **Count** register, for high-resolution counting. Which wouldn't be much good without . . .
- *CCRes* (3): Tells you how fast **Count** counts. It's a divider from the pipeline clock (if you read a value of "2," then **Count** increments every two cycles, at half the pipeline clock rate).

8.6 Instruction Encodings

All MIPS instructions defined by Release 2 of the MIPS32/64 ISA (and a judicious choice of different instructions defined by architectural variants) are listed in order of encoding in Table 8.6. Subsections 8.6.2 and 8.6.3 provide further notes on the material in this table.

Most MIPS manuals say there are only three instruction formats used. I daresay this corresponds to some reality in the original internal design of the chip, but it never looked like that to the user, to whom it appears that different instructions use the fields for quite different purposes. Newer instructions use more complex encodings.

Table 8.6 tells you the binary encoding and the mnemonic of the instruction in assembly code.

The column headed "ISA not MIPS32?" identifies instructions that are not implemented by all MIPS32-compliant CPUs: Entries in this column include "R2" for instructions not required until MIPS32/64 Revision 2, "MIPS64" for instructions only required for 64-bit CPUs, "EJTAG" for instructions associated with the debug unit, "3D" and "PS" for the related MIPS-3D and paired-single optional extensions to floating point, and "not in MIPS32/64" for instructions that were in MIPS I but are now obsolete. Occasionally, this last column will have the name of a specific CPU that offers a special instruction.

8.6.1 *Fields in the Instruction Encoding Table*

The following notes describe the fields in Table 8.6.

Field 31–26	The primary op-code "op," which is 6 bits long. Instructions that are having trouble fitting in 32 bits (like the "long" j and jal instructions or arithmetic with a 16-bit constant) have a unique "op" field. Other instructions come in groups that share an "op" value, distinguished by other fields.
Field 5–0	Subcode field used for the three-register arithmetical/logical group of instructions (major op-code zero).
Field 25–21	Yet another extended op-code field, this time used by coprocessor-type instructions.
s, t, w	Fields identifying source registers. Occasionally we'll use rs, rt when you might not be sure it was a general-purpose register.

o(b), offset	“o” is a signed offset that fits in a 16-bit field; “b” is a general-purpose base register whose contents are added to “o” to yield an address for a load or store instruction.
d	The destination register, to be changed by this instruction. Occasionally we’ll write rd to remind you it’s a general-purpose register.
shf	How far to shift, used in shift-by-constant instructions.
broffset	A signed 16-bit PC-relative word offset representing the distance in words (one word per instruction) to a label. Offset zero is the delay slot instruction after the branch, so a branch-to-self has an offset of -1.
target	A 26-bit <i>word</i> address to be jumped to (it corresponds to a 28-bit byte address, which is always word-aligned). The long jump j instruction is rarely used, so this format is pretty much exclusively for function calls (jal). The high-order 4 bits of the target address can’t be specified by this instruction and are taken from the address of the jump instruction. This means that these instructions can reach anywhere in the 256-MB region around the instructions’ location. To jump further, use a jr (jump register) instruction.
constant	A 16-bit integer constant for immediate arithmetic or logic operations. It’s interpreted as signed or unsigned according to the instruction context.
cs/cd	Coprocessor register as source or destination, respectively. Each coprocessor section of the instruction set may have up to 32 data registers and up to 32 control registers.
fr/fs/ft	Floating-point unit source registers.
fd	Floating-point destination register (written by the instruction).
N/M	Selector for FP condition code—“N” when it’s being read, and “M” when it’s being written by a compare instruction. The field is zero in older instruction sets that have only one FP condition code, and since older assembly code won’t have the field present, you can write the instruction without it and get the zeroth condition code.
hint	A hint for the prefetch instruction, described in section 8.5.8.
cachop	This is used with the cache instruction and encodes an operation to be performed on the cache entry discovered by the instruction’s address. See Table 4.2 in section 4.9.

TABLE 8.6 Machine Instructions in Order of Encoding[illegible]

continued

31	26 25	21 20 18 17 16 15	11 10 8 7 6 5	0	<i>Assembly name</i>	<i>ISA not MIPS32?</i>
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TABLE 8.6 *continued*

31	26	25	21	20	18	17	16	15	11	10	8	7	6	5	0	Assembly name	ISA not MIPS32?
0	s	t	d	0	43											sltu d,s,t	
0	s	t	d	0	44											dadd d,s,t	MIPS64
0	s	t	d	0	45											daddu d,s,t	MIPS64
0	s	t	d	0	46											dsub d,s,t	MIPS64
0	s	t	d	0	47											dsubu d,s,t	MIPS64
0	s	t	x		48											tge s,t	
0	s	t	x		49											tgeu s,t	
0	s	t	x		50											tlt s,t	
0	s	t	x		51											tltu s,t	
0	s	t	x		52											teq s,t	
0	s	t	x		54											tne s,t	
0	0	w	d	shf	56											dsll d,w,shf	MIPS64
0	0	w	d	shf	58											dsrl d,w,shf	MIPS64
1	0	w	d	shf	58											drotr d,w,shf	MIPS64R2
0	0	w	d	shf	59											dsra d,w,shf	MIPS64
0	0	w	d	shf	60											dsll32 d,w,shf	MIPS64
0	0	w	d	shf	62											dsrl32 d,w,shf	MIPS64
0	0	w	d	shf	63											dsra32 d,w,shf	MIPS64
1	s	0	broffset												bltz s,p		
1	s	1	broffset												bgez s,p		
1	s	2	broffset												bltzl s,p		
1	s	3	broffset												bgezl s,p		
1	s	8	constant												tgei s,j		
1	s	9	constant												tgeiu s,j		
1	s	10	constant												tlti s,j		

TABLE 8.6 *continued*

31	26 25	21 20181716 15	11 10 8 7 6 5	0	Assembly name	ISA not MIPS32?
1	s	11	constant		tltiu s,j	
1	s	12	constant		teqi s,j	
1	s	14	constant		tnei s,j	
1	s	16	broffset		bltzal s,p	
1	s	17	broffset		bgezal s,p	
1	s	18	broffset		bltzall s,p	
1	s	19	broffset		bgezall s,p	
1	b	31	o		synci, o(b)	R2
2	target				j target	
3	target				jal target	
4	s	t	broffset		beq s,t,p	
5	s	t	broffset		bne s,t,p	
6	s	0	broffset		blez s,p	
7	s	0	broffset		bgtz s,p	
8	s	d	(signed) const		addi d,s,const	
9	s	d	(signed) const		addiu d,s,const	
10	s	d	(signed) const		slti d,s,const	
11	s	d	(signed) const		sltiu d,s,const	
12	s	d	(unsigned) const		andi d,s,const	
13	s	d	(unsigned) const		ori d,s,const	
14	s	d	(unsigned) const		xori d,s,const	
15	0	d	(unsigned) const		lui d,const	

TABLE 8.6 *continued*

31	26	25	21	20	18	17	16	15	11	10	8	7	6	5	0	Assembly name	ISA not MIPS32?
16	0	t	cs	0	0											mfc0 t,cs	
16	1	t	cs	0	0											dmfc0 t,cs	MIPS64
16	2	t	cs	0	0											cfc0 t,cs	
16	4	t	cd	0	0											mtc0 t,cd	
16	5	t	cd	0	0											dmtc0 t,cd	MIPS64
16	10	xt	d	0	0											rdpgpr d,xt	R2
16	11	t	12	0	0											di t	R2
16	11	t	12	0	32											ei t	R2
16	14	t	xd	0	0											wrggpr xd,t	R2
16	16	0	0	0	1											tlbr	
16	16	0	0	0	2											tlbwi	
16	16	0	0	0	6											tlbwr	
16	16	0	0	0	8											tlbp	
16	16	0	0	0	16											rfe	MIPS I only
16	16	0	0	0	24											eret	
16	16	0	0	0	31											dret	MIPS II only
16	16	0	0	0	31											deret	EJTAG
16	16	0	0	0	33											standby	Vr4100
16	16	0	0	0	34											suspend	Vr4100
16	8	0	broffset													bc0f p	Not in MIPS32/64
16	8	1	broffset													bc0t p	Not in MIPS32/64
16	8	2	broffset													bc0f1 p	Not in MIPS32/64
16	8	3	broffset													bc0t1 p	Not in MIPS32/64

TABLE 8.6 *continued*

31	26	25	21	20	18	17	16	15	11	10	8	7	6	5	0	Assembly name	ISA not MIPS32?
17	0		t		fs		0		0							mfc1 t,fs	
17	1		t		fs		0		0							dmfc1 t,fs	MIPS64
17	2		t		cs		0		0							cfc1 t,cs	
17	3		t		fs		0		0							mfhc1 t,fs	R2
17	4		t		cs		0		0							mtc1 t,cs	
17	5		t		cs		0		0							dmtc1 t,cs	MIPS64
17	6		t		cs		0		0							ctc1 t,cs	
17	7		t		fs		0		0							mthc1 t,fs	R2
17	8	N	0		broffset											bclf N,p	
17	8	N	1		broffset											bclt N,p	
17	8	N	2		broffset											bclfl N,p	
17	8	N	3		broffset											bcltl N,p	
17	9	N	0		broffset											bclany2f N,p	3D
17	9	N	1		broffset											bclany2t N,p	3D
17	10	N	0		broffset											bclany4f N,p	3D
17	10	N	1		broffset											bclany4t N,p	3D
17	16	ft		fs	fd		0									add.s fd,fs,ft	
17	17	ft		fs	fd		0									add.d fd,fs,ft	
17	22	ft		fs	fd		0									add.ps fd,fs,ft	PS
17	16	ft		fs	fd		1									sub.s fd,fs,ft	
17	17	ft		fs	fd		1									sub.d fd,fs,ft	
17	22	ft		fs	fd		1									sub.ps fd,fs,ft	PS
17	16	ft		fs	fd		2									mul.s fd,fs,ft	
17	17	ft		fs	fd		2									mul.d fd,fs,ft	
17	22	ft		fs	fd		2									mul.ps fd,fs,ft	PS
17	16	ft		fs	fd		3									div.s fd,fs,ft	

TABLE 8.6 *continued*

31	26 25	21 20 18 17 16 15	11 10	8 7 6 5	0	Assembly name	ISA not MIPS32?
17	17	ft	fs	fd	3	div.d fd, fs, ft	
17	16	0	fs	fd	4	sqrt.s fd, fs	
17	17	0	fs	fd	4	sqrt.d fd, fs	
17	16	0	fs	fd	5	abs.s fd, fs	
17	17	0	fs	fd	5	abs.d fd, fs	
17	22	0	fs	fd	5	abs.ps fd, fs	PS
17	16	0	fs	fd	6	mov.s fd, fs	
17	17	0	fs	fd	6	mov.d fd, fs	
17	22	0	fs	fd	6	mov.ps fd, fs	PS
17	16	0	fs	fd	7	neg.s fd, fs	
17	17	0	fs	fd	7	neg.d fd, fs	
17	22	0	fs	fd	7	neg.ps fd, fs	PS
17	16	0	fs	fd	8	round.l.s fd, fs	MIPS64
17	17	0	fs	fd	8	round.l.d fd, fs	MIPS64
17	16	0	fs	fd	9	trunc.l.s fd, fs	MIPS64
17	17	0	fs	fd	9	trunc.l.d fd, fs	MIPS64
17	16	0	fs	fd	10	ceil.l.s fd, fs	MIPS64
17	17	0	fs	fd	10	ceil.l.d fd, fs	MIPS64
17	16	0	fs	fd	11	floor.l.s fd, fs	MIPS64
17	17	0	fs	fd	11	floor.l.d fd, fs	MIPS64
17	16	0	fs	fd	12	round.w.s fd, fs	
17	17	0	fs	fd	12	round.w.d fd, fs	
17	16	0	fs	fd	13	trunc.w.s fd, fs	
17	17	0	fs	fd	13	trunc.w.d fd, fs	
17	16	0	fs	fd	14	ceil.w.s fd, fs	
17	17	0	fs	fd	14	ceil.w.d fd, fs	
17	16	0	fs	fd	15	floor.w.s fd, fs	
17	17	0	fs	fd	15	floor.w.d fd, fs	

TABLE 8.6 *continued*

31	26 25	21 20 18 17 16 15	11 10	8 7 6 5	0	Assembly name	ISA not MIPS32?
17	16	N 0	fs	fd	17	movf.s fd,fs,N	
17	17	N 0	fs	fd	17	movf.d fd,fs,N	
17	22	N 0	fs	fd	17	movf.ps fd,fs,N	PS
17	16	N 1	fs	fd	17	movt.s fd,fs,N	
17	17	N 1	fs	fd	17	movt.d fd,fs,N	
17	16	t	fs	fd	18	movz.s fd,fs,t	
17	17	t	fs	fd	18	movz.d fd,fs,t	
17	22	t	fs	fd	18	movz.ps fd,fs,t	PS
17	16	t	fs	fd	19	movn.s fd,fs,t	
17	17	t	fs	fd	19	movn.d fd,fs,t	
17	22	t	fs	fd	19	movn.ps fd,fs,t	PS
17	16	0	fs	fd	21	recip.s fd,fs	
17	17	0	fs	fd	21	recip.d fd,fs	
17	16	0	fs	fd	22	rsqrt.s fd,fs	
17	17	0	fs	fd	22	rsqrt.d fd,fs	
17	22	0	fs	fd	24	addr.ps fd,fs	3D
17	22	0	fs	fd	26	mulr.ps fd,fs	3D
17	17	0	fs	fd	28	recip2.d fd,fs	3D
17	22	0	fs	fd	28	recip2.ps fd,fs	3D
17	17	0	fs	fd	29	recip1.d fd,fs	3D
17	22	0	fs	fd	29	recip1.ps fd,fs	3D
17	17	0	fs	fd	30	rsqrt1.d fd,fs	3D
17	22	0	fs	fd	30	rsqrt1.ps fd,fs	3D
17	17	0	fs	fd	31	rsqrt2.d fd,fs	3D
17	22	0	fs	fd	31	rsqrt2.ps fd,fs	3D

TABLE 8.6 *continued*

31	26 25	21 20 18 17 16 15	11 10	8 7 6 5	0	Assembly name	ISA not MIPS32?	
17	17	0	fs	fd	32	cvt.s.d fd,fs		
17	20	0	fs	fd	32	cvt.s.w fd,fs		
17	21	0	fs	fd	32	cvt.s.l fd,fs	MIPS64	
17	22	0	fs	fd	32	cvt.s.pu fd,fs	PS	
17	16	0	fs	fd	33	cvt.d.s fd,fs		
17	20	0	fs	fd	33	cvt.d.w fd,fs		
17	21	0	fs	fd	33	cvt.d.l fd,fs	MIPS64	
17	16	0	fs	fd	36	cvt.w.s fd,fs		
17	17	0	fs	fd	36	cvt.w.d fd,fs		
17	22	0	fs	fd	36	cvt.pw.ps fd,fs	3D	
17	16	0	fs	fd	37	cvt.l.s fd,fs	MIPS64	
17	17	0	fs	fd	37	cvt.l.d fd,fs	MIPS64	
17	16	0	fs	fd	38	cvt.ps.s fd,fs	PS	
17	20	0	fs	fd	38	cvt.ps.pw fd,fs	3D	
17	21	0	fs	fd	38	cvt.ps.pw.l fd,fs	PS	
17	22	0	fs	fd	40	cvt.s.pl fd,fs	PS	
17	22	0	fs	fd	44	pll.ps.ps fd,fs	PS	
17	22	0	fs	fd	45	plu.ps.ps fd,fs	PS	
17	22	0	fs	fd	46	pul.ps.ps fd,fs	PS	
17	22	0	fs	fd	47	puu.ps.ps fd,fs	PS	
17	16	ft	fs	M	0	48	c.f.s M,fs,ft	
17	17	ft	fs	M	0	48	c.f.d M,fs,ft	
17	22	ft	fs	M	0	48	c.f.ps M,fs,ft	PS
17	16	ft	fs	M	0	49	c.un.s M,fs,ft	
17	17	ft	fs	M	0	49	c.un.d M,fs,ft	
17	22	ft	fs	M	0	49	c.un.ps M,fs,ft	PS

TABLE 8.6 *continued*

31	26 25	21 20181716 15	11 10	8 7 6 5	0	Assembly name	ISA not MIPS32?	
17	16	ft	fs	M	0	50	c.eq.s M, fs, ft	
17	17	ft	fs	M	0	50	c.eq.d M, fs, ft	
17	22	ft	fs	M	0	50	c.eq.ps M, fs, ft	PS
17	16	ft	fs	M	0	51	c.ueq.s M, fs, ft	
17	17	ft	fs	M	0	51	c.ueq.d M, fs, ft	
17	22	ft	fs	M	0	51	c.ueq.ps M, fs, ft	PS
17	16	ft	fs	M	0	52	c.olt.s M, fs, ft	
17	17	ft	fs	M	0	52	c.olt.d M, fs, ft	
17	22	ft	fs	M	0	52	c.olt.ps M, fs, ft	PS
17	16	ft	fs	M	0	53	c.ult.s M, fs, ft	
17	17	ft	fs	M	0	53	c.ult.d M, fs, ft	
17	22	ft	fs	M	0	53	c.ult.ps M, fs, ft	PS
17	16	ft	fs	M	0	54	c.ole.s M, fs, ft	
17	17	ft	fs	M	0	54	c.ole.d M, fs, ft	
17	22	ft	fs	M	0	54	c.ole.ps M, fs, ft	PS
17	16	ft	fs	M	0	55	c.ule.s M, fs, ft	
17	17	ft	fs	M	0	55	c.ule.d M, fs, ft	
17	22	ft	fs	M	0	55	c.ule.ps M, fs, ft	PS
17	16	ft	fs	M	0	56	c.sf.s M, fs, ft	
17	17	ft	fs	M	0	56	c.sf.d M, fs, ft	
17	22	ft	fs	M	0	56	c.sf.ps M, fs, ft	PS
17	16	ft	fs	M	0	57	c.ngle.s M, fs, ft	PS
17	17	ft	fs	M	0	57	c.ngle.d M, fs, ft	PS
17	22	ft	fs	M	0	57	c.ngle.ps M, fs, ft	PS
17	16	ft	fs	M	0	58	c.seq.s M, fs, ft	
17	17	ft	fs	M	0	58	c.seq.d M, fs, ft	

TABLE 8.6 *continued*

31	26 25	21 20 18 17 16 15	11 10	8 7 6 5	0	Assembly name	ISA not MIPS32?	
17	22	ft	fs	M	0	58	c.seq.ps M, fs, ft	PS
17	16	ft	fs	M	0	59	c.ngl.s M, fs, ft	
17	17	ft	fs	M	0	59	c.ngl.d M, fs, ft	
17	22	ft	fs	M	0	59	c.ngl.ps M, fs, ft	PS
17	16	ft	fs	M	0	60	c.lt.s M, fs, ft	
17	17	ft	fs	M	0	60	c.lt.d M, fs, ft	
17	22	ft	fs	M	0	60	c.lt.ps M, fs, ft	PS
17	16	ft	fs	M	0	61	c.nge.s M, fs, ft	
17	17	ft	fs	M	0	61	c.nge.d M, fs, ft	
17	22	ft	fs	M	0	61	c.nge.ps M, fs, ft	PS
17	16	ft	fs	M	0	62	c.le.s M, fs, ft	
17	17	ft	fs	M	0	62	c.le.d M, fs, ft	
17	22	ft	fs	M	0	62	c.le.ps M, fs, ft	PS
17	16	ft	fs	M	0	63	c.ngt.s M, fs, ft	
17	17	ft	fs	M	0	63	c.ngt.d M, fs, ft	
17	22	ft	fs	M	0	63	c.ngt.ps M, fs, ft	PS
17	16	ft	fs	M	1	48	cabs.f.s M, fs, ft	3D
17	17	ft	fs	M	1	48	cabs.f.d M, fs, ft	3D
17	22	ft	fs	M	1	48	cabs.f.ps M, fs, ft	3D
17	16	ft	fs	M	1	49	cabs.un.s M, fs, ft	3D
17	17	ft	fs	M	1	49	cabs.un.d M, fs, ft	3D
17	22	ft	fs	M	1	49	cabs.un.ps M, fs, ft	3D
17	16	ft	fs	M	1	50	cabs.eq.s M, fs, ft	3D
17	17	ft	fs	M	1	50	cabs.eq.d M, fs, ft	3D
17	22	ft	fs	M	1	50	cabs.eq.ps M, fs, ft	3D

TABLE 8.6 *continued*

31	26 25	21 20 18 17 16 15	11 10	8 7 6 5	0	Assembly name	ISA not MIPS32?
17	16	ft	fs	M	1	51	cabs.ueq.s M, fs, ft 3D
17	17	ft	fs	M	1	51	cabs.ueq.d M, fs, ft 3D
17	22	ft	fs	M	1	51	cabs.ueq.ps M, fs, ft 3D
17	16	ft	fs	M	1	52	cabs.olt.s M, fs, ft 3D
17	17	ft	fs	M	1	52	cabs.olt.d M, fs, ft 3D
17	22	ft	fs	M	1	52	cabs.olt.ps M, fs, ft 3D
17	16	ft	fs	M	1	53	cabs.ult.s M, fs, ft 3D
17	17	ft	fs	M	1	53	cabs.ult.d M, fs, ft 3D
17	22	ft	fs	M	1	53	cabs.ult.ps M, fs, ft 3D
17	16	ft	fs	M	1	54	cabs.ole.s M, fs, ft 3D
17	17	ft	fs	M	1	54	cabs.ole.d M, fs, ft 3D
17	22	ft	fs	M	1	54	cabs.ole.ps M, fs, ft 3D
17	16	ft	fs	M	1	55	cabs.ule.s M, fs, ft 3D
17	17	ft	fs	M	1	55	cabs.ule.d M, fs, ft 3D
17	22	ft	fs	M	1	55	cabs.ule.ps M, fs, ft 3D
17	16	ft	fs	M	1	56	cabs.sf.s M, fs, ft 3D
17	17	ft	fs	M	1	56	cabs.sf.d M, fs, ft 3D
17	22	ft	fs	M	1	56	cabs.sf.ps M, fs, ft 3D
17	22	ft	fs	M	1	57	cabs.ngle.ps M, fs, ft 3D
17	16	ft	fs	M	1	58	cabs.seq.s M, fs, ft 3D
17	17	ft	fs	M	1	58	cabs.seq.d M, fs, ft 3D
17	22	ft	fs	M	1	58	cabs.seq.ps M, fs, ft 3D
17	16	ft	fs	M	1	59	cabs.ngl.s M, fs, ft 3D
17	17	ft	fs	M	1	59	cabs.ngl.d M, fs, ft 3D
17	22	ft	fs	M	1	59	cabs.ngl.ps M, fs, ft 3D
17	16	ft	fs	M	1	60	cabs.lt.s M, fs, ft 3D
17	17	ft	fs	M	1	60	cabs.lt.d M, fs, ft 3D
17	22	ft	fs	M	1	60	cabs.lt.ps M, fs, ft 3D

TABLE 8.6 *continued*

31	26	25	21	20	18	17	16	15	11	10	8	7	6	5	0	Assembly name	ISA not MIPS32?
17	16	ft	fs	M	1	61										cabs.nge.s M, fs, ft	3D
17	17	ft	fs	M	1	61										cabs.nge.d M, fs, ft	3D
17	22	ft	fs	M	1	61										cabs.nge.ps M, fs, ft	3D
17	16	ft	fs	M	1	62										cabs.le.s M, fs, ft	3D
17	17	ft	fs	M	1	62										cabs.le.d M, fs, ft	3D
17	22	ft	fs	M	1	62										cabs.le.ps M, fs, ft	3D
17	16	ft	fs	M	1	63										cabs.ngt.s M, fs, ft	3D
17	17	ft	fs	M	1	63										cabs.ngt.d M, fs, ft	3D
17	22	ft	fs	M	1	63										cabs.ngt.ps M, fs, ft	3D
18	0	t	cs	0	0										mfc2 t, cs		
18	1	t	cs	0	0										dmfc2 t, cs	MIPS64	
18	2	t	cs	0	0										cfc2 t, cs		
18	3	t	cs	0	0										mfhc2 t, cs	R2	
18	4	t	cs	0	0										mtc2 t, cs		
18	5	t	cs	0	0										dmtc2 t, cs	MIPS64	
18	6	t	cs	0	0										ctc2 t, cs		
18	7	t	cs	0	0										mthc2 t, cs	R2	
18	8	0	broffset												bc2f p		
18	8	1	broffset												bc2t p		
18	8	2	broffset												bc2fl p		
18	8	3	broffset												bc2tl p		
19	b	t	0	fd	0										lwxcl fd, t (b)		
19	b	t	0	fd	1										ldxcl fd, t (b)		
19	b	t	fs	0	8										swxc1 fs, t (b)		
19	b	t	fs	0	9										sdxcl fs, t (b)		
19	b	t	hint	0	15										prefix hint, t (b)	MIPS64 or R2	

TABLE 8.6 *continued*

31	26 25	21 20 18 17 16 15	11 10	8 7 6 5	0	Assembly name	ISA not MIPS32?
19	s	ft	fs	fd	30	alnv.ps fd,fs,ft,s	MIPS64 or R2
19	fr	ft	fs	fd	32	madd.s fd,fr,fs,ft	
19	fr	ft	fs	fd	33	madd.d fd,fr,fs,ft	
19	fr	ft	fs	fd	38	madd.ps fd,fr,fs,ft	PS
19	fr	ft	fs	fd	40	msub.s fd,fr,fs,ft	
19	fr	ft	fs	fd	41	msub.d fd,fr,fs,ft	
19	fr	ft	fs	fd	46	msub.ps fd,fr,fs,ft	PS
19	fr	ft	fs	fd	48	nmadd.s fd,fr,fs,ft	
19	fr	ft	fs	fd	49	nmadd.d fd,fr,fs,ft	
19	fr	ft	fs	fd	54	nmadd.ps fd,fr,fs,ft	PS
19	fr	ft	fs	fd	56	nmsub.s fd,fr,fs,ft	
19	fr	ft	fs	fd	57	nmsub.d fd,fr,fs,ft	
19	fr	ft	fs	fd	62	nmsub.ps fd,fr,fs,ft	PS
20	s	t	broffset			beql s,t,p	
21	s	t	broffset			bnel s,t,p	
22	s	0	broffset			blezl s,p	
23	s	0	broffset			bgtzl s,p	
24	s	d	(signed) const			daddi d,s,const	MIPS64
25	s	d	(signed) const			daddiu d,s,const	MIPS64
26	b	t	offset			ldl t,o(b)	MIPS64
27	b	t	offset			ldr t,o(b)	MIPS64

TABLE 8.6 *continued*

31	26	25	21	20	18	17	16	15	11	10	8	7	6	5	0	Assembly name	ISA not MIPS32?
28	s	t	0	0	0											madd s,t	
28	s	t	d	0	0											mad d,s,t	R3900
28	s	t	0	0	1											maddu s,t	
28	s	t	d	0	2											mul d,s,t	
28	s	t	0	0	4											msub s,t	
28	s	t	0	0	5											msubu s,t	
28	s	s	d	0	32											clz d,s	
28	s	s	d	0	33											clo d,s	
28	s	s	d	0	36											dclz d,s	
28	s	s	d	0	37											dclo d,s	
28	code				63											sdbbp code	EJTAG
29	target															jalx target	MIPS16e
31	s	t	sz	pos	0											ext t,s,pos,sz	R2
31	s	t	sz	pos	1											dextm t,s,pos,sz	MIPS64R2
31	s	t	sz	pos	2											dextu t,s,pos,sz	MIPS64R2
31	s	t	sz	pos	3											dext t,s,pos,sz	MIPS64R2
31	s	t	sz	pos	4											ins t,s,pos,sz	R2
31	s	t	sz	pos	5											dinsm t,s,pos,sz	MIPS64R2
31	s	t	sz	pos	6											dinsu t,s,pos,sz	MIPS64R2
31	s	t	sz	pos	7											dins t,s,pos,sz	MIPS64R2
31	0	t	d	2	32											wsbh d,t	R2
31	0	t	d	16	32											seb d,t	R2
31	0	t	d	24	32											seh d,t	R2
31	0	t	d	2	36											dsbh d,t	MIPS64R2
31	0	t	d	5	36											dshd d,t	MIPS64R2

TABLE 8.6 *continued*

31	26 25	21 20 18 17 16 15	11 10	8 7 6 5	0	Assembly name	ISA not MIPS32?
31	0	t	hwr	0	59	rdhwr t,hwr	R2
32	b	t	offset			lb t,o(b)	
33	b	t	offset			lh t,o(b)	
34	b	t	offset			lwl t,o(b)	
35	b	t	offset			lw t,o(b)	
36	b	t	offset			lbu t,o(b)	
37	b	t	offset			lhu t,o(b)	
38	b	t	offset			lwr t,o(b)	
39	b	t	offset			lwu t,o(b)	MIPS64
40	b	t	offset			sb t,o(b)	
41	b	t	offset			sh t,o(b)	
42	b	t	offset			swl t,o(b)	
43	b	t	offset			sw t,o(b)	
44	b	t	offset			sdl t,o(b)	MIPS64
45	b	t	offset			sdr t,o(b)	MIPS64
46	b	t	offset			swr t,o(b)	
47	b	op	offset			cache op,o(b)	
48	b	t	offset			ll t,o(b)	
49	b	ft	offset			l.s t,o(b)	
50	b	cd	offset			lwc2 cd,o(b)	
51	b	hint	offset			pref hint,o(b)	
52	b	t	offset			lld t,o(b)	MIPS64
53	b	ft	offset			l.d ft,o(b)	
54	b	cd	offset			ldc2 cd,o(b)	
55	b	t	offset			ld t,o(b)	MIPS64
56	b	t	offset			sc t,o(b)	

TABLE 8.6 *continued*

31	26 25	21 20 18 17 16 15	11 10 8 7 6 5	0	Assembly name	ISA not MIPS32?
57	b	ft	offset		s.s ft,o(b)	
57	b	ft	offset		swc1 ft,o(b)	
58	b	cs	offset		swc2 cs,o(b)	
60	b	t	offset		scd t,o(b)	MIPS64
61	b	ft	offset		s.d ft,o(b)	
61	b	ft	offset		sdc1 ft,o(b)	
62	b	cs	offset		sdc2 cs,o(b)	
63	b	t	offset		sd t,o(b)	MIPS64

8.6.2 Notes on the Instruction Encoding Table

- *Instruction aliases:* Mostly, we have suppressed all but one possible mnemonic for the same instruction, but occasionally we leave them in. Instructions such as **nop** and **l.s** are so ubiquitous that it seems simpler to include them than to leave them out.
- *Coprocessor instructions:* Instructions that were once defined but are no longer have been expunged. Coprocessor 3 was never used by any MIPS I CPU and is not compatible with a MIPS32/64 floating-point unit—and some of what would be standard coprocessor op-codes, including memory loads, have been recycled for different uses.

8.6.3 Encodings and Simple Implementation

If you look at the encodings of the instructions, you can sometimes see how the CPU is designed. Although there are variable encodings, fields that are required very early in the pipeline are encoded in a totally regular way:

- Source registers are always in the same place, so that the CPU can fetch two operands from the integer register file without any conditional decoding. In some instructions, both registers will not be needed, but since the register file is designed to provide two source values on every clock cycle, nothing has been lost.
- The 16-bit constant is always in the same place, permitting the appropriate instruction bits to be fed directly into the ALU's input multiplexer without conditional shifts.

8.7 Instructions by Functional Group

We've divided the instruction set into reasonable chunks, in this order:

- No-op
- Register/register moves: widely used, if not exciting; includes conditional moves
- Load constant: integer immediate values and addresses
- Arithmetical/logical instructions
- Integer multiply, divide, and remainder
- Integer multiply-accumulate
- Loads and stores
- Jumps, subroutine calls, and branches
- Breakpoint and trap
- CP0 functions: instructions for CPU control
- Floating point
- Limited user-mode access to “under the hood” features: **rdhwr** and **synci**

8.7.1 *No-op*

nop: The MIPS instruction set is rich in nops, since any instruction with **zero** as a destination is guaranteed to do nothing. The favored one is **sll zero, zero, 0**, whose binary encoding is a zero-valued word.

ssnop: Another no-op, whose encoding implies **sll zero, zero, 1**.

This instruction must not be issued simultaneously with any other, so is guaranteed to take at least one cycle period to run. This is irrelevant to simple-pipeline CPUs, but can be useful for enforced programmed delays on more sophisticated implementations.

8.7.2 *Register/Register Moves*

move: Usually implemented with an **or** with the **\$zero** register. A few CPUs—where for some reason adding is better supported than logical operations—use **addu**.

Conditional Move

Useful branch-minimizing alternative (see section 8.5.3).

movf, movt: Conditional move of integer register, testing floating-point condition code.

movn, movz: Conditional move of integer register, subject to state of another register.

8.7.3 *Load Constant*

dla, la: Macro instructions to load the address of some labeled location or variable in the program. You only need **dla** when using 64-bit pointers (which you'll only do in big UNIX-like systems). These instructions accept the same addressing modes as do all loads and stores (even though they do quite different things with them).

dli, li: Load constant immediate. **dli** is the 64-bit version, not supported by all toolchains, and is only needed to load unsigned numbers too big to fit in 32 bits. This is a macro whose length varies according to the size of the constant.

lui: Load upper immediate. The 16-bit constant is loaded into bits 16–31 of a register, with bits 32–63 (if applicable) set equal to bit 31 and bits 0–15 cleared. This instruction is one-half of the pair of machine instructions that load an arbitrary 32-bit constant. Assembly programmers will probably never write this explicitly; it is used implicitly for macros like **li** (load immediate), **la** (load address), and above all for implementing useful addressing modes.

8.7.4 *Arithmetical/Logical*

The arithmetical/logical instructions are further broken down into the following types:

Add

add, addi, dadd, daddi: Obscure and rarely used alternate forms of **addu**, which trap when the result would overflow. Probably of use to COBOL compilers.

addu, addiu, daddu, daddiu: Addition, with separate 32-bit and 64-bit versions. Here and throughout the instruction set, 64-bit versions of instructions are marked with a leading “d” (for doubleword); also, you don’t need to specify the “immediate” mnemonic—you just feed the assembler a constant. If the constant you need can’t be represented in the 16-bit field provided in the instruction, then the assembler will produce a sequence of instructions.

dsub, sub: Subtract variants that trap on overflow.

dsubu, subu: Regular 64- and 32-bit subtraction (there isn’t a subtract-immediate, of course, because the constant in add-immediate can be negative).

Miscellaneous Arithmetic

abs, dabs: Absolute value; expands to set and branch (or conditional move if there is one).

dneg, neg, dnegu, negu: Unary negate; mnemonics without U will trap on overflow.

Bitwise Logical Instructions

and, andi, or, ori, xor, xori, nor: Three-operand bitwise logical operations. Don't write the "immediate" types—the assembler will generate them automatically when fed a constant operand. Note that there's no **nori** instruction.

not: Two-operand instruction implemented with **nor**.

Shifts and Rotates

drol, dror, rol, ror: Rotate right and left; expand to a four-instruction sequence.

dsll, dsll32, dsllv: 64-bit (double) shift-left, bringing zeros into low bits. The three different instructions provide for different ways of specifying the shift amount: by a constant 0–31 bits, by a constant 32–63 bits, or by using the low 6 bits of the contents of another register. Assembly programmers should just write the **dsll** mnemonic.

dsra, dsra32, dsrav: 64-bit (double) shift-right arithmetic. This is "arithmetic" in that it propagates copies of bit 63—the sign bit—into high bits. That means it implements a correct division by a power of 2 when applied to signed 64-bit integer data. Always write the **dsra** mnemonic; the assembler will choose the instruction format according to how the shift amount is specified.

dsrl, dsrl32, dsrlv: 64-bit (double) shift-right logical. This is "logical" in that it brings zeros into high bits. Although there are three different instructions, assembly programmers should always use the **dsrl** mnemonic; the assembler will choose the instruction format according to how the shift amount is specified.

sll, sllv: 32-bit shift-left. You only need to write the **sll** mnemonic.

sra, srav: Shift-right arithmetic (propagating the sign bit). Always write **sra**.

srl, srlv: Shift-right logical (bringing zeros into high bits). Always write **srl**.

Set if...

slt, slti, sltiu, sltu: Hardware instructions, which write a 1 if the condition is satisfied and a 0 otherwise. Write **slt** or **sltu**.

seq, sge, sgeu, sgt, sgtu, sle, sleu, sne: Macro instructions to set the destination according to more complex conditions.

8.7.5 *Integer Multiply, Divide, and Remainder*

The integer multiply and divide machine instructions are unusual, because the MIPS multiplier is a separate unit not built into the normal pipeline and takes much longer to produce results than regular integer instructions. Machine instructions are available to fire off a multiply or divide, which then proceeds in parallel with the instructions.

The same multiply unit provides integer multiply-accumulate and multiply-add instructions (see section 8.7.6.)

As a result of being handled by a separate unit, multiply/divide instructions don't include overflow or divide-by-zero tests (they can't cause exceptions because they are running asynchronously) and don't usually deliver their results into general-purpose registers (it would complicate the pipeline by fighting a later instruction for the ability to write the register file). Instead, multiply/divide results appear in the two separate registers **hi** and **lo**. You can only access these values with the two special instructions **mfhi** and **mflo**. Even in the earliest MIPS CPUs, the result registers are interlocked: If you try to read the result before it is ready, the CPU will stall until the data arrives.

However, when you write the usual assembly mnemonics for multiply/divide, the assembler will generate a sequence of instructions that simulate a three-operand instruction and perform overflow checking. A **div** (signed divide) may expand into as many as 13 instructions. The extra instructions are usually placed between the **div**, which starts the divide, and the **mflo**, which retrieves the result. The extra instructions look inefficient, but they run in parallel with the hardware divider; the divide itself takes 7–75 cycles on most MIPS CPUs.

MIPS Inc.'s assembler will convert constant multiplication and division/remainder by constant powers of 2 into the appropriate shifts, masks, and so on. But the assembler found in most toolchains is likely to leave this to the compiler.

By a less-than-obvious convention, a multiply or divide written with the destination register **zero** (as in **div zero, s, t**) will give you the raw machine instruction.⁵ It is then up to you to fetch the result from **hi** and/or **lo** and to do any checking you need.

Following is the comprehensive list of multiply/divide instructions.

ddiv, ddivu, div, divu: Three-operand macro instruction for integer division, with 64-/32-bit and signed/unsigned options. All trap on divide-by-zero; signed types trap on overflow. Use destination **zero** to obtain just the divide-start instruction.

5. Some toolkits interpret special mnemonics, **mult** for multiplication and **divd** for division, for the machine instructions. However, specifying **zero** as the destination, though bizarre, is more portable.

ddivd, ddivdu, divd, divdu: Mnemonics for raw machine instruction provided by some toolchains. It is better to use **ddiv zero, . . .** instead.

divo, divou: Explicit name for divide with overflow check, but really just the same as writing **div, divu**.

dmul, mul: Three-operand 64-/32-bit multiply instruction. There is no overflow check; as a result, there doesn't need to be an unsigned version of the macro—the truncated result is the same for signed and unsigned interpretation. Assemblers will expand it into an equivalent macro if they know they are assembling for pre-MIPS32 CPUs that don't implement it.

mulo, mulou, dmulo, dmulou: Multiply macros that trap if the result overflows beyond what will fit in one general-purpose register.

dmult, dmultu, mult, multu: The machine instruction that starts off a multiply, with signed/unsigned and 32-/64-bit variants. The result never overflows, because there's 64 and 128 bits' worth of result, respectively. The least significant part of a result gets stored in **lo** and the most significant part in **hi**.

drem, dremu, rem, remu: Remainder operations, implemented as a divide followed by **mfhi**. The remainder is kept in the **hi** register.

mfhi, mflo, mthi, mtlo: Move from **hi** and so on. These are instructions for accessing the integer multiply/divide unit result registers **hi** and **lo**. You won't write the **mflo/mfhi** instructions in regular code if you stick to the synthetic **mul** and **div** instructions, which retrieve result data for themselves.

MIPS integer multiply, **mult** or **multu**, always produces a result with twice the bit length of the general-purpose registers, eliminating the possibility of overflow. The high-order and low-order register-sized pieces of the result are returned in **hi** and **lo**, respectively.

Divide operations put the result in **lo** and the integer remainder in **hi**. **mthi** and **mtlo** are used only when restoring the CPU state after an exception.

8.7.6 *Integer Multiply-Accumulate*

Some MIPS CPUs have various forms of integer multiply-accumulate instructions—none of them in a MIPS standard instruction set. All these instructions take two general-purpose registers and accumulate into **lo** and **hi**. As usual, “u” denotes an unsigned variant, but otherwise the mnemonic (and instruction code) is specific to a particular CPU implementation.

dmaddl16, maddl16: Specific to NEC Vr4100, these variants gain speed by only accepting 16-bit operands, making them of very limited use to a C compiler. **dmaddl16** accumulates a 64-bit result in the 64-bit **lo** register.

mad, madu: Found in Toshiba R3900, IDT R4640/4650, and QED CPUs, these take two 32-bit operands and accumulate a 64-bit result split between the **lo**

and **hi** registers. The Toshiba R3900 allows a three-operand version, **madd, s, t**, in which the accumulated value is also transferred to the general-purpose register **d**.

8.7.7 *Loads and Stores*

This subsection lists all the assembler's integer load/store instructions and anything else that addresses memory. Note the following points:

- There are separate instructions for the different data widths supported: 8 bit (byte), 16 bit (halfword), 32 bit (word), and 64 bit (doubleword).
- For data types smaller than the machine register, there's a choice of zero-extending ("u" suffix for unsigned) or sign-extending the operation.
- All the instructions listed here may be written with any addressing mode the assembler supports (see section 9.4).
- A store instruction is written with the source register first and the address register second to match the syntax for loads; this breaks the general rule that in MIPS instructions the destination is first.
- Machine load instructions require that the data be naturally aligned (halfwords on a two-byte boundary, words on a four-byte boundary, doublewords on an eight-byte boundary). But the assembler supports a complete set of macro instructions for loading data that may be unaligned, and these instructions have a "u" prefix (for unaligned).

All data structures that are declared as part of a standard C program will be aligned correctly. But you may meet unaligned data from addresses computed at run time, data structures declared using a nonstandard language extension, data read in from a foreign file, and so on.

- All load instructions deliver their result at least one clock cycle later in the pipeline than computational instructions. For any MIPS CPU, efficiency is maximized by filling the load delay slot with a useful but nondependent instruction.

In the oldest (MIPS I) CPUs the programmer was required to ensure a one-clock delay after a load: Assemblers recognizing MIPS I CPUs will automatically do that by inserting a **nop** if necessary.

Following is a list of the instructions.

lb, lbu: Load byte then sign-extend or zero-extend, respectively, to fill the whole register.

ld: Load doubleword (64 bits). This machine instruction is available only on 64-bit CPUs, but assemblers for 32-bit targets will often implement it as a macro instruction that loads 64 bits from memory into two consecutive integer registers. This is probably a really bad idea, but someone wanted some compatibility.

ldl, ldr, lwl, lwr, sdl, sdr, swl, swr: Load/store left/right, in word/doubleword versions. Used in pairs to implement unaligned load/store operations like **ulw**, though you can always do it for yourself (see section 8.5.1).

lh, lhu: Load halfword (16 bits), then sign-extend or zero-extend to fill the register.

ll, lld, sc, scd: Load-linked and store-conditional (32- and 64-bit versions); strange instructions for semaphores (see section 8.5.2).

lw, lwu: Load word (32 bits), then sign-extend or zero-extend to fill the register. **lwu** is found only in 64-bit CPUs.

pref, prefix: Prefetch data into the cache (see section 8.5.8). This is not available on MIPS III and earlier CPUs and may be a no-op. While **pref** takes the usual addressing modes, **prefix** adds a register+register mode implemented in a single instruction.

sb: Store byte (8 bits).

sd: Store doubleword (64 bits). This may be a macro (storing two consecutive integer registers into a 64-bit memory location) for 32-bit CPUs.

sh: Store halfword (16 bits).

sw: Store word (32 bits).

uld, ulh, ulhu, ulw, usd, ush, usw: Unaligned load/store macros. The doubleword and word versions are implemented as macro instructions using the special load left/load right instructions; halfword operations are built as byte memory accesses, shifts, and **ors**. Note that normal delay slot rules do not apply between the constituent load left/load right of an unaligned operation; the pipeline is designed to let them run head to tail. More on unaligned loads and how they're used in section 2.5.2.

Floating-Point Load and Store

l.d, l.s, s.d, s.s: Load/store double (64-bit format) and single (32-bit format). Alignment is required, and no unaligned versions are given here. On 32-bit CPUs, **l.d** and **s.d** are two-instruction macros that load/store two 32-bit chunks of memory into/from consecutive FP registers (see section 7.5). These instructions are also called **ldc1, ldc1, ldc1, and swc1** (load/store word/double to coprocessor 1), but don't write them like that.

ldxc1, lwxcl, sdxcl, swxc1: Base register + offset register addressing mode loads and stores. In the instruction **ldxc1 fd, i(b)**, the full address must lie in the same program memory region as is pointed to by the base register **b** or bad things might happen.

If your toolkit will accept syntax like **l.d fd, i(b)**, then use it.

8.7.8 *Jumps, Subroutine Calls, and Branches*

The MIPS architecture follows Motorola nomenclature for these instructions, as follows:

- PC-relative instructions are called “branch” and absolute-addressed instructions “jump”; the operation mnemonics begin with **b** or **j**.
- A subroutine call is “jump and link” or “branch and link,” and the mnemonics end ...**al**.
- All the branch instructions, even branch and link, are conditional, testing one or two registers. Unconditional versions can be and are readily synthesized—for example, **beq \$0, \$0, label**.

j: This instruction transfers control unconditionally to an absolute address. Actually, **j** doesn’t quite manage a 32-bit address: The top 4 address bits of the target are not defined by the instruction, and the top 4 bits of the current PC value are used instead. Most of the time this doesn’t matter; 28 bits still gives a maximum code size of 256 MB.

To reach a long way away, you must use the **jr** (jump to register) instruction, which is also used for computed jumps. You can write the **j** mnemonic with a register, but it’s quite popular not to do so.

jal, jalr: These implement a direct and indirect subroutine call. As well as jumping to the specified address, they store the return address (the instruction’s own address + 8) in register **ra**, which is the alias for **\$31**.⁶ Why add eight to the program counter? Remember that jump instructions, like branches, always execute the immediately following branch delay slot instruction, so the return address needs to be the instruction *after* the branch delay slot. Subroutine return is done with a jump to register, most often **jr ra**.

PC-relative subroutine calls can use the **bal**, **bgezal**, and **bltzal** instructions. The conditional branch-and-link instructions save their return address into **ra** even when the condition is false, which can be useful when doing a computation using the current instruction’s address.

b: Unconditional PC-relative (though relatively short-range) branch.

bal: PC-relative function call.

bc0f, bc0fl, bc0t, bc0tl, bc2f, bc2fl, bc2t, bc2tl: Branches that test the coprocessor 0 or coprocessor 2 condition bit, neither of which exist on most modern CPUs. On older CPUs, these test an input pin.

6. In fact the **jalr** machine instruction allows you to specify a register other than **\$31** to receive the return address, but this is seldom useful, and the assembler will automatically put in **\$31** if you do not specify one.

bc1f, bc1fl, bc1t, bc1tl: Branch on floating-point condition bit (multiple in later CPUs).

beq, beql, beqz, beqzl, bge, bge1, bgeu, bgeul, bgez, bgez1, bgt, bgt1, bgtu, bgtul, bgtz, bgtzl, ble, ble1, bleu, bleul, blez, blez1, blt, blt1, bltu, bltul, bltz, bltz1, bne, bne1, bnez, bnez1: A comprehensive set of two- and one-operand compare-and-branch instructions, most of them macros.

bgezal, bgezall, bltzal, bltzall: Raw machine instructions for conditional function calls, if you ever need to do such a thing.

8.7.9 *Breakpoint and Trap*

break: Causes an exception of type “break.” It is used in traps from assembler-synthesized code and by debuggers.

sdbbp: Breakpoint instruction that generates an EJTAG exception, as described in section 12.1.

syscall: Causes an exception type conventionally used for system calls.

teq, teqi, tge, tgei, tgeiu, tgeu, tlt, tlti, tltiu, tltu, tne, tnei: Conditional exception, testing various one- and two-operand conditions. These are for compilers and interpreters that want to implement runtime checks to array bounds and so on.

8.7.10 *CP0 Functions*

CP0 functions can be classified under the following types:

Move To/From

cfc0, ctc0: Move data in and out of CP0 control registers, of which there are none in any MIPS CPUs defined to date. But there may be such registers one day soon.

mfc0, mtc0, dmfc0, dmtc0: Move data between CP0 registers and general-purpose registers.

cfc2, ctc2, dmfc2, dmtc2, mfc2, mtc2: Instructions for coprocessor 2, if implemented. It has not often been done.

Special Instructions for CPU Control

eret: Return from exception (see Chapter 5).

dret: Return from exception (R6000 version). This instruction is obsolete and not described in this book.

rfe: Exception-end instruction from MIPS I—essentially obsolete. Curiously, **rfe** only restores the status register and relies on being placed in the branch delay slot of a **jr** instruction, which carries you back to the restart location.

cache: The polymorphic cache control instruction, as described in section 4.9.

sync: Memory access synchronizer for CPUs that might perform load/stores out of order (see section 8.5.9). Unlike all the other instructions in this section, it does not use CP0 encoding and is legal in a user-privilege program.

tlbp, tlbr, tlbwi, tlbwr: Instructions to control the TLB, or memory translation hardware (see section 6.3).

standby, suspend: Enter power-down mode (NEC Vr4100 CPUs).

8.7.11 *Floating Point*

Floating-point instructions are listed in section 8.3.

8.7.12 *Limited User-Mode Access to “Under the Hood” Features*

rdhwr (read hardware register): Allows a user-privilege program to read a few snippets of information that would normally only be visible to a kernel. See section 8.5.12.

synci: User-privilege instruction. It specifies an address like any load/store, and it affects the cache-line-sized, aligned piece of memory that holds the addressed byte. Use it when writing instructions for subsequent execution. **synci** does what is necessary to ensure that the CPU will correctly execute any instructions that the program previously wrote in this chunk of memory (often that means a write-back from the L1 D-cache and an invalidate of any previously held I-cache contents).

sync: Load/store barrier, listed here to emphasize that it’s not just a kernel-privileged instruction. It’s described in section 8.5.9.

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Reading MIPS Assembly Language

This chapter aims to help you read MIPS assembly code—specifically, that written for the MIPS version of the GNU *as* assembly program, since this is by far the most widely used assembler for MIPS these days.

Since this is just an introduction, we'll be working here with common 32-bit MIPS instructions that have been part of the ISA for many years, dating all the way back to MIPS I and MIPS II.

This chapter won't teach you how to write assembly code—that could easily fill another book, all by itself. The vast majority of readers will likely do all their programming in C, and will only occasionally run into an existing assembly file that they need to understand and perhaps modify in a small way; if your use of MIPS assembly goes much beyond this, you'll need to have a look at the documentation that comes with your MIPS toolchain.

Learning to read MIPS assembly means more than just gaining familiarity with the list of machine instructions. There are several reasons why this is so:

- MIPS assemblers provide a large number of extra macro instructions, so the assembly instruction set is much more extensive than the list of machine instructions. The list in the previous chapter includes all the macro instructions recognized by the GNU toolchain's assembler,¹ as well as the machine instructions that are recognized directly by the CPU hardware.
- MIPS assemblers recognize and interpret a set of special keywords called “directives” or “pseudo-ops” that are used to manage their behavior. For example, particular keywords are used to mark the start and end of program functions; to control instruction ordering and optimization; and so on.

1. At least, those recognized by a summer-2005 snapshot of the assembler as maintained by MIPS Technologies, which was perhaps a little ahead of the public archives at that time.

- It's almost universal practice (though not compulsory) to pass assembly source files as written by a human programmer through the C preprocessor before handing them to the assembler itself. Used appropriately, this can make for much more readable source files, in which items such as machine registers are referenced by names that convey meaning to the human programmer; the task of converting these names to the less readable forms required by the assembler itself is best left to the toolchain.

Many programmers find it helpful to follow a simple convention: an assembly source file that's fed to the preprocessor as input is given a name suffixed with ".S," and the preprocessed version generated as output is given the same name suffixed with ".s"; this makes it easier to manage the two forms of assembly source files.

Before you read too much further, you may find it useful to go back and refresh your memory of Chapter 2, which describes the low-level machine instruction set, data types, addressing modes, and conventional register usage. Even if you're already familiar with that material, it may still be helpful to insert an extra bookmark there so you can easily refer back to it as we continue.

9.1 A Simple Example

We'll use the same example as in Chapter 8: an implementation of the C library function `strcmp(1)`. But this time we'll include essential elements of assembly syntax and also show some hand-optimized and -scheduled code. The algorithm shown is somewhat cleverer than a naive `strcmp()` function; we'll start with this code—still in C—in a form that has all the operations separated out to make them easier to play with, as follows:

```
int
strcmp (char *a0, char *a1)
{
    char t0, t1;
    while (1) {
        t0 = a0[0];
        a0 += 1;
        t1 = a1[0];
        a1 += 1;
        if (t0 == 0)
            break;
        if (t0 != t1)
            break;
    }
    return (t0 - t1);
}
```

The execution time for the code in this initial form will suffer because of the two conditional branches (corresponding to the two `if()` statements) and the two loads (corresponding to the two array-indexing operations) that appear in each iteration of the loop; each of these branches and loads introduces a delay slot, and there just isn't enough work in the body of the loop to fill all them. Furthermore, as the code works its way along a pair of strings, it's forced to take a loop-closing branch on every pair of bytes compared (corresponding to the closing brace of the `while()` statement).

We can make useful improvements to this code before we even begin translating it into assembly. The biggest change is to unroll the loop so that it performs two comparisons per iteration; we can also move one of the loads down to the tail of the loop. With these changes, the delay slot for every load and branch can be filled with useful work:

```
int strcmp (char *a0, char *a1) {
    char t0, t1, t2;

    /* first load moved to loop end,
       so load for first iteration here */
    t0 = a0[0];

    while (1) {
        /* first byte */
        t1 = a1[0];
        if (t0 == 0)
            break;
        a0 += 2;
        if (t0 != t1)
            break;

        /* second byte */
        t2 = a0[-1]; /* we already incremented a0 */
        t1 = a1[1];  /* didn't increment a1 yet */

        if (t2 == 0)
            /* label t21 in assembler */
            return t2-t1;
        a1 += 2;
        if (t1 != t2)
            /* label t21 in assembler */
            return t2-t1;
        t0 = a0[0];
    }
    /* label t01 in assembler */
    return t0-t1;
}
```

Now that we've eliminated the basic inefficiencies from the code, let's translate it into MIPS assembly:

```
#include <mips/asm.h>
#include <mips/regdef.h>

LEAF(strcmp)
    .set      noreorder
    lbu       t0, 0(a0)
1:  lbu       t1, 0(a1)
    beq       t0, zero, .t01      # load delay slot
    addu      a0, a0, 2           # branch delay slot
    bne       t0, t1, .t01
    lbu       t2, -1(a0)         # branch delay slot
    lbu       t1, 1(a1)          # load delay slot
    beq       t2, zero, .t21
    addu      a1, a1, 2           # branch delay slot
    beq       t2, t1, 1b
    lbu       t0, 0(a0)          # branch delay slot

.t21: j       ra
    subu      v0, t2, t1         # branch delay slot

.t01: j       ra
    subu      v0, t0, t1         # branch delay slot
    .set      reorder
END(strcmp)
```

The comments in the example above help to clarify the way the instructions are scheduled; but before taking a closer look at that, we should explain the many new constructs that appear in the example above. Let's examine them in the order they appear:

- *#include*: This file takes advantage of the C preprocessor *cpp* to give mnemonic names to constants and of defining simple text-substitution macros. Here *cpp* is being used to put two header files in line before submitting the text to the assembler; *mips/asm.h* defines the macros **LEAF** and **END** (discussed further below), and *mips/regdef.h* defines the conventional register names like **t0** and **a0**, as described in section 2.2.1.
- *Macros*: We're using two macros defined in *mips/asm.h*, **LEAF** and **END**. Here is the basic definition for **LEAF**:

```
#define LEAF(name) \
    .text; \
    .globl name; \
    .ent    name; \
name:
```

LEAF is used to define a simple subroutine (one that calls no other subroutine and hence is a “leaf” on the calling tree—see section 11.2.9). Nonleaf functions have to do much more work saving variables, returning addresses, and so on. Unless you’re involved in very specialized programming, it’s unlikely that you’ll ever need to write a nonleaf function in assembly—it will almost certainly make better sense to write it in C, perhaps with a call to a leaf function that encapsulates any work that really needs to be coded in assembly. Note the following:

- **.text** tells the assembler that until further notice, it should direct any code that it subsequently produces into the section of the object file called “.text”; object files compiled from C use this same name for the section that holds all the code.
- **.globl** declares “name” as global, to be included in the module’s symbol table as a name that should be unique throughout the whole program. This mimics what the C compiler does to function names (unless they are marked `static`).
- **.ent** has no effect on the code produced but tells the assembler to mark this point as the beginning of a function called “name” and to use that information in debug records.
- **name** makes “name” a label for this point in the assembler’s output and marks the place where a subroutine call to function “name” will start.

END defines two more assembly items:

```
#define END(name) \
    .size name, .-name; \
    .end    name
```

- **.size** means that in the symbol table, “name” will now be listed with a size that corresponds to the number of bytes of instructions used.
- **.end** delimits the function for debug purposes.

■ **.set directives:**

These are used to tell the assembler how to do its work. By default, MIPS assemblers try to fill branch and load delay slots automatically by reordering the instructions around them (but don’t worry—the assembler never moves instructions around in ways that are unsafe; it leaves delay slots unfilled if it can’t find a safe rearrangement). Most of the time this is helpful, because it means you don’t have to worry about filling delay slots as you write your assembly files.

But what if we need to retain precise control over instruction ordering, as in the case of a heavily used library function? This is the purpose of the **.set noreorder** directive: It tells the assembler to suppress its reordering capabilities until the next time it encounters a corresponding **.set reorder** directive.

For the section of code enclosed between this pair of directives, we’re telling the assembler to put the op-codes it generates into the object file in the same order as the instructions are written.

- *Labels*: “1:” is a numeric label, which most assemblers will accept as a local label. You can have as many labels called “1” as you like in a program; a reference to “1f” (forward) will get the next one in sequence and “1b” (back) the previous one. That’s very useful.
- *Instructions*: You’ll notice some unexpected sequences, since the **.set noreorder** exposes the underlying branch delay slots, and leaves us to ensure (for efficiency) that load data is never used by the following instruction.

For example, note the use of register **t2** in the second half of the unrolled loop. It’s necessary to use a second register because the **lbu t2, -1(a0)** is in the delay slot of the preceding branch instruction and therefore must not overwrite **t0**—if the branch is taken, the code at its target will make use of the value in **t0**.

9.2 Syntax Overview

Working through the preceding example showed you how most of the important assembly directives are used in practice, and should also have helped you get used to the way MIPS instructions are written in assembly source files. Now we’ll summarize these things a bit more systematically. If you’ve used an assembler on a UNIX-like system before, then the main ideas should already be reasonably familiar.

9.2.1 *Layout, Delimiters, and Identifiers*

For this, you need to be familiar with C. But when you’re reading assembly code, note that:

- Assembly code is line oriented, and an end-of-line delimits an instruction or directive. You can have more than one instruction/directive on each line, however, as long as they are separated by semicolon (“;”) characters.
- All text from a “#” to the end of the line is a comment and is ignored. But don’t put a “#” in the leftmost column: The C preprocessor *cpp* treats such lines specially and gets confused, and you will probably want to use *cpp*. If you know your code is going to be run through *cpp*, you can use C-style comments: `/ * ... */`. These can span multiple lines if you like.
- Identifiers for labels and variables can be anything that’s legal in C but can also contain “\$” and “.”.

- In code you can use a number (decimal between 1 and 99) as a label. Conventional textual labels must be unique in the file, but you can use the same number as many times as you like. In a branch instruction, “1f” (forward) refers to the next “1:” label in the code, and “1b” (back) refers to the previous “1:” label. This device saves you inventing pointless names for little branches and loops. Reserve named labels for subroutine entry points or for exceptionally big jumps.
- It’s strongly recommended that you use MIPS conventional register names as summarized in Table 2.1; to do this, you should pass your source through the C preprocessor and `#include` a file probably called `mips/regdef.h`. If you choose not to use the preprocessor, remember that the assembler expects general-purpose register names to be written in the form “dollar-number,” such as `$3` to specify GP register 3, and that the GP register numbers run from 0 through 31.
- There is no direct analog of C’s “pointer” operator. But when the assembler is expecting a pointer-size value a label (or any other relocatable symbol) is replaced with its address. The identifier “.” (dot) represents the address of the current instruction or data declaration. You can even do some limited arithmetic with these things.
- Character constants and strings can be defined as in C.

9.3 General Rules for Instructions

MIPS assemblers typically allow some convenient shortcuts. You can provide fewer operands than the machine instruction needs, and the assembler will interpret that as a two-operand form. Or you can specify a constant, where the machine instruction needs a register, and the assembler will infer that you wanted the immediate variant of the instruction. This section summarizes the common cases.

9.3.1 *Computational Instructions: Three-, Two-, and One-Register*

MIPS computational machine instructions are three-register operations, that is, they are arithmetic or logical functions with two inputs and one output. For example:

$$d = s + t$$

is written as **addu** *d, s, t*.

We mentioned as well that any or all of the register operands may be identical. To produce a CISC-style two-operand instruction you just have to use the destination register as a source operand. The assembler will do this for you

automatically if you omit *s*: It will treat **addu** *d, s* in exactly the same way as **addu** *d, d, s*.

Unary operations like **neg**, **not** are always synthesized from one or more of the three-register instructions. The assembler expects a maximum of two operands for these instructions, so **negu** *d, s* is the same as **subu** *d, zero, s* and **not** *d* will be assembled as **nor** *d, zero, d*.

Probably the most common register-to-register operation is **move** *d, s*. The assembler implements this ubiquitous instruction as **or** *d, zero, s*.

9.3.2 *Immediates: Computational Instructions with Constants*

In assembly or machine language, a constant value encoded within an instruction is called an *immediate* value. Many of the MIPS arithmetical and logical operations have an alternative form that uses a 16-bit immediate in place of *t*. The immediate value is first sign-extended or zero-extended to 32 bits; the choice of how it's extended depends on the operation, but in general arithmetical operations sign-extend and logical operations zero-extend.

Although an immediate operand produces a different machine instruction from its three-register version (e.g., **addiu** instead of **addu**), there is no need for the programmer to write this explicitly. The assembler checks whether the final operand is a register or an immediate and chooses the correct machine instruction accordingly:

```
addu    $2, $4, 64    ⇒    addiu    $2, $4, 64
```

If an immediate value is too large to fit into the 16-bit field in the machine instruction, then the assembler helps out again. It automatically loads the constant into the *assembler temporary register* **at**/\$1 and then uses it to perform the operation:

```
addu    $4, 0x12345    ⇒    li      at, 0x12345
                                addu    $4, $4, at
```

Note the **li** (load immediate) instruction, which you won't find in the machine's instruction set; **li** is a heavily used macro instruction that loads an arbitrary 32-bit integer value into a register without the programmer having to worry about how it gets there—the assembler automatically chooses the best way to code the operation, according to the properties of the integer value.

When the 32-bit value lies between ± 32 K, the assembler can use a single **addiu** with **\$0**; when bits 16–31 are all zero, it can use **ori**; when bits 0–15 are all zero, it will use **lui**; and when none of these is possible, it will choose a **lui/ori** pair:

```
li      $3, -5          ⇒    addiu    $3, $0, -5
li      $4, 0x8000       ⇒    ori     $4, $0, 0x8000
```

```

li      $5, 0x120000    ⇒    lui      $5, 0x12
li      $6, 0x12345     ⇒    lui      $6, 0x1
                                ori      $6, $6, 0x2345

```

Assembly instructions that expand into multiple machine instructions are troublesome if you're using `.set noreorder` directives to take control of managing branch delay slots. If you write a multi-instruction macro in a delay slot, the assembler should warn you.

9.3.3 Regarding 64-Bit and 32-Bit Instructions

We saw earlier that the extension of the MIPS architecture to 64 bits (section 2.7.3) paid careful attention to ensuring that the behavior of MIPS32 programs remains unchanged, even if they're run on MIPS64 machines; in MIPS64 machines, the execution of MIPS32 instructions always leaves the upper 32 bits of any GP register set either to all ones or all zeros (reflecting the value of bit 31).

Many 32-bit instructions carry over directly to 64-bit systems—all bitwise logical operations, for example—but arithmetic functions don't. Adds, subtracts, shifts, multiplies, and divides all need new versions. The new instructions are named by prefixing the old mnemonic with **d** (double): For example, the 32-bit addition instruction **addu** is augmented by the new instruction **daddu**, which does full 64-bit-precision arithmetic. A leading “d” in an instruction mnemonic generally means “double.”

9.4 Addressing Modes

As noted previously, the hardware supports only one addressing mode, *base-reg+offset*, where *offset* is in the range -32768 to 32767 . However, the assembler will synthesize code to access data at addresses specified in various other ways:

- *Direct*: A data label or external variable name supplied by you
- *Direct+index*: An offset from a labeled location specified with a register
- *Constant*: Just a large number, interpreted as an absolute 32-bit address
- *Register indirect*: Just register+offset with an offset of zero

When these methods are combined with the assembler's willingness to do simple constant arithmetic at compile time and the use of a macro processor, you are able to do most of what you might want. Here are some examples:

<i>Instruction</i>	<i>Expands to</i>
<code>lw \$2, (\$3)</code>	<code>⇒ lw \$2, 0(\$3)</code>
<code>lw \$2, 8+4(\$3)</code>	<code>⇒ lw \$2, 12(\$3)</code>

lw	\$2, <i>addr</i>	⇒	lui	at, %hi(<i>addr</i>)
			lw	\$2, %lo(<i>addr</i>) (at)
sw	\$2, <i>addr</i>(\$3)	⇒	lui	at, %hi(<i>addr</i>)
			addu	at, at, \$3
			sw	\$2, %lo(<i>addr</i>) (at)

The symbol ***addr*** in the above examples can be any of the following:

- A relocatable symbol—the name of a label or variable (whether in this module or elsewhere)
- A relocatable symbol $\pm$ a constant expression (the assembler/linker can handle this at system build time)
- A 32-bit constant expression (e.g., the absolute address of a device register)

The constructs **%hi()** and **%lo()** represent the high and low 16 bits of the address. This is not quite the straightforward division into low and high half-words that it looks, because the 16-bit offset field of an **lw** is interpreted as signed. So if the ***addr*** value is such that bit 15 is a 1, then the **%lo(*addr*)** value will act as negative, and we need to increment **%hi(*addr*)** to compensate:

<i>addr</i>	%hi(<i>addr</i>)	%lo(<i>addr</i>)
0x1234.5678	0x1234	0x5678
0x1000.8000	0x1001	0x8000

The **la** (load address) macro instruction provides a similar service for addresses to that provided for integer constants by **li**:

la	\$2, 4(\$3)	⇒	addiu	\$2, \$3, 4
la	\$2, <i>addr</i>	⇒	lui	at, %hi(<i>addr</i>)
			addiu	\$2, at, %lo(<i>addr</i>)
la	\$2, <i>addr</i>(\$3)	⇒	lui	at, %hi(<i>addr</i>)
			addiu	\$2, at, %lo(<i>addr</i>)
			addu	\$2, \$2, \$3

In principle, **la** could avoid messing around with apparently negative **%lo()** values by using an **ori** instruction. But load/store instructions have a signed 16-bit address offset, and as a result the linker is already equipped with the ability to fix up addresses into two parts that can be added correctly. So **la** uses the add instruction to avoid the linker having to understand two different fix-up types.

9.4.1 *Gp-Relative Addressing*

A consequence of the way the MIPS instruction set is crammed into 32-bit operations is that accesses to compiled-in locations usually require at least two instructions, for example:

```
lw      $2, addr           ⇒      lui      at, %hi(addr)
                                   lw      $2, %lo(addr)(at)
```

In programs that make a lot of use of global or static data, this can make the compiled code significantly fatter and slower.

Early MIPS compilers introduced a fix for this, which has been carried into most MIPS toolchains. It's usually called *gp-relative addressing*. This technique requires the cooperation of the compiler, assembler, linker, and start-up code to pool all of the “small” variables and constants into a single memory region; then it sets register **\$28** (known as the *global pointer* or **gp** register) to point to the middle of this region. (The linker creates a special symbol, **_gp**, whose address is the middle of this region. The address of **_gp** must then be loaded into the **gp** register by the start-up code, before any load or store instructions are used.) So long as all the variables together take up no more than 64 KB of space, all the data items are now within 32 KB of the midpoint, so a load turns into:

```
lw      $2, addr           ⇒      lw      $2, addr - _gp(at)
```

The problem is that the compiler and assembler must decide which variables can be accessed via **gp** at the time the individual modules are compiled. The usual test is to include all objects of less than a certain size (eight bytes is the usual default). This limit can usually be controlled by the “**-G n**” compiler/assembler option; specifying “**-G 0**” will switch this optimization off altogether.

While it is a useful trick, there are some pitfalls to watch out for. You must take special care when writing assembly code to declare global data items consistently and correctly:

- Writable, initialized small data items must be put explicitly into the **.sdata** section.
- Global common data must be consistently declared with the correct size:


```
.comm  smallobj, 4
.comm  bigobj, 100
```
- Small external variables should also be explicitly declared:


```
.extern smalltext, 4
```
- Most assemblers will not act on a declaration unless it precedes the use of the variable.

In C, you must declare global variables correctly in all modules that use them. For external arrays, either omit the size, like this:

```
extern int extarray[];
```

or give the correct size:

```
int cmnarray[NARRAY];
```

Sometimes the way programs are run means this method can't be used. Some real-time operating systems (and many PROM monitors) are built with a separately linked chunk of code implementing the kernel, and applications invoke kernel functions with long-range subroutine calls. There's no cost-effective method by which you could switch back and forth between the two different values of `gp` that will be used by the application and OS, respectively. In this case either the applications or the OS (but not necessarily both) must be built with `-G 0`.

When the `-G 0` option has been used for compilation of any set of modules, it is usually essential that all libraries linked in with them should be compiled that way. If the linker is confronted with modules that disagree on whether a named variable should be put in the small or regular data sections, it's likely to give you peculiar and unhelpful error messages.

9.5 Object File and Memory Layout

This chapter concludes with a brief look at the way programs are typically laid out in system memory and notes some important points about the relationship between the memory layout and the object files produced by the toolchain. It's very useful to have a basic understanding of the way your code should appear after it's loaded into the system's memory, especially if you're going to face the task of getting MIPS code to run for the first time on newly developed system hardware.

The conventional code and data sections defined by MIPS conventions are illustrated (for ROMable programs) in Figure 9.1.

Within an assembly program the sections are selected as described in the groupings that follow.

.text, .rdata, and .data

Simply put the appropriate section name before the data or instructions, as shown in this example:

```
.rdata
msg:.asciiz "Hello world!\n"
```

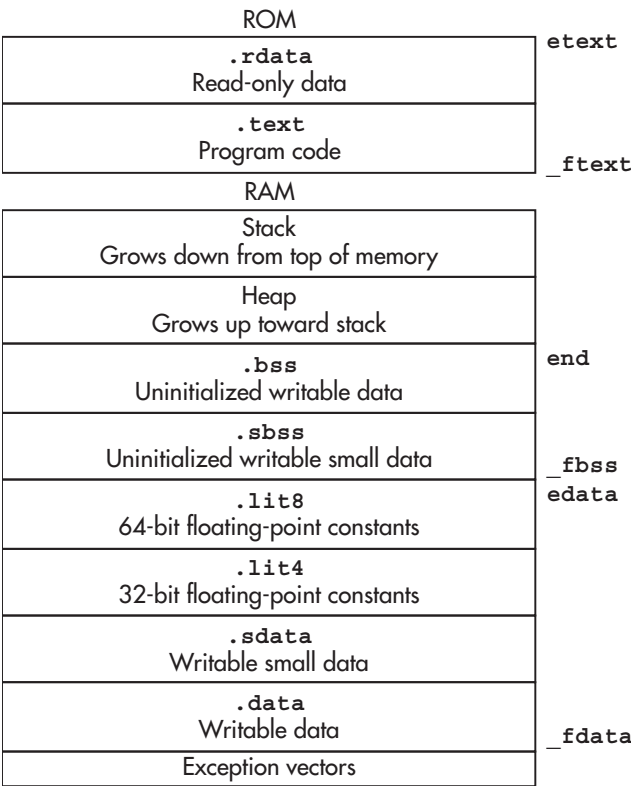


FIGURE 9.1 ROMable program’s object code segments and typical memory layout.

```
.data
table:
    .word    1
    .word    2
    .word    3

.text
func:sub    sp, 64
...
```

.lit4 and .lit8 Sections: Floating-Point Implicit Constants

You can’t write these section names as directives. They are read-only data sections created implicitly by the assembler to hold floating-point constants that

are given as arguments to the `li.s` or `li.d` macro instructions. Some assemblers and linkers will save space by combining identical constants.

`.lit4` and `.lit8` may be included in the “small data” region if the application is built to use gp-relative addressing.

.bss, .comm, and .lcomm Data

This section name is also not used as a directive. It is used to collect all static or global uninitialized data declared in C modules. It’s a feature of C that multiple same-named definitions in different modules are acceptable so long as not more than one of them is initialized, and the `.bss` section is often used for data that is not initialized anywhere. FORTRAN programmers would recognize this as what is called *common* data—that’s where the name of the directive comes from.

You always have to specify a size for the data (in bytes). When the program is linked, the item will get enough space for the *largest* size. If any module declares it in an initialized data section, all the sizes are used and that definition is used:

```
.comm    dbgflag, 4      # global common variable, 4 bytes
.lcomm   sum, 4          # local common variable, 8 bytes
.lcomm   array, 100      # local common variable, 100 bytes
```

“Uninitialized” is actually a misnomer. In C, static or global variables that are not explicitly initialized should be set to zero before the program starts—a job for the operating system or start-up code.

.sdata, Small Data, and .sbss

These sections are used as alternatives to the `.data` and `.bss` sections above by toolchains that want to separate out smaller data objects. Toolchains for MIPS processors do this because the resulting small-object section is compact enough to allow an efficient access mechanism that relies on maintaining a data pointer in a reserved register `gp`, as described in section 9.4.1.

Note that the `.sbss` is not a legal directive; the toolchain allocates a data item to the `.sbss` section if the item is declared with `.comm` or `.lcomm` and is of size smaller than the `-G` threshold value fed to the assembly program.

The implicit-constant sections `.lit4` and `.lit8` may be included in the small data region, according to the threshold setting.

When gp-relative addressing is used, `gp` will be initialized to point somewhere close to the midpoint of the “small data” region.

.section

Start an arbitrarily named section and supply control flags (which are object code specific and probably toolkit specific). See your toolkit manuals, and always use the specific section name directives for the common sections.

9.5.1 *Practical Program Layout, Including Stack and Heap*

The program layout illustrated in Figure 9.1 is suitable in most practical systems in which the code is stored in ROM and runs on a bare CPU (that is, without the services of any intermediate software such as an operating system). The read-only sections are likely to be located in an area of memory remote from the read/write sections.

The stack and heap are significant as areas of the system's address space, but it's important to understand that they're not known to the assembler or linker in the same way as, for example, the `.text` or `.data` sections. Typically, the stack and the heap are initialized and maintained by the runtime system. The stack is defined by setting the `sp` register to the top of available memory (aligned to an eight-byte boundary). The heap is defined by a global pointer variable used by functions such as `malloc()` functions; it's often initialized to the `end` symbol, which the linker has calculated as the highest location used by declared variables.

Special Symbols

Figure 9.1 also shows a number of special symbols that are automatically defined by the linker to allow programs to discover the start and end of their various sections. They are descended from conventions that grew up in UNIX-like OSs, and some are peculiar to the MIPS environment. Your toolkit might or might not define all of them; those marked with a ✓ in the following list are pretty certain to be there:

<i>Symbol</i>	<i>Standard?</i>	<i>Value</i>
<code>_ftext</code>		Start of text (code) segment
<code>etext</code>	✓	End of text (code) segment
<code>_fdata</code>		Start of initialized data segment
<code>edata</code>	✓	End of initialized data segment
<code>_fbss</code>		Start of uninitialized data segment
<code>end</code>	✓	End of uninitialized data segment

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Porting Software to the MIPS Architecture

Very few projects require absolutely all of their software to be created from scratch; the vast majority make use of at least some code that already exists—at the application level, in the operating system, or both. You may well find, however, that the existing code you’d like to use in your MIPS system was originally developed for some other microprocessor family. Of course, at a minimum, you’ll need to recompile the source code to create a new set of binaries for MIPS; but as we’ll see, the task may be more complicated than that. *Portability* refers to the ease with which a piece of software can be transferred successfully and correctly to a new environment, particularly a new instruction set. Porting a substantial body of software is rarely easy, and the level of difficulty tends to rise sharply if the software in question is (or includes) an OS or OS-related software such as device drivers.

High-level software (Linux application code or the like) will typically have been written with at least some notion of portability and will quite probably have been used in several environments already, so there’s a reasonable likelihood that you’ll be able to recompile it without having to make changes. Low-level software—perhaps a large portion of the source code, for some embedded systems—is more troublesome. Software that has been developed exclusively in just one particular environment is especially likely to present portability problems, since its creators may not have recognized any particular need to avoid or resolve them. The object of this chapter is to draw your attention to areas that are particularly likely to cause problems when you’re porting software to MIPS.

The parts of a system that drive the lowest-level hardware are inevitably nonportable; embedded systems are typically subject to significant design upgrades every couple of years or so, and it’s just not reasonable (and certainly not cost effective) to insist that the original hardware/software interfaces be preserved throughout such changes.

10.1 Low-Level Software for MIPS Applications: A Checklist of Frequently Encountered Problems

The following are problems that have come up fairly frequently:

- *Endianness*: The computer world is divided into little- and big-endian camps, and a gulf of incomprehension falls between them. Most MIPS CPUs can be set up to run either big-endian or little-endian; but even if you already know which way your MIPS system will be configured, it's strongly recommended that you make sure you understand this issue thoroughly. It's caught out many experienced developers before you, and it will catch out some more. Read about it in section 10.2.

- *Data layout and alignment in memory*: Your program may make unportable assumptions about the memory layout of data declared in C. It's almost always unportable to use C `struct` declarations to map input files or data received through a communication link, for example. Danger can lurk in a program that employs multiple views of private data with differently typed pointers or unions.

However, data layout goes together with a description of other conventions (for register use, argument passing, and stack handling) and you'll find that in the next chapter: If you need to take a peek ahead, it's in section 11.1.

- *Need for explicit cache management*: You may find that code you'd like to reuse was developed for a microprocessor that didn't implement caches at all, or one that used a CPU with caches that are "invisible" to software (almost all side effects of caching in PC-compatible processors are hidden by clever hardware, for instance). But most MIPS CPUs keep their hardware simple by letting some side effects remain visible and making software responsible for cache management; we'll describe what this means in section 10.3.

- *Memory access ordering and reordering*: In many modern embedded or consumer systems, data moving around the system may pass through a chain of subsystems as it moves from its source to its final destination. Those subsystems may themselves encapsulate a lot of complicated hardware, and may present you with unexpected problems. For example, pieces of information passed between the CPU and I/O devices may be forced to wait in queues, incurring variable amounts of delay; or they may be separated into several independent traffic streams, so the order in which they arrive at their respective destinations can't be guaranteed to match the order in which they were originally sent. Typical problems and solutions are discussed in section 10.4.

- *Writing it in C*: This is not so much a problem as an opportunity. But there are things you can do in C (and probably should do in preference to writing assembly code) that are fairly MIPS-specific. This section talks about inline assembly, using memory-mapped registers, and a ragbag of possible pitfalls using MIPS.

10.2 Endianness: Words, Bytes, and Bit Order

The word *endianness* was introduced to computer science by Danny Cohen (Cohen 1980). In an article of rare humor and readability, Cohen observed that computer architectures had divided up into two camps, based on an arbitrary choice of the way in which byte addressing and integer definitions are related in communications systems.

In Jonathan Swift’s *Gulliver’s Travels*, the “little-endians” and “big-endians” fought a war over the correct end at which to start eating a boiled egg. Swift was satirizing 18th-century religious disputes, and neither of his sides can see that their difference is entirely arbitrary. Cohen’s joke was appreciated, and the word has stuck. The problem is not just relevant to communications; it has implications for portability too.

Computer programs are always dealing with sequence and order of different types of data: iterating in order over the characters in a string, the words in an array, or the bits in a binary representation. C programmers live with a pervasive assumption that all these variables are stored in a memory that is itself visible as a sequence of bytes—`memcpy()` will copy any data type. And C’s I/O system models all I/O operations as bytes; you can also `read()` and `write()` any chunk of memory containing any data type.

So one computer can write out some data, and another computer can read it; suddenly, we’re interested in whether the second computer can understand what the first one wrote.

We understand that we need to be careful with padding and alignment (details in section 11.1). And it’s probably too much to expect that complex data types like floating-point numbers will always transfer intact. But we’d hope at least to see simple twos complement integers coming across OK; the curse of endianness is that they don’t. The 32-bit integer whose hexadecimal value was written as `0x1234.5678` quite often reads in as `0x7856.3412`—it’s been “byte-swapped.” To understand why, let’s go back a bit.

10.2.1 *Bits, Bytes, Words, and Integers*

A 32-bit binary integer is represented by a sequence of bits, with each bit having a different significance. The least significant bit is “ones,” then “twos,” then “fours”—just as a decimal representation is “ones,” “tens,” and “hundreds.” When your memory is byte-addressable, your 32-bit integer occupies four bytes.

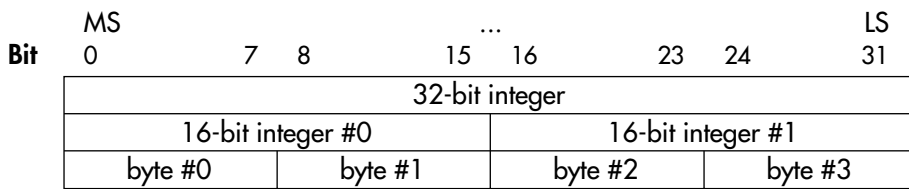


FIGURE 10.1 IBM (consistent big-endian) view.

There are two reasonable choices about how the integer and bitwise view tie up: Some computers put the least significant (LS) bits “first” (that is, in lower-addressed memory bytes) and some put the most significant (MS) bit first—and Cohen called them little-endian and big-endian, respectively. When I first got to know about computers in 1976, DEC’s minicomputers were little-endian and IBM mainframes were big-endian; neither camp was about to give way.

It’s worth stressing that the curse of choice only appeared once you could address bytes. Pioneering computers through to the late 1960s were generally organized around a single word size: Instructions, integers, and memory width were all the same word size. Such a computer has no endianness: It has word order in memory and bit order inside words, and those are unrelated.

Just as with opening a boiled egg, both camps have good arguments.

We’re used to writing decimals with the most significant digits to the left, and (reading left to right as usual) we say numbers that way: Shakespeare might have said “four and twenty,” but we say “twenty-four.” So if you write down numbers, it’s natural to put the most significant bits first. Bytes first appeared as a convenient way of packing characters into words, before memory was byte addressable. A 1970s vintage IBM programmer had spent most of his or her career poring over vast dump listings, and each set of characters represented a word, which was a number. Little-endian numbers look ridiculous. They were instinctive big-endians. But with numbers written MS to the left and byte addresses increasing in the same direction, it would have been inconsistent to have numbered the bits from right to left: So IBM labeled the MS bit of a word bit 0. Their world looked like Figure 10.1.

But it’s also natural to number the bits according to their arithmetic significance within integer data types—that is, to assign bit number n to the position that has arithmetic significance 2^n . It’s then consistent to store bits 0–7 in byte 0, and you’ve become a little-endian. Having words appear backward in dumps is a shame, but the little-endian view made particular sense to people who’d gotten used to thinking of memory as a big array of bytes. Intel, in particular, is little-endian. So its words, bytes, and bits look like Figure 10.2.

You’ll notice that these diagrams have exactly the same contents: only the MS/LS are interchanged, as well as the order of the fields. IBM big-endians see

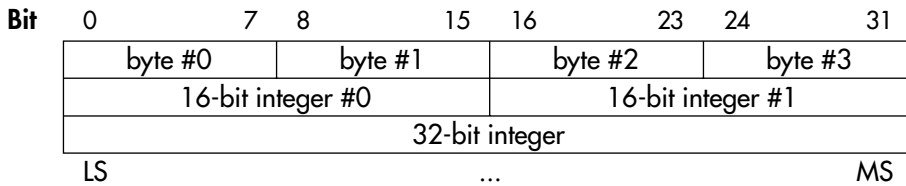


FIGURE 10.2 Consistent (Intel) little-endian view.

words broken into bytes, while little-endians see bytes built into words. Both these systems seemed unarguably right to different people: There's lots of merit in both, but you have to choose.

But let's get back to our observation about the problem above. Our mangled word started as `0x1234.5678`, which is `00010010 00110100 01010110 01111000` in binary. If you transfer it naively to a system with the opposite endianness, you'd surely expect to see all the bits reversed. In that case you'd receive the number `00011110 01101010 00101100 01001000`, which is hex `0x1E6A.2C48`. But we said we'd read hex `0x7856.3412`.

It's true that complete bit reversal could arise in some circumstances; there are communication links that send the MS bit first, and some that send the LS bit first. But sometime in the 1970s 8-bit bytes emerged as a universal base unit both inside computers and in computer communications systems (where they were called "octets"). Typically, communication systems build all their messages out of bytes, and only the lowest-level hardware engineers know which bit goes first.

Meanwhile, every microprocessor system got to use 8-bit peripheral controllers (wider controllers were reserved for the high-end stuff), and all those peripherals have 8-bit ports numbered 0 through 7, and the most significant bit is 7. Somehow, without a shot apparently fired, every byte was little-endian, and has been ever since.

Early microprocessor systems were 8-bit CPUs on 8-bit buses with 8-bit memory systems, so they had no endianness. Intel's 8086 was a 16-bit little-endian system. When Motorola introduced the 68000 microprocessor around 1978, they greatly admired IBM's mainframe architecture. Either in admiration for IBM or to differentiate themselves from Intel, they thought they should be big-endians too. But Motorola couldn't oppose the prevailing bits-within-bytes convention—every 8-bit Motorola peripheral would have had to be connected to a 68000 with its data bus bit-twisted. As a result, the 68000 family looks like Figure 10.3, with the bits and bytes *numbered in opposite directions*.

The 68000 and its successors went on to be used for most successful UNIX servers and workstations (notably with Sun). When MIPS and other RISCs emerged in the 1980s, their designers needed to woo system designers with the

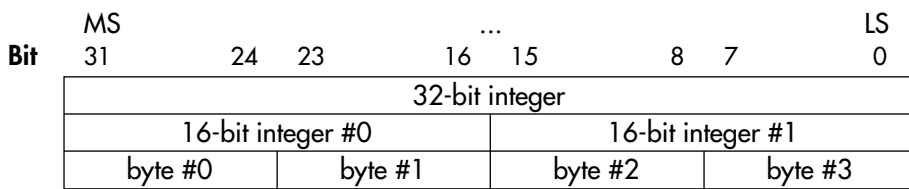


FIGURE 10.3 Inconsistent big-endian view as founded by the 68000 family.

right endianness, so they designed CPUs that could swing either way. But from the 68000 onward, big-endian has meant 68000-style big-endian, with bits and bytes going opposite ways. When you set up a MIPS CPU to be big-endian, it looks like Figure 10.3. And that’s where the trouble really starts.

One small difficulty comes when you read hardware manuals for your CPU and see register diagrams. Everyone’s convinced that registers are (first and foremost) 32-bit integers, so they’re invariably drawn with the MS bit (bit 31, remember) first. This has some consequences for programmers and hardware designers alike. That picture motivates the difference between “shift-left” and “shift-right” instructions, determines the bit-number arguments of bitfield instructions, and even refers to the labeling of the bitfields that make up MIPS instructions.

Once you get over that, there is serious software trouble when porting software or moving data between incompatible machines; there is hardware trouble when connecting incompatible components or buses. We’ll take the software and hardware problems separately.

10.2.2 *Software and Endianness*

Here’s a software-oriented definition of endianness: A CPU/compiler system where the lowest addressed byte of a multibyte integer holds the least significant bits is called little-endian; a system where the lowest addressed byte of a multibyte integer holds the most significant bits is called big-endian. You can very easily find out which sort of CPU you have by running a piece of deliberately nonportable code:

```
#include<stdio.h>

main ()
{
    union {
        int as_int;
        short as_short[2];
```

```

    char as_char[4];
} either;

either.as_int = 0x12345678;

if (sizeof(int) == 4 && either.as_char[0] == 0x78) {
    printf ("Little endian\n");
}
else if (sizeof(int) == 4 && either.as_char[0] == 0x12) {
    printf ("Big endian\n");
}
else {
    printf ("Confused\n");
}
}

```

Strictly speaking, software endianness is an attribute of the compiler toolchain, which could always—if it worked hard enough—produce the effect of either endianness. But on a byte-addressable CPU like MIPS with native 32-bit arithmetic it would be unreasonably inefficient to buck the hardware; thus we talk of the endianness of the CPU.

Of course, the question of byte layout within the address space applies to other data types besides integers; it affects any item that occupies more than a single byte, such as floating-point data types, text strings, and even the 32-bit op-codes that represent machine instructions. For some of these noninteger data types, the idea of arithmetic significance applies only in a limited way, and for others it has no meaning at all.

When a language deals in software-constructed data types bigger than the hardware can manage, then their endianness is purely an issue of software convention—they can be constructed with either endianness. I hope that modern compiler writers appreciate that it's a good idea to be consistent with the hardware's own convention.

Endianness and Program Portability

So long as binary data items are never imported into an application from elsewhere, and so long as you avoid accessing the same piece of data under two different integer types (as we deliberately did above), your CPU's endianness is invisible (and your code is portable). Modern C compilers will try to watch out for you: If you do this by accident, you'll probably get a compiler error or warning.

You may not be able to live within those limitations, however; you may have to deal with foreign data delivered into your system from elsewhere, or with memory-mapped hardware registers. For either of these, you need to know exactly how your compiler accesses memory.

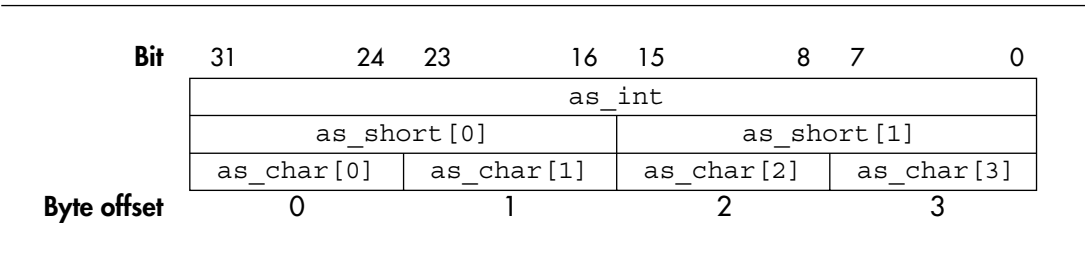


FIGURE 10.4 Typical big-endian’s picture.

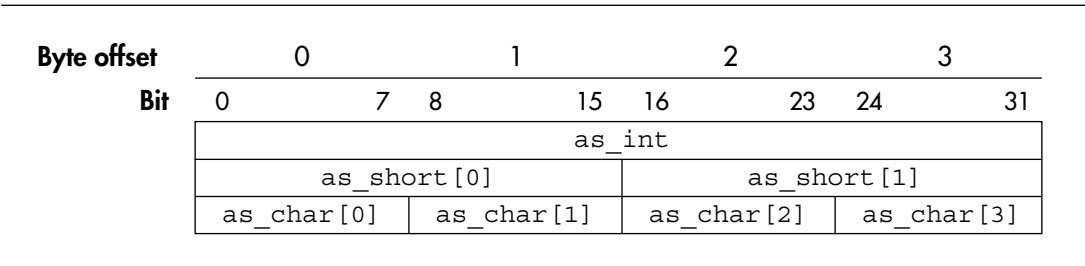


FIGURE 10.5 Little-endian’s picture.

This all seems fairly harmless, but experience shows that of all data-mapping problems, endianness is uniquely confusing. I think this is because it is difficult even to describe the problem without taking a side. The origin of the two alternatives lies in two different ways of drawing the pictures and describing the data; both are natural in different contexts.

As we saw above, big-endians typically draw their pictures organized around words. So that gives us a big-endian picture of the data structure we used in Figure 10.4. It would look a lot prettier with the IBM convention of labeling the MS bit as bit 0, but that’s no longer done.

But little-endians are likely to emphasize a software-oriented, abstract view of computer memory as an array of bytes. So the same data structure looks like Figure 10.5. Little-endians don’t think of computer data as primarily numeric, so they tend to put all the low numbers (bits, bytes, or whatever) on the left.

It’s very difficult to achieve a real grasp of endianness without drawing pictures, but many people find themselves struggling to set aside the conventions they’re used to; for example, if you’re used to numbering the bits from right to left, it can take a real effort of will to number them from left to right (a picture of a little-endian structure as drawn by someone with big-endian habits can look very illogical). This is the essence of the subject’s capacity to confuse: It’s difficult even to think about an unfamiliar convention without getting caught up in the ones you know.

10.2.3 *Hardware and Endianness*

We saw previously that a CPU's native endianness only shows up when it offers direct support both for word-length numbers and a finer-resolution, byte-sized memory system. Similarly, your hardware system acquires a recognizable endianness when a byte-addressed system is wired up with buses that are multiple bytes wide.

When you transfer multibyte data across the bus, each byte of that data has its own individual address. If the lowest-address byte in the data travels on the eight bus lines ("byte lane") with the lowest bit numbers, the bus is *little-endian*. But if the lowest-address byte in the data travels on the byte lane with the highest bit numbers, the bus is *big-endian*.

There's no necessary connection between the "native" endianness of a CPU and the endianness of its system interface considered as a bus. However, I don't know of any CPUs where the software and interface endianness are different, so we can talk about "the endianness of a CPU" and mean both internal organization and system interface.

Byte-addressable CPUs announce themselves as either big- or little-endian every time they transfer data. Intel and DEC CPUs are little-endian; Motorola 680x0 and IBM CPUs are big-endian. MIPS CPUs can be either, as configured from power-up; most other RISCs have followed the MIPS lead and chosen to make endianness configurable—a boon when updating an existing system with a new CPU.

Hardware engineers can hardly be blamed for connecting up different buses by matching up the bit numbers. But trouble strikes when your system includes buses, CPUs, or peripherals whose endianness doesn't match. In this case the choice is not a happy one; the system designer must choose the lesser of two evils:

- *Bit number consistent/byte sequence scrambled:* Most obviously, the designer can wire up the two buses according to their bit numbers, which will have the effect of preserving bit numbering within aligned "words." But since the relationship between bit numbers and bytes-within-words is different on the two buses, the two sides will see the sequence of bytes in memory differently.

Any data that is not of bus-width size and bus-width aligned will get mangled when transferred between the connected buses, with bytes swapped within each bus-width-sized unit. This looks and feels worse than the software problem. With wrong-endianness data in software, you have no problem finding data type boundaries; it's just that the data doesn't make sense. With this hardware problem the boundaries are scrambled too (unless the data are, by chance, aligned on bus-width "word" boundaries).

There's a catch here. If the data being passed across the interface is always aligned word-length integers, then bit-number-consistent wiring will

conceal the endianness difference, avoiding the need for software conversion of integers. But hardware engineers very rarely know exactly which data will be passed across an interface over the lifetime of a system, so be cautious.

- *Byte address consistent/integers scrambled:* The designer can decide to preserve byte addressing by connecting byte lanes that correspond to the same byte-within-word address, even though the bit-numbering of the data lines in the byte lane doesn't match at all. Then at least the whole system can agree on the data seen as an array of bytes.

However, there are presumably going to be components with mismatched *software* endianness in the system. So your consistent byte addressing is guaranteed to expose their disagreement about the representation of multibyte integers. And—in particular—even a bus-width-aligned integer (the “natural” unit of transfer) will appear byte-swapped when moved to the other endianness.

For most purposes, byte address scrambling is much more harmful, and we'd recommend “byte address consistent” wiring. When dealing with data representation and transfer problems, programmers will usually fall back on C's basic model of memory as an array of bytes, with other data types built up from that. When your assumptions about memory order don't work out, it's very hard to see what's going on.

Unfortunately, a bit number consistent/byte address scrambled connection looks much more natural on a schematic; it can be very hard to persuade hardware engineers to do the right thing.

Not every connection in a system matters. Suppose we have a 32-bit-wide memory system bolted directly to a CPU. The CPU's system interface may not include a byte-within-word address—the address bus does not specify address bits 1 and 0. Instead, many CPUs have four “byte enables,” which show that data is being transferred on particular byte lanes. The memory array is wired to the whole bus, and on a write the byte enables tell the memory array which of four possible byte locations within the word will actually get written. Internally, the CPU associates each of the byte lanes with a byte-within-word address, but that has no effect on the operation of the memory system. Effectively, the memory/CPU combination acts together and inherits the endianness of the CPU; where byte-within-word 0 actually goes in memory doesn't matter, so long as the CPU can read it back again.¹

It's very important not to be seduced by this helpful characteristic of a RAM memory into believing that there's no intrinsic endianness in a simple

1. Hardware-familiarized engineers will recognize that this is a consequence of a more general rule: It's a property of a writable memory array that it continues to work despite arbitrary permutations of the address and data lines to it. It doesn't matter where any particular data goes, so long as when you feed the matching read address into the array you get back the same data you originally wrote.

CPU/RAM system. You can spot the endianness of any transfer on a wide bus. Here's a sample list of conditions in which you can't just ignore the CPU's endianness when building a memory system:

- If your system uses firmware that's preprogrammed into ROM memory, the hardware address and byte lane connection assignments within the system need to match those assumed in the way the ROM was programmed, and the data contained in the ROM needs to match the CPU's configured endianness. In effect, the contents of the ROM are being delivered into your system from somewhere outside it. If the code is to be executed directly from the ROM, it's especially important to get the endianness right, because it's impossible for the CPU to apply any corrective software byte-swapping to the op-codes as it fetches them.
- When a DMA device gets to transfer data directly into memory, then its notions of ordering will matter.
- When a CPU interface does not in fact use byte enables, but instead issues byte-within-word addresses with a byte-width code (quite common for MIPS CPUs), then at least the hardware that decodes the CPU's read and write requests must know which endianness the CPU is using. This can be particularly tricky if the CPU allows endianness to be software-configured.

The next section is for you to tell your hardware engineer about how to set up a byte address consistent system—and even how to make that system configurable with the CPU, if some of your users might set up the MIPS CPU both ways.

Wiring Endianness-Inconsistent Buses

Suppose we've got a 64-bit MIPS CPU configured big-endian, and we need to connect it to a little-endian 32-bit bus such as PCI.

Figure 10.6 shows how we'd wire up the data buses to achieve the recommended outcome of consistent byte addresses as seen by the big-endian CPU and the little-endian bus.

The numbers called “byte lane” show the byte-within-bus-width part of the address of the byte data traveling there. Writing in the byte lane numbers is the key to getting one of these connections right.

Since the CPU bus is 64 bits wide and the PCI bus 32 bits, you need to be able to connect each half of the wide bus to the narrow bus according to the “word” address—that's address bit 2, since address bits 1 and 0 are the byte-within-32-bit-word address. The CPU's 64-bit bus is big-endian, so its high-numbered bits carry the lower addresses, as you can see from the byte lane numbers.

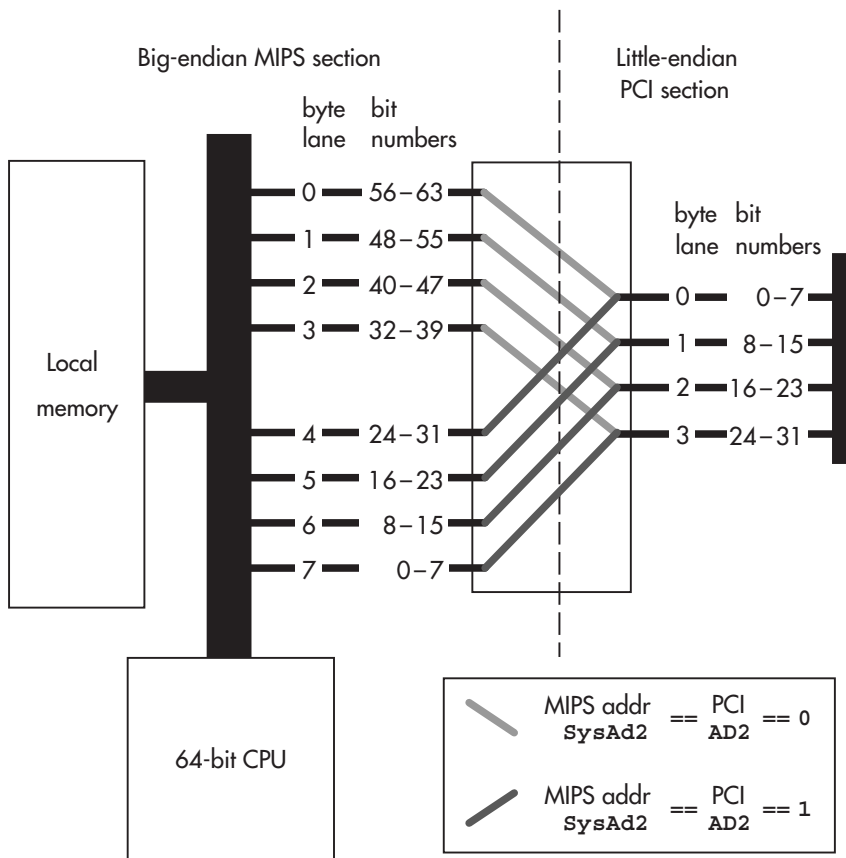


FIGURE 10.6 Wiring a big-endian CPU to a little-endian bus.

You may find yourself staring at the numbering of the connections around the bus switch for quite some time before they really make sense. Such are the joys of endianness.

And note that I only showed data. PCI is a multiplexed bus, and in some clock cycles those “byte lanes” are carrying an address. In address cycles, PCI bus wire 31 is carrying the most significant bit of the address. The address-time connection from your MIPS-based system should not be swapped.

Wiring an Endianness-Configurable Connection

Suppose you want to build a board or bus switch device that allows you to configure a MIPS CPU to run with either endianness. How can we generalize the advice above?

We'd suggest that, if you can persuade your hardware designer, you should put a programmable byte lane swapper between the CPU and the I/O system. The way this works is shown diagrammatically in Figure 10.7; note that this is only a 32-bit configurable interface and it's an exercise for you to generalize it to a 64-bit CPU connection.

We call this a *byte lane swapper*, not a byte swapper, to emphasize that it does not alter its behavior on a per-transfer basis, and in particular to indicate that it is not switched on and off for transfers of different sizes. There are circumstances where it can be switched on and off for transfers to different address regions—mapping some part of the system as bit number consistent/byte address scrambled—but that's for you to make work.

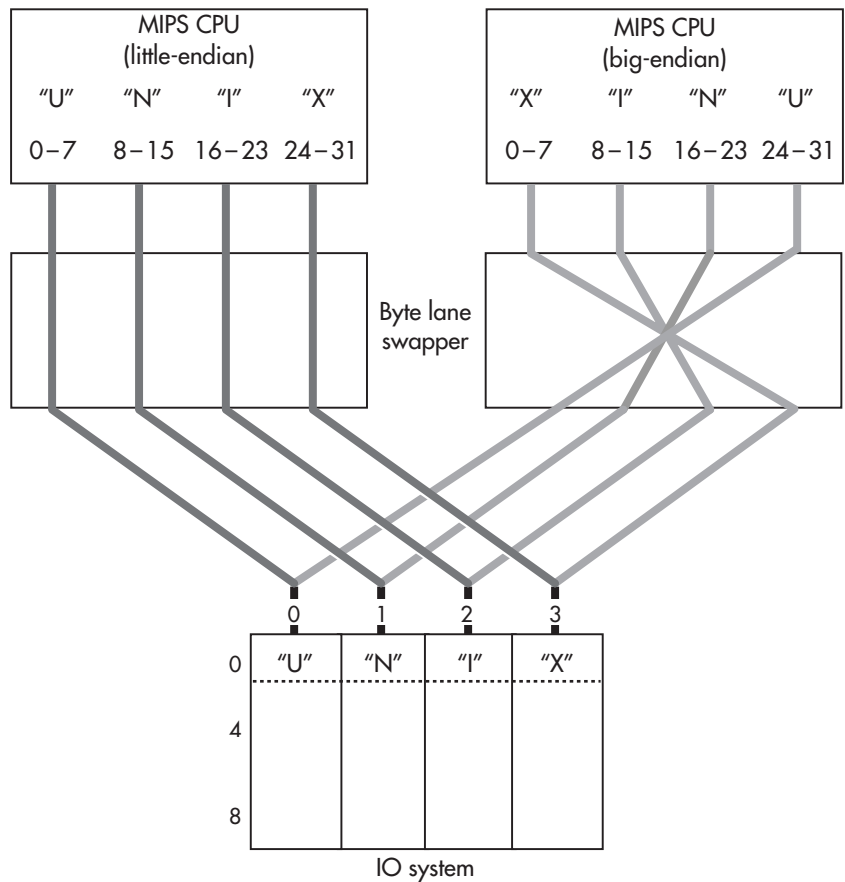


FIGURE 10.7 Byte lane swapper.

What a byte lane swapper *does* achieve is to ensure that whether your CPU is set up to be big- or little-endian, the relationship between the CPU and the now mismatched external bus or device can still be one where byte sequence is preserved.

You normally won't put the byte-lane swapper between the CPU and its local memory—this is just as well, because the CPU/local memory connection is fast and wide, which would make the byte swapper expensive.

As mentioned above, so long as you can decode the CPU's system interface successfully, you can treat the CPU/local memory as a unit and install the byte swapper between the CPU/memory unit and the rest of the system. In this case, the relationship between bit number and byte order inside the local memory changes with the CPU, but this fact is not visible from the rest of the world.

False Cures and False Prophets for Endianness Problems

Every design team facing up to endianness for the first time goes through the stage of thinking that the troubles reflect a hardware deficiency to be solved. It's never that simple. Here are a few examples.

- *Configurable I/O controllers*: Some newer I/O devices and system controllers can themselves be configured into big-endian and little-endian modes. You're going to have to read the manual very carefully before using such a feature, particularly if you mean to use it not as a static (design time) option but rather as a jumper (reset time) option.

It is quite common for such a feature to affect only bulk data transfers, leaving the programmer to handle other endianness issues, such as access to bit-coded device registers or shared memory control fields. Also, the controller designer probably didn't have the benefit of this book—and confusion about endianness is widespread.

- *Hardware that byte-swaps according to transfer type*: If you're designing in some byte-swap hardware, it seems appealing to try to solve the whole problem. If we just swapped byte data to preserve its addresses, but left words alone, couldn't we prevent the whole software problem? The answer is no, there aren't any hardware fixes for the software problem. For example, many of the transfers in a real system are of data cache lines. They may contain an arbitrary mixture of data sizes and alignments; if you think about it for a moment, you'll see that there simply isn't any way to know where the boundaries are, which means there's no way to determine the required swap configuration.

Conditional byte-swapping just adds confusion. Anything more than unconditional byte lane swapping is snake oil.

10.2.4 *Bi-endian Software for a MIPS CPU*

You may want to create binary code that will run correctly on MIPS CPUs with either endianness—probably for a particular board that may be run either way or to create a portable device driver that may run on boards of either configuration. It's a bit tricky, and you will probably only do a tiny part of your bootstrap code like this, but here are some guidelines.

The MIPS CPU doesn't have to do too much to change endianness. The only parts of the instruction set that recognize objects smaller than 32 bits are partial-word loads and stores. On a MIPS CPU with a 32-bit bus, the instruction:

```
lbu t0, 1(zero)
```

takes the byte at byte program address 1, loads it into the least significant bits (0 through 7) of register **t0**, and fills the rest of the register with zero bits. This description is endianness independent. However, in big-endian mode the data loaded into the register will be taken from bits 16–23 of the CPU data bus; in little-endian mode, the byte is loaded from bits 8–15 of the CPU data bus.

Inside the MIPS CPU, there's data-steering hardware that the CPU uses to direct all the active bytes in a transfer from their respective byte lanes at the interface to the correct positions within the internal registers. This steering logic has to accommodate all permutations of load size, address, and alignment (including the load/store left/right instructions described in section 8.5.1).

It is the change in the relationship between the active byte lane and the address on partial-word loads and stores that characterizes the MIPS CPU's endianness. When you reconfigure your MIPS CPU's endianness, it's that steering logic between data and register whose behavior changes.

Complementing the chip's configurability, most MIPS toolchains can produce code of either endianness, based on a command-line option.

If you set a MIPS CPU to the wrong endianness for its system, then a couple of things will happen.

First, if you change nothing else, the software will crash quickly, because on any partial-word write the memory system will pick up garbage data from the wrong part of the CPU bus. At the same time as reconfiguring the CPU, we'd better reconfigure the logic that decodes CPU cycles.²

If you fix that, you'll find that the CPU's view of byte addressing becomes scrambled with respect to the rest of the system; in terms of the description above, we've implicitly opted for a connection that keeps the bit numbers consistent, rather than the byte addresses.

Of course, data *written* by the CPU after a change of endianness will seem fine to the CPU itself; if we allow changes of endianness only at reset time, then volatile memory that is private to the CPU won't give us any trouble.

2. There are some CPU interfaces where partial-word transfers are signaled with independent byte lane enable signals, and in that case this problem doesn't happen.

	31	24 23	16 15	8 7	0
	r	e	m	E	
Byte address from BE CPU	0	1	2	3	
Byte address from LE CPU	3	2	1	0	
	c	n	e	g	
Byte address from BE CPU	4	5	6	7	
Byte address from LE CPU	7	6	5	4	
	x	x	\000	y	
Byte address from BE CPU	8	9	10	11	
Byte address from LE CPU	11	10	9	8	

FIGURE 10.8 Garbled string storage when mixing modes; see text.

Note also that the CPU's view of bit numbering within aligned bus-width words continues to match the rest of the system. This is the choice we described earlier as bit number consistent and that we suggested you should generally avoid. But in this particular case it has a useful side effect, because MIPS instructions are encoded as bitfields in 32-bit words. An instruction ROM that makes sense to a big-endian CPU will make sense to a little-endian CPU too, allowing us to share a bootstrap. Nothing works perfectly—in this case, any data in the ROM that doesn't consist of aligned 32-bit words will be scrambled. Many years ago, Algorithmics' MIPS boards had just enough bi-endian code in their boot ROM to detect that the main ROM program does not match the CPU's endianness and to print the helpful message:

```
Emergency - wrong endianness configured.
```

The word `Emergency` is held as a C string, null-terminated. You should now know enough to understand why the ROM start-up code contains the enigmatic lines:

```
.align 4
.ascii "remEcneg\000\000\000y"
```

That's what the string `Emergency` (with its standard C terminating null and two bytes of essential padding) looks like when viewed with the wrong endianness. It would be even worse if it didn't start on a four-byte-aligned location. Figure 10.8 (drawn from the point of view of a confirmed big-endian) shows what is going on.

You've seen that writing bi-endian code is possible, but be aware that when you're ready to load it into ROM, you'll be asking your tools to do something they weren't designed to handle. Typically, big-endian tools pack instruction

words into the bytes of a load file with the most significant bits first, and little-endian tools work the other way around. You'll need to think carefully about the result you need to achieve, and examine the files you generate to make sure everything went according to plan.

10.2.5 *Portability and Endianness-Independent Code*

By a fairly well-respected convention, most MIPS toolchains define the symbol `BYTE_ORDER` as follows:

```
#if BYTE_ORDER == BIG_ENDIAN
/* big-endian version... */
#else
/* little-endian version... */
#endif
```

So if you really need to, you can put in different code to handle each case. But it's better—wherever possible—to write endianness-independent code. Particularly in a well-controlled situation (such as when writing code for a MIPS system that may be initialized with the CPU in either mode), you can get rid of a lot of dependencies by good thinking.

All data references that pick up data from an external source or device are potentially endianness dependent. But according to how your system is wired, you may be able to produce code that works both ways. There are only two ways of wiring the wrong endianness together: One preserves byte addresses and the other bit numbers. For some particular peripheral register access in a particular range of systems, there's a good chance that the endianness change consistently sticks to one of these.

If your device is typically mapped to be byte address compatible, then you should program it strictly with byte operations. If ever, for reasons of efficiency or necessity, you want to transfer more than one byte at a time, you need to write endianness-conditional code that packs or unpacks that data.

If your device is compatible at the word (32-bit) level—for example, it consists of registers wired (by however devious and indirect a route) to a fixed set of MIPS data bus bits—then program it with bus-width read/write operations. That will be 32-bit or 64-bit loads and stores. If the device registers are not wired to MIPS data bus bits starting at 0, you'll probably want to shift the data after a read and before a write. For example, 8-bit registers on a 32-bit bus in a system originally conceived as big-endian are commonly wired via bits 31–24.

10.2.6 *Endianness and Foreign Data*

This chapter is about programming, not a treatise on I/O and communications, so we'll keep this section brief. Any data that is not initialized in

your code, chosen libraries, and OS is foreign. It may be data you read from some memory-mapped piece of hardware, data put into memory by DMA, data in a preprogrammed ROM that isn't part of your program, or you may be trying to interpret a byte stream obtained from an "abstract" I/O device under your OS.

The first stage is to figure out what this data looks like in memory; with C, that can usually be accomplished by mapping out what its contents are as an array of `unsigned char`. Even if you know your data and compiler well enough to guess which C structure will successfully map to the data, fall back to the array of bytes when something is not as you expect; it's far too easy to miss what is really going on if your data structure is incorrect.

Apart from endianness, the data may consist of data types that are not supported by your compiler/CPU; it may have similar types but with completely different encodings; it may have familiar data but be incorrectly aligned; or, falling under this section's domain, it may have the wrong endianness.

If the chain along which the data has reached you has preserved byte order at each stage, the worst that will happen is that integer data will be represented with an opposite order, and it's easy enough to build a "swap" macro to restore the two, four, or eight bytes of an integer value.

But if the data has passed over a bit number consistent/byte address scrambled interface, it can be more difficult. In these circumstances, you need to locate the boundaries corresponding to the width of the bus where the data got swapped; then, taking groups of bytes within those boundaries, swap them without regard to the underlying data type. If you do it right, the result should now make sense, with the correct byte sequence, although you may still need to cope with the usual problems in the data—including, possibly, the need to swap multibyte integer data again.

10.3 Trouble with Visible Caches

In section 4.6, you learned about the operations you can use to get your caches initialized and operating correctly. This section alerts you to some of the problems that can come up and explains what you can do to deal with them.

Most of the time, the caches are completely invisible to software, serving only to accelerate the system as they should. But especially if you need to deal with DMA controllers and the like, it can be helpful to think of the caches as independent buffer memories, as shown in Figure 10.9.³

It's important to remember that transfers between cache and memory always work with blocks of memory that fit the cache line structure—typically 16- or 32-byte-aligned blocks—so the cache may read or write a block because the

3. For MIPS CPUs with simple write-through data caches, the path labeled "write-back" in the figure doesn't exist.

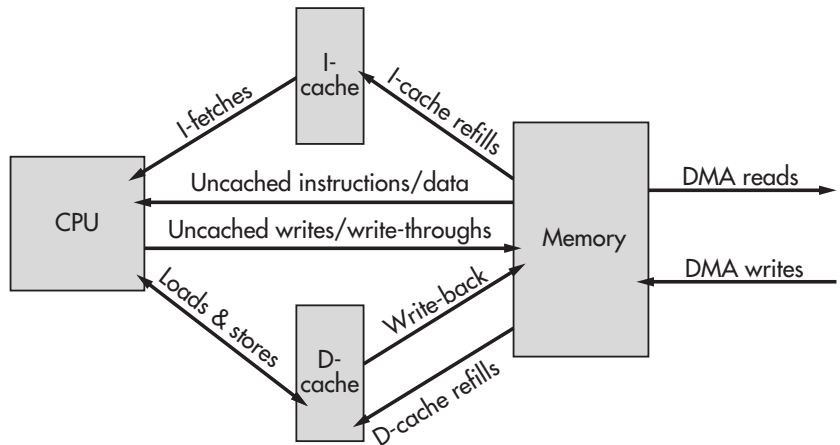


FIGURE 10.9 Data flow between CPU, memory, and caches.

CPU makes direct reference to just one of the byte addresses contained within it; even if the CPU never touches any of the other bytes in the cache line, it'll still include all the bytes in the next transfer of that cache line.

In an ideal system, we could always be certain that the state of the memory is up-to-date with all the operations requested by the CPU, and that every valid cache line contains an exact copy of the appropriate memory location. Unfortunately, practical systems can't always live up to this ideal. We'll assume that you initialize your caches after any reset, and that you avoid (rather than try to live with) the dreaded cache aliases described in section 4.12. Starting from those assumptions, how does a real system's behavior fall short of the ideal?

- *Stale data in the cache:* When your CPU writes to memory in cached space, it updates the cached copy (and may also write memory at the same time). But if a memory location is updated in any other way, any cached copies of its contents continue to hold the old value, and are now out-of-date. That can happen when a DMA controller writes data. Or, when the CPU writes new instructions for itself, the I-cache may continue to hold whatever was at the same location before. It's important for programmers to realize that the hardware generally doesn't deal with these conditions automatically.
- *Stale data in memory:* When the CPU writes some data into a (write-back) cache line, that data is not immediately copied to memory. If the data is read later by the CPU, it gets the cached copy and is fine; but if

something that isn't the CPU reads memory, it may get the old value. That can happen to an outbound DMA transfer.

The software weapons you have to fight problems caused by visible caches are a couple of standard subroutines that allow you to clean up cache/memory inconsistency, built on MIPS **cache** instructions. They operate on cache locations corresponding to a specified area of memory and can write-back up-to-date cache data or invalidate cache locations (or both).

Well, of course, you can always map data uncached. In fact, there are some circumstances when you will do just that: Some network device controllers, for example, have a memory-resident control structure where they read and write bytes and bit flags, and it's a lot easier if you map that control structure uncached. The same is true, of course, of memory-mapped I/O registers, where you need total control over what gets read and written. You can do that by accessing those registers through pointers in `kseg1` or some other uncached space; if you use cached space for I/O, bad things will happen!

If (unusually) you need to use the TLB to map hardware register accesses, you can mark the page translation as uncached. That's useful if someone has built hardware whose I/O registers are not in the low 512 MB of the physical memory space.

It's possible that you might want to map a memory-like device (a graphics frame buffer, perhaps) through cached space so as to benefit from the speed of the block reads and writes that the CPU only uses to implement cache refills and write-backs. But you'd have to explicitly manage the cache by invalidation and write-back on every such access. Some embedded CPUs provide strange and wonderful cache options that can be useful for that kind of hardware—check your manual.

10.3.1 *Cache Management and DMA Data*

This is a common source of errors, and the most experienced programmers will sometimes get caught out. Don't let that worry you too much: Provided you think clearly and carefully about what you're trying to achieve, you'll be able to get your caches to behave as they should while your DMA transfers work smoothly and efficiently.

When a DMA device puts data into memory, for example, on receipt of network data, most MIPS systems don't update the caches—even though some cache lines may currently be holding addresses within the region just updated by the DMA transfer. If the CPU subsequently reads the information in those cache lines, it'll pick up the old, stale version in the cache; as far as the CPU can tell, that's still marked valid, and there's no indication that the memory has a newer version.

To avoid this, your software *must* actively invalidate any caches' lines that fall within the address range covered by your DMA buffer, before there's any

chance that the CPU will try to refer to them again. This is much easier to manage if you round out all your DMA buffers so they start and end exactly at cache line boundaries.

For outbound transfers, before you allow a DMA device to transfer data from memory—such as a packet that you’re sending out via a network interface—you must make absolutely sure that none of the data to be sent is still just sitting in the cache. After your software has finished writing out the information to be transferred by DMA, it *must* force the write-back of all cache lines currently holding information within the address range that the DMA controller will use for the transfer. Only then can you safely initiate the DMA transfer.

On some MIPS CPUs, you can avoid the need for explicit write-back operations by configuring your caches to use write-through rather than write-back behavior, but this cure is really worse than the disease — write-through tends to be much slower overall and will also raise your system’s power consumption.

You really can get rid of the explicit invalidations and write-backs by accessing all the memory used for all DMA transfers via an uncached address region. This isn’t recommended either, because it’ll almost certainly degrade your system’s overall performance far more than you’d like. Even if your software’s access to the buffers is purely sequential, caching the DMA buffer regions will mean that information gets read and written in efficient cache-line-sized bursts rather than single transfers. The best general advice is to cache everything, with only the following exceptions:

- I/O device registers: Perhaps obvious, but worth pointing out. MIPS has no dedicated input/output instructions, so all device registers must be mapped somewhere in the address space, and very strange things will happen if you accidentally let them be cached.
- DMA descriptor arrays: Sophisticated DMA controllers share control/status information with the CPU using small *descriptor* data structures held in memory. Typically, the CPU uses these to create a long list of information to be transferred, and only then tells the DMA controller to begin its work. If your system uses descriptors, you’ll want to access the memory region that contains them through an uncached address region.

A portable OS like Linux must deal with a range of caches from the most sophisticated and invisible to the crude and simple, so it has a fairly well-defined API (set of stable function calls) for driver writers to use and some terse documentation on how to use them. See section 15.1.1.

10.3.2 ***Cache Management and Writing Instructions: Self-Modifying Code***

If your code ever tries to write *instructions* into memory, then execute them, you’ll need to make sure you allow for cache behavior.

This can surprise you on two levels. First, if you have a write-back D-cache, the instructions that your program writes out may not find their way to main memory until something triggers a write-back of the relevant cache lines. The instructions that your program wrote out could just be sitting in the D-cache at the time you try to execute them, and the CPU's fetches simply can't access them there. So the first step is to do write-back operations on the cache lines at which you write the instructions; that at least ensures they reach main memory.

The second surprise (regardless of which type of D-cache you have) is that even after writing out the new instructions to some region of main memory, your CPU's *I-cache* may still hold copies of the information that used to be held in those addresses. Before you tell the CPU to execute the newly written instructions, it's essential that your software first invalidates all the lines in the I-cache that contain information at the affected address range.

Of course, you could avoid the need for these explicit write-back and invalidate operations by writing and then executing the new instructions within an uncached address region; but that gives up the advantages of caching and is almost always a mistake.

The general-purpose cache management instructions described in section 4.9 are CP0 instructions, only usable by kernel-privilege software. That doesn't matter when cache operations are related to DMA operations, which are also entirely kernel matters. But it does matter with applications like writing instructions and executing them (think of a modern application using a "just-in-time" interpreted/translated language).

So MIPS32/64 provides the **synci** instruction, which does a D-side write-back and an I-side invalidate of one cache-line-sized piece of your new code. Find out how in section 8.5.11.

10.3.3 *Cache Management and Uncached or Write-Through Data*

If you mix cached and uncached references that map to the same physical range of addresses, you need to think about what this means for the caches. Uncached writes will update *only* the copy of a given address in main memory, possibly leaving what's now a stale copy of that location's contents in the D-cache—or the I-cache. Uncached loads will pick up whatever they find in main memory—even if that information is, in fact, stale with respect to an up-to-date copy present only in cache.

Careful use of cached and uncached references to the same physical region may be useful, or even necessary, in the low-level code that brings your system into a known state following a reset. But for running code, you probably don't want to do that. For each region of physical memory, decide whether your software should access it cached or uncached, then be absolutely consistent in treating it that way.

10.3.4 *Cache Aliases and Page Coloring*

There's more about the hardware origin of cache aliases in section 4.12. The problem occurs with L1 caches that are virtually indexed but physically tagged, and where the index range is big enough to span two or more page sizes. The index range is the size of one “set” of the cache, so with common 4-KB pages you can get aliases in an 8-KB direct-mapped cache or a 32-KB four-way set-associative cache.

The “page color” of a location is the value of those one or more virtual address bits that choose a page-sized chunk within the appropriate cache set. Two virtual pointers to the same physical data can produce an alias only if they have a different page color. So long as all pointers to the same data have the *same* color, all is well—all the data, even though at different virtual addresses, will be stored in the same physical portion of the cache and will be correctly identified by the (common) physical tag.

It's quite common in Linux (for example) for a physical page to be accessible at multiple virtual locations (shared libraries are routinely shared between programs at different virtual addresses).

Most of the time, the OS is able to overalign virtual address choices for shared data—the sharing processes may not use the same address, but we'll make sure their different virtual addresses are a multiple of, say, 64 KB apart, so the different virtual addresses have the same color. That takes up a bit more virtual memory, but virtual memory is fairly cheap.

It's easy to think that cache aliases are harmless so long as the data is “read-only” (it must have been written once, but that was before there were aliases to it): We don't care if there are multiple copies of a read-only page. But they're only mostly harmless. It is possible to tolerate aliases to read-only data, particularly in the I-cache: But you need to make sure that cache management software is aware that data that has been invalidated at one virtual address may still be cached at another.

With the widespread use of virtual-memory OSs (particularly Linux) in the embedded and consumer computing markets, MIPS CPUs are increasingly being built so that cache aliases can't happen. It's about time this long-lasting bug was fixed.

Whatever you need to do, the cache primitive operations required for a MIPS32/64 CPU are described in section 4.9.1.

10.4 Memory Access Ordering and Reordering

Programmers tend to think of their code executing in a well-behaved sequence: The CPU looks at an instruction, updates the state of the system in the appropriate ways, then goes on to the next instruction. But our program can run faster if we allow the CPU to break out of this purely sequential form of execution, so that operations aren't necessarily constrained to take place in strict

program order. This is particularly true of the read and write transactions performed at the processor's interface, triggered by its execution of load and store instructions.

From the CPU's point of view, a store requires only an outbound write request: Present the memory address and data, and leave the memory controller to get on with it. Practical memory and I/O devices are relatively slow, and in the time the write is completed the CPU may be able to run tens or hundreds of instructions.

Reads are different, of course: They require two-way communication in the form of an outbound request and an inbound response. When the CPU needs to know the contents of a memory location or a device register, there's probably not much it can do until the system responds with the information.

In the quest for higher performance, that means we want to make reads as fast as possible, even at the expense of making writes somewhat slower. Taking this thinking a step further, we can even make write requests wait in a queue, and pass any subsequent read requests to memory ahead of the buffered writes. From the CPU's point of view, this is a big advantage; by starting the read transaction immediately, it gets the response back as soon as possible. The writes will have to be done sometime, and the queue is of finite size: But it's likely that after this read is done there will be a period while the CPU is running from cache. And if the queue fills up, we'll just have to stop while some writes happen: That's certainly no worse than if we'd done the writes in sequence.

You can probably see a problem here: Some programs may write a location and then read it back again. If the read overtakes the write, we may get stale data from memory and our program will malfunction. Most of the time we can fix it with extra hardware that checks an outgoing read request against the addresses of entries in the write queue and doesn't allow the read to overtake a matching write.⁴

In systems where tasks that could be really concurrent (that is, they might be running on different CPUs) share variables, the problem of ordering reads and writes becomes more dangerous. It's true that much of the time the tasks have no expectation of mutual ordering. Ordering matters when the tasks are deliberately using shared memory for synchronization and communication, but in this case the software will be using carefully crafted OS synchronization operations (locks and semaphores, for example).

But there are some shared-memory communication tricks—often good, cheap, efficient ones—that don't need so many semaphores or locks but are disrupted by arbitrary cycle reordering. Suppose, for example, we have two

4. You could, in some circumstances, return data from the write buffer to fulfill the read. But we don't have to invent anything to allow the read to wait and then go to memory.

tasks: one is writing a data structure, the other is reading it. They use the data structure in turn, as shown in Figure 10.10.

For correct operation, we need to know that when the reader sees the updated value in the key field, we can guarantee that all the other updates will be visible to the reader as well.

Unless we discard all the performance advantages of decoupling reads and writes from the CPU, it's not practical for hardware to conceal all ordering issues from the programmer. The MIPS architecture provides the **sync** instruction for this purpose: You're assured that (for all participants in the shared memory) all accesses made before the **sync** will precede those made afterward. It's worth dwelling on the limited nature of that promise: It only relates to ordering, and only as seen by participants in uncached or cache-coherent memory accesses.

To make the example above reliable on a suitable system, the writer should include **sync** just before writing `keyfield`, and the reader should have a **sync** just after reading `keyfield`. See section 8.5.9 for details. But there's a lot more to this subject; if you're building such a system, you're strongly recommended to use an OS that provides suitable synchronization mechanisms, and read up on this subject.

Different architectures make different promises about ordering. At one extreme, you can require all CPU and system designers to contrive that all the writes and reads made by one CPU appear to be in exactly the same order from the viewpoint of another CPU: That's called "strongly ordered." There are weaker promises too (such as "all writes remain in order"); but the MIPS architecture takes the radical position that no guarantees are made at all.

<i>Writing task</i>	<i>Reading task</i>
<pre> ... /* update entries */ keyfield = WRITEDONE; sendsignaltoreader(); </pre>	<pre> keyfield = WAITINGFORWRITE; ... while (keyfield != WRITEDONE) { waitforsignalfromwriter(); } /* read entries */ ... </pre>

FIGURE 10.10 Tasks sharing a data structure.

10.4.1 *Ordering and Write Buffers*

Let's escape from the lofty theory and describe something rather more practical. The idea of holding outbound requests in a *write buffer* turns out to work especially well in practice because of the way store instructions tend to be bunched together. For a CPU running compiled MIPS code, it's typical to find that only about 10 percentage of the instructions executed are stores; but these accesses tend to come in bursts—for example, when a function prologue saves a group of register values.

Most of the time the operation of the write buffer is completely transparent to software. But there are some special situations in which the programmer needs to be aware of what's happening:

- *Timing relations for I/O register accesses:* This affects *all* MIPS CPUs. After the CPU executes a store to update an I/O device register, the outbound write request is liable to incur some delay in the write buffer, on its way to the device. Other events, such as inbound interrupts, may take place *after* the CPU executes the store instruction, but *before* the write request takes effect within the I/O device. This can lead to surprising behavior: For example, the CPU may receive an interrupt from a device “after” you have told it not to generate interrupts. To give another example: If an I/O device needs some software-implemented delay to recover after a write, you must ensure that the write buffer is empty before you start counting out that delay—ensuring also that the CPU waits while the write buffer empties. It's good practice to define a subroutine that does this job, and it's traditionally given the name `wbflush()`. See section 10.4.2 for hints on implementing it.
- *Reads overtaking writes:* The MIPS32/64 architecture permits this behavior, discussed above. If your software is to be robust and portable, it should not assume that read and write order is preserved. Where you need to guarantee that two cycles happen in some particular order, you need the **sync** instruction described in section 8.5.9.
- *Byte gathering:* Some write buffers watch for partial-word writes within the same memory word (or even writes within the same cache line) and will combine those partial writes into a single operation.
To avoid unpleasant symptoms when uncached writes are combined into a word-width, it's a good idea to map your I/O registers such that each register is in a separate word location (i.e., 8-bit registers should be at least four bytes apart).

10.4.2 *Implementing wbflush*

Most write queues can be emptied out by performing an uncached store to any location and then performing an operation that reads the same data back.

A write queue certainly can't permit the read to overtake the write—it would return stale data. Put a **sync** instruction between the write and the read, and that should be effective on any system compliant with MIPS32/64.

This is effective, but not necessarily efficient; you can minimize the overhead by loading from the fastest memory available. Perhaps your system offers something system-specific but faster. Use it after reading the following note!

CAUTION! Write buffers are often implemented within the CPU, but may also be implemented outside it; any system controller or memory interface that boasts of a *write-posting* feature introduces another level of write buffering to your system. Write buffers outside the CPU can give you just the same sort of trouble as those inside it. Take care to find out where all the write buffers are located in your system, and to allow for them in your programming.

10.5 Writing it in C

You probably already write almost everything in C or in C++. MIPS's lack of special I/O instructions means that I/O register accesses are just normal loads and stores with appropriately chosen addresses; that's convenient, but I/O register accesses are usually somewhat constrained, so you need to make sure the compiler doesn't get too clever. MIPS's use of large numbers of CP0 registers also means that OS code can benefit from well-chosen use of C `asm()` operations.

10.5.1 *Wrapping Assembly Code with the GNU C Compiler*

The GNU C Compiler ("GCC") allows you to enclose snippets of assembly code within C source files. GCC's feature is particularly powerful, but other modern compilers probably could support the example here. But their syntax is probably quite different, so we'll just discuss GCC here.

If you want low-level control over something that extends beyond a handful of machine instructions, such as a library function that carries out some clever computation, you'll really need to get to grips with writing pure MIPS assembly; but if you just want to insert a short sequence that consists of one or a few specific MIPS instructions, the `asm()` directive can achieve the desired result quite simply. Better still, you can leave it to the compiler to manage the selection of registers according to its own conventions.

As an example, the following code makes GCC use the three-operand form of multiply, available on more recent MIPS CPUs. If you just use the normal C language `*` multiplication operator, the work could end up being done by the

original form of the multiply instruction that accepts only two source operands, implicitly sending its double-length result to the **hi/lo** register pair.⁵

The C function `mymul()` is exactly like the three-operand **mul** and delivers the less significant half of the double-length result; the more significant half is simply discarded, and it's up to you to ensure that overflows are either avoided or irrelevant.

```
static int __inline__ mymul(int a, int b)
{
    int p;

    asm("mul %0, %1, %2"
        : "=r" (p)
        : "r" (a), "r" (b)
        );

    return p;
}
```

The function itself is declared `inline`, which instructs the compiler that a use of this function should be replaced by a copy of its logic (which permits local register optimization to apply). Adding `static` means that the function need not be published for other modules to use, so no binary of the function itself will be generated. It very often makes sense to wrap an `asm()` like this: You'd probably usually then put the whole definition in an include file. You could use a C preprocessor macro, but the inlined function is a bit cleaner.

The declarations inside the `asm()` parentheses tell GCC to emit a MIPS **mul** line to the assembler with three operands on the command line—one will be the output and two will be inputs.

On the line below, we tell GCC about operand `%0`, the product: first, that this value will be write-only (meaning that there's no need to preserve its original value) with the `"="` modifier; the `"r"` tells GCC that it's free to choose any of the general-purpose registers to hold this value. Finally, we tell GCC that the operand we wrote as `%0` corresponds to the C variable `p`.

On the third line of the `asm()` construct, we tell GCC about operands `%1` and `%2`. Again, we allow GCC to put these in any of the general-purpose registers, and tell it that they correspond to the C variables `a` and `b`.

At the end of the example function, the result we obtained from the multiply instruction is returned to the C caller.

GCC allows considerable control over the specification of the operands; you can tell it that certain values are both read and written and that certain hardware

5. At the time of writing there are versions of GCC in circulation that will use MIPS32's three-operand **mul** instruction—but this makes for a good example.

registers are left with meaningless values as a side effect of a particular assembly sequence. You can dig out the details from the MIPS-specific sections of the GCC manual.

10.5.2 *Memory-Mapped I/O Registers and “Volatile”*

Most of you will be writing code that accesses I/O registers in C—you certainly shouldn’t be using assembly code in the absence of any pressing need to do so, and since all I/O registers in MIPS must be memory-mapped, it is never difficult to access them from C. Having said that, you should keep in mind that as compilers advance, or if you make significant use of C++, it can become harder to predict exactly the low-level instruction sequences that’ll end up in your code. Here are some well-worn hints.

I might write a piece of code that is intended to poll the status register of a serial port and to send a character when it’s ready:

```
unsigned char *usart_sr = (unsigned char *) 0xBFF00000;
unsigned char *usart_data = (unsigned char *) 0xBFF20000;
#define TX_RDY 0x40

void putc (ch)
char ch;
{
    while ((*usart_sr & TX_RDY) == 0)
        ;
    *usart_data = ch;
}
```

I’d be upset if this sent two characters and then looped forever, but that would be quite likely to happen. The compiler sees the memory-mapped I/O reference implied by `*usart_sr` as a loop-invariant fetch; there are no stores in the `while` loop so it seems safe to pull the load out of the loop. Your compiler has recognized that your C program is equivalent to:

```
void putc (ch)
char ch;
{
    tmp = (*usart_sr & TX_RDY);

    while (tmp)
        ;
    *usart_data = ch;
}
```


You could prevent this particular problem by defining your registers as follows:

```
volatile unsigned char *usart_sr =
    (unsigned char *) 0xBFF00000;
volatile unsigned char *usart_data =
    (unsigned char *) 0xBFF20000;
```

A similar situation can exist if you examine a variable that is modified by an interrupt or other exception handler. Again, declaring the variable as `volatile` should fix the problem.

I won't guarantee that this will *always* work: The C bible describes the operation of `volatile` as implementation dependent. I suspect, though, that compilers that ignore the `volatile` keyword are implicitly not allowed to optimize away loads.

Many programmers have trouble using `volatile`. The thing to remember is that it behaves just like any other C type modifier—just like `unsigned` in the example above. You need to avoid syndromes like this:

```
typedef char * devptr;
volatile devptr mypointer;
```

You've now told the compiler that it must keep loading the pointer value from the variable `devptr`, but you've said nothing about the behavior of the register you're using it to point at. It would be more useful to write the code like this:

```
typedef volatile char * devptr;
devptr mypointer;
```

Once you've dealt with this, the most common reason that optimized code breaks will be that you have tried to drive the hardware too fast. There are often timing constraints associated with reads and writes of hardware registers, and you'll often have to deliberately slow your code to fit in.

What's the main lesson of this section? While it's easier to write and maintain hardware driver code in C than in assembly, it's important to use this option responsibly. In particular, you'll need to understand enough about the way the toolchain converts your high-level source code into low-level machine instructions to make sure you get the system behavior that you intended.

10.5.3 *Miscellaneous Issues When Writing C for MIPS Applications*

- *Negative pointers:* When running simple unmapped code on a MIPS CPU, all pointers are in the `kseg0` or `kseg1` areas, so any data pointer's 32-bit

value has the top bit set and looks “negative.” Unmapped programs on most other architectures are dealing with physical addresses, which are usually a lot smaller than 2 GB!

Such pointer values could cause trouble when pointer values are being compared, if the pointer were implicitly converted to a signed integer type. Any implicit conversions between integer and pointer types (quite common in C) should be made explicit and should specify an unsigned integer type (you should use `unsigned long` for this).

Most compilers will warn about pointer-to-integer conversions, though you may have to specify an option.

- *Signed versus unsigned characters:* In early C compilers, the `char` type used for strings was usually equivalent to `signed char`; this is consistent with the convention for larger integer values. However, as soon as you have to deal with character encodings using more than 7-bit values, this is dangerous when converting or comparing. Modern compilers usually make `char` equivalent to `unsigned char` instead.

If you discover that your old program depends on the default sign-extension of `char` types, good compilers offer an option that will restore the traditional convention.

- *Moving from 16-bit int:* A significant number of programs are being moved up from 16-bit x86 or other CPUs where the standard `int` is a 16-bit value. Such programs may rely, much more subtly than you think, on the limited size and overflow characteristics of 16-bit values. Although you can get correct operation by translating such types into `short`, that will be inefficient. In most cases you can let variables quietly pick up the MIPS `int` size of 32 bits, but you should be particularly aware of places where signed comparisons are used to catch 16-bit overflow.
- *Programming that depends on the stack:* Some kind of function invocation stack and data stack are implicit in C’s block structure. Despite the MIPS hardware’s complete lack of stack support, MIPS C compilers implement a fairly conventional stack structure. Even so, if your program thinks it knows what the stack looks like, it won’t be portable. If possible, don’t just replace the old assumptions with new ones: Two of the most common motivations for stack abuse are now satisfied with respectable and standards-conforming macro/library operations, which may tackle what your software was trying to do before:
 - `stdargs`: Use this include-file-based macro package to implement routines with a variable number of parameters whose type need not be predefined at compile time.
 - `alloca()`: To allocate memory at run time, use this library function, which is “on the stack” in the sense that it will be automatically

freed when the function allocating the memory returns. Some compilers implement `alloca()` as a built-in function that actually extends the stack; otherwise, there are pure-library implementations available. But don't assume that such memory is actually at an address with some connection with the stack.

MIPS Software Standards (ABIs)

For most of this book we’ve described the MIPS architecture from the perspective of how the hardware looks to a programmer. In this chapter, we’re going to describe some standards about how MIPS binary programs should be created to make them compatible with each other.

These standards are designed around characteristics of the hardware, but are often just arbitrary—it has to be done some way, and there’s an advantage if every toolchain does it the same. We’ve met one of those standards already: the register usage conventions described way back in section 2.2.

In this chapter, we’re going to look at how compilers represent data for MIPS programs, at argument passing for functions, and at the use of the stack. In all cases, we’ll draw all our examples from the C language, though essentially the same conventions apply to other languages. Data representation and function linkage are aspects of a formal standard called an ABI (Application Binary Interface), and we’re describing the conventions used in the ABI, sometimes called *o32*. But the ABI document also specifies the encoding of object files (formats like ELF used to hold binary programs and libraries), and you won’t find that in this book.

Linux requires more conventions to allow applications to be dynamically linked together out of incomplete programs and shared libraries. We’ll describe that in Chapter 16.

The organization of the data is profoundly affected by the CPU’s endianness, which was described at length in section 10.2.

The most important ABIs in MIPS history are:

- *o32*: Grew from traditional MIPS conventions (“o” for old), and described in detail here. *o32* is still pretty much universally used by embedded toolchains and for 32-bit Linux.
- *n64*: New formal ABI for 64-bit programs on 64-bit CPUs running under Silicon Graphics’ Irix operating system. SGI’s 64-bit model makes both pointers and C `long` integer types into 64-bit data items. However, *n64*

also changes the conventions for using registers and the rules for passing parameters; because it puts more arguments in registers, it improves performance slightly.

- *n32*: A partner ABI to *n64*, this is really for “32-bit” programs on 64-bit CPUs. It is mostly the same as *n64*, except for having pointers and the `C long` data type implemented as 32 bits. That can be useful—for applications where a 32-bit memory space is already spacious, 64-bit pointers represent nothing but extra overhead.

11.1 Data Representations and Alignment

When you define data in C, the data you get in memory is compilation-target dependent.¹ Moreover, while it’s “nonportable” to assume you have a particular layout, nonportable C is often a more maintainable way of defining a fixed data layout than anything else at your disposal.

The data layout chosen is constrained by what the hardware will do. MIPS CPUs can only load multibyte data that is “naturally” aligned—a four-byte quantity only from a four-byte boundary, and so on—but many CISC architectures don’t have this restriction. The MIPS compiler attempts to ensure that data lands in the right place; this requires far-reaching (and not always obvious) behaviors.

For the purposes of this section, memory is taken as an array of unsigned 8-bit quantities, whose index is the virtual address. For all known MIPS architecture CPUs, this corresponds to a C definition `unsigned char []`.

Like all the modern computers I know of, MIPS uses twos-complement representation for signed integers—so in any data size “-1” is represented by binary all-ones. The overwhelming advantage of twos-complement numbers is that the basic arithmetic operations (add, subtract, multiply, divide) have the same implementation for signed and unsigned data types.²

C integer data types come in *signed* and *unsigned* versions, which are always the same size and alignment. When you don’t specify which, you typically get a *signed int*, *long*, or *long long* but often an *unsigned char*.³

11.1.1 Sizes of Basic Types

Table 11.1 lists fundamental C data types and how they’re implemented for MIPS architecture CPUs. We’ll come back to the `long` and pointer types a bit later—their size changes according to which ABI you use.

1. Strictly speaking, it’s also compiler dependent, but in practice MIPS compilers all comply with the conventions described in this section.

2. At least, until the result has greater precision than the operands.

3. This is an ANSI C feature. In early C `char` was also signed by default. Most compilers allow you to change the default for `char` with a command-line flag—useful when recompiling old software.

TABLE 11.1 Data Types and Memory Representations

<i>C type</i>	<i>Assm Name</i>	<i>Size (bytes)</i>
char	byte	1
short	half	2
int	word	4
long long	dword	8
float	word	4
double	dword	8

The assembler does not distinguish storage definitions for integer and floating-point data types.

11.1.2 *Size of “long” and Pointer Types*

We left those out of the table, because they come out differently in the different ABIs. But they’re also always the same as something else.

So for o32, n32 (and any plausible ABI to be used on 32-bit CPUs), `long` is implemented just like an `int`; for the 64-bit n64 ABI `long` is implemented just like a `long long`.

And in all cases a pointer is always stored just like an `unsigned long`; the MIPS architecture always boasts a simple “flat” address space.

11.1.3 *Alignment Requirements*

All these primitive data types can only be directly handled by standard MIPS instructions if they are *naturally aligned*: that is, a two-byte datum starts at an address that is even (zero modulo 2), a four-byte datum starts at an address that is zero modulo 4, and an eight-byte datum starts at an address that is zero modulo 8.⁴

11.1.4 *Memory Layout of Basic Types and How It Changes with Endianness*

Figure 11.1 shows how each basic type is laid out in our byte-addressed memory; the arrangement is different for big-endian and little-endian software.

4. For MIPS32 CPUs using only 32-bit registers and data paths, eight-byte data types are not handled by any machine instruction, and the eight-byte alignment restriction is not strictly necessary. However, it is still imposed in all known ABIs.

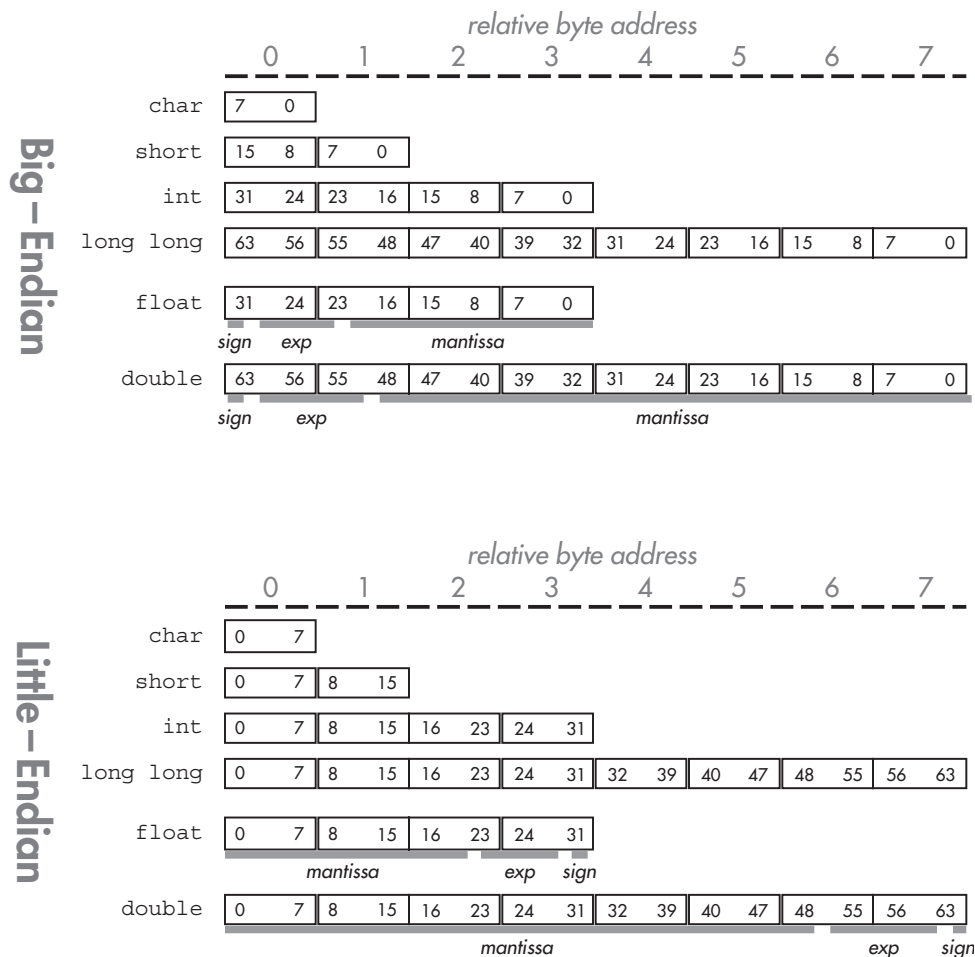


FIGURE 11.1 C data types in memory.

I’ve given in to the temptation to reverse the bits between the two endianness layouts. For memory addressing purposes, this is meaningless; bytes are indivisible 8-bit objects. However, reversing the bit numbers makes the bitwise depiction of the fields of floating-point numbers easier to absorb (and prettier).

Each of these data types is naturally aligned, as described previously. “Endianness” can be a troubling subject and is discussed at length in section 10.2.

11.1.5 *Memory Layout of Structure and Array Types and Alignment*

Complex types are built by concatenating simple types but inserting unused (“padding”) bytes between items, so as to respect the alignment rules.⁵

It’s worth giving a couple of examples. Here’s the byte offsets of data items in a `struct mixed`:

```
struct mixed {
    char c; /* byte 0 */
    /* bytes 1-7 are ``padding`` */
    double d; /* bytes 8-15 */
    short s; /* bytes 16-17 */
};
```

It’s worth stressing that the byte offsets of the fields of constructed data types (*other than those using C bitfields*, see section 11.1.6) are unaffected by endianness.

A data structure or array is aligned in memory to the largest alignment boundary required by any data type inside it. So a `struct mixed` will start on an eight-byte boundary; and that means that if you build an array of these structures you will need padding between each array element. C compilers provide for this by “tail padding” the structure to make it usable for an array, so `sizeof(struct mixed) == 24` and the structure should really be annotated:

```
struct mixed {
    char c; /* byte 0 */
    /* bytes 1-7 are ``padding`` */
    double d; /* bytes 8-15 */
    short s; /* bytes 16-17 */
    /* bytes 18-23 are ``tail padding`` */
};
```

Just to remind you: The size and alignment requirement of pointer and long data types can be four or eight, depending on whether you’re exploiting 64-bit operations.

11.1.6 *Bitfields in Structures*

C allows you to define structures that pack several short “bitfield” members into one or more locations of a standard integer type. This is a useful feature

5. Some compiler systems provide mechanisms to alter the alignment rules for particular data definitions. This allows you to model more possible data patterns with C data declarations, and the compiler will generate appropriate code (with some loss of efficiency) to handle the resulting unaligned basic data types. There are some hints in section 11.1.7.

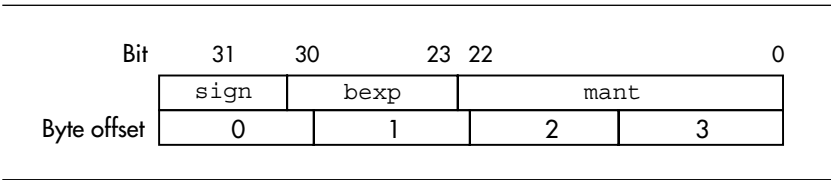


FIGURE 11.2 Bitfields from the big-endian viewpoint.

for emulation, for hardware interfacing, and perhaps for defining dense data structures, but it is fairly incomplete. Bitfield definitions are nominally CPU dependent (but so is everything) but also genuinely endianness dependent.

You may recall that in section 7.9.3 we used a bitfield structure to map the fields of a single-precision IEEE floating-point value (a C `float`) stored in memory. An FP single value is multibyte, so you should probably expect this definition to be endianness dependent. The big-endian version looked like this:

```
struct ieee754sp_konst {
    unsigned    sign:1;
    unsigned    bexp:8;
    unsigned    mant:23;
};
```

C bitfields are always packed—that is, the fields are not padded out to yield any particular alignment. But compilers reject bitfields that span the boundaries of the C type used to hold them (in the example, that’s an `unsigned`, which is short for `unsigned int`).

The structure and mapping for a big-endian CPU is shown in Figure 11.2 (using a typical big-endian’s picture); a little-endian version is shown in Figure 11.3.

The C compiler insists that, even for bitfields, items declared first in the structure occupy lower addresses: When you’re little-endian, you need to turn the declaration backward:

```
struct ieee754sp_konst {
    unsigned    mant:23;
    unsigned    bexp:8;
    unsigned    sign:1;
};
```

To see why that works, you can see from Figure 11.3 that in little-endian mode the compiler packs bits into structures starting from low-numbered bits.

Does this make sense? Certainly some; if you tried to implement bitfields in a less endianness-dependent way, then in the following example

Byte offset	0	1	2	3
	0		22	0
	mant			7
				sign

FIGURE 11.3 Bitfields from the little-endian viewpoint.

`struct fourbytes` would have a memory layout different from `struct fouroctets`- and that doesn't seem reasonable:

```
\nopagebreak
struct fourbytes {
    signed char a; signed char b; signed char c;
    signed char d;
}

struct fouroctets {
    int a:8; int b:8; int c:8; int d:8;
}
```

It's probably not surprising that the CPU's endianness shows up when looking inside a floating-point number; we said earlier that accessing the same data with different C types often showed up the CPU's nature. But it reminds us that endianness remains a real and pervasive issue, even when there's no foreign data or hardware to manage.

A field can only be packed inside one storage unit of its defined type; if we try to define a structure for a MIPS double-precision floating-point number, the mantissa field contains part of two 32-bit `int` storage units and can't be defined in one go. The best we can do is something like this:

```
struct ieee754dp_konst {
    unsigned sign:1;
    unsigned bexp:11;
    unsigned manthi:20; /* cannot get 52 bits into... */
    unsigned mantlo:32; /* .. a regular C bitfield */
};
```

You're permitted to leave out the name of the field definition, so you don't have to invent names for fields that are just there for padding.

With GNU C or other modern compilers, we could have used an `unsigned long long bitfield` and defined the double-precision floating-point register in one go. But the ANSI C specification does not require compilers to support `long long`.

The full alignment rules for bitfields are a little complicated:

- As we said above, a bitfield must reside entirely in a storage unit that is appropriate for its declared type. Thus, a bitfield never crosses its unit boundary.
- Bitfields can share a storage unit with other struct/union members, including members that are not bitfields (to pack together, the adjacent structure member must be of a smaller integer type).
- Structures generally inherit their own alignment requirement from the alignment requirement of their most demanding type. Named bitfields will cause the structure to be aligned (at least) as well as the type requires. Unnamed fields—regardless of their defined type—only force the storage unit or overall structure alignment to that of the smallest integer type that can accommodate that many bits.
- You might want to be able to force subsequent structure members to occupy a new storage unit. In some compilers you can do that with an *unnamed zero-width* field. Zero-width fields are otherwise illegal (or at least pointless).

You now know everything you need to map C data declarations to memory in a manner compatible with the various ABIs.

11.1.7 *Unaligned Data from C*

Sometimes the alignment rules that help make MIPS CPUs efficient are a nuisance, because you'd like to force your C structure to represent an exact byte-for-byte memory layout, perhaps to match some data from another application.

The GNU C compiler (and most other good ones) allows you to control the alignment used to find a location for any data, whether simple variable or complex structure. In the GNU C dialect, `__attribute__((packed))` in a declaration will cause all normal alignment requirements to be ignored and the data packed, while `__attribute__((aligned(16)))` insists on a 16-byte alignment—greater than that used for any standard data type by a MIPS compiler.⁶

By a judicious mixture of both attributes, you can set out a data structure any way you like (individual `__aligned__()` controls inside a packed structure alignment as specified).

The ANSI standard has a slightly less powerful and flexible syntax for data structures, though not for basic types (but because it's a standard, it's quite likely to work across different compilers): You use a preprocessor-like directive.

6. While the compiler will respect your wishes for a very large alignment, sometimes the linker or other later part of the toolchain may not do so—go carefully.

So if you bracket a data declaration starting with a `#pragma pack(2)` line, the compiler will use no more than a two-byte alignment for those items. Put in a `#pragma pack()` to restore normal behavior.

So, for example:

```
int unalignedload (ptr)
    void *ptr;
{
#pragma pack (1)
    /* define what you like here, with no assumptions about alignment */

    struct unaligned {
        int conts;
    } *ip;

#pragma pack ()
    /* back to default behavior */

    ip = (struct unaligned *) ptr;

    /* can now generate an unaligned load of int size */
    return ip->conts;
}
```

11.2 Argument Passing and Stack Conventions for MIPS ABIs

The compilers for C and all common compiled languages build programs out of many modules, compiled separately. To work together with each other and with the operating system, the compiled code of a module relies on conventions (enforced by compilers, and therefore mandatory for assembly language programmers) about register usage, stack construction, argument passing, and so on.

Some of you may not be writing in C or C++. If your programming is in Java or some other high-level language that is wholly or partially interpreted, then there will be nothing directly relevant to you here. I have written this section only in terms of C code (I don't understand other languages well enough, and in any case I can't figure out where I should stop).

This section will cover the stack, subroutine linkage, and argument passing, and how those are managed for MIPS code so as to do everything the programmer needs in a reasonably efficient way. Overall conventions about register use were introduced in section 2.2.1, and a description of how the compiler packages up data for a MIPS CPU was in section 11.1.

11.2.1 *The Stack, Subroutine Linkage, and Parameter Passing*

From the very start of MIPS' existence in the early 1980s, there was a set of conventions about how to pass arguments to functions (this is C-speak for “pass parameters to subroutines”) and about how to return values from functions.

These conventions follow logically from an underlying principle: All arguments are allocated space in a data structure on the stack, but the contents that belong to the first few stack locations are in fact passed in CPU registers—the corresponding memory locations are left undefined. In practice, this means that for most calls, the arguments are all passed in registers; however, the stack data structure (even when nothing is put in it) is the best starting point for understanding the process.

We'll describe the argument passing and stack features of the o32 standard in some detail, and then summarize the changes with n32 and n64 in section 11.2.8.

There has been much discussion about improving these standards: o32 is very old, and programming habits have changed since its invention. We hope something good comes out of all these discussions—but at the time of writing (Spring 2006) and for the foreseeable future, you will be using o32 compilers for embedded applications on 32-bit MIPS CPUs, and you won't lose a lot.

11.2.2 *Stack Argument Structure in o32*

The MIPS hardware does not directly support a stack, but the semantics of C pretty much mandate it. o32 has a stack that is grown downward, and the current stack bottom is kept in register **sp** (an alias for **\$29**). Any OS that is providing protection and security will make no assumptions about the user's stack, and the value of **sp** doesn't really matter except at the point where a function is called. But it is conventional to keep **sp** at or below the lowest stack location your function has used.

At the point where a function is called, **sp** must be eight-byte-aligned, matching the alignment of the largest basic types—a `long long` integer or a floating-point `double`. The eight-byte alignment is not required by 32-bit MIPS integer hardware, but it's essential for compatibility with CPUs with 64-bit registers, and thus part of the rules. Subroutines fit in with this by always adjusting the stack pointer by a multiple of eight.⁷

To call a subroutine according to the MIPS standard, the caller creates a data structure on the stack to hold the arguments, starting at the location that **sp** is pointing at. The first argument (leftmost in the C source) is lowest in memory. Each argument occupies at least one word (32 bits); 64-bit values like floating-point `double` and `long long` must be aligned on an eight-byte boundary (as are data structures that contain a 64-bit scalar field). The argument structure really does look like a C `struct`, but there are some more rules.

7. SGI's n32 and n64 standards call for the stack to be maintained with 16-byte alignment.

First, you should allocate a minimum of 16 bytes of argument space for any call, even if the arguments would fit in less.⁸

In the absence of a function prototype, C rules require that any simple integer argument smaller than an `int` (that is, any `char` or `short`) is “promoted” to an `int` and passed as a 32-bit object. That’s just done for arguments that are simple data types: It doesn’t apply to a partial-word field inside a `struct` argument.

11.2.3 *Using Registers to Pass Arguments*

Any arguments allocated to the first 16 bytes (four words) of the argument structure are passed in registers **a0–3** (**\$4–\$7**), and the caller can and does leave the first 16 bytes of the structure undefined. The stack-held structure must still be reserved; the called function is entitled to save **a0–3** back into memory if it needs to, and is entitled to do so completely blind, without knowing how many arguments there are, or of what type.

It’s inefficient to carry floating-point (FP) values in integer registers, so there’s a special test used to identify some function arguments that can be passed in FP registers instead.

The criteria for deciding when and how to use FP registers look peculiar. Old-fashioned C had no built-in mechanism for checking that the caller and callee agreed on the type of each argument to a function. To help programmers survive this, the caller converted arguments to fixed types: `int` for integer values and `double` for floating point. There was no way of saving a programmer who confused floating-point and integer arguments, but at least some possibilities for chaos were averted.

Modern C compilers use function prototypes available when the calling function is being compiled, and the prototypes define all the argument types. But even with function prototypes, there are routines—notably the familiar `printf()`—where the type of argument is unknown at compile time; `printf()` discovers the number and type of its arguments at run time.

MIPS made the following rules. Unless the first argument is a floating-point type, no arguments can be passed in FP registers. This is a kludge that ensures that traditional functions like `printf()` still work: Its first argument is a pointer, so all arguments are allocated to integer registers, and `printf()` will be able to find all its argument data regardless of the argument type. The rule is also not going to make common math functions inefficient, because they mostly take only FP arguments.

Where the first argument is a floating-point type, it will be passed in an FP register, and in this case so will any other FP types that fit in the first 16 bytes of the argument structure. Two `doubles` occupy 16 bytes, so only two FP registers are defined for arguments—**fa0** and **fa1**, or **\$f12** and **\$f14**. Evidently

8. Why? See section 11.2.3.

nobody thought that functions explicitly defined to have lots of single-precision arguments were frequent enough to make another rule.

One of the worst faults caused by the age of o32 is that its use of registers is compatible with the very earliest MIPS floating-point units, which used only the even-numbered registers to hold floating-point values. Double-precision values quietly extended into the adjacent odd-numbered register; the odd-numbered registers were used only when reading or writing FP values from memory, or from integer registers. o32's resulting register conventions do not quite prevent software from using all 32 registers in later CPUs, but they don't make for great efficiency.

Another peculiarity is that if you define a function that returns a structure type that is too big to be returned in the two registers normally used, then the return-value convention involves the invention of a pointer as the implicit first argument before the first (visible) argument (see section 11.2.7).

If you're faced with writing an assembly routine with anything but a simple and obvious calling convention, it's probably worth building a dummy function in C and compiling it with the "-S" option to produce an assembly file you can use as a template.

11.2.4 *Examples from the C Library*

Here is a code example:

```
thesame = strcmp("bear", "bearer", 4);
```

We'll draw out the argument structure and the registers separately (see Figure 11.4), though in this case no argument data goes into memory; later, we'll see examples where it does.⁹

There are fewer than 16 bytes of arguments, so they all fit in registers.

That seems a ridiculously complex way of deciding to put three arguments into the usual registers! But let's try something a bit more tricky from the math library:

```
double ldexp (double, int);

y = ldexp(x, 23); /* y = x * (2**23) */
```

Figure 11.5 shows the corresponding structure and register values.

9. After much mental struggle, I decided it was best to have the arguments ordered top to bottom in these figures. Because the stack grows down, that means memory addresses increase down the page, which is opposite from how I've drawn memory elsewhere in the book.

Stack Position	Contents	Register	Contents
sp+0	undefined	a0	address of "bear"
sp+4	undefined	a1	address of "bearer"
sp+8	undefined	a2	4
sp+12	undefined	a3	undefined

FIGURE 11.4 Argument passing for `strcmp()`, three non-FP operands.

Stack Position	Contents	Register	Contents
sp+0	undefined	\$f12	(double) x
sp+4		\$f13	
sp+8	undefined	a2	23
sp+12	undefined	a3	undefined

FIGURE 11.5 Argument passing for `ldexp()`: floating-point argument.

11.2.5 *An Exotic Example: Passing Structures*

C allows you to use structure types as arguments (it is much more common practice to pass pointers to structures instead, but the language supports both). To fit in with the MIPS rules, the structure being passed just becomes part of the argument structure, where its internal layout is exactly like its regular memory image. Inside a C structure, byte and halfword fields are packed together into single words of memory, so when we use a register to pass the data that conceptually belongs to the stack-resident structure, we have to pack the register with data to mimic the arrangement of data in memory.

So, if we have:

```
struct thing {
    char letter;
    short count;
    int value;
} = {"z", 46, 100000};

(void) processthing (thing);
```

then the arguments shown in in Figure 11.6 will be generated.

Stack Position	Contents	Register	Contents
sp+0	undefined	a0	"z" x 46
sp+4	undefined	a1	100000

FIGURE 11.6 Arguments when passing a structure type.

MIPS C structures are laid out with fields so their memory order matches the order of definition (though padded where necessary to conform to the alignment rules), so the placement of fields inside the register follows the byte order exposed by load/store instructions, which differ according to the CPU's endianness. The layout in Figure 11.6 is inspired by a big-endian CPU, when the `char` value in the structure should end up in the most significant 8 bits of the argument register but is packed together with the `short`.

If you really want to pass structure types as arguments, and they must contain partial-word data types, you should try this out and see whether your compiler gets it right.

11.2.6 *Passing a Variable Number of Arguments*

Functions for which the number and type of arguments are determined only at run time stress conventions to their limits. Consider this example:

```
printf ("length = %f, width = %f, num = %d\n", 1.414, 1.0, 12);
```

The rules above allow us to see that the argument structure and register contents will be as shown in Figure 11.7.

There are two things to note. First, the padding at `sp+4` is required to get correct alignment of the `double` values (the C rule is that floating-point arguments are always passed as double unless you explicitly decide otherwise with a `typedef` or function prototype). Note that padding to an eight-byte boundary can cause one of the standard argument registers to be skipped.

Second, because the first argument is not a floating-point value, the rules tell us not to use any FP registers for arguments. So the data for the second argument (coded as it would be in memory) is loaded into the two registers `a2` and `a3`.

This is much more useful than it looks!

The `printf()` subroutine is defined with the `stdarg.h` macro package, which provides a portable cover for the register and stack manipulation involved in accessing an unpredictable number of operands of unpredictable types. The `printf()` routine picks off the arguments by taking the address of the first or second argument and advancing through memory up the argument structure to find further arguments.

Stack Position	Contents	Register	Contents
sp+0	undefined	a0	format pointer
sp+4	undefined	a1	undefined
sp+8	undefined	a2	(double) 1.414
sp+12		a3	
sp+16	(double) 1.0		
sp+20			
sp+24	12		

FIGURE 11.7 Argument passing for `printf()`.

To make this work, we need to persuade the C compiler working on the `printf()` routine to store registers **a0** through **a3** into their shadow locations in the argument structure. Some compilers will see you taking the address of an argument and take the hint; ANSI C compilers should react to “...” in the function definition; others may need some horrible “pragma,” which will be decently concealed by the macro package.

Now you can see why it was necessary to put the `double` value into the integer registers; that way `stdarg` and the compiler can just store the registers **a0–a3** into the first 16 bytes of the argument structure, regardless of the type or number of the arguments.

11.2.7 *Returning a Value from a Function*

An integer or pointer return value will be in register **v0** (**\$2**). By MIPS convention, register **v1** (**\$3**) is reserved too, even though many compilers don’t use it. However, expect it to be used in 32-bit code for returning `long long` (64-bit integer) values. There seems to be some debate about whether a structure value that occupies five to eight bytes might be returned using both registers: GNU C always uses a pointer to return bigger-than-register structures, but the specification is ambiguous.

Any floating-point result comes back in register **\$f0** (implicitly using **\$f1** in a 32-bit CPU, if the value is double precision).

If a function is declared in C as returning a structure value that is too big to fit into the return registers **v0** and **v1**, something else has to be done. In this case, the caller makes room on its stack for an anonymous structure variable, and a pointer to that structure is prepended to the explicit arguments; the called function copies its return value to the template. Following the normal rules for arguments, the implicit first argument will be in register **a0** when the function is called. On return, **v0** points to the returned structure too.

11.2.8 *Evolving Register-Use Standards: SGIs n32 and n64*

For the purposes of this section (calling conventions and integer register usage) the n32 and n64 ABIs are identical.¹⁰ The n32/n64 ABIs are applicable only to MIPS III CPUs, which have 64-bit registers.

Despite the significant attempts to keep the register conventions similar, o32 and n32/n64 are deeply incompatible, and functions compiled in different ways will not link together successfully. The following points summarize the n32/n64 rules:

- Up to eight arguments can be passed in registers.
- Argument slots and therefore argument registers are 64 bits in size. Shorter integer arguments are promoted to a 64-bit size, exactly as they would if loaded into a register.
- The caller does not allocate stack space for arguments passed in registers.
- Any floating-point value that ends up occupying one of the first eight argument slots by itself is passed in an FP register. That even includes aligned `double` fields in arrays and structures, so long as the field isn't in a union and isn't a variable argument to `printf()` or a similar variable-argument function.
- n32 and n64 recognize 16-byte basic objects (such as `long double` floating-point), and such objects are 16-byte aligned. That also means that the stack must be realigned to a multiple of 16 bytes for each function's frame.

When life gets complicated (as when passing structures or arrays), the use of the registers is still figured out from a ghostly argument structure, even though there is now no stack space reserved for the first eight slots.

The n32/n64 conventions abandon o32's first-argument-not-FP kludge, which o32 uses to identify floating-point arguments as special cases for `printf()` and so on. The new conventions require that both caller and callee code be compiled with full knowledge of the number and type of arguments and therefore that they need function prototypes.

For a function like `printf()`, where the type of arguments is unknown at compile time, all the variable arguments are actually passed in integer registers.

The n32/n64 organization has a different set of register-use conventions; Table 11.2 compares the use of integer registers with the o32 system. There is only one material difference: Four registers that used to be regarded purely as temporaries should now be used to pass the fifth through the eighth arguments.

10. Under the n64 convention `long` and pointer types are compiled as 64-bit objects; with n32 only `long long` types are 64 bits.

TABLE 11.3 FP Register Usage with o32 and n32/n64 Conventions

Register number	o32 use	
\$f0, \$f2	Return values; f1 is used only for the native complex number data type used by FORTRAN, but not available in C	
\$f4, \$f6, \$f8, \$f10	Temporaries—functions can use without any need to save	
\$f12, \$f14	Arguments	
\$f16, \$f18	Temporaries	
\$f20, \$f22, \$f24, \$f26, \$f28, \$f30	Saved registers—functions must save and restore any of these registers they want to write, making them suitable for long-lived values that persist across function calls	
Register number	n32 use	n64 use
\$f0, \$f2	Return values— f2 is used only when returning a structure of exactly two floating-point values; this is a special case that deals with FORTRAN complex numbers	
\$f1, \$f3	Temporaries	
\$f4–\$f10		
\$f12–\$f19	Arguments	
\$f20–\$f23	Evens (from f20–\$f30) are temporary; odds (from f21–\$f31) are saved	Temporaries
\$f24–\$f31		Saved registers

maintained some vestiges of compatibility, the company decided instead to tear up most of the existing rules and start again.

In addition to the larger number of arguments that can be passed in registers, the n32/n64 standard doesn’t make any rules dependent on whether the first argument is a floating-point type. Instead, arguments are allocated to registers according to their position in the argument list. So, to repeat one of the examples used above:

```
double ldexp (double, int);

y = ldexp(x, 23); /* y = x * (2**23) */
```

In n64, the double argument will be in the FP register **f12** and the integer value 23 (“sign-extended” to 64 bits) will be in **a1**, the integer register that carries an argument in the second slot. No stack space will be reserved for these arguments.

Although n32/n64 can handle an arbitrary mix of floating-point and other values and still put any `double` types that are in the first eight arguments in FP registers, there are some careful rules. Any argument that is touched by a `union` (and that therefore might not really be a double) is excluded and so are any of the variable arguments of a variable-number-of-arguments function. Since the function itself and any callers must make the same decision (or things won't work), this depends on having correct function prototypes. Mostly, these days, that's a reasonable assumption.

11.2.9 *Stack Layouts, Stack Frames, and Helping Debuggers*

Figure 11.8 gives a diagrammatic view of the stack frame of a MIPS function. (We're back to having the stack growing down, with higher memory

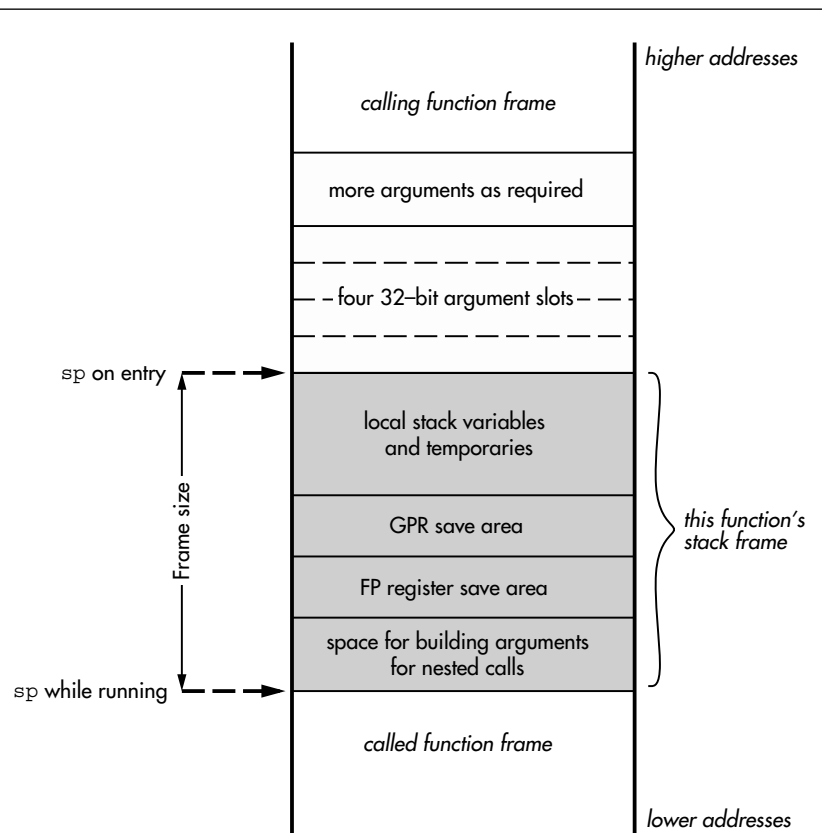


FIGURE 11.8 Stack frame for a nonleaf function.

at the top.) You should recognize the slots reserved for the first four words of the function's arguments as required by the traditional MIPS function calling convention—newer calling conventions will only provide any space they actually need.

The gray areas of the diagram show stack space used by the function itself; the white area, above the bold line, belongs to the caller. *All* the gray components of the stack frame are optional, and some functions need none of them; such a simple function does not need to do anything to the stack. We'll see some of those in the examples through the rest of the chapter.

Apart from the arguments (whose layout must be agreed with the caller), the stack structure is private to the function. The only reason we need a standard arrangement is for debugging and diagnostic tools, which want to be able to navigate the stack. If we interrupt a running program for debugging, we'd very much like to be able to run backward up the stack, displaying a list of the functions that have been called on the way to our breakpoint, and the arguments passed to those functions. Moreover, we'd like to be able to step the debugger context back up the stack a few positions and in that context to discover the value of variables—even if that piece of code was maintaining the variable's data in a register, as optimizing compilers should.

To perform this analysis, the debugger must know a standard stack layout and must be fed information that allows it to see the size of each stack frame component and the internal layout of each of those components. If a function somewhere up the stack saved the value of `$0` in order to use it, the debugger needs to know where to find the saved value.

In CISC architectures, there is often a complex function call instruction that maintains a stack frame similar to that in Figure 11.8 but with an additional frame pointer register that corresponds to the position marked “**sp** on entry” on our diagram. In such a CPU, the caller's frame pointer will be stored at some known stack position, allowing a debugger to skip up the stack by analyzing a simple linked list. But in a MIPS CPU, all this extra runtime work is eliminated; most of the time, a compiler knows how much to decrement the stack pointer at the head of a function and how much to increment it before return.

So in the minimal MIPS stack frame, where is a debugger to find out where data is stored? Some debuggers are quite heroic and will even interpret the first few instructions of a function to find how large the stack frame is and to locate the stored return address. But most toolchains pass at least some stack frame information in the object code, written there by suitable assembly directives.

Since the mixture of directives is quite dependent on the toolkit, it's worth defining prologue and epilogue macros that both save you from having to remember the details and make it easier to change to another toolkit if you need to. Most toolkits will come with some macros ready to use; you'll see simple ones called **LEAF** and **NESTED** used in the following examples.

We've made no attempt to provide details of the underlying assembly directives for debugging: sorry. Your best bet is probably to compile some code and study the assembly output of the compiler.

We can divide up functions into three classes and prescribe three different approaches, which will probably cover everything you need.

Leaf Functions

Functions that contain no calls to other functions are called *leaf functions*. They don't have to worry about setting up argument structures and can safely maintain data in the nonpreserved registers **t0–t7**, **a0–a3**, and **v0** and **v1**. They can use the stack for storage if they feel like it but can and should leave the return address in register **ra** and return directly to it.¹³

Most functions that you may write in assembly for tuning reasons or as convenience functions for accessing features not visible in C will be leaf functions; many of them will use no stack space at all. The declaration of such a function is very simple, for example:

```
#include <mips/asm.h>
#include <mips/regdef.h>

LEAF(myleaf)
    ...
    <your code goes here>
    ...
    j      ra
END(myleaf)
```

Most toolchains can pass your assembly source code through the C macro preprocessor before assembling it—UNIX-like tools decide based on the file-name extension. The files `mips/asm.h` and `mips/regdef.h` include useful macros (like **LEAF** and **END**, shown above) for declaring global functions and data; they also allow you to use the software register names, for example, **a0** instead of **\$4**.

The **LEAF** and **END** macros package together whatever you have to tell the assembler to help the debugger navigate your function: They don't contribute anything else except the function name.

Nonleaf Functions

Nonleaf functions are those that contain calls to other functions. Normally, the function starts with code (the function prologue) to reset **sp** to the low-water mark of its use inside the function; that low-water mark will be the base of the argument structure for the nested function call. We'll also need stack space to

13. Storing the return address somewhere else may work perfectly well, but the debugger won't be able to find it.

save the incoming values of any of the “saved” registers **s0–s8** that the function uses. Stack locations must also be reserved for **ra**, automatic (i.e., stack-based local) variables, and any further registers whose value this function needs to preserve over its own calls. (If the values of the argument registers **a0–a3** need to be preserved, they can be saved into their standard positions on the argument structure.)

Note that since **sp** is set only once (in the function prologue), all stack-held locations can be referenced by fixed offsets from **sp**.

To illustrate this, we will define a fairly trivial C module. But the function `nonleaf()` will take five arguments (so it will need to pass at least one argument on the stack), it will call another function with five arguments, and one of those arguments will be a pointer to a stack location. That should show you enough to get the feel for how the ABI works.

```
extern nested (int a, int b, int c, int d, int *e);

extern nonleaf (int a, int b, int c, int d, int e)
{
    nested(d, b, c, a, &e);
}
```

This will give you a stack structure similar to that shown in Table 11.4.

TABLE 11.4 Stack Layout for `nonleaf()`

	48	e
	44	(reserved for c/ a3)
	40	(reserved for b/ a2)
	36	(reserved for a/ a1)
sp on entry ⇒	32	(reserved for a/ a0)
	28	(pad to 8 bytes)
	24	saved ra
	20	(pad to 8 bytes)
	16	&e
	12	(reserved for a/ a3)
	8	(reserved for c/ a2)
	4	(reserved for b/ a1)
sp running ⇒	0	(reserved for d/ a0)

You can see where argument slots are reserved, but not used, because the corresponding arguments are in registers. You can also see where padding is introduced, because the lowest saved register and the bottom of the running stackframe must both be aligned to eight bytes.

Compile `nonleaf.c` to assembly code with the GNU C compiler and we get:

```
.file 1 "nonleaf.c"
.section .mdebug.abi32
.previous
.text
.align      2
.globl      nonleaf
.set        nomips16
.ent        nonleaf
nonleaf:
    .frame   $sp,32,$31    # vars= 0, regs= 1/0, args= 24, gp= 0
    .mask    0x80000000,-8
    .fmask    0x00000000,0
    .set      noreorder
    .set      nomacro

    addiu    $sp,$sp,-32
    sw       $31,24($sp)
    move     $2,$4
    addiu    $3,$sp,48
    sw       $3,16($sp)
    move     $4,$7
    jal      nested
    move     $7,$2

    lw       $31,24($sp)
    j        $31
    addiu    $sp,$sp,32

    .set     macro
    .set     reorder
    .end     nonleaf
    .ident   "GCC: (GNU) 3.4.4 mipssde-6.03.01-20051114"
```

We'll probably need to walk through that in sections.

```
.file 1 "nonleaf.c"
.section .mdebug.abi32
.previous
```

The start of the file is housekeeping information for very basic debug (if we'd compiled the file with the “-g” flag, we'd have got a lot more debugging information). No further explanation, sorry!

```
.text
.align      2
.globl      nonleaf
.ent        nonleaf
nonleaf:
```

Object code and linkage housekeeping. We're generating code so it will go in the **.text** section of the object file, and the function entry point name is a global symbol that is also an entry point. And there is the entry point label, too.

```
.frame      $sp,32,$31    # vars= 0, regs= 1/0, args= 24, gp= 0
.mask       0x80000000,-8
.fmask      0x00000000,0
```

Information about the stack. We'll generate a 32-byte stack frame for this function, as seen in Table 11.4. The only register to be saved is **\$31**, the return address. That shows up in the **.mask** as bit 31 of a bit map of saved registers, and that **-8** indicates the position of the stack block used for saving GP registers. **.fmask** does the same job for FP registers: There aren't any FP registers used here, so it's zero.

```
.set        noreorder
.set        nomacro
```

Those two are telling the assembler that the compiler is taking charge here. The assembler is not to move instructions around to fill branch delay slots, or to interpret “macro instructions” (ones that expand to more than one instruction in machine code).

It's about time we got on with some code:

```
addiu $sp,$sp,-32
sw     $31,24($sp)
```

Adjust the stack pointer and save the return address from the call to `nonleaf()`.

```
move    $2,$4          # remember 'a' for later
addiu   $3,$sp,48
sw      $3,16($sp)     # put &e in argument slot
move    $4,$7          # first argument is 'd'
jal     nested
move    $7,$2          # in delay slot, third argument is 'a'
```

Adjust the arguments and call `nested()`.

```
lw      $31,24($sp)
j       $31
addiu   $sp,$sp,32
```

Retrieve the return address and return. Restore the stack pointer in the delay slot.

```
.set    macro
.set    reorder
.end    nonleaf
.ident   "GCC: (GNU) 3.4.4 mipssde-6.03.01-20051114"
```

Re-enable all the assembly things you turned off, and include a string to identify the compiler build.

Frame Pointers for More Complex Stack Requirements

In the stack frames described above, the compiler has been able to manage the stack with just one reserved register, `sp`. Those of you who are familiar with other architectures will know that they often use two stack maintenance registers: a stack pointer (`sp`) to mark the stack low-water point and a frame pointer to point to the data structures created by the function prologue. However, so long as the compiler can allocate all the stack space needed by the function in the function prologue code, it should be able to decrement `sp` in the prologue and leave it pointing to a constant stack offset for the life of the function. If so, everything on the local stack frame is at a compile-time-known offset from `sp`, and no frame pointer is needed. But sometimes you want to mess with the stack pointer at run time: Figure 11.9 shows how MIPS allocates a frame pointer to cope with this need.

What leads to an unpredictable stack pointer? In some languages, and even in some extensions to C, dynamic variables can be created whose size varies at run time. And many C compilers can allocate stack space on demand through the

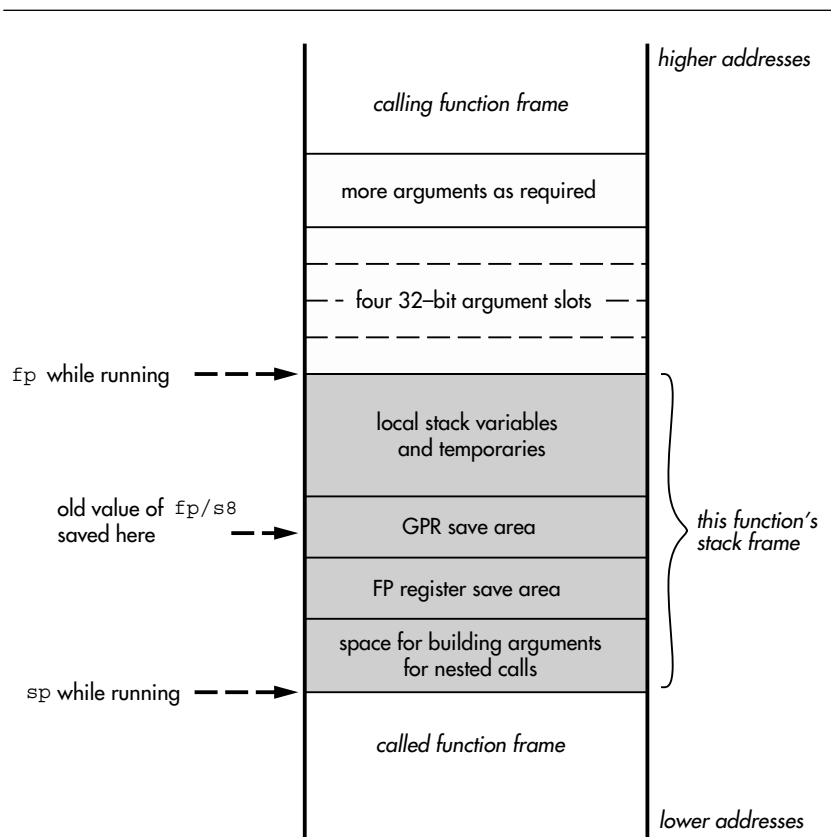


FIGURE 11.9 Stack frame using separate frame pointer register.

useful built-in function `alloca()`.¹⁴ In this case the function prologue grabs another register, `s8` (which has a regular alias of `fp`), and sets it to the incoming value of `sp`.

Since `fp` (in its other guise as `s8`) is one of the saved registers, the prologue must also save its old value, which is done just as if we were using `s8` as a subroutine variable. In a function compiled with a frame pointer, all local stack frame references are made via `fp`, so if the compiler needs to lower `sp` to make space for variables of runtime-computed size, it can go right ahead.

14. Actually, some implementations of `alloca()` don't just make space on the local stack, and some are pure library functions (which means that you never need go without `alloca()` for portability reasons). But compilers that implement `alloca()` using stack space make code that goes faster.

Note that if the function has a nested call that uses so many arguments that it needs to pass data on the stack, that will be done with relation to **sp**.

One ingenious feature of this trick is that neither the caller of a frame pointer function, nor anything called by it, sees it as anything special. Functions it calls are obliged to preserve the value of **fp** because it's a callee-saved register; and the callee-visible part of the stack frame looks like it should.

Assembly buffs may enjoy the observation that when you create space with `alloca()`, the address returned is actually a bit higher than **sp**, since the compiler has still reserved space for the largest argument structure required by any function call.

Some tools also employ an **fp**-based stack frame when the size of the local variables grows so large that some stack frame objects are too far from **sp** to be accessed in a single MIPS load/store instruction (with its ± 32 -KB offset limit).

11.2.10 *Variable Number of Arguments and stdargs*

If you need to build a new function that takes a variable number of arguments, use your toolkit's `stdarg.h` macro package (compulsory for ANSI compatibility). The macro package delivers the macros—or possibly functions—`va_start()`, `va_end()`, and `va_arg()`. To see how they're used, look at how the one package implements `printf()`:

```
int printf(const char *format, ... )
{
    va_list arg;
    int n;

    va_start(arg, format);
    n = vfprintf(stdout, format, arg);
    va_end(arg);

    return n;
}
```

Once we've called `va_start()`, we can extract any argument we like. So somewhere in the middle of the code that implements the format conversions for `printf()`, you'll see the following code used to pick up the next argument, supposing it to be a double-precision floating-point type:

```
...
d = va_arg(ap, double);
...
```

Never try to build an assembly function that takes a variable number of arguments—it isn't worth the portability hassle.

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Debugging MIPS Designs—Debug and Profiling Features

When you build a low-cost, low-power embedded device or consumer gizmo, it makes sense to squeeze out every feature that doesn't contribute to the final application.

Or does it?

In marketing terms, the way to make money out of these kinds of devices is to be there first: As the devices become familiar, prices drop rapidly (as has happened with DVD players, for example). Product lifetimes are short, so development timescales must be compressed.

There's a tension here: A cut-to-the-bone hardware system is a hostile base for development and test. The economics of SoC chips offers a way out. Most SoCs can accommodate a few percent more transistors used for predefined functions with no impact on production cost, because the transistor count is limited by the difficulty of designing circuits and making reasonably sure the chip will work first time. If the suppliers of SoC components can include development aids without passing on much risk to the SoC builder, everyone might win.

So modern MIPS CPUs come with a certain amount of hardware aimed to help developers. Different builders will resolve the competing requirements differently, so—unfortunately for developers and development tool creators—these features are optional and may not be present on your system. But it's a definite trend in the last few years to include more of them.

Developers benefit from three different kinds of information about their system, which can be reasonably divided into “debug,” “trace,” and “profile” information.

Debug—The Exact State of the System

To correct your logical errors about how the system should work, it's very helpful to be able to dig down and see exactly what is happening, sometimes at the lowest level (instruction by instruction).

This is the job of a software debugger. To be usable in a complex system, the debugger needs access to your software source, as well as binary, and must be able to correlate the two. Correlating debug events with source code is a large problem we shall ignore here, except to point out that such a debugger can't reside entirely in an embedded system under test, so the debugger needs some kind of host-target connection. At the CPU level, a debugger needs access to the program's memory and in-CPU state, as well as the ability to control execution by arranging traps to debugger software to happen at appropriate points.

MIPS specifications define an on-CPU debug help unit under the name of EJTAG, described below. But MIPS CPUs also retain pre-EJTAG features, which linger because they're more familiar and easier for some kinds of debug software to use, and we'll mention them too.

Trace—What Just Happened?

To make sense of where the system is, it's useful to have some kind of overview about how it progressed (for example) from "correct" to wrong behavior. The whole system state is unmanageably huge, and it's difficult to know which state will be important until afterward. But for a CPU, it's certainly interesting to look into the instructions it recently executed.

When the CPU is running as fast as billions of instructions per second, it's a challenge to save enough of that information to cover enough time at enough level of detail, and to get it off-chip fast enough to avoid either stopping the CPU or losing some trace information.

The system used in MIPS CPUs is called PDtrace and is hooked onto the EJTAG debug block.

Profile—Statistical Information about Behavior over a Period of Time

If debuggers are provided because programmers can't hold every possible detailed implication of their software in their heads, profilers are there because it's even more difficult to visualize the behavior of whole subsystems. The modular programming practices that make it possible to build a correct system insulate the programmer from knowledge of exactly what the CPU has to do to carry out tasks—but for a system of any complexity, that means you've only the vaguest idea whether pressing a button will lead to a satisfactory response or an unsettling wait.

A software tool that provides statistical information about the large-scale timing and behavior of a system is called a profiler. And profilers work by instrumenting the system, counting events over a particular time.

The EJTAG system includes an option called "PC sampling," which provides some useful information; but MIPS CPUs also have a number of

performance counters that are controlled by on-target software. The counters and their controls are provided as CP0 registers and described further below.

When a developer runs a system for debug, trace, or profiling, the hardware that provides the low-level information may itself change the behavior of the system. That isn’t helpful: It’s preferable if development assistance is, as far as possible, nonintrusive.

We said previously that transistors on your SoC devoted to development aids are extremely cheap and can be a bargain. What’s harder to provide at low cost is the connection to your development system, which carries the information back for analysis.

In older systems you typically linked the target and host with an extra serial port or some other piece of system I/O. But more often, it involved a tricky negotiation with the target system software to share the use of a connector and device that have some role in the final system.

That’s intrusive: We’d prefer a separate debug connection. Ideally (to reduce the impact on the system design, and to create a standard that will make the tools easier to build), it should come directly from the CPU. But the SoC with the CPU in it is usually at the heart of the system: Transistors may be cheap, but extra pins are costly.

But here we get lucky; someone else’s problem prompts a solution. Electronic systems are soldered together, and the soldering process has a significant defect rate. Modern electronic boards are automatically tested before being used in products, and a few pins on complex devices are reserved for that testing. The overwhelmingly chosen test connection standard is JTAG, so MIPS chose to connect the development subsystem by recycling the JTAG pins, which are otherwise unused at run time. That’s why the debug unit is called EJTAG.

However, the debug hardware is still useful to a more traditional debugger—one running on the target system and communicating using conventional devices of some kind. A debugger that is deeply enmeshed with a high-level operating system (for example) is likely to run substantially on the target. See section 12.1.13 for some notes on what has to be done for an on-target debugger to use the debug unit.

12.1 The “EJTAG” On-chip Debug Unit

The EJTAG unit is a loosely integrated collection of in-CPU resources used to build debug and trace tools. As we said above, EJTAG finds pins for a nonintrusive host connection (and gets its name) by recycling the JTAG pins already included in every SoC for chip test.¹

1. Though it can be quite useful to provide EJTAG with independent-of-hardware-test pins, if you can afford it.

Such a debug unit requires:

- Physical communications with the host—usually via some kind of *probe* device connected to the debug host by some general-purpose network or wire. The probe is attached to the SoC’s JTAG pins.
- The ability for the host/probe to “remote-control” the CPU. That’s done by directing the CPU to execute code from a magic *dmseg* memory region, where CPU reads and writes are serviced by the probe. *dmseg* is part of the special memory window *dseg*, which opens up in debug mode, as described in section 12.1.6.

Although the probe is wonderful for providing debug resources to a minimal system, the EJTAG resources are quite usable by a pure-software debugger running on the local CPU. We’ll come back to that later.

- A just-for-debug exception. In MIPS EJTAG, this is a special super-exception marked by a special debug mode flag, so you can use an EJTAG debugger to debug the whole system, even its own regular exception handlers. See section 12.1.4.
- A number of *EJTAG breakpoints*, hardware monitors that can be independently programmed to match addresses and even data.² When the CPU fetches an instruction or reads/writes a location, the fetch or load/store address and data are compared with any active breakpoints, causing a debug exception on a match.

There can be a lot of breakpoint control registers, and that would be difficult to fit into the CP0 register set, so they are memory-mapped into *dmseg* (the subregion used for EJTAG registers is called *drseg*).

- You can also get a debug exception from a debug breakpoint instruction **sdbbp**, an external signal *DINT*, or when the EJTAG probe wiggles a control bit it knows about.³
- For profiling rather than debugging, you may have the facility to instruct the EJTAG unit to take periodic snapshots of the address of the currently executing instruction (*PC sampling*) and make those samples available to the EJTAG probe, as described in section 12.1.12.

Powerful debug facilities can be built on these foundations.

2. Software engineers habitually use “breakpoint” to mean a debug-support instruction, and it’s often been more helpful to call the hardware monitors “watchpoints.” But all EJTAG documentation calls them breakpoints, so we’ll do the same.

3. Or as a rueful colleague notes, if you inadvertently bump the probe while debugging a system.

12.1.1 *EJTAG History*

Back in the stone age of microprocessors (before 1980) fine-grain debug could be available from an in-circuit emulator (ICE). An ICE was a plug-in replacement for your CPU with a large and proprietary cable leading to a rack full of electronics. The ICE executed instructions just like a CPU, but could be stopped and interrogated.

The ICE approach worked because microprocessors were expensive and fragile, so it was often worth attaching them through a chip socket rather than soldering them in—just as PC processors these days are so expensive that it makes sense not to solder them to the motherboard.

It was always difficult to get the complex circuitry of the ICE to run as fast as the simpler CPU chip, and as microprocessors got faster that got worse. ICE boxes started off expensive and got ever more so: Once they cost tens of thousands of dollars, lots of teams found ways of debugging systems without them. As the 1980s progressed, that undermined the market for ICE boxes.

So the ICE manufacturers needed something like EJTAG. With an on-chip debug unit, the critical paths (which might have made the CPU slower) were local, so the system could run at full speed. The probe you needed to make the connection and the complex host software that ran it were the sort of things they knew how to make. That was good, but they ran into a problem: They either had to sell the tiny probe and software for \$10,000, or change their business model to sell more units at lower prices. Technological change is easy, but institutional change is not; for the most part, the ICE manufacturers tried to keep their prices up and were bypassed.

It was LSI Logic that first incorporated on-chip debug units in their pioneering system-on-chip MIPS CPUs. With the help of some ICE manufacturers, they developed the EJTAG specification much as it is now. But these days probes cost less than \$1,000 and attach to the software debugger of your choice.

12.1.2 *How the Probe Controls the CPU*

The probe gets control when the CPU fetches an instruction from the dmseg region. It sounds like you have to wait for the CPU’s software to volunteer for this, but in fact:

- In response to your debugger start-up your probe will use its JTAG connection to set an internal flag, which moves the debug exception entry point to 0xFF20.0200—in dmseg.

The probe can cause a debug breakpoint directly or by setting up a hardware breakpoint and waiting for the software to trip it.

- Your probe can send an **EJTAGBOOT** command (more about commands in the next section) and then, after the next CPU reset, the CPU will read instructions from dmseg. In fact, a collection of features allows you to reboot the CPU and take complete control, even downloading new software via the JTAG link.

Once the probe is feeding the CPU with instructions, it can get the CPU to do anything. In particular, it can send it an instruction sequence that will read data (whether from a CPU register or from memory) and write it back to the probe via dmseg space.

12.1.3 *Debug Communications through JTAG*

Your host computer connects to the system under test through a small black box, the probe. The chip's JTAG pins give the probe access to special registers inside the CPU.

The JTAG standard was invented by chip-makers to make it easier to test circuit boards, and most complex chips already find room for the JTAG pins; subverting them for software debug is easier than fighting for new pins. In the 1970s (when JTAG was born) every extra transistor was a significant cost. So JTAG is designed to minimize the complexity of the logic that receives it, which is basically a shift register.⁴

There's a control/data enable input, a data-in line, and a data-out line. The control register is sent command codes, most of which just select between different data registers. JTAG command codes are called "instructions" but are often used interchangeably with the names of the data registers—a recipe for confusion when you're talking to software engineers.

But the net effect of this very simple hardware is that JTAG provides a way to read and/or write one of a number of registers internal to the EJTAG unit. The EJTAG unit provides a number of instructions the probe can use to deal with the CPU, and they're summarized in Table 12.1.

12.1.4 *Debug Mode*

Debug mode is a special CPU state, much like exception mode (but more so). The CPU goes into debug mode when it takes any debug exception—which can be caused by an **sddb** instruction, a hit on an EJTAG breakpoint register, from the external "debug interrupt" signal *DINT*, or single-stepping (the latter is peculiar and described briefly below). Debug mode state is visible and controllable through the CP0 register bit **Debug (DM)**—there's more on EJTAG's CP0 registers in section 12.1.7. Debug mode (like exception mode) implicitly disables all normal interrupts. The address map changes in debug mode to give you access to the dseg region, described below. Quite a lot of exceptions just won't happen in debug mode; those that do run peculiarly—see the next section.

4. For those who don't know, a shift register is a 1-bit-wide store accessed serially. As a bit is clocked in, all other bits are moved up one position to make space for it. In hardware, shift registers are often also wired in parallel (that is, the state of all the bits is reflected on a collection of wires).

TABLE 12.1 JTAG Instructions for the EJTAG Unit

<i>JTAG “instruction”</i>	<i>Description</i>
IDCODE	Reads out the MIPS CPU and revision—not very interesting for software, not described further here.
ImpCode	Reads bitfield showing which EJTAG options are implemented.
EJTAG_ADDRESS EJTAG_DATA	(Read/write) together, allow the probe to provide or accept data in response to instruction fetches and data reads/writes in the magic dmseg region described in section 12.1.6.
EJTAG_CONTROL	Package of flags and control fields for the probe to read and write.
EJTAGBOOT NORMALBOOT	The EJTAGBOOT instruction causes the next CPU reset to reboot from the probe; it’s controlled by EJTAG_CONTROL bits called ProbEn , ProbTrap , and EjtagBrk , but you’ll need to read a more detailed manual to find out more. The NORMALBOOT instruction reverts to the normal CPU bootstrap.
FASTDATA	Special access used to accelerate multiword data transfers with probe. The probe reads/writes the 33-bit register formed of EJTAG_CONTROL (PrAcc) with EJTAG_DATA .
TCBCONTROLA TCBCONTROLB TCBADDRESS	Access registers used to control “PDtrace” instruction trace output, if available. See section 12.3, but these JTAG-accessible registers are not detailed.

A CPU with a suitable probe attached can be set up so the debug exception entry point is in the dmseg region, running instructions provided by the probe itself. With no probe attached, the debug exception entry point is in the ROM (for an overview of all MIPS exception entry points, see Table 5.1.)

Exceptions in Debug Mode

A software debugger is likely to be coded to avoid causing exceptions (it will test addresses in software, for example, rather than risk address or TLB exceptions). So it’s reasonable that in debug mode, many conditions that would normally

cause an exception are ignored: interrupts, debug exceptions (other than that caused by executing **sdbbp**), and many others.

But a few exceptions occurring in debug mode are turned into *nested debug exceptions*—a facility that is probably mostly valuable to debuggers using the EJTAG probe.

On such a nested debug exception the CPU jumps to the debug exception entry point, remaining in debug mode. The **Debug (DExcCode)** field records the cause of the nested exception, and **DEPC** records the debug-mode-code restart address. This will not be survivable for the debugger unless it saved a copy of the original **DEPC** soon after entering debug mode, but it probably did that! To return from a nested debug exception, you don't use **deret** (which would inappropriately take you out of debug mode); you grab the address out of **DEPC** and use a **jr** instruction.

12.1.5 *Single-Stepping*

When the single-step bit **Debug (SSt)** is set and control returns from debug mode with a **deret**, the instruction selected by **DERET** will be executed in non-debug context;⁵ then a debug exception will be taken on the very next instruction to be fetched in sequence.

Since one instruction is run in normal mode, it can lead to a nondebug exception; in that case the “very next instruction in sequence” will be the first instruction of the exception handler, and you'll get a single-step debug exception whose **DEPC** points at the exception handler.

12.1.6 *The dseg Memory Decode Region*

EJTAG needs to use memory space both to accommodate its numerous breakpoint management registers (too many for CP0) and for its probe-mapped communication space. This memory space pops into existence near the top of the CPU's virtual address map when the CPU is in debug mode, as shown in Table 12.2.

In the table you can see that:

- *dseg*: Is the whole debug-mode-only memory area, which appears overlaying part of the upper kernel-accessible mapped area (kseg2) when the CPU is in debug mode. It's possible for debug-mode software to read the kseg2-mapped locations “underneath” by setting **Debug (LSNM)**—but a debug-friendly OS should probably avoid using this corner of the memory map for other purposes.

5. If **DERET** points to a branch instruction, both the branch and branch-delay instruction will be executed normally.

TABLE 12.2 EJTAG Debug Memory Region Map(dseg)

Virtual address	Region/subregions	Location/register	Virtual address		
0xC000.0000			0xE000.0000		
0xFF1F.FFFF			0xFF1F.FFFF		
0xFF20.0000	kseg2	dmseg	fastdata	0xFF20.0000	
0xFF20.000F				0xFF20.000F	
0xFF20.0010				0xFF20.0010	
0xFF20.0200			debug entry	0xFF20.0200	
0xFF2F.FFFF			0xFF2F.FFFF		
0xFF30.0000		dseg	DCR register	0xFF30.0000	
0xFF30.1000			IBS register	0xFF30.1000	
			<i>I-breakpoint #1 regs</i>		
0xFF30.1100			IBA1	0xFF30.1100	
0xFF30.1108			IBM1	0xFF30.1108	
0xFF30.1110			IBASID1	0xFF30.1110	
0xFF30.1118			IBC1	0xFF30.1118	
			<i>I-breakpoint #2 regs</i>		
0xFF30.1200			IBA2	0xFF30.1200	
0xFF30.1208			IBM2	0xFF30.1208	
0xFF30.1210			IBASID2	0xFF30.1210	
0xFF30.1218			IBC2	0xFF30.1218	
			<i>same for next two</i>		
			...		
0xFF30.2000			drseg	DBS register	0xFF30.2000
				<i>D-breakpoint #1 regs</i>	
0xFF30.2100				DBA1	0xFF30.2100
0xFF30.2108				DBM1	0xFF30.2108
0xFF30.2110				DBASID1	0xFF30.2110
0xFF30.2118				DBC1	0xFF30.2118
0xFF30.2120				DBV1	0xFF30.2120
0xFF30.2124				DBVHi1	0xFF30.2124
				<i>D-breakpoint #2 regs</i>	
0xFF30.2200				DBA2	0xFF30.2200
0xFF30.2208		DBM2		0xFF30.2208	
0xFF30.2210		DBASID2		0xFF30.2210	
0xFF30.2218		DBC2		0xFF30.2218	
0xFF30.2220		DBV2		0xFF30.2220	
0xFF30.2224		DBVHi2		0xFF30.2224	
0xFF30.2228				0xFF30.2228	
0xFF3F.FFFF		0xFF3F.FFFF			
0xFF40.0000		0xFF40.0000			
0xFFFF.FFFF		0xFFFF.FFFF			

- *dmseg*: Is the memory region where reads and writes are implemented by the probe. But if no active probe is plugged in, or if **DCR (PE)** is clear, then accesses here cause reads and writes to be handled like regular *kseg2* accesses.
- *drseg*: Is where the debug unit's main register banks are accessed. Accesses to *drseg* don't go off CPU. Registers in "drseg" are word-wide, and should be accessed only with word-wide (32-bit) loads and stores.
- *fastdata*: Is a corner of *dmseg* where probe-mapped reads and writes use a more JTAG-efficient block-mode probe protocol, reducing the amount of JTAG traffic and allowing for faster data transfer. There are no details about how it's done in this book.
- *debug entry*: Is the debug exception entry point. Because it lies in *dmseg*, the debug code can be implemented wholly in probe memory, allowing you to debug a system that has no physical memory reserved for debug. If there's no probe-supplied debug exception handler—a condition notified by the probe itself using the **EJTAG_CONTROL (ProbTrap)** bit—the debugger entry point will be at `0xBFC0.0480` (near the other uncached-space exception entry points).

12.1.7 EJTAG CP0 Registers, Particularly Debug

In normal circumstances (specifically, when not in debug mode), the only software-visible part of the debug unit is its set of three CP0 registers:

- **Debug** has configuration and control bits and is detailed below.
- **DEPC** keeps the restart address from the last debug exception (automatically used by the **deret** instruction).
- **DSAVE** is a CP0 register that is just 32 bits of read/write space. It's available for a debug exception handler that needs to save the value of a first general-purpose register, so that it can use that register as an address base to save all the others.

Debug is the most complicated and interesting. It has so many fields defined that we've taken them in three groups: debug exception cause bits in Figure 12.1, information about regular exceptions that are pending (they want to happen but can't because you're in debug mode) in Figure 12.2, and everything else. The "everything else" category includes the most important fields, as shown in Figure 12.3.

The fields in Figure 12.3 (the most important fields) are:

- DBD**: Tells you that the debug exception happened in a branch delay slot. Then **DEPC** points to the branch instruction, which is usually the right place to restart.

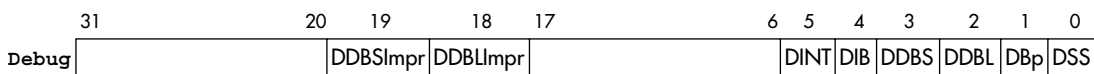


FIGURE 12.1 Exception cause bits in the debug register.

DM:	Debug mode—set on debug exception from user mode, cleared by deret . Not writable here.								
NoDCR:	Read-only—0 if the dseg region is implemented. It’s not clear what kind of EJTAG unit you’d have if this was 1.								
LSNM:	Set this to 1 if you want dseg to disappear, even in debug mode, so you can access the memory locations it otherwise overlays. This makes most of the EJTAG unit’s control system unavailable, so will probably only be done around a particular load/store.								
Doze:	Before the debug exception, CPU was in some kind of reduced power mode.								
Halt:	Before the debug exception, the CPU was stopped—probably asleep after a wait instruction.								
CountDM:	Set to 1 if, and only if, the Count register continues to run in debug mode. Sometimes this may be writable, so you get to choose: Otherwise this is a read-only bit that tells you what your CPU does.								
IEXI:	Set to 1 to defer imprecise exceptions. Set by default on entry to debug mode, cleared on exit, but writable. The deferred exception will come back when and if this bit is cleared: Until then you can see that it happened by looking at the pending bits shown in Figure 12.2.								
EJTAGver:	Read-only—tells you which revision of the specification this implementation conforms to. Known legal values are: <table> <tr><td>0</td><td>Version 2.0 and earlier</td></tr> <tr><td>1</td><td>Version 2.5</td></tr> <tr><td>2</td><td>Version 2.6</td></tr> <tr><td>3</td><td>Version 3.1</td></tr> </table>	0	Version 2.0 and earlier	1	Version 2.5	2	Version 2.6	3	Version 3.1
0	Version 2.0 and earlier								
1	Version 2.5								
2	Version 2.6								
3	Version 3.1								
DExcCode:	Cause of any nondebug exception you just handled from within debug mode—following first entry to debug mode, this field is undefined. The value will be one of those defined for Cause (ExcCode) and is listed in Table 3.2.								
NoSSt:	Read-only—reads 1 if single-step is not implemented. Single-step is usually available, so this usually reads zero.								
SSt:	Set this flag to 1 to enable single-step.								

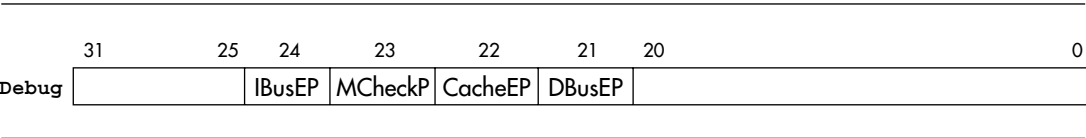


FIGURE 12.2 Debug register—exception-pending flags.

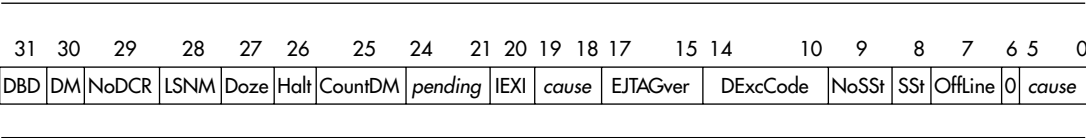


FIGURE 12.3 Fields in the EJTAG debug register.

OffLine: Often not implemented in single-threaded CPUs, in which case it just reads zero.

If implemented, you can set it to 1 to stop the CPU (or the CPU thread in a multithreading CPU) from running instructions except in debug mode. This is intended to allow a debugger to shut down some processors/threads in a multiprocessor/multithreading system, while still leaving those processors/threads to run in debug mode. The condition can only be cleared in debug mode, so if you do a **deret** with it set, nothing else will happen until and unless you get an externally initiated debug interrupt.

The fields in Figure 12.1 are:

- DDBSImpr, DDBLIImpr:** Imprecise store/load breakpoint, respectively—see section 12.1.11. **DEPC** probably points to an instruction some time later in sequence than the store/load that triggered the breakpoint. The debugger or user (or both) have to cope as best they can.
- DINT:** Debug interrupt caused by pulse on EJTAG pin.
- DIB:** We hit an instruction breakpoint.
- DDBS:** We hit a precise store breakpoint.
- DDBL:** We hit a precise load breakpoint.
- DBp:** We hit any sort of debug breakpoint (often found together with one of the above).
- DSS:** This was a single-step exception.

The “pending” flags (Figure 12.2) record exception conditions caused by instructions run in debug mode but that have not happened yet because they are imprecise and **Debug (IEXI)** is set. The exceptions remain pending and the

bits remain set until **Debug (IEXI)** is cleared explicitly or implicitly by a **deret**, when the exception is delivered and the corresponding pending bit is cleared.

The individual bits relate to different exception reasons:

- IBuseEP:** Bus error on instruction fetch.
- MCheckP:** Machine check (usually an illegal TLB update).
- CacheEP:** Parity/ECC error detected when reading from cache.
- DBusEP:** Bus error reported on some external read.

On any particular CPU, some of these bits will be always zero, because the corresponding exception is always precise (and precise exceptions are always dealt with immediately).

12.1.8 *The DCR (Debug Control) Memory-Mapped Register*

In addition to the CP0 **Debug** register, there are more EJTAG controls and flags in a register called **DCR**, which is memory-mapped through the debug-only drseg memory region at location 0xFF30.0000, so is only accessible if the CPU is in debug mode. The fields are in Figure 12.4.

Many of the fields in **DCR** are shared with the probe through one of the JTAG-accessible registers.

The fields are:

- ENM:** (Read-only) reports CPU endianness (1 == big).
- DB/IB:** (Read-only) 1 if EJTAG data/instruction hardware breakpoints are available, respectively. It seems hard to see why you’d build an EJTAG unit without any breakpoints.
- PCS, PCR:** **PCS** reads 1 if the “PC sampling” feature is available. When it is, **PCR** is a 3-bit field defining the sampling frequency as one sample every 2^{5+PCR} cycles. See section 12.1.12 for details.
- INTE/NMIE:** Set **DCR (INTE)** to zero to disable interrupts in nondebug mode (it’s a separate bit from the various nondebug-mode visible interrupt enables). The idea is that the debugger might want to run kernel subroutines (perhaps to discover OS-related information) without losing control because interrupts start up again.

	31	30	29	28		18	17	16	15		10	9	8		6	5	4		3		2		1	0
DCR	0	ENM		0		DB	IB		0		PCS		PRC		0	INTE	NMIE	NMIP		SRE		PE		

FIGURE 12.4 Fields in the memory-mapped **DCR** (debug control) register.

DCR (NMIE) Masks nonmaskable interrupt in nondebug mode (a nice paradox).

Both the **DCR (INTE, NMIE)** bits are “1” from reset.

NMIP: (Read-only) tells you that a nonmaskable interrupt is pending, and will happen when you leave debug mode (and according to **DCR (NMIE)** as above).

SRE: If implemented, write this zero to prevent a soft-reset.

PE: (Read-only) software-readable version of the probe-controlled enable bit. When set, debug exceptions are redirected to probe-controlled dmseg memory locations.

12.1.9 EJTAG Breakpoint Hardware

EJTAG breakpoints have hardware that monitors instruction fetch, load, and store accesses from the CPU. If you set a breakpoint, then an access that matches it for address (and possibly matches data, too, if so specified) will cause a debug exception.

Instruction and data breakpoints are separate. The EJTAG specification permits a unit to be built with no breakpoint hardware (though that wouldn’t be much use), but it seems common to have between two and six of them. The breakpoints:

- Work only on virtual addresses, not physical addresses. However, you can restrict the breakpoint to a single address space by specifying an ASID value to match. Where the operating system is using multiple address spaces, the debugger will have to work with the OS to make that useful.
- Use a bitwise address mask to permit a degree of fuzzy matching.
- Breakpoint hits can be made conditional on the data value read or write. But data-sensitive tests may lead to an exception that is *imprecise*—by the time the data is available for matching, it’s too late to prevent subsequent instructions from having some effect. See section 5.1 if you want to know more about precise and imprecise exceptions.

There are one overall I-side and one D-side control registers, mapped into the drseg region, accessible only when in debug mode at the addresses shown in Table 12.2. They’re called **IBS** and **DBS**. They are as shown in Figure 12.5.

	31	30	29	28	27	24	23		2	1	0
DBS/IBS	0	ASIDsup	NoSVM	NoLVM	BCN		0				BS1-0

FIGURE 12.5 Fields in the **IBS/DBS** (EJTAG breakpoint status) registers.

The fields in Figure 12.5 are:

- ASIDsup:** Is 1 if the breakpoints can use ASID matching to distinguish addresses from different address spaces. It’s hard to imagine any modern debug unit lacking that facility.
- NoSVM, NoLVM:** Are 1 if you *can’t* qualify a breakpoint by the data value on stores/load, respectively. Only valid in the **DBS** register.
- BCN:** The number of hardware breakpoints available. There are separate I- and D-side breakpoints.
- BS1–0:** Bitfields showing breakpoints that have been matched. Debug software has to clear down a bit after a breakpoint is detected. The size of the field depends on the number of breakpoints provided on either the I- or D-side.

Then, each EJTAG hardware breakpoint (“n” is 0–3 to select a particular breakpoint) is set up through four to six separate registers:

- IBCn, DBCn:** Breakpoint control register shown in Figure 12.6.
- IBAn, DBAn:** Breakpoint address.
- IBAMn, DBAMn:** Bitwise mask for breakpoint address comparison. A “1” in the mask marks an address bit that will be *excluded from* comparison, so set this register to zero for exact matching.
 Ingeniously, **IBAMn(0)** corresponds to the slightly bogus instruction address bit zero, used to track whether the CPU is running MIPS16 instructions, and allows you to determine that an instruction breakpoint may happen only in MIPS16 (or non-MIPS16) mode.
- IBASIDn, DBASIDn:** Specifies an 8-bit ASID, which may be compared against the current **EntryHi(ASID)** field to filter breakpoints so that they only happen to a program in the right address space (typically corresponding to one Linux process, for example). The ASID check can be enabled or disabled using **IBCn(ASIDuse)** or **DBCn(ASIDuse)**, respectively—see Figure 12.6 and its notes.
 The higher 24 bits of each of these registers are always zero.

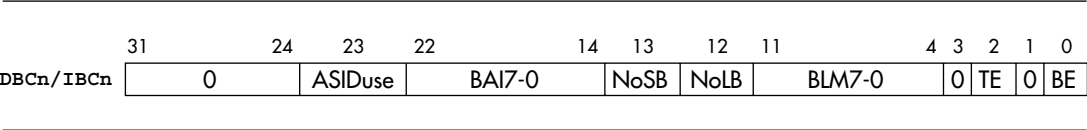


FIGURE 12.6 Fields in the hardware breakpoint control registers (**IBCn**, **DBCn**.)

DBVn, DBVHn: The value to be matched on load/store breakpoints. **DBCHn** defines bits 31–64 to be matched for 64-bit load/stores. A JTAG hardware breakpoint for a real 64-bit CPU would have 64-bit **DBVn** registers, so wouldn't need **DBVHn**—they're provided because some 32-bit CPUs implement 64-bit load or store for a double-precision coprocessor, perhaps a floating-point unit.

Note that you can disable data matching (to get an address-only data breakpoint) by setting the value byte-lane comparison mask **DBCn (BLM)** to all 1s.

So now let's look at the control registers in Figure 12.6.
The fields are:

ASIDuse: Set 1 to compare the ASID as well as the address.

The **BAI7–0**, **NoSB**, **NoLB**, and **BLM7–0** fields are applicable only to D-side (**DBCn**) breakpoints:

BAI7–0: Byte (lane) access ignore—which sounds mysterious. But this is really an *address* filter.

When you set a data breakpoint, you probably want to break on any access to the data of interest. You don't usually want to make the break conditional on whether the access is done with a load byte, load word, or even load-word-left: But the obvious way of setting up the address match for a breakpoint has that effect.

To make sure you catch any access to a location, you can use the address mask to disable subdoubleword address matching and then use **DBCn (BAI)** to mark the bytes of interest inside the doubleword: well, except that zero bits mark the bytes of interest, and 1 bits mark the bytes to ignore (hence the mnemonic).

The **DBCn (BAI)** bits are numbered by the byte-lane within the 32- or 64-bit data bus; so be careful, the relationship between the byte address of a datum and its byte lane is endianness-sensitive.

NoSB, NoLB: Set 0 to *enable*⁶ breakpoint on store/load, respectively.

BLM7–0: A per-byte mask for data comparison. A zero bit means compare this byte, a 1 bit means to ignore its value. Set this field all-1s to disable the data match.

The remaining fields are applicable to both I- and D-side breakpoints:

TE: Set 1 to use as trigger for PDtrace instruction tracing, as described in section 12.3.

6. “1-to-enable” would feel more logical. The advantage of using 0-to-enable here is that the zero value means you will break on either read or write, which is a better default than “never break at all.”

BE: Set 1 to activate breakpoint. This field resets to zero to avoid spurious breakpoints caused by random register settings: Don’t forget to set it!

12.1.10 *Understanding Breakpoint Conditions*

There are a lot of different fields and settings that are involved in determining when a hardware breakpoint detects its condition and causes an exception.

In all cases, there will be no break if you’re in debug mode already ... but then, for a break to happen, all the following must be true:

1. The breakpoint control register enable bit **IBAn (BE) /DBAn (BE)** is set.
2. The address generated by the program for instruction fetch, load, or store matches those bits of the breakpoint’s address register **IBAn/DBAn** for which the corresponding address-mask register bits in **IBAn/DBAn** are zero.
3. Either the **IBn (ASIDuse) /DBn (ASIDuse)** is zero (so we don’t care which address space we’re matching against) or the address-space ID of the running program, **EntryHi (ASID)**, is equal to the value in **IBASIDn/DBASIDn**.

That’s all for instruction breakpoints, but for data-side breakpoints you need to distinguish loads and stores and can distinguish accesses by subdoubleword address too:

4. It’s a load and **DBn (NoLB)** is zero, or it’s a store and **DBn (NoSB)** is zero.
5. The load or the store touches at least one byte within doubleword for which the corresponding **DBn (BAI)** bit is zero.

If you didn’t want to compare the load/store value, then you would set **DBn (BLM)** to all-ones, and you’re done. But if you also want to consider the value, then you have data compare conditions:

6. The data loaded or stored, as it would appear on the system bus, matches the 64-bit contents of **DBVHi**, with **DBVn** in each of those 8-bit groups for which the corresponding bit in **DBn (BLM)** is zero.

That’s it. It’s quite complicated, but logical.

12.1.11 *Imprecise Debug Breaks*

Instruction breakpoints, and data breakpoints filtering only on address conditions, break with **DEPC** pointing to the matching instruction. More accurately, such exceptions are “precise exceptions,” in the sense discussed in section 5.1.

Most exceptions in MIPS architecture CPUs are precise. But many MIPS CPUs optimize loads and stores by permitting the CPU to run on at least until it needs to use the data from a load, so data breakpoints that filter on the data value are *imprecise*. The debug exception will happen to whichever instruction (typically later in the instruction stream) is running when the hardware detects the match. The debugging software must cope as best it can.

12.1.12 *PC Sampling with EJTAG*

A valuable trick available with recent revisions of the EJTAG specification and probes, PC sampling provides a nonintrusive way to collect statistical information about the activity of a running system. You can find out whether your CPU offers this facility by looking at the appropriate bit in **DCR (PCS)**.

PC sampling hardware snapshots the current PC periodically and stores that value where it can be retrieved by a debug probe. It’s then up to software to construct a histogram of samples over a period of time, which (statistically) allows a programmer to see where the CPU has spent most cycles. Not only is this useful, but it’s also familiar: Systems have used intrusive interrupt-based PC sampling for many years, so there are tools (GNU/Linux `gprof`, for example) that can readily interpret this sort of data.

When PC sampling is configured into your CPU, it runs continuously. It doesn’t even stop when the CPU is hanging on a **wait** instruction (time spent waiting is still time you might want to measure). On a typical implementation, you might choose to sample as often as once per 32 cycles or as rarely as once per 4,096 cycles. Since it runs continuously, it’s a good thing that from reset the sampling period defaults to its maximum.

At every sampling point, the address of the instruction completing in that cycle (or if none completes, the address of the next instruction to complete) is deposited in a JTAG-accessible register. Sampling rate is controlled by a field in **DCR**.

The hardware stores not only 32 bits of the instruction address, but also the then-current ASID (so you can interpret the virtual PC) and an always written 1 “new” bit, which a probe can use to avoid double-counting the same sample.

12.1.13 *Using EJTAG without a Probe*

The EJTAG unit can be used by a conventional debugger, which runs entirely on the target system, and, in particular, it can be used to build debug facilities inside an OS kernel. But because it was invented primarily to be used via a probe, there are a few issues to work around:

- Handling debug exceptions: Without a probe plugged in and enabled, the debug exception entry point is in the ROM area of the uncached kseg1 region, at `0xBFC0.0480` (if you need to check for the presence of an enabled probe you can read **DCR (PE)**). You'll need to put code at that entry point that will transfer control into your debug exception handler, which may well be built into your OS kernel.
- Calling into debug mode inside the kernel: You can't just switch on debug mode: It's available only through a debug exception. But you need to be in debug mode to see the dmseg memory region, and thus to access things like breakpoint control registers. So for your software-only debugger, you will need a way of invoking debugger support routines in debug mode.

That requires a debug-exception version of a “system call” using a planted **sdbbp** instruction. The debug exception handler will need to distinguish those debugger system calls from other debug exceptions—most likely by looking at the return address in **DEPC**. Debugger system call arguments can be in general-purpose registers.

Similarly, you can't just switch off debug mode—to get out of debug mode you need to execute a **deret**.

In some cases you'll probably want to execute most of the debugger in ordinary kernel mode: So after a brief excursion into debug mode following a breakpoint match or debug instruction, the debug-mode software can patch up the normal CP0 registers to simulate a more conventional exception.

- Virtual-address-only breakpoints: Note that (unlike the non-EJTAG CP0 “watchpoints”) all EJTAG breakpoints operate on virtual addresses, optionally qualified by the address space ID (ASID).
- Handling debug breaks from exception mode or other “illegal” places: This looks like quite a problem when you first meet it. Debug breakpoints are exceptions, but because they're “super-exceptions” they don't play by the rules of the rest of the kernel. But it's really the same problem as that discussed previously. It's likely that a local debugger program using EJTAG will mostly not run in debug-exception mode. So the debug exception handler will run in exception mode while it saves state, then store the return address somewhere safe and use a faked-up **deret** to drop into normal high-privilege (kernel) mode.

Sometimes, though, a debug exception entry point can be from somewhere dangerous—most obviously, from code already running in normal exception mode. The debugger receiving such an exception has to tread carefully, and probably can't just invoke ordinary kernel-mode routines. Debugger authors need to figure out how far they want to go in allowing debug through exception.

12.2 Pre-EJTAG Debug Support—Break Instruction and CPO Watchpoints

The MIPS architecture existed without the EJTAG debug unit for quite a while, so it has more conventional debug facilities. Those include the **break** instruction, which simply causes an exception while having lots of uninterpreted bits that can be given significance to debug software.

But many CPUs also implement up to four hardware *watchpoints*, controlled by a handful of CP0 registers. Each watchpoint specifies a virtual address that may be checked against each instruction-fetch, load, or store operation, and can cause an exception if the load/store address matches.

Not all debuggers make use of the hardware watchpoints; some may ignore them altogether, and other debuggers use EJTAG facilities.

As compared with EJTAG breakpoints, the watchpoints don't have fuzzy address matching; they aren't ever data-sensitive; and CP0 watchpoints may work on both I- and D-side (there are flags to control which any particular watchpoint is sensitive to). And of course they cause just a normal exception when tripped, so may not be used to debug through exception handlers.

A watchpoint condition can match when the CPU is already in exception state (when **SR (EXL)**—or an equivalent bit for error or debug exceptions—is already set). It would cause unhelpful chaos to take the exception at this point, so it's deferred until later. The fact that the exception wanted to happen is recorded in the **Cause (WP)** bit, and a watchpoint exception will happen after the CPU returns to normal operation.

The watchpoint registers are shown in Figure 12.7.

Watchpoint addresses are maintained only to the nearest doubleword (eight bytes), so only address bits 31–3 need be matched. On 64-bit CPUs, **WatchLo** grows to 64 bits to be able to define a complete virtual address.

The other register fields are as follows:

WatchLo (I, R, W): Enables the watchpoint check—on instruction fetches if **WatchLo (I)** = 1, on loads if **WatchLo (R)** = 1, or on stores if **WatchLo (W)** = 1. Any combination of those bits is legal.

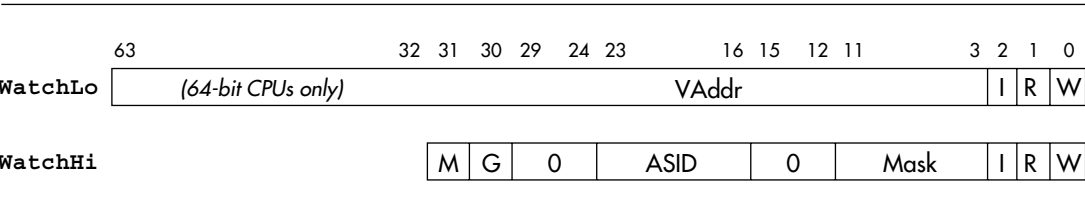


FIGURE 12.7 Layout of the **WatchLo** and **WatchHi** registers.

- WatchHi (M):** Read-only continuation bit—if this reads 1, then there is at least one more pair of watch registers available.
- WatchHi (G):** Global—set this bit 1 to match addresses regardless of the current ASID setting, see below.
- WatchHi (ASID):** Set this field (with **WatchHi (G)** zero) to match only accesses made in this particular address space. Operating systems using the MIPS TLB maintain their address space in **EntryHi (ASID)**, as described in Chapter 6.
- WatchHi (Mask):** Any “1” bit set in this field makes the corresponding bit of **WatchLo (VAddr)** a don’t-care, allowing you to trap accesses to an aligned power of 2 region of memory of any size from 8 to 2,048 bytes.
- WatchHi (I, R, W):** You read these fields after a watch exception to find out whether the exception was triggered by an I-fetch, read, or write, respectively. Software must clear these fields to make them useful again, and these bits are implemented as “write-one-to-clear”—that is, you clear **WatchHi (R)** by writing a value to **WatchHi**, which has a 1 in bit position 1. That’s done so that you can conveniently clear any of the bits you’ve seen by reading **WatchHi** and then writing the same value back again.

12.3 PDtrace

PDtrace is an add-on to the EJTAG debug unit (described earlier in this chapter) that can keep track of program execution for later reconstruction. The execution trace may be kept in an on-chip memory or played out in real time to a probe, using slightly exotic high-speed signaling techniques.

The earliest trace facilities record only the execution address (the PC), but PDtrace systems may also allow you to keep track of load/store addresses and even data values.

You don’t need very much data to trace execution. The analysis program is assumed to have a copy of the system’s complete binary, so when execution is sequential you only need know how far the CPU got through a sequence; when a conditional branch is met, you only need to know whether the condition was met or not. The trace needs to record a full address, though, on an instruction like jump-register.

Tracing to an on-chip memory is simple and fast; but practical on-chip trace buffers are very small, so you can only record a short period of program run time.

When you trace to a probe, there is likely to be a lot of trace memory (tens or hundreds of MBytes). But it requires a few pins on the package to connect

the trace data lines. When trace data is generated too fast for the link (which is likely to happen), you either have to slow the CPU until the link catches up or discard some of the trace.

EJTAG breakpoints are reused to provide fine-grain control for trace, switching the flow on and off dynamically when you hit particular breakpoint conditions.

PDtrace is almost exclusively of interest to probe suppliers and as such isn't documented further here. Manuals are available from MIPS Technologies Inc. (www.mips.com).

12.4 Performance Counters

Performance counters are there to let software and hardware engineers find out more about what the system is doing for tuning (and occasionally for debug). Each counter increments on a selected one of a number of events that your particular CPU chooses to instrument.

Modern CPUs typically have a couple of performance counters. Most can count some common useful things: elapsed cycles, instructions completed, cache misses, and so on. Bear in mind that CPU designs are difficult to build and validate, and that with “peripheral” functions like performance counting the events offered are likely to be those that are easy to count. What’s easy to count is not always exactly what seems natural to a software engineer, so always treat the event description with care and skepticism. Before you really believe what something is counting, it’s sensible to experiment with how the counter behaves in controlled conditions.

Each 32-bit counter is accompanied by a control register, which will be as shown in Figure 12.8, where:

- M:** A continuation bit; if it reads 1, then there’s another performance counter control/count pair after this one.
- W:** Reads 1 if your counter register is 64 bits. It probably won’t be!
- Event:** Determines which event to count—the list will depend entirely on your CPU, and you’ll need to read that manual.

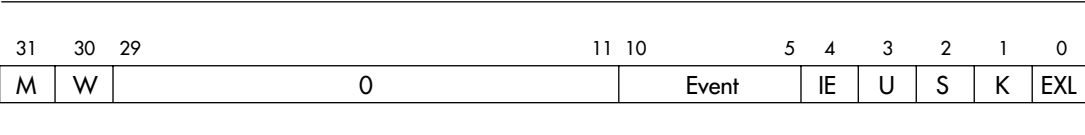


FIGURE 12.8 Layout of the **PerfCtl** register.

- IE:** Set to cause an interrupt when the counter “overflows” into its bit 31. This can be used to implement an extended count but is more often used (by presetting the counter appropriately) to notify software after a certain number of events have happened.
- U, S, K, EXL:** Count events in User mode, Supervisor mode, Kernel mode, and Exception mode (i.e., when **SR (EXL)** is set), respectively. You can set any combination of these bits.

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GNU/Linux from Eight Miles High

Why should a book about computer architecture devote several chapters to a particular operating system?

Well, this is what a CPU is for. A CPU “architecture” is the description of what a useful CPU does, and a useful CPU runs programs under the control of an operating system. Although many operating systems run on MIPS, the great thing about Linux is that it’s public. Anyone can download the source code, so anyone can see how it works.

Any operating system is just a bunch of programs. Ingenious programs, and—perhaps more than most software—built on a set of ideas that have been refined and figured out over the years. An operating system is supposed to be particularly reliable (doesn’t crash) and secure (doesn’t let some program do things the OS hasn’t been told to let it do).

Correct usage sees “Linux” as the name of the operating system kernel originally written by Linus Torvalds, a kernel whose subsequent history is, well, history. Most of the (much larger) rest of the system came from projects organized under the “GNU” banner of the Free Software Foundation. Everybody sometimes forgets and calls the whole thing “Linux.”

Both sides of this process emerged as a reaction to the seminal work on the UNIX operating system developed by Bell Laboratories in the 1970s. Probably because Bell saw it as of no commercial value, it distributed the software widely to academic institutions under terms that were then unprecedentedly “open.” But it wasn’t “open source”—many programmers worked on UNIX at university, only to find that their contributions were either lost or were now owned by Bell Labs (and their many successors). Frustration with this process eventually drove people to write “really free” replacements.

The last key part was the kernel. Kernels are quite difficult programs, but the delay was cultural: OS kernels were seen as something for academic groups, and those groups wanted to go beyond UNIX, not to recreate it. The post-UNIX

fashion was for a small, modular operating system built of clearly separated components, but no OS built on that basis ever found a significant user base.¹

Linux won out because it was a much more pragmatic project. Linus and his fellow developers wanted something that worked (on x86 desktops, in the first instance). When the Linux kernel was in competition with offshoots of the finally free BSD4.4 system, BSD protagonists insisted with some justification on their superior engineering. But the Linux community had arrived at an understanding of a far more “open” development style. Linux evolved quickly. Sometimes, it evolved quickly because Linux people were perfectly happy to adapt BSD code. It wasn’t long before Linux triumphed, and the engineering got better, too.

13.1 Components

To get to grips with any artifact you need to attach some good working meaning to the terms used by its experts, and you are particularly likely to be confused by terms you already know, but with not quite the same meaning. The UNIX/Linux heritage is long enough that there are lots of magic words:

- *Thread*: The best general definition of “thread” I know is “a set of computer instructions being run in the order specified by the programmer.” The Linux kernel has an explicit notion of a thread (for each thread there’s a `struct thread_struct`). It’s almost the same thing, but by the terms of my definition a low-level interrupt handler (for example) is a distinct thread that happens to have borrowed the environment of the interrupted thread to run with. Both definitions are valuable, and we’ll say “Linux thread” when necessary.

Linux loves threads (there are currently 134 on the desktop machine I’m typing this on). Most of those threads correspond to an active application program—but there are quite a few special-purpose threads that run only in the kernel, and some applications have multiple threads.

One of the kernel’s basic jobs is *scheduling*—picking which Linux thread to run next. See the bullet on the scheduler, below.

- *File*: A named chunk of data. In GNU/Linux, most of the interactions a program makes with the world beyond its process are done by reading and writing files. Files can just be things you write data to and get it back later. But there are also special files that lead to device drivers: Read one

1. Some claim that Windows/NT—and therefore most modern versions of Microsoft Windows—has a microkernel. That may be true, but it certainly lost any claims to be small or modular on its way to world domination.

of those and the data comes from a keyboard, write another and your data is interpreted as digital audio and sent out to a loudspeaker. The Linux kernel likes to avoid too many new system calls, so special `/proc` files are also used to allow applications to get information about the kernel.

- *User mode and system calls:* Linux applications run in user mode, the lower-privilege state of MIPS CPUs. In user mode, the software can't directly access the parts of the address space where the kernel lives, and all the locations it can address are mapped to pages the kernel has agreed to let the application play with. In user mode, you can't run the coprocessor-zero CPU control instructions.

GNU/Linux application code that runs in user mode is frequently referred to as *userland*.

To obtain any service from the kernel (most often, to read or write a file) the application makes a system call. A system call is a deliberately planted exception, interpreted by the kernel's exception handler. The exception switches to high-privilege mode.

Through the system call, Linux application threads run quite happily in the kernel in high-privilege mode (but of course they're running trusted code there). When it's done, the return from exception code involves an **eret**, which makes sure that the change back to user mode and the return to user mode code are done simultaneously.

- *Interrupt context:* Linux tries not to disable interrupts too much. When Linux is running, at any moment there's an active thread on a CPU:² So an interrupt borrows what appears to be the context of that thread until it finishes its business and returns. Code called from an interrupt handler is in interrupt context, and there are many things such code should not do. It can't do anything that might have to wait for some other software activity, for example. If your keyboard input routine is going to log all keystrokes to a file,³ then you can't do that by calling the file output routine from the interrupt handler.

There are decent ways to do that: You can get the keyboard interrupt to arrange to wake some Linux thread that obtains and logs the input, for example.

- *Interrupt service routine (ISR):* The lowest-level interrupt code in the device driver is generally called an ISR. In Linux you're encouraged to keep this code short: If there's lots of work to do, you can consider using some kind of "bottom half," as described in the next chapter.

2. Even if the kernel is waiting in a power-down mode, there's a thread that is executing the **wait** instruction.

3. Perhaps not a good idea from a security point of view, but still . . .

- *The scheduler*: A kernel subroutine. The OS maintains a list of threads that are ready to run (they're not blocked on an incomplete I/O transfer, for example), and that list is in priority order. The priority is dynamic, and is recalculated periodically—mostly to ensure that long-running computations don't hog the CPU and prevent it from responding to events. Applications can lower their own priority to volunteer for a life in the background but can't usually raise it.

After any interrupt is handled, the scheduler will be called. If the scheduler finds another thread is more worthy of running, it parks the current thread and runs the winner.

Older Linux kernels were not *preemptive*: once a thread was running in the kernel it was allowed to run until it either volunteered for rescheduling (by waiting on something) or until control was just about to pass back into userland—only then would the kernel contemplate a thread switch.

A nonpreemptive kernel is easier to program. Your kernel code sequence might have to worry about interrupt handlers running unexpectedly while it was in flight, but you knew it could never be unexpectedly caught halfway through something by some other mainstream kernel code. But it led to excessive delays and inadequate responsiveness.

The luxurious freedom from interference from parallel threads is lost when you have an SMP kernel (where two CPUs are simultaneously threading the same kernel). To make the SMP kernel work properly, hundreds of possible interactions need to be tracked down and protected with appropriate locks. The SMP locks are (in almost all cases) exactly where you need them to be to permit the scheduler to stop a running kernel thread and run another: That's called *kernel preemption*.

It's now an important kernel programming discipline to recognize code sequences where preemption must be temporarily inhibited. The macros used to mark the start and end of that code have definitions that change according to kernel configuration to work correctly on uniprocessor or SMP systems.

- *Memory map/address space*: The map of memory locations available to a particular Linux thread. The address space of a thread is defined through a `mm_struct`, pointed to by the thread.

For Linux OS ported to the MIPS architecture (hereinafter, "Linux/MIPS") on a 32-bit processor, the high half of the address space (addresses with bit 31 set) can be read and written only in kernel-privilege mode. The kernel code/data is normally in the corner of this, known as `kseg0`, which means the kernel itself does not depend on addresses translated through the TLB.

The user part of the address space is mapped differently for each application—only threads that collaborate in an explicitly multithreaded application share the user address space (i.e., they point to the same `mm_struct`). But all Linux threads share the same kernel map.

A thread running a conventional single-threaded application runs in an address space that is distinct from all other threads and is exactly what older UNIX-like systems called a “process.”

At any given time, much of an application’s address space may not in fact be mapped, or even not represented by any data present in physical memory at all. An attempt to access that will cause a TLB exception, which will be handled by the OS, which will load any missing data and set up an appropriate mapping before it returns to the application. That is, of course, *virtual memory*.

- *Thread group*: The collection of threads within the same memory map is called a thread group. Where a group has two or more members, those threads are cooperating to run the same program. The thread group is another good approximation in Linux to what is called a “process” in old UNIX systems.
- *High memory*: Physical memory above 512 MB (whether real read/write memory or memory-mapped I/O locations) is not directly accessible through the kseg0 (cached) or kseg1 (uncached) windows. On a 32-bit CPU physical addresses above the low 512 MB are “high memory” in the Linux sense and can only be accessed through TLB mappings. With a MIPS CPU, you can create a few permanent mappings by defining “wired” TLB entries, protected from replacement. But Linux tries to avoid using resources that will quickly run out, so mainstream kernel code avoids wired entries completely. For Linux/MIPS, high-memory mappings are maintained dynamically by TLB entries created on demand.
- *Libraries and applications*: Long ago, applications running on UNIX-like systems were monolithic pieces of code, which were loaded as required. You built them by compiling some source code and gluing in some library functions—prebuilt binaries provided with your toolchain.

But there are two things wrong with that. One is that the library code is often bigger than the application that attaches to it, bloating all the programs. The other is that if a supplier fixes a bug in a library function, you don’t get full benefit from the fix until every software maintainer rebuilds his or her application.

Instead, the application is built without the library functions. The names of the missing libraries are built into the application, so the loader can find the required libraries and stitch them in when the application is loaded. So long as the library continues to provide identical functions, everything should be fine (there’s a library version-tracking system to allow libraries to evolve functionally, too, but that’s beyond our scope).

That carries a penalty. When you link a program at load time out of pieces (each of which may get separately updated), the exact address of the components is unpredictable at build time. You can’t predict in

advance which locations will be available for loading a particular library. The runtime loader can do no better than to load each library in the next space available, so even the starting address for a library is unpredictable. A library binary has to be *position-independent code* or *PIC*—it must run correctly wherever its code and data are positioned in virtual address space.

We'll discuss PIC code and the construction of application memory spaces in Chapter 16.

13.2 Layering in the Kernel

From one point of view, the kernel is a set of subroutines called from exception handlers. The raw post-exception “exception mode” environment on a MIPS CPU is all-powerful and very low-overhead but tricky to program. So with each entry to the kernel you get something like a foreshortened bootstrap process, as each “layer” constructs the environment necessary for the next one. Moreover, as you exit from the kernel you pass through the same layers again, in reverse order, passing briefly through exception mode again before the final **eret** which returns you to userland.

Different environments in the kernel are built by more or less elaborate software which makes up for the limitations of the exception handler environment. Let's list a few starting at the bottom, as the kernel is entered:

13.2.1 *MIPS CPU in Exception Mode*

Immediately after taking an exception, the CPU has **SR (EXL)** set—it's in exception mode. Exception mode forces the CPU into kernel-privilege mode and disables interrupts, regardless of the setting of other **SR** bits. Moreover, the CPU cannot take a nested exception in exception mode except in a very peculiar way.⁴

The first few instructions of an exception handler usually save the values of the CPU's general-purpose registers, whose values are likely to be important to the software that was running before the exception. They're saved on the kernel stack of the process that was running when the interrupt hit. It's in the nature of MIPS that the store operations that save the register require you to *use* at least one general-purpose register first, which is why the registers called **k0** and **k1** are reserved for the use of exception handlers.

The handler also saves the values of some key CP0 registers: **SR** will be changed in the next section of the exception handler, but the whole at-exception value should be kept intact for when we return. Once that's done, we're ready to leave exception mode by changing **SR**, though we are going to leave interrupts disabled.

4. There are some cunning tricks in MIPS history that exploit the peculiar behavior of an exception from exception mode—but Linux doesn't use any of them.

A CISC CPU like an x86 has no equivalent of exception mode; the work done in MIPS exception mode is done by hardware (really by invisible microcode). An x86 arrives at an interrupt or trap handler with registers already saved.

The software run in MIPS exception mode can be seen as producing a virtual machine that looks after saving the interrupted user program's state immediately after an exception and then restores it while preparing for the **eret**, which will take us back again. Programmers need to be very careful what they do in exception mode. Exceptions are largely beyond the control of the software locks that make the kernel thread-safe, so exception code may only interact very carefully with the rest of the kernel.

In the particular case of the exception used to implement a system call, it's not really necessary to save GP registers at all (so long as the exception handler doesn't overwrite the **s0–s8** “saved” registers, that is). In a system call or any noninterrupt exception, you can call straight out to code running in thread context.

Some particularly simple exception handlers never leave exception mode. Such code doesn't even have to save the registers (it just avoids using most of them). An example is the “TLB refill” exception handler described in section 14.4.8.

It's also possible—though currently unusual—to have an interrupt handler that runs briefly at exception level, does its minimal business, and returns. But such an interrupt handler has no real visibility at the OS level, and at some point will have to cause a Linux-recognized interrupt to get higher-level software working on its data.

13.2.2 *MIPS CPU with Some or All Interrupts Off*

As we'll see in the next chapter, an interrupt routine exits exception mode but continues to run with at least some interrupts disabled.

Running with all interrupts disabled is a costly but effective way of getting a single CPU to be nonpreemptive (the longest time software spends with interrupts disabled determines your worst-case interrupt latency, and every device driver with a real-time constraint must budget for it). And of course it doesn't prevent re-entrance where there's a second CPU at work.

The simplest, shortest kind of ISR may opt to run to completion without ever re-enabling interrupts—Linux can support this and calls it a *fast interrupt handler*. You get that behavior by setting the flag `SA_INTERRUPT` when registering the ISR. But most run for a while with higher-priority interrupts enabled.

Potentially, you can get a stack of interrupts interrupting interrupts. Infinite recursion (and stack overflow and an inevitable crash) can't happen because Linux makes sure you can stack up at most one entry at each distinct interrupt level. The amount of data saved at each level must be small enough that the

maximum stack of interrupt save information will not overfill a thread's kernel stack.

13.2.3 *Interrupt Context*

After an interrupt, even after the interrupt handler has re-enabled most interrupts and built a full C environment, interrupt code is still limited because it's borrowing the state (and kernel stack) of whichever thread happened to be interrupted.

Servicing an interrupt is someone's business, certainly, but it has no systematic relationship with the thread that is executing when the interrupt happens. An interrupt borrows the kernel stack of its victim thread and runs parasitically on that thread's environment. The software is in *interrupt context*, and to prevent unreasonable disruption, interrupt-context code is restricted in what it can do.

One vital job done by the kernel is the scheduler, which determines which thread the OS should run next. The scheduler is a subroutine, called by a thread; in some cases it's called by a thread in interrupt context. Once the interrupt-context part of an interrupt handler can get to the point where the hardware's immediate needs are met, it can (and often does) schedule a thread that will complete the interrupt-handling job, this time in thread context.

13.2.4 *Executing the Kernel in Thread Context*

You can arrive in the kernel in thread context either when an application has made a voluntary system call or a forced call for resources on a virtual memory exception (and the system call or VM exception has emerged from its lower layers), or as a result of a reschedule—which is, in turn, always either caused by an interrupt or by another thread voluntarily rescheduling itself because it's waiting for some event.

System calls are a sort of “subroutine call with security checks.” But a range of other exceptions—notably virtual memory maintenance exceptions—are very much the same, even though the application didn't know this particular system call was necessary until it got the exception.

Not every thread is an application thread. Special threads with no attached application can be used to schedule work in the kernel in process context for device management and other kernel functions.

Thread context is the “normal” state of the kernel, and much effort is spent making sure that most kernel execution time is spent in this mode. An interrupt handler's “bottom half” code, which is scheduled into a work queue (see section 14.1), is in thread context, for example.

How Hardware and Software Work Together

Let me tell you a story . . .
Well, in fact let me tell you several stories illustrating how the MIPS hardware provides the low-level features that prop up the Linux kernel.

- *The life and times of an interrupt:* What happens when some piece of hardware signals the CPU that it needs attention? Interrupts can cause chaos when an interrupt-scheduled piece of code interrupts another code halfway through some larger operation, so that leads on to a section on threads, critical regions, and atomicity.
- *What happens on a system call:* After a userland application program calls a kernel subroutine, what happens and how is that kept secure?
- *How addresses get translated in Linux/MIPS:* A fairly long story of virtual memory and how the MIPS hardware serves the Linux memory map.

14.1 The Life and Times of an Interrupt

It all starts when something happens in the real world: Maybe you tapped a key on the keyboard.

The device controller hardware picks up the data and activates an interrupt signal. That wends its way—probably via various pieces of hardware outside the CPU, which are all different and not interesting right now—until it shows up as the activation of one of the CPU’s interrupt signals, most often one of *Int0-5**.

The CPU hardware polls those inputs on every clock cycle. The Linux kernel sometimes disables interrupts, but not often; most often, the CPU is receptive. The CPU responds by taking an interrupt exception the next time it’s ready to fetch an instruction.

It's sometimes difficult to get across just how fast this happens. A 500-MHz MIPS CPU presented with an interrupt will fetch the first instruction of the exception handler (if in cache) within 3 to 4 clocks: That's 8 nanoseconds. Seasoned low-level programmers know that you can't depend on an 8-ns interrupt latency in any system, so we'll discuss below what gets in the way.

As the CPU fetches instructions from the exception entry point, it also flips on the exception state bit **SR (EXL)**, which will make it insensitive to further interrupts and puts it in kernel-privilege mode. It will go to the general exception entry point, at `0x8000.0180`.¹

The general exception handler inspects the **Cause** register and in particular the **Cause (ExcCode)** field: That has a zero value, showing that this was an interrupt exception. The **Cause (IP7-2)** field shows which interrupt input lines are active, and **SR (IM7-2)** shows which of these are currently sensitized. There ought to be at least one active, enabled interrupt; the software calculates a number corresponding to the number of the active interrupt input that will become Linux's `irq` number.²

Before calling out into something more like generic code, the main exception handler saves all the integer register values³—they belong to the thread that was just interrupted and they will need to be restored before that thread is allowed to continue, somewhere in the system's future. The values are saved on the kernel stack of the thread that was running when the interrupt happened, and the stack pointer is set accordingly. Some key CP0 register values are saved too; those include **SR**, **EPC**, and **Cause**.

With the **SR (EXL)** bit set, we're not really ready to call routines in the main kernel. In this state we can't take another exception,⁴ so we have to be careful to steer clear of any mapped addresses, for example.

Now that we have saved registers and established a stack we can change **SR**, clearing **SR (EXL)** but also clearing **SR (IE)** to avoid taking a second interrupt—after all, the interrupt signal we responded to is still active.

Now we're ready to call out to `do_IRQ()`. It's a machine-dependent routine but written in C, with a fairly well standardized flow. `do_IRQ()` is passed a structure full of register values as a parameter. Its job is to selectively disable this interrupt in particular and acknowledge any interrupt management hardware that needs a confirmation that the interrupt has been serviced.

-
1. Of course, it can't be that simple; some MIPS CPUs now provide a dedicated interrupt-exception entry point, and some will even go to a different entry point according to which interrupt signal was asserted. But we'll keep it simple—many OSs still do.
 2. In many cases, the exception handler may consult system-dependent external registers to refine the information about the interrupt number.
 3. Some MIPS CPUs can avoid this overhead for very low level interrupt handlers using shadow registers, as described in section 5.8.6.
 4. This is not quite inevitable. The MIPS architecture has some carefully designed corners that make it possible to nest exceptions in carefully controlled conditions. But Linux doesn't use them.

Assuming there is a device ISR (interrupt service routine) registered on this irq number, `do_IRQ()` calls on to `handle_IRQ_event()`. It's undesirable to run for long with all interrupts disabled; some other device out there may need fast response to its interrupt. But we're now into an area where it all depends on the nature of the device whose interrupt we're handling. If this handler is known to be very short and efficient (`SA_INTERRUPT` set), we can just go ahead and run it with all interrupts disabled; if it's going to take longer, then we'll re-enable interrupts in general—`do_IRQ()` has disabled the particular interrupt we're handling.

Multiple interrupt handlers can be registered for this one irq number; if so, they're called in turn.

Once the interrupt handlers are finished, `handle_IRQ_event()` disables interrupts again and returns (that's what `do_IRQ()` expects). After some more opportunity for machine-dependent tidying up, that calls `ret_from_intr()` to unwind the interrupt exception.

Before finally returning from the interrupt, we check whether something that happened in an ISR means that we should call the scheduler. Linux has a global flag, `needs_resched`, for this purpose; the code is not quite so simple as that because rescheduling from a thread interrupted in kernel mode (kernel preemption) may cause trouble. In fact, if the interrupted thread was executing with kernel privilege—as shown by the saved value of **SR (KSU)**—we won't reschedule if the global `preempt_count` is nonzero.

If the system doesn't schedule another thread, we just carry on and return from the interrupt. We restore all the saved register values. In particular, we restore the postexception value of **SR** to take us back into exception mode, and we restore **EPC**, which holds the restart location. Then we just run a MIPS **eret**.

If the system *does* schedule another thread, then our interrupted thread is parked exactly as it is, teetering on the edge of returning from the interrupt. When our interrupt-victim thread is selected to run again, it will burst out, back to whatever it was doing when interrupted.

Some device drivers just grab a little data from the device and send it up to the next level; others may do quite a lot of processing at interrupt time. But ISRs should not run for too long in interrupt context. Even after we've re-enabled other interrupts, the driver has unconditionally seized the attention of the CPU without the OS scheduler getting to apply its policies. Where there's more than a few tens of instructions' worth of work to do in the ISR, it would be better to defer that interrupt processing until the scheduler can decide whether there may be something more vital to do.

Linux traditionally splits the ISR into a “top half,” run at interrupt time, and a “bottom half,” deferred until after a rescheduling opportunity.⁵ Modern

5. The name could be confusing. Linux used to have a subsystem called “bottom half” and with functions whose names started “bh,” but that's obsolete and was finally omitted from the 2.6 kernel. But we still use “bottom half” as a generic term for work moved out of the interrupt handler.

Linux kernels have a system called *softirq*, which can arrange to run one of 32 functions; a couple of very demanding devices have their own *softirq* bottom halves, but most share a more flexible system called *tasklets*, built on top of *softirqs*.

Both of these are implemented in machine-independent code and are ways to arrange that a secondary interrupt handler will run soon after the real ISR returns.

A tasklet is ultimately called from the interrupt handler and can't do anything that might cause the unrelated interrupt-victim thread to sleep. Tasklet code must not do anything that might need waiting for. But there's always more than one way to do it. An ISR can also put extra driver tasks on a *work queue*. The work queue is serviced by a regular kernel thread, and work queue functions can do anything, even operations that might make the thread sleep.

Since this is all machine-independent, we won't dwell on the details here; it's described well in Love, 2004.

14.1.1 *High-Performance Interrupt Handling and Linux*

Compared with a lightweight OS, there's a fair amount of extra overhead in a Linux ISR. Linux unconditionally saves all registers and makes several levels of function calls (for example, isolating interrupt controller handling in `do_IRQ()`). It's possible to do useful work in a MIPS interrupt handler that never leaves exception mode (that is, it keeps **SR(EXL)** set)—but that's not directly supported by conventional Linux. It's possible to build a MIPS OS that is careful never to disable all interrupts for more than a handful of instructions, but Linux relies on disabling all interrupts at the CPU to avoid interrupts at uncomfortable places.

There are several efforts out there to make Linux more responsive to interrupts. While reducing the worst-case interrupt latency—the time it takes from the hardware interrupt being asserted to the ISR being entered—is important, it's not the whole job. To be responsive, the system also has to be prompt about scheduling the thread that is concerned with the input or output.

One project (the “low-latency patches” project) is evolutionary, aiming to improve average performance by a patient and diligent process of refining kernel locking/preemption control. Some of the low-latency work is in the mainstream 2.6+ Linux kernel.

Other projects (various kinds of “real-time” Linux) are revolutionary, commonly jacking up the Linux OS so that it doesn't see real interrupts at all. Interrupts are handled underneath by a sort of RTOS, of which the whole Linux kernel is but one thread. When Linux disables interrupts, it doesn't really disable hardware interrupts; it just prevents Linux ISRs from being scheduled by the RTOS layer.

It's worth noting that these two approaches are complementary—there's no reason not to pursue both.

14.2 Threads, Critical Regions, and Atomicity

A thread in Linux has to be aware that the data it's manipulating may also be getting attention from other, concurrent, activity. One of the main difficulties encountered in implementing a multitasking OS is finding and protecting all such sequences. The concurrent activity can come from:

- *Code run by another CPU*: In a real multiprocessor system
- *Code run in interrupt context*: That is, code run on this same CPU as a result of an interrupt
- *Code run by another thread that preempts this one*: More subtle, this is code run by some other kernel thread which (no doubt as a result of some interrupt) got scheduled over ours

The problem arises when a sequence of operations on data will be self-consistent when completed, but transiently creates some inconsistent state that other software cannot safely interpret or operate on. What software wants is the ability to mark a data transformation as *atomic*: Any outside view should either see the whole change made, or none of it. Linux provides a small range of simple atomic operations: Those operations can be specially implemented for a particular architecture, if the architecture offers some particularly neat way of doing them. But to make complex operations atomic, Linux uses locks.⁶ A lock is a ticket to operate on a particular chunk of data, and so long as all software accessing the data cooperates, only one thread will get to work on it at once. Any contending accessor will be held when they try to *acquire the lock* for the data.

The piece of code that does the want-to-be atomic sequence of operations on the data is called a *critical region*. So what we'll do is to get anyone wanting to access contended data to do something like this:

```
acquire_lock(contended_data_marker)
/* do your stuff in the critical region */
release_lock(contended_data_marker)
```

Every time this code is called, there's some time spent in those acquire and release operations, however they are implemented. When contention actually happens, the thread arriving at the acquire point and finding the lock currently unavailable must be held up, and the other thread arriving at the release must somehow pass the message on that it's now OK to continue.

6. A lock is a particularly simple form of what is called a *semaphore* in the more general case—but it probably helps to start with Linux's simpler and more minimal definition.

When the contender is known to be running on a separate CPU and the critical region consists of a handful of instructions, it may make most sense to get the temporarily frustrated acquirer to spin in a loop watching for the condition to clear. That's a *spinlock*, and Linux provides them for SMP.

When the contender could be a separate thread on the same CPU, you can't spin (while the waiting thread spins, the thread holding the lock could not be rescheduled, never gets to finish and release the lock, so we're all deadlocked forever). So you need a more heavyweight lock, where a thread that fails to acquire the lock marks itself as not currently runnable and calls the OS scheduler. Moreover, the `release_lock()` routine must arrange that any other thread waiting for the lock is told it can have another go. The code that suspends and wakes Linux threads is not CPU dependent. But the code that tests the lock state and sets the lock in the common uncontended case depends on an atomic test-and-set operation, whose implementation is done with special MIPS tricks, described below.

Where the contender might be in interrupt context (called, ultimately, from the interrupt entry point and borrowing the context of some randomly chosen interrupted thread to do so), Linux depends on disabling the relevant interrupt during the critical region. And since it's difficult to mask just one interrupt, that often means disabling *all* interrupts around the critical region. As mentioned previously, code called in interrupt context is expected to be disciplined and avoid doing things that will cause trouble. In fact, interrupt handlers are expected to have only simple, stylized interactions with the rest of the kernel.

What does the MIPS architecture bring to help implement simple atomic operations and locks?

14.2.1 *MIPS architecture and atomic operations*

MIPS has the load-linked/store-conditional pair. You use them to implement an arbitrary read-modify-write sequence on a variable (just read using `ll` and write using `sc`). The sequence is not atomic, in itself. But the store will do nothing unless it is certain that the sequence did run atomically, and `sc` returns a value the software can test. If the test shows the `sc` failed, the software can retry the RMW sequence until it finally does succeed.

These instructions were invented primarily for multiprocessor systems, as an alternative to a guaranteed-atomic RMW operation. They're a good choice there, because they avoid the overhead of a system-wide atomic transaction, which can stop every memory access in a large multiprocessor to make sure none of those accesses touch the data of interest.

The implementation is fairly simple. `ll` sets a per-CPU *link bit* and (for multiprocessor or hardware multithreading) records the load address so the CPU can monitor accesses to it. `sc` will succeed—doing the store and returning a “1” value—only if the link bit is still set. The CPU must clear the link bit if

it detects that the variable is (or may have been) updated by some unrelated software. In a multiprocessor system, the external-access detection is implemented by the snooping logic that keeps the caches coherent. But within a single CPU, the link is broken by any exception.⁷

Linux's `atomic_inc(&mycount)` uses the instructions to do an atomic increment of any integer variable:

```
atomic_inc:
    ll    v0, 0(a0)                # a0 has pointer to 'mycount'
    addu  v0, 1
    sc    v0, 0(a0)
    beq   v0, zero, atomic_inc
    nop
    jr    ra
    nop
```

If you emerge from this routine, you emerge having added 1 to `mycount` in a sequence that—it is guaranteed—was not interfered with by any SMP partner CPU, a local interrupt allowing other software to run, or another thread running on a hardware-multithreaded machine. It is possible for the routine to spin for a period of time if it is continually frustrated by external writes or interrupts, but—because the sequence between the `ll` and `sc` is very short—it's extraordinarily unlikely to have to try as many as three times.

You can probably see from the above that the `ll/sc` test is not suitable for use on very complicated operations, where the load and store are far apart. If you wait long enough, all links are broken.

14.2.2 *Linux Spinlocks*

Spinlocks use an atomic test-and-set operation to implement the simplest, cheapest locking primitive that protects multiple CPUs from each other. It's an exercise for the reader to track down how `ll/sc` can be used to construct an efficient spinlock.

Unlike the atomic operations themselves, spinlocks are quite useless on uniprocessor (that is, single-threaded uniprocessor) systems.

Historically, Linux kernels were built to be SMP-safe before kernel preemption was permitted. “Kernel preemption” is when an interrupt causes a thread running in the kernel to be descheduled and another one run. It's quite possible to build a Linux system in which a thread in the kernel runs either until its time-slice expires, it sleeps voluntarily (waiting some event), or it attempts to return

7. In most implementations, the link bit is in fact cleared by the `eret` instruction, which is essential to return from the exception.

to user mode. In fact, standard Linux kernels before 2.6 did that and were still more responsive than a contemporary Windows system with a background task running. But kernel preemption is better still, if you can make it work properly.

Along the road to the 2.6 release of the kernel, George Anzinger observed that almost all the critical regions that need software protection from kernel preemption also needed protection from SMP concurrency, so these regions are already delimited by spinlock operations. Spinlocks are macros, so they can be disabled (and will compile away to nothing) for a uniprocessor system. When a 2.6+ kernel is built with the `CONFIG_PREEMPT` configuration defined, the spinlock operations gain the side effect of disabling preemption between their acquire and release.

Well, that was nearly it. “Almost all” is a dangerous phrase. Just occasionally, a piece of data manipulation is inherently SMP-safe (most obviously when it addresses per-CPU data) so it doesn’t need spinlocks, but it does need defense against preemption, since other kernel code on the same CPU is accessing the same data. Making the 2.6 kernel reliable with kernel preemption still required a fair amount of patient combing through code and even more patient debugging.

14.3 What Happens on a System Call

A system call is “just” a subroutine implemented by the kernel for a user-space program. Some system calls just return information known in the kernel but not outside (the accurate time of day, for example).

But there are two reasons why it’s more difficult than that. The first has to do with security—the kernel is supposed to be robust in the face of faulty or malicious application code.

The second reason has to do with stability. The Linux kernel should be able to run applications built for it, but should also be able to run any application built for the same or a compatible architecture at any time in the past. A system call, once defined, may only be removed after a lot of argument, work, and waiting.

But back to security. Since the system call will run in kernel mode, the entry to kernel mode must be controlled. For the MIPS architecture that’s done with a software-triggered exception from the **syscall** instruction, which arrives in the kernel exception handler with a distinctive code (8 or “Sys”) in the CPU’s **Cause (ExcCode)** register field. The lowest-level exception handler will consult that field to decide what to do next and will switch into the kernel’s system call handler.

That’s just a single entry point: The application sets a numeric argument to select which of several hundred different syscall functions should be called. Syscall values are architecture-specific: The MIPS system call number for a function may not be the same as the x86 number for the same function.

System call arguments are passed in registers, as far as possible. It's good to avoid unnecessary copies between user- and kernel-space. In the 32-bit MIPS system:

- The syscall number is put in `v0`.
- Arguments are passed as required by the “o32” ABI. Most of the time, that means that up to four arguments are passed in the registers `a0`–`a3`; but there are corner cases where something else happens, and it's all explained in section 11.2.1.⁸

Although the kernel uses something like o32 to define how arguments are passed into system calls, that does not mean that userland programs are obliged to use o32. Good programming practice requires that userland calls to the OS should always pass through the C runtime library or its equivalent, so that the system call interface need only be maintained in one place. The library can translate from any ABI to the o32-like syscall standard, and that's done when running 32-bit software on 64-bit MIPS Linux.

- The return value of the syscall is usually in `v0`. But the `pipe(2)` system call returns an array of two 32-bit file descriptors, and it uses `v0` and `v1`—perhaps one day another system call will do the same.⁹

All system calls that can ever fail return a summary status (0 for good, nonzero for bad) in `a3`.

- In keeping with the calling conventions, the syscall preserves the values of those registers that o32 defines as surviving function calls.

Making sure the kernel implementation of a system call is safe involves a healthy dose of paranoia. Array indexes, pointers, and buffer lengths must be checked. Not all application programs are permitted to do everything, and the code for a system call must check the caller's “capabilities” when required (most of the time, a process running for the root super-user can do anything).

The kernel code running a system call is in process context—the least restrictive context for kernel code. System call code can do things that may require the thread to sleep while some I/O event happens, or can access virtual memory that may need to be paged in.

When copying data in and out, you're dependent on pointers provided by the application, and they may be garbage. If we use a bad pointer, we'll get an exception, which could crash the kernel—not acceptable.

So copying inputs from user space or outputs to user space can be done safely with functions `copy_to_user()`/`copy_from_user()`, provided for that

8. Many of the darkest corners of argument passing don't affect the kernel, because it doesn't deal in floating-point values or pass or return structures by value.

9. I hear a chorus of “Over our dead bodies!” from the kernel maintainers.

purpose. If the user did pass a bad address, that will cause a kernel exception. The kernel, though, maintains a list of the functions¹⁰ that are trusted to do dangerous things: `copy_to_user()` and `copy_from_user()` are on that list. When the exception handler sees that the exception restart address is in one of these functions, it returns to a carefully constructed error handler. The application will be sent a rude signal, but the kernel survives.

Return from a system call is done by a return through the end of the **syscall** exception handler, which ends with an **eret**. The important thing is that the change back to user mode and return to the user instruction space happen together.

14.4 How Addresses Get Translated in Linux/MIPS Systems

Before getting down to how it's done, we should get an overview of the job.

Figure 14.1 sketches out the memory map seen by a Linux thread on a 32-bit Linux/MIPS system.¹¹ The map has to fit onto what the hardware does, so user-accessible memory is necessarily in the bottom half of the map.

Useful things to remember:

- *Where kernel is built to run:* The MIPS Linux kernel's code and data are built to run in `kseg0`; virtual addresses from `0x8000.0000` upward. Addresses in this region are just a window onto the low 512 Mbytes of physical memory, requiring no TLB management.
- *Exception entry points:* In most MIPS CPUs to date, these are hard-wired near the bottom of `kseg0`. The latest CPUs may provide the **EBase** register to allow them to be relocated (see section 3.3.8), mainly so that multiple memory-sharing CPUs can use different exception handlers without the trouble of special-casing memory decoding. In the Linux kernel, even when there are multiple CPUs, they should all run the same exception-handling code, so this is unlikely to be used for Linux.
- *Where user programs are built to run:* MIPS Linux applications (which are run in low-privilege “user mode”) have virtual addresses from 0 through `07FFF.FFFF`. Addresses in this region are accessible in user mode and translated through the TLB.

The main program of an application is built to run starting somewhere near zero. Not quite zero—one or more pages from virtual address zero are never mapped, so that an attempt to use a null pointer will be caught as a memory-management error. The library components of an application, though, are loaded incrementally into user space at load

10. It's actually a list of instruction address ranges.

11. The 64-bit Linux/MIPS map is built on the same principles, but the more complicated hardware map—see Figure 2.2—makes it relatively confusing.

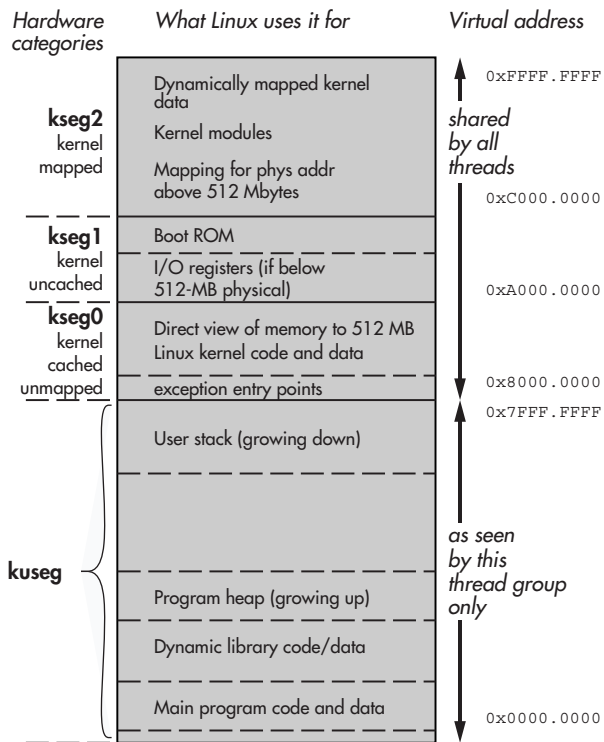


FIGURE 14.1 Memory map for a Linux thread.

time or even later. That's possible because a library component is built to be position-independent (see Chapter 16) and is adjusted to fit the place that the loader finds for it.

- **User stack and heap:** An application's stack is initially set to the top of user-accessible memory (about 2 GB up in virtual space) and grows down. The OS detects accesses to not-yet-mapped memory near the lowest stack entries it has allocated and automatically maps more pages to let the stack grow.

Meanwhile, new shared libraries or explicit user data allocated with `malloc()` and its descendants are growing up from the bottom of the user space. So long as the sum of all these remains less than 2 GB, all is well: This restriction is rarely onerous in any but the biggest servers.

- **Physical memory up to 512 MB:** Can be accessed cached through `kseg0` and uncached through `kseg1`. Historically, the Linux kernel assumed it had direct access to all the physical memory of the machine. For smaller

MIPS systems that use 512 MB or less of physical memory range, that's true; in this case, all memory is directly accessible (cached) in `kseg0` and (uncached) in `kseg1`.

- *Physical memory over 512 MB is “high memory”*: Now, 512 MB is no longer enough, even for embedded systems. Linux has an architecture-independent concept of *high memory*—physical memory that requires special, architecture-dependent handling, and for a 32-bit Linux/MIPS system physical memory above 512 MB is high memory. When we need to access it, we'll create appropriate translation entries and have them copied into the TLB on demand.

Early MIPS CPUs sought applications in UNIX workstations and servers, so the MIPS memory-management hardware was conceived as the minimum hardware that could hope to provide memory management for BSD UNIX. The BSD UNIX system was the first UNIX OS to provide real paged virtual memory on the DEC VAX minicomputer. The VAX was in many ways the model for the 32-bit paged-translation virtual-memory architectures that have dominated computing ever since; perhaps it's not surprising that there are some echoes of the VAX memory-management organization in MIPS. But this is a RISC, and the MIPS hardware does much less. In particular, many problems that the VAX (or an x86) solves with microcode are left to software by the MIPS system.

In this chapter, we'll start close to where MIPS started, with the requirements of a basic UNIX-like OS and its virtual memory system; but this time the OS is Linux.

We'll show how the essence of the MIPS hardware is a reasonable response to that requirement. For real nuts and bolts details, refer to Chapter 6.

14.4.1 *What's Memory Translation For?*

Memory translation hardware (while we're being general we'll call it MMU for *memory management unit*) serves several distinct purposes:¹²

- *Hiding and protection*: User-privilege programs can only access data whose program address is in the `kuseg` memory region (lower program addresses). Such a program can only get at the memory regions that the OS allows.

Moreover, each page can be individually specified as writable or write protected; the OS can even stop a program from accidentally overwriting its code.

- *Allocating contiguous memory to programs*: With the MMU, the OS can build contiguous program space out of physically scattered pages of

12. Given just how much the MMU contributes, it's remarkable that a fairly workable minimal Linux system (uClinux) can be built without it—but that's another story.

memory, allowing us to allocate memory from a simple pool of fixed-size pages.

- *Extending the address range:* Some CPUs can't directly access their full potential physical memory range. MIPS32 CPUs, despite their genuine 32-bit architecture, arrange their address map so that the unmapped address space windows kseg0 and kseg1 (which don't depend on the MMU tables to translate addresses) are windows onto the first 512 MB of physical memory. If you need a memory map that extends further, you must go through the MMU.
- *Making the memory map suit your program:* With the MMU, your program can use the addresses that suit it. In a big OS, there may be many copies of the same program running, and it's much easier for them all to be using the same program addresses.
- *Demand paging:* Programs can run as if all the memory resources they needed were already allocated, but the OS can actually give them out only as needed. A program accessing an unallocated memory region will get an exception that the OS can process; the OS then loads appropriate data into memory and lets the program continue.

In theory (and sometimes in computer science textbooks), it says that demand paging is useful because it allows you to run a larger program than will fit into physical memory. This is true, if you've got a lot of time to spare, but in reality if a program really needs to use more memory than is available it will keep displacing bits of itself from memory and will run very, very slowly.

But demand paging is still very useful, because large programs are filled with vast hinterlands of code that won't get run, at least on this execution. Perhaps your program has built-in support for a data format you're not using today; perhaps it has extensive error handling, but those errors are rare. In a demand-paged system, the chunks of the program that aren't used need never be read in. And it will start up fast, which will please an impatient user.

- *Relocation:* The addresses of program entry points and predeclared data are fixed at program compile/build time—though this is much less true for position-independent code, used for all Linux's shared libraries and most applications. The MMU allows the program to be run anywhere in physical memory.

The essence of the Linux memory manager's job is to provide each program with its own memory space.

Well, really Linux has separate concepts: Threads are what is scheduled, and address spaces (memory maps) are the protection units. Many threads in a thread group can share one address space. The memory translation system is obviously interested in address spaces and not in threads. Linux's

implementation is that all threads are equal, and all threads have memory-management data structures. Threads that run in the same address space, in fact, share most of those data structures.

But for now it will be simpler if we just consider one thread per address space, and we can use the older word “process” for them both.

If the memory management job is done properly, the fate of each process is independent of the others (the OS protects itself too): A process can crash or misbehave without bringing down the whole system. This is obviously a useful attribute for a university departmental computer running student programs, but even the most rigorous commercial environment needs to support experimental or prototype software alongside the tried and tested.

The MMU is not just for big, full virtual memory systems; even small embedded programs benefit from relocation and more efficient memory allocation. Any system in which you may want to run different programs at different times will find it easier if it can map the program’s idea of addresses onto whatever physical address space is readily available.

Multitasking and separation between various tasks’ address spaces used to be only for big computers, then migrated into personal computers and small servers, and are increasingly common in the vanishingly small computers of consumer devices.

However, few non-Linux embedded OSs use separate address spaces. This is probably not so much because this would not be useful, but is due to the lack of consistent features on embedded CPUs and their available operating systems. And perhaps because once you add separate address spaces to your system you’re too close to reinventing Linux to make sense!

This is an unexpected bonanza for the MIPS architecture. The minimalism that was so necessary to make the workstation CPU simple in 1986 is a great asset to the embedded systems of the early 21st century. Even small applications, beset by rapidly expanding code size, need to use all known tricks to manage software complexity; and the flexible software-based approach pioneered by MIPS is likely to deliver whatever is needed. A few years ago it was hard to convince CPU vendors addressing the embedded market that the MMU was worth including; now (in 2006) Linux is everywhere.

14.4.2 *Basic Process Layout and Protection*

From the OS’s point of view, the low part of memory (kuseg) is a safe “sandbox” in which the user program can play all it wants. If the program goes wild and trashes all its own data, that’s no worry to anyone else.

From the application’s point of view, this area is free for use in building arbitrarily complicated private data structures and to get on with the job.

Inside the user area, within the program’s sandbox, the OS provides more stack to the program on demand (implicitly, as the stack grows down). It will also provide a system call to make more data available, starting from the highest

predeclared data addresses and growing up—systems people call this a *heap*. The heap feeds library functions, such as `malloc()`, that provide your program with chunks of extra memory.

Stack and heap are supplied in chunks small enough to be reasonably thrifty with system memory but big enough to avoid too many system calls or exceptions. However, on every system call or exception, the OS has a chance to police the application's memory consumption. An OS can enforce limits that make sure the application doesn't get so large a share of memory as to threaten other vital activities.

A Linux thread keeps its identity inside the OS kernel, and when it runs in process context in the kernel (as in a system call) it's really just calling a careful subroutine.

The operating system's own code and data are of course not accessible to user space programs. On some systems, this is done by putting them in a completely separate address space; on MIPS, the OS shares the same address space, and when the CPU is running at the user-program privilege level, access to these addresses is illegal and will trigger an exception.

Note that while each process's user space maps to its own private real storage, the privileged space is shared. The OS code and resources are seen at the same address by all processes—an OS kernel is a multithreaded but single-address-space system inside—but each process's user space addresses access its own separate space. Kernel routines running in process context are trusted to cooperate safely, but the application need not be trusted at all.

We mentioned that the stack is initialized to the highest permissible user addresses: That's done so we can support programs that use a great deal of memory. The resulting wide spread of addresses in use (with a huge hole in between) is one characteristic of this address map with which any translation scheme must cope.

Linux maps application code as read-only to the application, so that code can be safely shared by threads in different address spaces—after all, it's common to have many processes running the same application.

Many systems share not just whole applications but chunks of applications accessed through library calls (shared libraries). Linux does that through shared libraries, but we'll pick up that story later.

14.4.3 *Mapping Process Addresses to Real Memory*

Which mechanisms are needed to support this model?

We want to be able to run multiple copies of the same application, and they'd better use different underlying copies of the data. So during program execution, application addresses are mapped to physical addresses according to a scheme fixed by the OS when the program is loaded.

Although it would be possible for the OS to rush around patching all the address translation information whenever we switched contexts from one process to another, it would be very inefficient. Instead, we award each active

thread’s memory map an 8-bit number called the *address space ID* or ASID. When the hardware finds a translation entry for the address, it looks for a translation entry that matches the current ASID as well as the address,¹³ so the hardware can hold translations for different spaces without getting them mixed up. Now all the software needs to do when scheduling a thread in a different address space is to load a new ASID into **EntryHi (ASID)**.

The mapping facility also allows the OS to discriminate between different parts of the user address space. Some parts of the application space (typically code) can be mapped read-only, and some parts can be left unmapped and accesses trapped, meaning that a program that runs amok is likely to be stopped earlier.

The kernel part of the process’s address space is shared by all processes. The kernel is built to run at particular addresses, so it mostly doesn’t need a flexible mapping scheme and can take advantage of the kseg0 region, leaving memory mapping resources for application use. Some dynamically generated kernel areas are conveniently built from mapped memory (for example, memory used to hold kernel loadable modules for device drivers and so on is mapped).

14.4.4 *Paged Mapping Preferred*

Many exotic schemes have been tried for mapping addresses. Until the mid-1980s or so, the industry kept thinking there must be a better solution than fixed-size pages. Fixed-size pages have nothing to do with the needs or behavior of application programs, after all, and no top-down analysis will ever invent them.

But if the hardware gives programs what they want (chunks of memory in whichever size is wanted at that moment), available memory rapidly becomes fragmented into awkward-sized pieces. When we finally make contact with aliens, their wheelbarrows will have round wheels and their computers will probably use fixed-size pages.

So all practical systems map memory in *pages*—fixed-size chunks of memory. Pages are always a power of 2 bytes big. Small pages might take too much management (big tables of translations, many translations needed for a program); big pages may be inefficient, since quite a lot of small objects end up occupying a whole page. A total of 4 KB is the overwhelmingly popular compromise for Linux.¹⁴

With 4-KB pages, a program/virtual address can be simply partitioned thus:

nn	12	11	0
Virtual page number (VPN)			Address within page

13. Actually, we’ll see that some translation entries can be ASID-independent or “global.”
14. The MIPS hardware would be quite happy with bigger pages: 16 KB is the next directly supported size and is probably a better trade-off for many modern systems. But the page size is pervasive, with tentacles into the way programs are built; so we almost all still use 4 KB.

The address-within-page bits don't need to be translated, so the memory management hardware only has to cope with translating the high-order addresses (traditionally called virtual page number or VPN), into the high-order bits of a physical address (a physical frame number, or PFN—nobody can remember why it's not PPN).

14.4.5 *What We Really Want*

The mapping mechanism must allow a program to use a particular address within its own process/address space and translate that efficiently into a real physical address to access memory.

A good way to do this would be to have a table (the *page table*) containing an entry for each page in the whole virtual address space, with that entry containing the correct physical address. This is clearly a fairly large data structure and is going to have to be stored in main memory. But there are two big problems.

The first is that we now need two references to memory to do any load or store, and that's obviously hopeless for performance. You may foresee the answer to this: We can use a high-speed cache memory to store translation entries and go to the memory-resident table only when we miss in the cache. Since each cache entry covers 4 KB of memory space, it's plausible that we can get a satisfactorily low miss rate out of a reasonably small cache. (At the time this scheme was invented, memory caches were rare and were sometimes also called "lookaside buffers," so the memory translation cache became a translation lookaside buffer or TLB; the acronym survives.)

The second problem is the size of the page table; for a 32-bit application address space split into 4-KB pages, there are a million entries, which will take at least 4 MB of memory. We really need to find some way to make the table smaller, or there'll be no memory left to run the programs.

We'll defer any discussion of the solution for this, beyond observing that few real programs use anything like the 4 Gbytes addressable with 32 bits. More modest programs have huge holes in their program address space, and if we can invent some scheme that avoids storing all the "nothing here" translation entries corresponding to the holes, then things are likely to get better.

We've now arrived, in essence, at the memory translation system DEC figured out for its VAX minicomputer, which has been extremely influential in most subsequent architectures. It's summarized in Figure 14.2.

The sequence in which the hardware works is something like this:

- A virtual address is split into two, with the least significant bits (usually 12 bits) passing through untranslated—so translation is always done in pages (usually 4 KB).
- The more significant bits, or VPN, are concatenated with the currently running thread's ASID to form a unique page address.

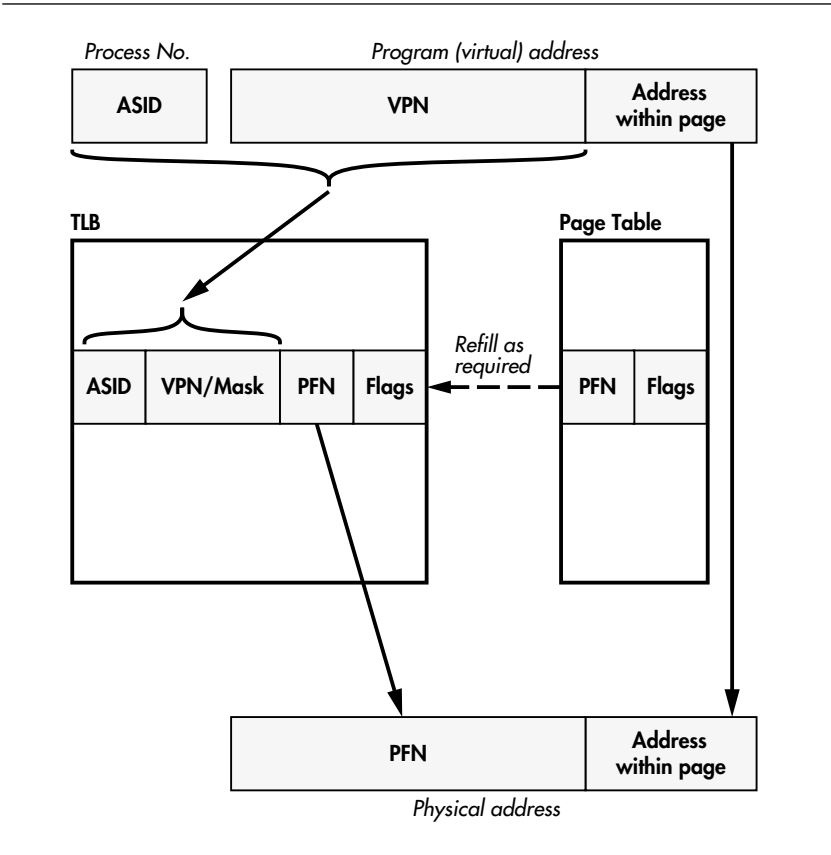


FIGURE 14.2 Desirable memory translation system.

- We look in the TLB (translation cache) to see if we have a translation entry for the page. If we do, it gives us the high-order physical address bits and we've got the address to use.
The TLB is a special-purpose store and can match addresses in various useful ways. It may have a global flag bit that tells it to ignore the value of ASID for some entries, so that these TLB entries can be used to map some range of virtual addresses for *every* thread.
Similarly, the VPN may be stored with some mask bits that cause some parts of the VPN to be excluded from the match, allowing the TLB entry to map a larger range of virtual addresses.
Both of these features are available in MIPS MMUs (there's no variable-size pages in some very old MIPS CPUs, though).
- There are usually extra bits (flags) stored with the PFN that are used to control which kind of access is allowed—most obviously, to permit reads

but not writes. We'll discuss the MIPS architecture's flags in the next section.

- If there's no matching entry in the TLB, the system must locate or build an appropriate entry (using main-memory-resident page table information) and load it into the TLB and then run the translation process again. In the VAX minicomputer, this process was controlled by microcode and seemed to the programmer to be completely automatic. If you build the right format of page table in memory and point the hardware at it, all memory translation just works.

14.4.6 *Origins of the MIPS Design*

The MIPS designers wanted to figure out a way to offer the same facilities as the VAX with as little hardware as possible. The microcoded TLB refill was not acceptable, so they took the brave step of consigning this part of the job to software.

That means that apart from a register to hold the current ASID, the MMU hardware is simply a high-speed, fixed-size table of translations. System software can (and usually does) use the hardware as a cache of entries from some kind of comprehensive memory-resident page table, so it makes sense to call the hardware table a TLB. But there's nothing in the TLB hardware to make it a cache, except this: When presented with an address it can't translate, the TLB triggers a special exception (*TLB refill*) to invoke the software routine. Some care is taken with the details of the TLB design, the associated control registers, and the refill exception to help the software to be efficient.

The MIPS TLB has always been implemented on chip. The memory translation step is required even for cached references, so it's very much on the critical path of the machine. That meant it had to be small, particularly in the early days, so it makes up for its small size by being clever.

It's basically a genuine *associative memory*. Each entry in an associative memory consists of a key field and a data field; you present the key and the hardware returns the data of any entry the key matches. Associative memories are wonderful, but they are expensive in hardware. MIPS TLBs have had between 32 and 64 entries; a store of this size is manageable as a silicon design.

All contemporary CPUs use a TLB in which each entry is doubled up to map two consecutive VPNs to independently specified physical pages. The paired entries double the amount of memory that can be mapped by the TLB with only a little extra logic, without requiring any large-scale rethinking of TLB management.

You will see the TLB referred to as being fully associative; this emphasizes that all keys are really compared with the input value in parallel.¹⁵

15. The common 32-entry paired TLB would be correctly, if pedantically, described as a 32-way set-associative store, with two entries per set.

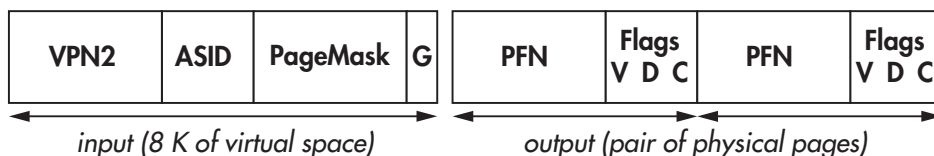


FIGURE 14.3 TLB entry fields.

The TLB entry is shown schematically in Figure 14.3. For the moment, we'll assume that pages are 4 Kbytes in size. The TLB's key—the input value—consists of three fields:

- **VPN2**: The page number is just the high-order bits of the virtual address—the bits left when you take out the 12 low bits that address the byte within page. The “2” in VPN2 emphasizes that each virtual entry maps 8 Kbytes because of the doubled output field. Bit 12 of the virtual address selects either the first or the second physical-side entry of the pair.
- **PageMask**: Controls how much of the virtual address is compared with the VPN and how much is passed through to the physical address; a match on fewer bits maps a larger region. A “1” bit causes the corresponding address bit to be ignored. Some MIPS CPUs can be set up to map as much as 16 MB with a single entry. The most significant ignored bit is used to select the even or odd entry.
- **ASID**: Marks the translation as belonging to a particular address space, so this entry will only be matched if the thread presenting the address has **EntryHi (ASID)** set equal to this value.

The **G** bit, if set, disables the ASID match, making the translation entry apply to all address spaces (so this part of the address map is shared between all spaces). The ASID is 8 bits: The OS-aware reader will appreciate that even 256 is too small an upper limit for the number of simultaneously active processes on a big UNIX system. However, it's a reasonable limit so long as “active” in this context is given the special meaning of “may have translation entries in the TLB.” OS software has to recycle ASIDs where necessary, which will involve purging the TLB of translation entries for any processes being downgraded from active. It's a dirty business, but so is quite a lot of what OSs have to do; and 256 entries should be enough to make sure it doesn't have to be done so often as to constitute a performance problem.

For programming purposes, it's easiest if the **G** bit is kept in the kernel's page tables with the output-side fields. But when you're translating, it belongs to the input side. On MIPS32/64 CPUs, the two output-side values are AND-ed together to produce the value that is used, but the

realistic outcome is that you must make sure the **G** bit is set the same in both halves.

The TLB's output side gives you the physical frame number and a small but sufficient bunch of flags:

- *Physical frame number (PFN)*: This is the physical address with the low bits cut off (the low 12 bits if this is representing a 4-Kbyte page).
- *Write control bit (D)*: Set 1 to allow stores to this page to happen. The “D” comes from this being called the dirty bit; see the next section for why.
- *Valid bit (V)*: If this is 0, the entry is unusable. This seems pretty pointless: Why have a record loaded into the TLB if you don't want the translation to work? There are two reasons. The first is that the entry translates a pair of virtual pages, and maybe only one of them ought to be there. The other is that the software routine that refills the TLB is optimized for speed and doesn't want to check for special cases. When some further processing is needed before a program can use a page referred to by the memory-held table, the memory-held entry can be left marked invalid. After TLB refill, this will cause a different kind of trap, invoking special processing without having to put a test in every software refill event.
- *Cache control (C)*: This 3-bit field's primary purpose is to distinguish cacheable (3) from uncached (2) regions.
But that leaves six other values, used for two somewhat incompatible purposes: In shared-memory multiprocessor systems, different values are used to hint whether the memory is shared (when hardware will have to work hard to keep any cached data consistent across the whole machine). In “embedded” CPUs, different values select different local cache management strategies: write-through versus write-back, for example. See your CPU manual.

Translating an address is now simple, and we can amplify the description above:

- *CPU generates a program address*: This might be an instruction fetch, a load, or store—and it's one with an address that does not lie in the special unmapped regions of the MIPS address space.
The low 13 bits are separated off, and the resulting VPN2, together with the current ASID (**EntryHi (ASID)**), is looked up in the TLB. The match is affected by the the **PageMask** and by **G** fields of the various TLB entries.
- *TLB matches key*: If there's no matching entry, we'll take a TLB refill exception. But where there is a match, that entry is selected. Bit 12 of the virtual address selects which physical-side half we'll use.

The PFN from the TLB is glued to the low-order bits of the program address to form a complete physical address.

- *Valid?* The V and D bits are consulted. If it isn't valid, or a store is being attempted with D unset, the CPU takes an exception. As with all translation traps, the **BadVAddr** register will be filled with the offending program address; as with any TLB exception, the TLB **EntryHi** register will be preloaded with the VPN of the offending address.

The **BadVPN2** fields of the convenience registers **Context** (and **XContext** on 64-bit CPUs) will be preloaded (in part, as advertised) with the appropriate bits of the virtual address we failed to translate during a TLB refill exception. But the specification is less definite about how those address fields behave in other exceptions. It's probably a good idea to stick to **BadVAddr** in other exceptions.

- *Cached?* If the C bit is set, the CPU looks in the cache for a copy of the physical location's data; if it isn't there, it will be fetched from memory and a copy left in the cache. Where the C bit is clear, the CPU neither looks in nor refills the cache.

Of course, the number of entries in the TLB permits you to translate only a relatively small number of program addresses—a few hundred KB worth. This is far from enough for most systems. The TLB is almost always going to be used as a software-maintained cache for a much larger set of translations.

When a program address lookup in the TLB fails, a *TLB refill* trap is taken.¹⁶ System software has the following job:

- It figures out whether there is a correct translation; if not, the trap will invoke the software that handles address errors.
- If there is a correct translation, it constructs a TLB entry that will implement it.
- If the TLB is already full (and it almost always is full in running systems), the software selects an entry that can be discarded.
- The software writes the new entry into the TLB.

See section 14.4.8 for how this is done in Linux.

14.4.7 *Keeping Track of Modified Pages (Simulating “Dirty” Bits)*

An operating system that provides a page for an application program to use often wants to keep track of whether that page has been modified since the OS last obtained it (perhaps from disk or network) or saved a copy of it.

16. Should this be called a “TLB miss” (which is what just happened) or a “TLB refill” (which is what we're going to do to sort it out)? I'm afraid we probably use both terms in MIPS documentation.

Unmodified (“clean”) pages may be quietly discarded, since they can easily be recovered from a file system if they’re ever needed again.

In OS parlance the modified pages are called “dirty,” and the OS must take care of them until the application program exits or the dirty page is cleaned by being saved away to backing store. To help out with this process, it is common for CISC CPUs to maintain a bit in the memory-resident page table indicating that a write operation to the page has occurred. The MIPS CPU does not support this feature, even in the TLB entries. The D bit of the page table (found in the **EntryLo** register) is a write-enable and of course is used to flag read-only pages.

So here’s the trick:

- When a writable page is first loaded into memory, you mark its page table entry with D clear (leaving it read-only).
- When any write is attempted to the page, a trap will result; system software will recognize this as a legitimate write but will use the event to set a “modified” bit in the memory resident tables—which, since it’s in the **EntryLo (D)** position, permits future writes to be done without an exception.
- You will also want to set the D bit in the TLB entry so that the write can proceed (but since TLB entries are randomly and unpredictably replaced, this would be useless as a way of remembering the modified state).

14.4.8 *How the Kernel Services a TLB Refill Exception*

MIPS’s TLB refill exception has always had its own unique entry point (at least as long as the CPU wasn’t already in exception mode; in Linux, that would be a fatal error, and isn’t considered further).

When the exception routine is called, the hardware has set up **EntryHi (VPN2)** to the page number of the location we just couldn’t translate: **EntryHi** is set exactly to what is required to create a new TLB entry to map the offending address.

The hardware has also set up a bunch of other address-related fields, but we’re not going to use any of them.

In particular, we’re not going to use the convenient “MIPS standard” way of using **Context** to find the relevant page table entry, which was described in section 6.2.4. The MIPS standard way requires that a (notionally very long) linear page table is constructed in `kseg2` (it won’t really take up excessive space, because `kseg2` is mapped and the vast empty spaces of the table would never be mapped to real memory). But Linux really does not like to have the unique-to-thread-group kernel mappings that would require.

Instead, Linux’s TLB refill page tables are organized as a three-level hierarchy of tables (called “global,” “middle,” and just “PTE”). But cunning use of C macros allows the middle level to disappear completely without changing the

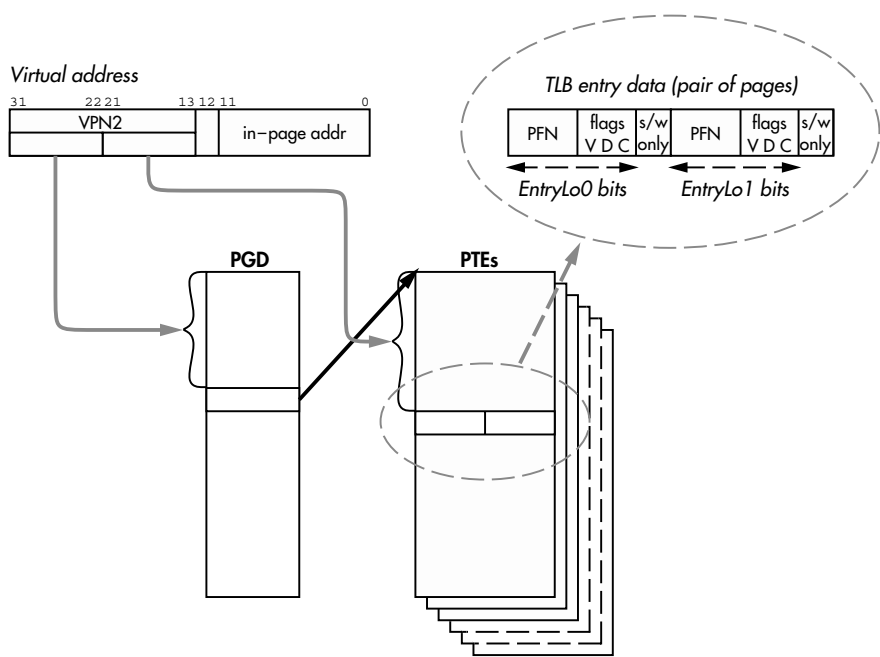


FIGURE 14.4 Linux two-level page table (32-bit MIPS design).

code—a two-level structure is enough for 32-bit MIPS.¹⁷ So you can find the data for any TLB entry you want—if it exists at all—by following the links, as shown in Figure 14.4.

This structure is reasonably economical of kernel memory: A largish Linux program with a 50-MB virtual address space has about 12-K 4-KB pages, each needing four bytes of PTE: That makes 48 KB or 12 pages. That’s an underestimate, because the address space components have holes in them, but a reasonable-size thread group’s mapping needs take only 15 or so pages. Kernel (kseg2) mappings take more, but they’re common to all memory maps, so there is only one set of PTEs for kernel mappings.

Most 32-bit CPUs are restricted to a 32-bit physical address, too: Where that’s so, there are six unused high-order bits in the **EntryLo0–1** registers. Linux recycles those to remember some software state about the page table entries.

This does mean that the TLB refill handler for this 32-bit needs to do two index calculations and three loads. It’s certainly longer than the tiny instruction

17. However, all three are required for the extended virtual memory space available with 64-bit MIPS.

sequence of the original magic MIPS scheme (where the index calculation is done by magic by the way the **Context** register works, and the only loads are from the final page table), but it's not bad.

You may have noticed that this description is very particular to a certain sort of MIPS CPU. Do all Linux systems have a unique kernel? Well, no: But in the current Linux/MIPS kernel certain critical routines—including the TLB miss exception handler—are binaries generated by table-driven kernel software during start-up and tailored to the requirements of this particular CPU.

Here, for example, is the TLB miss handler autogenerated for a system based on a 32-bit MIPS 24-K core:

```
tlb_refill:
    # (1) get base of PGD into k1
    lui      k1, %hi(pgd_current)

    # get miss virtual address (VA)
mfc0      k0, c0_badvaddr # (2) moved up to save a cycle
lw        k1, %lo(pgd_current)(k1)
srl       k0, k0, 24
sll       k0, k0, 2      # (3) shift and mask VA for PGD index
addu      k1, k1, k0      # got a pointer to the correct PGD entry

    # get VA again, this time from Context register
mfc0      k0, c0_context # (4) moved up to save a cycle
lw        k1, 0(k1)      # OK, read the PTE pointer

srl       k0, k0, 0x1     # (5) Context register designed for 2x64-bit,
                        # entry, ours are half that size
andi      k0, k0, 0xff8   # (6) mask out higher VPN bits etc

addu      k1, k1, k0      # pointer to VPE entry

    # load the TLB entries
lw        k0, 0(k1)      # (7) will lose a cycle here
lw        k1, 4(k1)

srl       k0, k0, 0x6     # (8) shift-out software-only bits
mtc0      k0, c0_entrylo0
srl       k1, k1, 0x6     # same for other half
mtc0      k1, c0_entrylo1

ehb                          # (9) wait for CP0 values to be ready
tlbwr                      # (10) write into TLB (somewhere)
eret                      # (11) back to user
```


Notes on the TLB Refill Code

- (1) In an exception handler, the software convention allows us to use two registers (**k0–1**) with impunity. All other general-purpose registers are still holding application data, and we can't touch them.
- (2, 4, 5) The code is made a little harder to read by the places where one sequence has been interleaved with the next. That's done for efficiency: A 24-K CPU load requires the register base to be ready one clock early, so if you compute the base in the previous instruction you'll get a one-clock stall. At (2) and (4) we found a nondependent instruction to interleave, but at (5) there isn't one.
- (3) As you can see from Figure 14.4, we want to use the top bits of the virtual address with the missing translation (which is in **BadVaddr**) to index the PGD. Since the PGD has pointers (four bytes) in each entry, we need to shift the index value by two to get a byte offset. You can't just do a shift-right by 22 bits, because you might generate a nonword-aligned pointer, which is illegal for a MIPS load-word.
- (4) See (2, 4, 5).
- (5) With only two registers to play with, we can't keep the original virtual address. So now we get it again, but this time from the **Context** register, which has the low bits of the "VPN2" field we need. For a 64-bit CPU—where we'd need to keep two 64-bit entries to fill the **EntryLo0–1** pair—it's just right, with the VPN2 number shifted up by four positions to index a table of 16-byte entries. But Linux/MIPS for MIPS32 has eight-byte entries in the PTE table; so we shift it right one place.
- (6) The high bits of **Context** are unwanted, so we mask them out. Even above the page number, they're not always zero: SMP Linux systems use the **Context (PTEBase)** field, which consists of plain read/write bits, to store a CPU ID.
- (7) See (2, 4, 5).
- (8) On systems like this with 32-bit physical addresses, the **EntryLo** register only has 26 meaningful bits. Linux uses the other six for software flags, which we shift away.
- (9) This execution hazard barrier ensures that the instructions following it are delayed until the writes to the CP0 **EntryLo0–1** registers are complete. In many CPUs, the write to the CP0 register from the **mtc0** instruction happens in a late pipeline stage.
- (10) **tlbwr** writes the complete TLB entry to a "random" place in the TLB. In fact, it gets the index from **Random**, which just counts continually round TLB entries, but the timing of TLB misses is random enough that this works well.

Note that the TLB contents comprise four CP0 registers: We’ve just loaded **EntryLo0-1**, **EntryHi** was set up automatically by the TLB miss exception hardware, and **PageMask** is maintained by Linux/MIPS at a constant value meaning 4-KB pages.

- (11) **eret** takes us back to the instruction that suffered the TLB miss, and it should do better this time. Note that **eret** is also a CP0 hazard barrier, and it won’t allow the returned-to user instruction to be fetched until the **tlbwr** is complete.

14.4.9 *Care and Maintenance of the TLB*

The MIPS TLB is “just” a collection of translation entries. Entries get into it purely by the refill handler’s action. Most entries get overwritten in the same way, of course. But sometimes the kernel needs to change a page table entry, and then it’s important to invalidate any copy in the TLB.

For a single entry, we can look up the TLB entry using the virtual address + ASID of the original entry, doing a **tlbp**: If the matching entry is there, we can overwrite that page’s entry with an invalid one.

Sometimes, though, you need to do surgery on a larger scale. The ASID mechanism allows the TLB to hold entries for 256 different memory maps, but a Linux system will commonly run more than 256 processes between start-up and shutdown. So there are times when a whole batch of translations must be rescinded, because the ASID is going to be recycled for a new process and the old entries would be damagingly wrong.

There’s no elegant way of doing that. You just have to iterate through all the TLB entries (index values go in the **Index** register), read each entry out into the usual registers, check the ASID value retrieved into **EntryHi (ASID)**, and invalidate any that match the victim ASID value.

14.4.10 *Memory Translation and 64-Bit Pointers*

When the MIPS architecture was invented, 32-bit CPUs had been around for a while, and the largest programs’ data sets were already moving up toward 100 MB—the address space had only 6 bits or so to spare.¹⁸ There was therefore every reason to be reasonably careful with the 32-bit space and not to reduce it by profligate fragmentation; this is why application programs (running with user privilege) keep 31 bits’ worth of addressing for themselves.

When the MIPS III instruction set introduced 64-bit registers in 1991, it was leading the industry and, as we discussed in section 2.7, MIPS was probably four to six years ahead of real pressure on a 32-bit address boundary. The doubling of register size only had to yield a few bits of extra address space to

18. Historically, application program demand for memory space seems to have grown at about $\frac{3}{4}$ bit per year.

be remarkably future-proof; it's been more important to be cautious about the potentially exploding size of OS data structures than to make efficient use of all address space.

The limitations to the practical address space resulting from the basic 64-bit memory map are not going to be reached for a while; they theoretically permit the mapped user and other spaces to grow to 61 bits without any reorganization. But so far, a 40-bit user virtual space has been ample. Most other 64-bit Linux systems have 8-Kbyte pages, but that would be annoying for MIPS. The MIPS TLB can map either 4 Kbyte or 16 Kbyte in a single entry, but not 8 Kbyte, so 64-bit MIPS kernels use 4-Kbyte pages, with 16-Kbyte pages a desirable option in some brave future.

If you take a look back at Figure 14.4 and imagine a set of intermediate (PMD) tables between the PGD and PTEs, you can comfortably resolve a 40-bit virtual address in a three-level table. We'll leave details to those enthusiastic enough to read the source code.

MIPS Specific Issues in the Linux Kernel

Much of the Linux kernel is written in portable C, and a great deal of it is portable to a clean architecture like MIPS with no further trouble. We looked in the last chapter at the obvious machine-dependent code around exceptions and memory management. This chapter is looking at other places where MIPS-specific code is needed.

The first two sections deal with cases where most MIPS CPUs have traded off programming convenience for hardware simplicity: first, that MIPS caches often require software management and, second, that the MIPS CP0 (CPU control) operations sometimes require explicit care with pipeline effects. We'll take a quick look at what you need to know about MIPS for a multiprocessor (SMP) Linux system. And the last section is a glimpse at the use of heroic assembly code to speed up a heavily used kernel routine.

15.1 Explicit Cache Management

In the x86 CPUs, where Linux was born and grew up, the caches are mostly invisible, with hardware keeping everything just as if you were talking directly to memory.

Not so MIPS systems, where many MIPS cores have caches with no extra “coherence” hardware of any kind. Linux systems must deal with troubles in several areas.

15.1.1 *DMA Device Accesses*

DMA controllers write memory (leaving cache contents out-of-date) or read it (perhaps missing cached data not yet written back). On some systems—particularly x86 PCs—the DMA controllers find some way to tell the hardware cache controller about their transfers, and the cache controller automatically

invalidates or writes back cache contents as required to make the whole process transparent, just as though the CPU was reading and writing raw memory. Such a system is called “I/O-cache coherent” or more often just “I/O coherent.”

Few MIPS systems are I/O-cache coherent. In most cases, a DMA transfer will take place without any notification to the cache logic, and the device driver software must manage the caches to make sure that no stale data in cache or memory is used.

Linux has a DMA API that exports routines to device drivers that manage DMA data flow (many of the routines become null in an I/O coherent system). You can read about it in the documentation provided with the Linux kernel sources, which includes `Documentation/DMA-API.txt`. In fact, if you’re writing or porting a device driver, you should read that.

When a driver asks to allocate a buffer, it can choose:

- “*Consistent*” memory: Linux guarantees that “consistent” memory is I/O coherent, possibly at some cost to performance. On a MIPS CPU this is likely to be uncached, and the cost to performance is considerable. But consistent buffers are the best way to handle small memory-resident control structures for complex device controllers.
- *Using nonconsistent memory for buffers*: Since consistent memory will be uncached for many MIPS systems, it can lead to very poor performance to use it for large DMA buffers.

So for most regular DMA, the API offers calls with names like `dma_map_xx()`. They provide buffers suitable for DMA, but the buffers won’t be I/O coherent unless the system makes universal coherence cheap.

The kernel memory allocator makes sure the buffer is in a memory region that DMA can reach, segregates different buffers so they don’t share the same cache lines, and provides you with an address in a form usable by the DMA controller.

Since this is not coherent, there are calls that operate on the buffer and do the necessary cache invalidation or write-back operations before or after DMA: They are called `dma_sync_xx()`, and the API includes instructions on when and how to call these functions.

For genuinely coherent hardware, the “sync” functions are null.

The language of the API documentation is unfortunate here. There is a little-used extension to the API whose function names contain the word “noncoherent,” but you should *not* use it unless your system is really strange. A regular MIPS system, even though it is not I/O coherent, can and should work fine with drivers using the standard API.

This is all moderately straightforward by OS standards. But many driver developers are working on machines that manage this in hardware, where the “sync” functions are just stubs. If they forget to call the right sync function at

the right moment, their software will work: It will work until you port it to a MIPS machine requiring explicit cache management. So be cautious when taking driver code from elsewhere. The need to make porting more trouble-free is the most persuasive argument for adding some level of hardware cache management in future CPUs.

If you're interested in peeking into the Linux implementation to see how the cache synchronization functions are built from MIPS instructions, use section 4.6 as a reference.

15.1.2 *Writing Instructions for Later Execution*

A program that writes instructions for itself can leave the instructions in the D-cache but not in memory, or can leave stale data in the I-cache where the instructions ought to be.

This is *not* a kernel-specific problem: In fact, it's more likely to be met in applications such as the "just-in-time" translators used to speed up language interpreters. It's beyond the scope of this book to discuss how you might fix this portably, but any fix for MIPS will be built on the **synci** instruction. That's the ideal: **synci** was only defined in 2003 with the second revision of the MIPS32/64 specifications, and many CPUs without the instruction are still in use. On such CPUs there must be a special system call to do the necessary D-cache write-back and I-cache invalidation using privileged **cache** instructions. You can find out the details of how **synci** works in section 8.5.11.

15.1.3 *Cache/Memory Mapping Problems*

Virtual caches (real ones with virtual index and tagging) seem a wonderful free ride, since the whole cache search process can start earlier and run in parallel with page-based address translation.

A plain virtual cache must be emptied out whenever there's a memory map change, which is intolerable unless the cache is very small. But if you use the ASID to extend the virtual address, entries from different processes are disambiguated.

OS programmers know why virtual caches are a bad idea: The trouble with virtual caches is that the data in the cache can survive a change to the page tables. In general, the virtual cache ought to be checked after any mapping is rescinded. That's costly, so OS engineers try to minimize updates, miss some corner case, and end up with bugs.

In a heroic attempt to make Linux work successfully even with virtual caches, the kernel provides a set of rules and function calls that should be provided as part of the port to an architecture with troublesome caches. They're the functions with names starting `flush_cache_xxx()` described in the kernel documentation `Documentation/cachetlb.txt`. I don't like the word "flush" to describe cache operations: It's been used to mean too many things. So note

carefully that in the Linux kernel a “cache flush” is something you do to get rid of cache entries that relate to obsolete memory mappings. In a system where all caches are physically indexed and tagged, none of these calls needs to do anything.

Fortunately, virtual D-caches are rare on MIPS CPUs. Some recent CPUs have virtual I-caches: Implement the “flush” functions as described in the documentation and you should be all right. But L1 caches with physical tags but virtual indexes are common on MIPS CPUs. They solve the problems described in this section, but they lead to a different problem called a “cache alias”—read on.

15.1.4 *Cache Aliases*

We’re now getting to something more pernicious. MIPS CPU designers were among the first to realize that the benefits of using the virtual address to index their cache could be combined with the benefit of using the physical address to tag it. This can lead to cache aliases—for more explanation, see section 4.12.

The R4000 CPU was the first to use virtually indexed caches. As originally conceived, the CPU always came with an L2 cache (the cache memory was off-chip, but the L2 controller is included with the CPU), and it used the L2 cache to detect L1-cache aliases. If you loaded an alias to a line that was already present in the L1, the CPU generated an exception, which could be used to clean up.

But the temptation to produce a smaller, cheaper R4000 variant by omitting the L2 cache memory chips and the pins that wired them up proved too strong. Contemporary UNIX systems had a fairly stylized way of using virtual memory, which meant that you could control memory allocation to avoid ever loading an alias. In retrospect we can see that generating aliases is a bug, and the careful memory management was a workaround for it. But it worked, and people forgot, and it became a feature.

There are basically two ways to deal with cache aliases.

The first is to try to ensure that whenever a page is shared, all the virtual references to it have the same “page color” (that means that the references may be different, but the difference between them is a multiple of the cache set size). Any data visible twice in same-color pages will be stored at the same cache index and handled correctly. It’s possible to ensure that all user-space mappings of a page are of the same color—more on this in section 10.3.4.

But unlike the old BSD systems, Linux provides features where correct page coloring is impossible. Those will be cases where you have both a user-space and kernel mapping to the same page (in many cases, on a MIPS kernel, the kernel “mapping” will be a `kseg0` address). So the MIPS port has special code to detect those cases and clean out any old alias mappings.

The Cache/TLB documentation (that’s `Documentation/cachetlb.txt`, as mentioned in the section above) makes a heroic attempt to deal with cache aliases as “just another symptom” of virtual caches in general. It provides some notes on how to configure the kernel to do what it can on page coloring and how to handle kernel/user-space aliases.

The work of fixing around cache aliases is never finished. As the years go past, Linux is expected to offer more exciting facilities. Programmers struggle to fix the places where aliases can break legitimate code, but as new facilities are added to the OS, it breaks again. I think the CPU designers are now recognizing that relying on software workarounds for aliases creates maintenance problems: Increasingly, new CPUs have at least the D-cache alias-free.

MIPS was a very influential architecture, so some of its competitors faithfully copied its mistakes;¹ as a result, cache aliases are found in other architectures too.

15.2 CP0 Pipeline Hazards

As we've seen in some of the examples above, CP0 operations which are dependent on some CP0 register value need to be done carefully: The pipeline is not always hidden for these OS-only operations. The OS is where all the CP0 operations happen, of course.

On a modern CPU (compliant to Release 2 of the MIPS32/MIPS64 specifications), hazard barrier instructions (`eret`, `jalr.hb`, and `jr.hb`) are available, as described in section 3.4.

On older CPUs, only entry into exception and the `eret` instruction are guaranteed to act as CP0 hazard barriers. So where you're writing a code sequence where something depends on the correct completion of a CP0 operation, you may need to pad the code with a calculated number of no-op instructions (your CPU may like you to use the `ssnop` rather than a plain old `nop`—read the manual).

The CPU manual for an older CPU should describe the (CPU-specific) maximum duration of any hazard and let you calculate how many instruction times might pass before you are safe.

15.3 Multiprocessor Systems and Coherent Caches

In the 1990s, MIPS CPUs were used by SGI and others to build cache-coherent multiprocessor systems. SGI were not exactly pioneers in this area, but the MIPS R4000 was one of the first microprocessors designed from the outset to fit into such a system, and SGI's large MIPS multiprocessor server/supercomputers were very successful products.

However, little about this technology is really specific to MIPS, so we'll be brief.

The kind of multiprocessor system we're describing here is one in which all the CPUs share the same memory and, at least for the main read/write memory,

1. If imitation is the sincerest form of flattery, imitation of an architecture's mistakes must be tantamount to hero-worship.

the same physical address space—that’s an “SMP” system (for Symmetric MultiProcessing—it’s “symmetric” because all CPUs have the same relationship with memory and work as peers).

The preferred OS structure for SMP systems is one where the CPUs cooperate to run the same Linux kernel. Such a Linux kernel is an explicitly parallel program, adapted to be executed by multiple CPUs simultaneously. The SMP version of the scheduler finds the best next job for any CPU that calls it, effectively sharing out threads between the pool of available CPUs. That’s a big upgrade for an OS kernel that started life on x86 desktops, but the problems posed by multiple CPUs running in parallel are similar to those posed by multitasking a single CPU.

Back to the hardware. That memory may be built as one giant lump shared by all CPUs, or distributed so that a CPU is “closer” to some memory regions than others. But all multiprocessor systems have the problem that the shared memory has lower performance than a private one: Its total bandwidth is divided between the CPUs, but—usually more important—the logic that regulates sharing puts delays into the connection and increases memory latency. Performance would be appalling without large, fast CPU caches.

The trouble with the caches is that while the CPUs execute independently a lot of the time, they rely on shared memory to coordinate and communicate with each other. Ideally, the CPU’s individual caches should be invisible: Every load and store should have exactly the same outcome as if the CPUs were directly sharing memory (but faster)—such caches are called “coherent.” But with simple caches (as featured on many MIPS CPUs), that doesn’t happen—CPUs with a cached copy of data won’t see any change in memory made by some other CPU, and CPUs may update their cached copy of data without writing back to memory and giving other CPUs a chance to see it.

What you need is a cache that keeps a copy of memory data in the usual way, but which automatically recognizes when some other CPU wants to access the same data, and deals with the situation (most of the time, by quietly invalidating the local copy). Since relatively few references are to shared code, this can be very efficient.

It took a while to figure out how to do it. But the system that emerged is something like this: Memory sharing is managed in units that correspond to cache lines (typically 32 bytes). Each cache line may be copied in any number of the caches so long as they’re reading the data; but a cache line that one CPU is writing is the exclusive property of one cache.

In turn, that’s implemented within each CPU cache by maintaining a well-defined state for each resident cache line. The bus protocol is organized by specifying a state machine for each cached line, with state transitions caused by local CPU reads and writes, and by messages sent between the caches. Coherent systems can be classified by enumerating the permitted states, leading to terms like MESI, MOSI, and MOESI (from states called modified, owned, exclusive, shared, and invalid). Generally, simpler protocols lead to more expensive sharing.

The first practical systems used a single broadcast bus connecting all caches and main memory, which had the advantage that all participants could see all transactions (simultaneously, and all in the same order—which turns out to make things much simpler). Much of the intercache communication could be achieved by letting the caches “snoop” on cache-memory refill and write-back operations. That’s why coherent caches are sometimes called “snoopy caches.”

Single-bus systems don’t scale well either to more CPUs or to very high frequencies. Modern systems use more complex networks of cache-to-cache and cache-to-memory connections; that means you can’t rely on snooping and have no one place in the system where you can figure out in which order different requests happened. Much more complicated . . .

At the time of writing in 2006, this technology is migrating to the smallest systems, where multiple processors on a system-on-chip (SoC) share memory. Chip-level multiprocessing (CMP) is more attractive than you might think, because it increases compute power without very high clock rates. The only known practical design and test methods for SoC don’t deliver very high frequencies—and, in any case, a 3-GHz processor taking 70 W of electric power and dissipating it as heat is hardly practical in a consumer device. It’s hard to build anything on an SoC that behaves like a single snoopable bus. In contemporary (2006) SoCs the state of the art is a small cluster of CPUs, whose mutual coherency is maintained by a piece of logic that couples those CPUs quite tightly together, implementing a single ordering point. Future SoCs may use more loosely coupled networks of CPUs, which will need more complicated cache coherency protocols.

It was difficult to get cache-coherent SMP hardware to work, and then to work efficiently. Those struggles produced a fair amount of folklore about how to do things well, and it’s worth getting an introduction to it before you work with a robust multi-CPU application like the Linux kernel. Here are some headline issues:

- *Selecting pages that don’t need coherent management:* Cache coherency looks after itself, but performance will be better if the OS can tell the hardware about pages that are *not* shared and don’t need coherence management. Read-only pages need no management—they come from memory, and caches may make as many copies of them as required. No page is entirely read-only, of course: Data had to be written sometime and somewhere. But in Linux, instruction pages come close enough to be worth special handling. The kernel’s own instructions were safely written before anything was cached, so they’re OK. Application instruction pages are generally read in from a file, and at that point all caches must have any copies invalidated; but after that they’re OK too.

Quite a lot of data is perthread. Process stack is always perthread, and for single-threaded applications, so is all the user-space data. Unfortunately, while there’s only one CPU interested in a single-threaded process at any point in time, an application can migrate from CPU to CPU

when it's rescheduled (and if you want a multiprocessor system to work well, you need few constraints on migration). It's a matter of judgment whether the benefit of more relaxed coherency management on known single-threaded data pages is worthwhile, even though you'll need to do some error-prone, manual cache-cleaning when a thread migrates.

- *Atomicity, critical regions, and multiprocessor locks:* SMP systems in which multiple CPUs run the same code and negotiate shared data structures don't work without some attention to making data reads and updates atomic in the sense found in section 14.2.

The MIPS architecture's **ll/sc** (load-linked/store-conditional) instructions are its primitive building blocks for atomicity and locks and were designed to scale well to large multiprocessors.

- *Uncertain memory ordering and memory barriers:* In a cache-coherent multiprocessor, when any CPU writes a memory location, another CPU reading that location will (sometime a bit later) see it updated. But once you get away from the systems built round a single common bus, it's hard to make sure even that reads and writes stay in the same relative order. Most of the time that doesn't matter—the threads running on the different CPUs have to cope with the fact that they don't know much about how other threads are progressing. But software using shared memory to communicate between threads is always vulnerable.

This problem was discussed in section 10.4, which describes the role of the MIPS **sync** instruction as a memory barrier.

sync has another life. In many CPUs it has additional or stronger CPU-family-specific semantics such as “all data has cleared the system interface,” or “wait until all my load and store queues are empty.” Read your CPU manual to find out.

15.4 Demon Tweaks for a Critical Routine

It's worth looking quickly at some optimized Linux code to get a feel for how far it's worth taking architecture- or CPU-specific tuning.

The `clear_page()` routine is heavily used in the Linux kernel. Pages filled with zero are directly required for the “uninitialized” portion of application data spaces, and it's also used as a way of cleaning ex-application pages to avoid accidental leakage of data from one application to another (which would violate the security policy).

This implementation of `clear_page()` uses several ideas. It unrolls loops—each iteration of the loop clears a whole cache line (in this case, that's 32 bytes, eight words).

It also uses the MIPS-specific **pref** operation to avoid waiting on cache traffic. If prefetch really does need to read data from memory, that's going to

take a long time compared with the CPU execution rate: So we'll prefetch a long way ahead. In this case, "a long way" is 512 bytes, 16 cache lines.

But if the CPU can understand it, the prefetch hint is the specialized "prefetch for store" version, which makes a cache entry for a line without actually reading any data in (it relies on the programmer's promise to overwrite the whole cache line)—see section 8.5.8.

Here's the function with added commentary. Bear in mind that `PAGE_SIZE` is usually 4,096.

```
#define PREF_AHEAD 512

clear_page:
    # the first loop (main_loop) stops short so as not to prefetch off
    # end of page
    addiu a2, a0, PAGE_SIZE - PREF_AHEAD

main_loop:
    # the prefetch. Bring in cache line, but with luck we won't read
    # memory. But if all this CPU offers is a simple prefetch, that
    # should work too.
    pref Pref_PrepereForStore, PREF_AHEAD(a0)

    # now we're going to do eight stores
    sw    zero, 0(a0)
    sw    zero, 4(a0)
    sw    zero, 8(a0)
    sw    zero, 12(a0)
    addiu a0, a0, 32    # some CPUs choke on too many writes at
                        # full-speed, so increment the loop pointer
                        # in the middle to give it a break.
    sw    zero, -16(a0)
    sw    zero, -12(a0)
    sw    zero, -8(a0)
    bne   a2, a0, main_loop
    sw    zero, -4(a0)    # last store in the branch delay slot

    # the second (end) loop does the rest, and has no prefetch which
    # would overrun.

    addiu a2, a0, PREF_AHEAD
end_loop:
    sw    zero, 0(a0)
    sw    zero, 4(a0)
    sw    zero, 8(a0)
```

```
sw    zero, 12(a0)
addiu a0, a0, 32
sw    zero, -16(a0)
sw    zero, -12(a0)
sw    zero, -8(a0)
bne   a2, a0, end_loop
sw    zero, -4(a0)

jr    ra
nop
```

Linux Application Code, PIC, and Libraries

A GNU/Linux thread running an application is reasonably called a *program*. Most such applications are bolted together out of previously independent parts at load time, a process that depends on a rather radical sort of position-independent code (the PIC of our title). Before we start on it, note that nothing in the Linux kernel obliges you to compile and build userland program binaries in any particular way: The kernel makes very few assumptions about what is going on in user-space.

GNU/Linux programs typically include large quantities of independently compiled library code, which is shared between multiple processes (running the same or different programs). There are great advantages in having the application “main program” link to the libraries *dynamically* when the program is run, rather than building library code into an independent monolithic binary when you compile it: To take just two of many arguments, dynamic linking greatly reduces the size of program binaries both on disk and in memory, and it means you can fix bugs in libraries just by supplying a new library.

Programs and libraries on disk reside in *object files*, which contain the precompiled binary, initialized data, and some management information (including symbol tables and relocation records) that will be required to join up the program and its libraries.

There’s a sketch of the virtual memory image of a typical application in Figure 16.1.

The program consists of multiple quasi-independent *link units*¹—one for the main program and one for each library it uses.

1. There’s no consensus on what to call one of the individual binaries making up a dynamically linked program. What we’ve called a “link unit” has been called a dynamic shared object (DSO), an object, or a module. But that’s nothing to do with a C++ object; and the word “module” is already used to mean what is built in one go by the compiler. So in this section I’m going to use the neutral phrase “link unit” to remind you that such a binary is the final thing produced at link time. The library link unit files are usually marked with a “.so” suffix.

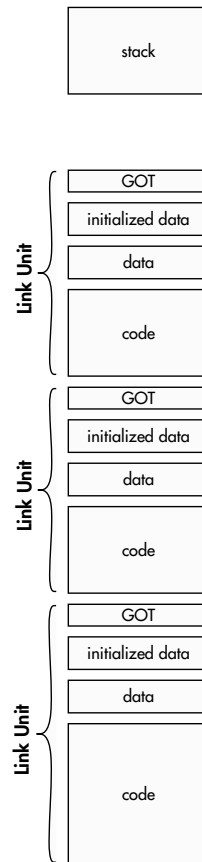


FIGURE 16.1 Linux program memory image.

Each link unit is loaded into a number of chunks of memory (segments), many of which have initial data provided by chunks of the object file (sections). But some chunks of memory—notably, those that hold the stack—are created by the loader.

In Linux, program code is shared (page by page) between all active programs that incorporate the same link unit. Among the sharers may be programs with quite different memory maps; the memory image of the code itself must work regardless of where the code is located in virtual memory. That is, the code must be *position-independent code* or PIC. It's not uncommon to describe code as “position-independent” just because the branch and call instructions are all PC-relative: But Linux PIC code is radically position-independent, because its data may be anywhere, too. All code and data references must go on working regardless of where the link unit's segments end up in memory.

Consequently, when one link unit's code calls a function or refers to a piece of data in another link unit, the function or data address (today's actual value) has to be computed by the running program. That's where the GOT (global offset table) comes in.

Each piece of code has a GOT, defined in the object file. By the time the dynamic loader has done its work, the GOT contains a pointer² to each external function or data structure referred to by any part of the link unit's code. The object file of the link unit has no knowledge about these external addresses: It has a symbol table and tracks GOT entries by a mapping between symbols and indexes. So you'll quite often read that the GOT has an entry for each "external symbol"—which is true, but can be confusing when you're staring at a loaded binary. The *dynamic loader* reads the object files of both link units, allocates both link units some memory space, then matches symbols to find the right external addresses to put in the GOT.

We just said that the kernel didn't care how applications were built, and in fact the dynamic loader is not part of the kernel. It is itself a Linux link unit, and the main program refers to it pretty much like any other shared library. PIC library-sharing applications start up by running the dynamic loader, which then patches together the rest of the VM image before calling back to the application's `main()`.

16.1 How Link Units Get into a Program

There are three ways to glue a new link unit into a program:

- *Brought in when the program is loaded:* Each link unit, starting with the main program (the executable), includes a list of other link units that are to be loaded into virtual memory before `main()` is called. That doesn't mean the programs are actually read into memory, or even that virtual memory mappings set up: This is still a virtual memory OS, and much of the program code or data won't appear until it's referenced.
- *Loaded on first reference to subroutine:* The dynamic loader defers loading some link units ("lazy binding"), provided the link unit is a pure library whose interface consists entirely of function calls and with no externally visible data.

The loader does that by setting GOT entries relating to a lazily bound link unit so they call back to a function in the dynamic loader—a function whose job is to load the missing link unit, fix up the GOT, and then return control (carefully) to the newly loaded link unit's function.

2. That's potentially an oversimplification. You might put anything in the GOT, so long as it permits the link unit's code to call external functions or refer to data.

Link units that are candidates for lazy binding are identified in the object file.

- *Loaded explicitly:* The `dlopen()` routine finds a library link unit and glues it into the program, building a data structure that includes pointers to the functions the new program will export.

Such a link unit is working very much like a C++ object (the function pointers are reminiscent of C++ methods)—that’s why link units are sometimes called dynamic shared objects.

Most of the time, it doesn’t matter in which order you load the various link units: It will change the memory layout, but each link unit will work wherever it ends up.

But it’s possible to build programs where the same symbol is provided by different link units.³ When that happens, the link unit that loaded first “wins” and provides the variable.

16.2 Global Offset Table (GOT) Organization

The program build process (I refer to compilation and build-time linking) defines at least one GOT for each link unit, though it’s perfectly legal for there to be more than one.⁴ Each function can find its GOT, because it’s at a known offset from the function entry point (remember, a link unit really is brought into memory as one rigid piece, so internal offsets are as the compiler made them).

Each entry in the GOT is—effectively—an absolute pointer to a piece of data or function entry point⁵ defined (implicitly or explicitly) as external by any function in the link unit using this particular GOT. GOT entries depend on the layout of link units within the program’s address space and are computed by the dynamic loader as link units are loaded.

Because the binary image of the code (and read-only data) of a link unit is really shared in an OS like Linux, we can’t do any address-map dependent fixups on the code when loading it. But GOTs are part of the data, and each program has its own copy. The pointers in the GOT may be different for each program that uses the shared link unit.

-
3. Sometimes this might just be an error causing the load to fail, but some symbols are specifically marked as permitting multiple definitions (“first definition loaded wins”) with some safety restrictions.
 4. The use of multiple GOTs—one per C module, usually—is the preferred way to deal with large link units that overflow the 128-KB GOT size (if the GOT grows bigger than this, compilers and assemblers have to generate longer code sequences to retrieve pointers from the GOT).
 5. In some limited circumstances the compiler might “know” the offset between two data items: In that case, it can make do with one GOT entry for the two items.

GOT accesses are frequent, so in PIC code the compiler/linker synthesizes code that keeps the **gp** register pointing to each function's GOT. The register **gp** is a good choice for the table base, since this use is analogous to the GP-relative data idea that led to **gp** being reserved. Old-fashioned GP-relative data is inherently incompatible with PIC, so there's no great conflict.

The GOT pointer must be computed and loaded into **gp** at the beginning of each function. That's slightly tricky—this is PIC code, so we don't know at compile time where the GOT will be. But the GOT is part of the same link unit, so the offset from the function's own entry point to the GOT *is* known at compile time.

It makes this computation easier if a remote function “knows” that it's been called by address, since it then has a register that already contains the address of the function, and that saves effort in computing the pointer to its own GOT. The Linux/MIPS convention is that all PIC calls use **t9** to hold the address of the called function.

Then a function prologue can compute GP in a single step (not necessarily a single instruction, though):

```
entrypoint:
    addu gp, t9, GOT - entrypoint
```

Code using a GOT entry to access data in a different link unit knows *only* the slot of the GOT where it should find the pointer: The GOT slots are allocated when the link unit is linked together, but the pointer stored in the GOT is not fixed until the other link unit is loaded at program load time. So the compiler must generate code that does a double-indirect, loading a pointer to the external data:

```
load:
    lw t1, gp(MYSYMBOL_INDEX)
    lw t1, (t1)

store:
    lw t1, gp(MYSYMBOL_INDEX)
    sw t2, (t1)

call:
    lw t9, gp(MYFUNCTION_INDEX)
    jalr t9
```

PIC code runs slower than code using fixed addresses, but the benefits of shared libraries are important enough to make this a price well worth paying. We mentioned a couple at the beginning of the chapter, but only to get you

motivated. The advantages are typical of those resulting from any form of late binding. A more full list might include:

- *Smaller binaries:* As above, your system binaries don't need to each carry a copy of the C library. Large and complex libraries often dwarf simple applications.
- *Fix a library and you've fixed all dependent applications:* Some of the most complex code in the system is in the libraries, so it's good to have the same update cycle as for individual binaries, even though individual systems may have different sets of applications installed.
- *Creates new options in providing legacy interfaces:* When you want to obsolete some library or kernel function, you can provide a compatibility library to reimplement it using the new features.
- *Library layer insulates applications from kernels:* This allows applications and kernels to evolve separately.

In the end, dynamic library code is a somewhat higher-level development environment than is offered by statically linked C/C++.

There is a downside, apart from the execution overhead of doing all external calls and data references through the GOT. With dynamic loading, applications and libraries are entangled in complex webs of dependencies: It makes installing new software tricky, and removing existing software from a system even trickier. A robust library search and version-labeling system is essential. Earlier Windows systems notoriously suffered "DLL hell," where different applications had incompatible requirements of the same library.

Modern systems (Windows and Linux alike) seem to have constructed dependency database systems that keep track of dependencies robustly. Dynamically linked code is here to stay, and the memory economy means it is important to any embedded system with a significant amount of application software.



MIPS Multithreading

Multithreading hardware was 2004–2005’s “Next Big Thing” in computers. Not so in 2006: Virtualization looks like this year’s NBT, a crown it last held around 1975–1980. The crest of the multithreading hype wave has gone past, and multithreading still looks pretty important.

So let’s attempt to define what it is, and then why it’s useful. We’ll use “multithreading” without the hyphen, and use the abbreviation “MT” to refer to the MIPS32/64 architecture extension (the MT ASE).

A.1 What Is Multithreading?

A thread is a sequence of instructions executed in the order the programmer intended.

That seems a bit simple, really, though it leaves a “thread” as a thing that is probably pretty tangled (there are muddled paths following the call tree of subroutines and loops within loops). But when a conventional CPU takes an interrupt, that isn’t the same thread (at least, not until and unless the interrupt handler returns). And what operating systems variously call “processes,” “tasks,” or “jobs” are all separate threads.

You only need to define a thread when you intend to have more than one (otherwise it’s just the program the CPU is running). It’s easy to see why you might want more than one thread on a timeshared computer where different users were impatient to get their programs run.

Sometimes you want a computer to run a single program, but one that must reconcile competing needs to respond. Some problems are inherently concurrent. You can (in theory) write an explicit program using some kind of multiway select call to wait for one of a number of events, figure out the event, and call the appropriate subroutine. Or maybe you can invent a concurrent programming language.

But explicitly concurrent programs are very hard to write, and no concurrent language ever took off. In the 1970s, research into practical ways to design, write, and test programs that dealt with concurrent events arrived at a conclusion. Researchers homed in on the idea of using multiple simple threads that influence each other only through simple, stylized interfaces.

Furious internecine wars were fought over whether those interfaces should be based on message passing or something simpler like semaphores: The wars have long since been abandoned, and everybody won. Modern OSs contain a variety of communication mechanisms, though different programming communities seem to favor different subsets.

The word *thread* emerged as a term distinct from “process” in order to free the thread from other baggage: UNIX processes each had their own address space, but threads were things that might share an address space. The POSIX standard 1003.1 (known as “PThreads”) proposed a rich cross-OS programming interface for threads that explicitly share code and data, and PThreads has been substantially accepted by a wide range of modern operating systems.

Which Resources Do You Need to Run Two Threads at Once?

Two complete CPUs will do it, of course. But that seems a lot: What’s the minimum you need?

You can get a good clue about this by looking at the information kept about each thread by a multithreading OS (on a conventional CPU). Look at a thread when it’s not running, and you’ll find a PC (the address the thread will execute from when it next starts up) and saved values of all the programmer-visible registers. For user-privilege programs on MIPS, that’s the general-purpose registers and the `hi/lo` accumulator of the multiplier. The OS will need to keep some kind of identifier as to which thread is running, as well as a flag to say whether it’s currently holding kernel privilege (as it will be when executing a system call). A MIPS OS maintaining multiple address spaces will need the ASID value too, because it’s used by the hardware when translating addresses.

That’s about it: PC, general-purpose registers, thread ID, kernel mode, ASID. An OS calls this the *thread context*, and at a minimum each thread must have its own copy of those registers and fields.

There’s another angle you can take. If you want someone to buy your multithreading hardware, it helps if your hardware can run existing software. Dual-processor x86 PC hardware (particularly the closely related x86 small-server hardware) has been around for a few years, and Microsoft Windows, Linux, and other UNIX OSs already know how to exploit that hardware. So when Intel introduced multithreading in its “Hyperthreading” CPUs, it augmented the hardware so that a system with one multithreading CPU would run software built for a dual-CPU system, unchanged. To do the same thing for MIPS, you’d need to provide separate copies of every register to each “virtual processor.”

That’s two different approaches to multithreading. Which should MIPS adopt? Before we come to that, let’s take a belated look at why it’s worth doing at all.

A.2 Why Is MT Useful?

Conventional CPUs in 2005–2006 are hitting the “memory wall.” The time taken to run an instruction has gone down much faster than the time taken to access memory. How much faster? Intel’s 32-bit x86 CPUs have gone from running at 16 MHz in 1985 to about 3 GHz in 2005—200 times faster in 20 years. During that time memory access time has dropped too, but by rather less than a factor of four, from about 180 ns in 1985 to 50 ns now.¹

Roughly speaking, that means a memory read that stopped a CPU for four cycles in 1985 now stops it for 150 cycles. Vast improvements in caching and clever CPU designs that run as far as possible ahead of the returning data have covered up the problem, but the scope for further improvement is limited. It’s normal for a CPU to spend well over half its time idle, waiting for data from memory.

Embedded systems don’t run at anything like 3 GHz, but they have simpler caches and CPU designs. Contemporary embedded systems, too, tend to have CPUs that are stalled out 50 percentage of the time.

There’s another reason why multithreading hardware might be useful for embedded systems. If you have some very demanding I/O service requirement, a thread dedicated to that requirement can respond instantly. You can “park” the thread reading some value from the I/O system, and as soon as the data is ready it will leap back to life. There’s no interrupt overhead and—more importantly—potentially no OS overhead. Who says the I/O thread must run the standard OS code?

So multithreading might be a good thing. How do we do it? And is it a good bargain? As usual for embedded systems the biggest cost factors are silicon area and power consumption.

A.3 How to Do Multithreading for MIPS

RISC CPUs like to leave the software in charge, rather than moving policy decisions into hardware. RISC principles favor flexible and general-purpose solutions. That motivates a number of features of MIPS MT:

- *Expose thread registers to each other:* This gives the OS total control over multithreading. `mfttr` and `mttr` are privileged instructions (“move

1. Memory *bandwidth* has risen at a much higher rate, but “Money can buy bandwidth. Latency requires bribing God.”—in this case, technology stands in for money.

from/to thread register”) that allow OS software to get at the registers of a different TC. They’re the basis of all cross-thread initialization and maintenance.

- *Start up single-threaded:* This respects the principle of least surprise for bootstrap software. OS or bootstrap software can wake more threads when it feels ready.
- *Provide both sorts of multithreading:* Both minimal-overhead thread engines and complete “virtual CPUs” can be mixed and matched. The basic MT feature is the minimal thread engine, known as a *TC* (inspired by “thread context”). If we stopped there, everything in the CPU beyond the immediate thread context would be shared.

Instead, we permit MT CPUs to replicate everything required to make a MIPS32/64-compliant CPU. One of these CPU-like sets of registers and other resources is called a *VPE*: One or more TCs affiliate to the VPE and share its registers and resources, and the TC(s) in the VPE make a “virtual processing element,” which leads to the acronym. A CPU can implement more than one VPE to produce something that looks (to software) like multiple CPUs.

MIPS Technologies’ first MT product, the 34-K CPU, can have up to five TCs distributed at will between two VPEs.

Many things can be shared between the VPEs while leaving them software-equivalent to a MIPS32 CPU, of course: This is not a real dual CPU, it only looks like one. The caches, main pipeline, control logic, arithmetic/logic units, and system interface are all shared. The TLB can either be hard-partitioned between the VPEs, or shared (the shared-TLB version involves some localized changes to the OS).

Each instruction being run by a machine has a TC number, which selects its particular context. Whenever the instruction accesses some state—reads or writes a general-purpose register, for example—it uses its TC number to extend the register-number field, which is already defined inside the instruction. An instruction sees a different set of registers depending on the TC number: It’s very simple, and it just works.

MIPS MT is not quite that simple, because of the TC/VPE trick mentioned above. So this instruction might be for TC #5 (it uses general-purpose registers from the fifth bank) or VPE #1 (it gets CP0 registers from the first bank). Again, this should just work. What’s more complicated, of course, is to get those CPU resources working, which can’t simply be reduced to registers. But that’s not architecture, it’s implementation; you’ll have to read CPU manuals for that stuff.

The MT specification doesn’t require it, but effective MT CPUs need to be able to change threads very quickly, or time lost in thread changes will eat away at the potential throughput gain. Other things being equal, it will usually be best to mix threads on as fine grain as is possible: In a single-issue CPU, that means

issuing an instruction from a different thread in each clock period so long as more than one thread is ready.

New CP0 Registers for MT

They come in three groups: those provided for each TC, those provided for each VPE, and a couple that exist per-CPU. The latter are registers that provide an inventory of the CPU's resources (how many TCs? how many VPEs? and so on) and allow you to share them out; they are not described further here.

The per-TC registers include:

- **TCHalt**: A 1-bit register, write 1 to halt the target. With a target TC halted, its state is stable and it's safe to write its registers.
- **TCRestart**: The thread "PC"—when the thread is halted, this is the address of the first instruction it will run when it next runs. When not halted, it may not make much sense.
- **TCStatus**: Per-TC "legacy" fields—kernel/user status, ASID, and instruction set options flags (such as the one that enables floating point). There's also a flag to indicate that the instruction the thread stopped before is in a branch delay slot (in which case **TCRestart** points at the preceding branch).
- **TCBind**: Contains the ID of the VPE to which this TC is affiliated (writable in a 34-K family CPU), and a read-only ID for this TC.
- **TCContext**: A pure software read/write register, typically used to store an OS-specific thread ID.

The per-VPE register **VPEControl** is for control fields you might reasonably change during the running of an OS, while **VPEConf0–1** have configuration fields you'd most likely set once and then leave alone.

Exceptions and Interrupts

An exception in a single-threaded MIPS architecture CPU is usually quite disruptive in the pipeline and is commonly implemented by discarding a lot of execution state (pipelines get flushed and instructions discarded). An exception on a MIPS MT machine happens within a thread context—and other threads (at least those on separate VPEs) expect to continue undisturbed. So you'd expect there to be some difficulties when we redefine exceptions on a multithreading machine.

Bear in mind that an OS is a program (a set of threads, in fact). It's likely not to matter which TC happens to execute some part of it. As we observed at the beginning of this appendix, each exception handler is a thread in its own

right. So to run an exception handler, we need to borrow a TC to interrupt its thread and run this one instead.

There are two types of exceptions. Interrupts are asynchronous—they happen for reasons unconnected with any particular instruction. But most other exceptions are synchronous—they’re associated with a particular instruction. That’s what we’ll look at first.

Synchronous exception handlers are run by the TC whose instruction caused the exception. The TC immediately ceases work on its normal thread and starts fetching instructions from the appropriate exception handler.

The MIPS MT ASE requires that once a TC enters exception state, all the other TCs within the same VPE are suspended. None of the other TC’s instructions may be executed until the exception handler leaves exception mode: that is, until **SR (EXL)** is cleared either by the **eret** at the end of the exception handler or as a result of the exception handler branching off to some less restrictively coded part of the OS. It’s already good practice for exception handlers to minimize the amount of time spent in exception mode, and most OS code does that.

But if your application needs to maximize concurrency, you should consider minimizing exceptions—you may be able to use a thread blocked on an ITC access or **yield** condition instead. And, of course, arrange that exception handlers (as soon as they can) save the state necessary that they can drop back out of exception mode.

MIPS MT and Interrupts

In the MIPS architecture, interrupt management is by CP0 registers (in particular, **Cause** and **SR**). Those registers are replicated per-VPE, not per-TC, so interrupt masking and steering are managed per-VPE. Even the interrupt signal wiring into the core is per-VPE.

Each interrupt input may be connected to just one VPE or to all of them, so which VPE receives an interrupt comes down to how the system is wired up, but possibly also to the software configuration. If you connect and unmask an interrupt on multiple VPEs, any number of them may take the interrupt exception—you probably don’t want that to happen, so either don’t connect or don’t enable some of them.

The interrupt exception may be taken by any available TC associated with the VPE.

The MIPS architecture already provides multiple ways to refuse an interrupt exception. An interrupt to any thread from this VPE can be prevented by exception mode, a global interrupt-enable flag (which may be zero), and per-interrupt mask bits: that’s **SR (EXL)**, **SR (IE)**, and **SR (IM)**—and that list is not exhaustive.

The MIPS MT specification adds yet another reason not to take an interrupt. You can now set a new per-TC CP0 register field **TCStatus (IXMT)** (interrupt exempt), which will prevent the particular TC from being used for an interrupt exception.

Thread Priority Hints

Some application developers express interest in being able to steer the provision of CPU bandwidth to one thread rather than another. The MIPS MT specification provides for this, though it doesn't say how it should be done. A scheduling policy manager (often PM) is a piece of logic outside the CPU, customizable for a particular application, that generates a 2-bit priority for each TC. Inside the core, a runnable higher-priority TC will always be preferred over a runnable lower-priority one.

It's too early to comment on how successful developers will be at using this scheme.

User-Privilege Dynamic Thread Creation—the Fork Instruction

An interesting but commercially unexplored application of multithreading hardware is to provide another way of discovering and exploiting parallelism in sequential algorithms. For example, a loop might be optimized by having two threads running its odd- and even-numbered iterations, respectively.

If that's to be done under the aegis of a protected OS, this requires a very efficient mechanism for firing off a new thread on demand. MIPS MT's **fork** instruction provides such a mechanism.

So **fork d, s, t** is a user-privileged instruction. If all is well, it starts a thread on a ready-to-fork TC, which starts execution at the instruction pointed to by **s**. The child thread's **d** receives the value from the parent's **t**.

The OS enables this by maintaining a pool of ready-to-fork but otherwise idle TCs (they're distinguished by a flag bit **TCStatus (DA)**). After the new thread has done its work in the application, it terminates and returns the TC to the pool using a **yield \$0** instruction.²

fork is a forward-looking feature: At the time of writing, no OS yet supports such a pool of TCs. It's there so that the MIPS MT architecture can be well placed as multithreading ideas spread.

A.4 MT in Action

It's early days yet, but what are people doing with multithreading and, particularly, with MIPS MT?

Broadly speaking, there are two different kinds of systems being explored and/or developed. One seeks to use modified standard SMP OS code (perhaps quite lightly modified) to provide something well suited to provide multithreading efficiency to a relatively conventional multiple-thread workload; the other seeks to build “underneath” the OS to provide a low-level, ultra-responsive “real-time” environment for novel applications.

2. **yield** fed with a nonzero value from its source register operand has a quite distinct application.

SMP Linux

It's possible to port an SMP Linux to an MT CPU with two or more TCs in the role of the SMP system's CPUs. If you've got an application that already uses the Linux implementation of the PThreads threading API (it's called *NPTL*), then you can just drop that application onto such a Linux and have it exploit multithreading.

SMP systems need locks and semaphores that work between the CPUs, and for MIPS systems, those are built using the load-linked/store-conditional instruction pair (**ll/sc**, described in section 8.5.2). So this wouldn't work unless the MT CPU preserved the **ll/sc** semantics between TCs: It does. The 34-K family hardware does that by keeping track of an address used by an **ll** issued by any TC, and breaking the link if any other TC's store modifies an address sharing the same doubleword. The link bit is also broken if some management software takes over the TC and reschedules it by writing its **TCRestart** register.

It's very easy to do the SMP port if the TCs are in separate VPEs, because a TC alone in a VPE, by design, looks like a separate CPU. But MIPS Technologies has also demonstrated it using TCs in the same VPE. There are some difficulties here but they appear not to cause much loss of performance.

It's important to note, though, that only some applications benefit directly. You need an application that uses the Linux environment, exploits threading, and where for significant periods of time there are not only two threads running, but two threads that divide up the CPU's efforts fairly evenly between them.

Highly Responsive Programming with MT

Another approach being taken by MT pioneers is to use dedicated TCs to provide close-to-hardware intelligence, whose response time can be independent of whatever might be going on in the OS running on another TC in the other VPE. Such a system promises the best of both worlds: a convenient programming environment for the high-level code, but instant response to events that make the hardware simpler at the low end.

Low-end code on such a system is much in need of efficient synchronization mechanisms. Even an I/O read is snail-paced compared with the execution speed of a microprocessor. MIPS MT defines two mechanisms: *yield-on-signal* and *gating storage*.

The **yield** instruction, as hinted above, has a double life. When it's not doing the opposite of **fork** to provide user-privilege thread termination, it provides a way to cause a thread to wait for a hardware event, defined as the assertion of an external signal sampled into the CPU.

The 34-K CPU can sample any of 15 external signals. A **yield** instruction waits until one of any selection of them is asserted, based on an input argument register whose value is interpreted as a 15-bit vector: A "1" bit makes it sensitive

to the correspondingly numbered input signal. A low-level thread waiting on a **yield** instruction starts instantly when one of its signals is asserted.

But you might want a synchronization mechanism that also transfers data and is useful between software threads. In that case, the gating storage interface helps. MIPS MT defines gating storage as a special physical address region where both load and store instructions become blocking. A thread that reads or writes a gating storage location blocks on that load or store until data is successfully transferred. Moreover, an incomplete operation can be terminated (causing any waiting thread to receive an exception).

The gating storage interface is suitable for hardware interfaces where a high-speed data flow may be interrupted by error or end-of-block conditions.

The 34-K CPU design is licensed together with a gating storage system called *ITC* (for interthread communication memory): It provides software-to-software communication using “empty/full” storage or short FIFOs. For more information, look at the CPU manuals from MIPS Technologies.

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Other Optional Extensions to the MIPS Instruction Set

B.1 MIPS16 and MIPS16e ASEs

MIPS16 is an optional instruction set extension that can reduce the size of binary programs by 30 percent to 40 percent. MIPS16e is the name for the slightly enhanced version of the original MIPS16 instruction set used on CPUs that already conform to MIPS32/64. MIPS16 is targeted at applications where code size is a major concern—which mostly means very low cost systems. While it will only be used in certain implementations, it is a multivendor standard: MIPS Technologies, LSI, NEC, and Philips have all produced CPUs that support MIPS16.

We said back in section 1.2 that what makes MIPS binaries larger than those for other architectures is not that MIPS instructions do less work, but that they are larger—each is four bytes in size, in contrast to a typical average of about three bytes for some CISC architectures.

MIPS16 adds a mode in which the CPU decodes fixed-size 16-bit instructions. Most MIPS16 instructions expand to one regular MIPS32/64 instruction, so it's clear that this will be a rather restricted subset of instructions. The trick is to make the subset sufficient to encode enough of the program efficiently to make a substantial impact on the overall program size.

Of course, 16-bit instructions don't make this a 16-bit instruction set; MIPS16 CPUs are real MIPS CPUs with either 32- or 64-bit registers and operations that work on the whole of those registers.

MIPS16 is far from a complete instruction set—there's neither CPU control nor floating-point instructions, for example.¹ But that's OK, because every

1. MIPS did not invent the idea of providing an alternate half-size version of just part of the instruction set; Advanced RISC Machine's Thumb version of its ARM CPU was out first.

MIPS16 CPU must also run a complete MIPS ISA. You can run a mixture of MIPS16 and regular MIPS code; every function call or jump-register instruction can change the mode.

In MIPS16 it's convenient and effective to encode the mode as the least significant bit of an instruction address. MIPS16 instructions have to be even byte aligned, so bit 0 has no role as part of the instruction pointer; instead, every jump to an odd address starts MIPS16 execution, and every jump to an even address returns to regular MIPS. The target address of the MIPS subroutine-call instruction **jal** is always word aligned, so a new instruction, **jalx**, hides the mode change in the instruction.

To crush the instruction to half size we allocate only 3 bits to choose a register for most instructions, allowing free access to only eight general-purpose registers; also, the 16-bit constant field found in many MIPS instructions gets squeezed, often to 5 bits. Many MIPS16 instructions only specify two registers, not three; in addition, there are some special encodings described in the next section.

B.1.1 *Special Encodings and Instructions in the MIPS16 ASE*

MIPS16 has two particular weaknesses that will add size back to the program; the 5-bit immediate field is inadequate to build constants, and there's not enough address range on load/store operations. Three new kinds of instructions and one extra convention help out.

extend is a special MIPS16 instruction consisting of a 5-bit code and an 11-bit field, which is concatenated with the immediate field in the following instruction to allow an instruction pair to encode a 16-bit immediate. It appears as an instruction prefix in assembly language.

Loading constants takes extra instructions even in regular MIPS and would be a huge burden in MIPS16; it's quicker to put the constants into memory and load them. MIPS16 adds support for loads relative to the instruction's own location (PC-relative loads), allowing constants to be embedded in the code segment (typically, just before the start of a function). These are the only MIPS16 instructions that don't correspond exactly to a normal MIPS instruction—MIPS has no PC-relative data operations.

Many MIPS load/stores are directed at the stack frame, and **\$29/sp** is probably the most popular base register. MIPS16 defines a group of instructions that implicitly use **sp**, allowing us to encode a function's stack frame references without needing a separate register field.

MIPS load instructions always generate a full 32-bit address. Since load-word instructions are only valid for an address that is a multiple of four, the two least significant bits are wasted. MIPS16 loads are scaled: The address offset is shifted left according to the size of the object being loaded/stored, increasing the address range available from the instruction.

As an additional escape mechanism, MIPS16 defines instructions that allow it to do an arbitrary move between one of the eight MIPS16-accessible registers and any of the 32 MIPS general registers.

B.1.2 *The MIPS16 ASE Evaluated*

MIPS16 is not a suitable language for assembly coding, and we're not going to document it here. It's intended for compilers. Most programs compiled with MIPS16 seem to shrink to 60 percent to 70 percent of their MIPS size, which is more compact than 32-bit CISC architectures, similar to ARM's Thumb and reasonably competitive with pure 16-bit CPUs.

There's no such thing as a free lunch, however; a MIPS16 program will probably compile into 40 percent to 50 percent more instructions than would be required for MIPS. That means that running a program through the CPU core will take 40 percent to 50 percent more clock cycles. However, low-end CPUs are usually largely memory limited, not core limited, and the smaller MIPS16 program requires less bandwidth to fetch and will promote lower cache miss rates. Where the caches are small and program memory is narrow, MIPS16 will close the gap on and possibly overhaul regular MIPS code.

Because of the performance loss, MIPS16 code is not attractive in computers with large memory resources and wide buses. That's why it's an optional extension.

At the upper end of its range, MIPS16 will find itself in competition with software compression techniques. A regular MIPS program compressed into ROM storage with a general-purpose file compression algorithm will be smaller than the unencoded MIPS16 equivalent and a little larger than the compressed MIPS16 equivalent;² if your system has enough volatile memory to be able to use ROM as a file system and to decompress code into RAM for execution, software decompression of a full ISA will most likely give you better overall performance.

There's also a clear trend toward writing systems that make extensive use of code written in a byte-coded interpreted language (Java or its successors) for the bulk of code that is not time critical. That kind of intermediate code is very small, much more efficient than any machine binary; if only the interpreter and a few performance-critical routines are left in the native machine ISA, a tighter instruction set encoding will only affect a small part of the program. Of course, interpreters (particularly for Java) may themselves be quite large, but the inexorable increase in application complexity should soon cause that to dwindle in importance.

I expect to see MIPS16 continuing in use in a small range of power-, size-, and cost-constrained systems. It was worth inventing and is worth maintaining, because some of these systems are likely to be produced in huge volumes.

2. Tighter encodings have less redundancy for a compression algorithm to exploit.

B.2 The MIPS DSP ASE

The MIPS DSP ASE aims to overcome the perceived deficiencies of conventional instruction sets when confronted by multimedia applications. Jobs like encoding/decoding audio for soft modem or streaming applications or image/video compression/decompression use mathematically based algorithms that were once seen as the preserve of special-purpose digital signal processors. At the computational level, multimedia tasks like this often involve the repeated application of the same operation to large vectors or arrays of data.

Inside a register-based machine, one good way to accelerate vector operations is to pack multiple data items into a single machine register and perform a register-to-register instruction that does the same job on each field of each of its registers. This is a very explicit form of parallel processing called SIMD (for “single instruction, multiple data”). The DSP ASE includes quad-byte and paired-half SIMD operations.

It’s not just about vectors, and the DSP ASE has a bundle of other features:

- *Fixed-point fractional data types*: It is not yet economical (in terms of either chip size or power budget) to use floating-point calculations in these contexts. DSP applications use fixed-point fractions. Such a fraction is just a signed integer but interpreted as the value of that integer divided by some power of 2. A 32-bit fractional format where the implicit divisor is 2^{16} (65,536) would be referred to as a Q15.16 format; that’s because there are 16 bits devoted to fractional precision and 15 bits to the whole number range (the highest bit does duty as a sign bit and isn’t counted).

With this notation, Q31.0 is a conventional signed integer, and Q0.31 is a fraction representing numbers between -1 and 1 (well, nearly 1). It turns out that Q0.31 is the most popular 32-bit format for DSP applications, since it won’t overflow when multiplied (except in the corner case where -1×-1 leads to the just-too-large value 1). Q0.31 is often abbreviated to Q31.

The DSP ASE provides support for Q31 and Q15 (signed 16-bit) fractions.

- *Saturating arithmetic*: It’s not practicable to build in overflow checks to DSP algorithms—they need to be too fast. Clever algorithms may be built to be overflow-proof; but not all can be. Often the least worst thing to do when a calculation overflows is to make the result the most positive or most negative representable value. Arithmetic that does this is called *saturating*—and quite a lot of operations in the DSP ASE saturate (in many cases there are saturating and nonsaturating versions of what is otherwise the same instruction).

- *Multiplying fractions:* If you multiply two Q31 fractions by reusing a full-precision integer multiplier, then you'll get a 64-bit result, which consists of a Q62 result with (in the very highest bit) a second copy of the sign bit. This is a bit peculiar, so it's more useful if you always do a left-shift-by-1 on this value, producing a Q63 format (a more natural way to use 64 bits). Q15 multiply operations that generate a Q31 value have to do the shift-left too.
- *Rounding:* Some fractional operations implicitly discard less significant bits. But you get a better approximation if you bump the result by one when the discarded bits represent more than a half of the value of a 1 in the new LS position. That's what the DSP ASE means by *rounding*.
- *Multiply-accumulate sequences with choice of four accumulators:* (With fixed-point types, sometimes saturating).

Modern MIPS32 CPUs like MIPS Technologies' 24-K CPU already have quite a slick integer multiply-accumulate operation, but it's not so efficient when used for fractional and saturating operations.

The sequences are made more usable by having four 64-bit result/accumulator registers, **ac0–3**. The old MIPS multiply-divide unit has just one, accessible as the **hi/lo** registers. The new **ac0** is just the old **hi/lo**, with a new name.

The DSP instruction set is nothing like the regular and orthogonal MIPS32 instruction set. It's a collection of special-case instructions, in many cases aimed at the known hot-spots of important algorithms.

For a summary of the instructions see the programmer's manuals for the 34-K or 24-K CPUs, published by MIPS Technologies.

B.3 The MDMX ASE

MDMX is a much older attempt than the DSP ASE at providing DSP-like SIMD media operations within a MIPS instruction set. MDMX was developed and promoted by SGI and announced in 1997.

There are substantial differences of approach. MDMX chose to handle media data in the 64-bit registers of the floating-point unit, while the DSP ASE uses the general-purpose registers and much of it is built as an extension to the integer multiply unit. MDMX's choice is probably better for ultimate performance and programming convenience, but it costs more silicon.

The introduction of the DSP ASE provides another reason to doubt that MDMX will achieve significant use in real applications, so we won't describe it further here. If there's ever a third edition of this book, perhaps this is a mistake we'll have to correct.

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MIPS Glossary

\$f0–\$f31 registers: The 32 general-purpose 32-bit floating-point registers. In MIPS I (32-bit) CPUs, only even-numbered registers can be used for arithmetic (the odd-numbered registers hold the low-order bits of 64-bit, double-precision numbers).

\$n register: One of the CPU's 32 general-purpose registers.

a0–a3 register: Aliases for CPU registers \$4–\$7, conventionally used for passing the first four words of the arguments to a function.

ABI (application binary interface): A particular standard for building program binaries that in turn is supposed to guarantee correct execution on a conforming environment. MIPS32 CPUs generally use a standard known as o32, while for 64-bit operation there's a choice of n32 and n64. Many embedded systems rely on only a small subset of the whole ABI.

accumulator: A register specially designated to capture the result of repeated add/subtract operations. In the MIPS architecture, the multiply unit result registers `hi/lo` form an accumulator.

address regions: Refers to the division of the MIPS program address space into regions called `kuseg`, `kseg0`, `kseg1`, and `kseg2`. See under individual region names.

address space: The whole range of addresses as seen by the application program. Programs running under a protected OS have their addresses checked for validity and translated. Since such an OS can run many applications concurrently, there are many address spaces.

Alchemy Semiconductor: A 1999 start-up company seeded by key StrongARM designers leaving DEC, with a mission to make low-power, medium-performance, highly integrated MIPS microprocessors. Their success was predicated on an explosion in “pocket computers,” which didn't really happen.

In 2002 Alchemy became part of AMD and in 2006 part of Raza. The AU1xxx product line is still available for embedded applications.

Algorithmics: The U.K. company, specializing in MIPS technology and tools, of which I am a partner.

alignment: Positioning of data in a memory with respect to byte-address boundaries. Data items are called *naturally aligned* if they start on an address that is zero modulo their own size. MIPS CPUs require that their machine-supported data objects are naturally aligned; hence, words (four bytes) must be on four-byte boundaries, and a floating-point `double` datum must be on an eight-byte boundary.

alloca: C library function returning a memory area that will be implicitly freed on return from the function from which the call is made.

Alpha: The range of RISC CPUs made by Digital Semiconductor; it is the nearest relative to MIPS.

ALU (arithmetic/logic unit): A term applied to the part of the CPU that does computational functions.

AMD: A long-standing microelectronics company prominent around 2004 for building better x86 CPUs than Intel. They're in here because they have a MIPS embedded CPU line, acquired as part of Alchemy Semiconductor (see above).

analyzer: See *logic analyzer*.

Apache group (SVR4.2): An industry group of suppliers of MIPS-architecture UNIX systems that cooperated on a standard version of the UNIX System V Release 4.2 operating system and the *MIPS ABI standard*.

API (application program interface): A procedural interface to some software function, typically presented as standardized C/C++ function calls. It's a great idea, usually spoiled by the fact that capturing useful behavior behind clean interfaces is a very difficult art.

architecture: See *instruction set architecture*.

archive: Alternative name for an *object code library*.

argument: In C terminology, a value passed to a function. Often called a parameter in other languages. C arguments are parameters passed by value, if that helps.

ARM: The U.K.-based company that licenses microprocessor cores to the ARM architecture. The ARM architecture is almost RISC, though its initial design trade-offs favored circuit simplicity over pipeline convenience. However, its very simplicity helped ARM (the company) to be the first viable supplier of 32-bit synthesizable core CPUs.

ASCII: Used very loosely for the character encoding used by the C language.

ASE: An application-specific extension to the MIPS architecture. These are optional extensions defined in add-ons to the MIPS32/64 base architecture. The presence of an ASE is marked by a flag defined in MIPS32/64, usually a field in one of the **Config** CP0 registers.

ASIC (application-specific integrated circuit): A chip specially designed or adapted for use in a particular circuit.

ASIC-core CPU: See *core CPU*.

ASID: The address space ID maintained in the CPU register **EntryHi**. Used to select a particular set of address translations: Only those translations whose own ASID field matches the current value will produce valid physical addresses.

assembler, assembly code: Assembler code (sometimes called assembly code or assembly language) is the human-writable form of a computer's machine instructions. The assembler is the program that reads assembly language and translates it to an executable program, probably through an interim *object code*.

associative store: A memory that can be looked up by presenting part of the stored data. It requires a separate comparator for each data field, so large associative stores use up prodigious amounts of logic. The MIPS TLB, if fitted, uses a fully associative memory with between 32 and 64 entries.

associativity: See *cache, set-associativity*.

asynchronous logic: An asynchronous circuit is one that is not organized around the transitions of one or a few clock signals. See *synchronous logic*.

ATMizer: A component made by LSI Logic for ATM network interfacing; it that has an internal MIPS CPU as just one component inside.

atomic, atomically, atomicity: In computer science jargon, a group of operations is atomic when either all of them happen or none of them do.

backtrace: See *stack backtrace*.

BadVAddr register: CPU control register that holds the value of an address that just caused a trap for some reason (misaligned, inaccessible, TLB miss, etc.).

barrier, hazard: See *hazard barrier*.

barrier, memory: See entry on the **sync** instruction.

BAT: An (almost obsolete) option for reusing program memory regions in a MIPS CPU that has no memory-mapping hardware (TLB).

bcopy: C library function to copy the contents of a chunk of memory.

benchmark: A program that can be run on many different computers, with a view to finding something about their relative performance. Benchmarks have evolved from fragments of code intended to measure the speed of some very

specific task to large suites, which should give some guidance as to the speed at which a system handles common applications.

Berkeley UNIX: See *BSD UNIX*.

BEV (boot exception vectors): A bit in the CPU status register that causes traps to go through a pair of alternate vectors located in uncached (kseg1) memory. The locations are close to the reset-time start point so that they can both conveniently be mapped to the same read-only memory.

bias: See *floating-point bias*.

BiCMOS: A particular technology for building chips, mixing dense and cool CMOS transistors for internal logic with faster and electrically quieter bipolar transistors for interface. It was in vogue for CPUs in the late 1980s, but nobody used it with any great success.

big-endian: Describes an architecture where the most significant part of a multibyte integer is stored at the lowest byte address; see section 10.2.

bitfield: A part of a word that is interpreted as a collection of individual bits.

block size: See *cache line size*.

blocking: An operation that stops execution until it completes, the opposite of *nonblocking*.

bootstrap: A program or program fragment that is responsible for starting up from a condition where the state of the CPU or system is uncertain.

branch: In the MIPS instruction set, a PC-relative jump.

branch and link: A PC-relative subroutine call.

branch delay slot: The position in the memory-stored instruction sequence immediately following a jump/branch instruction. The instruction in the branch delay slot is always executed before the instruction that is the target of the branch. It is sometimes necessary to fill the branch delay slot with a **nop** instruction.

branch optimization: The process (carried out by the compiler, assembler, or programmer) of adjusting the memory sequence of instructions so as to make the best use of branch delay slots.

branch penalty: Many CPUs pause momentarily after taking a branch, because they have fetched instructions beyond the branch into their pipeline and must backtrack and refill the pipeline. This delay (in clock cycles) is called the branch penalty. It's zero on short-pipeline MIPS chips, but the two-clock-cycle branch penalty on the long-pipeline R4000 was a significant factor in reducing its efficiency.

branch prediction: A CPU implementation technique where simple logic working at the front end of the pipeline identifies branch instructions and guesses the branch target. The fetch stages of the pipeline then follow the guess, rather than always prefetching sequentially. The CPU needs some mechanism to back out of the guess when it turns out to be wrong.

It turns out that branch prediction works very well, for an acceptable level of complexity, and it's employed in all but the simplest 32-bit CPUs.

BrCond3-0: CPU inputs that are directly tested by the coprocessor conditional branch instructions.

breakpoint: When debugging a program, a breakpoint is an instruction position where the debugger will take a trap and return control to the user. Implemented by pasting a break instruction into the instruction sequence under test.

Broadcom Corporation: A prominent manufacturer of chips for computer networking, mentioned here because own it and produces the SB12xx range of 64-bit MIPS CPUs, typically integrated with many useful networking interfaces. The SB-12xx designs originated with *SiByte*, see below.

BSD UNIX: The Berkeley Software Distribution was a variant of the UNIX operating system originally on the DEC VAX minicomputer. BSD was the first variant of UNIX to feature a full 32-bit virtual memory kernel and was the basis for Sun's original OS. After a protracted argument, BSD version 4.4 was released as open source (but the argument went on long enough to spark the development of Linux as a successor kernel). It's most modern direct descendant is Apple's OS X.

BSS: In a compiled C program, the chunk of memory that holds variables declared but not explicitly initialized. Corresponds to a segment of object code. Nobody seems to be able to remember what BSS ever stood for.

Most C compilation systems use the “.bss” name for the data area to which global variables are assigned that have not been explicitly initialized.

burst read cycles: MIPS CPUs (except for some very early parts) refill their caches by fetching more than one word at a time from memory (four words is common) in a burst read cycle.

byte: An 8-bit datum.

byte order: Used to emphasize the ordering of items in memory by byte address. This seems obvious, but it can get confusing when considering the constituent parts of words and halfwords.

byte-swap: The action of reversing the order of the constituent bytes within a word. This may be required when adapting data acquired from a machine of different endianness.

C preprocessor: The program `cpp`, typically run as the first pass of the C compiler. `cpp` is responsible for textual substitutions and omissions. It processes

comments and the useful directives that start with a “#”, like `#define`, `#include`, and `#ifdef`. Despite its association with C, it is a general-purpose macro language, which can be, and often is, used with other languages. In this book, its important non-C application is to preprocess assembly language programs.

C++: A compiled language retaining much of the syntax and appearance of C, but offering a range of object-oriented extensions.

cache: A small auxiliary memory, located close to the CPU, that holds copies of some data recently read from memory. MIPS caches are covered extensively in Chapter 4.

cache, direct-mapped: A direct-mapped cache has, for any particular location in memory, only one slot where it can store the contents of that location. Direct-mapped caches are particularly liable to become inefficient if a program happens to make frequent use of two variables in different parts of memory that happen to require the same cache slot; however, direct-mapped caches are simple, so they can run at high clock rates. Perhaps more importantly, a direct-mapped cache requires many less wires to connect its controller and storage, which makes direct-mapped caches a natural choice for an off-chip cache.

Direct-mapped caches are now rarely encountered in even the simplest CPUs.

cache, duplicate tags: In cache-coherent multiprocessors, the bus interface controller must often look at the CPU’s cache—specifically, at the cache tags—to check whether a particular bus transaction should interact with the data currently in the cache. Such accesses are costly, either in delays to the CPU if the bus interface time-slices the tags with the CPU or in hardware and interlocks if the tags are dual ported. It’s often cheaper to keep a second copy of the cache information the bus interface is interested in, updated in parallel with the main cache—the events that cause either to change are bus-visible anyway. The duplicate tags don’t need to be perfect to be useful; if they allow the bus interface to avoid accessing the CPU’s tags in a high proportion of cases, they’ll still make the system more efficient.

cache, physical-addressed: A cache that is accessed entirely by using physical (translated) addresses. Early MIPS CPU caches, and all MIPS L2 caches, are like this.

cache, set-associative: A cache that has more than one place in the cache where data from a particular memory location may be stored. You’ll commonly see two-way set-associative caches, which means there are two cache slots available for any particular memory data. In effect, there are two caches, searched simultaneously, so the system can cope with a situation where two frequently accessed items are sitting at the same cache index.

A set-associative cache requires wider buses than a direct-mapped cache and cannot run quite as fast. Early RISCs used direct-mapped caches to save pins

on the external cache. Although the wide buses are not much of a problem for on-chip caches, some early integrated CPUs still had direct-mapped caches to boost the clock frequency. These days, set-associative on-chip caches are usually preferred for their lower miss rate.

cache, snooping: In a cache, snooping is the action of monitoring the bus activity of some other device (another CPU or DMA master) to look for references to data that are held in the cache. Originally, the term “snooping” was used for caches that could *intervene*, offering their own version of the data where it was more up-to-date than the memory data that would otherwise be obtained by the other master; the word has come to be used for any cache that monitors bus traffic.

cache, split: A cache that has separate caches for instruction fetches and for data references.

cache, write-back: A D-cache in which CPU write data is kept in the cache but not (for the time being) sent to main memory. The cache line is marked as “dirty.” The data is written back to main memory either when that line in the cache is needed for data from some other location or when the line is deliberately targeted by a write-back operation.

cache, write-through: A D-cache in which every write operation is made simultaneously to the cache (if the access hits a cached location) and to memory. The advantage is that the cache never contains data that is not already in memory, so cache lines can be freely discarded.

Usually, the data bound for memory can be stored in a *write buffer* while the memory system’s (relatively slow) write cycle runs, so the CPU does not have to wait for the memory write to finish.

Write-through caches work very well as long as the memory cycles fast enough to absorb writes at something a little higher than the CPU’s average write rate. That’s rare in modern systems.

cache aliases: In a virtual-memory OS, you can sometimes have the same data mapped at different locations. This can happen with data shared between two tasks’ distinct address spaces or with data for which there is a separate application and kernel view.

Now, many MIPS CPUs use program (virtual) addresses to index the cache—it saves time to be able to start the cache search in parallel with translating the address. But if different program addresses can access the same data, we could end up with the same data in the cache at two locations—a cache alias. If we then start writing the locations, that’s going to go horribly wrong.

Cache aliases turn out to be avoidable. The paged address translation used in MIPS CPUs means that at least 12 low-order addresses are unchanged by translation, and it turns out that you only use about 15 low-order address bits to index the biggest likely cache. Kernel software needs to be careful, when

generating multiple different addresses for a page, that the pages are allocated to program addresses where bits 12–15 are the same.

cache coherency: The name for that state of grace where the contents of your cache will always deliver precisely what your program and the rest of the system has stored into the cache/memory combination. Many complex techniques and hardware tricks are deployed in the search for coherency; older MIPS CPUs for “servers” (like R4000SC or R10000) have clever features in the cache for this. But such technology is not much used outside the world of large server computers, as yet.

cache flush: A somewhat ambiguous term; it is worth avoiding. It is never quite clear whether it means write-back, invalidate, or both.

cache hit: What happens when you look in the cache and find what you were looking for.

cache index: The cache index is the part of the address that is used to select the cache location in each set.

cache invalidation: Marks a line of cache data as no longer to be used. There’s always some kind of valid bit in the control bits of the cache line for this purpose. It is an essential part of initialization for a MIPS CPU.

cache line size: Each cache slot can hold one or more words of data, and the chunk of data marked with a single address tag is called a line. Big lines save tag space and can make for more efficient refill, but big lines waste space by loading more data you don’t need.

The best line size tends to increase as you get further from the CPU and for big cache miss penalties. MIPS I CPUs always had one-word data cache lines, but later CPUs tend to favor four or eight words.

cache miss: What happens when you look in the cache and don’t find what you are looking for.

cache miss penalty: The time the CPU spends stalled when it misses in the cache, which depends on the system’s memory response time.

cache profiling: Measuring the cache traffic generated when a particular program runs, with a view to rearranging the program in memory to minimize the number of cache misses. It is not clear how practicable this is, except for very small programs or sections of program.

cache refill: The memory read that is used to obtain a cache line of data after a cache miss. This is first read into the cache, and the CPU then restarts execution, this time hitting in the cache.

cache set: One chunk of a set-associative cache.

cache simulator: A software tool used for cache profiling.

cache tag: The information held with the cache line that identifies the main memory location of the data.

cache write-back: The process that takes the contents of a cache line and copies them back into the appropriate block of main memory. It's usually performed conditionally, because cache lines have a "dirty" flag, which remembers when they've been written since being fetched from memory.

cacheable: Used of an address region or a page defined by the memory translation system.

CacheErr register: CPU control (coprocessor 0) register in R4000 CPUs and descendants, full of information for analyzing and fixing cache parity/ECC errors.

cacheop: The MIPS architecture provides a multipurpose **cache** instruction used to initialize and manipulate cache contents.

callee: In a function call, the function that is called.

caller: In a function call, the function where the call instruction is found and where control returns afterward.

Cause register: CPU control register that, following a trap, tells you which kind of trap it was. **Cause** also shows you which external interrupt signals are active.

Cavium Networks: Cavium's network-specialized computer chips, launched in 2006, are called Octeon. Each chip contain multiple MIPS64 CPUs.

ceiling: A floating-point-to-integer conversion, rounded to the nearest integer that is as least as positive. Implemented by the MIPS instruction **ceil**.

char: C name for a small quantity used to hold character codes. In MIPS CPUs (and practically always, nowadays) this is a single byte.

CISC: An acronym used to refer to non-RISC architectures. In this book, we mean architectures like the DEC VAX, Motorola 680x0, and Intel x86 (32-bit version). All these instruction sets were invented before the great RISC discovery, and all are much harder than a RISC CPU to execute fast.

clock cycle: The period of the CPU's clock signal. For a RISC CPU, this is the rate at which successive pipeline stages run.

CMOS: The transistor technology used to make all practical MIPS CPUs. CMOS chips are denser and use less power per transistor than any other kind, so they are favored for leading-edge integration. With CPUs, the ability to put a lot of circuitry into a small space has proven to be the key performance factor, so all fast CPUs are now CMOS.

coherency: See *cache coherency*.

Compare register: CPU control register provided on CPUs for implementing a timer. Some very old MIPS CPUs did not provide this, but probably not any you'll ever meet.

Config register: CPU control register for configuring basic CPU behavior, standardized on MIPS32/64 CPUs. It has been somewhat standardized since MIPS III.

console: The putative I/O channel on which messages can be sent for the user and user input can be read.

const: C data declaration attribute, implying that the data is read-only. It will often then be packed together with the instructions.

Context register: CPU control register seen only on CPU types with a TLB. Provides a fast way to process TLB misses on systems using a certain arrangement of page tables.

context switch: The job of changing the software environment from one task to another in a multitasking OS.

coprocessor: Some part of the CPU, or some other closely coupled machine part, that executes some particular set of reserved instruction encodings. This is a MIPS architecture concept that has succeeded in separating optional or implementation-dependent parts of the instruction set and thus reducing the changes to the mainstream instruction set. It's been fairly successful, but the nomenclature has caused a lot of confusion.

coprocessor condition: Every coprocessor subset of special instructions in the MIPS architecture gets a single bit for communicating status to the integer CPU, tested by a **bcxt/bcxf** instruction. See Chapter 3.

coprocessor conditional branches: The MIPS architecture's coprocessors can provide one or more condition bits, and all coprocessors get branches such as **bc1t label** that branch according to the sense of the condition.

coprocessor 0: The (rather fuzzily defined) bits of CPU function that are connected with the privileged control instructions for memory mapping, exception handling, and suchlike.

core CPU: A microprocessor designed to be built in as one component of an ASIC, making what is sometimes called a "system on a chip." MIPS CPUs are increasingly being used as cores.

CorExtend ASE: An instruction set extension, optional on some of MIPS Technologies' CPU cores, that makes it relatively easy for chip designers to add and use new, application-specific, computational instructions.

Count register: Continuously running timer register, available in R4000-like CPUs and some earlier ones.

cpp: The C preprocessor program.

CPU core: See *core CPU*.

CSE (common subexpression elimination): Perhaps the single most important optimization step for an optimizing compiler, this requires logic to detect when the compiler is repeating a computation it already carried out with the same data. Then it can consider storing the first result and reusing it. It's particularly important because other compiler transformations can lead to replicating calculations, and it's much simpler to tidy up with CSE than to build somewhat-similar logic into other optimizations.

cycle: Clock cycle.

D-cache: Data cache (MIPS CPUs always have separate instruction and data caches).

D-TLB: Some MIPS processors have tiny separate translation caches fed from the main TLB to avoid a resource conflict when translating both instruction and data addresses. Most MIPS CPUs have an I-side I-TLB, many faster CPUs also have the D-TLB. Its operation is invisible to software, other than an occasional extra clock spent fetching main TLB entries.

data dependencies: The relationship between an instruction that produces a value in a register and a subsequent instruction that wants to use that value.

data path swapper: See *byte-swap*.

data/instruction cache coherency: The job of keeping the I-cache and D-cache coherent. No MIPS CPU does this for you; it is vital to invalidate I-cache locations whenever you write or modify an instruction stream. See *cache coherency*.

debug mode: A special CPU state, much like exception mode, associated with the **Debug (DM)** register bit. You get into debug mode through a debug exception, and exit it when the debugger program runs a **deret**. See section 12.1.4.

debug probe: A small box of electronics that connects both to your software-development host computer and to your CPU's *EJTAG* unit and allows you to debug software on a target system. Such a debugger requires no resources built into the target beyond the *EJTAG* unit and interface, so it can be used on the most stripped-down embedded device.

debugger: A software tool for controlling and interrogating a running program.

DEC: Digital Equipment Corporation, the most successful manufacturer of minicomputers through the 1980s. Their PDP-11 and VAX computers inspired UNIX. They had a short but significant involvement with MIPS CPUs from 1989.

DECstation: DEC's trade name for the MIPS-architecture workstations they produced between 1989 and 1993.

delayed branches: See *branch delay slot*.

delayed loads: See *load delay slot*.

demand paging: A process by which a program is loaded incrementally. It relies on an OS and underlying hardware that can implement virtual memory—references to thus-far-unloaded parts of the program are caught by the OS, which reads in the relevant data, maps it so that the program will see it in the right place, and then returns to the program, re-executing the reference that failed. It's called paging because the unit of memory translation and loading is a fixed-size block called a page.

denormalized: A floating-point number is denormalized when it is holding a value too small to be represented with the usual precision. The way the IEEE 754 standard is defined means that it is quite hard for hardware to cope directly with denormalized representations, so MIPS CPUs always trap when presented with them or asked to compute them.

dereferencing: A fancy term for following a pointer and obtaining the memory data it points at.

diagnostics: Short for “diagnostic tests,” referring to software written not to get a job down, but to try to cause, detect, and diagnose problems in the computer and attached hardware. Many features are added to highly integrated chips to provide handles for diagnostic software.

direct mapped: See *cache, direct-mapped*.

directive: One of the terms used for the pieces of an assembly program that don't generate machine instructions but that tell the assembler what to do—for example, `.globl`. They're also called pseudo-ops.

dirty: In a virtual memory system, this describes the state of a page of memory that has been written to since it was last fetched from or written back to secondary storage. Dirty pages must not be lost.

disassembler: A program that takes a binary instruction sequence in memory and produces a readable listing in assembly mnemonics.

displacement: A value used in an address calculation by being added to some “base” address. In the common MIPS case, almost every load/store address is formed from a base address in a register and a 16-bit signed displacement encoded into the instruction.

DMA (direct memory access): An external device transferring data to or from memory without CPU intervention.

double: C and assembly language name for a double-precision (64-bit) floating-point number.

doubleword: The preferred term for a 64-bit data item (not used for floating-point) in MIPS architecture descriptions.

download: The act of transferring data from host to target (in case of doubt, host tends to mean the machine to which the user is connected).

DRAM: Used sloppily to refer to large memory systems (which are usually built from DRAM components). Sometimes used less sloppily to discuss the typical attributes of memories built from DRAMs.

dseg, drseg, dmseg: Virtual memory regions (overlapping what would otherwise be part of kseg2), which are accessible only in debug mode and are used to map EJTAG debug unit resources for debugger code.

DSP (digital signal processor): A particular style of microprocessor aimed at applications that process a stream of data, each data item representing some kind of sampled value ultimately derived from an analog-to-digital converter. DSPs focus on speed at certain popular analog algorithms: They stress multiply-accumulate performance. Compared with a general-purpose processor, they often lack precision, easy programming in high-level language, and the facilities to build basic OS facilities. But the definition of DSP is not much more firm than that of RISC.

DSP ASE: An extension to the MIPS32/64 architecture that adds a large number of instructions to help multimedia applications; many of its new operations are *SIMD*, *saturating*, and/or are some kind of *multiply-accumulate*.

duplicate tags: See *cache*, *duplicate tags*.

dword: The MIPS assembly name for a 64-bit integer datum, or doubleword.

dynamic linking: A term for the process by which an application finds and binds to a subroutine library at run time, immortalized by Microsoft as DLLs. Most Linux/MIPS systems are based on dynamic linking for user-space applications.

dynamic variables: An old-fashioned programmer's term for variables (like those defined inside C functions) that are really or notionally kept on the stack.

ECC (error-correcting code): Stored data is accompanied by check bits that are not only effective in diagnosing corruption but permit errors (supposed to affect only a small number of bits) to be rectified. Some MIPS CPUs use an ECC that adds 8 check bits to each 64-bit doubleword for data both in the caches and on the memory bus (and probably in memory too, though that's a system design decision).

ECL (emitter-coupled logic): An electrical standard for deciding whether a signal represents a one or a zero. ECL allows faster transitions and less noise

susceptibility than the more common standard TTL, but with a penalty in higher power consumption. It's now pretty much obsolete. The name describes the transistor implementation originally used in this sort of chip.

EIC (external interrupt controller) mode: A distinct way of signaling interrupts to a MIPS CPU available as an option in MIPS32/64. It provides for up to 63 immediately distinguishable interrupts (very useful for embedded systems, which often have very large numbers of interrupt sources), but at the cost of requiring the system designer to hardwire interrupt priority. See *IPL*, and for more details see section 5.8.5.

EJTAG: A specification for an on-core debug unit for MIPS32/64 CPUs, usable by software but even more usable with a *debug probe* connected through the JTAG (test interface) pins. See section 12.1.

ELF (executable and linking format): An object code format defined for UNIX standardization and mandated by the MIPS ABI standard.

emacs: The Swiss Army knife of text editors and the essential tool for real programmers, emacs (frighteningly) runs the Lisp program of your choice every time you hit a key. It is indescribably customizable, so with any job you do you get small and valuable contributions from numerous people who went before you. This book was written with it.

embedded: Describes a computer system that is part of a larger object that is not (primarily) seen as a computer. That could be anything from a video games box to a glass furnace controller.

emulator: See *in-circuit emulator*; *software instruction emulators*.

endianness: Whether a machine is big-endian or little-endian. See Chapter 10.

endif: (`#endif`) The end of a piece of code conditionally included by the magic of `cpp`. See also `#ifdef`.

EntryHi/EntryLo register: CPU control registers implemented only in CPUs with a TLB. Used to stage data to and from TLB entries.

EPC (exception program counter) register: CPU control register telling you where to restart the CPU after an exception.

epilogue: See *function epilogue*.

EPROM (erasable programmable read-only memory): The device most commonly used to provide read-only code for system bootstrap; used sloppily here to mean the location of that read-only code.

errno: The name of the global variable used for reporting I/O errors in most C libraries.

ErrorEPC register: R4x00 and later CPUs detect cache errors, and, to allow them to do so even if the CPU is halfway through some critical (but regular)

exception handler, the cache-error system has its own separate register for remembering where to return to. See section 4.9.3.

ExcCode: The bitfield in the **Cause** register that contains a code showing what type of exception just occurred.

exception: In the MIPS architecture, an exception is any interrupt or trap that causes control to be diverted to one of the two trap entry points.

exception, IEEE: See *floating-point (IEEE) exception*. Alas, this is a different animal from a MIPS exception.

exception vector, exception entry point: The (virtual) memory address where the CPU hardware starts executing after an exception. “Exception vector” is a misuse, really, derived from architectures where entry points are software-redefinable through a memory-resident table. In MIPS CPUs, the exception handler entry points may only be changed—all at once—by altering a single base address in the **EBase** register.

exception victim: On an exception, the victim is the first instruction in sequence *not* to be run (yet) as a result of the exception. For exceptions that are caused by the CPU’s own activity, the victim is also the instruction that led to the exception. It’s also normally the point to which control returns after the exception; but this is not always the case, because of the effect of branch delays.

exceptional value: A floating-point value outside the normal range of representation (an infinity, NaN for invalid value, and so on). Any calculation producing such a value could have led to an IEEE 754 exception, if software had decided to enable the exception.

executable: Describes a file of object code that is ready to run.

exponent: Part of a floating-point number. See Chapter 7.

extended floating point: Not provided by the MIPS hardware, this usually refers to a floating-point format that uses more than 64 bits of storage (80 bits is popular).

extern: C data attribute for a variable that is defined in another module.

FastMath: See entry for *Intrinsity*.

fault, faulting: See *page fault*.

FCC (floating-point unit condition code): MIPS I through MIPS III CPUs have only one; higher-numbered ISAs have eight.

FCSR register: The floating-point control/status register, see Chapter 7.

FIFO (first-in, first-out): A queue that temporarily holds data, in which the items have to come out in the same order they went in.

fixup: In object code, this is the action of a linker/locator program when it adjusts addresses in the instruction or data segments to match the location at which the program will eventually execute.

flag: Used here (and often in computer books) to mean a single-bit field in a control register.

floating-point bias: An offset added to the exponent of a floating-point number held in IEEE format to make sure that the exponent is positive for all legitimate values.

floating-point condition code/flag: A single bit set by FP compare instructions, which are communicated back to the main part of the CPU and tested by **bc1t** and **bc1f** instructions.

floating-point emulation trap: A trap taken by the CPU when it cannot implement a floating-point (coprocessor 1) operation. A software trap handler can be built that mimics the action of the FPU and returns control, so that application software need not know whether FPU hardware is installed or not. The software routine is likely to be 50–300 times slower.

floating-point (IEEE) exception: The IEEE 754 standard for floating-point computation considers the possibility that the result can be “exceptional”—a catch-all term for various kinds of results that some users may not be happy with. The standard requires that conforming CPUs allow each type of exception to be caught—and then it gets confusing, because the MIPS mechanism for catching events in general is also called exception.

floating-point unit (FPU): The name for the part of the MIPS CPU that does floating-point math. Historically, it was a separate chip.

foo: The ubiquitous name for a junk or worthless file.

FORTRAN: Early computer language favored for scientific and numerical uses, where its reasonable portability outweighed its appalling flaws.

FP: Floating point.

fp (frame pointer) register: A CPU general-purpose register (**\$30**) sometimes used conventionally to mark the base address of a *stack frame*.

fpcond: Another name for the FP condition bit (also known as coprocessor 1 condition bit).

FPU: Floating-point unit.

fraction, fractional part: Part of a floating-point value. (Also called the *mantissa*.) See Chapter 7.

frame, framesize: See *stack frame*.

Free Software Foundation: The Lone Rangers of free software, FSF is the (loose) organization that keeps the copyright of GNU software.

fully associative: See *associative store*.

function: The C language name for a subroutine, which we use through most of this book.

function epilogue: In assembly code, the stereotyped sequence of instructions and directives found at the end of a function and concerned with returning control to the caller.

function inlining: An optimization offered by advanced compilers, where a function call is replaced by an interpolated copy of the complete instruction sequence of the called function. In many architectures this is a big win (for very small functions) because it eliminates the function-call overhead. In the MIPS architecture the function-call overhead is negligible, but inlining is still sometimes valuable because it allows the optimizer to work on the function in context.

function prologue: In assembly language, a stereotyped set of instructions and directives that start a function, saving registers and setting up the stack frame.

gcc: The usual name for the *GNU C compiler*.

gdb: The GNU debugger, partner to GNU C.

global: Old-fashioned programmer's name for a data item whose name is known and whose value may be accessed across a whole program. Sloppily extended to any named data item that is awarded its own storage location—and that should properly be called static.

global pointer: The MIPS **gp** register, used in some MIPS programs to provide efficient access to those C data items defined as *static* or *extern* that live at a fixed program address. See section 2.2.1.

globl: Assembly declaration attribute for data items or code entry points that are to be visible from outside the module.

GNU: The name of the Free Software Foundation's project to provide freely redistributable versions for all the components of a UNIX-like OS (with the possible exception of the kernel itself).

GNU C compiler: Free product of an extraordinary interaction between maverick programmer and Free Software Foundation leading light Richard Stallman and a diverse collection of volunteers from all over the world. GNU C is the best compiler for MIPS targets unless you're using a Silicon Graphics workstation.

GOT (global offset table): An essential part of the dynamic linking mechanism used by Linux/MIPS and originally developed under the name MIPS/ABI. See section 16.2.

gp register: CPU register **\$28**, often used as a pointer to program data. Program data that can be linked within ± 32 K of the pointer value can be loaded with a single instruction. Not all toolchains, nor all runtime environments, support this.

halfword: MIPS architecture name for a 16-bit data type.

hazard: Also called *pipeline hazard*. A hazard occurs where an instruction sequence may be unreliable, because the effect of a producer instruction may in fact not apply reliably to a consumer instruction that is later in program sequence (typically because the producer's effect appears late in the pipeline).

It's only a hazard when the CPU architecture does not insist that the hardware make the effect invisible: It's possible to define an architecture with no hazards, at the cost of implementation complexity and (perhaps) loss of performance. In the very oldest MIPS CPUs, there was one hazard visible in user-privilege programs: An attempt to use data in the instruction immediately following the load might fail. Modern MIPS CPUs have no hazards in user-privilege code.

But even new CPUs have hazards associated with CPU control, and those are discussed in section 3.4.

hazard barrier: An instruction that, when placed between the producer and consumer of a hazard, ensures that the consumer sees the producer's effect fully formed.

heap: Program data space allocated at run time.

Heinrich, Joe: Esteemed author of the definitive *MIPS User's Manual*, from which almost all official MIPS ISA manuals are derived.

Hennessy, John: MIPS's intellectual father and founding parent, Professor Hennessy led the original MIPS research project at Stanford University.

hi, lo: the result registers of the MIPS integer multiply unit. Taken together they act as an accumulator for integer multiply-accumulate operations.

hit: See *cache hit*.

I-cache: Instruction cache (MIPS CPUs always have separate instruction and data caches). The I-cache is called upon when the CPU reads instructions.

ICE: See entry for *in-circuit emulator*.

ICU: Interrupt control unit.

idempotent: A mathematician's term for an operation that has the same effect when done twice as done once (and hence also the same effect when done nine

times or 99). Stirring your coffee is an idempotent operation, but adding sugar isn't.

When a pipelined CPU takes an exception, and subsequently returns to the interrupted task, it's difficult to make sure that everything gets done exactly once; if you can make some of the operations idempotent, the system can survive a spuriously duplicated operation. All MIPS branch instructions, for example, are idempotent.

IDT: Integrated Device Technology, Inc., pioneers of MIPS CPUs for embedded applications through the 1980s and 1990s. IDT continues to thrive, but CPUs are no longer such a significant part of its business.

IEEE: An acronym for the U.S. Institute of Electrical and Electronics Engineers. This professional body has done a lot to promulgate standards in computing. Its work is often more practicable, sensible, and constructive than that of other standards bodies.

IEEE 754 floating-point standard: An industry standard for the representation of arithmetic values. This standard mandates the precise behavior of a group of basic functions, providing a stable base for the development of portable numeric algorithms.

ifdef, ifndef: `#ifdef` and `#endif` bracket conditionally compiled code in the C language. This feature is actually affected by the *C preprocessor* and so can be used in other languages, too.

immediate: In instruction set descriptions, an immediate value is a constant that is embedded in the code sequence. In assembly language, it is any constant value.

implementation: Used in opposition to “architecture.” In this book, it most often means we're talking about how something is done in some particular CPU.

in-circuit emulator (ICE): Originally, this meant a device that replaced a CPU chip with a module that, as well as being able to exactly imitate the behavior of the CPU, provides some means to control execution and examine CPU internals. Microprocessor ICE units are inevitably based on a version of the microprocessor chip (often a higher-speed grade).

It is often possible to do development without an ICE—and they are expensive and can prove troublesome.

But these days the acronym is sometimes reused for a *debug probe* (see above), which provides similar facilities in a different way.

index, cache: See *cache index*.

index register: CPU control register used to define which TLB entry's contents will be read into or written from **EntryHi/EntryLo**.

Indy: A relatively low-cost family of workstations made by SGI (Silicon Graphics Inc.) through the 1990s using R4000, R4600, and R5000 64-bit MIPS CPUs.

inexact: Describes a floating-point calculation that has lost precision. Note that this happens very frequently on the most everyday calculations; for example, the number $\frac{1}{3}$ has no exact representation. IEEE 754 compliance requires that MIPS CPUs can trap on an inexact result, but nobody ever turns that trap on.

infinity: A floating-point data value standing in for any value too big (or too negative) to represent. IEEE 754 defines how computations with positive and negative versions of infinity should behave.

inline, inlined, inlining: See *function inlining*.

instruction scheduling: The process of moving instructions around to exploit the CPU's pipelining for maximum performance. On a simple pipelined MIPS CPU, that usually comes down to making the best use of delay slots. This is done by the compiler and (sometimes) by the assembler.

instruction set architecture (ISA): The functional description of the CPU, which defines exactly what it does with any legitimate instruction stream (but does not have to define how it is implemented). MIPS32/64 is the main ISA described in this book.

instruction synthesis by assembly: The MIPS instruction set omits many useful and familiar operations (such as an instruction to load a constant outside the range ± 32 K). Most assemblers for the MIPS architecture will accept instructions (sometimes called macro instructions) that they implement with a short sequence of machine instructions.

int: The C name for an integer data type. The language doesn't define how many bits are used to implement an `int`, and this freedom is intended to allow compilers to choose something that is efficient on the target machine.

interlock: A hardware feature where the execution of one instruction is delayed until something is ready. There are few interlocks in the MIPS architecture.

interrupt: An external signal that can cause an exception (if not masked).

interrupt mask: A bit-per-interrupt mask, held in the CPU status register, that determines which interrupt inputs are allowed to cause an interrupt at any given time.

interrupt priority: See *IPL*.

interruptible: Generally used of a piece of program where an interrupt can be tolerated (and where the programmer has therefore allowed interrupts to occur).

Intrinsity Semiconductor: Intrinsity announced, built, and demonstrated the FastMath MIPS32 CPU (with an attached array coprocessor) at up to 2 GHz—making it the fastest MIPS CPU to date. However, the company was probably focused more on silicon design methodology than on MIPS CPUs as such.

invalidation: See *cache invalidation*.

IPL (interrupt priority level): In many architectures, the interrupt inputs have built-in priority; an interrupt will not take effect during the execution of an interrupt handler at equal or higher priority. Historically, MIPS hardware did none of this, leaving priority to the software.

But MIPS32 CPUs using the EIC optional interrupt system communicate and implement interrupt priority in hardware. See section 5.8.5.

Irix: The operating system (based on UNIX System V) on the Silicon Graphics workstations/servers.

ISA: Instruction set architecture.

ISR (interrupt service routine): Another name for the software invoked by an interrupt exception.

issue, instruction: When talking about computer implementations, issue is the point at which some CPU resources get used to begin doing the operations associated with some instruction.

I-TLB: A tiny hardware table duplicating information from the *TLB* that is used for translating instruction addresses without having to fight the hardware that is translating data addresses. It was called the “micro-TLB” or “uTLB” in early MIPS CPUs. It is not visible to software, unless you’re counting time so carefully that you notice the one-clock pause in execution when an I-fetch has to access the main TLB.

JPEG: A standard for compressing image data.

JTAG: A standard for connecting electronic components to implement test functions. The JTAG signals are intended to be daisy-chained through all the active components in a design, allowing one single point of access for everything. That never quite happened, because small single-function chips don’t usually support JTAG; but it remains an indispensable part of production testing of circuit boards.

More relevantly to this book, the JTAG signals provide a workable way of connecting an *EJTAG* on-chip debug unit to a *debug probe*.

jump and link (jal) instruction: MIPS instruction set name for a function call, which puts the return address (the link) into **ra**.

k0 and k1 registers: Two general-purpose registers that are reserved, by convention, for the use of trap handlers. It is difficult to contrive a trap handler that does not trash at least one register.

kernel: The smallest separately compiled unit of an operating system that contains task scheduling functions. Some OSs (like Linux) are monolithic, with big kernels that do a lot; some are modular, with small kernels surrounded by helper tasks.

kernel privilege: For a protected CPU, a state where it's allowed to do anything. That's usually how it boots up; and in small systems or simple operating systems, that's how it stays.

Kernighan, Brian: Coauthor (with Denis Ritchie) of the *C Programming Handbook*, and generally held responsible for systematizing the C language. No programmer should ever read another book about C.

kludge: An engineer's derogatory expression for a quick-and-dirty fix.

kseg0, kseg1: The unmapped address spaces (actually, they are mapped in the sense that the resulting physical addresses are in the low 512 MB). kseg0 is for cached references and kseg1 for uncached references. Standalone programs, or programs using simple OSs, are likely to run wholly in kseg0/kseg1.

KSU, KU: The kernel/user privilege field in the status register (described in section 3.3.)

kuseg: The low half of the MIPS program address space, which is accessible by programs running with user privileges and always translated (in CPUs equipped with a TLB). See Figure 2.1.

L1, L2, L3 cache: Alternative names for primary, secondary, and tertiary caches, respectively.

latency: The delay attributable to some unit or other. Memory read latency is the time taken for memory to deliver some data and is generally a much more important (and more neglected) parameter than bandwidth.

leaf function: A function that itself contains no other function call. This kind of function can return directly to the **ra** register and typically uses no stack space.

level sensitive: An attribute of a signal (particularly an interrupt signal). MIPS interrupt inputs are level sensitive; they will cause an interrupt any time they are active and unmasked.

library: See *object code library*.

line size: See *cache line size*.

linker, link-loader: A program that joins together separately compiled object code modules, resolving external references.

Linux: Properly this is the name of the OS kernel of the “GNU/Linux” operating system. More in Chapter 13.

little-endian: An architecture where the least significant part of a multibyte integer is stored at the lowest byte address; see section 10.2.

ll, load-linked: MIPS32/64's load-linked, store-conditional instruction pair provides an efficient way of building *atomic operations* and other OS building blocks for both multiprocessor and uniprocessor systems. See section 8.5.5.

LLAddr register: A CPU control (coprocessor 0) register in R4000 and later CPUs, with no discernible software use outside diagnostics. It holds an address from a previous load-linked (**ll**) instruction.

lo, hi registers: Dedicated output registers of the integer multiply/divide unit. These registers are interlocked—an attempt to copy data from them into a general-purpose register will be stalled until the multiply/divide can complete.

load delay slot: In the very first commercial MIPS CPUs, the load delay was not just a penalty, it was a rule: The hardware did not interlock the load value, and it was software's responsibility never to use load data in the immediately following instruction.

The instruction position just after a load was called the load delay slot by analogy with the branch delay slot (the latter, though, is and has always been part of the architecture).

The compiler, assembler, or programmer may move code around to try to make best use of load delay slots, but in the old CPUs sometimes you just had to put a no-op there.

load/store architecture: Describes an ISA like MIPS, where memory data can be accessed only by explicit load and store instructions. Many other architectures define instructions (e.g., a stack push, or arithmetic on a memory variable) that implicitly access memory.

load-to-use penalty, load delay: In most MIPS CPUs, it's impossible to provide load data early enough that it can be used without delay by the next instruction in sequence (even on a cache hit, there's typically at least a one-instruction pause if you try to do that).

The delay in CPU cycles that results from using a loaded value immediately is called the load-to-use delay. Folklore suggests that a load-to-use delay of one cycle is relatively harmless, two is troublesome, and more than two is probably a mistake.

loader: A program that takes an object code module and assigns fixed program addresses to instructions and data in order to make an executable file.

local variable: A named data item accessible only within the module currently being compiled/assembled.

locality of reference: The tendency of programs to focus a large number of memory references on a small subset of memory locations (at least in the short term). It's what makes caches useful.

logic analyzer: A piece of test equipment that simultaneously monitors the logic level (i.e., as 1 or 0) of many signals. It is often used to keep a list of the addresses of accesses made by a microprocessor.

long: C's extra-precision integer; for programs compatible with 32-bit MIPS CPUs, it is 32 bits in size, just like an `int`. For 64-bit MIPS programs it may well be 64 bits, but it will generally be the same length as a pointer.

loop unrolling: An optimization used by advanced compilers. Program loops are compiled to code that can implement several iterations of the loop without branching out of line. This can be particularly advantageous on implementations (rare for MIPS) where a long pipeline and instruction prefetching make taking branches relatively costly. Even on the MIPS architecture, however, loop unrolling can help. By intermingling code from different loop iterations, you can improve instruction scheduling.

LRU (least recently used): The optimal replacement algorithm to use when maintaining a cache is to recycle the space of the least recently used entry. Maintaining LRU state is complex for large sets, though, so strict LRU is rarely used for sets of more than four items.

LS: Least significant.

LSI: LSI Logic Corporation, which makes MIPS CPUs—these days, mostly as ASIC core components to be integrated by their customers into systems on a chip.

macro: A “word” in a computer language that will be replaced by some predefined textual substitution before compilation/assembly. More specifically, it's something defined in a C preprocessor `#define` statement.

madd: See *multiply-add*.

mainframe: An old name for a large and expensive computer, rich in storage, used for data-intensive applications.

mantissa: Part of the representation of a floating-point number. (Also called *fraction* or *fractional part*.) See Chapter 7.

mapped: Term used to describe a range of addresses generated by a program that will be translated in some nontrivial way before appearing as physical addresses.

mask: A bitfield used to select part of a data structure with a bitwise logical “and” operation.

MDMX ASE: An extension to MIPS32/64 (originally older than those). MDMX uses the FP registers to represent small arrays of integers (of length 8 or 16 bits) and provides arithmetic- and graphics-oriented operations that do the same thing simultaneously to all the integers in the array. It's somewhat like x86's MMX extension.

This kind of operation is thought to be useful for accelerating common tasks in audio and video processing (multimedia). But these days, you're probably more likely to see the *DSP ASE*, which supports the same kind of applications but uses GP registers.

`memcpy()`: A function from the standard C library for copying blocks of data.

memory barrier: See **sync** entry.

micro-TLB, uTLB: Old name for the *I-TLB*.

microcode: Many CPUs consist of a low-level core (the microengine) programmed with a wide, obscure machine language (microcode). Instructions from the official ISA are implemented by microcode subroutines.

During the 1970s, microcode was the favored way of managing the complexity of a CPU design. As better design tools were developed in the 1980s, particularly better circuit simulators, it became possible to go back to implementing ISA operations directly in hardware. But many CPUs (particularly CISCs) still use microcode for complicated or obscure instructions.

minicomputer: A computer “the size of an icebox,” from the 1970s or 1980s. Such computers could be sold at relatively low prices in the 1970s, within the budget of individual university departments or small businesses. First UNIX and then the whole open source movement started on minicomputers, particularly DEC's PDP-11.

MiniRISC: An LSI Logic trade name for a series of MIPS CPU cores optimized for small size.

Minix: A small and simple OS that retains some of UNIX's features, written by Andrew Tannenbaum. It's been most influential as an OS that can be taught in introductory OS courses. Notably, Linux was written by Linus Torvalds out of frustration at Minix's limited ambition.

MIPS: In *See MIPS Run*, we use this as the name of the architecture. MIPS is a registered trademark in the U.S. and other countries for architectures, synthesizable cores, hardware and software created by MIPS Technologies, Inc.

MIPS/ABI: The latest standard for MIPS applications, supported by all UNIX system vendors using the MIPS architecture in big-endian form.

MIPS Computer Systems, Inc.: The organization that initially commercialized and promoted the MIPS architecture. Sometimes sloppily used to include its successor, the MIPS Technologies group within Silicon Graphics.

MIPS silicon vendor: Any company that has built and sold MIPS CPUs or components containing MIPS CPUs of somewhat original design (that is, excluding those who have only incorporated someone else's core CPU). The roll call includes LSI Logic, IDT, Performance Semiconductor, NEC, Siemens, Toshiba, NKK, Philips, QED, Sony, SGI, PMC-Sierra, SiByte/Broadcom,

Alchemy/AMD, Cavium, and Raza. No other CPU architecture could produce a list of anything like this length.

MIPS System VR3, RISC/OS: These are all ways of referring to the same basic operating system, a derivative of UNIX System V Release 3. These ports are also one of the roots of *Irix*.

MIPS Technologies, Inc.: The MIPS CPU core company, authors of the MIPS32 and MIPS64 architectures, and guardian of all versions of the MIPS architecture from 1999 to date. MIPS Technologies was formed by a demerger of SGI's MIPS CPU operation.

After a brief flirtation with high-end “hard core” MIPS64 designs, MIPS Technologies has focused on a range of *synthesizable* core designs. These cores are now found in a wide range of embedded systems.

MIPS UMIPS 4.3BSD: MIPS Computers' first operating system, a port of Berkeley's BSD4.3 version of UNIX. It's historically important, since some MIPS architecture features are oriented specifically to BSD's requirements.

MIPSEB, MIPSEL: These are the words you use to request big-endian and little-endian output (respectively) from most MIPS compiler toolchains.

misaligned: Unaligned.

MMU (memory-management unit): The only memory-management hardware provided in the MIPS architecture is the *TLB*, which can translate program addresses from any of up to 64 pages into physical addresses.

MS: Most significant.

multiply-add, multiply-accumulate: A single instruction that multiplies two numbers together and then performs an addition sum. It's a multiply-accumulate if there is a dedicated (perhaps oversized) accumulator for the value, and a multiply-add if the final result can be put in a general-purpose register.

Such instructions are often a powerful and effective way of encoding numerical algorithms, particularly for floating point. Not all pre-MIPS32 CPUs implement integer multiply-accumulate; those that do are mostly compatible with MIPS32/64's definition.

MIPS32/64 CPUs with floating-point hardware have a full set of floating-point multiply-add instructions.

The *DSP ASE* adds some specialized multiply-accumulate instructions.

multiprocessor: A system with multiple processing elements; in practice we'll use it only as a synonym for *SMP*.

multitasking: Describes an OS that supports multiple *threads* of control. At the most mundane level, a thread is characterized by a stack and a next-instruction address, so there needs to be some scheduler in the OS that picks which task to run next and makes sure that all tasks make progress.

multithreading hardware: A CPU that can run instructions from multiple threads in hardware (simultaneously, or at least thoroughly overlapped in the pipeline). The MIPS32/64 MT ASE specifies a way of doing this for MIPS and is described in Appendix A.

NaN (not a number): A special floating-point value defined by IEEE 754 as the value to be returned from operations presented with illegal operands.

naturally aligned: A datum of length n is naturally aligned if its base address is zero mod n . A word is naturally aligned if it is on a four-byte boundary; a halfword is naturally aligned if it is on a two-byte boundary; see also *alignment*.

NEC: The electronics giant was a leading supplier of MIPS CPU chips in the 1990s.

nested exception/interrupt: What happens when you get a MIPS exception while still executing the exception handler from the last one. This is sometimes OK.

NKK: The semiconductor division of a large Japanese trading company, NKK produced some MIPS CPUs (mostly as a second source for *IDT*) during the early 1990s.

NMI (nonmaskable interrupt): Available (both as an input signal and as an event) on R4000 and subsequent components. On MIPS CPUs, it's not quite clear whether it's a nonmaskable interrupt or a very soft reset; there's no real difference.

noat, nomacro, noreorder: Assembly language controls, which provide the programmer with a way of disabling some more complicated things the assembler does and that are not always welcome (the corresponding names without the "no" switch the features back on again).

.set noat prevents the assembler from translating assembly code into binary sequences relying on the **at/\$1** register.

.set nomacro prevents the assembler from translating a single assembly statement into more than one instruction.

.set noreorder prevents the assembler from juggling the code sequence to move useful instructions into the branch delay slot.

nonblocking: An operation that you might expect to stop execution until it completes, but that somehow avoids doing so. So a nonblocking load is one where the CPU runs on past the load instruction, with the load data retrieved in the background. The load data is destined to return to a register, and access to that register for future instructions is interlocked, so the program will block on the first use of the loaded data, unless it's already arrived by then.

nonleaf function: A function that somewhere calls another function. Normally, the compiler will translate them with a function prologue, which saves

the return address (and possibly other register values) on a stack, and a function epilogue, which restores these values.

nonvolatile memory: Applied to any memory technology that retains data with the system power off.

nop, no-op: No operation. On MIPS **nop** is actually an alias for **sllv zero, zero, zero**, which doesn't have much effect; its binary code is all zeros.

normalize: The action of converting a floating-point value to the normalized form by shifting the mantissa and modifying the exponent. The IEEE standard for all except very small numbers is a normalized representation.

nullified: Applied to an instruction that, although it has been issued and will continue through the pipeline, will not be allowed to have any effect—its write-back is suppressed and it's not allowed to cause an exception. Where the instruction is one that “never should have been executed” (perhaps it was a result of incorrect *speculation*), this is sometimes referred to as a “pipeline bubble.”

In general, instructions never have any irrevocable effect until late in the pipeline, usually somewhere around where load data from a cache hit should arrive. In early simple-pipeline CPUs, instructions are only nullified when they follow an instruction that gets an exception. Later CPUs use the trick more widely—for example, to implement the “likely” variants of branch instructions.

NVRAM: Nonvolatile RAM, used rather generically to refer to any writable storage that is preserved while the system is powered down.

objdump: Typical name for a utility program that decodes and prints information from an object file.

object code: A special file format containing compiled program source and data in a form that can be quickly and easily converted to executable format.

object code library: A file holding several (separately compiled) modules of object code, together with an index showing which public function or variable names are exported by each module. The system linker can accept libraries as well as simple object modules, and will link only those modules from the library that are required to satisfy external references from the supplied modules.

octal: Base 8 notation for binary numbers, traditionally written with a leading zero. In fact, an integer written with a leading zero will most likely be interpreted as octal by the assembler.

OOO, out-of-order: An implementation technique for CPUs. OOO CPUs fetch instructions in order, but instruction execution proceeds (so long as critical queues don't fill up) in data-flow order, as operands become available. To provide the correct, sequential, semantics such CPUs must ensure that completed instructions retire in order and that all side effects of any instructions in flight can be undone.

That sounds very complicated, but data-flow order allows many more instructions to be executed in parallel, and the resulting performance gains are so high that this technique is universally used by very high end CPUs. OOO implementations for the embedded or synthesizable core CPU space have not happened yet, but it seems inevitable that they will. The MIPS architecture's clean register-to-register organization and lack of condition codes make OOO a little easier.

op-code: The field of the binary representation of an instruction that is constant for a given instruction mnemonic, excluding register selectors, embedded constants, and so on.

operand: A value used by an operation.

optimizer: The part of a compiler that transforms one correct representation of a program into a different equivalent representation that (it is to be hoped) is either smaller or likely to run faster.

OS: Operating system.

overflow: When the result of an operation is too big to be represented in the output format.

padding: Spaces left in memory data structures and representations that are caused by the compiler's need to align data to the boundaries the hardware prefers.

page: A chunk of memory, usually some power-of-two bytes in size and naturally aligned, that is the unit of memory handled by the address translation system. Most MIPS operating systems deal in 4-KB fixed-size pages, but the hardware is sometimes capable of mixing translations with a number of different page sizes.

page color: The color of a page is the value of 1 to 4 low bits of the virtual page address. In a virtually indexed, physically tagged cache (common in MIPS CPUs) data with the same physical address but whose virtual addresses have different colors will use different cache locations, producing a *cache alias*.

However, aliases can be avoided if all mappings to the same physical address have the same color. In many of the most common situations in which you have multiple mappings, it is fairly natural for the OS to maintain the same page color. Unfortunately, in some more obscure situations, it's more difficult.

page fault: An OS term meaning an event where a program accesses a location in a page for which there is no valid physical page translation assigned; in such an OS a page fault is resolved by fetching the appropriate contents, allocating physical memory, setting up the translation, and restarting the program at the offending instruction.

page table: A typical OS's TLB miss exception handler keeps a large number of page translations (more or less in ready-to-use TLB entry format) indexed by the high-order bits of the virtual address. Such a structure is called a page table.

paged: A memory management system (such as MIPS) where fixed-size pages (in MIPS they are 4 KB in size) are mapped; high bits are translated while the low bits (11 bits for MIPS) are passed through unchanged.

PageMask register: Register used in the MIPS memory management system; see Chapter 6.

parameter: When talking about subroutines, some programmers talk about passing parameters to subroutines and some (following the C programming manual) talk about passing arguments to functions. They're talking about the same thing.

parity: The simplest error check. A redundant "check bit" is added to a byte or other data item and set so that the total number of 1 bits (including the parity bit) is made odd (odd parity) or even (even parity).

partial-word, partial-doubleword: A piece of data less than a whole word or doubleword, but one that the hardware can transfer as a unit. In some MIPS CPUs this can be between one and seven bytes.

Patterson, David: From the MIPS point of view, he is Professor Hennessy's sidekick and coauthor (see Hennessy and Patterson). Outside the MIPS field, David Patterson is probably just as famous, having led the Berkeley RISC project from which the SPARC descended.

PC (program counter): Shorthand for the address of the instruction currently being executed by a CPU.

PC relative: An instruction is PC relative if it uses an address that is encoded as an offset from the instruction's own location. PC-relative branches within modules are convenient, because they need no fixing when the entire module is shifted in memory; this is a step toward *PIC* (full position-independent code).

PCI: I/O bus invented for PCs about 1993 and now a universal way of gluing I/O controllers to computers.

PDP-11: The world's favorite minicomputer in the 1970s, made by DEC. It was vastly influential, because good design decisions and superb documentation made it the best thing for programmers to play with. In particular, Ken Thompson played with it and made UNIX, and history.

PDtrace: An extension of the *EJTAG* debug unit that can collect trace information, a sequence of addresses in a compressed form that allows external software to reconstruct the whole execution path of a program. It's a challenge to store a reasonable amount of trace information in real time. See section 12.3.

peephole optimization: A form of optimization that recognizes particular patterns of instruction sequence and replaces them with shorter, simpler patterns. Peephole optimizations are not terribly important for RISCs, but they are very important to CISCs, where they provide the principal mechanism by which compilers can exploit complex instructions.

PFN (physical frame number): The high-order part of the physical address, which is the output of the paged MMU.

Philips: The diverse European electronic giant has designed and made its own MIPS cores and continues to use MIPS cores.

physical address: The address that appears on the outer pins of your CPU and that is passed on to main memory and the I/O system. Not the same as the program address (virtual address).

physical cache: Short for “cache that is physically indexed and physically tagged,” meaning that the physical (translated) address is used for both these functions.

PIC: See *position-independent code*.

pipeline: The critical architectural feature by which several instructions make progress in parallel; see section 1.1.

pipeline concealment by assembly: MIPS assembly language does not usually require the programmer to take account of the pipeline, even though the machine language does. The assembler moves code around or inserts **nops** to prevent unwanted behavior.

pipeline hazard: See *hazard*.

pipeline stall: See *stall*.

pipestage: Specifically, one of the (often five) phases of the MIPS pipeline. More generally, in any pipelined design a pipestage is the logic that affects an instruction between any two clock-edge boundaries.

pixie, pixprof: Historically influential *profiling* tools provided with early MIPS UNIX systems. Used to measure and present the instruction-by-instruction behavior of programs at high speed. Pixie works by translating the original program binary into a version that includes metering instructions that count the number of times each basic block is executed (a basic block is a section of code delimited by branches and/or branch targets).

pixprof takes the huge indigestible array of counts produced by a pixie run and munches them down into useful statistics. One day, perhaps, these tools or similar ones will be available with other toolkits.

PlayStation: Sony’s 1995 games machine, driven by a 32-bit MIPS microprocessor.

PMC-Sierra: PMC-Sierra is a broad-range networking device company. It acquired MIPS CPU specialists QED in 2000 and have since built a number of MIPS products. The QED-derived line of high-performance custom CPUs continued to the dual-64-bit R9000X2 CPU, and further products use CPU cores from MIPS Technologies.

porting/portability/portable: Adapting a program designed to work on one computer to work on another. A readily ported program is portable, and you can rate programs according to their portability.

position-independent code (PIC): Code that can execute correctly regardless of where it is positioned in program address space—notably, this is required by Linux/MIPS applications. See Chapter 16.

A weaker form of PIC can be produced by simply making sure all references are *PC relative*.

POSIX: An IEEE standard for the programming interface provided by a compliant operating system. The whole standard is somewhat cumbersome, but some subsets (such as the multithreading standard POSIX 1003, known as PThreads) have become de facto standards.

PostScript: A computing language as well as a digital way of representing a printed page. A truly brilliant idea, originally from Xerox Parc, which failed to take over the world mostly because Adobe Systems, Inc., thought it would make more money by keeping it out of the mass market.

pragma: The C compiler `#pragma` directive is used to select compiler options from within the source code. It's blessed by the ANSI C standard, but it is ugly and inflexible: GCC tends to evolve so that useful options become language extensions.

precise exception: Following an exception, all instructions earlier in instruction sequence than the instruction referenced by **EPC** are completed, whereas all instructions later in instruction sequence appear never to have happened. The MIPS architecture offers precise exceptions.

precision: (Of a data type), the number of bits available for data representation.

preemption: Preemption is when you stop what you're doing because something more important has come up. In a computer system, the ultimate cause of "something important coming up" must be an interrupt, so a preemptive OS is one that permits a full reschedule on any interrupt. Old versions of Linux (but none since 2.6) were not preemptive: If an interrupt happened while code was running in the kernel, no consequent reschedule would be done until the interrupted code paused voluntarily or returned voluntarily to user mode.

prefetch: An operation that initiates a fetch of the addressed data into a cache, but without delaying the running program. If the programmer can arrange to

do prefetches some way in advance of when the data is really loaded, performance may be considerably improved.

preprocessor: See *C preprocessor*.

PRId register: CPU control register (read-only) that tells you the type and revision number of your CPU. You shouldn't rely on it for much.

primary (L1) cache: In a system with more than one level of cache, this is the cache closest to the CPU.

privilege level: CPUs capable of running a secure OS must be able to operate at two different privilege levels. All known MIPS OSs use just two: kernel and user. (The *supervisor*-privilege level is universally ignored.)

User-privilege programs are not allowed to interfere with each other or with the privileged kernel programs; the privileged programs have just got to work properly.

privilege violation: A program trying to do what it's not allowed to, which will cause an exception. The OS must then decide which punishment to mete out.

probe, debug: See *debug probe*.

process: A UNIX word for the chunk of computation that corresponds to a word on the command line or a single application; it combines a thread of control, a program binary to run, and its own address space in which it can run safely.

profiling: Running a program with some kind of instrumentation to derive information about its resource usage and running. That instrumentation can be pure software, obtained by a periodic interrupt with software assistance, or pure hardware (as in *PDtrace*, see above).

program address: The software engineer's view of addresses, as generated by the program. Also known as virtual address.

prologue: See *function prologue*.

PROM (programmable read-only memory): Used sloppily to mean any read-only program memory. Originally it meant those that are programmable after manufacture, but hard-programmed ("mask") ROMs are now rare in systems of a size that might use MIPS CPUs.

protected OS: An operating system that runs tasks at a low privilege level, where they can be prevented from doing destructive things.

PTE: A page table entry. On MIPS that's the software-maintained data used to construct an entry for the hardware TLB.

PTEBase: Part of the MIPS **Context** or **XContext** registers and typically loaded with a pointer to an in-memory page table of translations ready to be loaded into the TLB.

PThreads: See *POSIX*.

QED: Quantum Effect Devices, Inc., the most prolific MIPS CPU design group of the 1990s.

quad-precision (128-bit) floating point: Not supported by MIPS hardware, but defined in the n64 ABI and implemented in software by some compilers. In C, this is called a `long double`.

R2000, R3000: The first commercial MIPS CPUs, packaged to use external static RAMs as L1 cache.

ra register: CPU register `$31`, conventionally used for the return address from subroutines. This use is supported by the ISA, in that it is used by the `jal` instruction (whose 26-bit target address field leaves it no room to specify which register should receive the return address value).

RAM (random access memory): Computer memory that can be both read and written. See *ROM*.

Random register: A CPU control register present only if there is a TLB. It increments continually and autonomously and is used for pseudorandom replacement of TLB entries.

Raza Microelectronics, Inc.: A relative newcomer to MIPS companies, Raza makes the XLR family of multithreaded, multicore MIPS64 CPUs for near-network applications, which are probably the highest throughput MIPS devices currently available. Definitely worth watching!

read priority: Because of the write buffer, the CPU package may simultaneously want to do a read and a (delayed) write. It is possible, and can boost performance, to do the read first. If the CPU is always waiting for the read data, the condition is called read priority. But it causes coherency problems when the location being read is affected by a pending write, so few MIPS CPUs tried it (LSI's LR33000 was an exception).

register renaming: A technique for implementing high-performance computers that permits instructions to be executed out of their normal sequence without this sequence being visible to the programmer. Used (heroically) in the MIPS R10000.

relocatable object module: A chunk of object code that still contains the necessary information and records for a program to be able to find and alter all the offsets and hidden addresses that tie the module to a particular location in memory.

relocation: The process of patching binary object code to make it runnable at a different location in memory.

renormalization: After a floating-point calculation, the number is probably no longer *normalized*. Renormalization is the process of making it so again.

reset: Used in this book for the event that happens when you activate the Reset input to the CPU; this happens at power-on or system reinitialization.

RISC (reduced instruction set computer): Generic term used in this book for a class of CPU architectures designed for easy pipelining. They were introduced in the second half of the 1980s.

RMW (read-modify-write): A frequently encountered sequence of actions on a storage location of any kind. A common fault in any system with multiple threads occurs when an RMW sequence is interrupted between the read and write, so the interrupted thread may use an out-of-date value and overwrite an interrupting thread's value. That is, an RMW sequence often needs to be *atomic*.

ROM (read-only memory): A storage device that can't be written. (More often these days, it means it can't be written in normal operation—there's often some offline or exceptional means by which it can be reprogrammed.)

rounding mode: Defines the exact behavior of floating-point operations. Configurable through the floating-point status/control register (see Chapter 7).

RTOS (real-time operating system): A much-abused acronym. Effectively, every OS simpler than Linux is described as an RTOS. A simple OS really built to help a system work to deadlines is probably called a “hard real-time” OS these days.

s0–s9 registers: A collection of CPU general-purpose registers (\$16–\$23 and \$30) conventionally used for variables of function scope. They must be saved by any function that modifies them.

sandbox: A safely fenced off set of resources (disk, filespace, memory, CPU time) within which untrusted programs can be safely run. One of the Internet's best pieces of jargon.

SandCraft Inc.: A MIPS CPU design house, founded in 1996, that worked on full-custom CPU designs. It was formed partly by a core of engineers who worked on the R4300 project at SGI. SandCraft worked on CPUs including NEC's Vr54xx family. SandCraft was acquired in 2003 by Raza Microelectronics.

saturating arithmetic: Operations where overflow is handled by generating the nearest representable value when the unlimited-precision result is outside the range of the result format. For signed operations, the truncated result will be either the most positive or most negative representable number, while for unsigned operations, a result that would be negative is replaced by zero.

For calculations that might overflow, but it's impractical to code a software check, this is a much less bad option than the “natural” wrap-around.

sc, store conditional: see *load-linked*.

scalar: A simple variable (as distinct from an array or data structure). By analogy, a CPU that operates on single chunks of data at a time is called scalar. This term was originally used to distinguish such a CPU from a vector processor, which can operate on a whole chunk at a time.

scheduler: In a multitasking system, the scheduler is the program that decides which task to run next.

SDE-MIPS: The Algorithmics toolkit for developing programs for MIPS targets, built around GNU C.

SDRAM, DDR DRAM (synchronous DRAM): Bulk memory chips with various kinds of high-bandwidth data interface based on serial data transmission along data wires.

secondary (L2) cache: In a system with more than one level of cache, this is the cache second closest to the CPU.

section: The name for the chunks used to separate the code, various kinds of data, debug information, and so on from a program and to carry them through the object code. Eventually, you get to decide where in memory each section ends up.

segment: See *kseg0*, *kseg1*.

segmentation: An obsolete approach to memory translation and protection, where program addresses are modified by being added to a base address. It was used in the x86, but it hasn't been needed since the 386.

semaphore: A powerful organizing concept for designing robustly cooperating multitasking or multiprocessing systems; see section 5.8.4.

.set, assembly language controls: See *noat*, etc.

set, cache: See *cache set*.

set-associative: See *cache*, *set-associative*.

SGI (Silicon Graphics, Inc.): Dominant supplier of MIPS-powered computers and guardians of the MIPS architecture during the 1990s. The company is insolvent at the time of writing.

short: In C, the name for an integer data type at least as big as a *char* and no larger than an *int*. In 32- and 64-bit architectures, a *short* seems always to be a 16-bit integer.

SiByte: A CPU design company that created a 64-bit MIPS CPU in the late 1990s. It was acquired by Broadcom, which still uses the SB-12xx CPUs.

signal: A kind of primitive interrupt that is fed to regular programs in a UNIX-like OS. Improved in Berkeley UNIX and codified by the POSIX working group

to represent a reasonably clean and simple way of communicating simple events in a multitasking system.

silicon vendor: In the MIPS world, one of the companies making and selling MIPS CPUs.

SIMD (single instruction multiple data): SIMD instructions repeat the same operation on multiple data items. Modern SIMD instructions typically obtain their multiple operands as “slices” of one wide register.

SmartMIPS ASE: An instruction set extension aimed at “smart cards”—very low-power CPUs whose biggest task is encryption/decryption. It adds a small number of computational instructions and some memory-management tweaks to provide more fine-grain protection for these security-orientated applications.

SMP (symmetric multiprocessing): The most successful form of large-scale multi-CPU system, in which a number of CPUs effectively sharing memory (typically through *coherent* caches) run the same OS kernel. Linux/MIPS readily runs on suitable SMP system hardware.

snooping, snoop: See *cache, snooping*.

SoC (system on a chip): A complex component built out of multiple subsystems. Increasingly, that may be multiple subsystems licensed from different components.

soft reset: In digital electronics, reset is that ubiquitous signal asserted to get everything back to a starting condition. For a CPU, it represents an instant roll of the karmic wheel—death and rebirth in a few milliseconds. Sometimes you’d rather reset your CPU in a way that allows it to remember something of its past life—that’s a “soft reset.” See section 5.9.

software instruction emulators: A program that emulates the operation of a CPU/memory system. It can be used to check out software too low level to be compatible with a debugger.

software interrupts: Interrupts invoked by setting bits in the **Cause** register; the software interrupt typically happens only when those bits are unmasked. See section 5.8.

Sony: Consumer electronics company that used MIPS chips in its PlayStations.

source-level debugger: A debugger that interprets current program state in terms of the source program (instruction lines, variable names, data structures). Source-level debuggers need access to source code, so when working with embedded system software, the debugger must run on the host and obtains its information about the program state from a simple debug monitor running on the target.

sp register/stack pointer: CPU register **\$29**, used by convention as a stack pointer.

SPARC: Sun Microsystems' RISC CPU architecture. Derived fairly directly from the University of California at Berkeley RISC project, whereas MIPS came out of Stanford University. Stanford (on the San Francisco peninsula) is private and somewhat conservative; Berkeley (across the bay) is public and radical. There's a lot of social history in microprocessor design.

sparse address space: Some OS tactics (notably, using an object's address as a long-term handle) work only if you have a much larger address space than you really need, so you can afford to spread things out thinly and allocate space recklessly as a sparse address space. No sparse-address OS has been commercially successful yet.

speculative execution: A CPU implementation technique in which the CPU executes instructions before it really knows it should (most commonly, while it's still figuring out whether or not a conditional branch should have happened). Used in high-end MIPS CPUs from R10000 on and increasingly in higher-end embedded CPUs.

spinlock: A form of low-level semaphore for use by an SMP system. Its characteristic is that a thread that is blocked on the semaphore just goes into a busy loop checking the lock variable. That makes sense when a lock is typically acquired for only a small number of cycles.

SR (Status) register: CPU status register, one of the privileged control registers. Contains control bits for any modes the CPU respects. See section 3.3 for details.

stack: The last-in, first-out data structure used to record the execution state of CPUs that are running the most interesting languages.

stack argument structure: A conceptual data structure used in section 11.2.1 to explain how arguments are passed to functions according to the MIPS convention.

stack backtrace: A debugger function that interprets the state of the program stack to show the nest of function calls that has got to the current position. Depends wholly on strict stack usage conventions, which assembly programs must notate with standard directives.

stack frame: The piece of stack used by a particular function.

stack underrun: An error that occurs when you try to pop more off a stack than was ever put on it.

stale data: Term used for data lying about that has been superseded by a more recent write. It could be data in memory where a CPU's cached copy has been updated but has not yet been written back; it could be data in a cache where the

memory contents have been replaced by a DMA device and the cache has not yet been invalidated. Using stale data is a bug.

stall: Condition in which the pipeline is frozen (no instruction state is advanced) while the CPU waits for some resource to do its thing.

standalone software: Software operating without the benefit of any kind of operating system or standard runtime environment.

Stanford: The San Francisco–area university where the MIPS academic project was run by Professor Hennessy and from where the MIPS company was born.

static variable: C terminology for a data item that has a compile-time fixed place in memory.

Status register: Another name for the **SR** register, see above.

stdarg: ANSI-approved C macro package that hides the implementation-dependent details of how to provide for functions with a variable number of arguments or arguments whose type can only be determined at run time (or both).

strcmp: C library function that compares two (null-terminated) strings.

strcpy: C library function that copies a (null-terminated) string.

supercomputer: Colloquially, a computer built for performance on numerical, compute-intensive applications, essentially without regard for cost. That's often achieved with relatively exotic architecture features such as vector floating-point instructions.

superpipelined CPU: If pipelining is a good thing, perhaps it can be made better by cranking up the clock rate and breaking down individual execution stages into smaller pieces, each of which can fit into the shrunken clock cycle—that's superpipelining.

The MIPS R4000 CPU was slightly superpipelined, breaking each of the I-fetch and D-cache access stages into two and removing half clock cycles to get an eight-stage pipeline. However, the longer pipeline leads to increases in the branch penalty (which can be mitigated by *branch prediction*) and the load-to-use penalty.

The R4000 established that over a wide range of RISC-like architectures, you shouldn't really use more than five pipeline stages unless you include branch prediction and tackle the load-to-use penalty in some way.

superscalar: A CPU implementation that can issue more than one instruction at the same time. The ideal is that enough pipeline stages should be duplicated to allow a significant number of instructions to execute alongside each other, giving you “two for the price of one.”

It turns out that, until you embrace the complexity of an *OOO* organization, superscalar CPUs offer very modest performance gains on compiled code. They

may do better with code sequences carefully tailored to the implementation by heroic programming.

But some classes of instructions (floating point, for example) are rather cheap to issue and run in parallel, so the trick is cost effective. It has, however, been more popular than it has deserved.

supervisor-privilege level: Intermediate privilege level between kernel and user. It's optional in MIPS32/64, implemented on many MIPS CPUs, but has never been used. See section 3.3.1.

swapper: See *byte-swapper*.

sync, memory synchronization barrier: An instruction that allows a programmer to indicate where the order of reads and writes in a program really matters. Any read or write preceding the **sync** instruction in program order must be carried out before any read or write following the **sync**.

synchronous logic: Means logic organized as “combinatorial” (pure logic) stages interposed between successive registers, where each register stores information on the transitions of a regular global clock signal. The compilers that can turn Verilog code into working chip logic are, for most purposes, restricted to Verilog code that maps onto synchronous logic, and that's how all known synthesizable CPUs work.

The reliable synthesis of working *asynchronous logic* is a frustratingly difficult goal, which could produce big gains in performance versus power consumption.

synthesized instructions: See *instruction synthesis by assembly*.

syscall (system call): An instruction that exists to produce an exception. The exception acts as a secure subroutine call into the kernel. The **syscall** instruction has a spare field, uninterpreted by the hardware, that software can use to encode different system call types. However, Linux doesn't use it—it prefers to distinguish system calls using a value passed in a GP register.

t0–t9 register/temporaries: CPU registers **\$8–\$15** and **\$24–\$25**, conventionally used as temporaries; any function can use these registers. The downside is that the values can't be guaranteed to survive any function call.

TagHi, TagLo registers: CP0 registers in MIPS32/64; they are staging posts for cache tag contents. See section 4.9.

TC: In a MIPS MT (multithreading) CPU, a TC is the hardware component that runs a thread. It has a PC and a set of GP registers.

temporary register: See **t0**.

tertiary (L3) cache: A third level of cache between L2 and memory.

thrashing: Collapse of a heuristic optimization characterized by a repeated cycle of failure. “Cache thrashing” is a specific case in which two locations in

frequent use by a program compete for the same cache storage, repeatedly displacing each other and making the cache ineffective. Much software effort has been expended in vain trying to construct code that is less susceptible to cache thrashing, but the problem more or less disappears with *set-associative* caches.

thread: A thread is a part of a program that runs in the sequence the programmer intended. It's the primitive component of a software *multitasking* OS, a *multiprocessor* system, or *multithreading* CPU hardware.

timer: As a facility for CPUs, a constant-rate counter with some mechanism to cause an interrupt when the counter reaches some specified value.

TLB (translation lookaside buffer): The associative store that translates program to physical page numbers. When the TLB doesn't contain the translation entry you need, the CPU takes an exception, and it is up to system software to load an appropriate entry before returning to re-execute the faulting reference. See Chapter 6.

TLB, wired entries: The first few TLB entries may be defended from the usual random replacement policy of the TLB refill handler: Just set the **wired** register above zero and the **random** register will never take a value the same or lower than **wired**. Very old MIPS CPUs lack the **wired** register and always reserved eight entries.

TLB invalid exception: The exception taken when a TLB entry matches the address but is marked as not valid.

TLB miss: The exception taken when no TLB entry matches the program address. Most TLB miss exceptions use a unique, dedicated exception entry point. This was done because this is by far the most common trap in a hard-working operating system, and it saves time to avoid the code that must work out which kind of trap has occurred.

Very old MIPS CPUs used the dedicated exception entry only for TLB misses from user mode: MIPS32/64 CPUs use it for TLB misses other than misses from exception mode (a rare but useful special case).

TLB modified exception: The exception taken when a TLB entry matches a store address, but the entry is not flagged as writable.

TLB probe, *t1bp*: An instruction used to submit a program address to the TLB to discover whether it would be translated by any existing entry.

TLB refill: The process of adding a new entry to the TLB following a miss.

toolchain, toolkit: The complete set of tools required to produce runnable programs starting from source code (compiler, assembler, linker, librarian, etc.).

Toshiba: Japanese chip maker and MIPS licensee. Toshiba has not been prominent as a supplier of CPU components for general sale, but is notable for the

“Emotion Engine” heart of the Sony PlayStation 2: That’s a 64-bit floating-point vector processor, which wouldn’t have disgraced a 1980 supercomputer.

translated address or address region: A MIPS program (virtual) address that is destined to be translated through the TLB (or to cause an error). This includes the kuseg region, where all user-privilege software must run, as well as the mapped kernel-privilege region kseg2. The 64-bit CPUs have more translated regions.

translation lookaside buffer: See *TLB*.

trap: An exception caused by some internal event affecting a particular instruction.

trunc: The floating-point instruction **trunc** rounds a floating-point number to the next integer toward zero.

TTL: An acronym for transistor-transistor logic, this is a signaling convention that enables you to decide whether an electrical signal represents 1, 0, or something in between and undefined. TTL is based on the habits of some early 5V logic families. TTL signaling has commonly been used in all microprocessor systems at least up to the late 1990s; its most likely replacement is a slight modification to fit in with 3.3V power supplies.

two-way set-associative: See *cache, set-associative*.

UART (universal asynchronous receiver/transmitter): A serial port controller.

Ultrix: DEC’s trade name for their BSD-family operating system running on MIPS-based DECstation computers. Ultrix ran in the DEC-friendly little-endian mode, making it binary incompatible with contemporary MIPS workstations and servers.

UMIPS: See *MIPS UMIPS 4.3BSD*.

unaligned access exception: Trap caused by a memory reference (load/store word or halfword) at a misaligned address.

unaligned data: Data stored in memory but not guaranteed to be on the proper alignment boundary. Unaligned data can only be accessed reliably by special code sequences, see section 8.5.1.

uncacheable: Memory areas where CPU reads and writes never search through or affect the cache. True of the region kseg1 or translated address regions where the TLB entry is flagged as uncached.

uncached: A CPU read/write that doesn’t search through or write to the cache.

underflow: What happens when a floating-point operation should produce a result that is too small to represent properly. See also *denormalized*.

unified cache: A cache that is searched and updated for all CPU accesses, both instruction fetches and data references. Essentially all MIPS CPUs use separate primary I- and D-caches, but most L2 and L3 caches are unified.

unimplemented instruction exception: Exception taken when the CPU does not recognize the instruction code; it is also used when it cannot successfully complete a floating-point instruction and wants the software emulator to take over.

union: A C declaration of an item of data that is going to have alternative interpretations with different data types. If you store data of one type and read it back as the other type, the result is highly unportable, in an interesting sort of way.

uniprocessor: A CPU that doesn't share its memory with another.

UNIX-like: A system something of the manner of real UNIX, but without any implication as to copyright or formal standardization. Includes Linux, OSs from the various OpenBSD and FreeBSD groups, and commercial OSs like Sun's Solaris or SGI's Irix.

unmapped, untranslated: Refers to the *kseg0*, *kseg1* address spaces.

unrolled loop: A loop in a program, transformed by arranging that (most of the time) the work of more than one iteration of the loop is done between jumps. It can often make programs go faster; it's sometimes done automatically by clever compilers.

user space: The space of user-privilege-accessible addresses (*kuseg*).

userland: Linux name for the parts of the system that are outside the kernel: libraries and basic application. The userland for the GNU/Linux OS comes mostly from the GNU project.

user-privilege level: The lowest privilege state for a MIPS CPU, where only the regular instruction set is usable and program addresses must stay inside *kuseg*. An operating system can prevent user-privilege programs from interfering with each other or the OS.

tlbmiss exception: A *TLB miss* from user mode. On old MIPS CPUs, the dedicated TLB miss exception entry point was only used for exceptions from user mode.

v0–v1 registers: CPU registers *\$2–\$3*, conventionally used to hold values being returned by functions.

varargs: An old but now deprecated version of *stdarg*.

VAX: DEC's groundbreaking 32-bit minicomputer architecture, definitely not a RISC. The first minicomputer to support virtual memory (hence the "V").

vector, vector processor: A processor that has instructions that perform the same operation on a whole collection of data at a time, mostly floating-point operations. This is an example of parallel processing characterized as single instruction, multiple data (SIMD); it was the first kind of parallel processing to be useful. Number-crunching supercomputers depend on vector processing for their speed.

vectorizable: A program source that is amenable to automatic loop optimization to exploit vector or other SIMD operations is called vectorizable. In practice, programs usually need to be fairly carefully written to allow that kind of optimization.

Verilog: A programming language used to describe logic designs for simulation or “synthesis” to real hardware. Most practical modern reusable logic designs are written in Verilog.

virtual address: See *program address*.

virtual memory (VM): A way of running an application without actually giving it all the memory it thinks it needs, but in such a way that it doesn’t know the difference. You do this by arranging that an attempt to access something that isn’t really there causes the operating system to be called in. The OS finds the required memory (be it code or data), changes the mapping so the application will find it, and then restarts the application at the instruction that led to the bad access. Bigger OSs (UNIX-like or modern Windows) always use virtual memory.

VMS: The operating system DEC developed for the VAX minicomputer.

void: A data type used to tidy up C programs, indicating that no value is available.

volatile: An attribute of declared data in either C or assembly. A volatile variable is one that may not simply behave like memory (i.e., does not simply return the value last stored in it). In the absence of this attribute, optimizers may assume that it is unnecessary to reread a value; and if the variable represents a memory-mapped I/O location you are polling, this will be a mistake.

VPE: In MIPS MT (multithreading), a VPE contains one or more TCs (which run the programs) and provides a complete set of CP0 registers and the other resources that make up what looks like a complete MIPS32/64-compatible CPU.

VPN (virtual page number): The part of a program (virtual) address that gets translated. The low-order bits of the program address (which is the address within a page, usually a 4-KB page) pass unchanged through to the physical address.

VxWorks: A real-time OS kernel used in embedded applications, written and sold by Wind River Systems, Inc.

WatchHi, WatchLo register: Coprocessor 0 registers that implement a data watchpoint, available in some R4000-style CPUs.

watchpoint: A debugger feature that will cause the running program to be suspended and control passed back to the user whenever an access is made to the specified address. NEC's Vr4300 CPU has one of these.

wbflush: A standard name for a routine/macro that ensures that any queued external write cycles have left the CPU.

Whitechapel: A briefly flowering UK-based UNIX workstation company that shipped the first MIPS desktop computers in 1987.

workstation: Used here to mean a desktop computer running a UNIX-like OS.

wraparound: Some memory systems (including the MIPS cache when isolated) have a property that accesses beyond the memory array size; they simply wrap around and start accessing the memory again at the beginning.

write buffer: A FIFO store that keeps both the address and data of a CPU write cycle (usually up to four of each). The CPU can continue execution while the writes are carried out as fast as the memory system will manage. A write buffer is particularly effective when used with a write-through cache.

write-back cache: See *cache*, *write-back*.

write-through cache: See *cache*, *write-through*.

XContext register: Coprocessor 0 register associated with the TLB (memory management hardware). Provides a fast way to process TLB misses on systems using a certain arrangement of page tables for 64-bit-addressed virtual memory regions.

zero register: CPU register \$0, which is very special: Regardless of what is written to it, it always returns the value zero.

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Books and Articles

Cohen, D. “On Holy Wars and a Plea for Peace.” *USC/ISI IEN* 137, April 1, 1980. To find this paper online look up “endianness” in Wikipedia (www.wikipedia.org).

Heinrich, J. *MIPS R4000 User's Manual*. Englewood Cliffs, NJ: Prentice Hall, 1993.

The bible of the MIPS III architecture; lots of details, though this can make it difficult to locate information. It also takes a rather strict view in covering only those areas that are truly general to the MIPS architecture, omitting anything implementation-specific. At the time of this writing, the book is available for download as a PDF document (see the list of online resources below).

Hennessy, J., and D. Patterson. *Computer Architecture: A Quantitative Approach*. 3rd edition. San Francisco: Morgan Kaufmann, 2002.

Outside the MIPS-specific field, this is the only book worth having on modern computer architecture. Its sole defect is its size, but in this case it's worth it. Older editions still do a fine job of presenting the essential concepts, and used copies are likely to be available at reasonable prices; but if you can justify the cost of the third edition, you'll find the updates well worthwhile. Before you buy, check for availability of subsequent editions.

Kernighan, B., and D. Richie. *The C Programming Language*. 2nd edition. Englewood Cliffs, NJ: Prentice Hall, 1988.

This is the book to have if you want to learn more about C. You should probably get the updated ANSI edition now, although regrettably it is somewhat fatter.

Love, R. *Linux Kernel Development*. Carmel, IN: Sams Publishing, 2003.

A well-written and modestly sized high-level guide to the Linux kernel, readable both with and without the source code. It's good at explaining why things are done as they are.

Rosenberg, J. *How Debuggers Work: Algorithms, Data Structures, and Architecture*. New York; John Wiley & Sons, 1996.

Unless you're developing debug tools, you probably won't need to understand them to the level covered in this book; all the same, it can help you to become a more effective user of the tools you have.

Sweazey, P., and A. J. Smith. "A Class of Compatible Cache-Consistency Protocols and Their Support by the IEEE Future Bus." *Proceedings of the 13th International Symposium on Computer Architecture*, 1986.

Tanenbaum, A., and A. S. Woodhull. *Operating Systems Design and Implementation*. 3rd edition. Englewood Cliffs, NJ: Prentice Hall, 2006.

This book is useful as an introduction to operating systems. The Minix operating system described in the book is also of historical significance, having provided (in an earlier version) the initial impetus for the creation of Linux.

Online Resources

The URLs noted below have all been checked and confirmed to be valid as this book goes to press, but as with any printed information about the Web, they're likely to change. If you're unable to access a URL, use your favorite search engine to find out whether it may have moved and also to check for new sources of information about MIPS and Linux that may have appeared since this book went to press.

The MIPS architecture has been around for quite a long time, and you may find other useful reading by looking around the Web.

Advanced Micro Devices (AMD): The Web site at www.amd.com has information about the Au1000 through Au1550 MIPS processors, originally designed by Alchemy Semiconductor before they were acquired by AMD. Search Google for `alchemy site:amd.com` to find the right page.

Broadcom Corporation: The Web site at www.broadcom.com has information about the BCM series MIPS processors designed by SiByte before they were bought out by Broadcom. Search Google for `'communications processors' site:broadcom.com` to locate the relevant pages.

GNU C compiler has a home page at gcc.gnu.org. Online manuals for current and older versions are available at gcc.gnu.org/onlinedocs.

Linux MIPS port: The group responsible for this maintains a homepage at www.linux-mips.org. This Web site always offers the most up-to-date source code for the MIPS port of the Linux kernel; furthermore, there's helpful supporting information about porting and building the kernel. The Web site can also help you find out more about processors, systems, toolchains, and so forth, and has sections containing historical and background information.

MIPS Technologies: The guardians of the MIPS architecture make their living from licensing core CPUs and are on the Web at www.mips.com. You'll find numerous links to other companies involved in MIPS.

PMC-Sierra Inc.: The Web site at www.pmc-sierra.com has information about PMC-Sierra's MIPS devices, including many of the processor designs originally created by QED Inc., which was subsequently acquired by PMC-Sierra.

The Linux Devices Web site: This is at www.linuxdevices.com and offers a range of news and information about the use of Linux in embedded systems.

The Linux Kernel Archives: These are at www.kernel.org. You can always download the source code for the latest version of the Linux kernel here (this Web site serves all architectures for which ports exist). Recent versions are usually preserved here too, and you may find this helpful if the latest version hasn't yet been ported to your hardware.

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